Professional Data Engineer Questions

[202610]

Question #1 Topic 1

Your company built a TensorFlow neutral-network model with a large number of neurons and layers. The model fits well for the training data. However, when tested against new data, it performs poorly. What method can you employ to address this?

A. Threading

B. Serialization

C. Dropout Methods **Most Voted**

D. Dimensionality Reduction

Correct Answer: C

Community vote distribution

C (100%)

Comments

henriksoder24 **Highly Voted** 3 years, 7 months ago

Answer is C. Bad performance of a model is either due to lack of relationship between dependent and independent variables used, or just overfit due to having used too many features and/or bad features. A: Threading parallelisation can reduce training time, but if the selected featuers are the same then the resulting performance won't have changed B: Serialization is only changing data into byte streams. This won't be useful. C: This can show which features are bad. E.g. if it is one feature causing bad performance, then the dropout method will show it, so you can remove it from the model and retrain it. D: This would become clear if the model did not fit the training data well. But the question says that the model fits the training data well, so D is not the answer.

samdhimal **Highly Voted** 3 years, 2 months ago

C. Dropout Methods Dropout is a regularization technique that can be used to prevent overfitting of the model to the training data. It works by randomly dropping out a certain percentage of neurons during training, which helps to reduce the complexity of the model and prevent it from memorizing the training data. This can improve the model's ability to generalize to new data and reduce the risk of poor performance when tested against new data.

samdhimal 3 years, 2 months ago

A. Threading: it's not a method to address overfitting, it's a technique to improve the performance of the model by parallelizing the computations using multiple threads. B. Serialization: it's a technique to save the model's architecture and trained parameters to a file, it's helpful when you want to reuse the model later, but it doesn't address overfitting problem. D. Dimensionality Reduction: it's a technique that can be used to reduce the number of features in the data, it's helpful when the data contains redundant or irrelevant features, but it doesn't address overfitting problem directly.

3244fd8 5 months, 2 weeks ago

Selected Answer: C

The model fits the training data well but performs poorly on new data → this is overfitting, especially with a large/deep network. Dropout is a regularization technique that randomly deactivates neurons during training, reduces co-adaptation, and improves generalization. Threading only affects runtime, Serialization is for saving models, and Dimensionality Reduction isn't the primary fix here since the model already fits the training set well. Therefore: C. Dropout Methods.

gunnerski 6 months, 3 weeks ago

Selected Answer: C

Only option which fix overfitting

israndroid 7 months ago

Selected Answer: C

Answer is C. For Neural Networks Dropout method of regularization thecnique, helps to prevent the overfit of the model, the main idea is remove/turn/off random % of neurons at training.

MikeFR 10 months, 1 week ago

Selected Answer: C

Droput method

su787 11 months, 2 weeks ago

Selected Answer: C

Dropout method

monyu 1 year ago

Selected Answer: C

Correct answer is C. A - Is not Threading because it is used to accelerate the training in order to reduce training time. B - Is not Serialization because it transforms (serializes into bytes) the training data but does not increase or change the original nature. D - Is not dimensionality reduction because the model fits the training data.

Ahamada 1 year, 1 month ago

Selected Answer: C

Dropout methods is the solution here to resolve overfitting issue

Git3 9 months, 2 weeks ago

okay, like in linear overfitting?

onschamekh 1 year, 2 months ago

Selected Answer: C

Dropout is a specific technique to prevent overfitting by randomly disabling a certain percentage of neurons during training. This helps the network avoid relying too heavily on a subset of neurons, thereby improving its ability to generalize to new data.

jithinlife 1 year, 3 months ago

Selected Answer: C

Can we expect similar questions like this in GCP exam as well?

SamuelTsch 1 year, 5 months ago

Selected Answer: C

It occurs overfitting problem. A general idea is to simplify the model. A GENERALIZATION related method should be used.

rtcpost 1 year, 6 months ago

Selected Answer: C

C. Dropout Methods Dropout is a regularization technique commonly used in neural networks to prevent overfitting. It helps improve the generalization of the model by randomly setting a fraction of the neurons to zero during each training iteration, which prevents the network from relying too heavily on specific neurons. This, in turn, can lead to better performance on new, unseen data.

rocky48 1 year, 6 months ago

Selected Answer: C

A: Threading parallelisation can reduce training time, but if the selected featuers are the same then the resulting performance won't have changed B: Serialization is only changing data into byte streams. This won't be useful. C: This can show which features are bad. E.g. if it is one feature causing bad performance, then the dropout method will show it, so you can remove it from the model and retrain it. D: This would become clear if the model did not fit the training data well. But the question says that the model fits the training data well. So, C is the answer.

trashbox 1 year, 11 months ago

Selected Answer: C

Dropout Methods are useful to prevent a TensorFlow model from overfitting

azmiozgen 2 years, 8 months ago

Selected Answer: C

Answer is C. Dropout methods are used to mitigate overfitting. Hence, it is commonly used in training phase and it's beneficial for test-time performance.

dgteixeira 2 years, 9 months ago

Selected Answer: C

Answer is C

https://zhangxuetu.online

[202610]

Question #2 Topic 1

You are building a model to make clothing recommendations. You know a user's fashion preference is likely to change over time, so you build a data pipeline to stream new data back to the model as it becomes available. How should you use this data to train the model?

A. Continuously retrain the model on just the new data.

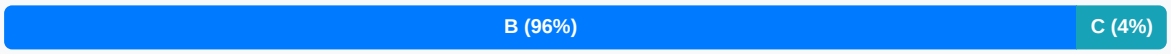
B. Continuously retrain the model on a combination of existing data and the new data. **Most Voted**

C. Train on the existing data while using the new data as your test set.

D. Train on the new data while using the existing data as your test set.

Correct Answer: B

Community vote distribution



Comments

serg3d **Highly Voted** 5 years, 10 months ago

I think it should be B because we have to use a combination of old and new test data as well as training data

dambilwa 5 years, 9 months ago

Yes - The training set should be shuffled well to represent data across all scenarios

jagadamba **Highly Voted** 5 years, 9 months ago

B, as we need to train the data with new data, so that it will keep learning, and as well as used for test

gunnerski 6 months, 3 weeks ago

Selected Answer: B

You always need to retrain on such combination

israndroid 7 months ago

Selected Answer: B

Answer is B: Fow new scenarios, we must need use old and new data for training, for continuos learning of patterns, etc.

Ahamada 1 year, 1 month ago

Selected Answer: B

Answer is B, need to take acount both data to train

cqrm3n 1 year, 2 months ago

Selected Answer: B

We should continuously retrain existing data with latest data to balance the need to adapt to new changes while not overriding historical knowledge. This is called Continuous Retraining where the model is periodically updated with latest data so that recommendations remain accurate over time.

jaimecalderon 1 year, 3 months ago

Selected Answer: B

Continuously retrain the model on a combination of existing data and the new data.

SamuelTsch 1 year, 5 months ago

Selected Answer: B

From my point of view, we should take both datasets.

rocky48 2 years, 5 months ago

Selected Answer: B

Option A is not recommended because retraining the model on just new data will cause the model to lose the information it has learned from the historical data. Option C and D are not recommended because they are using the new data as test set and this approach will lead to a model that is overfitting and not generalize well to new users. So answer is B

rajkinz 2 years, 5 months ago

Answer is C. It is time sensitive data so latest data should be used for testing. Reference: https://cloud.google.com/automl-tables/docs/prepare#ml-use

rtcpost 2 years, 5 months ago

Selected Answer: B

This approach allows the model to benefit from both the historical data (existing data) and the new data, ensuring that it adapts to changing preferences while retaining knowledge from the past. By combining both types of data, the model can learn to make recommendations that are up-to-date and relevant to users' evolving preferences.

Websurfer 2 years, 7 months ago

Selected Answer: B

train on old and new data

AmmarFasih 2 years, 10 months ago

Selected Answer: B

Option B is the right answer. Since the questions states the models needs to be updated since the clothing preference changes. Hence we need the new data to be utilized for training/ updating model.

bha11111 3 years ago

Selected Answer: B

Have verified this

jin0 3 years, 1 month ago

there are two point first when retraining second what data. I think retraining should be occur when the model could not predict well in this case there is monitoring metric should be needed first but no one said, second what data? in this case I think the answer is A. because when the model could not predict well it means the data variance and bias are changed so, it's no make sense what is combination new data with old data because the data being not be changed is not necessary anymore..

jin0 3 years, 1 month ago

And the questions should explain in detail.. whether it's deep learning or tree based machine learning model.. and how large of new dataset is.. I think

Morock 3 years, 1 month ago

Selected Answer: C

The trend keep changing, so must mix new and old data...

samdhimal 3 years, 2 months ago

Selected Answer: B

B. Continuously retrain the model on a combination of existing data and the new data. This approach will help to ensure that the model remains up-to-date with the latest fashion preferences of the users, while also leveraging the historical data to provide context and improve the accuracy of the recommendations. Retraining the model on a combination of existing and new data will help to prevent the model from being overly influenced by the new data and losing its ability to generalize to users with different preferences. Option A is not recommended because retraining the model on just new data will cause the model to lose the information it has learned from the historical data. Option C and D are not recommended because they are using the new data as test set and this approach will lead to a model that is overfitting and not generalize well to new users.

rocky48 2 years, 5 months ago

Nice explanation bro.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #3 Topic 1

You designed a database for patient records as a pilot project to cover a few hundred patients in three clinics. Your design used a single database table to represent all patients and their visits, and you used self-joins to generate reports. The server resource utilization was at 50%. Since then, the scope of the project has expanded. The database must now store 100 times more patient records. You can no longer run the reports, because they either take too long or they encounter errors with insufficient compute resources. How should you adjust the database design?

A. Add capacity (memory and disk space) to the database server by the order of 200.

B. Shard the tables into smaller ones based on date ranges, and only generate reports with prespecified date ranges.

C. Normalize the master patient-record table into the patient table and the visits table, and create other necessary tables to avoid self-join. **Most Voted**

D. Partition the table into smaller tables, with one for each clinic. Run queries against the smaller table pairs, and use unions for consolidated reports.

Correct Answer: C

Community vote distribution

C (100%)

Comments

MaxNRG **Highly Voted** 4 years, 4 months ago

C is correct because this option provides the least amount of inconvenience over using pre-specified date ranges or one table per clinic while also increasing performance due to avoiding self-joins. A is not correct because adding additional compute resources is not a recommended way to resolve database schema problems. B is not correct because this will reduce the functionality of the database and make running reports more difficult. D is not correct because this will likely increase the number of tables so much that it will be more difficult to generate reports vs. the correct option. https://cloud.google.com/bigquery/docs/best-practices-performance-patterns https://cloud.google.com/bigquery/docs/reference/standard-sql/query-syntax#explicit-alias-visibility

gord_nat 1 year, 3 months ago

Why are we assuming the database in question is BigQuery? There are several other RDBMS options in GCP. .. Also, why was the original db pushed to prod without being normalized first? Typically, the normalized db is released to prod. When the data set becomes larger, you would add partitioning. The scenario being presented is unrealistic.

balseron99 **Highly Voted** 5 years, 2 months ago

A is incorrect because adding space won't solve the problem of query performance. B is incorrect because there is nothing related to the report generation which is specified and sharding tables on date ranges is not a good option as it will create many tables. C is CORRECT because the statement says "the scope of the project has expanded. The database must now store 100 times more patient records". As the data increases there would be difficulty in managing the tables and querying it. Hence creating different table is correct as per the need. D is Incorrect as it Partitions on each clinic. We have to adjust the database design so that it performs optimally when generating reports. Also nothing is specified for generation of reports in the required statement.

Nanto90 8 months, 4 weeks ago

Selected Answer: C

Because you can optimize the storage of each table; furthermore, you will avoid self-joins.

Ahamada 1 year, 1 month ago

Selected Answer: C

answer is C, the problem here is the self-join (avoid self-join if possible) on a Denormalized table. So the solution is to Normalize

cqrm3n 1 year, 2 months ago

Selected Answer: C

Normalizing the database into separate Patients and Visits tables, along with creating other necessary tables, is the best solution for handling the increased data size while ensuring efficient query performance and maintainability. This approach addresses the root problem instead of applying temporary fixes.

SamuelTsch 1 year, 5 months ago

Selected Answer: C

C is the most suitable solution for this situation. It provides a better way for scalability and monitoring. B has a constraint on predefined date range, which is usually not suitable for reporting.

rocky48 2 years, 5 months ago

Selected Answer: C

Normalization is a technique used to organize data in a relational database to reduce data redundancy and improve data integrity. Breaking the patient records into separate tables (patient and visits) and eliminating self-joins will make the database more scalable and improve query performance. It also helps maintain data integrity and makes it easier to manage large datasets efficiently. Options A, B, and D may provide some benefits in specific cases, but for a scenario where the project scope has expanded significantly and there are performance issues with self-joins, normalization (Option C) is the most robust and scalable solution.

rtcpost 2 years, 5 months ago

Selected Answer: C

Normalization is a technique used to organize data in a relational database to reduce data redundancy and improve data integrity. Breaking the patient records into separate tables (patient and visits) and eliminating self-joins will make the database more scalable and improve query performance. It also helps maintain data integrity and makes it easier to manage large datasets efficiently. Options A, B, and D may provide some benefits in specific cases, but for a scenario where the project scope has expanded significantly and there are performance issues with self-joins, normalization (Option C) is the most robust and scalable solution.

vaga1 2 years, 10 months ago

Selected Answer: C

"100 times more patient records"immediately brings to create a patient dimensional table to save space on disk if a generical relational database is mentioned.

maurilio_cardoso_multiedro 3 years ago

C - https://cloud.google.com/bigquery/docs/best-practices-performance-patterns

bha11111 3 years ago

Selected Answer: C

C- This is correct have verified from different sources

Morock 3 years, 1 month ago

Selected Answer: C

Should be C. Basic ER design...

GCPpro 3 years, 2 months ago

c - is the correct one.

testoneAZ 3 years, 2 months ago

C should be the correct answer

Brillianttyagi 3 years, 3 months ago

Selected Answer: C

C- Is the correct answer!

Arkon88 4 years, 1 month ago

Selected Answer: C

C - based on Google documentation, self-join is an anti-pattern: https://cloud.google.com/bigquery/docs/best-practices-performance-patterns

ch1nczyk 4 years, 1 month ago

Selected Answer: C

Correct

https://zhangxuetu.online

[202610]

Question #4 Topic 1

You create an important report for your large team in Google Data Studio 360. The report uses Google BigQuery as its data source. You notice that visualizations are not showing data that is less than 1 hour old. What should you do?

A. Disable caching by editing the report settings. **Most Voted**

B. Disable caching in BigQuery by editing table details.

C. Refresh your browser tab showing the visualizations.

D. Clear your browser history for the past hour then reload the tab showing the virtualizations.

Correct Answer: A

Community vote distribution



Comments

Khaled_Rashwan **Highly Voted** 3 years, 1 month ago

A. Disable caching by editing the report settings. By default, Google Data Studio 360 caches data to improve performance and reduce the amount of queries made to the data source. However, this can cause visualizations to not show data that is less than 1 hour old, as the cached data is not up-to-date. To resolve this, you should disable caching by editing the report settings. This can be done by following these steps: Open the report in Google Data Studio 360. Click on the "File" menu in the top left corner of the screen. Select "Report settings" from the dropdown menu. In the "Report settings" window, scroll down to the "Data" section. Toggle off the "Enable cache" option. Click the "Save" button to apply the changes. Disabling caching ensures that the data shown in the visualizations is always up-to-date, but it may increase the query load on the data source and affect the report's performance. Therefore, it's important to consider the trade-off between performance and data accuracy when making this change.

3244fd8 5 months, 2 weeks ago

Selected Answer: A

A. Disable caching by editing the report settings.

Ahamada 1 year, 1 month ago

Selected Answer: A

By default there's a cach in google data studio (looker studio). So you should disable it

SamuelTsch 1 year, 5 months ago

Selected Answer: A

To ensure that Google Data Studio visualizations show the most recent data, you should disable caching within Data Studio's report settings.

rocky48 2 years, 5 months ago

Selected Answer: A

A. Disable caching by editing the report settings. By default, Google Data Studio 360 caches data to improve performance and reduce the amount of queries made to the data source. However, this can cause visualizations to not show data that is less than 1 hour old, as the cached data is not up-to-date.

rtcpost 2 years, 5 months ago

Selected Answer: A

Disabling caching in the report settings will ensure that the visualizations are not using cached data and will reflect the most up-to-date information from your Google BigQuery data source. This will allow your report to show data that is less than 1 hour old. Caching is often used for performance optimization, but it can result in delays in displaying real-time or near-real-time data, so disabling it is the appropriate action in this case.

Websurfer 2 years, 7 months ago

Selected Answer: A

disable caching in report setting will get the issue resolved

Morock 3 years, 1 month ago

Selected Answer: D

The solution from the site is perfect.

PolyMoe 3 years, 2 months ago

Selected Answer: A

what is relevant here is to uncache Data Studio

samdhimal 3 years, 2 months ago

A. Disable caching by editing the report settings. Data Studio 360 uses caching to speed up report loading times. When caching is enabled, Data Studio 360 will only show the data that was present in the data source at the time the report was loaded. To ensure that the visualizations in your report are always up-to-date, you should disable caching by editing the report settings. This will force Data Studio 360 to retrieve the latest data from the data source (in this case BigQuery) every time the report is loaded. Option B is incorrect as it would only disable caching in BigQuery, but it wouldn't affect the caching in Data Studio 360, so the visualizations would still not show the latest data. Option C and D will not help as the data is not being updated in Data Studio 360, it's just the cache that needs to be updated.

korntewin 3 years, 2 months ago

I'm confused, is it possible that the cache is in BigQuery level? and the looker just get the cache from bigquery

ejlp 3 years, 4 months ago

Selected Answer: A

Based on the doc, you can refresh the report using refresh button on the report, not the browser's refresh button. So the answer is A.

maksi 3 years, 4 months ago

On my opinion it's C, because Data Studio doesn't support the Real-Time dashboard updates, so it means that if caching will be disabled, user will be forced to update the dashboard manually, otherwise report will be stuck on the data from the last update. According to documentation https://support.google.com/looker-studio/answer/7020039?hl=en#zippy=%2Cin-this-article, if we want to keep the data fresh we need to setup caching with minimum value of 15 minites - it means that data in the report will be updated automatically wvery 15 minutes, if cache will be disabled completely then the report will be stuck until we will manually update it. So, tbh for me it doesn't make sense to disable cache.

viks1122 3 years, 5 months ago

C. Since the data is not always stale. When it is, click on refresh button. Document also says the same Refresh report data manually Report editors can refresh the cache at any time: View or edit the report. In the top right, click More options. and then click RefreshRefresh data . This refreshes the cache for every data source added to the report.

beowulf_kat 3 years, 5 months ago

Refreshing the web browser does not refresh the data behind the viz's in Data Studio. You have to click the 'refresh data source' button.

nicholascz 3 years, 5 months ago

Selected Answer: A

https://support.google.com/looker-studio/answer/7020039?hl=en#zippy=%2Cin-this-article

kennyloo 3 years, 5 months ago

A. should be correct. after disabled the cache, it will retrieve data every time.

max_c 3 years, 5 months ago

Selected Answer: C

Same question from a Cloud guru and answer was C. The wording is slightly different in the documentation but still, the idea is that you can trigger a manual refresh

Lestrang 3 years, 2 months ago

The option is to refresh the browser tab not inside data studio itself. incorrect.

https://zhangxuetu.online

[202610]

Question #5 Topic 1

An external customer provides you with a daily dump of data from their database. The data flows into Google Cloud Storage GCS as comma-separated values
(CSV) files. You want to analyze this data in Google BigQuery, but the data could have rows that are formatted incorrectly or corrupted. How should you build this pipeline?

A. Use federated data sources, and check data in the SQL query.

B. Enable BigQuery monitoring in Google Stackdriver and create an alert.

C. Import the data into BigQuery using the gcloud CLI and set max_bad_records to 0.

D. Run a Google Cloud Dataflow batch pipeline to import the data into BigQuery, and push errors to another dead-letter table for analysis. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Radhika7983 **Highly Voted** 5 years, 4 months ago

The answer is D. An ETL pipeline will be implemented for this scenario. Check out handling invalid inputs in cloud data flow https://cloud.google.com/blog/products/gcp/handling-invalid-inputs-in-dataflow ParDos . . . and don’ts: handling invalid inputs in Dataflow using Side Outputs as a “Dead Letter” file

jkhong 3 years, 3 months ago

The sources you've provided cannot be accessed. Here is an updated best practice. https://cloud.google.com/architecture/building-production-ready-data-pipelines-using-dataflow-developing-and-testing#use_dead_letter_queues

nadavw 1 year, 7 months ago

https://cloud.google.com/dataflow/docs/guides/write-to-bigquery#:~:text=It%27s%20a%20good%20practice%20to%20send%20the%20errors%20to%20a%20dead%2Dletter%20queue%20or%20table%2C%20for%20later%20processing.%20For%20more%20information%20about%20this%20pattern%2C%20see%20BigQueryIO%20dead%20letter%20pattern. It's a good practice to send the errors to a dead-letter queue or table, for later processing. For more information about this pattern, see BigQueryIO dead letter pattern.

rocky48 **Highly Voted** 1 year, 6 months ago

Selected Answer: D

Option A is incorrect because federated data sources do not provide any data validation or cleaning capabilities and you'll have to do it on the SQL query, which could slow down the performance. Option B is incorrect because Stackdriver monitoring can only monitor the performance of the pipeline, but it can't handle corrupted or incorrectly formatted data. Option C is incorrect because using gcloud CLI and setting max_bad_records to 0 will ignore the corrupted or incorrectly formatted data and continue the load process, this will lead to incorrect analysis. Answer D. Run a Google Cloud Dataflow batch pipeline to import the data into BigQuery, and push errors to another dead-letter table for analysis.

3244fd8 5 months, 2 weeks ago

Selected Answer: D

D. Run a Google Cloud Dataflow batch pipeline to import the data into BigQuery, and push errors to another dead-letter table for analysis.

willyunger 1 year ago

Selected Answer: D

"you want to keep the data"

Ahamada 1 year, 1 month ago

Selected Answer: D

You should transform the raw data by eliminate the error before analyse.

rtcpost 1 year, 6 months ago

Selected Answer: D

Google Cloud Dataflow allows you to create a data pipeline that can preprocess and transform data before loading it into BigQuery. This approach will enable you to handle problematic rows, push them to a dead-letter table for later analysis, and load the valid data into BigQuery. Option A (using federated data sources and checking data in the SQL query) can be used but doesn't directly address the issue of handling corrupted or incorrectly formatted rows. Options B and C are not the best choices for handling data quality and error issues. Enabling monitoring and setting max_bad_records to 0 in BigQuery may help identify errors but won't store the problematic rows for further analysis, and it might prevent loading any data with issues, which may not be ideal.

samdhimal 1 year, 6 months ago

D. Run a Google Cloud Dataflow batch pipeline to import the data into BigQuery, and push errors to another dead-letter table for analysis. By running a Cloud Dataflow pipeline to import the data, you can perform data validation, cleaning and transformation before it gets loaded into BigQuery. Dataflow allows you to handle corrupted or incorrectly formatted rows by pushing them to another dead-letter table for analysis. This way, you can ensure that only clean and correctly formatted data is loaded into BigQuery for analysis.

samdhimal 3 years, 2 months ago

Option A is incorrect because federated data sources do not provide any data validation or cleaning capabilities and you'll have to do it on the SQL query, which could slow down the performance. Option B is incorrect because Stackdriver monitoring can only monitor the performance of the pipeline, but it can't handle corrupted or incorrectly formatted data. Option C is incorrect because using gcloud CLI and setting max_bad_records to 0 will ignore the corrupted or incorrectly formatted data and continue the load process, this will lead to incorrect analysis.

hamza101 2 years, 8 months ago

for Option C i think when setting max_bad_records to 0 this will prevent the loading to be achieved since the condition will cut off the loading if we have at least 1 corrupted row

RT_G 1 year, 6 months ago

Selected Answer: D

All other options only alert or error out bad data. As the question requires, option D sends bad data to the dead letter table for further analysis while valid data is loaded to the table

vaga1 2 years, 10 months ago

Selected Answer: D

Agreed: D

odiez3 3 years ago

D because you need Transform the data

Morock 3 years, 1 month ago

Selected Answer: D

D. The question is asking pipeline, then let's build a pipeline.

Besss 3 years, 5 months ago

Selected Answer: D

Agreed: D

Dip1994 3 years, 8 months ago

The correct answer is D

Arkon88 4 years, 1 month ago

Selected Answer: D

Correct - D (as we need to create Pipeline) which possible via 'D'

MaxNRG 4 years, 4 months ago

Looks like D, with C you will not import anything, stackdriver alerts will not help you with this and with federated resources you won't know what happened with those bad records. D is the most complete one.
https://cloud.google.com/blog/products/gcp/handling-invalid-inputs-in-dataflow

anji007 4 years, 5 months ago

Ans: D

nickozz 4 years, 6 months ago

D seems to be correct. explained here how combined wth Pub/Sub, this can be achieved.
https://cloud.google.com/pubsub/docs/handling-failures

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #6 Topic 1

Your weather app queries a database every 15 minutes to get the current temperature. The frontend is powered by Google App Engine and server millions of users. How should you design the frontend to respond to a database failure?

A. Issue a command to restart the database servers.

B. Retry the query with exponential backoff, up to a cap of 15 minutes. **Most Voted**

C. Retry the query every second until it comes back online to minimize staleness of data.

D. Reduce the query frequency to once every hour until the database comes back online.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Radhika7983 **Highly Voted** 5 years, 4 months ago

Correct answer is B. App engine create applications that use Cloud SQL database connections effectively. Below is what is written in google cloud documnetation. If your application attempts to connect to the database and does not succeed, the database could be temporarily unavailable. In this case, sending too many simultaneous connection requests might waste additional database resources and increase the time needed to recover. Using exponential backoff prevents your application from sending an unresponsive number of connection requests when it can't connect to the database. This retry only makes sense when first connecting, or when first grabbing a connection from the pool. If errors happen in the middle of a transaction, the application must do the retrying, and it must retry from the beginning of a transaction. So even if your pool is configured properly, the application might still see errors if connections are lost. reference link is https://cloud.google.com/sql/docs/mysql/manage-connections

llamaste **Highly Voted** 5 years, 8 months ago

https://cloud.google.com/sql/docs/mysql/manage-connections#backoff

3244fd8 5 months, 1 week ago

Selected Answer: B

Retry the query with exponential backoff, up to a cap of 15 minutes.

willyunger 1 year ago

Selected Answer: B

Exponential backoff avoids swamping the server. Higher rates may only make problem worse. Front-end should not have option to restart DB.

cqrm3n 1 year, 2 months ago

Selected Answer: B

We should use exponential backoff because it reduces load on the failing database, optimizes retry timing and is the industry best practice. Exponential backoff is a retry strategy where the wait time between retry increase exponentially. By gradually increasing the retry interval, the system avoids wasting resources on immediate retries when the database is likely still down.

rtcpost 1 year, 6 months ago

Selected Answer: B

Exponential backoff is a commonly used technique to handle temporary failures, such as a database server becoming temporarily unavailable. This approach retries the query, initially with a short delay and then with increasingly longer intervals between retries. Setting a cap of 15 minutes ensures that you don't excessively burden your system with constant retries. Option C (retrying the query every second) can be too aggressive and may lead to excessive load on the server when it comes back online. Option D (reducing the query frequency to once every hour) would result in significantly stale data and a poor user experience, which is generally not desirable for a weather app. Option A (issuing a command to restart the database servers) is not a suitable action for a frontend component and might not address the issue effectively. Database server restarts should be managed as a part of the infrastructure and not initiated by the frontend.

samdhimal 1 year, 6 months ago

correct answer -> Retry the query with exponential backoff, up to a cap of 15 minutes. If your application attempts to connect to the database and does not succeed, the database could be temporarily unavailable. In this case, sending too many simultaneous connection requests might waste additional database resources and increase the time needed to recover. Using exponential backoff prevents your application from sending an unresponsive number of connection requests when it can't connect to the database.

samdhimal 3 years, 2 months ago

Exponential backoff with a cap is a common technique used to handle temporary failures, such as database outages. In this approach, the frontend will retry the query with increasing intervals (e.g., 1s, 2s, 4s, 8s, etc.) up to a maximum interval (in this case, 15 minutes), this will help to avoid overwhelming the database servers with too many requests at once, and minimize the impact of the failure on the users. Option A, is not recommended because it's not guaranteed that restarting the database servers will fix the problem, it could be a network or a configuration problem and it could cause more downtime. Option C is not recommended because it could cause too many requests to be sent to the server, overwhelming the database and causing more downtime. Option D is not recommended because reducing the query frequency too much would result in stale data, and users will not receive the most up-to-date information.

RT_G 2 years, 4 months ago

Selected Answer: B

Retries with exponential backoff seems like the most efficient option in this scenario

rocky48 2 years, 4 months ago

Selected Answer: B

Correct answer is B

gudguy1a 2 years, 6 months ago

Selected Answer: B

good answer, good answer @radhika7983.

Datardp 2 years, 9 months ago

B is anser

vaga1 2 years, 10 months ago

Selected Answer: B

I agree with the exponential backoff technique, even thoght I do not see why 15 minutes should be a desired choice.

vaga1 2 years, 10 months ago

I guess that when u have failed after 15 minutes, your app must go through a serious review before being used again, since it is not able to provide the updated results as quickly as desired.

yafsong 3 years, 3 months ago

Truncated exponential backoff is a standard error-handling strategy for network applications. In this approach, a client periodically retries a failed request with increasing delays between requests

hiromi 3 years, 4 months ago

Selected Answer: B

B is right

shiv14 4 years, 1 month ago

Selected Answer: B

According to the documentation

deep_ROOT 4 years, 2 months ago

B is Correct; this question appeared in Cloud Architect exam also

MaxNRG 4 years, 4 months ago

B, backoff is a standard error handling strategy for network applications in which a client periodically retries a failed request with increasing delays between requests. Clients should use truncated exponential backoff for all requests to Cloud Storage that return HTTP 5xx and 429 response codes, including uploads and downloads of data or metadata.

https://zhangxuetu.online

[202610]

Question #7 Topic 1

You are creating a model to predict housing prices. Due to budget constraints, you must run it on a single resource-constrained virtual machine. Which learning algorithm should you use?

A. Linear regression **Most Voted**

B. Logistic classification

C. Recurrent neural network

D. Feedforward neural network

Correct Answer: A

Community vote distribution

A (100%)

Comments

Radhika7983 **Highly Voted** 5 years, 5 months ago

Correct answer is A. A tip here to decide when a liner regression should be used or logistics regression needs to be used. If you are forecasting that is the values in the column that you are predicting is numeric, it is always liner regression. If you are classifying, that is buy or no buy, yes or no, you will be using logistics regression.

Anirkent 5 years, 2 months ago

Liner Regression is correct but this is one aspect of the question, how does it relates to resource constrained machines? or that could be just a distraction?

muzammilnxs 5 years, 1 month ago

Neural Networks(Feed Forward or Recurrent) require resource intensive machines(i.e GPU's) whereas Linear regression can be done on ordinary CPU's

willyunger 1 year ago

Selected Answer: A

linear regression to predict numeric with least cost. for classifying with options use logistics.

SamuelTsch 1 year, 5 months ago

Selected Answer: A

the keyword here is running it on a single resource-constrained virtual machine. linear regression is a simple and efficient algorithm that is well-suited for predicting continuous values.

rtcpost 1 year, 6 months ago

Selected Answer: A

Linear regression is a simple and resource-efficient algorithm for predicting continuous values like housing prices. It's computationally lightweight and well-suited for single machines with limited resources. It doesn't require the extensive computational power or specialized hardware that more complex algorithms like neural networks (options C and D) might need. Option B (Logistic classification) is used for binary classification tasks, not for predicting continuous values like housing prices, so it's not the right choice in this context.

AshishDhamu 2 years, 1 month ago

Selected Answer: A

Linear regression is used for continous distribution.

Fazan456 2 years, 2 months ago

Selected Answer: A

Here, due to budget constraints, we're utilizing a single resource-constrained virtual machine, operating in a minimal resource environment. Linear regression emerges as the appropriate algorithm. It's a lightweight predictive model that suits our resource limitations

RT_G 2 years, 4 months ago

Selected Answer: A

Linear regression will be used since the prediction requires forecasting prices involving numeric values and is computationally less resource intensive

rocky48 2 years, 4 months ago

Selected Answer: A

Correct answer is A

AmmarFasih 2 years, 10 months ago

Selected Answer: A

Correct Answer is A. Since linear regression is used to predict a numeric value. While logistic regression is used to classify among the binary scenario. Further option C and D are advance ML options and not cost and resource effective for the current situation.

Zosby 3 years, 1 month ago

predict housing prices = linear regresssion

JJJJim 3 years, 1 month ago

Selected Answer: A

must be A. Though C can do it, linear regression is the better practice.

lukas_xls 3 years, 3 months ago

Selected Answer: A

Must be A

rowan_ 3 years, 7 months ago

A for sure. B is for classification. Neural nets can accomplish the task but they take WAY too many resources

samdhimal 4 years, 2 months ago

correct answer -> Linear Regression Linear regression is a statistical method that allows to summarize and study relationships between two continuous (quantitative) variables: One variable, denoted X, is regarded as the independent variable. The other variable denoted y is regarded as the dependent variable. Linear regression uses one independent variable X to explain or predict the outcome of the dependent variable y. Whenever you are told to predict some future value of a process which is currently running, you can go with a regression algorithm.

samdhimal 3 years, 2 months ago

Linear regression is a simple and computationally efficient algorithm that can be used to predict a continuous target variable based on one or more input variables. It is particularly well-suited for resource-constrained environments, as it requires minimal computational resources and can be run on a single virtual machine. Linear regression is a good fit for this problem as it is a supervised learning algorithm that can be used for regression problems, and it's not computationally expensive. Option B is not recommended as Logistic classification is a supervised learning algorithm that is used for classification problems, not regression problems. Option C and D are not recommended as Recurrent Neural Network (RNN) and Feedforward Neural Network (FNN) are computationally expensive and may require significant computational resources and memory to run on a single virtual machine.

MaxNRG 4 years, 4 months ago

A as Supervised learning using Regression can help build a model to predict house prices. Option B is wrong as Classification would not help to solve the problem. Options C & D are wrong as they would need more resources.

anji007 4 years, 5 months ago

Ans: A

StefanoG 4 years, 7 months ago

Ok the right answer is A, but the question is why? Then: - B not because we are make forecasting and not classifying - C and D not because this solution need more nodes, then more VM. Right?

https://zhangxuetu.online

[202610]

Question #8 Topic 1

You are building new real-time data warehouse for your company and will use Google BigQuery streaming inserts. There is no guarantee that data will only be sent in once but you do have a unique ID for each row of data and an event timestamp. You want to ensure that duplicates are not included while interactively querying data. Which query type should you use?

A. Include ORDER BY DESK on timestamp column and LIMIT to 1.

B. Use GROUP BY on the unique ID column and timestamp column and SUM on the values.

C. Use the LAG window function with PARTITION by unique ID along with WHERE LAG IS NOT NULL.

D. Use the ROW_NUMBER window function with PARTITION by unique ID along with WHERE row equals 1. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Ender_H **Highly Voted** 3 years, 6 months ago

I personally don't think any answer is correct, D is the closest one but it's missing a "ORDER BY timestamp DESC" to ensure to get only the latest record based in the timestamp

ndimu 1 year, 4 months ago

the idea is you can have multiple events occurring at the same time so the only way to distinct is id

Davijde13 3 years, 2 months ago

The question mention only duplicated data and nothing about taking only the latest ones. Therefore I assume there is no need to always take the latest, we should ensure we take only one record for each ID.

daghayeghi **Highly Voted** 5 years ago

D: https://cloud.google.com/bigquery/streaming-data-into-bigquery#manually_removing_duplicates

joqu 6 months, 3 weeks ago

Selected Answer: D

D but it needs ORDER BY and should use QUALIFY instead of WHERE

willyunger 1 year ago

Selected Answer: D

D is closest, as there will always be at least 1 row for each ID. Would have rather used SELECT DISTINCT.

RT_G 1 year, 6 months ago

Selected Answer: D

D ensures data is partitioned by the unique id and only one record is picked thereby ensuring results are de-duplicated

rtcpost 1 year, 6 months ago

Selected Answer: D

This approach will assign a row number to each row within a unique ID partition, and by selecting only rows with a row number of 1, you will ensure that duplicates are excluded in your query results. It allows you to filter out redundant rows while retaining the latest or earliest records based on your timestamp column. Options A, B, and C do not address the issue of duplicates effectively or interactively as they do not explicitly remove duplicates based on the unique ID and event timestamp.

Radhika7983 1 year, 6 months ago

Correct answer is D. Group by column us used to check for the duplicates where you can have the count(*) for each of the unique id column. If the count is greater than 1, we will know duplicate exists.The easiest way to remove duplicates while streaming inserts is to use row_number. Use GROUP BY on the unique ID column and timestamp column and SUM on the values will not remove duplicates. I also executed LAG function and LAG function will return NULL on unique id when no previous records with same unique id exist. Hence LAG is also not an option here.

MaxNRG 1 year, 6 months ago

D is correct because it will just pick out a single row for each set of duplicates. A is not correct because this will just return one row. B is not correct because this doesn't get you the latest value, but will get you a sum of the same event over time which doesn't make too much sense if you have duplicates. C is not correct because if you have events that are not duplicated, it will be excluded.

Zosby 3 years, 1 month ago

Correct D

Morock 3 years, 1 month ago

Selected Answer: D

Row number gives the unique number ranking based on target column.

odacir 3 years, 3 months ago

Selected Answer: D

It's the only valid option, try it your self with examples in QB.

Mamta072 3 years, 9 months ago

Ans is D as Row number is the clause to fetch unique record from duplicate

Arkon88 4 years, 1 month ago

Answer: D

samdhimal 4 years, 2 months ago

correct answer -> Use the ROW_NUMBER window function with PARTITION by unique ID along with WHERE row equals 1. You can use the ROW_NUMBER() to turn non-unique rows into unique rows and then delete the duplicate rows. Reference: https://www.mysqltutorial.org/mysql-window-functions/mysql-row_number-function/

samdhimal 1 year, 6 months ago

When you are using BigQuery streaming inserts, there is no guarantee that data will only be sent once. However, you can use the ROW_NUMBER window function to ensure that duplicates are not included while interactively querying

data. By using a PARTITION BY clause on the unique ID column, you can assign a unique number to each row within a result set, based on the order specified in the timestamp column. Then, a WHERE clause can be used to select only the row with the number 1. This will return the first row for each unique ID based on the timestamp column, which will ensure that duplicates are not included in your query results.

samdhimal 3 years, 2 months ago

Option A is not recommended because it will only return the first row based on the timestamp column, it doesn't consider the unique ID, so you could have multiple rows with the same timestamp, and you will get one of them arbitrarily. Option B is not recommended because it's used for aggregation, it doesn't return the first row for each unique ID based on the timestamp column. Option C is not recommended because it's used for comparing rows, it doesn't return the first row for each unique ID based on the timestamp column.

nofaruccio 4 years, 4 months ago

Sorry, but IMHO no response is correct, because, in addition to making the ID field unique, it occurs consider the record with most recent timestamp

anji007 4 years, 5 months ago

Ans: D

lbhhoya82 5 years ago

Correct : D

https://zhangxuetu.online

[202610]

Question #9 Topic 1

Your company is using WILDCARD tables to query data across multiple tables with similar names. The SQL statement is currently failing with the following error:

```
# Syntax error : Expected end of statement but got "-" at [4:11]
SELECT age
FROM
     bigquery-public-data.noaa_gsod.gsod
WHERE
     age != 99
     AND_TABLE_SUFFIX = '1929'
ORDER BY
     age DESC
```

Which table name will make the SQL statement work correctly?

A. 'bigquery-public-data.noaa_gsod.gsod'

B. bigquery-public-data.noaa_gsod.gsod*

C. 'bigquery-public-data.noaa_gsod.gsod'*

D. 'bigquery-public-data.noaa_gsod.gsod*` Most Voted

Correct Answer: D

Community vote distribution



Comments

Ender_H Highly Voted 3 years, 6 months ago

None, the actual `bigquery-public-data.noaa_gsod.gsod*` with back ticks at the beginning and at the end.

Davijde13 3 years, 2 months ago

I suspect there has been some typo with copy-paste of the option D

jitvimol 2 years, 3 months ago

yes, I see from another source that actually ans D has to be backtick. Probably a problem when this web do data ingestion.

af17139 5 months, 2 weeks ago

Selected Answer: D

A very common reason for SQL statements with wildcard tables to fail in BigQuery is not enclosing the table name pattern in backticks (`)

Setna 6 months, 2 weeks ago

Selected Answer: D

Focus only on what the question asks. This question contains several errors in the query, but the question addresses only one of them. The table name. Option D.

fassil 11 months, 3 weeks ago

Selected Answer: D

just read this guys : https://cloud.google.com/bigquery/docs/querying-wildcard-tables

Mariaantonirajc 1 year, 3 months ago

Selected Answer: B

B This is the correct syntax. The wildcard * is outside any quotes or string delimiters. This tells BigQuery to query all tables that match the pattern gsod* within the noaa_gsod dataset.

sravi1200 1 year, 3 months ago

Selected Answer: B

Big Query does not use quotations while fetching data from the table : example: select * from project-id.dataset_name.table_name; is the syntax

vaga1 1 year, 6 months ago

Selected Answer: D

let's forget the fact that in BQ is used ` instead than ' which retrieves an error in any case. ` is called backquote, backtick, or left quote while ' is simply an apostrophe. Let's consider ' to be ` in every answer, since moderators could have not been aware of such when they had received the question.

vaga1 2 years, 10 months ago

Who used BQ knows that the backquote is necessary only for the project name, while it can be used for the whole string, and necessary only when the project name contains special (special in this specific context) characters. - is a special character. so `bigquery-public-data`.noaa_gsod.gsod1940 would have worked too. The question now turns out to be `bigquery-public-data`.noaa_gsod.gsod* still works or due to the * presence we need to write `bigquery-public-data.noaa_gsod.gsod*` ? I personally do not remember, and I do not have a BQ at my disposal at the moment. But I know for sure that `bigquery-public-data.noaa_gsod.gsod*` works while `bigquery-public-data`.noaa_gsod.gsod* is not in the options.

rtcpost 1 year, 6 months ago

Selected Answer: D

Option D (assuming to have backticks) Refer: https://cloud.google.com/bigquery/docs/querying-wildcard-tables The following query is NOT valid because it isn't properly quoted with backticks: ``` #standardSQL /* Syntax error: Expected end of statement but got "-" at [4:11] */ SELECT max FROM # missing backticks bigquery-public-data.noaa_gsod.gsod* WHERE max != 9999.9 # code for missing data AND _TABLE_SUFFIX = '1929' ORDER BY max DESC ```

RT_G 1 year, 6 months ago

Selected Answer: D

Reference: https://cloud.google.com/bigquery/docs/querying-wildcard-tables The wildcard table name contains the special character (*), which means that you must enclose the wildcard table name in backtick (`) characters. For example, the following query is valid because it uses backticks: #standardSQL /* Valid SQL query */ SELECT max

FROM `bigquery-public-data.noaa_gsod.gsod*` WHERE max != 9999.9 # code for missing data AND _TABLE_SUFFIX = '1929' ORDER BY max DESC

ABKR1300 1 year, 6 months ago

Few might go with the Option B which will be a blunder because of the below reason. While querying the tables or views with the name, it is optional to surround with the backticks. But while querying the list of tables with Wild card character, it is must to surround with the backticks. We can get the Syntax error: Expected end of input but got "*" with the below query SELECT * FROM bigquery-public-data.noaa_gsod.gsod* WHERE _TABLE_SUFFIX = "2024" So, option D might be the correct one, provided if there is a typo.

Chintu_573 1 year, 9 months ago

Selected Answer: B

IN option D, there is differert ' ` on first and last. That's why right option is second.

dsyouness 1 year, 10 months ago

Selected Answer: B

bigquery-public-data.noaa_gsod.gsod* also works

RT_G 2 years, 4 months ago

Selected Answer: D

Agree with others - Option D

axantroff 2 years, 5 months ago

Selected Answer: D

D. 'bigquery-public-data.noaa_gsod.gsod*` is the right answer with 1 typo

Pavaan 2 years, 10 months ago

Answer is 'D' Reference : https://cloud.google.com/bigquery/docs/wildcard-table-reference Enclose table names with wildcards in backticks The wildcard table name contains the special character (*), which means that you must enclose the wildcard table name in backtick (`) characters.

Melampos 2 years, 11 months ago

Selected Answer: B

bigquery-public-data.noaa_gsod.gsod* works

hkhnhan 3 years ago

Selected Answer: B

should be B, the backtick at D answer is wrong ' instead of `

https://zhangxuetu.online

[202610]

Question #10 Topic 1

Your company is in a highly regulated industry. One of your requirements is to ensure individual users have access only to the minimum amount of information required to do their jobs. You want to enforce this requirement with Google BigQuery. Which three approaches can you take? (Choose three.)

A. Disable writes to certain tables.

B. Restrict access to tables by role. **Most Voted**

C. Ensure that the data is encrypted at all times.

D. Restrict BigQuery API access to approved users. **Most Voted**

E. Segregate data across multiple tables or databases. **Most Voted**

F. Use Google Stackdriver Audit Logging to determine policy violations.

Correct Answer: BDE

Community vote distribution



Comments

samdhimal **Highly Voted** 4 years, 2 months ago

correct option -> B. Restrict access to tables by role. Reference: https://cloud.google.com/bigquery/docs/table-access-controls-intro correct option -> D. Restrict BigQuery API access to approved users. ***Only approved users will have access which means other users will have minimum amount of information required to do their job.*** Reference: https://cloud.google.com/bigquery/docs/access-control correct option -> F. Use Google Stackdriver Audit Logging to determine policy violations. Reference: https://cloud.google.com/bigquery/docs/table-access-controls-intro#logging A. Disable writes to certain tables. ---> Read is still available(not minimal access) C. Ensure that the data is encrypted at all times. ---> Data is encrypted by default. E. Segregate data across multiple tables or databases. ---> Normalization is of no help here.

samdhimal 1 year, 6 months ago

I was WRONG. I am not sure why s o many upvotes lol. I think this is the correct answer: B. Restrict access to tables by role. D. Restrict BigQuery API access to approved users. E. Segregate data across multiple tables or databases. Restrict access to tables by role: You can use BigQuery's access controls to restrict access to specific tables based on user roles. This allows you to ensure that users can only access the data they need to do their job. Restrict BigQuery API access to approved users: By using Cloud Identity and Access Management (IAM) you can control who has access to the BigQuery API, and what actions they are allowed to perform. This will help to ensure that only authorized users can access the data. Segregate data across multiple tables or databases: You can use multiple

tables or databases to separate different types of data, so that users can only access the data they need. This will prevent users from seeing data they shouldn't have access to.

samdhimal 3 years, 2 months ago

Option A is incorrect because disabling writes to certain tables would prevent users from updating the data which is not in line with the goal of providing access to the minimum amount of information required to do their jobs. Option C is incorrect because while data encryption is important for security it doesn't specifically help with providing users access to the minimum amount of information required to do their jobs. Option F is incorrect because while Google Stackdriver Audit Logging can help to determine policy violations it does not help to enforce the access controls and segregation of data.

directtoking 1 year, 3 months ago

If there is "Restrict access to tables by role", then what is the requirement for "Segregate data across multiple tables or databases"?

IsaB Highly Voted 5 years, 7 months ago

Yes. Access control on table level is now possible in BigQuery : https://cloud.google.com/bigquery/docs/table-access-controls-intro

axantroff 2 years, 5 months ago

Thanks. For me, this type of answer is more valuable because even as time passes, I can revisit existing solutions and ideas and refresh the concepts of the initial question. It helped me on the ACE exam

pippococah24 8 months, 1 week ago

Selected Answer: BDE

F makes sense, but only once there has been a violation. I'd go with BDE, even though normalization across different datasets might not solve the problem.

cqrm3n 1 year, 2 months ago

Selected Answer: BDE

B - Use IAM to define granular permissions. D - Only authorised users or systems can query or manipulate BigQuery data. E - By segregating data into different tables or datasets, specific permissions can be assigned to each data subset.

NicolasN 1 year, 6 months ago

Selected Answer: BDE

I disagree with [F]. It's too late for a "highly regulated industry" to detect access violations by audit logs. [E] is a more reasonable answer, since it is a kind of row-level security, especially the times when BigQuery row-level security wasn't available. It is a practice still recommended (even with row-level sec. available) for the extreme scenario that: (Through repeated observation of query duration when querying tables with row-level access policies,) "a user could infer the values of rows that otherwise might be protected by row-level access policies" "If you are sensitive to this level of protection, we recommend using separate tables to isolate rows with different access control requirements, instead." Source: https://cloud.google.com/bigquery/docs/best-practices-row-level-security#limit-side-channel-attacks

Asheesh1909 1 year, 6 months ago

Selected Answer: BDE

E seems more sensible to be as the question concentrates more on table access restriction than access violation, policy violations can only be determined through stackdriver how ever we cant restrict the access to tables.Probably option E should be considered as, by segregating the data into diferrent tables , we can restrict access to tables.

odacir 3 years, 3 months ago

E says segregate across multiple tables or databases, this is not the pattern of BigQuery, in BQ there is only one database, and you can organize your data in datasets...

rtcpost 1 year, 6 months ago

Selected Answer: BDE

B. Restrict access to tables by role: You can define roles in BigQuery and grant specific permissions to these roles to control who can access particular tables. D. Restrict BigQuery API access to approved users: You can control access to the BigQuery API and, consequently, to the underlying data by ensuring that only approved users or services can make API requests. E. Segregate data across multiple tables or databases: You can separate data into different tables or databases based on user access requirements, which allows you to limit users' access to specific data sets. These approaches, when used together, can help you enforce data access controls in a regulated environment. Options A, C, and F are also important considerations but are not direct methods for enforcing fine-grained access control to specific data.

SatyamKishore 1 year, 6 months ago

B. Restrict access to tables by role. Use IAM roles and permissions to control access to specific datasets or tables based on the user's role. D. Restrict BigQuery API access to approved users. Limit API access to only those users or services that need it, ensuring that unauthorized users cannot interact with the data. E. Segregate data across multiple tables or databases. Organize data in a way that separates sensitive information, allowing more granular control over who has access to specific datasets. These options directly contribute to enforcing the principle of least privilege, ensuring users can only access the data necessary for their roles.

MaxNRG 2 years, 3 months ago

Selected Answer: BDE

You want to enforce this requirement with Google BigQuery -> BDE

RT_G 2 years, 4 months ago

Selected Answer: BDF

BDF. We are fairly unanimous with options B and D. I'm going with F because it does help identifying policy violations which is also one aspect to be considered when designing access controls. Option D only indicates segregating into multiple tables and databases which may or may not help with controlling access leaving it open-ended for the architect to decide.

rocky48 2 years, 4 months ago

Selected Answer: BDF

In Google BigQuery, you can organize and segregate data across multiple tables within the same dataset, but you cannot directly segregate data into separate databases. BigQuery uses a flat namespace structure where data is organized into datasets and tables within those datasets. Datasets are the highest level of organization within BigQuery. So i'm sticking with BDF

RheaZzang 2 years, 5 months ago

Selected Answer: BDE

B. Restrict access to tables by role. D. Restrict BigQuery API access to approved users. E. Segregate data across multiple tables or databases.

AnonymousPanda 2 years, 7 months ago

Selected Answer: BDF

BDF as per other answers

nescafe7 2 years, 8 months ago

Selected Answer: BDF

Regarding E or F, opinions seem to be divided into two parts. I think E is insufficient because it seems that appropriate conditions must be additionally described for table or dataset separation. F is also emphasized in Google's official textbook. You need to ensure that it is operating well as set up through monitoring. So, BDF!

Liting 2 years, 8 months ago

Selected Answer: DEF

Why B is correct? Access control can only be applied on dataset and views, not on partitions and tables. => So it is not possible to restrict access to table, but only to dataset. Can someone help me understand why in this scenario B is correct?

FP77 2 years, 8 months ago

I was thinking the same thing. I thought dataset access gave you access to all tables within it, and that you couldn't restrict access on the table level.

KK0202 2 years, 9 months ago

Selected Answer: BDF

Option E says "...or databases". The data housing service in question in BigQuery and the context is to design that support BigQuery access delegation. Seems random to include moving to another database as an option. If it did not mention databases and stopped at just tables, then E would also be the right option

Oleksandr0501 2 years, 11 months ago

B. Restrict access to tables by role: This approach can be used to control access to tables based on user roles. Access controls can be set at the project, dataset, and table level, and roles can be customized to provide granular access controls to different groups of users. D. Restrict BigQuery API access to approved users: This approach involves using IAM (Identity and Access Management) to control access to the BigQuery API. Access can be granted or revoked at the project or dataset level, and policies can be customized to control access based on user roles, IP addresses, and other factors. E. Segregate data across multiple tables or databases: This approach involves breaking down large datasets into smaller, more manageable tables or databases. This helps to ensure that individual users have access only to the minimum amount of information required to do their jobs, and reduces the risk of data breaches or policy violations.

https://zhangxuetu.online

[202610]

Question #11 Topic 1

You are designing a basket abandonment system for an ecommerce company. The system will send a message to a user based on these rules:

➪ No interaction by the user on the site for 1 hour

Has added more than $30 worth of products to the basket

▪

➪ Has not completed a transaction

You use Google Cloud Dataflow to process the data and decide if a message should be sent. How should you design the pipeline?

A. Use a fixed-time window with a duration of 60 minutes.

B. Use a sliding time window with a duration of 60 minutes.

C. Use a session window with a gap time duration of 60 minutes. **Most Voted**

D. Use a global window with a time based trigger with a delay of 60 minutes.

Correct Answer: C

Community vote distribution

C (100%)

Comments

vetaal **Highly Voted** 3 years, 4 months ago

There are 3 windowing concepts in dataflow and each can be used for below use case 1) Fixed window 2) Sliding window and 3) Session window. Fixed window = any aggregation use cases, any batch analysis of data, relatively simple use cases. Sliding window = Moving averages of data Session window = user session data, click data and real time gaming analysis. The question here is about user session data and hence session window. Reference: https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines

cqrm3n 1 year, 2 months ago

Selected Answer: C

The answer is C because session window is specifically designed to handle use cases where activity is grouped by gaps. A. Fixed-time window divides data into non-overlapping, equally-size intervals but do not track gaps in user activity. B. Sliding-time window process overlapping intervals and are better suited for periodic aggregation. D. Global windows process all data over the pipeline's lifetime and rely on custom triggers to handle time-based logic. It is technically possible but unnecessarily complex so no.

rtcpost 1 year, 6 months ago

Selected Answer: C

C. Use a session window with a gap time duration of 60 minutes. A session window with a gap time duration of 60 minutes is appropriate for capturing user sessions where there has been no interaction on the site for 1 hour. It allows you to group user activity within a session, and when the session becomes inactive for the defined gap time, you can evaluate whether the user added more than $30 worth of products to the basket and has not completed a transaction. Options A and B (fixed-time window and sliding time window) might not capture the specific session-based criteria of inactivity and user interaction effectively. Option D (global window with a time-based trigger) is not suitable for capturing user sessions and checking inactivity based on a specific time duration. It's more appropriate for cases where you need a single global view of the data.

RT_G 2 years, 4 months ago

Selected Answer: C

Session window since the question specifically talks about a specific user for a fixed duration.

rocky48 2 years, 4 months ago

Selected Answer: C

Session window = user session data, click data and real time gaming analysis.

imran79 2 years, 5 months ago

The basket abandonment system needs to determine if a user hasn't interacted with the site for 1 hour, has added products worth more than $30, and hasn't completed a transaction. Therefore, the pipeline should account for periods of user activity and inactivity. A session-based windowing approach is appropriate here. The right choice is: C. Use a session window with a gap time duration of 60 minutes. Session windows group data based on periods of activity and inactivity. If there's no interaction for the duration of the gap time (in this case, 60 minutes), a new window is started. This would help identify users who haven't interacted with the site for the specified duration, fulfilling the requirement for the basket abandonment system.

MikkelRev 2 years, 6 months ago

Selected Answer: C

session windows can divide a data stream representing user activity
https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#session-windows

Chesternut999 3 years ago

Selected Answer: C

C - The best option for this use case.

bha11111 3 years ago

Selected Answer: C

Session window is used for these type of scenario

samdhimal 3 years, 1 month ago

C. Use a session window with a gap time duration of 60 minutes. A session window would be the most appropriate option to use in this case, as it would allow you to group events into sessions based on time gaps. In this case, the gap time of 60 minutes could be used to define a session, and if there is no interaction from the user for 60 minutes, a new session would be created. By using a session window, you can track the behavior of the user during each session, including the products added to the basket, and determine if the conditions for sending a message have been met (i.e., the user has added more than $30 worth of products to the basket and has not completed a transaction).

kennyloo 3 years, 5 months ago

Only C is feasible for this question

AWSandeep 3 years, 7 months ago

Selected Answer: C

C. Use a session window with a gap time duration of 60 minutes.

https://zhangxuetu.online

[202610]

Question #12 Topic 1

Your company handles data processing for a number of different clients. Each client prefers to use their own suite of analytics tools, with some allowing direct query access via Google BigQuery. You need to secure the data so that clients cannot see each other's data. You want to ensure appropriate access to the data.
Which three steps should you take? (Choose three.)

A. Load data into different partitions.

B. Load data into a different dataset for each client. Most Voted

C. Put each client's BigQuery dataset into a different table.

D. Restrict a client's dataset to approved users. Most Voted

E. Only allow a service account to access the datasets.

F. Use the appropriate identity and access management (IAM) roles for each client's users. Most Voted

Correct Answer: BDF

Community vote distribution



Comments

saurabh1805 Highly Voted 5 years, 7 months ago

My vota also goes for B,D,F

sumanshu Highly Voted 4 years, 11 months ago

Some voted for 'E' i.e. E. Only allow a service account to access the datasets. Not sure why ? if we gave access ONLY to service account - Does not it mean - we need to access BigQuery using Some Code (by mentioning Service account credentials there) OR using some other resource like VM) In this case - i think person can't even access the Big Query Service via UI (if we give access only to Service account). Correct me if there is option on UI as well

awssp12345 4 years, 9 months ago

yes, that is precisely why we need to eliminate E.

vosang5299 11 months, 2 weeks ago

Selected Answer: BDF

B,D,F is the correct answer as per Google best practices

cqrm3n 1 year, 2 months ago

Selected Answer: BDF

A is wrong because partitions do not provide access boundaries between the clients. All partitions within a table are accessible to anyone with access to the table. C is wrong because tables within the same dataset share the same access controls. E is wrong because service accounts are typically used for automated or backend processes, not client-specific access.

Nirca 1 year, 6 months ago

Selected Answer: BDF

B, D, F! C - is technically wrong . tables are being logically stored in a single dataset. A - Partitioning data is for improving performance. once you SQL (select) the table, you can not control the data being selected for the developer.

jin0 3 years, 1 month ago

For C. What if thinking about that there are tables by clients? such as customer_clients_a table and giving IAM from each table to users??..

samdhimal 1 year, 6 months ago

Selected Answer: BDF

B. Load data into a different dataset for each client. D. Restrict a client's dataset to approved users. F. Use the appropriate identity and access management (IAM) roles for each client's users. By loading each client's data into a separate dataset, you ensure that each client's data is isolated from the data of other clients. Restricting access to each client's dataset to only approved users, as specified in D, further enhances data security by ensuring that only authorized users can access the data. By using appropriate IAM roles for each client's users, as specified in F, you can grant different levels of access to different clients and their users, ensuring that each client has only the level of access required for their specific needs.

certs4pk 1 year, 4 months ago

so, we r assuming there is no 'common' data shared by different clients! if yes, will B still ba correct option?

suku2 1 year, 6 months ago

Selected Answer: BDF

B. Load data into a different dataset for each client. D. Restrict a client's dataset to approved users. F. Use the appropriate identity and access management (IAM) roles for each client's users.

rtcpost 1 year, 6 months ago

Selected Answer: BDF

B. Load data into a different dataset for each client: Organize the data into separate datasets for each client. This ensures data isolation and simplifies access control. D. Restrict a client's dataset to approved users: Implement access controls by specifying which users or groups are allowed to access each client's dataset. This restricts data access to approved users only. F. Use the appropriate identity and access management (IAM) roles for each client's users: Assign IAM roles based on client-specific requirements to manage permissions effectively. IAM roles help control access at a more granular level, allowing you to tailor access to specific users or groups within each client's dataset. These steps ensure that each client's data is separated, and access is controlled based on client-specific requirements. Options A, C, and E, while important in other contexts, are not sufficient on their own to ensure client data isolation and access control in a multi-client environment.

philli1011 2 years, 2 months ago

My Vote is BDF. I was thinking BEF but the question shows that the Big Query warehouse will be accessed by both direct users and other applications, as preferred by each customer.

SoloLeveling 2 years, 2 months ago

Selected Answer: BDF

agreed B,D,F

RT_G 2 years, 4 months ago

Selected Answer: BDF

Agree with others

imran79 2 years, 5 months ago

the answers are B, D, and F. To ensure that clients cannot see each other's data and have appropriate access, you would want to: Segregate the data by client. Restrict access to each client's data. Use proper identity and access management techniques.

Chi_Wang 2 years, 6 months ago

Selected Answer: BDF

B,D,F is the answer

elitedea 3 years ago

BDF is right

Nirca 3 years, 2 months ago

B, D, F! C - is technically wrong . tables are being logically stored in a single dataset. A - Partitioning data is for improving performance. once you SQL (select) the table, you can not control the data being selected for the developer.

DeeData 3 years, 3 months ago

Please why is DEF not correct?

Kyr0 3 years, 3 months ago

Selected Answer: BDF

Agree BDF

https://zhangxuetu.online

[202610]

Question #13 Topic 1

You want to process payment transactions in a point-of-sale application that will run on Google Cloud Platform. Your user base could grow exponentially, but you do not want to manage infrastructure scaling.
Which Google database service should you use?

A. Cloud SQL

B. BigQuery

C. Cloud Bigtable

D. Cloud Datastore **Most Voted**

Correct Answer: D

Community vote distribution



Comments

DeepakKhattar **Highly Voted** 5 years, 10 months ago

Initially, thinking D is the best answer but when question is re-re-read, A seems to be correct answer for following reasons 1. Is payment TRANSACTION -- DB should able to perform full blown transaction (updating inventory, sales info etc, though not specified) , not just ATOMIC which DataStore provides 2. Its point-of-sale application, not ONLINE STORE where HIGH number of concurrent users ordering stuff. 3. User Base could grow exponentially - again more users does mot mean concurrent users and more processing power. Its only about storage. 4. Do not want to Manage infrastructure scaling. - Cloud SQL can scale in terms of storage. 5. CloudStore is poor selection for OLTP application - Each property is index - so higher latency Not sure, during exam 2 min is enough to think on various point.. I may be wrong or wrong path ... lets brainstrom..

canon123 4 years, 4 months ago

CloudSql does not auto scale.

BigQuery 4 years, 3 months ago

https://cloud.google.com/architecture/elastically-scaling-your-mysql-environment#objectives Please read. It can be configured for autoscaling.

hendrixlives 4 years, 3 months ago

That link explains how to set MySQL autoscaling with Google Compute Engine instances (you install and manage MySQL on the VM). This can not be applied to Cloud SQL (managed service). In Cloud SQL, only the storage can be automatically increased, and changing the Cloud SQL instance size requires a manual edit of the instance type.

MisuLava 3 years, 5 months ago

yes. and that is ok since this is a point of sale. an exponential increase in number of clients still means reduced parallel processing (how many customers can buy in the very same time) so an increase in memory and CPU is very unlikely to be necessary. yes, an exponential increase in the number of customers means more memory, and more storage, which in cloud SQL increases automatically.

nkunwar 3 years, 6 months ago

C SQL doesn't AUTO SCALE, you need to manually edit , Please show where does it says AUTO SCALING

Blobby 4 years, 7 months ago

Can't online be considered PoS? CloudSQL does have constraints for scaling and Google seem to specifically be selling Datastore for transactional use cases so going with D:
https://cloud.google.com/datastore/docs/concepts/transactions

Blobby 4 years, 6 months ago

Based on a re-read of the above comments and other later questions agree with A. pls ignore my first answer.

jvg637 Highly Voted 6 years ago

D seems to be the right one. Cloud SQL doesn't automatically scale

BigQuery 4 years, 3 months ago

Cloud SQL does scale automatically. THERE IS A SETTING WHERE YOU DEFINE INCREASE MEMORY SPACE WHEN IT REACHED 70%. https://cloud.google.com/sql/docs/features#features_3 Here it say's -> Fully managed SQL Server databases in the cloud. -> Custom machine types with up to 624 GB of RAM and 96 CPUs. -> Up to 64 TB of storage available, with the ability to automatically increase storage size as needed.

hendrixlives 4 years, 3 months ago

Storage scale is automatic (e.g. you begin with a 50GB disk and it grows automatically as needed), but the instance size (CPU/memory) will be the same. The questions states that the user base may increase exponentially. Even if you have enough disk space to store all your user data, the increase in users will cause problems if your instance (CPU/memory) is too small, since the instance will not be able to process all the queries at the required speed.

imsaikat50 3 years, 3 months ago

I believe the key point is it's a POS, not an e-commerce. Keeping that in mind, exponential user increase in POS might not mean concurrent user increase, which could be a huge consideration in case of it is being e-commerce. I would rather go with 'Cloud SQL' as the best answer.

fassil 11 months, 3 weeks ago

Selected Answer: D

You don't need to go far, guys. Cloud SQL does not support auto scale; cut the shit here based on the requirement—the question specifically says "you do not want to manage infrastructure scaling."

Parandhaman_Margan 1 year ago

Selected Answer: D

POS app needing scalable DB without managing infrastructure. Cloud Datastore or Firestore? But options are Cloud SQL, BigQuery, Bigtable, Datastore. Since it's for transactions and scale, maybe Cloud Datastore (D) or Cloud Bigtable. But Cloud Bigtable is for high throughput. The question says not to manage scaling. Cloud Datastore is serverless. So D.

monyu 1 year ago

Selected Answer: D

B and C are discarded Since we are dealing with transactional data. A is discarded since we require to deal with Infrastructure scaling in case the base of users increase and also (not necessarily) the concurrent transactions. Users

base is gonna grow and we DO NOT WANT TO DEAL with infrastructure scaling. D is the most appropriated

cqrm3n 1 year, 2 months ago

Selected Answer: D

The answer is D, Cloud Datastore (now Firestore in Datastore mode), because it supports auto scaling and low latency reads and writes. Cloud SQL is not the correct answer because it requires more active management for scaling.

GHill1982 1 year, 5 months ago

Selected Answer: C

For handling payment transactions in a point-of-sale application with potential exponential growth and without the need to manage infrastructure scaling, Cloud Bigtable would be the best choice.

Radhika7983 1 year, 6 months ago

D seems to be the answer. This is what I think based on my analysis below. POS is OLTP system but now a days NOSQL with ACID properties also are used for OLTP, Cloud sql is good for relational database and it would have been an option here but it clearly says that "you do not want to manage infrastructure scaling". In cloud SQL, which is managed service and not server less, you need to manually do vertical scaling(scale up and scale down). Hence I believe CLOUD SQL is not the option here. I also tried creating a datastore using google cloud console and it gives 2 options now that is cloud firestore in native mode and cloud firestore in data store mode. automatic scaling is available in both where there is no manual scaling up or down is required. Also, both firestore in native and datastore provides ACID properties. Also, firestore is now optimized for OLTP. Please see below https://cloud.google.com/solutions/building-scalable-apps-with-cloud-firestore Though the question only talks about datastore, I am just providing additional information. Considering all what I read through, D is the answer.

awssp12345 4 years, 9 months ago

I agree. I think people are missing the part of the question that mentions they don't want to maintain the DB.

awssp12345 4 years, 9 months ago

This should be accepted and the highest voted answer.

kishanu 1 year, 6 months ago

Selected Answer: D

D is the hero here. Though Cloud SQL has an upper hand when it comes to transactions(OLTP), it does not autoscale its computing capabilities as compared to datastore. Do visit: https://cloud.google.com/datastore/docs/concepts/overview#what_its_good_for

hendrixlives 1 year, 6 months ago

Selected Answer: D

D is correct: Datastore (currently Firestore in native or datastore mode). It is a fully managed and serverless solution that allows for transactions and will autoscale (storage and compute) without the need to manage any infrastructure. A is wrong: Cloud SQL is fully a managed transactional DB, but only the storage grows automatically. As your user base increases, you will need to increase the CPU/memory of the instance, and to do that you must edit the instance manually (and the questions specifically says "you do not want to manage infrastructure scaling") B is wrong: Bigquery is OLAP (for analytics). NoOps, fully managed, autoscales and allows transactions, but it is not designed for this use case. C is wrong: Bigtable is a NoSQL database for massive writes, and to scale (storage and CPU) you must add nodes, so it is completely out of this use case.

kuik 4 years, 3 months ago

May be some history can help to decide which is best answer. Datastore built by Google uses BigTable as it's storage, while the company who built FireStore uses Cloud Spanner as it's storage. Google decided that they like the FireStore

technology and acquired it. If Cloud Spanner is an option I would choose it. So, D for me, although it's json storage format, but the Cloud Spanner it uses as storage fits all the requirements.

axantroff 1 year, 6 months ago

Selected Answer: D

B - not an option C - lack of ACID transactions A - lack of resource automatic scalability D - (correct, IMHO) support ACID, suitable for OLPT and scalable enough

rocky48 1 year, 6 months ago

Selected Answer: D

B - not an option C - lack of ACID transactions A - lack of resource automatic scalability D - (correct, IMHO) support ACID, suitable for OLPT and scalable enough

TVH_Data_Engineer 1 year, 6 months ago

Selected Answer: D

Cloud Datastore (now part of Google Cloud Firestore in Datastore mode) is designed for high scalability and ease of management for applications. It is a NoSQL document database built for automatic scaling, high performance, and ease of application development. It's serverless, meaning it handles the scaling, performance, and management automatically, fitting your requirement of not wanting to manage infrastructure scaling. Cloud SQL, while a fully-managed relational database service that makes it easy to set up, manage, and administer your SQL databases, is not as automatically scalable as Datastore. It's better suited for applications that require a traditional relational database.

baimus 1 year, 6 months ago

Selected Answer: A

This actually is A. I was initially in the D camp, and have spent considerable time reading about it now (circa 1 hour). D is explicitly not suitable for payment transactions, as Datastore supports ACID transactions, but only within entity groups, which are small, localized sets of data. This restriction means that transactions are not suitable for scenarios requiring multi-entity consistency across the entire database. The only two products recommended for payments in the Google ecosystem are Cloud Spanner and Cloud SQL. Is Cloud SQL managed? I'd say it wasn't really, due to the need to configure instances, but that is trumped by the fact it is the only choice suitable for a payment transaction system.

SatyamKishore 1 year, 7 months ago

Cloud Datastore is a fully managed, NoSQL document database that is highly scalable and designed to automatically handle large increases in traffic without requiring manual intervention. It's well-suited for applications with a rapidly growing user base.

iooj 1 year, 8 months ago

Selected Answer: D

Firestore extension of Datastore can handle acid transactions and allows autoscaling

jamalkhan 1 year, 9 months ago

Selected Answer: A

A. Requires Transactions.

https://zhangxuetu.online

[202610]

Question #14 Topic 1

You want to use a database of information about tissue samples to classify future tissue samples as either normal or mutated. You are evaluating an unsupervised anomaly detection method for classifying the tissue samples. Which two characteristic support this method? (Choose two.)

A. There are very few occurrences of mutations relative to normal samples. **Most Voted**

B. There are roughly equal occurrences of both normal and mutated samples in the database.

C. You expect future mutations to have different features from the mutated samples in the database. **Most Voted**

D. You expect future mutations to have similar features to the mutated samples in the database.

E. You already have labels for which samples are mutated and which are normal in the database.

Correct Answer: AC

Community vote distribution



Comments

jvg637 **Highly Voted** 6 years ago

I think that AD makes more sense. D is the explanation you gave. In the rest, A makes more sense, in any anomaly detection algorithm it is assumed a priori that you have much more "normal" samples than mutated ones, so that you can model normal patterns and detect patterns that are "off" that normal pattern. For that you will always need the no. of normal samples to be much bigger than the no. of mutated samples.

BigQuery 4 years, 3 months ago

Guys its A & C. Anomaly detection has two basic assumptions: ->Anomalies only occur very rarely in the data. (a) ->Their features differ from the normal instances significantly. (c) link -> https://towardsdatascience.com/anomaly-detection-for-dummies-15f148e559c1#:~:text=Unsupervised%20Anomaly%20Detection%20for%20Univariate%20%26%20Multivariate%20Data.&text=Anomaly%20detection%20has%20two%20basic,from%20the%20normal%20instances%20significantly.

szefco 4 years, 3 months ago

I don't agree on C. Anomaly detection assumes "Their features differ from the NORMAL INSTANCES significantly" and in the C option you have: "You expect future mutations to have different features from the MUTATED SAMPLES IN THE DATABASE". IMHO Answer D fits better: "D. You expect future mutations to have similar features to the mutated samples in the database." - in other words: Expect future anomalies to be similar to the anomalies that we already have in database

AmitK121981 1 year, 3 months ago

as per chatGPT, it can be different (C) - thats how unsupervised anomaly detection works - as long as they are different than "normal' tissues , they would be detected

jvg637 Highly Voted 6 years ago

A instead of B: "anomaly detection (also outlier detection[1]) is the identification of rare items, events or observations which raise suspicions by differing significantly from the majority of the data

BigDataBB 2 weeks, 2 days ago

Selected Answer: AC

The D answer is correct for supervisioned method. With an unsupervised anomaly detection method the correct answer is C

assiafella 2 months, 3 weeks ago

Selected Answer: AC

I think it's A & C, the anomaly detection algorithm train on "normal" data and anything that deviates from that is classified "mutated", and so there is no commun caracterstics or patterns for mutated data, this is why the statement C is correct

vosang5299 11 months, 2 weeks ago

Selected Answer: AD

A. There are very few occurrences of mutations relative to normal samples. This is a strong characteristic supporting unsupervised anomaly detection. Anomaly detection methods often work well when the "normal" class is the majority and anomalies are rare outliers. The mutated samples, being few, would likely appear as anomalies relative to the cluster of normal samples. D. You expect future mutations to have similar features to the mutated samples in the database. This characteristic supports unsupervised anomaly detection. If future mutations are expected to share similar characteristics with the existing mutated samples (even if they are rare), an anomaly detection method could potentially learn the pattern of the normal samples and flag anything significantly different (including the mutated samples) as an anomaly.

Parandhaman_Margan 1 year ago

Selected Answer: AC

Unsupervised anomaly detection. Characteristics are few anomalies and future mutations different. So A and C.

LP_PDE 1 year, 2 months ago

Selected Answer: AC

AC. Answer C. Unsupervised anomaly detection methods are particularly useful when you don't have labeled examples of the anomalies you're trying to detect. Why not D, if future mutations are similar to existing ones, a supervised model trained on labeled examples of known mutations would likely be more accurate in classifying new samples.

cqrm3n 1 year, 2 months ago

Selected Answer: AC

The answer should be A and C. Unsupervised anomaly detection is useful when labels are unavailable, or when anomalies are rare and distinct. Hence A is definitely correct because anomaly detection excels when anomalies are rare compared to normal data. I think C is correct because by adding new mutation data that is similar to the existing mutation data, the model will learn in a broader sense on what constitutes to 'mutation', and it leads to a better generalization. If the new data is too similar to the existing mutation data (answer D), the model might overfit to those specific examples. However, the new data should still share some fundamental characteristic to the existing data so that the model can recognize them as belonging to the same anomaly category.

kumar34 1 year, 3 months ago

Selected Answer: AC

I think it's A&C. For an anomaly detection model, the ratio of normal vs abnormal is expected to be high. 'C' because the model is expected to be adaptive meaning the model detects the abnormal features that can be different from the abnormal features currently being trained on.

SamuelTsch 1 year, 5 months ago

Selected Answer: AC

The keyword is unsupervised anomaly detection. So A is correct. We think and should ensure the majority of data represents 'normal'. Unsupervised methods are good for detecting unknown patterns. Thus C could be correct.

SamuelTsch 1 year, 5 months ago

I correct my answer. AD should be better. Unsupervised method is usually used for grouping the data. So, if the future mutations have similar features to the mutated samples, our trained model should group it into anomalies even though no label exists.

hendrixlives 1 year, 6 months ago

Selected Answer: AD

AD: to use unsupervised anomaly detection the anomalies a) must be rare b) they must differ from the NORMAL. So... A: mutated samples must be scarce compared to normal tissue. D: yes, we expect the future mutated samples to have similar features to the mutated samples currently in the database. Why not C? If I train my model with mutated samples with specific characteristics, I do not expect it to find different mutations. In the future, when new mutations appear, I would retrain my model including those new samples.

MaxNRG 1 year, 6 months ago

Selected Answer: AD

Anomaly detection has two basic assumptions: *Anomalies only occur very rarely in the data. *Their features differ from the normal instances significantly. Anomaly detection involves identifying rare data instances (anomalies) that come from a different class or distribution than the majority (which are simply called “normal” instances). Given a training set of only normal data, the semi-supervised anomaly detection task is to identify anomalies in the future. Good solutions to this task have applications in fraud and intrusion detection. The unsupervised anomaly detection task is different: Given unlabeled, mostly-normal data, identify the anomalies among them. https://www.science.gov/topicpages/u/unsupervised+anomaly+detection A because “Unsupervised anomaly detection techniques detect anomalies in an unlabeled test data set under the assumption that the majority of the instances in the data set are normal”, B is for Supervised anomaly detection https://en.wikipedia.org/wiki/Anomaly_detection

gudiking 1 year, 6 months ago

Selected Answer: AD

A - anomaly detection is used for detecting rare events, meaning it is expected that there are much less of those than of normal ones. D - you expect the future mutations to be similar to the mutations you already have, so that you can detect them (pattern recognition)

jkhong 1 year, 6 months ago

Selected Answer: AD

A makes sense C and D compares future mutations to mutated samples in database The question is pretty badly worded… If we were to run a full unsupervised anomaly detection over the entire dataset, C and D will be true, since some future mutations may be similar to current mutations and some will be significantly different to current mutations. The question is suggesting "labelling" tissue samples using unsupervised anomaly detection, and subsequently using the labels with a supervised algorithm to classify future samples. If this interpretation of the question is correct, then D makes sense

korntewin 1 year, 6 months ago

Selected Answer: AD

The answer should be AD. A, anomaly should have a little amount, if there are many samples then we should do classification instead, because unsupervised will give a lot of false positive. D, the future anomaly should be of the same distribution as present anomaly! or else our anomaly detection will not be generalize to the future feature.

samdhimal 1 year, 6 months ago

A. There are very few occurrences of mutations relative to normal samples. This characteristic is supportive of using an unsupervised anomaly detection method, as it is well suited for identifying rare events or anomalies in large amounts of data. By training the algorithm on the normal tissue samples in the database, it can then identify new tissue samples that have different features from the normal samples and classify them as mutated. D. You expect future mutations to have similar features to the mutated samples in the database. This characteristic is supportive of using an unsupervised anomaly detection method, as it is well suited for identifying patterns or anomalies in the data. By training the algorithm on the mutated tissue samples in the database, it can then identify new tissue samples that have similar features and classify them as mutated.

azmiozgen 1 year, 6 months ago

Selected Answer: AD

D should be correct. You expect future samples will correlate with the training samples. That's the whole point of learning procedure. If you do not expect that they have similar features, then why would you use features in the training samples in the first place? A is also correct, since anomaly labels would be seen rarely.

https://zhangxuetu.online

[202610]

Question #15 Topic 1

You need to store and analyze social media postings in Google BigQuery at a rate of 10,000 messages per minute in near real-time. Initially, design the application to use streaming inserts for individual postings. Your application also performs data aggregations right after the streaming inserts. You discover that the queries after streaming inserts do not exhibit strong consistency, and reports from the queries might miss in-flight data. How can you adjust your application design?

A. Re-write the application to load accumulated data every 2 minutes.

B. Convert the streaming insert code to batch load for individual messages.

C. Load the original message to Google Cloud SQL, and export the table every hour to BigQuery via streaming inserts.

D. Estimate the average latency for data availability after streaming inserts, and always run queries after waiting twice as long. **Most Voted**

Correct Answer: D

Community vote distribution

D (55%) A (23%) B (22%)

Comments

noob_master **Highly Voted** 1 year, 6 months ago

Selected Answer: D

Answer: D. The only that describe a way to resolve the problem, with buffering the data. (the question is possible old, the best approach would be Pub/Sub + Dataflow Streaming + Bigquery for streaming data instead near-real time)

MaxNRG **Highly Voted** 4 years, 4 months ago

B. Streams data into BigQuery one record at a time without needing to run a load job: https://cloud.google.com/bigquery/docs/reference/rest/v2/tabledata/insertAll Instead of using a job to load data into BigQuery, you can choose to stream your data into BigQuery one record at a time by using the tabledata.insertAll method. This approach enables querying data without the delay of running a load job: https://cloud.google.com/bigquery/streaming-data-into-bigquery The BigQuery Storage Write API is a unified data-ingestion API for BigQuery. It combines the functionality of streaming ingestion and batch loading into a single high-performance API. You can use the Storage Write API to stream records into BigQuery that become available for query as they are written, or to batch process an arbitrarily large number of records and commit them in a single atomic operation. Committed mode. Records are available for reading immediately as you write them to the stream. Use this mode for streaming workloads that need minimal read latency. https://cloud.google.com/bigquery/docs/write-api

Abhi16820 4 years, 4 months ago

IN THIS ALSO BIGQUERY HAS A BUFFER WHICH IT TAKES SLOWLY ANS INSERTS INTO REAL THING, WHAT YOU SAID IS HELPFULL IN REMOVING THE APPLICATION PART

MarcoDipa 4 years, 3 months ago

could you please argue?

henry1991_bg 1 week, 2 days ago

Selected Answer: D

2 minutes is ambigous. It could be only D.

BigDataBB 2 weeks, 2 days ago

Selected Answer: A

As written in GCP documentation the best approach to this kind of scenario is accumulate data ad save in batch mode: https://docs.cloud.google.com/bigquery/docs/optimize-load-jobs#batch_loading > Perform fewer, larger writes instead of many small ones. > Minimize separate write jobs to your final production table each day. > Instead of sending data to BigQuery each time new information becomes available, collect and load it into BigQuery using a single large job. So the answer is A

29d063d 3 months, 1 week ago

Selected Answer: A

Option A Works: Batch loading (even frequent batches like every 2 minutes) provides strong consistency Once a batch load completes, the data is immediately and fully available for queries

nats828 6 months, 1 week ago

Selected Answer: A

A. Loading data in batches (using batch loads instead of streaming inserts) ensures strong consistency in BigQuery. Batch loads are available for querying immediately after the load job completes. This approach is recommended when you need consistent query results right after data ingestion. Answer D (which is most voted) is a workaround, not a solution

Annie00000 9 months, 2 weeks ago

Selected Answer: A

Why this is best: Instead of inserting each message individually using streaming (which has consistency delay), you batch the data and load it using BATCH LOADS, which are strongly consistent once complete. This reduces the chance of missing in-flight data in queries and avoids unnecessary complexity. Loading in intervals (e.g. every 2 minutes) also helps control costs and improves query reliability.

ajit_j 9 months, 3 weeks ago

Selected Answer: B

not D because we want data in report in near real-time.

Rav761 1 year, 3 months ago

Selected Answer: A

To address the issue of strong consistency and ensure your reports do not miss in-flight data after streaming inserts, you should re-write the application to load accumulated data every 2 minutes (option A). Here's why: By accumulating and loading data in 2-minute intervals, you can balance between real-time data processing and ensuring data consistency. This approach allows you to process the data in manageable batches, reducing the likelihood of inconsistencies that might occur with individual streaming inserts. It maintains a near real-time analysis capability while allowing enough time for all in-flight data to be captured and accurately represented in your reports. This adjustment should help improve the reliability of your data analysis and reporting.

imrane1995 1 year, 4 months ago

Selected Answer: A

Accumulating data and loading it periodically (e.g., every 2 minutes) via batch inserts ensures strong consistency for queries. Batch loads in BigQuery allow you to avoid the latency issues inherent to streaming inserts and guarantee data availability for queries.

GHill1982 1 year, 5 months ago

Selected Answer: A

For maintaining data consistency while handling high throughput streaming inserts and subsequent aggregations in Google BigQuery, the best approach is to re-write the application to load accumulated data every 2 minutes.

fire558787 1 year, 6 months ago

"D" seems to use the typical approximate terminology of a wrong answer. "estimate the time" (how do you do that? do you do that over different times of the day?) and "wait twice as long" (who tells you that there are not a lot of cases when lag is twice as long?). Instead, "A" seems good. You don't need to show the exact results, but an approximation thereof, but you still want consistency. So an aggregation of the data every 2 minutes is a viable thing.

Parth_P 1 year, 6 months ago

Selected Answer: D

D is correct. The problem requirement is doing analytics on real-time data. You cannot do batch processing because the business requires it to be real-time even if it makes your job simpler, so B is incorrect. Other options are not streaming.

jkhong 1 year, 6 months ago

Selected Answer: D

There are assumptions over the quality of data acceptable. If slight variations of the analytics against actual can be accepted, then D would be a good choice. Many people chose B, but this also requires some form of waiting for the late data to arrive. I think a combination of D and B can be applied, but for an intial fix, delaying the aggregation queries with D seems to make more sense. If the variance is small and the some late data leakage is acceptable, and we can remain as D. If problems arise, we can always proceed to attempt B

korntewin 1 year, 6 months ago

Selected Answer: D

The streaming mode may be in pending mode or buffered mode where the streaming data is not immediately available before committing or flushing. Thus, we need to wait before the data will be available. Or else we need to switch to commited mode (which is not present in the choices).

musumusu 1 year, 6 months ago

Answer: D What to learn or look for 1. In-Flight data = (Real Time data, i.e still in streaming pipeline and not landed in BigQuery) 2. Dataflow (assume in best case) streaming pipeline is running to send data to Bigquery. Why not option B: change streaming to batch upload is not business requirement, we have to stuck to streaming and real time analysis. Option D: make bigquery run after waiting for sometime (twice here), How will you do it? - there is not setting in BQ to do it, right!. So, adjust it in your pipeline (dataflow) - For example, add Fixed window, and you want to execute aggregation query after 2 min. Code ```pipeline.apply(...) .apply(Window. <TableRow>into(FixedWindows.of(Duration.standardMinutes(2)))) .apply(BigQueryIO.writeTableRows() .to("my_dataset.my_table") ```

philli1011 2 years, 2 months ago

Answer: D I agree with the first part of the D answer, but for the second part, I don't know how they came about the 2 mins, is it from a calculation?

https://zhangxuetu.online

[202610]

Question #16 Topic 1

Your startup has never implemented a formal security policy. Currently, everyone in the company has access to the datasets stored in Google BigQuery. Teams have freedom to use the service as they see fit, and they have not documented their use cases. You have been asked to secure the data warehouse. You need to discover what everyone is doing. What should you do first?

A. Use Google Stackdriver Audit Logs to review data access. **Most Voted**

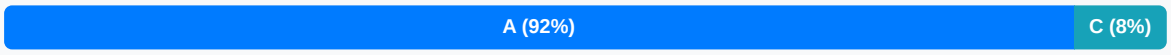
B. Get the identity and access management IIAM) policy of each table

C. Use Stackdriver Monitoring to see the usage of BigQuery query slots.

D. Use the Google Cloud Billing API to see what account the warehouse is being billed to.

Correct Answer: A

Community vote distribution



Comments

Radhika7983 **Highly Voted** 5 years, 4 months ago

Table access control is now possible in big query. However, before even checking table access control permission which is not set by the company as a formal security policy yet, we need to first understand by looking at the big query immutable audit logs as who is accessing what DAT sets and tables. Based on the information, access control policy at dataset and table level can be set. So the correct answer is A

Cloud_Student **Highly Voted** 5 years, 8 months ago

A - need to check first who is accessing which table

henry1991_bg 1 week, 2 days ago

Selected Answer: A

First step is a general diagnostic.

cqrm3n 1 year, 2 months ago

Selected Answer: A

Stackdriver Audit Logs is now called Cloud Audit logs. To secure a data warehouse, the first step is to understand how the datasets are being accessed and used. Cloud Audit logs can track data access as it provides a detailed log of all data access operations.

MaxNRG 1 year, 6 months ago

A is correct because this is the best way to get granular access to data showing which users are accessing which data. B is not correct because we already know that all users already have access to all data, so this information is

unlikely to be useful. It will also not show what users have done, just what they can do. C is not correct because slot usage will not inform security policy. D is not correct because a billing account is typically shared among many people and will only show the amount of data queried and stored
https://cloud.google.com/bigquery/docs/reference/auditlogs/#mapping-audit-entries-to-log-streams
https://cloud.google.com/bigquery/docs/monitoring#slots-available

rocky48 1 year, 6 months ago

Selected Answer: A

A. Use Google Stackdriver Audit Logs to review data access. Reviewing the audit logs provides visibility into who is accessing your data, when they are doing so, and what actions they are taking within BigQuery. This is crucial for understanding current data usage and potential security risks. Option B (getting the IAM policy of each table) is important but more focused on controlling access rather than discovering what everyone is currently doing. Option C (using Stackdriver Monitoring to see query slots usage) can help with monitoring and optimizing your BigQuery usage but doesn't provide a comprehensive view of what users are doing with the data. Option D (using the Google Cloud Billing API) is more related to tracking billing information rather than understanding what users are doing with the data.

rtcpost 1 year, 6 months ago

Selected Answer: A

To begin securing your data warehouse in Google BigQuery and gain insights into what everyone is doing with the datasets, the first step you should take is: A. Use Google Stackdriver Audit Logs to review data access. Reviewing the audit logs provides visibility into who is accessing your data, when they are doing so, and what actions they are taking within BigQuery. This is crucial for understanding current data usage and potential security risks. Option B (getting the IAM policy of each table) is important but more focused on controlling access rather than discovering what everyone is currently doing. Option C (using Stackdriver Monitoring to see query slots usage) can help with monitoring and optimizing your BigQuery usage but doesn't provide a comprehensive view of what users are doing with the data. Option D (using the Google Cloud Billing API) is more related to tracking billing information rather than understanding what users are doing with the data.

NeoNitin 1 year, 6 months ago

A. Use Google Stackdriver Audit Logs to review data access. In this scenario, you have been asked to secure the data warehouse in Google BigQuery. To do that, you first need to understand what everyone is doing with the data, i.e., who is accessing it and what actions they are performing. Google Stackdriver Audit Logs can provide you with a detailed record of all the data access and actions taken by users in Google BigQuery. It's like having a logbook that keeps track of who enters the library, which books they read, and what they do with the books. C just give how many people accessing the same dataset at given time C. Another tool you have is called "Stackdriver Monitoring." It helps you see how many people are using the library at the same time. It's like knowing how many readers are in the library at any given moment.

RT_G 1 year, 6 months ago

Selected Answer: A

A - Since the question is to discover what everyone is doing. Also the question has indicated that no security policies have been implemented.

rtcpost 1 year, 6 months ago

Selected Answer: A

A. Use Google Stackdriver Audit Logs to review data access. Reviewing the audit logs provides visibility into who is accessing your data when they are doing so, and what actions they are taking within BigQuery. This is crucial for understanding current data usage and potential security risks.

phillil011 2 years, 2 months ago

A is the answer. But recently, I think Dataplex is used for data governance .

imran79 2 years, 5 months ago

A. Use Google Stackdriver Audit Logs to review data access. Stackdriver Audit Logs provide detailed logs on who accessed what resources and when, including data in BigQuery. Reviewing these logs will give you insight into which users and service accounts are accessing datasets, what operations they are performing, and when these accesses occur. This would be a crucial first step in understanding current usage and subsequently in crafting a security policy.

suku2 2 years, 6 months ago

Selected Answer: A

Stackdriver audit logs is where we will view which datasets are being accessed by whom

bha11111 3 years ago

Selected Answer: A

In order to take a decision you need to analyze the access lofs

niketd 3 years, 1 month ago

"Discover what everyone is doing" will happen through Audit logs, hence correct answer is A

Nirca 3 years, 3 months ago

Selected Answer: A

"...to secure the data warehouse" is to list all tables/views/Mviews VS. who is accessing these objects. Slot info is not relevant.

fedebos8 3 years, 4 months ago

Selected Answer: A

A is correct.

https://zhangxuetu.online

[202610]

Question #17 Topic 1

Your company is migrating their 30-node Apache Hadoop cluster to the cloud. They want to re-use Hadoop jobs they have already created and minimize the management of the cluster as much as possible. They also want to be able to persist data beyond the life of the cluster. What should you do?

A. Create a Google Cloud Dataflow job to process the data.

B. Create a Google Cloud Dataproc cluster that uses persistent disks for HDFS.

C. Create a Hadoop cluster on Google Compute Engine that uses persistent disks.

D. Create a Cloud Dataproc cluster that uses the Google Cloud Storage connector. Most Voted

E. Create a Hadoop cluster on Google Compute Engine that uses Local SSD disks.

Correct Answer: D

Community vote distribution

D (100%)

Comments

MaxNRG Highly Voted 4 years, 4 months ago

D is correct because it uses managed services, and also allows for the data to persist on GCS beyond the life of the cluster. A is not correct because the goal is to re-use their Hadoop jobs and MapReduce and/or Spark jobs cannot simply be moved to Dataflow. B is not correct because the goal is to persist the data beyond the life of the ephemeral clusters, and if HDFS is used as the primary attached storage mechanism, it will also disappear at the end of the cluster's life. C is not correct because the goal is to use managed services as much as possible, and this is the opposite. E is not correct because the goal is to use managed services as much as possible, and this is the opposite.

certs4pk 1 year, 4 months ago

B is incorrect bcoz, it did not say 'off cluster' persistent HDFS discs???

Radhika7983 Highly Voted 5 years, 4 months ago

The correct answer is D. Here is the explanation to why Data proc and why not Data flow. When a company wants to move their existing Hadoop jobs on premise to cloud, we can simply move the jobs in cloud data prod and replace hdfs with gs:// which is google storage. This way you are keeping compute and storage separately. Hence the correct answer is D. However, if the company wants to complete create a new jobs and don't want to use the existing Hadoop jobs running on premise, the option is to create new data flow jobs.

suku2 1 year, 6 months ago

Selected Answer: D

D. Create a Cloud Dataproc cluster that uses the Google Cloud Storage connector. Dataproc clusters can be created to lift and shift existing Hadoop jobs Data stored in Google Cloud Storage extends beyond the life of a Dataproc cluster.

imran79 1 year, 6 months ago

D. Create a Cloud Dataproc cluster that uses the Google Cloud Storage connector. Here's why: Cloud Dataproc allows you to run Apache Hadoop jobs with minimal management. It is a managed Hadoop service. Using the Google Cloud Storage (GCS) connector, Dataproc can access data stored in GCS, which allows data persistence beyond the life of the cluster. This means that even if the cluster is deleted, the data in GCS remains intact. Moreover, using GCS is often cheaper and more durable than using HDFS on persistent disks.

certs4pk 1 year, 4 months ago

what if option B said, 'off cluster' persistent HDFS disks?

rtcpost 1 year, 6 months ago

Selected Answer: D

D. Create a Cloud Dataproc cluster that uses the Google Cloud Storage connector. Google Cloud Dataproc is a managed Hadoop and Spark service that allows you to easily create and manage Hadoop clusters in the cloud. By using the Google Cloud Storage connector, you can persist data in Google Cloud Storage, which provides durable storage beyond the cluster's lifecycle. This approach ensures data is retained even if the cluster is terminated, and it allows you to reuse your existing Hadoop jobs. Option B (Creating a Dataproc cluster that uses persistent disks for HDFS) is another valid choice. However, using Google Cloud Storage for data storage and processing is often more cost-effective and scalable, especially when migrating to the cloud. Options A, C, and E do not take full advantage of Google Cloud's services and the benefits of cloud-native data storage and processing with Google Cloud Storage and Dataproc.

fahadminhas 1 year, 9 months ago

Option D is incorrect, as it would not provide persistent HDFS storage within cluster itself. Rather B should be the correct answer.

kshehadyx 2 years, 6 months ago

Correct D

bha11111 3 years ago

Selected Answer: D

Hadoop --> Dataproc Persistent storage after the processing --> GCS

samdhimal 3 years, 2 months ago

Selected Answer: D

D Seems right. Cloud storage can be used to achieve data storage even after the life of cluster.

korntewin 3 years, 2 months ago

Selected Answer: D

The answer is D! Dataproc have no need for use to manage the infra and cloudstorage also no need for us to manage too!

Nirca 3 years, 3 months ago

Selected Answer: D

D. Create a Cloud Dataproc cluster that uses the Google Cloud Storage connector.

assU2 3 years, 4 months ago

Seems like it is D. https://cloud.google.com/dataproc/docs/concepts/dataproc-hdfs Never saw they mentioned persistent disks, although they are not deleted with the clusters...

assU2 3 years, 4 months ago

although: By default, when no local SSDs are provided, HDFS data and intermediate shuffle data is stored on VM boot disks, which are Persistent Disks.

assU2 3 years, 4 months ago

and it says that only VM Boot disks are deleted when the cluster is deleted.

achafill 3 years, 5 months ago

Selected Answer: D

Correct Answer : D

nkunwar 3 years, 6 months ago

Selected Answer: D

Dataproc cluster set up will be ephemeral to run HDFS Jobs and can be killed after Job execution killing persistent storage with cluster

crisimenjivar 3 years, 7 months ago

Anwer: D

Asheesh1909 3 years, 9 months ago

Isn't it A and D both dataflow for reusable jobs and gcs for data peraistance?

kmaiti 3 years, 11 months ago

Selected Answer: D

Two key points: Managed hadoop cluster - dataproc Persistent storage: GCS (dataproc uses gcs connector to connect to gcs)

https://zhangxuetu.online

[202610]

Question #18 Topic 1

Business owners at your company have given you a database of bank transactions. Each row contains the user ID, transaction type, transaction location, and transaction amount. They ask you to investigate what type of machine learning can be applied to the data. Which three machine learning applications can you use? (Choose three.)

A. Supervised learning to determine which transactions are most likely to be fraudulent.

B. Unsupervised learning to determine which transactions are most likely to be fraudulent. **Most Voted**

C. Clustering to divide the transactions into N categories based on feature similarity. **Most Voted**

D. Supervised learning to predict the location of a transaction. **Most Voted**

E. Reinforcement learning to predict the location of a transaction.

F. Unsupervised learning to predict the location of a transaction.

Correct Answer: BCD

Community vote distribution



Comments

jvg637 **Highly Voted** 6 years ago

BCD makes more sense to me. Its for sure not unsupervised, since locations are in the data already. Reinforcement also doesn't fit, as there no AI and no interactions with data from the observer.

sergio6 4 years, 7 months ago

D make sense, but i have a doubt: location is a discrete value (no regression), so a multiclass classification model should be applied ... to predict locations?

hellofrnds 4 years, 5 months ago

yes. multiclass classification model should be applied

StefanoG **Highly Voted** 1 year, 6 months ago

Selected Answer: BCD

As wrote by RP123 B - Not labelled as Fraud or not. So Unsupervised. C - Clustering can be done based on location, amount etc. D - Location is already given. So labelled. Hence supervised.

henry1991_bg 1 week, 2 days ago

Selected Answer: CDE

BASIC ML

af17139 5 months, 2 weeks ago

Selected Answer: ABC

No sense in predicting the location in a dataset that already contains it.

Parandhaman_Margan 1 year ago

Selected Answer: ABC

The three most applicable machine learning applications for analyzing bank transactions are

Bulleen 1 year, 6 months ago

BCD makes sense, but I now agree that BCE is the correct answer. Say the model predict a location, guessing US or Sweden are both wrong when the answer is Canada. But US is closer, the distance from the correct location can be used to calculate a reward. Through reinforcement learning (E) the model could guess a location with better accuracy than supervised (D).

anji007 1 year, 6 months ago

Ans: B, C and D i) Fraudulent transaction, is nothing but anomaly detection which falls under Unsupervised. ii) All transactions can be categorized using type etc - clustering algorithm. iii) Using location as a label, supervised classification can be developed to predict location.

ler_mp 1 year, 6 months ago

Selected Answer: BCD

BCD makes more sense. B, C should not be controversial. For D vs E, in this use case D fits better than usage of reinfocement learning

Kyr0 1 year, 6 months ago

Selected Answer: BCD

BCD makes more sens to me

musumusu 1 year, 6 months ago

Anwer: BCD Things to understand: Supervised learning will only predict the column that is labeled. In this case, there is not Fraud or not Fraud column inside which he will train on. So Option A, wrong. option D: Supervised learning for column (transaction location) is possible as column exist to train on. Option C: Custering N-type is possible and also an unsupervised learning to make cluster of similar pattern. Option B: Its a weaker point here, User should be able to know which clusters are fraud in history. As it doesn't give enough information about past analysis whether user knows potential frauds or not. Ignore this option, if question asked for 2 right options only.

iooj 1 year, 8 months ago

Selected Answer: ABC

Why would you need to predict a location...

Roulle 1 year, 8 months ago

Selected Answer: ACD

C and D are good for sureCliquez pour utiliser cette solution et E, F wrong for sure. Then, to choose between A and B. Both options indicate that we know which transactions are fraudulent and which are not. Indeed, in order to use unsupervised classification to determine the characteristics of fraudulent transactions, we must already know which ones are fraudulent, either because all transactions in the dataset are fraudulent, or because a variable allows us to identify them. If all transactions were fraudulent, this would probably have been specified in the statement. It is therefore more likely that the "type of transaction" variable can be used to distinguish fraudulent transactions from

others. In this case, we have a target variable to predict, enabling us to build interpretable supervised models to understand the typology of fraudulent transactions. I therefore opt for A, C and D

TVH_Data_Engineer 2 years, 4 months ago

Options B, E, and F are not as suitable for the given scenario: B. Unsupervised learning to determine which transactions are most likely to be fraudulent. Unsupervised learning, while useful for anomaly detection, might not be as effective for fraud detection without labeled data indicating which transactions are fraudulent. E. Reinforcement learning to predict the location of a transaction. Reinforcement learning is more suitable for scenarios where an agent learns to make decisions through trial and error, which doesn't seem to align with predicting transaction locations. F. Unsupervised learning to predict the location of a transaction. Unsupervised learning typically doesn't involve predicting specific values (like location) without labeled data for training. In summary, A, C, and D are the most appropriate machine learning applications for investigating the provided bank transactions dataset.

rocky48 2 years, 4 months ago

Selected Answer: BCD

Answer: BCD

Waqasghaloo 2 years, 6 months ago

Location is already given as attribite so what value is served with predicting location?

youare87 2 years, 7 months ago

A, B: Data features without the definition of fraudulent, so we can not obtain the answer even if using the unsupervise learning. C: Kmeans solve this. D: logistic regression. Just put the location into target. E: Give the positive reward when the model predicts correct location. F: Same as C. Use all features but locations, and use similarity to predict new data.

xiaofeng_0226 2 years, 7 months ago

Selected Answer: BCD

Absolutely

https://zhangxuetu.online

[202610]

Question #19 Topic 1

Your company's on-premises Apache Hadoop servers are approaching end-of-life, and IT has decided to migrate the cluster to Google Cloud Dataproc. A like-for- like migration of the cluster would require 50 TB of Google Persistent Disk per node. The CIO is concerned about the cost of using that much block storage. You want to minimize the storage cost of the migration. What should you do?

A. Put the data into Google Cloud Storage. **Most Voted**

B. Use preemptible virtual machines (VMs) for the Cloud Dataproc cluster.

C. Tune the Cloud Dataproc cluster so that there is just enough disk for all data.

D. Migrate some of the cold data into Google Cloud Storage, and keep only the hot data in Persistent Disk.

Correct Answer: A

Community vote distribution



Comments

anji007 **Highly Voted** 1 year, 6 months ago

Ans: A B: Wrong eVM wont solve the problem of larger storage prices. C: May be, but nothing mentioned in terms of what to tune in the question, also this is like-for-like migration so tuning may not be part of the migration. D: Again, this is like-for-like so need to define what is hot data and which is cold data, also persistent disk costlier than cloud storage.

henry1991_bg 1 week, 2 days ago

Selected Answer: A

The company is coming to an end

29d063d 3 months, 1 week ago

Selected Answer: A

Ans A Best Practice: Use Cloud Storage as the primary storage layer for Dataproc, with minimal Persistent Disk only for temporary processing and shuffle data. This is the cloud-native architecture Google recommends.

Bugnumber1 7 months ago

Selected Answer: A

First I selected D, it just made sense right? It's faster to access data on the persistent disk. But then there are a couple of things to take into account: - The question wants you to minimise cost. You only do this 100% on "A", "D" is still costing you a bit more. - Performance, speed of access... is never mentioned. Common sense? Yes, but no one is asking you about that. - In general Google Cloud's approach to Big Data architecture is to separate compute and storage, because of all the improvements and resilience GCS has. - "D" is not very direct, has additional effort, and

you could consider that cost. - People mention the "like for like" section of the question. I'd argue like for like is both having a persistent disk, or having all data in one point. I'd not really consider it a point. So all in all, A. It's just simpler and cheaper. Also, Gemini said so ;)

fassil 11 months, 3 weeks ago

Selected Answer: A

A like-for-like migration to Cloud Dataproc that replicates on-premises Hadoop would require each node to have 50 TB of persistent disk, which is costly. Instead, you can minimize storage costs by leveraging Google Cloud Storage (GCS). Cloud Dataproc seamlessly integrates with GCS through the Hadoop connector, allowing you to store your data cost-effectively in Cloud Storage and run ephemeral clusters that read data directly from GCS. This approach eliminates the need for each node to carry 50 TB of expensive persistent disk storage while still supporting your Hadoop workload.

Parandhaman_Margan 1 year ago

Selected Answer: D

. Migrate some of the cold data into Google Cloud Storage, and keep only the hot data in Persistent Disk. Google Cloud Storage (GCS) is a cost-effective alternative to Persistent Disk for storing less frequently accessed ("cold") data. Hot data that requires fast access can remain on Persistent Disk, reducing storage costs while maintaining performance. Cloud Dataproc supports HDFS-to-GCS integration, allowing Hadoop jobs to access data in GCS seamlessly.D

Vullibabu 2 years, 2 months ago

You are most of the people looking at like for like migration would require 50TB persistent storage but missing to look at CIO concern about cost of block storage...considering CIO concern the option here is cloud storage... moreover that is recommended as well ..

imran79 2 years, 5 months ago

Option A: Put the data into Google Cloud Storage. This is the best option. Google Cloud Dataproc is designed to work well with Google Cloud Storage. Using GCS instead of Persistent Disk can save money, and GCS offers advantages such as higher durability and the ability to share data across multiple clusters.

emmylou 2 years, 6 months ago

I have seen this question in other places and I believe that you store the older data in Cloud Storage and retain processing data in persistent disk. D

hxy8 2 years, 6 months ago

Answer: D Question: A like-for- like migration of the cluster would require 50 TB of Google Persistent Disk per node. which means Persistent is still required.

suku2 2 years, 6 months ago

Google Cloud Storage is designed for 11 9's availability. So it is also kind of persistent storage. Also, it is a Google product, hence recommended. https://cloud.google.com/storage/docs/availability-durability#key-concepts

GHOST1985 2 years, 7 months ago

the question is talking about block storage , GCS is object storage !

hjava_111 2 years, 8 months ago

Selected Answer: A

GCS is cost-effective and also Google's recommendation!

bha11111 3 years ago

Selected Answer: A

Minimize cost then GCS

Nirca 3 years, 2 months ago

Selected Answer: A

A - is the right answer.

DGames 3 years, 3 months ago

Selected Answer: A

A - dataproc - storage - cost effective is cloud storage

devaid 3 years, 5 months ago

Selected Answer: A

Cloud Storage

sankar_s 3 years, 10 months ago

Selected Answer: A

Cloud Storage is google recommended one

https://zhangxuetu.online

[202610]

Question #20 Topic 1

You work for a car manufacturer and have set up a data pipeline using Google Cloud Pub/Sub to capture anomalous sensor events. You are using a push subscription in Cloud Pub/Sub that calls a custom HTTPS endpoint that you have created to take action of these anomalous events as they occur. Your custom HTTPS endpoint keeps getting an inordinate amount of duplicate messages. What is the most likely cause of these duplicate messages?

A. The message body for the sensor event is too large.

B. Your custom endpoint has an out-of-date SSL certificate.

C. The Cloud Pub/Sub topic has too many messages published to it.

D. Your custom endpoint is not acknowledging messages within the acknowledgement deadline. Most Voted

Correct Answer: D

Community vote distribution

D (87%) B (13%)

Comments

jvg637 Highly Voted 6 years ago

The Answer should be D. The custom endpoint is not acknowledging the message, that is the reason for Pub/Sub to send the message again and again. Not B.

MauryaSushil Highly Voted 5 years, 12 months ago

D : Doubt should be only between B & D. But B is not possible because if SSL is expired then endpoint URL will not receive any messages forget about duplicates. So It should be D for Duplicates.

henry1991_bg 1 week, 2 days ago

Selected Answer: D

Can't be problem of the sub

29d063d 3 months, 1 week ago

Selected Answer: D

D. Your custom endpoint is not acknowledging messages within the acknowledgement deadline Why This Causes Duplicate Messages How Pub/Sub Delivery Works: Pub/Sub delivers a message to your HTTPS endpoint Your endpoint must return an HTTP success status (200, 201, 202, or 204) within the acknowledgement deadline (default is 10 seconds) If Pub/Sub doesn't receive acknowledgement in time, it assumes delivery failed Pub/Sub redelivers the message, causing duplicates

atnafu2020 1 year, 6 months ago

D Why are there too many duplicate messages? Pub/Sub guarantees at-least-once message delivery, which means that occasional duplicates are to be expected. However, a high rate of duplicates may indicate that the client is not acknowledging messages within the configured ack_deadline_seconds, and Pub/Sub is retrying the message delivery. This can be observed in the monitoring metrics pubsub.googleapis.com/subscription/pull_ack_message_operation_count for pull subscriptions, and pubsub.googleapis.com/subscription/push_request_count for push subscriptions. Look for elevated expired or webhook_timeout values in the /response_code. This is particularly likely if there are many small messages, since Pub/Sub may batch messages internally and a partially acknowledged batch will be fully redelivered.

ganesh2121 1 year, 6 months ago

D is correct As per google docs- When you do not acknowledge a message before its acknowledgement deadline has expired, Pub/Sub resends the message. As a result, Pub/Sub can send duplicate messages. Use Google Cloud's operations suite to monitor acknowledge operations with the expired response code to detect this condition

Radhika7983 1 year, 6 months ago

The correct answer is D. Look for the link https://cloud.google.com/pubsub/docs/faq Why are there too many duplicate messages? Pub/Sub guarantees at-least-once message delivery, which means that occasional duplicates are to be expected. However, a high rate of duplicates may indicate that the client is not acknowledging messages within the configured ack_deadline_seconds, and Pub/Sub is retrying the message delivery. This can be observed in the monitoring metrics pubsub.googleapis.com/subscription/pull_ack_message_operation_count for pull subscriptions, and pubsub.googleapis.com/subscription/push_request_count for push subscriptions. Look for elevated expired or webhook_timeout values in the /response_code. This is particularly likely if there are many small messages, since Pub/Sub may batch messages internally and a partially acknowledged batch will be fully redelivered. Another possibility is that the subscriber is not acknowledging some messages because the code path processing those specific messages fails, and the Acknowledge call is never made; or the push endpoint never responds or responds with an error.

MaxNRG 1 year, 6 months ago

D as the Cloud Pub/Sub will deliver duplicate messages only if there has been no acknowledgement from the subscriber. Refer GCP documentation - Cloud Pub/Sub FAQs - Duplicates: https://cloud.google.com/pubsub/docs/faq#duplicates Why are there too many duplicate messages? Cloud Pub/Sub guarantees at-least-once message delivery, which means that occasional duplicates are to be expected. However, a high rate of duplicates may indicate that the client is not acknowledging messages within the configured ack_deadline_seconds, and Cloud Pub/Sub is retrying the message delivery.

crismo04 1 year, 6 months ago

Selected Answer: D

If the answer had been B the messages had not reached the HTTP page since the push subscription requires an SSL certificate (https://cloud.google.com/pubsub/docs/push#:~:text=Endpoint%20URL%20(required)), so the answer must be D. I think it is due to a bad response due to overcharge in the HTTPS page, which is sending a non correct status code (https://cloud.google.com/pubsub/docs/push#:~:text=VPC%20Service%20Controls.-,Receive%20messages,-When%20Pub/Sub)

rtcpost 1 year, 6 months ago

Selected Answer: D

In Google Cloud Pub/Sub, when you use a push subscription, messages are delivered to the specified endpoint (in this case, your custom HTTPS endpoint). The acknowledgment deadline is the time given to the endpoint to acknowledge that it has received and processed the message. If the acknowledgment is not received within the deadline, Pub/Sub may consider the message as unacknowledged and may attempt redelivery, which can lead to

duplicate messages. You should ensure that your custom HTTPS endpoint acknowledges messages within the acknowledgment deadline to prevent duplicate messages from being sent. Additionally, it's essential to handle messages in an idempotent way, so even if duplicates do occur, the action taken by your endpoint doesn't have unintended consequences.

petergjohnson 1 year, 6 months ago

Selected Answer: B

After re-reading the question it seem to me that it is asking for a root cause. It is possible that the most common cause of this symptom is and expired certificate. Once expired duplicates would be received for every message.

VictorBa 1 year, 10 months ago

Selected Answer: D

Agree with previous explanations regarding validity of D

searching4alicense 2 years, 1 month ago

D - If a message has not been acknowledged within its acknowledgement deadline, Dataflow attempts to maintain the lease on the message by repeatedly extending the acknowledgement deadline to prevent redelivery from Pub/Sub. However this is best effort and there is a possibility that messages may be redelivered. This can be monitored using metrics listed here. https://cloud.google.com/blog/products/data-analytics/handling-duplicate-data-in-streaming-pipeline-using-pubsub-dataflow

philli1011 2 years, 2 months ago

D should be the answer. If acknowlegement is not received back to pub/sub , pub/sub may resend meassages.

imran79 2 years, 5 months ago

The correct answer is: D. Your custom endpoint is not acknowledging messages within the acknowledgement deadline.

emmylou 2 years, 6 months ago

if there were an out of date certificate then nothing would get through. D

FP77 2 years, 8 months ago

Selected Answer: D

It should be D https://cloud.google.com/pubsub/docs/troubleshooting#dupes

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #21 Topic 1

Your company uses a proprietary system to send inventory data every 6 hours to a data ingestion service in the cloud. Transmitted data includes a payload of several fields and the timestamp of the transmission. If there are any concerns about a transmission, the system re-transmits the data. How should you deduplicate the data most efficiency?

A. Assign global unique identifiers (GUID) to each data entry. **Most Voted**

B. Compute the hash value of each data entry, and compare it with all historical data.

C. Store each data entry as the primary key in a separate database and apply an index.

D. Maintain a database table to store the hash value and other metadata for each data entry.

Correct Answer: A

Community vote distribution



Comments

dg63 **Highly Voted** 5 years, 9 months ago

The best answer is "A". Answer "D" is not as efficient or error-proof due to two reasons 1. You need to calculate hash at sender as well as at receiver end to do the comparison. Waste of computing power. 2. Even if we discount the computing power, we should note that the system is sending inventory information. Two messages sent at different can denote same inventory level (and thus have same hash). Adding sender time stamp to hash will defeat the purpose of using hash as now retried messages will have different timestamp and a different hash. if timestamp is used as message creation timestamp than that can also be used as a UUID.

retax 5 years, 5 months ago

If the goal is to ensure at least ONE of each pair of entries is inserted into the db, then how is assigning a GUID to each entry resolving the duplicates? Keep in mind if the 1st entry fails, then hopefully the 2nd (duplicate) is successful.

MarcoDipa 4 years, 3 months ago

Answer is D. Using Hash values we can remove duplicate values from a database. Hash values will be same for duplicate data and thus can be easily rejected. Obviously you won't check hash for timestmp. D is better thatn B because maintaning a different table will reduce cost for hash computation for all historical data

Mathew106 2 years, 8 months ago

Why can't it be A, where the GUID is a hash value? Why do we need to store the hash with the metadata in a separate database to do the deduplication?

ralf_cc 4 years, 9 months ago

A - In D, same message with different timestamp will have different hash, though the message content is the same.

omakin 4 years, 8 months ago

Strong Answer is A - in another question on the gcp sample questions: the correct answer to that particular question was "You are building a new real-time data warehouse for your company and will use BigQuery streaming inserts. There is no guarantee that data will only be sent in once but you do have a unique ID for each row of data and an event timestamp. You want to ensure that duplicates are not included while interactively querying data. Which query type should you use?" This means you need a "uniqueid" and timestamps to properly dedupe a data.

Tanzu 4 years, 2 months ago

U need a uniqueid but in this scenario, there is none. So u have to calculate by hashing w/ some of the fields in the dataset. A is assigning guid in processing side will not solve the issue. Cause u will assign diff. ids...

cetanx 3 years, 2 months ago

Answer - D Key statement is "Transmitted data includes a payload of several fields and the timestamp of the transmission." So the timestamp is appended to message while sending, in other words that field is subject to change if message is retransmitted. However, adding a GUID doesn't help much because if message is transmitted twice you will have different GUID for both messages but they will be the same/duplicate data. You can simply calculate a hash based on not all data but from a select of columns (with the payload of several fields AND definitely by excluding the timestamp). By doing so, you can assure a different hash for each message.

MaxNRG 4 years, 2 months ago

agreed, the key here is "payload of several fields and the timestamp"

MaxNRG 4 years, 2 months ago

"payload of several fields and the timestamp of the transmission"

BigDataBB 4 years, 1 month ago

Hi Max, I also think that the hash value would be worng because the timestamp is part of payload and is not written that the hash value is generated without the ts; but it also not written if GUID is linked or not with sending. Often this is a point where the answer is vague. Because don't specify if the GUID is related to the data or to the send.

emmylou 2 years, 6 months ago

If you add a unique ID aren't you by definition not getting a duplicate record. Honestly I hate all these answers.

billalltf 1 year, 10 months ago

You can add a function or condition that verifies if the global unique id already exists or just do a deduplication later

medeis_jar Highly Voted 4 years, 2 months ago

Selected Answer: A

Transmitted data includes fields and timestamp of transmission. So, hash value changes with re-transmission ==> Option B & D are wrong.

Tanzu 4 years, 2 months ago

U dont have to put timestamp in your hash algorithm.

JK007 4 years ago

But this question is not saying how you are calculating the hash algo. It says hash of "data entry" which includes other fields and timestamp field. So taking hash of the entire data entry the hash value will be different as timestamp will be different each time.

exnaniantwort 4 years, 2 months ago

The clearest explanation why D is wrong.

Tanzu 4 years, 2 months ago

Hashing is inevitable. So only b or d is appropriate. The main diff. b/w them is persistence. At least 6+6 hours, u need to persist the hashes in either storage or a table. The is no persistence in b. So d is more ok then ..

arwa_eiad 3 months ago

Selected Answer: D

Why not the others: A. GUIDs: You can only assign GUIDs if the source system supports it. Here, the system retransmits without changing the payload, so GUIDs won't help unless added upstream. B. Compute hash and compare with all historical data: This is inefficient because you'd need to scan all previous entries every time. C. Store each data entry as the primary key: Payloads can be large and variable, making them unsuitable as primary keys. Also, indexing on large text fields is costly.

lmch 3 months, 2 weeks ago

Selected Answer: D

Option D correctly specifies that the hash value must be stored in a database table (or a high-speed key/value store like Redis/Memcached) with an index applied to it. This allows the ingestion service to perform a single, fast primary key lookup using the hash.

Pulkit2706 6 months, 3 weeks ago

Selected Answer: A

The correct answer is: Assign global unique identifiers (GUID) to each data entry. This method allows highly efficient deduplication because each entry carries a unique ID, making it easy to identify and ignore duplicates during ingestion or later analysis. Assigning a GUID to each entry ensures that every record—even across re-transmissions—has a unique identifier. At ingestion, the deduplication system can simply filter by GUID, drastically reducing the need for costly comparisons or hash storage. This approach avoids computational overhead and is scalable for frequent data transmissions. Alternative methods like comparing hashes with all historical data, indexing the full data payload, or maintaining tables of hashes and metadata are less efficient due to increased computational, storage, or operational complexity. Using GUIDs is the simplest and most scalable solution for deduplication in this scenario.

Bugnumber1 7 months ago

Selected Answer: D

This question is bullsh*t, because the correct answer would be "A", but for that you have to imagine that this GUID gets added to the original message. Instead, it says "assign". For me that sounds like when the message arrives, you assign it an ID, which is utterly useless for deduplication. D is not efficient, but is the most efficient of the bunch here, plus, it's not that different from what "A" does. In fact, once you do "A", you still have to store that in a database and look it up, but it doesn't say that anywhere in the answer. Me? I believe "A" it's a trap. But as we'll never get official confirmation, feel free to believe it's just a question where you assume "A" has all the magic things that make deduplication happen.

1479 7 months, 1 week ago

Selected Answer: B

Maintain a table with hash can be painfull and we can write it in hystorical data.

Surabhi20 7 months, 2 weeks ago

Selected Answer: A

timestamp may be changed while re-transmitting the same message

jsg 8 months, 1 week ago

Selected Answer: D

Answer-D , A- Assign GUIDs: This only works if the source system can guarantee unique IDs per original transmission. But if re-transmissions recreate the GUID, or if the system doesn't assign one, this fails.

Annie00000 9 months, 2 weeks ago

Selected Answer: A

Assigning a GUID (Globally Unique Identifier) at the SOURCE system to each payload ensures: Idempotency: You can identify if a record has already been processed. Efficiency: A simple lookup on the GUID is much faster than comparing hashes or checking entire payloads. Scalability: Works well even when millions of records are transmitted.

1479 7 months, 1 week ago

When you compare GUID, your system will hash it first ... Compute can only copare digits

22c1725 10 months, 2 weeks ago

Selected Answer: B

Use hashing (answer B) for efficient, scalable, and fast deduplication — it's a well-established best practice in streaming and batch data pipelines.

1479 7 months, 1 week ago

Why maintain an hastable as you can add it directly in a new column ? B better than D. GUID is poorly designed as it need to hash the GUID's to compare it ... waste of compute

22c1725 10 months, 2 weeks ago

Selected Answer: D

Even If you defien GUID. How will the source know about it? he hold no such ID.

fassil 11 months, 3 weeks ago

Selected Answer: D

A is incorrect. how can you find duplicates if you assign a unique id to every record? The answer is D.

Mo5454545454 12 months ago

Selected Answer: D

The most efficient way to deduplicate your inventory data would be: D. Maintain a database table to store the hash value and other metadata for each data entry. This approach is optimal because: It creates a lightweight reference table that stores just the hash values and essential metadata (like timestamps) rather than the full payload data Hash values can be quickly compared to identify duplicates without expensive full-data comparisons The metadata can help with auditing and troubleshooting transmission issues This solution scales well as your data volume grows Option A (using GUIDs) doesn't address the retransmission scenario well, as new GUIDs might be generated each time. Option B requires comparing against all historical data, which becomes increasingly inefficient over time. Option C creates unnecessary storage overhead by using entire data entries as primary keys when only a hash value is needed for comparison.

Parandhaman_Margan 1 year ago

Selected Answer: D

Deduplicate data with retransmissions. Use a database table with hash

Abizi 1 year ago

Selected Answer: A

most obvious answer

Rav761 1 year, 3 months ago

Selected Answer: D

Option D: Maintain a database table to store the hash value and other metadata for each data entry. This approach is efficient and scalable. By storing a computed hash value (as a compact representation of the data) along with metadata, deduplication can be performed by comparing new entries with the stored hashes. This minimizes storage requirements and improves lookup efficiency.

https://zhangxuetu.online

[202610]

Question #22 Topic 1

Your company has hired a new data scientist who wants to perform complicated analyses across very large datasets stored in Google Cloud Storage and in a
Cassandra cluster on Google Compute Engine. The scientist primarily wants to create labelled data sets for machine learning projects, along with some visualization tasks. She reports that her laptop is not powerful enough to perform her tasks and it is slowing her down. You want to help her perform her tasks.
What should you do?

A. Run a local version of Jupiter on the laptop.

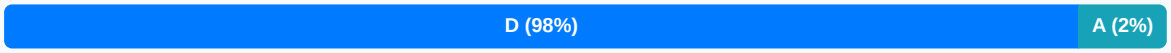
B. Grant the user access to Google Cloud Shell.

C. Host a visualization tool on a VM on Google Compute Engine.

D. Deploy Google Cloud Datalab to a virtual machine (VM) on Google Compute Engine. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

Rajokkiyam **Highly Voted** 5 years, 6 months ago
Answer should be D.

rickywck **Highly Voted** 5 years, 6 months ago
Obviously the answer is D

Abizi 1 year ago
Selected Answer: D
D is the right answer

VictorBa 1 year, 4 months ago
Selected Answer: D
Google Cloud Datalab is a powerful interactive tool for data exploration, analysis, and machine learning.

trashbox 1 year, 4 months ago
Selected Answer: D
My answer is Google Cloud Datalab, but since that service has already been discontinued, I question whether a problem like this would actually be asked on the actual exam.

Surabhi20 7 months, 2 weeks ago

The intention is to build the logic which will be ofcourse always helpful for any exam

GCanteiro 1 year, 8 months ago

Selected Answer: D

D sounds good for me

TVH_Data_Engineer 1 year, 9 months ago

Selected Answer: A

Hash Value for Deduplication: By computing a hash value for each data entry, you create a unique identifier based on the content of the data. This allows you to efficiently identify duplicates, as entries with identical content will have the same hash value. Storing Hash Value and Metadata: Maintaining a database table that includes the hash value and other relevant metadata (like the timestamp of transmission) allows for quick lookups and comparisons. This way, when new data is received, you can check if an entry with the same hash value already exists. Assign global unique identifiers (GUID) to each data entry: While GUIDs are unique, they do not inherently identify duplicate content. Two transmissions of the same data would have different GUIDs.

Surabhi20 7 months, 2 weeks ago

Wrong question palette

simpa17 1 year, 1 month ago

You mistakenly answered the question above haha

axantroff 1 year, 10 months ago

Selected Answer: D

D sounds good for me

RT_G 1 year, 10 months ago

Selected Answer: D

Agree with D

rtcpost 1 year, 11 months ago

Selected Answer: D

D. Deploy Google Cloud Datalab to a virtual machine (VM) on Google Compute Engine. Google Cloud Datalab is a powerful interactive tool for data exploration, analysis, and machine learning. By deploying it to a VM on Google Compute Engine, you can provide her with a robust and scalable environment where she can work with large datasets, create labeled datasets, and perform data analyses efficiently. Option A (running a local version of Jupyter on her laptop) might not be sufficient for very large datasets, and her laptop's limited power could still be a bottleneck. Option B (granting access to Google Cloud Shell) is useful for running command-line tools but may not provide the interactive and visual capabilities she needs. Option C (hosting a visualization tool on a VM on Google Compute Engine) is beneficial for visualization tasks but does not cover the full spectrum of data analysis and machine learning tasks that Google Cloud Datalab offers.

gudguy1a 2 years ago

Selected Answer: D

D - as it is a FULL set up, not a shell that is needed...

sergiomujica 2 years, 1 month ago

Nowadays it should be similar to D, deploy a Vertex workbench

yash12 2 years, 1 month ago

As per Options , Correct Answer should be D. ie Datalab However Datalab is no longer used in GCP (Deprecated in Sep2022), It is Vertex AI or Deep Learning VM Images

HeoMaTo 2 years, 1 month ago

Selected Answer: D

I think. Answer is D

Acocado 2 years, 2 months ago

Datalab is deprecated. This question should appear in the exam.

Acocado 2 years, 2 months ago

typo- should NOT appear in the exam

axantroff 1 year, 11 months ago

Good point - https://cloud.google.com/datalab/deprecation-notice. Google recommends using Vertex AI Workbench instead

dgteixeira 2 years, 3 months ago

Selected Answer: D

Should be D, because Cloud shell alone does not provide access to what they need. Nowadays is Vertex AI, but still, correct answer is D

Maurilio_Cardoso 2 years, 4 months ago

Selected Answer: D

Google Cloud Datalab is now Vertex AI. So, letter D make more sense.

https://zhangxuetu.online

[202610]

Question #23 Topic 1

You are deploying 10,000 new Internet of Things devices to collect temperature data in your warehouses globally. You need to process, store and analyze these very large datasets in real time. What should you do?

A. Send the data to Google Cloud Datastore and then export to BigQuery.

B. Send the data to Google Cloud Pub/Sub, stream Cloud Pub/Sub to Google Cloud Dataflow, and store the data in Google BigQuery. **Most Voted**

C. Send the data to Cloud Storage and then spin up an Apache Hadoop cluster as needed in Google Cloud Dataproc whenever analysis is required.

D. Export logs in batch to Google Cloud Storage and then spin up a Google Cloud SQL instance, import the data from Cloud Storage, and run an analysis as needed.

Correct Answer: B

Community vote distribution

B (100%)

Comments

DataExpert **Highly Voted** 4 years, 9 months ago

B is more correct than other options so B is the answer. But if this is actual use case you have to deal with use Cloud BigTable instead of bigquery. So the pipeline will be like this. IOT-Devices -> Cloud Pub/Sub -> Cloud BigTable -> Cloud Data Studio (For real-time analytics)

Abizi 1 year ago

Selected Answer: B

B is the correct answer

regal_2010 1 year, 5 months ago

Selected Answer: B

Answer is B

axantroff 1 year, 10 months ago

Selected Answer: B

In short, B is less complex and more recommended other than D

rtcpost 1 year, 11 months ago

Selected Answer: B

B. Send the data to Google Cloud Pub/Sub, stream Cloud Pub/Sub to Google Cloud Dataflow, and store the data in Google BigQuery. Here's why this approach is preferred: Google Cloud Pub/Sub allows for efficient ingestion and real-time data streaming. Google Cloud Dataflow can process and transform the streaming data in real-time. Google BigQuery is a fully managed, highly scalable data warehouse that is well-suited for real-time analysis and querying of large datasets.

GCP_PDE_AG 2 years, 1 month ago

Obviously B.

Maurilio_Cardoso 2 years, 4 months ago

Selected Answer: B

PubSub for queue in real time, Dataflow for processing (pipeline) and Bigquery for analyses.

bha11111 2 years, 6 months ago

Selected Answer: B

B is correct

DGames 2 years, 9 months ago

Selected Answer: B

GCP recommend best practice for streaming data pipeline as option B - pub/sub, dataflow & Bigquery

Nirca 2 years, 9 months ago

Selected Answer: B

B. Send the data to Google Cloud Pub/Sub, stream Cloud Pub/Sub to Google Cloud Dataflow, and store the data in Google BigQuery.

gitaexams 2 years, 10 months ago

B aris tqve yleebo

devaid 2 years, 11 months ago

Selected Answer: B

B of course

Dip1994 3 years, 1 month ago

B is the correct answer

noob_master 3 years, 3 months ago

Selected Answer: B

Answer: B Deafult ETL streaming process: Pub/Sub + Dataflow + Bigquery.

nexus1_ 3 years, 3 months ago

Definitely B

vw13 3 years, 5 months ago

Selected Answer: B

B is the only option for real time process & analysis

devric 3 years, 5 months ago

The most appropriate is B but BQ can't solve Analyzing data in RT.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #24 Topic 1

You have spent a few days loading data from comma-separated values (CSV) files into the Google BigQuery table CLICK_STREAM. The column DT stores the epoch time of click events. For convenience, you chose a simple schema where every field is treated as the STRING type. Now, you want to compute web session durations of users who visit your site, and you want to change its data type to the TIMESTAMP. You want to minimize the migration effort without making future queries computationally expensive. What should you do?

A. Delete the table CLICK_STREAM, and then re-create it such that the column DT is of the TIMESTAMP type. Reload the data.

B. Add a column TS of the TIMESTAMP type to the table CLICK_STREAM, and populate the numeric values from the column TS for each row. Reference the column TS instead of the column DT from now on.

C. Create a view CLICK_STREAM_V, where strings from the column DT are cast into TIMESTAMP values. Reference the view CLICK_STREAM_V instead of the table CLICK_STREAM from now on.

D. Add two columns to the table CLICK STREAM: TS of the TIMESTAMP type and IS_NEW of the BOOLEAN type. Reload all data in append mode. For each appended row, set the value of IS_NEW to true. For future queries, reference the column TS instead of the column DT, with the WHERE clause ensuring that the value of IS_NEW must be true.

E. Construct a query to return every row of the table CLICK_STREAM, while using the built-in function to cast strings from the column DT into TIMESTAMP values. Run the query into a destination table NEW_CLICK_STREAM, in which the column TS is the TIMESTAMP type. Reference the table NEW_CLICK_STREAM instead of the table CLICK_STREAM from now on. In the future, new data is loaded into the table NEW_CLICK_STREAM. **Most Voted**

Correct Answer: E

Community vote distribution



Comments

jvg637 **Highly Voted** 6 years ago

"E" looks better. For D, the database will be double in size (which increases the storage price) and the user has to spend some more days reloading all the data.

jkhong 3 years, 3 months ago

Also D doesn't make sense since we're filtering IS_NEW to true to only consider future data, which disregards our previously loaded data

assU2 3 years, 4 months ago

"You want to minimize the migration effort without making future queries computationally expensive." Nothing about storage price.

StelSen Highly Voted 5 years, 2 months ago

I would go for Option.E Reason: 1. Question says, I want to change data type to the TIMESTAMP. So there is change required 2. To minimize migration effort without making future queries computationally expensive: No view required as I will run this query again and again. In order to convert string to Timestamp we need to use CAST function for sure. Simple copy of numeric value won't work. https://cloud.google.com/bigquery/docs/reference/standard-sql/conversion_rules Option-A: I don't want to redo this and spent few days again Option-B: Copying numeric value won't work. Option-C: View is expensive to run again and again. Option-D: To me this seems onetime data load which took 3 days. So, the IS_NEW seems give us reduntant data as we are adding appended mode. Option-E: Technically it will work, although it will add more space (new table) ,cost. But no choice, because Option-B didn't mention Cast function. Else I would have gone for "B"

Bugnumber1 7 months ago

Selected Answer: E

It's E. C looks good because it's very easy and straightforward. But every time you query that view, a transformation will take place, making queries expensive, and it will be forever like this as the original table doesn't change. One time effort of loading the data into a new table with a small query. Heck you can even rename it right after if you don't want to change the reference of the ingestion.

Parandhaman_Margan 1 year ago

Selected Answer: C

Changing DT to TIMESTAMP. Create a view casting the string. Answer C.

Abizi 1 year ago

Selected Answer: E

I was hesitating between C and E, but E seems to be the good one

LP_PDE 1 year, 2 months ago

Selected Answer: C

Both options C (creating a view) and E (creating a new table) avoid reloading the original data. However, C is the better choice for minimizing effort and maintaining performance. Views don't store any data themselves. They simply act as a layer on top of the existing table. This means you avoid the cost of storing duplicate data, which can be significant for large tables.

56d02cd 9 months, 1 week ago

But the cast of column DT from string to timestamp is executed on every query against the view making it more expensive, which is what we awant to avoid. I think E is the better solution.

Nittin 1 year, 7 months ago

Selected Answer: C

Create a view no data migration easy to do but computational efficient queries not sure (?)

mark1223jkh 1 year, 10 months ago

E. It recreates the table one time and everything is fixed. Next time you load, load to the new table, you can delete the previous one. Definitely not C. The question says I have to minimize future query effort, which literally means "don't create a view that converts from STR to TIMESTAMP for every row."

suwalsageen12 1 year, 10 months ago

Selected Answer: E

Option E is correct. The question is asking to consider the Query cost for future. This is a one time job to fix the Timestamp column. no views were created.

teka112233 1 year, 11 months ago

Selected Answer: E

Why Option E is the best choice: It modifies the schema with minimal data movement. The original table remains untouched for potential future needs. Future data loads can directly go to the new table with the desired schema. Queries referencing the new table (NEW_CLICK_STREAM) will benefit from the optimized data type for timestamp operations.

GYORK 2 years ago

Selected Answer: C

minimizing effort is key.

TVH_Data_Engineer 2 years, 3 months ago

Selected Answer: C

A view in Google BigQuery is a virtual table defined by a SQL query. By creating a view that casts the DT column as a TIMESTAMP, you can transform the data format without altering the underlying data in the CLICK_STREAM table. This means you don't have to reload any data, thereby minimizing migration effort.

apoio.certificacoes.closer 1 year, 3 months ago

Depends. If you create a materialized view, that tracks. If it's not a materialized view, the underlying query will run every time there's a query against the view.

axantroff 2 years, 4 months ago

Selected Answer: E

Good point about the logical views and the desire to reduce costs. I would vote for E

mk_choudhary 2 years, 5 months ago

The best way to minimize the migration effort without making future queries computationally expensive is to create a view and reference it instead of the table. This is because views are materialized when they are queried, so they do not incur any additional overhead. So the answer is (C).

brokeasspanda 2 years, 4 months ago

C doesn't say materialized view, there's a difference with a regular view so it'll be slower and more expensive on every call to that view.

rtcpost 2 years, 5 months ago

Selected Answer: E

Option "E" It avoids the need to delete and recreate the entire CLICK_STREAM table, which is time-consuming and requires reloading all data. It allows you to use a simple query to cast the existing DT column as TIMESTAMP values and store the results in a new table, NEW_CLICK_STREAM. You can gradually migrate to the new data format, and your future queries will be able to utilize the TIMESTAMP data type for more efficient processing.

sergiomujica 2 years, 7 months ago

Option D duplicates, not a good solution

NeoNitin 2 years, 8 months ago

E. E. You can use a special command to change the time on the old cards to the better type "TIMESTAMP" and create a new box called "NEW_CLICK_STREAM." From now on, you'll look at the new box whenever you want to know the

time. It's like having a new and better box to keep things tidy and organized. So, the best way to change the time on the little cards to the better type "TIMESTAMP" is option E. It's like using magic to create a new box and making sure everything is still easy to find and work with. It's a clever way to keep track of time and make your website even better!

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #25 Topic 1

You want to use Google Stackdriver Logging to monitor Google BigQuery usage. You need an instant notification to be sent to your monitoring tool when new data is appended to a certain table using an insert job, but you do not want to receive notifications for other tables. What should you do?

A. Make a call to the Stackdriver API to list all logs, and apply an advanced filter.

B. In the Stackdriver logging admin interface, and enable a log sink export to BigQuery.

C. In the Stackdriver logging admin interface, enable a log sink export to Google Cloud Pub/Sub, and subscribe to the topic from your monitoring tool.

D. Using the Stackdriver API, create a project sink with advanced log filter to export to Pub/Sub, and subscribe to the topic from your monitoring tool. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

jvg637 **Highly Voted** 5 years, 6 months ago

I would choose D. A and B are wrong since don't notify anything to the monitoring tool. C has no filter on what will be notified. We want only some tables.

MaxNRG **Highly Voted** 3 years, 10 months ago

D as the key requirement is to have notification on a particular table. It can be achieved using advanced log filter to filter only the table logs and create a project sink to Cloud Pub/Sub for notification. Refer GCP documentation - Advanced Logs Filters: https://cloud.google.com/logging/docs/view/advanced-queries A is wrong as advanced filter will help in filtering. However, there is no notification sends. B is wrong as it would send all the logs and BigQuery does not provide notifications. C is wrong as it would send all the logs.

suwalsageen12 1 year, 4 months ago

D is the correct answer because: - we need to advance filtering to filter the logs for the specific table - we need to use monitoring tool for notification.

axantroff 1 year, 10 months ago

Selected Answer: D

Good point by MaxNRG about reducing the number of logs sending to Pub/Sub

ruben82 1 year, 11 months ago

Theorically Pub/Sub could filters log to forward the right ones to the correct topic. https://cloud.google.com/pubsub/docs/subscription-message-filter So C could be accepted, but It's better if filtering is performed earlier, so in this case D is more performing

rtcpost 1 year, 11 months ago

Selected Answer: D

D. Using the Stackdriver API, create a project sink with an advanced log filter to export to Pub/Sub, and subscribe to the topic from your monitoring tool. This approach allows you to set up a custom log sink with an advanced filter that targets the specific table and then export the log entries to Google Cloud Pub/Sub. Your monitoring tool can subscribe to the Pub/Sub topic, providing you with instant notifications when relevant events occur without being inundated with notifications from other tables. Options A and B do not offer the same level of customization and specificity in targeting notifications for a particular table. Option C is almost correct but doesn't mention the use of an advanced log filter in the sink configuration, which is typically needed to filter the logs to a specific table effectively. Using the Stackdriver API for more advanced configuration is often necessary for fine-grained control over log filtering.

suku2 2 years ago

Selected Answer: D

D makes sense.

GCP_PDE_AG 2 years, 1 month ago

D should be the answer

Mathew106 2 years, 2 months ago

Selected Answer: D

A and B mention nothing about notifications and C would push all data. It's D.

bha11111 2 years, 6 months ago

Selected Answer: D

D makes sense

Jackalski 2 years, 9 months ago

Selected Answer: D

"advanced log filter" is the key word here, all other options push all data ...

Jasar 2 years, 10 months ago

Selected Answer: D

D is the best choice

alecuba16 3 years, 5 months ago

Selected Answer: D

Using the Stackdriver API, create a project sink with advanced log filter to export to Pub/Sub, and subscribe to the topic from your monitoring tool.

devric 3 years, 5 months ago

Selected Answer: D

D. Option B doesn't make sense

samdhimal 3 years, 8 months ago

correct answer -> Using the Stackdriver API, create a project sink with advanced log filter to export to Pub/Sub, and subscribe to the topic from your monitoring tool. Option C is also most likely right answer but it doesn't have the filter. We don't want all the tables. We only want one. So the correct answer is D. Logging sink - Using a Logging sink, you

can direct specific log entries to your business logic. In this example, you can use Cloud Audit logs for Compute Engine which use the resource type gce_firewall_rule to filter for the logs of interest. You can also add an event type GCE_OPERATION_DONE to the filter to capture only the completed log events. Here is the Logging filter used to identify the logs. You can try out the query in the Logs Viewer. Pub/Sub topic – In Pub/Sub, you can create a topic to which to direct the log sink and use the Pub/Sub message to trigger a cloud function. Reference: https://cloud.google.com/blog/products/management-tools/automate-your-response-to-a-cloud-logging-event

santoshindia 3 years, 8 months ago

Selected Answer: D

explained by MaxNRG

medeis_jar 3 years, 9 months ago

Selected Answer: D

as explained by MaxNRG

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #26 Topic 1

You are working on a sensitive project involving private user data. You have set up a project on Google Cloud Platform to house your work internally. An external consultant is going to assist with coding a complex transformation in a Google Cloud Dataflow pipeline for your project. How should you maintain users' privacy?

A. Grant the consultant the Viewer role on the project.

B. Grant the consultant the Cloud Dataflow Developer role on the project.

C. Create a service account and allow the consultant to log on with it.

D. Create an anonymized sample of the data for the consultant to work with in a different project. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

jvg637 **Highly Voted** 6 years ago

The Answer should be B. The Dataflow developer role will not provide access to the underlying data.

cleroy 6 years ago

Remember he's an external consultant. You need to create a service account for him, you can't grant before that... I think C is correct in this case.

Rajuuu 5 years, 8 months ago

Service account is between applications and non human entry.

Tanzu 4 years, 2 months ago

u can enable a service account as user so that externals can use to login

Tanzu 4 years, 2 months ago

u can enable a service account as user so that externals can use to login. but the problem is service account is about login. not the minimum resources to do the dataflow related staffs. so C is not enough !. so the answer should be B. if the question was about "doing the 1st thing", then yeah may be creating a service account could be the 1st thing.

VincentMenzel 2 years, 8 months ago

Im not sure how you expect the consultant to implement a pipeline without having access to any data that is being processed. Having test data is a prerequisite.

ThorstenStaerk 2 years, 11 months ago

and now? For seeing test data, (D) would be right. And the system tells me (C) is the right answer. What shall I click in the exam?

willymac2 3 years, 9 months ago

The answer should be D. You do not need any DataFlow permission to implement a pipeline. If needed, you can test using the DirectRunner which runs locally: ttps://cloud.google.com/dataflow/docs/concepts/access-control#example_role_assignment

willymac2 3 years, 9 months ago

Sorry I did a wrong copy/paste on the link, I wanted to send: https://cloud.google.com/dataflow/docs/concepts/security-and-permissions#security_and_permissions_for_local_pipelines https://cloud.google.com/dataflow/docs/guides/setting-pipeline-options#LocalExecution

Anirkent Highly Voted 4 years, 5 months ago

Not sure who it could be anything apart from D. if I put myself in developer shoes then without seeing the data how can I develop any logic let alone the complex one. and if I have access thru any means (ex. service account) then I can just print the logs and see the data in the logs anyway. So option D appears to be the only option.

brokeTechBro 6 months, 3 weeks ago

Selected Answer: D

A. Viewer role: Still exposes all resources and metadata in the project, too broad. B. Dataflow Developer role: Lets them run pipelines on the real project (and potentially access private data). C. Service account sharing: A security anti-pattern; violates identity/accountability. D. Anonymized sample in a separate project: Safest and compliant option—consultant can code and test transformations without ever seeing sensitive data.

brokeTechBro 7 months, 1 week ago

Selected Answer: D

I think only D really covers user's privacy

Annie00000 9 months, 2 weeks ago

Selected Answer: D

You’re dealing with sensitive user data (possibly PII or other regulated content). Principle of least privilege and data minimization are key. External consultants should never be given access to production data unless absolutely necessary — and in this case, it's not. Creating an anonymized or synthetic dataset lets them do their work (e.g., test code or build transformations) without compromising real user data. Using a separate project also provides better audit boundaries and isolation.

theRafael7 9 months, 4 weeks ago

Selected Answer: D

D because the data is anonymized and that satisfies the privacy part of the question. It also isolates the consultant's work from your actual work. This is standard for privacy and the consultant's work will not impact your actual project.

Abizi 1 year ago

Selected Answer: D

logical answer for me

baimus 1 year, 6 months ago

Selected Answer: D

The answer cannot be B, because B is too retrictive, it can only create and manage dataflow jobs, but cannot view data. I acknowledge that is secure, but no consultant can do the job without seeing representative test data. D is the

only one that provides enough to do the job, while still remaining totally private.

mouthwash 1 year, 6 months ago

D cannot be the answer because the question clearly states the developer has to work in your project. Creating another project is not in scope and is a waste of time. Correct answer is B. Developer role has developer rights only, no view rights.

iooj 1 year, 8 months ago

Selected Answer: D

A. Grant the consultant the Viewer role on the project. This role provides read-only access to all resources in the project, which could expose sensitive data to the consultant, violating privacy principles. B. Grant the consultant the Cloud Dataflow Developer role on the project. This role allows the consultant to create and manage Dataflow jobs but does not give them access the underlying data, it is not sufficient, the developer still needs data. - C. Create a service account and allow the consultant to log on with it. Allowing the consultant to log on with a service account could grant them access to sensitive data if the service account has broad permissions. This approach does not address the need to limit data exposure. - D. Create an anonymized sample of the data for the consultant to work with in a different project. - this fits the requrements

Shash_88 2 years ago

D B is a good option to maintain privacy of sensitive data, but he also need some test data to validate the transformation logic right, so creating sample data and allow him to test in another project seems good.

hamzad_basha 2 years, 1 month ago

Selected Answer: B

Data flow data privacy rules cant allow the developer to see what the data, He/she just designs the pipelines and the flow as the interdependent tasks for the composer

MaxNRG 2 years, 3 months ago

Selected Answer: B

B as the Dataflow developer role would help provide the third-party consultant access to create and work on the Dataflow pipeline. However, it does not provide access to view the data, thus maintaining user's privacy. Refer GCP documentation - Dataflow roles: https://cloud.google.com/dataflow/docs/concepts/access-control#roles Option A is wrong as it would not allow the consultant to work on the pipeline. Option C is wrong as the consultant cannot use the service account to login. Option D is wrong as it does not enable collaboration.

Jconnor 2 years, 3 months ago

C and A will not maintain user's privacy so out. B without data will be enough. D will give a good sample data, maintain privacy and the consultant will help creating the dataflow pipe for the project as requested. so D.

axantroff 2 years, 4 months ago

Selected Answer: D

I follow the corresponding logic choosing between B and D: Yes, with the Dataflow Developer role it is possible to execute and manipulate Dataflow jobs, but do we need to execute it? Based on my understanding we only need to ask for help to write it down. Is it possible without having access to test the data? I don't think so. At the same time, we need to perform an anonymization on it. So the answer D is more appropriate for me

rocky48 2 years, 4 months ago

Selected Answer: D

By creating an anonymized sample of the data, you can provide the consultant with a realistic dataset that doesn't contain sensitive or private information. This way, the consultant can work on the project without direct access to sensitive data, reducing privacy risks. Options A and B involve granting the consultant access to the project, which

may expose sensitive data, even if they have limited permissions. Option C involves creating a service account, but it doesn't address the need to anonymize the data or provide a separate, safe environment for the consultant to work with. Option D provides a controlled environment that allows the consultant to work effectively while maintaining data privacy.

rtcpost 2 years, 5 months ago

Selected Answer: D

D. Create an anonymized sample of the data for the consultant to work within a different project. By creating an anonymized sample of the data, you can provide the consultant with a realistic dataset that doesn't contain sensitive or private information. This way, the consultant can work on the project without direct access to sensitive data, reducing privacy risks. Options A and B involve granting the consultant access to the project, which may expose sensitive data, even if they have limited permissions. Option C involves creating a service account, but it doesn't address the need to anonymize the data or provide a separate, safe environment for the consultant to work with. Option D provides a controlled environment that allows the consultant to work effectively while maintaining data privacy.

https://zhangxuetu.online

[202610]

Question #27 Topic 1

You are building a model to predict whether or not it will rain on a given day. You have thousands of input features and want to see if you can improve training speed by removing some features while having a minimum effect on model accuracy. What can you do?

A. Eliminate features that are highly correlated to the output labels.

B. Combine highly co-dependent features into one representative feature. **Most Voted**

C. Instead of feeding in each feature individually, average their values in batches of 3.

D. Remove the features that have null values for more than 50% of the training records.

Correct Answer: B

Community vote distribution



Comments

anji007 Highly Voted 3 years, 11 months ago

Ans: B A: correlated to output means that feature can contribute a lot to the model. so not a good idea. C: you need to run with almost same number, but you will iterate twice, once for averaging and second time to feed the averaged value. D: removing features even if it 50% nulls is not good idea, unless you prove that it is not at all correlated to output. But this is nowhere so can remove.

pamepadero Highly Voted 3 years, 7 months ago

Trying to find a reason why it is B and not D, found this and it seems the answer is D. https://cloud.google.com/architecture/data-preprocessing-for-ml-with-tf-transform-pt1 Feature selection. Selecting a subset of the input features for training the model, and ignoring the irrelevant or redundant ones, using filter or wrapper methods. This can also involve simply dropping features if the features are missing a large number of values.

Dayashankar_H_A 3 years, 4 months ago

Yes. But nearly 50% of the non-null data still seems to be a lot to ignore.

sofiane_kihal 1 year, 2 months ago

Selected Answer: B

I think the best option is B. D could be an option but what if the feature is very correlated to the result ?

mark1223jkh 1 year, 4 months ago

B: combine the dependent features. It is more like PCA (principal component analysis). D: could be the answer, but what if that feature is very important, or how often do you get a feature with more than 50% NULL values?

axantroff 1 year, 10 months ago

Selected Answer: B

I am not into ML, to be honest, so I will rely on community opinion and choose B

rtcpost 1 year, 11 months ago

Selected Answer: B

B. Combine highly co-dependent features into one representative feature. Combining highly correlated features into a single representative feature can reduce the dimensionality of your dataset, making the training process faster while preserving relevant information. This approach often helps eliminate redundancy in the input data. Option A (eliminating features that are highly correlated to the output labels) can be counterproductive, as you want to maintain features that are informative for your prediction task. Removing features that are correlated with the output may reduce model accuracy. Option C (averaging feature values in batches of 3) is not a common technique for reducing dimensionality, and it could lead to loss of important information. Option D (removing features with null values for more than 50% of training records) can help reduce the dimensionality and may be useful if you have a large number of features with missing data, but it may not necessarily address co-dependency among features.

suku2 2 years ago

Selected Answer: B

B. Combine highly co-dependent features into one representative feature. This is the best choice.

WillemHendr 2 years, 4 months ago

"D" is wrong, and very dangerous. For instance, it might represent modern measurements only installed in <50% of weather stations, but very very precise and valuable. Nulls are not a problem for models, out-of-the-box or with transformations models can handle nulls just fine.

jin0 2 years, 7 months ago

Selected Answer: D

wrong question. there are two answers B, D

jin0 2 years, 7 months ago

B. Combine highly co-dependent features into one representative feature. -> Explainable feature should be dependent from each other feature. expecially not deep leanring. so, in this case normally eliminated or combined D. Remove the features that have null values for more than 50% of the training records. -> it's too large null data in the feature. normally the feature should be removed because it's too hard to fill up replacing data

AzureDP900 2 years, 9 months ago

Answer is Combine highly co-dependent features into one representative feature. A: correlated to output means that feature can contribute a lot to the model. so not a good idea. C: you need to run with almost same number, but you will iterate twice, once for averaging and second time to feed the averaged value. D: removing features even if it 50% nulls is not good idea, unless you prove that it is not at all correlated to output. But this is nowhere so can remove.

jin0 2 years, 7 months ago

But, if there are null datas more than 50% then, it should be eliminated because there are two ways to train the model. first, remove records containing having null but in this case there are too many records should be removed and second, replace null to other data but in this case cause of it's too large data having null then It's literally hard to replace. so normally the feature having too many null data should be removed. So, there are two answer in this question B, D I think

Thasni 2 years, 10 months ago

I have a doubt, instead of combining highly corelated features why cant we remove corelated features which may give much more simplified dataset?

noob_master 3 years, 3 months ago

Selected Answer: B

Answer: B Data that is co-dependent is high corelated is some kind of reduldant information in some cases. If the features x1, x2 and x3 are x2 = x1 + 1 and x3 = 2*x1, for example, x2 and x3 are reduldant because can be explained with x1 feature, so can be excluded of the the model. Other option is to group this features. There is a lot of ways to resolve, but the main ideia is to use data engineer in co-depedent features to reduce the number of features in the model

Ishiske 3 years, 3 months ago

Selected Answer: B

This method is called Data Engineering, that you combine two or more values to get a custom info, this will avoid that the model read an extra column on the training and probably increase it's accuracy.

Yad_datatonic 3 years, 4 months ago

Answer: B

alecuba16 3 years, 5 months ago

Selected Answer: B

Co-dependent -> correlated -> correlated info = already present info in other variable.

exnaniantwort 3 years, 8 months ago

Selected Answer: B

B null values can have many meanings and need different approach to handle, otherwise it causes inaccurate model, so not D

ZIMARAKI 3 years, 8 months ago

Selected Answer: B

For me the best option is B

https://zhangxuetu.online

[202610]

Question #28 Topic 1

Your company is performing data preprocessing for a learning algorithm in Google Cloud Dataflow. Numerous data logs are being are being generated during this step, and the team wants to analyze them. Due to the dynamic nature of the campaign, the data is growing exponentially every hour.
The data scientists have written the following code to read the data for a new key features in the logs.

```
BigQueryIO.Read
     .named("ReadLogData")
     .from("clouddataflow-readonly:samples.log_data")
```

You want to improve the performance of this data read. What should you do?

A. Specify the TableReference object in the code.

B. Use .fromQuery operation to read specific fields from the table. **Most Voted**

C. Use of both the Google BigQuery TableSchema and TableFieldSchema classes.

D. Call a transform that returns TableRow objects, where each element in the PCollection represents a single row in the table.

Correct Answer: B

Community vote distribution

B (88%) D (12%)

Comments

arthur2385 **Highly Voted** 3 years, 1 month ago

B BigQueryIO.read.fromQuery() executes a query and then reads the results received after the query execution. Therefore, this function is more time-consuming, given that it requires that a query is first executed (which will incur in the corresponding economic and computational costs).

maxdataengineer **Highly Voted** 2 years, 11 months ago

Since we want to be able to analyze data from a new ML feature (column) we only need to check values from that column. By doing a fromQuery(SELECT featueColum FROM table) we are optimizing costs and performance since we are not checking all columns. https://cloud.google.com/bigquery/docs/best-practices-costs#avoid_select_

maxdataengineer 2 years, 11 months ago

The answer is B

cetanx 2 years, 4 months ago

According to Chat GPT, it is also B In general, if your "primary goal is to reduce the amount of data read and transferred", and the downstream processing mainly focuses on a subset of fields, using .fromQuery to select specific fields would be a good choice. On the other hand, if you need to simplify downstream processing and optimize resource utilization, transforming data into TableRow objects might be more suitable.

MaxNRG 1 year, 9 months ago

Selected Answer: B

B as BigQueryIO.read.from() directly reads the whole table from BigQuery. This function exports the whole table to temporary files in Google Cloud Storage, where it will later be read from. This requires almost no computation, as it only performs an export job, and later Dataflow reads from GCS (not from BigQuery). BigQueryIO.read.fromQuery() executes a query and then reads the results received after the query execution. Therefore, this function is more time-consuming, given that it requires that a query is first executed (which will incur in the corresponding economic and computational costs). https://stackoverflow.com/questions/54413681/bigqueryio-read-vs-fromquery

axantroff 1 year, 10 months ago

Selected Answer: B

B works for me

pue_dev_anon 1 year, 10 months ago

Selected Answer: B

We are trying to optimize reading each row is not optimal, we want columns

rtcpost 1 year, 11 months ago

Selected Answer: B

B. Use the .fromQuery operation to read specific fields from the table. Using the .fromQuery operation allows you to specify the exact fields you need to read from the table, which can significantly improve performance by reducing the amount of data that needs to be processed. This is particularly important when dealing with large and growing datasets. Option A (specifying the TableReference object) provides information about the table but doesn't inherently improve the performance of reading specific fields. Option C (using Google BigQuery TableSchema and TableFieldSchema classes) is related to specifying the schema of the data but doesn't directly address improving the performance of reading specific fields. Option D (calling a transform that returns TableRow objects) is more about how the data is processed after it's read, not how it's initially read from BigQuery.

emmylou 2 years ago

When I have a different answer then the "Correct Answer", I run it through AI and it keeps saying ExamTopics is wrong. Is there any way to know if I am going to pass or fail this exam?

axantroff 1 year, 10 months ago

AI is just a LLM model, not a silver bullet at all

suku2 2 years ago

Selected Answer: B

Since the requirement is to read the data for a *new* key features in the logs, it makes sense to select limited columns, which are required rather than using .from() method which exports the entire BigQuery table. B makes sense here.

gudguy1a 2 years ago

Selected Answer: B

SHOULD be B. Not quite sure how D is the correct answer (Red herring....?) when you want to improve the query, which is .fromQuery and NOT transform and PCollection....

odiez3 2 years, 2 months ago

Answer Is D, imagine that you dont have permission on BQ AND you cant see the table info or anything else about the table you only aré working whit dataflow the only way Is transform the data using apache beam

Mathew106 2 years, 2 months ago

Selected Answer: B

I have seen people explain why B is not right because it doesn't optimize performance but only cost, which is not true, or because fromQuery is still not performant. I think it's B because no other option is more performant, even if you claim it's not good. As for option D, the transform given by the description is already a transform that provides as output a PCollection of TableRow objects. So how would that be any different?
https://beam.apache.org/releases/javadoc/2.1.0/org/apache/beam/sdk/io/gcp/bigquery/BigQueryIO.html

theseawillclaim 2 years, 2 months ago

Why should it be D? "fromQuery()" allows us to read only the columns we want, I see no point in using a Transform for each row of a "SELECT *", which, moreover, is a bad BQ Practice.

jkh_goh 2 years, 8 months ago

Selected Answer: B

Does BigQuery have a pCollections? I thought it's unique to Apache Beam i.e. Cloud Dataflow

kelvintoys93 2 years, 10 months ago

Guys, how is B the answer? Like all the justifications given here, BigQueryIO.read.fromQuery() is time consuming and the question asked for a better performance solution.

Lestrang 2 years, 8 months ago

That part is the docs trying to explain the side effects of using it, however, the part that is important to us is the fact that it reads from a query. "Read" reads the whole table. If we specify a query we can say select col1 only, which makes it all more efficient.

gcm7 2 years, 11 months ago

Selected Answer: B

reading only relevant cols

devaid 2 years, 11 months ago

Selected Answer: D

Answer is D, apparently.

Kowalski 3 years ago

Answer is Use .fromQuery operation to read specific fields from the table. BigQueryIO.read.from() directly reads the whole table from BigQuery. This function exports the whole table to temporary files in Google Cloud Storage, where it will later be read from. This requires almost no computation, as it only performs an export job, and later Dataflow reads from GCS (not from BigQuery). BigQueryIO.read.fromQuery() executes a query and then reads the results received after the query execution. Therefore, this function is more time-consuming, given that it requires that a query is first executed (which will incur in the corresponding economic and computational costs). Reference:
https://cloud.google.com/bigquery/docs/best-practices-costs#avoid_select_

https://zhangxuetu.online

[202610]

Question #29 Topic 1

Your company is streaming real-time sensor data from their factory floor into Bigtable and they have noticed extremely poor performance. How should the row key be redesigned to improve Bigtable performance on queries that populate real-time dashboards?

A. Use a row key of the form <timestamp>.

B. Use a row key of the form <sensorid>.

C. Use a row key of the form <timestamp>#<sensorid>.

D. Use a row key of the form >#<sensorid>#<timestamp>. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

sumanshu **Highly Voted** 4 years, 3 months ago

Vote for 'D' - Store multiple delimited values in each row key. (But avoid starting with Timestamp) "Row keys to avoid" https://cloud.google.com/bigtable/docs/schema-design

sumanshu 4 years, 2 months ago

A is not correct because this will cause most writes to be pushed to a single node (known as hotspotting) B is not correct because this will not allow for multiple readings from the same sensor as new readings will overwrite old ones. C is not correct because this will cause most writes to be pushed to a single node (known as hotspotting) D is correct because it will allow for retrieval of data based on both sensor id and timestamp but without causing hotspotting.

Surabhi20 7 months, 2 weeks ago

Agree, this is the correct justification for the options

samdhimal **Highly Voted** 3 years, 8 months ago

A. Use a row key of the form <timestamp>. ---> Incorrect, because google says don't use a timestamp by itself or at the beginning of a row key. B. Use a row key of the form <sensorid>. --->Incorrect, because google says Include a timestamp as part of your row key. C. Use a row key of the form <timestamp>#<sensorid>. ---> Incorrect, because google says don't use a timestamp by itself or at the beginning of a row key. D. Use a row key of the form >#<sensorid>#<timestamp>. ---> Correct answer, because of option A,B,C reasons. - Timestamp isn't by itself, neither at the beginning. - Timestamp is included. Reference: https://cloud.google.com/bigtable/docs/schema-design#row-keys

vosang5299 11 months, 2 weeks ago

Selected Answer: D

D is correct

axantroff 1 year, 10 months ago

Selected Answer: D

Looks like D is the best option Reference: https://cloud.google.com/bigtable/docs/schema-design#time-based

mark1223jkh 1 year, 4 months ago

Thank you that is right.

rtcpost 1 year, 11 months ago

Selected Answer: D

D. Use a row key of the form <sensorid>#<timestamp>. By using the sensor ID as the prefix in the row key, you can achieve better distribution of data across Bigtable tablets. This can help balance the workload and prevent hotspots in the table. Additionally, placing the timestamp after the sensor ID allows you to perform range scans for a specific sensor and retrieve data efficiently within a time frame. Option C (using a row key of the form <timestamp>#<sensorid>) can work for some use cases but may not be as efficient for range scans when you want to retrieve data for a specific sensor within a time range. Option A (using a row key of the form <timestamp>) may lead to hotspots and inefficient range scans because it doesn't consider sensor IDs. Option B (using a row key of the form <sensorid>) is not optimal because it doesn't allow for efficient time-based filtering and could lead to uneven data distribution in Bigtable.

AzureDP900 2 years, 9 months ago

D is right Best practices of bigtable states that rowkey should not be only timestamp or have timestamp at starting. It's better to have sensorid and timestamp as rowkey. Reference: https://cloud.google.com/bigtable/docs/schema-design

Nirca 2 years, 9 months ago

Selected Answer: D

#<sensorid>#<timestamp> ------> low cardinality # high cardinality This is current Bigtable Best Practice (to avoid Hotspots on the inserts)

maxdataengineer 2 years, 11 months ago

Selected Answer: D

Discard: A -> timestamp unique id could not be unique in the case that sensors transmit data at the same time. B -> sensorId repeated id for messages coming from the same sensor C -> a bad performance choice D -> BEST CHOICE. Each time BigTable looks for data in a table it does a scan and sort operations. By starting each unique id by sensorId it will make it easier to group and sort data since it has the lowest cardinality https://cloud.google.com/bigtable/docs/schema-design#general-concepts

John_Pongthorn 3 years ago

as I look at https://cloud.google.com/bigtable/docs/schema-design#row-keys asia#india#bangalore asia#india#mumbai they didn't have # ahead of this first value. asia#india#bangalore OR #asia#india#bangalore Are both valid?

crisimenjivar 3 years, 1 month ago

ANSWER: D

som_420 3 years, 3 months ago

Selected Answer: D

Answer is D

anji007 3 years, 11 months ago

Ans: D

naga 4 years, 7 months ago

Correct D

NamitSehgal 4 years, 9 months ago

Should be D Reverse of timestamp even better but no options for that. Also changing sensor ID if they are in sequential to hash or changing data to bits even better. Idea is not to use timestamp or sequential ID as first key.

Tanzu 3 years, 8 months ago

reverse TS or hashing is not always first choice or better. never.

Radhika7983 4 years, 10 months ago

The correct answer is D. Refer to the link https://cloud.google.com/bigtable/docs/schema-design for Big table schema design. C is not the right answer becuase Timestamps If you often need to retrieve data based on the time when it was recorded, it's a good idea to include a timestamp as part of your row key. Using the timestamp by itself as the row key is not recommended, as most writes would be pushed onto a single node. For the same reason, avoid placing a timestamp at the start of the row key. For example, your application might need to record performance-related data, such as CPU and memory usage, once per second for a large number of machines. Your row key for this data could combine an identifier for the machine with a timestamp for the data (for example, machine_4223421#1425330757685).

arghya13 4 years, 11 months ago

answer would be D to avoid hotspoting..

ch3n6 5 years, 3 months ago

correct: D why not C? Using the timestamp by itself as the row key is not recommended, as most writes would be pushed onto a single node. For the same reason, avoid placing a timestamp at the start of the row key. https://cloud.google.com/bigtable/docs/schema-design#row-keys

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #30 Topic 1

Your company's customer and order databases are often under heavy load. This makes performing analytics against them difficult without harming operations.
The databases are in a MySQL cluster, with nightly backups taken using mysqldump. You want to perform analytics with minimal impact on operations. What should you do?

A. Add a node to the MySQL cluster and build an OLAP cube there.

B. Use an ETL tool to load the data from MySQL into Google BigQuery. **Most Voted**

C. Connect an on-premises Apache Hadoop cluster to MySQL and perform ETL.

D. Mount the backups to Google Cloud SQL, and then process the data using Google Cloud Dataproc.

Correct Answer: B

Community vote distribution



Comments

HectorLeon2099 **Highly Voted** 5 years, 3 months ago

It is a GOOGLE exam. The answer won't be on-premise or OLAP cubes even if it is the easiest. The answer is B

Preetmehta1234 2 years, 1 month ago

That’s so true! This should be the first logic for elimination

Tanzu 4 years, 2 months ago

choose dataproc over hadoop cluster chose bigquery over all.. there is no special customer requirement that gonna drive us to hadoop or dataproc.

cetanx 3 years, 2 months ago

Answer - B mysql dump: This utility creates a logical backup and a flat file containing the SQL statements that can be run again to bring back the database to the state when this file was created. So this file can easily be processed by an ETL tool and loaded into BQ.

[Removed] **Highly Voted** 6 years ago

Answer: D Description: Easy and it won’t affect processing

Alexej_123 5 years, 2 months ago

I think it is B and not D: 1) There are no info regarding date freshness required for analytics. So nightly backup might be not enough as a source because it will only provide info one tie a day. 2) Dataproc is recommended as easiest way for migration of hadoop processes. SO no reason to use Dataproc for designing a new analytics processes. 3) The solution is really very limited if you will extend it in the future and add new data sources or create new aggregate

tables. Where they should be created? 4) There is no info on which version is on prem MySQL database (I am not an expert in MySql) but I can imagine there might be compartibility issue for backup / restore between different releases

hellofrnds 4 years, 5 months ago

" Dataproc makes open source data and analytics processing fast, easy, and more secure in the cloud ". Please refer this google link. https://cloud.google.com/blog/products/data-analytics/genomics-data-analytics-with-cloud-pt2

sergio6 4 years, 5 months ago

The link titles "Genomics analysis with Hail, BIGQUERY, and Data Proc", the solution describes the use of Bigquery to do analytics

stepbroooooo 2 weeks, 1 day ago

Selected Answer: D

D is correct, because you want minimal impact on operation, while B will put pressure on the source database.

Bugnumber1 7 months ago

Selected Answer: B

Keyword is analysis, so Bigquery. What are you analysing with Dataproc?

22c1725 10 months, 2 weeks ago

Selected Answer: D

It point that it shouldn't affect processing

vosang5299 11 months, 2 weeks ago

Selected Answer: B

B is correct

willyunger 1 year ago

Selected Answer: D

Option D has no impact on operations, uses backups which are already there. Option B with ETL could impact MySQL performance.

Juanesdelacruz97 1 year, 2 months ago

I think it's B, today BigQuery has multiple connectors that can allow an easy connection to external data sources without impacting the database itself, even if the database was in a SQL instance, MySQL, Federated queries could be used. In my opinion it's B

Augustax 1 year, 2 months ago

Selected Answer: D

Since the question mentions the nightly backup, why we cannot use it? ETL reduces the impact of the source system but still some impacts. D doesn't add any additional impact.

mark1223jkh 1 year, 10 months ago

Answer B: I don't know why people are choosing D. It is two steps, first cloudsql and then dataproc, a lot of overhead. BigQuery is just perfect fit.

0725f1f 2 years ago

Selected Answer: D

This won’t affect processing

TVH_Data_Engineer 2 years, 4 months ago

Selected Answer: B

Based on these considerations, option B is likely the best approach. By using an ETL tool to load data from MySQL into Google BigQuery, you're leveraging BigQuery's strengths in handling large-scale analytics workloads without impacting the performance of the operational databases. This option provides a clear separation of operational and analytical workloads and takes advantage of BigQuery's fast analytics capabilities.

axantroff 2 years, 4 months ago

Selected Answer: B

Do not spend much time on in - just B

rocky48 2 years, 4 months ago

Selected Answer: B

Answer is B - Use an ETL tool to load the data from MySQL into Google BigQuery. * Google BigQuery is a serverless, highly scalable data warehouse that can handle large-scale analytics workloads without impacting your MySQL cluster's performance. * Using an ETL (Extract, Transform, Load) tool to transfer data from MySQL to BigQuery allows you to maintain a separate analytics environment, ensuring that your operational database remains unaffected. Option C (connecting an on-premises Apache Hadoop cluster to MySQL and performing ETL) introduces complexity and may not be as scalable as a cloud-based solution. Option D (mounting backups to Google Cloud SQL and processing the data using Google Cloud Dataproc) could be an option for historical data analysis but might not be the best choice for real-time analytics while the MySQL cluster is under heavy load. Additionally, the backups need to be restored and processed, which might introduce some delay.

mk_choudhary 2 years, 5 months ago

It's GOOGLE exam where choosing the GCP service shall be first preference. Now notice the problem statement "perform analytics with minimal impact on operations" BigQuery is right option for analytic as well as Cloud SQL does provide easy export to GCS where we can query from BigQuery without loading into BQ to save storage cost.

rtcpost 2 years, 5 months ago

Selected Answer: B

B. Use an ETL tool to load the data from MySQL into Google BigQuery. * Google BigQuery is a serverless, highly scalable data warehouse that can handle large-scale analytics workloads without impacting your MySQL cluster's performance. * Using an ETL (Extract, Transform, Load) tool to transfer data from MySQL to BigQuery allows you to maintain a separate analytics environment, ensuring that your operational database remains unaffected. Option C (connecting an on-premises Apache Hadoop cluster to MySQL and performing ETL) introduces complexity and may not be as scalable as a cloud-based solution. Option D (mounting backups to Google Cloud SQL and processing the data using Google Cloud Dataproc) could be an option for historical data analysis but might not be the best choice for real-time analytics while the MySQL cluster is under heavy load. Additionally, the backups need to be restored and processed, which might introduce some delay.

melligeri 2 years, 5 months ago

Selected Answer: B

The question clearly says there is load on MYSQL already so doing analytics on it is bad idea. Its bad to run analytics on MYSQL but still a better option to run etl with it to load it to BigQuery.

https://zhangxuetu.online

[202610]

Question #31 Topic 1

You have Google Cloud Dataflow streaming pipeline running with a Google Cloud Pub/Sub subscription as the source. You need to make an update to the code that will make the new Cloud Dataflow pipeline incompatible with the current version. You do not want to lose any data when making this update. What should you do?

A. Update the current pipeline and use the drain flag. **Most Voted**

B. Update the current pipeline and provide the transform mapping JSON object.

C. Create a new pipeline that has the same Cloud Pub/Sub subscription and cancel the old pipeline.

D. Create a new pipeline that has a new Cloud Pub/Sub subscription and cancel the old pipeline.

Correct Answer: A

Community vote distribution



Comments

VishalB **Highly Voted** 5 years, 8 months ago

Correct Option : A Explanation:-This option is correct as the key requirement is not to lose the data, the Dataflow pipeline can be stopped using the Drain option. Drain options would cause Dataflow to stop any new processing, but would also allow the existing processing to complete

VishalB 5 years, 8 months ago

Option C & D are incorrect as Cancel Option will lead to loose the data Option B is very Close, since the new Code make pipeline incompatible by providing transform mapping JSON file you can handle this

sergio6 4 years, 5 months ago

A is incorrect because updating pipeline does not include any drain flag

Tanzu 4 years, 2 months ago

two steps.. 1st drain the job w/ sdk or console. then, update the pipeline. cause it is OK to update a job while in draining

maxdataengineer 3 years, 5 months ago

Yes but the compatibility problem will still be there, stopping the pipeline does not solve that

Tanzu 4 years, 2 months ago

There are 5 update scenarios in a job in update-a-pipeline context. a- changing transform name (requires mapping) , adding a new step (no need for mapping) b- windowing or triggering (only for minor changes, otherwise don't do that) c- coders (don't do that d- schema (adding or required to nullable is possible) other scenarios not possible e- stateful operations none of them are relevant, here. cause there is no specific detail, secondly incompatible w. new pipeline.

and mostly if in compatible only a has a solve. but not for all cases. so, drain == no data loss (ingesting, buffered and in-flight data) is the only scenario.

maxdataengineer 3 years, 5 months ago

As you said, Drain stops the pipeline but it does not solve the compatibility issue. The pipeline will not be able to be updated which is the core problem of the question.

assU2 3 years, 4 months ago

You do not want to lose any data when making this update - is the core problem. You are doing it ANYWAY.

sergio6 4 years, 7 months ago

C and D are incorrect because canceling the old pipeline can cause data loss https://cloud.google.com/dataflow/docs/guides/stopping-a-pipeline A is incorrect because updating pipeline does not include any drain flag https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline

sergio6 4 years, 5 months ago

B is correct: Update the current pipeline and provide the transform mapping JSON object. Dataflow always performs a compatibility check between the old and new job and without the mapping (necessary as old and new are incompatible) it would give an error and the old job would continue to be executed https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#Mapping https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#CCheck

Tanzu 4 years, 2 months ago

new pipeline is incompatible means, compatibility check will fail. so you wil not be able to update as new pipeline. that's why B cannot be valid answer here in this context.

maxdataengineer 3 years, 5 months ago

B is a way to solve compatibility issues

Tanzu 4 years, 2 months ago

drain is in the guide ...stopping-a-pipeline. Just ...updating-a-pipeline is not enough to evaluate this question. that's why drainnin is not a flag in a pipeline update. it is a process about how to stop a pipeline w/o data loss ! data in dataflow is in 3 stages. ingestion data, buffered data and in-flight data which is processing by old pipeline.

BigQuery 4 years, 3 months ago

To all the New Guys Here. Please don't get confused with all the people's fight over here. Just google the question and you will get the correct ans in many website. Still I recommend to refer this website for question. for this Particular problem ans is A. Reason is here --> https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#python have time to read the full page when to use Update using Json mapping and when to use Drain. (you will have question following for Drain option though). Thumb rule is this, # If any major change to windowing transformation (like completely changing window fn from fixed to sliding) in Beam/Dataflow/you want to stop pipeline but want inflight data --> use Drain option. # For all other use cases and Minor changing to windowing fn (like just changing window time of sliding window) --> Use Update with Json mapping. In this case it is Code change to new version. so, Update with Json mapping. Simple as that. All the Best Guys.

BigQuery 4 years, 3 months ago

SORRY I MEANT TO SAY ANS IS 'B'. In this case it is Code change to new version. so, Update with Json mapping.

anji007 4 years ago

Its clearly mentioned in the question that pipeline in compatible, if it is so you can not update with JSON mapping. Only way is to stop the pipeline with Drain and replace it with a new one. So the closest answer is A only.

maxdataengineer 3 years, 5 months ago

JSON Mapping is a way to solve compatibility issues when updating

IKGx1iGetOWGSjAQDD2x3 Highly Voted 5 years, 3 months ago

Answer is D. * A and B are not possible, since the new job is not compatible. * C might lead to lost data. * D might lead to data being processed twice, but no data will be lost. Better would usually be to drain and start a new pipeline.

b2aaace 5 months ago

Selected Answer: B

B is correct: Update the current pipeline and provide the transform mapping JSON object.

leticiaarj 9 months, 2 weeks ago

Selected Answer: D

This is the Google Cloud recommended way to transition between streaming pipeline versions without data loss: 1. Create a new subscription on the same Pub/Sub topic. 2. New subscription starts receiving messages from the moment of creation. 3. Create a new pipeline with this new subscription. 4. After validation, cancel the old pipeline and subscribe if necessary.

abhaya2608 1 year ago

Selected Answer: B

Please refer the google doc link below, https://cloud.google.com/dataflow/docs/guides/stopping-a-pipeline#drain

Parandhaman_Margan 1 year ago

Selected Answer: D

Create a new pipeline that has a new Cloud Pub/Sub subscription and cancel the old pipeline

dans_puts 1 year, 3 months ago

Selected Answer: D

D. Create a new pipeline that has a new Cloud Pub/Sub subscription and cancel the old pipeline: By creating a new subscription, the new pipeline will consume messages independently of the old pipeline. This ensures no data is lost as messages published to Pub/Sub are delivered to all subscriptions. Once the new pipeline is verified to be running as expected, the old pipeline can be safely canceled.

Smakyel79 1 year, 4 months ago

Why option D is better for this case - In this scenario: the pipeline is incompatible with the old one; running the two pipelines concurrently ensures no data loss and allows for easier debugging of the new pipeline; a new subscription ensures the old pipeline can finish processing its messages while the new pipeline starts fresh

Smakyel79 1 year, 4 months ago

Draining a pipeline stops it in an orderly manner but does not address incompatibility issues. Once the pipeline is drained, no more data is processed, and the new pipeline starts fresh. This can lead to data loss if there are messages in Pub/Sub that the drained pipeline didn't process.

philli1011 2 years, 2 months ago

Option: C using draining will stop the subscription totally while allowing the existing data to complete processing. While the pipeline is stopped, will lose streaming data. The best option is to create a new pipeline that is connected to the same subscription, then we can apply drain to the old pipeline and end it. That way we will capture all the streaming data.

MaxNRG 2 years, 3 months ago

Selected Answer: A

Drain flag: This flag allows the pipeline to finish processing all existing data in the Pub/Sub subscription before shutting down. This ensures no data is lost during the update. Current pipeline: Updating the current pipeline minimizes disruption and avoids setting up entirely new infrastructure. Incompatible changes: Even with incompatible changes, the drain flag ensures existing data is processed correctly.

MaxNRG 2 years, 3 months ago

While other options might work in some cases, they have drawbacks: B. Transform mapping JSON: This option is mainly for schema changes and doesn't guarantee data completion before shutdown. C. New pipeline, same subscription: This risks duplicate processing of data if both pipelines run concurrently. D. New pipeline, new subscription: This loses the current pipeline's state and potentially data, making it impractical for incompatible changes. Therefore, the most reliable and data-safe approach is to update the current pipeline with the drain flag for seamless transition and data integrity. Remember, always test updates in a staging environment before deploying to production.

TVH_Data_Engineer 2 years, 3 months ago

Selected Answer: C

Same Cloud Pub/Sub Subscription: By using the same Cloud Pub/Sub subscription for the new pipeline, you ensure that no messages are lost during the transition. Pub/Sub manages message delivery, ensuring that unacknowledged messages (those that haven't been processed by your old pipeline) will be available for the new pipeline to process. Creating a New Pipeline: Since the update makes the new pipeline incompatible with the current version, it's necessary to create a new pipeline. Attempting to update the current pipeline in place (options A and B) would not be feasible due to compatibility issues. Cancel the Old Pipeline: Once the new pipeline is up and running and processing messages, you can safely cancel the old pipeline. This ensures a smooth transition with no data loss.

JOKKUNO 2 years, 3 months ago

In order to make an update to a Google Cloud Dataflow streaming pipeline without losing any data, the recommended approach is: A. Update the current pipeline and use the drain flag. Explanation: The drain flag is designed to allow the current pipeline to finish processing any remaining data before shutting down. This helps ensure that no data is lost during the update process. By updating the current pipeline and using the drain flag, you allow the pipeline to complete its current processing before the update takes effect, minimizing the risk of data loss. This approach is a safe way to transition from the old version to the new version without interrupting data processing.

axantroff 2 years, 4 months ago

I would vote for A because of the structure of the exam, but there are other options worth considering as well

RT_G 2 years, 4 months ago

Selected Answer: C

My answer is C. Chatted with ChatGPT and narrowed down on this option. Let me know your thoughts on this perspective. Option C - By using the existing subscription, you can ensure that the data flow remains uninterrupted, and there is no loss of data during the transition from the old pipeline to the new one. Creating a new pipeline that uses the same Cloud Pub/Sub subscription allows for a seamless transition without any interruptions to the data flow. This approach ensures that the new pipeline can continue to consume data from the same subscription as the old pipeline, thereby maintaining data continuity throughout the update process.

rocky48 2 years, 4 months ago

Selected Answer: A

Correct Option : A Explanation:-This option is correct as the key requirement is not to lose the data, the Dataflow pipeline can be stopped using the Drain option.

mk_choudhary 2 years, 5 months ago

It should be B Drain will stop the existing job only and it does not suffice the updated schema. In order to bring updated schema into effect, updated JSON mapping need to be applied.

https://zhangxuetu.online

[202610]

Question #32 Topic 1

Your company is running their first dynamic campaign, serving different offers by analyzing real-time data during the holiday season. The data scientists are collecting terabytes of data that rapidly grows every hour during their 30-day campaign. They are using Google Cloud Dataflow to preprocess the data and collect the feature (signals) data that is needed for the machine learning model in Google Cloud Bigtable. The team is observing suboptimal performance with reads and writes of their initial load of 10 TB of data. They want to improve this performance while minimizing cost. What should they do?

A. Redefine the schema by evenly distributing reads and writes across the row space of the table. **Most Voted**

B. The performance issue should be resolved over time as the site of the BigDate cluster is increased.

C. Redesign the schema to use a single row key to identify values that need to be updated frequently in the cluster.

D. Redesign the schema to use row keys based on numeric IDs that increase sequentially per user viewing the offers.

Correct Answer: A

Community vote distribution

A (100%)

Comments

IsaB **Highly Voted** 5 years, 6 months ago

I hate it when I read the question, than I think oh easy and I KNOW the answer, then I look at the choices and the answer I thought of is just not there at all... and I realize I absolutely have no idea :'D

MaxNRG **Highly Voted** 4 years, 4 months ago

A as the schema needs to be redesigned to distribute the reads and writes evenly across each table. Refer GCP documentation - Bigtable Performance: https://cloud.google.com/bigtable/docs/performance The table's schema is not designed correctly. To get good performance from Cloud Bigtable, it's essential to design a schema that makes it possible to distribute reads and writes evenly across each table. See Designing Your Schema for more information. https://cloud.google.com/bigtable/docs/schema-design Option B is wrong as increasing the size of cluster would increase the cost. Option C is wrong as single row key for frequently updated identifiers reduces performance Option D is wrong as sequential IDs would degrade the performance. A safer approach is to use a reversed version of the user's numeric ID, which spreads traffic more evenly across all of the nodes for your Cloud Bigtable table.

09878d5 1 year, 8 months ago

Selected Answer: A

B is a Lie C and D are actually not recommended A is correct as it will help in even distribution of load and avoid hotspots

JOKKUNO 2 years, 3 months ago

Improving performance in Google Cloud Bigtable involves optimizing the schema design to distribute the load efficiently across the clusters. Given the scenario, the best option would be: A. Redefine the schema by evenly distributing reads and writes across the row space of the table. Explanation: Distributing reads and writes evenly across the row space helps prevent hotspots and ensures that the load is spread evenly, avoiding performance bottlenecks. Google Cloud Bigtable's performance is influenced by how well the data is distributed across the tablet servers, and evenly distributing the load can lead to better performance. This approach aligns with best practices for designing scalable and performant Bigtable schemas.

axantroff 2 years, 4 months ago

Selected Answer: A

The comment from hilel_eth totally makes sense to me. I would go with A

hkris909 2 years, 7 months ago

Guys, how relevant are these questions, as of Aug 14, 2023 Could we clear the PDE exam with these set of questions?

roty 2 years, 3 months ago

HEY DID U CLEAR THE EXAM

FP77 2 years, 8 months ago

Selected Answer: A

A is the only one that makes sense and is correct

Mathew106 2 years, 8 months ago

I understand why it could be A. But why not B also? Is it because of the typo saying BigDate instead of BigTable?

Adswerve 2 years, 11 months ago

Selected Answer: A

A to avoid hot-spotting https://cloud.google.com/bigtable/docs/schema-design

Brillianttyagi 3 years, 3 months ago

Selected Answer: A

A - Make sure you're reading and writing many different rows in your table. Bigtable performs best when reads and writes are evenly distributed throughout your table, which helps Bigtable distribute the workload across all of the nodes in your cluster. If reads and writes cannot be spread across all of your Bigtable nodes, performance will suffer. https://cloud.google.com/bigtable/docs/performance#troubleshooting

hilel_eth 3 years, 3 months ago

Selected Answer: A

A good way to improve read and write performance in a database system like Google Cloud Bigtable is to redefine the schema of the table so that reads and writes are evenly distributed across the row space of the table. This can help reduce bottlenecks in processing capacity and improve efficiency in table management. In addition, by evenly distributing read and write operations, it can prevent the accumulation of operations in one part of the table, which can improve the overall performance of the system.

Arkon88 4 years ago

Selected Answer: A

A is correct https://cloud.google.com/bigtable/docs/performance#troubleshooting If you find that you're reading and writing only a small number of rows, you might need to redesign your schema so that reads and writes are more evenly distributed.

samdhimal 4 years, 2 months ago

correct answer -> Redefine the schema by evenly distributing reads and writes across the row space of the table. Make sure you're reading and writing many different rows in your table. Bigtable performs best when reads and writes are evenly distributed throughout your table, which helps Bigtable distribute the workload across all of the nodes in your cluster. If reads and writes cannot be spread across all of your Bigtable nodes, performance will suffer. If you find that you're reading and writing only a small number of rows, you might need to redesign your schema so that reads and writes are more evenly distributed. Reference: https://cloud.google.com/bigtable/docs/performance#troubleshooting

anji007 4 years, 5 months ago

Ans: A

sumanshu 4 years, 8 months ago

Vote for A

timolo 4 years, 9 months ago

Correct is A: https://cloud.google.com/bigtable/docs/performance#troubleshooting Make sure you're reading and writing many different rows in your table. Bigtable performs best when reads and writes are evenly distributed throughout your table, which helps Bigtable distribute the workload across all of the nodes in your cluster. If reads and writes cannot be spread across all of your Bigtable nodes, performance will suffer.

naga 5 years, 1 month ago

Correct A

https://zhangxuetu.online

[202610]

Question #33 Topic 1

Your software uses a simple JSON format for all messages. These messages are published to Google Cloud Pub/Sub, then processed with Google Cloud
Dataflow to create a real-time dashboard for the CFO. During testing, you notice that some messages are missing in the dashboard. You check the logs, and all messages are being published to Cloud Pub/Sub successfully. What should you do next?

A. Check the dashboard application to see if it is not displaying correctly.

B. Run a fixed dataset through the Cloud Dataflow pipeline and analyze the output. Most Voted

C. Use Google Stackdriver Monitoring on Cloud Pub/Sub to find the missing messages.

D. Switch Cloud Dataflow to pull messages from Cloud Pub/Sub instead of Cloud Pub/Sub pushing messages to Cloud Dataflow.

Correct Answer: B

Community vote distribution



Comments

[Removed] Highly Voted 5 years ago

Answer: C Description: Stackdriver can be used to check the error like number of unack messages, publisher pushing messages faster

tprashanth 4 years, 8 months ago

B. Stack driver monitoring is for performance, not logging of missing data.

mikey007 4 years, 8 months ago

https://cloud.google.com/pubsub/docs/monitoring

ritinhabb 2 years, 9 months ago

Exactly!

snamburi3 4 years, 4 months ago

All messages are being published to Cloud Pub/Sub successfully. so Stackdriver might not help.

kubosuke 3 years, 6 months ago

messages sent successfully to Topic, but not Subscription. in this case, if Dataflow cannot handle messages correctly it might not return acknowledgments to the Pub/Sub, and these errors can be seen from Monitoring.
https://cloud.google.com/pubsub/docs/monitoring#monitoring_exp

Tanzu 3 years, 2 months ago

to be more precise, first to publisher, - then forwards to topic, and persistance for a while - then forwards to subscribe, - then to subscription.. - then acknowledgement happens so in every steps, there is possibly for errors.

jkhong 2 years, 3 months ago

PubSub doesn't forward from subscriber to subscription. A topic sends it over to subscription first, then to subscriber

[Removed] Highly Voted 5 years ago

Should be B

rtcpost 1 year, 5 months ago

Selected Answer: B

B. Run a fixed dataset through the Cloud Dataflow pipeline and analyze the output. * By running a fixed dataset through the Cloud Dataflow pipeline, you can determine if the problem lies within the data processing stage. This allows you to identify any issues with data transformation, filtering, or processing in your pipeline. * Analyzing the output from this fixed dataset will help you isolate the problem and confirm whether it's related to data processing or the dashboard application.

ruben82 1 year, 5 months ago

You must know what kind of data causes errors. I think, the first step is to get erroneous data and then test with sample of it.

imran79 1 year, 5 months ago

B. Run a fixed dataset through the Cloud Dataflow pipeline and analyze the output. If this results in the expected output, then the problem might be with the dashboard application (Option A), and that should be checked next.

WillemHendr 1 year, 9 months ago

Selected Answer: B

"...to find the missing messages" Up to that remark, Monitoring was a valid option as well. But missing messages cannot be found with monitoring. It is simply not possible to find the exact missing message. I read this remark as a test if you know what is, and what isn't possible with monitoring.

izekc 1 year, 11 months ago

Selected Answer: B

here is to determine next step. Not better way to optimize the workload. So B is the correct next step

Jarek7 1 year, 11 months ago

B is not the next step. The next step is between pub/sub ands dataflow(C). B will not help with it at all. However it could show the issue if it is the pipeline or the view. But also it could not show it - you have no idea why some messages are not shown, so most probably it wouldnt get you any info. Definitely next step is to chek if the issue is between pubsub and dataflow. Then you coud go with B.

Adswerve 1 year, 11 months ago

Selected Answer: D

Pull subscription is the correct one. Push subscription means Dataflow cannot keep up with the topic.

Jarek7 1 year, 11 months ago

It could be the issue. But C would reveal it if this is the real issue - if you will not check stackdriver, you cannot be sure if you really resolved the issue, as even if it seems to be working properly after switch to pull you cannot be sure if it is because of some other temporal factor.

midgoo 2 years, 1 month ago

Selected Answer: B

If the Dataflow does not have the expected output, it is either wrong at the input or at the pipelines. The chance that the issue is at the input (PubSub) is very low. For this case, it is likely the pipelines got some mistakes (e.g. JSON parsing failed). So we should follow B to debug the pipelines (using snapshot as test dataset for example)

ploer 2 years, 1 month ago

Selected Answer: B

The most efficient solution would be to run a fixed dataset through the Cloud Dataflow pipeline and analyze the output (Option B). This will allow you to determine if the issue is with the pipeline or with the dashboard application. By analyzing the output, you can see if the messages are being processed correctly and determine if there are any discrepancies or missing messages. If the issue is with the pipeline, you can then debug and make any necessary updates to ensure that all messages are processed correctly. If the issue is with the dashboard application, you can then focus on resolving that issue. This approach allows you to isolate and identify the root cause of the missing messages in a controlled and efficient manner.

Lestrang 2 years, 2 months ago

Selected Answer: B

I've just skimmed over the Stackdriver docs, yes guys, it helps you check the number and age of messages that were not received/acknowledged, excellent, hurray. So first off, c will not give us the missing messages, it will give us the count and age. that means that c is inherently incorrect. Additionally, will knowledge of the number of messages make resolving the problem any easier? No, it is just confirming what we already know. Meanwhile, approach B, will allow us to see HOW and WHY it is missing some messages, which is the step that proceeds the fix.

PolyMoe 2 years, 2 months ago

Selected Answer: C

Here is ChatGPT answer : It's always a good practice to start by checking the logs and monitoring tools to see if there is any indication of an issue with the messages being published to Cloud Pub/Sub. In this case, you should use Google Stackdriver Monitoring to investigate if the missing messages have been published or not. You can also run a fixed dataset through the Cloud Dataflow pipeline to see if the pipeline is processing the messages correctly. If there is no issue found on the Cloud Pub/Sub and Cloud Dataflow, then you can check the dashboard application to see if it is not displaying the messages correctly. As a last resort, you can switch Cloud Dataflow to pull messages from Cloud Pub/Sub instead of Cloud Pub/Sub pushing messages to Cloud Dataflow.

Lestrang 2 years, 2 months ago

I provided it with the question as input but added the metrics available in Stackdriver, here is the response: B. Run a fixed dataset through the Cloud Dataflow pipeline and analyze the output. If messages are being published successfully to Cloud Pub/Sub but are missing in the dashboard, the issue is likely to be with the Cloud Dataflow pipeline that processes the messages. To find the root cause of the problem, you should run a fixed dataset through the pipeline and analyze the output. This will allow you to see if the pipeline is correctly processing all messages, and identify any processing errors that might be causing messages to be lost. The output can be compared to the expected results to identify any discrepancies and resolve the issue.

Jarek7 1 year, 11 months ago

I'm tired with these responses about what chatGPT says. Most probably you've used the free 3.5 version which is absolute disaster regarding being all knowing oracle. BTW in this case I wouldn't believe even GPT4. It is a difficult question that needs a specific knowledge and experience which might be not available in the GPT training data. You cannot use any GPT up to 4 as an argument in such cases.

axantroff 1 year, 4 months ago

Exactly. Sometimes it is total garbage

desertlotus1211 2 years, 2 months ago

The question is really not asking for a solution to the problem, per se - but more of what would the next step in the situation to triage the issue.... Answer would be B over D. Answer D would be the recommended solution IF the question asked to rectify/fixed the issue. Thoughts?

izekc 1 year, 11 months ago

Agree with u

Prakzz 2 years, 3 months ago

Selected Answer: D

D. Dataflow must PULL the data to process it in real-time. Missing messages in the dashboard, means that the Pub/Sub to Dataflow was misconfigured as PUSH.

hasoweh 2 years, 2 months ago

Pull will lead to latency as new data will not be streamed upon arrival, but instead will only be passed on when Dataflow makes a pull request. So if data comes in at time 0:01 but pull requests are only happening every 10 seconds, we have 9 second delay. Push will automatically push the data to any subscribers as soon as the data comes, and thus is closer to real-time.

Krish6488 2 years, 3 months ago

Selected Answer: B

To me, B sounds more logical for the below reason. Option C would have been ideal because any debugging starts with checking the logs, however the option says, check stackdriver for missing messages. Had it been, check stackdriver to figure out the number of undelivered messages, C would have been more suitable. Given the slight bit of dodginess in option c, I would go with B

Nirca 2 years, 3 months ago

Selected Answer: B

Why checking Pub/Sub again when this is already verified to be fine according to the question. Shouldn't you be checking the next stage in the flow which is Dataflow? Option - B

DGames 2 years, 3 months ago

Selected Answer: B

Answer - B. Because already we know message is missing so better to test with fixed dataset and check code .

Atnafu 2 years, 4 months ago

C
https://cloud.google.com/pubsub/docs/monitoring#:~:text=the%20specific%20metrics.-,Monitor%20message%20backlog,information%20about%20this%20metric%2C%20see%20the%20relevant%20section%20of%20this%20document.,-Create%20alerting%20policies

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #34 **Topic 1**

Flowlogistic Case Study -

Company Overview -
Flowlogistic is a leading logistics and supply chain provider. They help businesses throughout the world manage their resources and transport them to their final destination. The company has grown rapidly, expanding their offerings to include rail, truck, aircraft, and oceanic shipping.

Company Background -
The company started as a regional trucking company, and then expanded into other logistics market. Because they have not updated their infrastructure, managing and tracking orders and shipments has become a bottleneck. To improve operations, Flowlogistic developed proprietary technology for tracking shipments in real time at the parcel level. However, they are unable to deploy it because their technology stack, based on Apache Kafka, cannot support the processing volume. In addition, Flowlogistic wants to further analyze their orders and shipments to determine how best to deploy their resources.

Solution Concept -
Flowlogistic wants to implement two concepts using the cloud:
✑ Use their proprietary technology in a real-time inventory-tracking system that indicates the location of their loads
✑ Perform analytics on all their orders and shipment logs, which contain both structured and unstructured data, to determine how best to deploy resources, which markets to expand info. They also want to use predictive analytics to learn earlier when a shipment will be delayed.

Existing Technical Environment -
Flowlogistic architecture resides in a single data center:
✑ Databases
8 physical servers in 2 clusters
- SQL Server `" user data, inventory, static data
3 physical servers
- Cassandra `" metadata, tracking messages
10 Kafka servers `" tracking message aggregation and batch insert
✑ Application servers `" customer front end, middleware for order/customs
60 virtual machines across 20 physical servers
- Tomcat `" Java services
- Nginx `" static content
- Batch servers
✑ Storage appliances
- iSCSI for virtual machine (VM) hosts
- Fibre Channel storage area network (FC SAN) `" SQL server storage
- Network-attached storage (NAS) image storage, logs, backups
✑ 10 Apache Hadoop /Spark servers

- Core Data Lake
- Data analysis workloads
✑ 20 miscellaneous servers
- Jenkins, monitoring, bastion hosts,

Business Requirements -
Build a reliable and reproducible environment with scaled panty of production.
▪

✑ Aggregate data in a centralized Data Lake for analysis
✑ Use historical data to perform predictive analytics on future shipments
✑ Accurately track every shipment worldwide using proprietary technology
✑ Improve business agility and speed of innovation through rapid provisioning of new resources
✑ Analyze and optimize architecture for performance in the cloud
✑ Migrate fully to the cloud if all other requirements are met

Technical Requirements -
✑ Handle both streaming and batch data
✑ Migrate existing Hadoop workloads
✑ Ensure architecture is scalable and elastic to meet the changing demands of the company.
✑ Use managed services whenever possible
✑ Encrypt data flight and at rest
✑ Connect a VPN between the production data center and cloud environment

SEO Statement -
We have grown so quickly that our inability to upgrade our infrastructure is really hampering further growth and efficiency. We are efficient at moving shipments around the world, but we are inefficient at moving data around. We need to organize our information so we can more easily understand where our customers are and what they are shipping.

CTO Statement -
IT has never been a priority for us, so as our data has grown, we have not invested enough in our technology. I have a good staff to manage IT, but they are so busy managing our infrastructure that I cannot get them to do the things that really matter, such as organizing our data, building the analytics, and figuring out how to implement the CFO' s tracking technology.

CFO Statement -
Part of our competitive advantage is that we penalize ourselves for late shipments and deliveries. Knowing where out shipments are at all times has a direct correlation to our bottom line and profitability. Additionally, I don't want to commit capital to building out a server environment.
Flowlogistic wants to use Google BigQuery as their primary analysis system, but they still have Apache Hadoop and Spark workloads that they cannot move to
BigQuery. Flowlogistic does not know how to store the data that is common to both workloads. What should they do?

A. Store the common data in BigQuery as partitioned tables.

B. Store the common data in BigQuery and expose authorized views.

C. Store the common data encoded as Avro in Google Cloud Storage. **Most Voted**

D. Store he common data in the HDFS storage for a Google Cloud Dataproc cluster.

Correct Answer: C

Community vote distribution



Comments

rtcpost **Highly Voted** 2 years, 5 months ago

Selected Answer: C

C. Store the common data encoded as Avro in Google Cloud Storage. This approach allows for interoperability between BigQuery and Hadoop/Spark as Avro is a commonly used data serialization format that can be read by both systems. Data stored in Google Cloud Storage can be accessed by both BigQuery and Dataproc, providing a bridge between the two environments. Additionally, you can set up data transformation pipelines in Dataproc to work with this data.

iooj 1 year, 8 months ago

Selected Answer: C

in BigQuery we can use BigLake tables based on Avro for historical data, and Spark stored procedures

dhvanil 1 year, 9 months ago

Data lake,fully managed, data analytics. Stores structured and unstructured data are keywords,so answer is GCS, OPTION C

JOKKUNO 2 years, 3 months ago

Given the scenario described for Flowlogistic's requirements and technical environment, the most suitable option for storing common data that is used by both Google BigQuery and Apache Hadoop/Spark workloads is: C. Store the common data encoded as Avro in Google Cloud Storage.

nescafe7 2 years, 8 months ago

Selected Answer: D

To simplify the question, Apache Hadoop and Spark workloads that cannot be moved to BigQuery can be handled by DataProc. So the correct answer is D.

Mathew106 2 years, 8 months ago

Selected Answer: B

B is the right answer. Common data will lie in BigQuery but will be accessible via the views with SQL in Hadoop workloads.

midgoo 3 years, 1 month ago

Selected Answer: B

C should be the correct answer. However, please note that Google just released the BigQuery Connector for Hadoop, so if they ask the same question today, B will be the correct answer. A could be correct too, but I cannot see why it has to be partitioned

res3 2 years, 9 months ago

If you check the https://cloud.google.com/dataproc/docs/concepts/connectors/bigquery, it unloads the BQ data to GCS, utilizes it, and then deletes it from the GCS. Storing common data twice (at BQ and GCS) will not be the best option compared to 'C' (using GCS as the main common dataset).

korntewin 3 years, 2 months ago

Selected Answer: C

I would vote for C as it can be used for analysis with Bigquery. Furthermore, Hadoop workload can also be transferred to dataproc connected to GCS.

DGames 3 years, 3 months ago

Selected Answer: B

Answer B look ok , because in question they want to store common data which can use by both workload, and using big query and primary analytical tool that would be best option and easy to analysis common data.

kelvintoys93 3 years, 4 months ago

"Perform analytics on all their orders and shipment logs, which contain both structured and unstructured data" - BigQuery cant take unstructured data so A and B are out. Storing data in HDFS storage is never recommended unless latency is a requirement, so D is out. That leaves us with GCS. Answer is C

tunstila 3 years, 3 months ago

I thought you can now store unstructured data in BigQuery via the object tables announced during Google NEXT 2022... If that's possib;e, does that make B a better choice?

drunk_goat82 3 years, 4 months ago

Selected Answer: C

BigQuery can use federated queries to connect to the avro data in GCS while running spark jobs on it. If you duplicate the date you have to manage both data sets.

wan2three 3 years, 4 months ago

A They wanted BigQuery. And connector is all you need to perform Hadoop or spark. Hadoop migration can be done using dataproc.

wan2three 3 years, 4 months ago

Also apparently they want all data at one place and want bigQ

gudiking 3 years, 4 months ago

Selected Answer: C

C as it can be used as an external table from BigQuery and with the Cloud Storage Connector it can be used by the Spark workloads (running in Dataproc)

solar_maker 3 years, 4 months ago

Selected Answer: C

C, as both capable of AVRO, but the customer does not know what they want to do with the data yet.

Leelas 3 years, 4 months ago

Selected Answer: D

In Technical requirements it Was clearly mentioned that they need to Migrate existing Hadoop Cluster for which Data Proc Cluster is a replacement.

vishal0202 3 years, 6 months ago

C is ans...avro data can be accessed by spark as well

ducc 3 years, 7 months ago

Selected Answer: C

The answer is C

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #35 **Topic 1**

Flowlogistic Case Study -

Company Overview -
Flowlogistic is a leading logistics and supply chain provider. They help businesses throughout the world manage their resources and transport them to their final destination. The company has grown rapidly, expanding their offerings to include rail, truck, aircraft, and oceanic shipping.

Company Background -
The company started as a regional trucking company, and then expanded into other logistics market. Because they have not updated their infrastructure, managing and tracking orders and shipments has become a bottleneck. To improve operations, Flowlogistic developed proprietary technology for tracking shipments in real time at the parcel level. However, they are unable to deploy it because their technology stack, based on Apache Kafka, cannot support the processing volume. In addition, Flowlogistic wants to further analyze their orders and shipments to determine how best to deploy their resources.

Solution Concept -
Flowlogistic wants to implement two concepts using the cloud:
✑ Use their proprietary technology in a real-time inventory-tracking system that indicates the location of their loads
✑ Perform analytics on all their orders and shipment logs, which contain both structured and unstructured data, to determine how best to deploy resources, which markets to expand info. They also want to use predictive analytics to learn earlier when a shipment will be delayed.

Existing Technical Environment -
Flowlogistic architecture resides in a single data center:
✑ Databases
8 physical servers in 2 clusters
- SQL Server `" user data, inventory, static data
3 physical servers
- Cassandra `" metadata, tracking messages
10 Kafka servers `" tracking message aggregation and batch insert
✑ Application servers `" customer front end, middleware for order/customs
60 virtual machines across 20 physical servers
- Tomcat `" Java services
- Nginx `" static content
- Batch servers
✑ Storage appliances
- iSCSI for virtual machine (VM) hosts
- Fibre Channel storage area network (FC SAN) `" SQL server storage
- Network-attached storage (NAS) image storage, logs, backups
✑ 10 Apache Hadoop /Spark servers

- Core Data Lake
- Data analysis workloads
✑ 20 miscellaneous servers
- Jenkins, monitoring, bastion hosts,

Business Requirements -
✑ Build a reliable and reproducible environment with scaled panty of production.
✑ Aggregate data in a centralized Data Lake for analysis
✑ Use historical data to perform predictive analytics on future shipments
✑ Accurately track every shipment worldwide using proprietary technology
✑ Improve business agility and speed of innovation through rapid provisioning of new resources
✑ Analyze and optimize architecture for performance in the cloud
✑ Migrate fully to the cloud if all other requirements are met

Technical Requirements -
✑ Handle both streaming and batch data
✑ Migrate existing Hadoop workloads
✑ Ensure architecture is scalable and elastic to meet the changing demands of the company.
✑ Use managed services whenever possible
✑ Encrypt data flight and at rest
✑ Connect a VPN between the production data center and cloud environment

SEO Statement -
We have grown so quickly that our inability to upgrade our infrastructure is really hampering further growth and efficiency. We are efficient at moving shipments around the world, but we are inefficient at moving data around. We need to organize our information so we can more easily understand where our customers are and what they are shipping.

CTO Statement -
IT has never been a priority for us, so as our data has grown, we have not invested enough in our technology. I have a good staff to manage IT, but they are so busy managing our infrastructure that I cannot get them to do the things that really matter, such as organizing our data, building the analytics, and figuring out how to implement the CFO' s tracking technology.

CFO Statement -
Part of our competitive advantage is that we penalize ourselves for late shipments and deliveries. Knowing where out shipments are at all times has a direct correlation to our bottom line and profitability. Additionally, I don't want to commit capital to building out a server environment.
Flowlogistic's management has determined that the current Apache Kafka servers cannot handle the data volume for their real-time inventory tracking system.
You need to build a new system on Google Cloud Platform (GCP) that will feed the proprietary tracking software. The system must be able to ingest data from a variety of global sources, process and query in real-time, and store the data reliably. Which combination of GCP products should you choose?

A. Cloud Pub/Sub, Cloud Dataflow, and Cloud Storage **Most Voted**

B. Cloud Pub/Sub, Cloud Dataflow, and Local SSD

C. Cloud Pub/Sub, Cloud SQL, and Cloud Storage

D. Cloud Load Balancing, Cloud Dataflow, and Cloud Storage

Correct Answer: A

Community vote distribution



Comments

jvg637 Highly Voted 5 years, 6 months ago

I would say A. I think Pub/Sub can't directly send data to Cloud SQL.

Dhamsl Highly Voted 3 years ago

This site make me feel that it intends to make users to be involved in discussion by suggesting wrong answer

billalltf 1 year, 4 months ago

Selected Answer: A

A is right answer

JOKKUNO 1 year, 10 months ago

Given the requirements for ingesting data from global sources, processing and querying in real-time, and storing the data reliably for the real-time inventory tracking system, the most suitable combination of Google Cloud Platform (GCP) products is: A. Cloud Pub/Sub, Cloud Dataflow, and Cloud Storage Explanation: Cloud Pub/Sub: It is a messaging service that allows you to asynchronously send and receive messages between independent applications. Cloud Dataflow: It can handle both streaming and batch data, making it suitable for real-time processing of data from various sources. Cloud Storage: Cloud Storage can be used to store the processed and analyzed data reliably. It provides scalable, durable, and globally accessible object storage, making it suitable for storing large volumes of data.

rtcpost 1 year, 11 months ago

Selected Answer: A

A. Cloud Pub/Sub, Cloud Dataflow, and Cloud Storage Here's why this combination is suitable: Cloud Pub/Sub: It is used for ingesting real-time data from various global sources. It's a messaging service that can handle large volumes of data and is highly scalable. Cloud Dataflow: It's a stream and batch data processing service that allows you to process and analyze the data in real-time. It can take data from Pub/Sub and perform transformations or aggregations as needed. Cloud Storage: It provides reliable storage for the data. You can store the processed data in Cloud Storage for further analysis, and it is a scalable and durable storage solution. Option B is not ideal because Local SSDs are not a suitable storage option for persisting data that needs to be reliably stored. Option C includes Cloud SQL, which is not typically used for ingesting and processing real-time data. Option D includes Cloud Load Balancing, which is not relevant to the use case of ingesting and processing data for the inventory tracking system.

Vipul1600 2 years, 2 months ago

Since Cloud SQL is fully managed service & Dataflow is serverless hence we should opt for dataflow as it is thumb rule for google that we should choose serverless product over fully managed service.

Mathew106 2 years, 2 months ago

Selected Answer: A

The technical requirements mention that the pipeline should handle both streaming and batch data. The solution should include DataFlow and not Cloud SQL. the answer is A.

niketd 2 years, 7 months ago

Selected Answer: A

Pub/Sub to scale streaming data, Dataflow to processes both structured and unstructured data and cloud storage to store common data

PolyMoe 2 years, 8 months ago

Selected Answer: A

Option B. Cloud Pub/Sub, Cloud Dataflow, and Local SSD is not a good option as Local SSD is not a scalable solution and could not handle large amount of data Option C. Cloud Pub/Sub, Cloud SQL, and Cloud Storage is not a good option as Cloud SQL is a relational database and is not suitable for real-time processing and querying large amounts of data Option D. Cloud Load Balancing, Cloud Dataflow, and Cloud Storage is not a good option as Cloud Load Balancing is used for distributing traffic across multiple instances, it doesn't handle data processing and storage.

PolyMoe 2 years, 8 months ago

Selected Answer: A

A. Cloud Pub/Sub, Cloud Dataflow, and Cloud Storage is the best combination of GCP products for the use case described. Cloud Pub/Sub can be used to ingest data from a variety of global sources, as it allows for easy integration with external systems through its publish-subscribe messaging model. Cloud Dataflow can be used to process and query the data in real-time, as it is a fully managed service for creating data pipelines that can handle both batch and streaming data. Cloud Storage can be used to store the data reliably, as it is a fully managed object storage service that can handle large amounts of data and is highly durable and available.

jkh_goh 2 years, 8 months ago

Selected Answer: A

Answer is A. Cloud Dataflow for batch + streaming, Cloud Pub/Sub for streaming ingestion, Cloud Storage for long term data storage.

Jay_Krish 2 years, 10 months ago

Are scenario based questions still in the latest exam?? Are these still relevant?

kastuarr 2 years, 11 months ago

Selected Answer: C

Existing inventory data is in SQL, data ingested from Kafka will need to update inventory at some point. Existence of SQL in current estate indicates SQL must be present in the Cloud estate

Megmang 3 years, 1 month ago

Selected Answer: A

Answer is clearly option A.

ratnesh99 3 years, 2 months ago

Answer A : because Cloud Sql not suitable for Global

CedricLP 3 years, 5 months ago

Selected Answer: A

Only A can manage a lot's of data. Target is Cloud Storage (obviously not SSD) Input is Pub/Sub to replace Kafka Cloud SQL + Storage has no sense in this context

Arkon88 3 years, 7 months ago

Selected Answer: A

A. Cloud Pub/Sub, Cloud Dataflow, and Cloud Storage as explained by JayZeeLee : B is incorrect, because local SSD wouldn't satisfy the needs. C is incorrect, because one of the requirements is 'Global', Cloud SQL is well suited for regional applications. Cloud Spanner is a better suit in that regard. D is incorrect, because Load Balancer is for web traffic, not messages.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

Question #36 Topic 1

Flowlogistic Case Study -

Company Overview -
Flowlogistic is a leading logistics and supply chain provider. They help businesses throughout the world manage their resources and transport them to their final destination. The company has grown rapidly, expanding their offerings to include rail, truck, aircraft, and oceanic shipping.

Company Background -
The company started as a regional trucking company, and then expanded into other logistics market. Because they have not updated their infrastructure, managing and tracking orders and shipments has become a bottleneck. To improve operations, Flowlogistic developed proprietary technology for tracking shipments in real time at the parcel level. However, they are unable to deploy it because their technology stack, based on Apache Kafka, cannot support the processing volume. In addition, Flowlogistic wants to further analyze their orders and shipments to determine how best to deploy their resources.

Solution Concept -
Flowlogistic wants to implement two concepts using the cloud:
Use their proprietary technology in a real-time inventory-tracking system that indicates the location of their loads
▪

✑ Perform analytics on all their orders and shipment logs, which contain both structured and unstructured data, to determine how best to deploy resources, which markets to expand info. They also want to use predictive analytics to learn earlier when a shipment will be delayed.

Existing Technical Environment -
Flowlogistic architecture resides in a single data center:
✑ Databases
8 physical servers in 2 clusters
- SQL Server `" user data, inventory, static data
3 physical servers
- Cassandra `" metadata, tracking messages
10 Kafka servers `" tracking message aggregation and batch insert
✑ Application servers `" customer front end, middleware for order/customs
60 virtual machines across 20 physical servers
- Tomcat `" Java services
- Nginx `" static content
- Batch servers
✑ Storage appliances
- iSCSI for virtual machine (VM) hosts
- Fibre Channel storage area network (FC SAN) `" SQL server storage
- Network-attached storage (NAS) image storage, logs, backups
✑ 10 Apache Hadoop /Spark servers

- Core Data Lake
- Data analysis workloads
✑ 20 miscellaneous servers
- Jenkins, monitoring, bastion hosts,

Business Requirements -
✑ Build a reliable and reproducible environment with scaled panty of production.
✑ Aggregate data in a centralized Data Lake for analysis
✑ Use historical data to perform predictive analytics on future shipments
✑ Accurately track every shipment worldwide using proprietary technology
✑ Improve business agility and speed of innovation through rapid provisioning of new resources
✑ Analyze and optimize architecture for performance in the cloud
✑ Migrate fully to the cloud if all other requirements are met

Technical Requirements -
Handle both streaming and batch data
▪
✑ Migrate existing Hadoop workloads
✑ Ensure architecture is scalable and elastic to meet the changing demands of the company.
✑ Use managed services whenever possible
✑ Encrypt data flight and at rest
✑ Connect a VPN between the production data center and cloud environment

SEO Statement -
We have grown so quickly that our inability to upgrade our infrastructure is really hampering further growth and efficiency. We are efficient at moving shipments around the world, but we are inefficient at moving data around. We need to organize our information so we can more easily understand where our customers are and what they are shipping.

CTO Statement -
IT has never been a priority for us, so as our data has grown, we have not invested enough in our technology. I have a good staff to manage IT, but they are so busy managing our infrastructure that I cannot get them to do the things that really matter, such as organizing our data, building the analytics, and figuring out how to implement the CFO' s tracking technology.

CFO Statement -
Part of our competitive advantage is that we penalize ourselves for late shipments and deliveries. Knowing where out shipments are at all times has a direct correlation to our bottom line and profitability. Additionally, I don't want to commit capital to building out a server environment.
Flowlogistic's CEO wants to gain rapid insight into their customer base so his sales team can be better informed in the field. This team is not very technical, so they've purchased a visualization tool to simplify the creation of BigQuery reports. However, they've been overwhelmed by all the data in the table, and are spending a lot of money on queries trying to find the data they need. You want to solve their problem in the most cost-effective way. What should you do?

A. Export the data into a Google Sheet for virtualization.

B. Create an additional table with only the necessary columns.

C. Create a view on the table to present to the virtualization tool. **Most Voted**

D. Create identity and access management (IAM) roles on the appropriate columns, so only they appear in a query.

Correct Answer: C

Community vote distribution



Comments

Radhika7983 **Highly Voted** 5 years, 4 months ago

Answer is C. A logical view can be created with only the required columns which is required for visualization. B is not the right option as you will create a table and make it static. What happens when the original data is updated. This new table will not have the latest data and hence view is the best possible option here.

jin0 3 years, 1 month ago

I don't think so because in question they worried about spending money for query but, using view could not make money safe because logical view scan all of the data in the table. so, for saving money for query then Answer B is more suitable

mark1223jkh 1 year, 10 months ago

The point is reducing the number of columns, not caching. Yes, it will query the table, but with only the necessary columns the view has.

Dan137 **Highly Voted** 3 years, 7 months ago

Selected Answer: B

I go with B becase according to views documentation: https://cloud.google.com/bigquery/docs/views-intro#view_pricing "BigQuery's views are logical views, not materialized views. Because views are not materialized, the query that defines the view is run each time the view is queried. Queries are billed according to the total amount of data in all table fields referenced directly or indirectly by the top-level query. For more information, see query pricing."

Mathew106 2 years, 8 months ago

Google materialized views and you will find there are also materialized views. However, I do believe that the answer should mention "materialized" and not just view.

cqrm3n 1 year, 1 month ago

Selected Answer: C

Answer is C: A BigQuery view solves this problem by: Simplifying the dataset → Shows only the relevant columns/data. Improving query efficiency → Reduces query costs by filtering unnecessary data. Enhancing usability → Sales teams get only the insights they need without dealing with complex queries. Reducing costs → Since views don’t store data separately, they don’t incur additional storage costs.

SamuelTsch 1 year, 5 months ago

Selected Answer: C

Answer C contains both requirements - minimal columns by creating view of the table and visualization tool for report

rocky48 2 years, 4 months ago

Selected Answer: C

Answer: C

rtcpost 2 years, 5 months ago

Selected Answer: C

C. Create a view on the table to present to the virtualization tool. Creating a view in BigQuery allows you to define a virtual table that is a subset of the original data, containing only the necessary columns or filtered data that the sales team requires for their reports. This approach is cost-effective because it doesn't involve exporting data to external tools or creating additional tables, and it ensures that the sales team is working with the specific data they need without running expensive queries on the full dataset. It simplifies the data for non-technical users while keeping the data in BigQuery, which is a powerful and cost-efficient data warehousing solution. Options A (exporting to Google Sheet) and B (creating an additional table) might introduce data redundancy and maintenance overhead, and they don't provide the same level of control and security as creating a view. Option D (IAM roles) doesn't address the issue of simplifying the data for the sales team; it's more focused on access control.

Mathew106 2 years, 8 months ago

Selected Answer: C

C. You won't pay for storage for the view, and it will only include the necessary columns. Even if we assume that we don't talk about a materialized view, a logical view query can use the cache as much as a table query. So a new table does not have any benefit over a view, even if the view is logical.

abi01a 2 years, 11 months ago

The answer is C

kplam 2 years, 11 months ago

Answer is C

lucaluca1982 3 years ago

Selected Answer: B

B it is more cost-effective and efficient approach to handle reports

bha11111 3 years ago

Selected Answer: C

C is correct

Booqq 3 years ago

C —— view is better than another table to keep data consistent

JJJJim 3 years, 1 month ago

Selected Answer: C

Answer is C, creating view tables can easy and flexible to do the most cost-effetive way.

PolyMoe 3 years, 2 months ago

Selected Answer: C

The appropriate solution is C, creating a view on the table, by selecting the relevant columns only (and not by creating another, static, table)

dconesoko 3 years, 2 months ago

Selected Answer: B

Providing that the question was explicit in B and D about the selection of the appropriate columns it quite intriguing that it did not mention the selection of the appropriate column for the view. We can definitely build a view which might just present the same data or something much complex, thus i vote for B

GCPpro 3 years, 2 months ago

C is the correct answer

Isaga 3 years, 2 months ago

Selected Answer: C

I mean C

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #37 **Topic 1**

Flowlogistic Case Study -

Company Overview -
Flowlogistic is a leading logistics and supply chain provider. They help businesses throughout the world manage their resources and transport them to their final destination. The company has grown rapidly, expanding their offerings to include rail, truck, aircraft, and oceanic shipping.

Company Background -
The company started as a regional trucking company, and then expanded into other logistics market. Because they have not updated their infrastructure, managing and tracking orders and shipments has become a bottleneck. To improve operations, Flowlogistic developed proprietary technology for tracking shipments in real time at the parcel level. However, they are unable to deploy it because their technology stack, based on Apache Kafka, cannot support the processing volume. In addition, Flowlogistic wants to further analyze their orders and shipments to determine how best to deploy their resources.

Solution Concept -
Flowlogistic wants to implement two concepts using the cloud:
✑ Use their proprietary technology in a real-time inventory-tracking system that indicates the location of their loads
✑ Perform analytics on all their orders and shipment logs, which contain both structured and unstructured data, to determine how best to deploy resources, which markets to expand info. They also want to use predictive analytics to learn earlier when a shipment will be delayed.

Existing Technical Environment -
Flowlogistic architecture resides in a single data center:
✑ Databases
8 physical servers in 2 clusters
- SQL Server `" user data, inventory, static data
3 physical servers
- Cassandra `" metadata, tracking messages
10 Kafka servers `" tracking message aggregation and batch insert
✑ Application servers `" customer front end, middleware for order/customs
60 virtual machines across 20 physical servers
- Tomcat `" Java services
- Nginx `" static content
- Batch servers
✑ Storage appliances
- iSCSI for virtual machine (VM) hosts
- Fibre Channel storage area network (FC SAN) `" SQL server storage
- Network-attached storage (NAS) image storage, logs, backups
✑ 10 Apache Hadoop /Spark servers

- Core Data Lake
- Data analysis workloads
✑ 20 miscellaneous servers
- Jenkins, monitoring, bastion hosts,

Business Requirements -
✑ Build a reliable and reproducible environment with scaled panty of production.
✑ Aggregate data in a centralized Data Lake for analysis
✑ Use historical data to perform predictive analytics on future shipments
✑ Accurately track every shipment worldwide using proprietary technology
✑ Improve business agility and speed of innovation through rapid provisioning of new resources
✑ Analyze and optimize architecture for performance in the cloud
✑ Migrate fully to the cloud if all other requirements are met

Technical Requirements -
✑ Handle both streaming and batch data
✑ Migrate existing Hadoop workloads
✑ Ensure architecture is scalable and elastic to meet the changing demands of the company.
✑ Use managed services whenever possible
✑ Encrypt data flight and at rest
✑ Connect a VPN between the production data center and cloud environment

SEO Statement -
We have grown so quickly that our inability to upgrade our infrastructure is really hampering further growth and efficiency. We are efficient at moving shipments around the world, but we are inefficient at moving data around. We need to organize our information so we can more easily understand where our customers are and what they are shipping.

CTO Statement -
IT has never been a priority for us, so as our data has grown, we have not invested enough in our technology. I have a good staff to manage IT, but they are so busy managing our infrastructure that I cannot get them to do the things that really matter, such as organizing our data, building the analytics, and figuring out how to implement the CFO' s tracking technology.

CFO Statement -
Part of our competitive advantage is that we penalize ourselves for late shipments and deliveries. Knowing where out shipments are at all times has a direct correlation to our bottom line and profitability. Additionally, I don't want to commit capital to building out a server environment.
Flowlogistic is rolling out their real-time inventory tracking system. The tracking devices will all send package-tracking messages, which will now go to a single
Google Cloud Pub/Sub topic instead of the Apache Kafka cluster. A subscriber application will then process the messages for real-time reporting and store them in
Google BigQuery for historical analysis. You want to ensure the package data can be analyzed over time.
Which approach should you take?

A. Attach the timestamp on each message in the Cloud Pub/Sub subscriber application as they are received.

B. Attach the timestamp and Package ID on the outbound message from each publisher device as they are sent to Clod Pub/Sub. **Most Voted**

C. Use the NOW () function in BigQuery to record the event's time.

D. Use the automatically generated timestamp from Cloud Pub/Sub to order the data.

Correct Answer: B

Community vote distribution



Comments

Manue **Highly Voted** 3 years, 11 months ago

"However, they are unable to deploy it because their technology stack, based on Apache Kafka, cannot support the processing volume." Sure man, Kafka is not performing, let's use PubSub instead hahaha...

ralf_cc 3 years, 9 months ago

lol this is a vendor exam...

sfsdeniso 2 years, 4 months ago

google send via pub sub web indexes twice a day a whole internet is being sent via pub sub

humza **Highly Voted** 3 years, 8 months ago

Answer: B A. There is no indication that the application can do this. Moreover, due to networking problems, it is possible that Pub/Sub doesn't receive messages in order. This will analysis difficult. B. This makes sure that you have access to publishing timestamp which provides you with the correct ordering of messages. C. If timestamps are already messed up, BigQuery will get wrong results anyways. D. The timestamp we are interested in is when the data was produced by the publisher, not when it was received by Pub/Sub.

rtcpost 1 year, 5 months ago

Selected Answer: B

B. Attach the timestamp and Package ID on the outbound message from each publisher device as they are sent to Cloud Pub/Sub. Here's why this approach is the most suitable: By attaching a timestamp and Package ID at the point of origin (publisher device), you ensure that each message has a clear and consistent timestamp associated with it from the moment it is generated. This provides a reliable and accurate record of when each package-tracking message was created, which is crucial for analyzing the data over time. This approach allows you to maintain the chronological order of events as they occurred at the source, which is important for real-time reporting and historical analysis. Option A suggests attaching the timestamp in the Cloud Pub/Sub subscriber application. While this can work, it introduces a potential delay and the risk of timestamps not being accurate if there are issues with message processing. Option C, using the NOW() function in BigQuery, records the time when the data is ingested into BigQuery, which may not reflect the actual time of the event.

JJJJim 2 years, 1 month ago

Selected Answer: B

Answer is B, attach the timestamp and ID is necessary to analyze data easily.

nidmed 2 years, 11 months ago

Selected Answer: B

Answer: B

Arkon88 3 years ago

Selected Answer: B

we need package ID + Timestamp so B

davidqianwen 3 years, 2 months ago

Selected Answer: B

Answer: B

exnaniantwort 3 years, 2 months ago

Selected Answer: B

agree with humza

sraakesh95 3 years, 2 months ago

Selected Answer: D

https://cloud.google.com/pubsub/docs/reference/rest/v1/PubsubMessage

Chelseajcole 3 years, 5 months ago

It is about processing time and event time.. Answer is B.

Tanzu 3 years, 2 months ago

not just timing, but also package-id .. cause they are sending 1 topic in gcp instead of to many in kafka. that means there must be added some additional critical data too.

anji007 3 years, 5 months ago

Ans: B A: Adding timestamp as they received is not a better option, messages may not arrive in order at the receiver/ subscriber, could be due to connectivity or network. B: Timestamp should be added here. C: Doesn't make sense at all. D: Ordering should be based on the order how messages are generated at the publisher but not as per order they reach the pub/sub.

sumanshu 3 years, 8 months ago

Vote for B

funtoosh 4 years, 1 month ago

Better if the publisher attached the package ID and Timestamp as packages can come in an Asynchronous fashion.

naga 4 years, 1 month ago

Correct B

Radhika7983 4 years, 4 months ago

The answer is B. JSON representation { "data": string, "attributes": { string: string, ... }, "messageId": string, "publishTime": string, "orderingKey": string } In the attribute, we can have package id and timestamp.

snamburi3 4 years, 4 months ago

D. "PublishTime- It must not be populated by the publisher in a topics.publish call." https://cloud.google.com/pubsub/docs/reference/rest/v1/PubsubMessage

Tanzu 3 years, 2 months ago

you can add timestamp as data , not as processTime.

snamburi3 4 years, 4 months ago

We can order the messages: https://cloud.google.com/pubsub/docs/ordering

daghayeghi 4 years ago

but no one can guarantee that finally the message will receive in order, because during the transmission of message the order of them will change. then using an identifier and timestamp would be a must.

haroldbenites 4 years, 7 months ago

B Correct

https://zhangxuetu.online

掌学兔
ZHANGXUETU

Question #38 Topic 1

MJTelco Case Study -

Company Overview -
MJTelco is a startup that plans to build networks in rapidly growing, underserved markets around the world. The company has patents for innovative optical communications hardware. Based on these patents, they can create many reliable, high-speed backbone links with inexpensive hardware.

Company Background -
Founded by experienced telecom executives, MJTelco uses technologies originally developed to overcome communications challenges in space. Fundamental to their operation, they need to create a distributed data infrastructure that drives real-time analysis and incorporates machine learning to continuously optimize their topologies. Because their hardware is inexpensive, they plan to overdeploy the network allowing them to account for the impact of dynamic regional politics on location availability and cost.
Their management and operations teams are situated all around the globe creating many-to-many relationship between data consumers and provides in their system. After careful consideration, they decided public cloud is the perfect environment to support their needs.

Solution Concept -
MJTelco is running a successful proof-of-concept (PoC) project in its labs. They have two primary needs:
✑ Scale and harden their PoC to support significantly more data flows generated when they ramp to more than 50,000 installations.
✑ Refine their machine-learning cycles to verify and improve the dynamic models they use to control topology definition.
MJTelco will also use three separate operating environments `" development/test, staging, and production `" to meet the needs of running experiments, deploying new features, and serving production customers.

Business Requirements -
✑ Scale up their production environment with minimal cost, instantiating resources when and where needed in an unpredictable, distributed telecom user community.
✑ Ensure security of their proprietary data to protect their leading-edge machine learning and analysis.
✑ Provide reliable and timely access to data for analysis from distributed research workers
✑ Maintain isolated environments that support rapid iteration of their machine-learning models without affecting their customers.

Technical Requirements -
✑ Ensure secure and efficient transport and storage of telemetry data
✑ Rapidly scale instances to support between 10,000 and 100,000 data providers with multiple flows each.
✑ Allow analysis and presentation against data tables tracking up to 2 years of data storing approximately 100m records/day
✑ Support rapid iteration of monitoring infrastructure focused on awareness of data pipeline problems both in telemetry flows and in production learning cycles.

CEO Statement -
Our business model relies on our patents, analytics and dynamic machine learning. Our inexpensive hardware is organized to be highly reliable, which gives us cost advantages. We need to quickly stabilize our large distributed data pipelines to meet our reliability and capacity commitments.

CTO Statement -
Our public cloud services must operate as advertised. We need resources that scale and keep our data secure. We also need environments in which our data scientists can carefully study and quickly adapt our models. Because we rely on automation to process our data, we also need our development and test environments to work as we iterate.

CFO Statement -
The project is too large for us to maintain the hardware and software required for the data and analysis. Also, we cannot afford to staff an operations team to monitor so many data feeds, so we will rely on automation and infrastructure. Google Cloud's machine learning will allow our quantitative researchers to work on our high-value problems instead of problems with our data pipelines.
MJTelco's Google Cloud Dataflow pipeline is now ready to start receiving data from the 50,000 installations. You want to allow Cloud Dataflow to scale its compute power up as required. Which Cloud Dataflow pipeline configuration setting should you update?

A. The zone

B. The number of workers

C. The disk size per worker

D. The maximum number of workers **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Radhika7983 **Highly Voted** 4 years, 10 months ago

The correct answer is D. Please look for the details in below https://cloud.google.com/dataflow/docs/guides/specifying-exec-params We need to specify and set execution parameters for cloud data flow . Also, to enable autoscaling, set the following execution parameters when you start your pipeline: --autoscaling_algorithm=THROUGHPUT_BASED --max_num_workers=N The objective of autoscaling streaming pipelines is to minimize backlog while maximizing worker utilization and throughput, and quickly react to spikes in load. By enabling autoscaling, you don't have to choose between provisioning for peak load and fresh results. Workers are added as CPU utilization and backlog increase and are removed as these metrics come down. This way, you're paying only for what you need, and the job is processed as efficiently as possible.

jvg637 **Highly Voted** 5 years, 6 months ago

D. The maximum number of workers answers to the scale question

cqrm3n 1 year, 1 month ago

Selected Answer: D

The answer is D because Google Dataflow is serverless and auto scales based on demand. To allow it to scale up compute power dynamically, we need to set the maximum number of workers.

I__SHA1234567 1 year, 6 months ago

Selected Answer: D

Cloud Dataflow dynamically scales the number of workers based on the amount of data being processed and the processing requirements. By updating the maximum number of workers, you allow Dataflow to scale up the compute power as needed to handle the workload efficiently. This ensures that the pipeline can adapt to changes in data volume and processing demands.

rtcpost 1 year, 11 months ago

Selected Answer: D

D. The maximum number of workers By increasing the maximum number of workers, you ensure that Cloud Dataflow can scale its compute power to handle the increased data processing load efficiently.

vaga1 2 years, 4 months ago

Selected Answer: D

dataflow auto-scales, then if it is not scaling is because it has reached the maximum number of workers that have been set.

abi01a 2 years, 5 months ago

A is the correct answer. Dataflow is Serverless. Specify your Region, autoscaling and other 'knobing' activities that are 'under the hood' will be taken care for you. Remember the company cannot afford to staff an Operations team to monitor data feeds so rely on ...

bha11111 2 years, 6 months ago

Selected Answer: D

this is correct

GCPpro 2 years, 8 months ago

D . is the correct answer

jkh_goh 2 years, 8 months ago

Answer A provided is definitely wrong. Who comes up with these answers?

Ender_H 3 years ago

Selected Answer: D

Correct Answer: D A: The zone has nothing to do with scaling computer power. B: The key word here is, "Scale its compute power up AS REQUIRED", with this answer, the number of workers would immediately scale the computer power. C: we need to scale compute power, not storage D: is the correct answer, changing the Number of Maximum workers will allow Dataflow to add up to that number of workers if required. https://cloud.google.com/dataflow/docs/reference/pipeline-options#resource_utilization

sraakesh95 3 years, 8 months ago

Selected Answer: D

@Radhika7983

medeis_jar 3 years, 9 months ago

Selected Answer: D

The correct answer is D. https://cloud.google.com/dataflow/docs/guides/specifying-exec-params We need to specify and set execution parameters for cloud data flow . Also, to enable autoscaling, set the following execution parameters when you start your pipeline: --autoscaling_algorithm=THROUGHPUT_BASED --max_num_workers=N

maurodipa 3 years, 10 months ago

Answer is A: Dataflow is serverless, so no need to specify neither the number of workers, nor the max number of workers. https://cloud.google.com/dataflow

Jarek7 2 years, 5 months ago

Have you ever use it? You pay for workers processing, so you specify max number of workers. Here is the doc: https://cloud.google.com/sdk/gcloud/reference/dataflow/jobs/run

anji007 3 years, 11 months ago

Ans: D

sumanshu 4 years, 2 months ago

Vote for D

Lodu_Lalit 4 years, 6 months ago

D, thats because scalability is directly corerlated to max number of workers, size determines the speed of functioning

https://zhangxuetu.online

掌学兔
ZHANGXUETU

Question #39 Topic 1

MJTelco Case Study -

Company Overview -
MJTelco is a startup that plans to build networks in rapidly growing, underserved markets around the world. The company has patents for innovative optical communications hardware. Based on these patents, they can create many reliable, high-speed backbone links with inexpensive hardware.

Company Background -
Founded by experienced telecom executives, MJTelco uses technologies originally developed to overcome communications challenges in space. Fundamental to their operation, they need to create a distributed data infrastructure that drives real-time analysis and incorporates machine learning to continuously optimize their topologies. Because their hardware is inexpensive, they plan to overdeploy the network allowing them to account for the impact of dynamic regional politics on location availability and cost.
Their management and operations teams are situated all around the globe creating many-to-many relationship between data consumers and provides in their system. After careful consideration, they decided public cloud is the perfect environment to support their needs.

Solution Concept -
MJTelco is running a successful proof-of-concept (PoC) project in its labs. They have two primary needs:
✑ Scale and harden their PoC to support significantly more data flows generated when they ramp to more than 50,000 installations.
✑ Refine their machine-learning cycles to verify and improve the dynamic models they use to control topology definition.
MJTelco will also use three separate operating environments `" development/test, staging, and production `" to meet the needs of running experiments, deploying new features, and serving production customers.

Business Requirements -
✑ Scale up their production environment with minimal cost, instantiating resources when and where needed in an unpredictable, distributed telecom user community.
✑ Ensure security of their proprietary data to protect their leading-edge machine learning and analysis.
✑ Provide reliable and timely access to data for analysis from distributed research workers
Maintain isolated environments that support rapid iteration of their machine-learning models without affecting their customers.
▪

Technical Requirements -
✑ Ensure secure and efficient transport and storage of telemetry data
✑ Rapidly scale instances to support between 10,000 and 100,000 data providers with multiple flows each.
✑ Allow analysis and presentation against data tables tracking up to 2 years of data storing approximately 100m records/day
✑ Support rapid iteration of monitoring infrastructure focused on awareness of data pipeline problems both in

telemetry flows and in production learning cycles.

CEO Statement -
Our business model relies on our patents, analytics and dynamic machine learning. Our inexpensive hardware is organized to be highly reliable, which gives us cost advantages. We need to quickly stabilize our large distributed data pipelines to meet our reliability and capacity commitments.

CTO Statement -
Our public cloud services must operate as advertised. We need resources that scale and keep our data secure. We also need environments in which our data scientists can carefully study and quickly adapt our models. Because we rely on automation to process our data, we also need our development and test environments to work as we iterate.

CFO Statement -
The project is too large for us to maintain the hardware and software required for the data and analysis. Also, we cannot afford to staff an operations team to monitor so many data feeds, so we will rely on automation and infrastructure. Google Cloud's machine learning will allow our quantitative researchers to work on our high-value problems instead of problems with our data pipelines.
You need to compose visualizations for operations teams with the following requirements:
✑ The report must include telemetry data from all 50,000 installations for the most resent 6 weeks (sampling once every minute).
✑ The report must not be more than 3 hours delayed from live data.
✑ The actionable report should only show suboptimal links.
✑ Most suboptimal links should be sorted to the top.
✑ Suboptimal links can be grouped and filtered by regional geography.
✑ User response time to load the report must be <5 seconds.
Which approach meets the requirements?

A. Load the data into Google Sheets, use formulas to calculate a metric, and use filters/sorting to show only suboptimal links in a table.

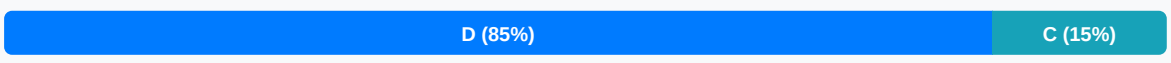
B. Load the data into Google BigQuery tables, write Google Apps Script that queries the data, calculates the metric, and shows only suboptimal rows in a table in Google Sheets.

C. Load the data into Google Cloud Datastore tables, write a Google App Engine Application that queries all rows, applies a function to derive the metric, and then renders results in a table using the Google charts and visualization API.

D. Load the data into Google BigQuery tables, write a Google Data Studio 360 report that connects to your data, calculates a metric, and then uses a filter expression to show only suboptimal rows in a table. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

itche_scratche Highly Voted 5 years, 5 months ago

D; dataflow doesn't connect to datastore, and not really for reporting. BQ, and data studio is a better choice.

ckanaar 2 years ago

Dataflow does connect to Datastore, D is still the right answer though.

arnabbis4u Highly Voted 5 years, 4 months ago

Correct D

AdriHubert 10 months, 2 weeks ago

Selected Answer: D

It is now named looker studio

rtcpost 1 year, 11 months ago

Selected Answer: D

D. Load the data into Google BigQuery tables, write a Google Data Studio 360 report that connects to your data, calculates a metric, and then uses a filter expression to show only suboptimal rows in a table. Here's why this option is the most suitable: Google BigQuery is a powerful data warehouse for processing and analyzing large datasets. It can efficiently handle the telemetry data from all 50,000 installations. Google Data Studio 360 is designed for creating interactive and visually appealing reports and dashboards. Using Google Data Studio allows you to connect to BigQuery, calculate the required metrics, and apply filters to show only suboptimal links. It can provide real-time or near-real-time data updates, ensuring that the report is not more than 3 hours delayed from live data. Google Data Studio can also be used to sort and group suboptimal links and display them based on regional geography. With the right design, you can ensure that user response time to load the report is less than 5 seconds. This approach leverages Google's cloud services effectively to meet the specified requirements.

mark1223jkh 1 year, 4 months ago

Is Google Data studio 360 a product now?

theseawillclaim 2 years, 2 months ago

Selected Answer: D

Why bother with a custom GAE app when you have Data Studio?

DGames 2 years, 9 months ago

Selected Answer: C

Its think answer would be C because of telemetry data and response time is <5 second that force me to think about datastore,

willymac2 3 years, 3 months ago

I believe the answer is C. First requirement is that it must be a visualisation with, so A and B do not work (create a table and a spreadsheet). Now the second constraint which I believe is important is that the report MUST load in less than 5 seconds. But we do not know how complex the metric computation is, thus I cannot assume that we can compute it when we want to load the report, making me think that it be must be pre-computed. Thus option D cannot work as it create the metric AFTER querying the data (we are also not sure if we can really compute it in a query).

gudguy1a 2 years ago

Ummm, sorry @willymac2, but you have to account for size and growth which datastore cannot scale to. Then, you have to worry about sub-second response time and datastore cannot do that as well as BigQuery...

Raj0123 3 years, 4 months ago

Answer D

CedricLP 3 years, 5 months ago

Selected Answer: D

DataStudio and BQ are the simpliest way to do it

devric 3 years, 5 months ago

Selected Answer: D

They also can activate BI Engine feature to improve the response time.

sraakesh95 3 years, 8 months ago

Selected Answer: D

D: Usually when a reporting tool is involved for GCP, DataStudio mostly goes by default due to it's no cost analytics and BigQuery joins it due to it's OLAP nature and the wonderful integration provided by GCP for these 2

medeis_jar 3 years, 9 months ago

Selected Answer: D

as explained by JayZeeLee

JayZeeLee 3 years, 11 months ago

D. A and B are incorrect, because Google Sheets are not the best fit to handle large amount of data. C may work, but it requires building an application which equates to more work. D is more efficient, therefore a better option.

wubston 2 years, 10 months ago

I can't think of a single compelling reason to go with anything but D, given the scope definition in the question brief.

Chelseajcole 3 years, 11 months ago

Visualization = Data Studio 360

Chelseajcole 3 years, 11 months ago

Next question give you the answer: Question #40. They using Data Studio 360 and Bigquery as source

anji007 3 years, 11 months ago

Ans: D

sumanshu 4 years, 2 months ago

Vote for D

zosoabi 4 years, 4 months ago

just check the next question (#40) to get an idea about correct answer

https://zhangxuetu.online

掌学兔
ZHANGXUETU

Question #40 Topic 1

MJTelco Case Study -

Company Overview -
MJTelco is a startup that plans to build networks in rapidly growing, underserved markets around the world. The company has patents for innovative optical communications hardware. Based on these patents, they can create many reliable, high-speed backbone links with inexpensive hardware.

Company Background -
Founded by experienced telecom executives, MJTelco uses technologies originally developed to overcome communications challenges in space. Fundamental to their operation, they need to create a distributed data infrastructure that drives real-time analysis and incorporates machine learning to continuously optimize their topologies. Because their hardware is inexpensive, they plan to overdeploy the network allowing them to account for the impact of dynamic regional politics on location availability and cost.
Their management and operations teams are situated all around the globe creating many-to-many relationship between data consumers and provides in their system. After careful consideration, they decided public cloud is the perfect environment to support their needs.

Solution Concept -
MJTelco is running a successful proof-of-concept (PoC) project in its labs. They have two primary needs:
✑ Scale and harden their PoC to support significantly more data flows generated when they ramp to more than 50,000 installations.
✑ Refine their machine-learning cycles to verify and improve the dynamic models they use to control topology definition.
MJTelco will also use three separate operating environments `" development/test, staging, and production `" to meet the needs of running experiments, deploying new features, and serving production customers.

Business Requirements -
✑ Scale up their production environment with minimal cost, instantiating resources when and where needed in an unpredictable, distributed telecom user community.
✑ Ensure security of their proprietary data to protect their leading-edge machine learning and analysis.
Provide reliable and timely access to data for analysis from distributed research workers
▪

✑ Maintain isolated environments that support rapid iteration of their machine-learning models without affecting their customers.

Technical Requirements -
✑ Ensure secure and efficient transport and storage of telemetry data
✑ Rapidly scale instances to support between 10,000 and 100,000 data providers with multiple flows each.
✑ Allow analysis and presentation against data tables tracking up to 2 years of data storing approximately 100m records/day
✑ Support rapid iteration of monitoring infrastructure focused on awareness of data pipeline problems both in

telemetry flows and in production learning cycles.

CEO Statement -
Our business model relies on our patents, analytics and dynamic machine learning. Our inexpensive hardware is organized to be highly reliable, which gives us cost advantages. We need to quickly stabilize our large distributed data pipelines to meet our reliability and capacity commitments.

CTO Statement -
Our public cloud services must operate as advertised. We need resources that scale and keep our data secure. We also need environments in which our data scientists can carefully study and quickly adapt our models. Because we rely on automation to process our data, we also need our development and test environments to work as we iterate.

CFO Statement -
The project is too large for us to maintain the hardware and software required for the data and analysis. Also, we cannot afford to staff an operations team to monitor so many data feeds, so we will rely on automation and infrastructure. Google Cloud's machine learning will allow our quantitative researchers to work on our high-value problems instead of problems with our data pipelines.
You create a new report for your large team in Google Data Studio 360. The report uses Google BigQuery as its data source. It is company policy to ensure employees can view only the data associated with their region, so you create and populate a table for each region. You need to enforce the regional access policy to the data.
Which two actions should you take? (Choose two.)

A. Ensure all the tables are included in global dataset.

B. Ensure each table is included in a dataset for a region. Most Voted

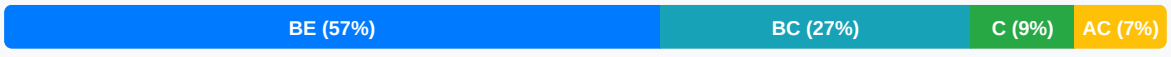
C. Adjust the settings for each table to allow a related region-based security group view access.

D. Adjust the settings for each view to allow a related region-based security group view access.

E. Adjust the settings for each dataset to allow a related region-based security group view access. Most Voted

Correct Answer: BE

Community vote distribution



Comments

shanjin14 Highly Voted 4 years, 10 months ago

C is correct starting 2020, as BigQuery come with table level access control
https://cloud.google.com/blog/products/data-analytics/introducing-table-level-access-controls-in-bigquery

samstar4180 4 years, 7 months ago

Yes, the correct answer should be BC - since we can have table-level access and each region represents a table.

hendrixlives Highly Voted 4 years, 3 months ago

Selected Answer: BE

B/E: Even if now BQ offers table level access control, since the number of tables can be expected to be high, controlling access at the dataset level would ease operations. That is why I would still go for E instead of C.

exnaniantwort 4 years, 2 months ago

1 table for each region you can just expect there is a limited number of regions this number should be manageable not E

devric 3 years, 12 months ago

But the option B means that you will generate a data set per region also. So the permissions can be assigned at the dataset level. If you pick the option A also, then you should assign the permissions over tables but... what about views? You should assign permissions to them also. That's why I pick B and E.

willyunger 1 year ago

Selected Answer: BE

Less effort to assign access by dataset, not by table or view.

iooj 1 year, 8 months ago

Selected Answer: BE

C doesn't make sense, if you have already selected B

suwalsageen12 1 year, 10 months ago

Selected Answer: BE

If I choose Option E, then Option C and D are eliminated because, Once you provide the access level in dataset for the use group, it applies for both Table and view. Now for remaining options A and B. The Question itself has stated to include the table region wise in dataset, So Option A is eliminated. So, B and E are Correct answers.

niru12376 2 years, 1 month ago

BE... If we've already split the tables into regions via datasets then why give regional access through tables again.. If not, then AC but definitely not BC.. Please help

rocky48 2 years, 4 months ago

Answer : BE B. Ensure each table is included in a dataset for a region. This means that you should organize your data in BigQuery into separate datasets, one for each region. Each dataset contains the tables specific to that region. This ensures that data is segregated by region. E. Adjust the settings for each dataset to allow a related region-based security group view access.

rtcpost 2 years, 5 months ago

Selected Answer: BE

B. Ensure each table is included in a dataset for a region. This means that you should organize your data in BigQuery into separate datasets, one for each region. Each dataset contains the tables specific to that region. This ensures that data is segregated by region. E. Adjust the settings for each dataset to allow a related region-based security group view access. You should set the access controls at the dataset level in BigQuery. This means configuring access permissions for each dataset based on regional security groups. This way, you can enforce the regional access policy to the data, ensuring that users from different regions can only access the data associated with their region. Option A is not necessary because you don't need to include all the tables in a global dataset. Segregating data into region-specific datasets is a better approach for enforcing access controls. Options C and D are not typical actions in BigQuery. Access control and permissions are usually managed at the dataset level, and you can grant access to specific groups at that level.

Mathew106 2 years, 8 months ago

It's B and E or B and C. However, B and E makes some more sense because if you have one dataset for each region and they need to access the data for each region then why not allow them access to the whole dataset? What if you want to add other supplementary tables later? If you did that on a table level you would have to add access to every table separately. Still, I think both are valid because we don't have any extra requirement, but B E makes more sense.

Jarek7 2 years, 11 months ago

Selected Answer: C

The intended answer was for sure BE. If C or D would be the right answers there is absolutely no reason to do B, right? Why should you put each table into separate dataset if you then set the accesss on table/view level? What is more the question is about tables not views, so I have no idea why would anybody take D. The issue is that this question is out of date and now the right answer would be sole C.

Oleksandr0501 2 years, 11 months ago

The two actions that should be taken are B and E. B. Ensure each table is included in a dataset for a region: By creating separate datasets for each region and including only the tables associated with that region, you can enforce the regional access policy. E. Adjust the settings for each dataset to allow a related region-based security group view access: By adjusting the settings for each dataset to allow only the related region-based security group view access, you can ensure that employees can only view data associated with their region. A is incorrect because including all tables in a global dataset would not enforce the regional access policy. C is incorrect because adjusting the settings for each table is not a scalable solution, especially as the number of tables grows. D is incorrect because adjusting the settings for each view does not ensure that employees can only view data associated with their region.

sjtesla 2 years, 11 months ago

Selected Answer: BC

B: Location is on dataset level: https://cloud.google.com/bigquery/docs/datasets#dataset_limitations C: IAM can be set on table level

Lestrang 3 years, 2 months ago

Guys, there are 2 possible combinations If you think that each table represents a region, then they should all be in a global dataset and you should apply table access control to them. So A+C Otherwise you would put each table in a regional dataset, and apply access control to the dataset. Why would you create a dataset for the purpose of controlling regional access, and then only apply the controls to a table inside it? that is not extensible in the future. Anyway create dataset+access control for dataset (B+E) is also valid. Which to choose? I dont know.

PolyMoe 3 years, 2 months ago

Selected Answer: BE

First put tables in region-dedicated dataset (B) Then, ensure access control at dataset level (by creating region-based security groups) (E)

korntewin 3 years, 2 months ago

Selected Answer: AC

I would vote for AC. As we already split the table for each region, why do we need to split the dataset per region? Furthermore, the access control will be provided to the users based on table level anyway.

korntewin 3 years, 2 months ago

Oh, the location should be specified in the dataset level! Then, the dataset should be splitted by region, my bad!

MisuLava 3 years, 5 months ago

Selected Answer: BE

if you create table-level access control and grant it to different groups for different tables, what is the point of putting tables in different datasets and different regions? So i choose BE

svkds 3 years, 6 months ago

Selected Answer: BC

BigQuery come with table level access control. Since we can have table-level access and each region represents a table, B & C is correct answer.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #41 Topic 1

MJTelco Case Study -

Company Overview -
MJTelco is a startup that plans to build networks in rapidly growing, underserved markets around the world. The company has patents for innovative optical communications hardware. Based on these patents, they can create many reliable, high-speed backbone links with inexpensive hardware.

Company Background -
Founded by experienced telecom executives, MJTelco uses technologies originally developed to overcome communications challenges in space. Fundamental to their operation, they need to create a distributed data infrastructure that drives real-time analysis and incorporates machine learning to continuously optimize their topologies. Because their hardware is inexpensive, they plan to overdeploy the network allowing them to account for the impact of dynamic regional politics on location availability and cost.
Their management and operations teams are situated all around the globe creating many-to-many relationship between data consumers and provides in their system. After careful consideration, they decided public cloud is the perfect environment to support their needs.

Solution Concept -
MJTelco is running a successful proof-of-concept (PoC) project in its labs. They have two primary needs:
✑ Scale and harden their PoC to support significantly more data flows generated when they ramp to more than 50,000 installations.
✑ Refine their machine-learning cycles to verify and improve the dynamic models they use to control topology definition.
MJTelco will also use three separate operating environments `" development/test, staging, and production `" to meet the needs of running experiments, deploying new features, and serving production customers.

Business Requirements -
✑ Scale up their production environment with minimal cost, instantiating resources when and where needed in an unpredictable, distributed telecom user community.
✑ Ensure security of their proprietary data to protect their leading-edge machine learning and analysis.
✑ Provide reliable and timely access to data for analysis from distributed research workers
✑ Maintain isolated environments that support rapid iteration of their machine-learning models without affecting their customers.

Technical Requirements -
Ensure secure and efficient transport and storage of telemetry data
▪

✑ Rapidly scale instances to support between 10,000 and 100,000 data providers with multiple flows each.
✑ Allow analysis and presentation against data tables tracking up to 2 years of data storing approximately 100m records/day
✑ Support rapid iteration of monitoring infrastructure focused on awareness of data pipeline problems both in

telemetry flows and in production learning cycles.

CEO Statement -
Our business model relies on our patents, analytics and dynamic machine learning. Our inexpensive hardware is organized to be highly reliable, which gives us cost advantages. We need to quickly stabilize our large distributed data pipelines to meet our reliability and capacity commitments.

CTO Statement -
Our public cloud services must operate as advertised. We need resources that scale and keep our data secure. We also need environments in which our data scientists can carefully study and quickly adapt our models. Because we rely on automation to process our data, we also need our development and test environments to work as we iterate.

CFO Statement -
The project is too large for us to maintain the hardware and software required for the data and analysis. Also, we cannot afford to staff an operations team to monitor so many data feeds, so we will rely on automation and infrastructure. Google Cloud's machine learning will allow our quantitative researchers to work on our high-value problems instead of problems with our data pipelines.
MJTelco needs you to create a schema in Google Bigtable that will allow for the historical analysis of the last 2 years of records. Each record that comes in is sent every 15 minutes, and contains a unique identifier of the device and a data record. The most common query is for all the data for a given device for a given day.
Which schema should you use?

A. Rowkey: date#device_id Column data: data_point **Most Voted**

B. Rowkey: date Column data: device_id, data_point

C. Rowkey: device_id Column data: date, data_point

D. Rowkey: data_point Column data: device_id, date

E. Rowkey: date#data_point Column data: device_id

Correct Answer: A

Community vote distribution



Comments

itche_scratche **Highly Voted** 5 years, 11 months ago
None, rowkey should be Device_Id+Date(reverse)

Rajuuu 5 years, 8 months ago
A is a better option then other ..though not perfect as you mentioned.

sraakesh95 4 years, 2 months ago

Totally agree if we have to avoid hotspotting! , but, incase we need to choose one of the options below, would you be going for A?

sumanshu 4 years, 8 months ago

For READ operation it's is correct. i.e. Date#Device (so that data read from single node) - For write operation it should be DeviceID#Date (so that data write via multiple nodes)

Ankit267 5 years, 8 months ago

A - Key should be less granular item first to more granular item, there are more devices than date key (every 15 min)

jvg637 Highly Voted 6 years ago

think is A, since “The most common query is for all the data for a given device for a given day”, rowkey should have info for both devcie and date.

michaelkhan3 4 years, 6 months ago

Google specifically mentions that it's a bad idea to use a timestamp at the start of a rowkey https://cloud.google.com/bigtable/docs/schema-design#row-keys-avoid The answer really should be Device_id#Timestamp but with the answers we were given you would be better off leaving the timestamp out all together

Whoswho 3 years, 3 months ago

I remember seeing it as well. the answer should be A. (reversed)

wan2three 3 years, 4 months ago

but it didnt say cant use date, date and timestamp are different

FP77 2 years, 7 months ago

The date is even worse than timestamp for the problem of hot-spotting

9a58d2c 1 month, 3 weeks ago

Selected Answer: C

the only option that follows the correct design principle is the C; the one that starts with device_id. If device_id#date were an option, it would be perfect.

Ronn27 1 year, 3 months ago

Selected Answer: A

Its very confusing but what I found is timebucket concept and day can be used instead of timestamp. https://cloud.google.com/bigtable/docs/schema-design-time-series#time-buckets

cloud_rider 1 year, 4 months ago

Selected Answer: C

The correct option should be device_id#Date as it will distribute the load while writing and also be performant while reading. C is the second best option in my understanding as device Id will ensure that data sent by all the devices on a day is distributed between nodes and will not create hotspot.

SamuelTsch 1 year, 5 months ago

Selected Answer: A

I would go to date#device_id. However, i don't find this combination. A should be then chosen.

cmira123 1 year, 7 months ago

A-https://cloud.google.com/bigtable/docs/schema-design-time-series?hl=es-419#use_tall_and_narrow_tables

Lenifia 1 year, 8 months ago

Selected Answer: A

showed up in my exam. picked A. passed the exam. still not sure it's correct though

39405bb 1 year, 10 months ago

A. Rowkey: date#device_id Column data: data_point Explanation: Optimized for Most Common Query: The most common query is for all data for a given device on a given day. This schema directly matches the query pattern by including both date and device_id in the row key. This enables efficient retrieval of the required data using a single row key prefix scan. Scalability: As the number of devices and data points increases, this schema distributes the data evenly across nodes in the Bigtable cluster, avoiding hotspots and ensuring scalability. Data Organization: By storing data points as column values within each row, you can easily add new data points or timestamps without modifying the table structure.

mark1223jkh 1 year, 10 months ago

Answer C: https://cloud.google.com/bigtable/docs/schema-design#time-based:~:text=Don%27t%20use%20a%20timestamp%20by%20itself%20or%20at%20the%20beginning%20of%20a%20row%20key%2C

0725f1f 2 years ago

Selected Answer: C

c without any doubt

philli1011 2 years, 1 month ago

The right answer should be Reverse A, but since we don't have that, the best answer is C.

gise 2 years, 2 months ago

Selected Answer: C

C. This schema is best suited for historical analysis of device data over time when the most common query is to retrieve all data for a **specific device** on a **given day**. * **Row Key as `device_id`:** This allows for efficient retrieval of all data points related to a particular device in a single operation. Bigtable sorts data lexicographically by row key, so all data for a single device will be stored together. * **Column with `date` and `data_point`:** - Using `date` as a column name or part of the column qualifier allows you to quickly filter and retrieve data for specific date ranges. - Storing `data_point` as the column value provides the actual data associated with each timestamp. **Example:** With this schema, a query to get all data for `device_12345` on `2023-12-20` would efficiently target the specific row key `device_12345` and fetch the relevant columns (with dates around `2023-12-20`).

JonFrow 2 years, 3 months ago

C - the answer should the right answer. Key is "all the data for a given device for a given day" as in, Device first, and all the data + data points after. This has nothing to do with Date-based search.

rocky48 2 years, 4 months ago

Selected Answer: A

A - Key should be less granular item first to more granular item, there are more devices than date key (every 15 min)

imran79 2 years, 5 months ago

the closest match to this in the provided options is: C. Rowkey: device_id Column data: date, data_point Thus, option C would be the best choice from the given option

kenwilliams 2 years, 10 months ago

Selected Answer: A

It all comes down to the most common query

FP77 2 years, 7 months ago

Exactly "all the data for a given device for a given day" That's why the answer is C. You start by selecting the device and then the date. This solution is not prone to hot-spotting, yours is.

https://zhangxuetu.online

[202610]

Question #42 Topic 1

Your company has recently grown rapidly and now ingesting data at a significantly higher rate than it was previously. You manage the daily batch MapReduce analytics jobs in Apache Hadoop. However, the recent increase in data has meant the batch jobs are falling behind. You were asked to recommend ways the development team could increase the responsiveness of the analytics without increasing costs. What should you recommend they do?

A. Rewrite the job in Pig.

B. Rewrite the job in Apache Spark. **Most Voted**

C. Increase the size of the Hadoop cluster.

D. Decrease the size of the Hadoop cluster but also rewrite the job in Hive.

Correct Answer: B

Community vote distribution

B (100%)

Comments

jvg637 **Highly Voted** 5 years, 6 months ago

I would say B since Apache Spark is faster than Hadoop/Pig/MapReduce

Trocinek 1 year, 6 months ago

But it requires much more memory causing it more expensive, which is not what we're aiming for here..

ler_mp **Highly Voted** 2 years, 9 months ago

Wow, a question that does not recommend to use Google product

axantroff 1 year, 10 months ago

Selected Answer: B

Just a regular Spark. B

DataFrame 1 year, 10 months ago

C. I think it should be C because intent of asking question is to realize the problem of on-prem auto-scaling not the optimization that we achieve using spark in-memory features. Its GCP exam they want to highlight if hadoop cluster commodity hard doesn't increase when data increases then it can create problem unlike GCP. Hence migrate to GCP.

itsmynickname 2 years, 2 months ago

None. Being a GCP exam, it must be either Dataflow or BigQuery :D

KHAN0007 2 years, 5 months ago

I would like to take a moment to thank you all guys You guys are awesome!!!

Whoswho 2 years, 9 months ago

looks like he's trying to spark the company up.

itsmynickname 2 years, 2 months ago

It seems he's not well paid.

Krish6488 2 years, 9 months ago

Selected Answer: B

Both Pig & Spark requires rewriting the code so its an additional overhead, but as an architect I would think about a long lasting solution. Resizing Hadoop cluster can resolve the problem statement for the workloads at that point in time but not on longer run. So Spark is the right choice, although its a cost to start with, it will certainly be a long lasting solution

Mamta072 3 years, 3 months ago

Ans is B . Apache spark.

alecuba16 3 years, 5 months ago

Selected Answer: B

SPARK > hadoop, pig, hive

kped21 3 years, 7 months ago

B - Apache Spark

luamail 2 years, 5 months ago

https://www.ibm.com/cloud/blog/hadoop-vs-spark

kped21 3 years, 8 months ago

B Spark for optimization and processing.

sraakesh95 3 years, 8 months ago

Selected Answer: B

B: Spark is suitable for the given operation is much more powerful

medeis_jar 3 years, 9 months ago

Selected Answer: B

as explained by pr2web

pr2web 3 years, 9 months ago

Selected Answer: B

Ans B: Spark is a 100 times faster and utilizes memory, instead of Hadoop Mapreduce's two-stage paradigm.

MaxNRG 3 years, 10 months ago

B as Spark can improve the performance as it performs lazy in-memory execution. Spark is important because it does part of its pipeline processing in memory rather than copying from disk. For some applications, this makes Spark extremely fast.

MaxNRG 3 years, 10 months ago

With a Spark pipeline, you have two different kinds of operations, transforms and actions. Spark builds its pipeline used an abstraction called a directed graph. Each transform builds additional nodes into the graph but spark doesn't execute the pipeline until it sees an action. Spark waits until it has the whole story, all the information. This allows

Spark to choose the best way to distribute the work and run the pipeline. The process of waiting on transforms and executing on actions is called, lazy execution. For a transformation, the input is an RDD and the output is an RDD. When Spark sees a transformation, it registers it in the directed graph and then it waits. An action triggers Spark to process the pipeline, the output is usually a result format, such as a text file, rather than an RDD.

MaxNRG 3 years, 10 months ago

Option A is wrong as Pig is wrapper and would initiate Map Reduce jobs Option C is wrong as it would increase the cost. Option D is wrong Hive is wrapper and would initiate Map Reduce jobs. Also, reducing the size would reduce performance.

kastuarr 2 years, 11 months ago

Wont Option B increase the cost ? Cost of re-writing the job in Spark + Cost of additional memory ?

anji007 3 years, 11 months ago

Ans: B Spark performs better than MapReduce due to in memory processing.

https://zhangxuetu.online

[202610]

Question #43 Topic 1

You work for a large fast food restaurant chain with over 400,000 employees. You store employee information in Google BigQuery in a Users table consisting of a FirstName field and a LastName field. A member of IT is building an application and asks you to modify the schema and data in BigQuery so the application can query a FullName field consisting of the value of the FirstName field concatenated with a space, followed by the value of the LastName field for each employee. How can you make that data available while minimizing cost?

A. Create a view in BigQuery that concatenates the FirstName and LastName field values to produce the FullName. **Most Voted**

B. Add a new column called FullName to the Users table. Run an UPDATE statement that updates the FullName column for each user with the concatenation of the FirstName and LastName values.

C. Create a Google Cloud Dataflow job that queries BigQuery for the entire Users table, concatenates the FirstName value and LastName value for each user, and loads the proper values for FirstName, LastName, and FullName into a new table in BigQuery.

D. Use BigQuery to export the data for the table to a CSV file. Create a Google Cloud Dataproc job to process the CSV file and output a new CSV file containing the proper values for FirstName, LastName and FullName. Run a BigQuery load job to load the new CSV file into BigQuery.

Correct Answer: A

Community vote distribution



Comments

[Removed] **Highly Voted** 6 years ago

Answer will be A because when you create View it does not store extra space and its a logical representation, for rest of the option you need to write large code and extra processing for dataflow/dataproc

beowulf_kat 3 years, 4 months ago

I agree that A is correct. Also, I think B is wrong as the UPDATE statement is used to update values in existing columns, not to create a new column.

ovokpus 3 years, 4 months ago

Of course, you use UPDATE after creating the new column, that is what the option said

funtoosh 5 years, 1 month ago

cannot be 'A'as it clearly says that you need to change the schema and data.

exnaniantwort 4 years, 2 months ago

Your primary task is to "make data available". Changing the schema is just the request from the member "A member of IT is building an application and ***asks you to modify the schema and data*** in BigQuery". You don't have to follow it if it does not make sense.

exnaniantwort 4 years, 2 months ago

There is always different application requirement to use different format. That way you will just creating more and more redundant columns in different formats. That is tedious.

YorelNation 3 years, 7 months ago

A yes, That make a lot of sense and also if you update the table only once with UPDATE if there is a new employee it will not be up to date with the new column, if the app use a view it will be up to date every time it query. But in any case the cost will not be minimized.

[Removed] 6 years ago

Because views are not materialized, the query that defines the view is run each time the view is queried. Queries are billed according to the total amount of data in all table fields referenced directly or indirectly by the top-level query

lgdantas 5 years, 7 months ago

Wouldn´t "total amount of data in all table fields referenced directly or indirectly by the top-level query" be FirstName and LastName?

lollo1234 4 years, 10 months ago

You're right, BigQuery bills on number of bytes processed, regardless of them being materialized. If you don't create a new column and use a view instead, you will probably have a small performance hit but query costs would be the same and storage cost wouldn't increase (unlike storing a new column)

yoshik 4 years, 6 months ago

You are asked to modify the schema and data. By using a view, the underlined table remains intact.

HarshKothari21 3 years, 6 months ago

good catch, yoshik.

alecuba16 3 years, 10 months ago

Views are cached the same as regular tables are, so I don't get the point of billing. It will cost the same as query to a regular table.

ovokpus 3 years, 4 months ago

the point of billing is extra storage costs for a new concatenated column

cloud_rider 1 year, 4 months ago

This was try in Oracle era, BQ prunes the query before running, so having a view as an intermediate layer does not have any impact, unless there is a heavy filtering happening within the view definition.

BhupiSG Highly Voted 5 years ago

Correct: B BigQuery has no quota on the DML statements. (Search Google - does bigquery have quota for update). Why not C: This is a one time activity and SQL is the easiest way to program it. DataFlow is way overkill for this. You will need to find an engineer who can develop DataFlow pipelines. Whereas, SQL is so much more widely known and easier. One of the great features about BigQuery is its SQL interface. Even for BigQueryML services.

DGames 3 years, 3 months ago

But you need to maintain table means regularly you have to execute the update query whenever new data comes.

lollo1234 4 years, 10 months ago

I will also add that B would imply changing upstream workloads to write the new field every time a records gets added

lollo1234 4 years, 10 months ago

DML statements don't increase costs, but storing a new column does. I see A is correct (also see my comment above)

exnaniantwort 4 years, 2 months ago

Exactly. Cost is the reason to reject B. How come so many people vote for this wrong option?

ler_mp 3 years, 2 months ago

Storage is cheap compared to computation

henry1991_bg 1 day, 11 hours ago

Selected Answer: A

A is better because B will make a query for each employeer

willyunger 1 year ago

Selected Answer: A

Minimal cost: no extra space, no cost to set up, no need to write code, rest of applications see no change, no need to offload/reprocess/reload (although batch load is free).

LP_PDE 1 year, 2 months ago

Selected Answer: B

I would say A but since it specifically says "modify" then the answer is B.

iooj 1 year, 8 months ago

E. Say to the IT specialist to take care of it on the app side... B would work for historical data if we had an underlying change made to automate the concatenation for new records. It is not clear, so I would say A is a quick solution.

Ramanaiah 1 year, 11 months ago

Selected Answer: B

Requirement is to be able to filter on full name. So, you would be querying all data unless you have materialized full name column.

philli1011 2 years, 2 months ago

Definitely A

axantroff 2 years, 4 months ago

Selected Answer: A

The question might be outdated, but I would like to offer my perspective: 1. Ideally, I would opt for a materialized view to avoid updating pipelines 2. In 2023, I see no concerns regarding the costs involved in storing denormalized data for analytical needs 3. Regarding this question I would choose option A, although the concern about extra costs due to recalculations is valid for me

LaxmanTiwari 2 years, 3 months ago

Did u pass the exam ?

steghe 2 years, 4 months ago

Answer should be A 'cos the First request is: make that data available.

alihabib 2 years, 7 months ago

Its A ..... "asked to change schema" is a trick to test your skills. Better to make use of MV's if anyhow the application is gonna query repeatedly. MV's will rebuild itself, if query invalidates from cache results

nescafe7 2 years, 8 months ago

Selected Answer: A

In the case of B, the data pipeline that adds new employee information must also be modified, which is not the correct answer in terms of cost minimization.

Mathew106 2 years, 8 months ago

Selected Answer: A

It's A. If you add a column to the table, you will be billed every time you query that new column. The same way you would be billed with the view created by A. B,C and D create a new column. A does not create a new column. It just provides the interface for the application to access the data. B,C and D will have to be rerun to compute the column value of new customers. A is done only once, costs 0 for storage, and is charged about the same as all the others when it comes to compute because even if you choose B C and D you would have to query the data in the end anyway.

autumn2005 2 years, 8 months ago

Selected Answer: C

modify the schema

theseawillclaim 2 years, 8 months ago

Can you code a script for a BQ Column? I don't think it's "B", it is pretty tricky

KC_go_reply 2 years, 9 months ago

Selected Answer: A

Everything but A) new view is wrong. B) sounds okay, but introduces a new column which means more storage, thus increasing cost. C) Dataflow is obvious overkill for a simple task such as concatenating two strings. D) Starting up a Dataproc cluster just for string concatenation is super overkill.

vaga1 2 years, 10 months ago

Selected Answer: A

if a new field is only necessary for one project, and it is only the concatenation of two existing fields, it is ok to create a view that gets used for a specific task.

https://zhangxuetu.online

[202610]

Question #44 Topic 1

You are deploying a new storage system for your mobile application, which is a media streaming service. You decide the best fit is Google Cloud Datastore. You have entities with multiple properties, some of which can take on multiple values. For example, in the entity 'Movie' the property 'actors' and the property 'tags' have multiple values but the property 'date released' does not. A typical query would ask for all movies with actor=<actorname> ordered by date_released or all movies with tag=Comedy ordered by date_released. How should you avoid a combinatorial explosion in the number of indexes?

A. Manually configure the index in your index config as follows:

```
Indexes:
-kind: Movie
    Properties:
    -name: actors
    name: date_released
-kind: Movie
    Properties:
    -name: tags
    name: date_released
```

Most Voted

B. Manually configure the index in your index config as follows:

```
Indexes:
    -kind: Movie
        Properties:
        -name: actors
        -name: tags
-name: date_published
```

C. Set the following in your entity options: exclude_from_indexes = 'actors, tags'

D. Set the following in your entity options: exclude_from_indexes = 'date_published'

Correct Answer: A

Community vote distribution



Comments

Wasss123 Highly Voted 2 years ago

Selected Answer: A

Correct answer is A Read in reference : https://cloud.google.com/datastore/docs/concepts/indexes#index_limits n this case, you can circumvent the exploding index by manually configuring an index in your index configuration file: indexes: - kind: Task properties: - name: tags - name: created - kind: Task properties: - name: collaborators - name: created This reduces the number of entries needed to only (|tags| * |created| + |collaborators| * |created|), or 6 entries instead of 9

jkhong 1 year, 9 months ago

Selected Answer: A

you can circumvent the exploding index by manually configuring an index in your index configuration file: https://cloud.google.com/datastore/docs/concepts/indexes#index_limits

Krish6488 1 year, 9 months ago

Selected Answer: D

Tempted to go with D as the syntax in Option A seems incorrect. D is still a possible answer because one of the ways to get rid of index errors is to remove the entities that are causing the index to explode. In this case its date_released and hence D appears right to me

DGames 1 year, 9 months ago

Selected Answer: A

Option B & D reject because mention date_publised in question date_released is column Option C also not correct, I would go with option A.

Ender_H 2 years ago

Selected Answer: D

Correct Answer D: This is the way the DB is typically queried: - movies with actor=<actorname> ordered by date_released - movies with tag=Comedy ordered by date_released so it seems that we need indices in actor,tag and date_released for sorting. A: this would be the correct answer, however, the format is incorrect, the correct format would be '- name: date_released' correctly indented. B: This seems to be unnecessary, since typically actor and tag are not queried together. also, there is a clear indentation issue C: We don't want to ignore actor and tag, we need those indices. D: If we leave datastore to automatically create the indices and if we specify that the 'date_released' property needs to be excluded from indices, then we would have less indices (but maybe slower queries when ordering them, but hey, how many 'comedies' there could be in the world)

Ender_H 2 years ago

Findings for this answer: Indices, if not defined, will be automatically created: "By default, a Datastore mode database automatically predefines an index for each property of each entity kind. These single property indexes are suitable for simple types of queries." source: https://cloud.google.com/datastore/docs/concepts/indexes In the index limits section we see this: "a Datastore mode database creates an entry in a predefined index for every property of every entity except those you have explicitly declared as excluded from your indexes." source: https://cloud.google.com/datastore/docs/concepts/indexes#index_limits

Ender_H 2 years ago

And here is the correct way to configure indices: https://cloud.google.com/datastore/docs/tools/indexconfig so this would be the best answer: indexes: - kind: Movie properties: - name: actors - name: date_released direction: asc. <This could be left out, it defaults to direction: asc if excluded> - kind: Movie properties: - name: tag - name: date_released direction: asc. <This could be left out, it defaults to direction: asc if excluded>

Hm92730 2 years ago

What do people think about C? The question is asking how to avoid a combinatorial explosion in the number of indexes. It says "You have entities with multiple properties, some of which can take on multiple values". Put this with the below text from the documentation for Datastore indexes, it seems they're looking for "exclude the properties that will cause combinatorial explosion" which would be C. "The situation becomes worse in the case of entities with multiple properties, each of which can take on multiple values. To accommodate such an entity, the index must include an entry for every possible combination of property values. Custom indexes that refer to multiple properties, each with multiple values, can "explode" combinatorially, requiring large numbers of entries for an entity with only a relatively small number of possible property values."[1] [1] https://cloud.google.com/datastore/docs/concepts/indexes#index_limits

soichirokawa 2 years ago

B. is correct To avoid combinatoric explosion of indexes. "Two queries of the same form but with different filter values use the same index." https://cloud.google.com/datastore/docs/concepts/indexes

Wasss123 2 years ago

Correct answer is A In the same reference you provided In this case, you can circumvent the exploding index by manually configuring an index in your index configuration file: indexes: - kind: Task properties: - name: tags - name: created - kind: Task properties: - name: collaborators - name: created This reduces the number of entries needed to only (|tags| * |created| + |collaborators| * |created|), or 6 entries instead of 9

AWSandeep 2 years ago

Selected Answer: A

A. Manually configure the index in your index config as follows:

https://zhangxuetu.online

[202610]

Question #45 Topic 1

You work for a manufacturing plant that batches application log files together into a single log file once a day at 2:00 AM. You have written a Google Cloud
Dataflow job to process that log file. You need to make sure the log file in processed once per day as inexpensively as possible. What should you do?

A. Change the processing job to use Google Cloud Dataproc instead.

B. Manually start the Cloud Dataflow job each morning when you get into the office.

C. Create a cron job with Google App Engine Cron Service to run the Cloud Dataflow job. **Most Voted**

D. Configure the Cloud Dataflow job as a streaming job so that it processes the log data immediately.

Correct Answer: C

Community vote distribution

C (100%)

Comments

captainbu **Highly Voted** 2 years, 8 months ago

Selected Answer: C

C was correct but nowadays you'd schedule a Dataflow job with Cloud Scheduler:
https://cloud.google.com/community/tutorials/schedule-dataflow-jobs-with-cloud-scheduler

Radhika7983 **Highly Voted** 4 years, 10 months ago

Answer is C. https://cloud.google.com/appengine/docs/flexible/nodejs/scheduling-jobs-with-cron-yaml

grshankar9 1 year, 2 months ago

Selected Answer: C

App Engine Cron is limited to scheduling tasks within your App Engine application, whereas Cloud Scheduler can trigger actions on various Google Cloud services like Cloud Functions, Pub/Sub topics, or external HTTP endpoints.

axantroff 1 year, 10 months ago

Selected Answer: C

Service was renamed, but the answer is still - C

imran79 1 year, 11 months ago

C. Using the Google App Engine Cron Service to run the Cloud Dataflow job allows you to automate the execution of the job. By creating a cron job, you can ensure that the Dataflow job is triggered exactly once per day at a specified time. This approach is automated, reliable, and fits the requirement of processing the log file once per day.

itsmynickname 2 years, 2 months ago

C. For a modern solution, Cloud Scheduler

Maurilio_Cardoso 2 years, 4 months ago

Selected Answer: C

Currently, Cloud Scheduler takes over the scheduling functions.

jin0 2 years, 7 months ago

I don't understand why that dataflow is used for processing? even though it should be processed once per a day?? is it more suitable for processing by using Dataproc instead?

mark1223jkh 1 year, 4 months ago

Actually, google recommends Dataflow over Dataproc for both batch and streaming. Dataproc is only recommended if you are coming from hadoop, spark, ...

Ender_H 3 years ago

Selected Answer: C

Correct Answer: C. A: Dataproc is a managed Apache Spark and Apache Hadoop service, makes no sense to use it B: This might sound as the cheapest, but is highly error prone, besides, anyone in charge of this has a salary and I doubt is a low one. C: This is the easiest/fastest/cheapest way to trigger job runs, you can even set retry attempts. source: https://cloud.google.com/appengine/docs/flexible/nodejs/scheduling-jobs-with-cron-yaml. D: Setting this would be much more expensive than the cron-job

noob_master 3 years, 3 months ago

Selected Answer: C

Answer: C

anji007 3 years, 11 months ago

Ans: C

Chelseajcole 3 years, 11 months ago

I know probably this question is testing on if you know corn.yaml and its function in App Engine. But why B will be more expensive? Human capital cost? Let's say if hiring a person click the button will be cheaper than launch an app engine, should we reconsider B?

AmirN 3 years, 3 months ago

Would you rather pay someone $100,000 a year to click 'run' on jobs all day, or have them automate it and do more cutting edge work? This would be opportunity cost.

Chelseajcole 4 years ago

Scheduling Jobs with cron.yaml Free applications can have up to 20 scheduled tasks. Paid applications can have up to 250 scheduled tasks.

sumanshu 4 years, 3 months ago

Vote for 'C'

naga 4 years, 7 months ago

Correct C

haroldbenites 5 years, 1 month ago

C Correct

https://zhangxuetu.online

[202610]

Question #46 Topic 1

You work for an economic consulting firm that helps companies identify economic trends as they happen. As part of your analysis, you use Google BigQuery to correlate customer data with the average prices of the 100 most common goods sold, including bread, gasoline, milk, and others. The average prices of these goods are updated every 30 minutes. You want to make sure this data stays up to date so you can combine it with other data in BigQuery as cheaply as possible.
What should you do?

A. Load the data every 30 minutes into a new partitioned table in BigQuery.

B. Store and update the data in a regional Google Cloud Storage bucket and create a federated data source in BigQuery **Most Voted**

C. Store the data in Google Cloud Datastore. Use Google Cloud Dataflow to query BigQuery and combine the data programmatically with the data stored in Cloud Datastore

D. Store the data in a file in a regional Google Cloud Storage bucket. Use Cloud Dataflow to query BigQuery and combine the data programmatically with the data stored in Google Cloud Storage.

Correct Answer: B

Community vote distribution



Comments

mmarulli **Highly Voted** 6 years ago
this is one of the sample exam questions that google has on their website. The correct answer is B

nadavw 1 year, 7 months ago
B - since it seems that not all data is in BigQuery but the analysis is done using BigQuery so federated query is the optimal approach

BhupiSG **Highly Voted** 5 years ago
Correct B As per google docs on BigQuery: Use cases for external data sources include: Loading and cleaning your data in one pass by querying the data from an external data source (a location external to BigQuery) and writing the cleaned result into BigQuery storage. Having a small amount of frequently changing data that you join with other tables. As an external data source, the frequently changing data does not need to be reloaded every time it is updated.

BigDataBB 1 week, 6 days ago
Selected Answer: B

The right answer is B becasue the minimum partition time ramnge is 1 hour:
https://docs.cloud.google.com/bigquery/docs/partitioned-tables#ingestion_time

jatinbhatia2055 1 year, 3 months ago

Selected Answer: A

BigQuery is a powerful data warehouse designed for analyzing large datasets efficiently. Partitioning tables allows you to manage large datasets by splitting them into segments based on a key, such as time. By creating a partitioned table and updating it every 30 minutes, you can load the new price data directly into the correct partitions. BigQuery's partitioned tables optimize both the storage and querying cost because BigQuery only scans the relevant partitions when querying, minimizing the amount of data read and hence reducing costs. Partitioning by time (e.g., timestamp or date columns) is particularly effective for datasets with periodic updates (like price data) since each batch of data will be loaded into the corresponding partition.

SamuelTsch 1 year, 5 months ago

Selected Answer: B

Actually, in this question, I think B is the most suitable. C, D are somehow overkill. A due to the minimum partition granularity. However, with B, the data could not be previewd also it is not possible to estimate the cost.

Pennepal 1 year, 11 months ago

D. Store the data in a file in a regional Google Cloud Storage bucket. Use Cloud Dataflow to query BigQuery and combine the data programmatically with the data stored in Google Cloud Storage. Here's why this approach is ideal: Cost-Effective Storage: Cloud Storage offers regional storage classes that are cost-effective for frequently accessed data. Storing the price data in a regional Cloud Storage bucket keeps it readily available. Cloud Dataflow for Updates: Cloud Dataflow is a managed service for building data pipelines. You can create a Dataflow job that runs every 30 minutes to: Download the latest economic data file from Cloud Storage. Process and potentially transform the data as needed. Load the updated data into BigQuery. BigQuery Integration: BigQuery seamlessly integrates with Cloud Dataflow. The Dataflow job can directly load the processed data into a BigQuery table for further analysis with your customer data.

TVH_Data_Engineer 2 years, 3 months ago

Selected Answer: A

BigQuery supports partitioned tables, which allow for efficient querying and management of large datasets that are updated frequently. By loading the updated data into a new partition every 30 minutes, you can ensure that only relevant partitions are queried, reducing the amount of data processed and thereby minimizing costs. What's wrong with B ? While creating a federated data source in BigQuery pointing to a Google Cloud Storage bucket is feasible, it might not be the most efficient for data that is updated every 30 minutes. Querying federated data sources can sometimes be more expensive and less performant than querying data stored directly in BigQuery.

Melampos 2 years, 11 months ago

Selected Answer: D

Federated queries let you send a query statement to Cloud Spanner or Cloud SQL databases not to cloud storage

sid_is_dis 2 years, 9 months ago

Is you are right about "federated queries", but the option B says about "federated data source". These are different concepts

Abhilash_pendyala 2 years, 11 months ago

ChatGPT says partitioned tables is the best approach, The answers here are quite contrasting with that answer, Even i thought it has to be option A, I am so confused now? Any proper straight forward answer ?

musumusu 3 years, 1 month ago

Answer B: Uploading data into staging tables/ external tables or federated source in BQ is the best approach. Option A is also good approach, anyone can explain about his part what is wrong about this?

yoga9993 3 years ago

we can't implement A, it's because biquery partition table can only be done minimun in range 1 hour, the requirement said it must be update every 30 minutes, so A is imposible option as the minimum partition is in hour level

AzureDP900 3 years, 2 months ago

B is right

Krish6488 3 years, 3 months ago

Selected Answer: B

Discounting A due to limitations on partitions Discounting C because datastore does not fit into the nature of data we are talking about and federation between BQ and datastore it an overkill Between B and D, updating the price file on GCS and joining BQ tables and external tables sourcing data from GCS is most cost optimal way for this use case

ler_mp 3 years, 2 months ago

D is also overkill for this use case, so I'd pick B

jkhong 3 years, 3 months ago

Selected Answer: B

Consideration: As cheaply as possible. Make sure data stays up to date. Initially chose A. But in actuality there is no need to maintain or store past data so storage of past data and partitioning doesn't seem like a key requirement. Instead we can connect just to a single Cloud Storage file, either by: i. replace previous prices with latest prices ii. store previous prices in GCS if required to be retained

DGames 3 years, 3 months ago

Selected Answer: B

B is most inexpensive approach.

odacir 3 years, 3 months ago

Selected Answer: B

The technical requirement is having frequently access info to join with other BQ data, as cheap as possible. B fits perfectly. Corner cases for external data sources: • Avoiding duplicate data in BigQuery storage • Queries that do not have strong performance requirements • Small amount of frequently changing data to join with other tables in BigQuery https://cloud.google.com/blog/products/gcp/accessing-external-federated-data-sources-with-bigquerys-data-access-layer

assU2 3 years, 4 months ago

Selected Answer: D

I would say D, regional Google Cloud Storage bucket - cheap. A - not cheap B - NoSQL database for your web and mobile applications C - Federated queries let you send a query statement to Cloud Spanner or Cloud SQL databases And we need to combine data in DQ with data from bucket

MisuLava 3 years, 5 months ago

according to this : https://cloud.google.com/bigquery/docs/external-data-sources Federated queries don't work with Cloud Storage. how can it be B ?

cloudmon 3 years, 4 months ago

Correct, it cannot be B because BQ federated queries only work with Cloud SQL or Spanner

gudiking 3 years, 4 months ago

It seems to me that they do: https://cloud.google.com/bigquery/docs/external-data-cloud-storage

https://zhangxuetu.online

[202610]

Question #47 Topic 1

You are designing the database schema for a machine learning-based food ordering service that will predict what users want to eat. Here is some of the information you need to store:

- The user profile: What the user likes and doesn't like to eat
- The user account information: Name, address, preferred meal times
- The order information: When orders are made, from where, to whom

The database will be used to store all the transactional data of the product. You want to optimize the data schema. Which Google Cloud Platform product should you use?

A. BigQuery

B. Cloud SQL **Most Voted**

C. Cloud Bigtable

D. Cloud Datastore

Correct Answer: B

Community vote distribution



Comments

jvg637 **Highly Voted** 6 years ago

You want to optimize the data schema + Machine Learning --> Bigquery. So A

yoshik 4 years, 6 months ago

BigQuery is a datawarehouse, not a transactional db. You need to store transactional data as a requirement.

alexmirmao 4 years, 5 months ago

In my opinion transactional data doesnt mean transactions they could be grouped so there is no need to write register by register.

yoshik 4 years, 5 months ago

In other questions they talk about 'transactional log data' when referring to past transactions, but you could be right, agree. In that case ok A BigQuery. Nevertheless, the question is formulated ambiguously.

alecuba16 3 years, 8 months ago

Biquery Supports transactions: https://cloud.google.com/bigquery/docs/reference/standard-sql/transactions , but indeed is not a good DB for OLTP. But I would said or CloudSQL or BigQuery

itche_scratche **Highly Voted** 5 years, 11 months ago

A; ML should be BQ; stored all transactional data (not use for transactional) so should BQ

GeeBeeEl 5 years, 4 months ago

Do you have a link to back this up?

1479 7 months, 1 week ago

Selected Answer: A

BQ for storing ML tables

1479 7 months, 1 week ago

Selected Answer: A

bq storing for machine learning

56d02cd 9 months, 1 week ago

Selected Answer: D

The docs for Datastore call out user profiles as a use case.
https://cloud.google.com/appengine/docs/legacy/standard/go111/datastore#what_its_good_for

22c1725 10 months ago

Selected Answer: A

"for a machine learning-based" This is the why. So, the objective of storege here is not (ACID) opeartion at all. there is only two requirment here: Store the data. Do machine learning. Cloud SQL is perfect for ACID opeartion which the question didn't hint. You only want to store the data, improve data schame for ML. I think alot of people here are misunderstanding the question.

abhaya2608 1 year ago

Selected Answer: A

Cloud SQL doesn't support ML so the right answer will be BigQuery

dcruzado 1 year, 1 month ago

Selected Answer: B

Transactional db -> CloudSQL

dcruzado 1 year, 1 month ago

Selected Answer: B

Since they need transactional data i would say B However thinking on machine learning is better A

cqrm3n 1 year, 1 month ago

Selected Answer: B

The answer should be Cloud SQL because it is a relational database suitable for transactional data. BigQuery is for analytics and querying - not suitable for transactional workload. Bigtable is for unstructured and time series data. Datastore is a nosql document database for semi structured data.

Yad_datatonic 1 year, 2 months ago

Selected Answer: A

For a machine learning-based food ordering service that requires optimised storage of transactional data, Google Cloud BigQuery is a suitable choice

grshankar9 1 year, 2 months ago

Selected Answer: A

ithin Google Cloud, the database that most readily allows for data schema optimization is BigQuery; it provides features like schema auto-detection, columnar storage, and the ability to manually define your schema to tailor it for efficient querying and analysis of large datasets.

manikolbe 1 year, 2 months ago

Selected Answer: B

Cloud SQL is the best choice for your application as it provides relational database management and is optimized for storing transactional data with SQL querying capabilities. It is well-suited for managing user profiles, account information, and orders, ensuring data integrity, and supporting complex queries necessary for the food ordering service.

Ronn27 1 year, 2 months ago

Selected Answer: B

Use BigQuery for analyzing aggregated data (e.g., predicting food trends or training ML models). Use Cloud Bigtable for large-scale real-time recommendation engines if needed in the future. Use Firestore for dynamic, semi-structured data with real-time updates if you need flexibility over transactional consistency. Cloud SQL strikes the right balance for this use case due to its support for structured data, transactions, and easy integration with other GCP services. So B. CloudSQL is the right answer

sravi1200 1 year, 3 months ago

Selected Answer: B

Cloud SQL can store transactional data not Big Query. Big Query is an analytical service.

DGames 1 year, 3 months ago

Selected Answer: A

Easy implement data schema + Machine Learning model in Big Query

julydev82 1 year, 3 months ago

Selected Answer: B

database will be used to storage all transactional data.... I think that you need a relational database for that, then federated tables to bigquery to analysis.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #48 Topic 1

Your company is loading comma-separated values (CSV) files into Google BigQuery. The data is fully imported successfully; however, the imported data is not matching byte-to-byte to the source file. What is the most likely cause of this problem?

A. The CSV data loaded in BigQuery is not flagged as CSV.

B. The CSV data has invalid rows that were skipped on import.

C. The CSV data loaded in BigQuery is not using BigQuery's default encoding. **Most Voted**

D. The CSV data has not gone through an ETL phase before loading into BigQuery.

Correct Answer: C

Community vote distribution



Comments

YAS007 **Highly Voted** 3 years, 1 month ago

Answer : C : " If you don't specify an encoding, or if you specify UTF-8 encoding when the CSV file is not UTF-8 encoded, BigQuery attempts to convert the data to UTF-8. Generally, your data will be loaded successfully, but it may not match byte-for-byte what you expect." https://cloud.google.com/bigquery/docs/loading-data-cloud-storage-csv#details_of_loading_csv_data

saurabh1805 **Highly Voted** 4 years, 1 month ago

C is correct answer, Refer below link for more informaiton. https://cloud.google.com/bigquery/docs/loading-data-cloud-storage-csv#details_of_loading_csv_data

desertlotus1211 1 year ago

Selected Answer: B

The byte-to-byte mismatch is more consistent with invalid rows being skipped during the load process (due to format or parsing issues), rather than an encoding issue. Answer B

samdhimal 1 year, 8 months ago

SITUATION: - Your company is loading comma-separated values (CSV) files into Google BigQuery. - Data is fully imported successfully. PROBLEM: - Imported data is not matching byte-to-byte to the source file. Reason?

samdhimal 1 year, 8 months ago

A. The CSV data loaded in BigQuery is not flagged as CSV. Since BigQuery support multiple formats it could be that maybe avro or json was selected. But the file import was successful hence csv was selected. Either manually or it was

left as is since the default file type is csv. Lastly, this is WRONG. B. The CSV data has invalid rows that were skipped on import. -> Since the data was successfully imported there were no invalid rows. Hence, This is wrong answer too.

samdhimal 1 year, 8 months ago

C. The CSV data loaded in BigQuery is not using BigQuery's default encoding. -> "BigQuery supports UTF-8 encoding for both nested or repeated and flat data. BigQuery supports ISO-8859-1 encoding for flat data only for CSV files." Source: https://cloud.google.com/bigquery/docs/loading-data Default BQ Encoding: UTF-8 This is probably the correct answer because if the csv file encoding was not UTF-8 and instead it was ISO-8859-1 then we would have to tell bigquery that orelse it will assume it is UTF-8. Hence, Imported data is not matching byte-to-byte to the source file. CORRECT ANSWER!

samdhimal 1 year, 8 months ago

D. The CSV data has not gone through an ETL phase before loading into BigQuery. -> ETL means Extract, Transform and Load and this is actually very important content for Cloud Data Engineers. Look into it if interested! But getting back to the topic: ETL is usually required when the source format and target format are different. You need to extract source file and the transform it before loading the data to fit the target. This is also not a viable option. Also Data is imported successfully and the question doesn't mention anything regarding ETL.

medeis_jar 2 years, 9 months ago

Selected Answer: C

A is not correct because if another data format other than CSV was selected then the data would not import successfully. B is not correct because the data was fully imported meaning no rows were skipped. C is correct because this is the only situation that would cause successful import. D is not correct because whether the data has been previously transformed will not affect whether the source file will match the BigQuery table.

MaxNRG 2 years, 10 months ago

Selected Answer: C

C is correct because this is the only situation that would cause successful import. A is not correct because if another data format other than CSV was selected then the data would not import successfully. B is not correct because the data was fully imported meaning no rows were skipped. D is not correct because whether the data has been previously transformed will not affect whether the source file will match the BigQuery table. https://cloud.google.com/bigquery/docs/loading-data#loading_encoded_data

NicolasN 1 year, 9 months ago

Exactly⬆ The updated link (Dec. 2022) and the quote: https://cloud.google.com/bigquery/docs/loading-data-cloud-storage-csv#encoding "If you don't specify an encoding, or if you specify UTF-8 encoding when the CSV file is not UTF-8 encoded, BigQuery attempts to convert the data to UTF-8. Generally, your data will be loaded successfully, but it may not match byte-for-byte what you expect."

anji007 2 years, 11 months ago

Ans: C

sumanshu 3 years, 3 months ago

Vote for 'C'

sumanshu 3 years, 2 months ago

A is not correct because if another data format other than CSV was selected then the data would not import successfully. B is not correct because the data was fully imported meaning no rows were skipped. C is correct because this is the only situation that would cause successful import. D is not correct because whether the data has been previously transformed will not affect whether the source file will match the BigQuery table.

naga 3 years, 7 months ago

Correct C

haroldbenites 4 years, 1 month ago

C is correct

https://zhangxuetu.online

[202610]

Question #49 Topic 1

Your company produces 20,000 files every hour. Each data file is formatted as a comma separated values (CSV) file that is less than 4 KB. All files must be ingested on Google Cloud Platform before they can be processed. Your company site has a 200 ms latency to Google Cloud, and your Internet connection bandwidth is limited as 50 Mbps. You currently deploy a secure FTP (SFTP) server on a virtual machine in Google Compute Engine as the data ingestion point. A local SFTP client runs on a dedicated machine to transmit the CSV files as is. The goal is to make reports with data from the previous day available to the executives by 10:00 a.m. each day. This design is barely able to keep up with the current volume, even though the bandwidth utilization is rather low.
You are told that due to seasonality, your company expects the number of files to double for the next three months. Which two actions should you take? (Choose two.)

A. Introduce data compression for each file to increase the rate file of file transfer.

B. Contact your internet service provider (ISP) to increase your maximum bandwidth to at least 100 Mbps.

C. Redesign the data ingestion process to use gsutil tool to send the CSV files to a storage bucket in parallel. Most Voted

D. Assemble 1,000 files into a tape archive (TAR) file. Transmit the TAR files instead, and disassemble the CSV files in the cloud upon receiving them. Most Voted

E. Create an S3-compatible storage endpoint in your network, and use Google Cloud Storage Transfer Service to transfer on-premises data to the designated storage bucket.

Correct Answer: CD

Community vote distribution

CD (79%) AC (17%) DE (3%)

Comments

Toto2020 Highly Voted 5 years, 3 months ago

E cannot be: Transfer Service is recommended for 300mbps or faster https://cloud.google.com/storage-transfer/docs/on-prem-overview Bandwidth is not an issue, so B is not an answer Cloud Storage loading gets better throughput the larger the files are. Therefore making them smaller with compression does not seem a solution. -m option to do parallel work is recommended. Therefore A is not and C is an answer. https://medium.com/@duhroach/optimizing-google-cloud-storage-small-file-upload-performance-ad26530201dc That leaves D as the other option. It is true you cannot user tar directly with gsutil, but you can load the tar file to Cloud Storage, move the file to a Compute Engine instance with Linux, use tar to split files and copy them back to Cloud

Storage. Batching many files in a larger tar will improve Cloud Storage throughput. So, given the alternatives, I think answer is CD

awssp12345 4 years, 9 months ago

This should be the correct answer.

musumusu 3 years, 1 month ago

50mbps is so slow, why you think bandwidth is ok! For parallel upload you need good internet ?

Booqq 3 years, 1 month ago

normally the solutions are Google Cloud Services based, as it's a vendor exam

Jarek7 2 years, 11 months ago

They have 20.000 files 4kb each per hour, so bandwith needed for it is far below 1mbps. 50mbps is enough to upload all day generated data in about 5 minutes.

vholti 4 years, 5 months ago

D is incorrect. gsutil with -m option uses multiprocessing/multithreading. It means it will copy the file in parallel. The benefit of multiprocessing/multithreading is significantly high when working with large number of files, instead of file size. The important point of multiprocessing/multithreading is sending multiple files in parallel. Hence file size doesn't give impact to gsutil with -m option. Gsutil with -m option doesn't split a big file into multiple chunks and transfer it in parallel. So in my opinion the answer is A and C.

vholti 4 years, 5 months ago

Here is the docs which support my opinion:
https://cloud.google.com/storage/docs/gsutil/addlhelp/TopLevelCommandLineOptions

Mathew106 2 years, 8 months ago

We have small files of 4KB and no issues with bandwidth. It's not an issue that -m does not split files. Our problem is with total volume.

Mathew106 2 years, 8 months ago

As far as I understand compression is not something we want here because bandwidth is not an issue and compressed files will need to be decompressed on the cloud. On top of that if we want to load those files later in BigQuery to create the report we know that we cannot load compressed csv files in parallel. gsutil makes the most sense because it will be used to load all new files in parallel. I answered D as well because I thought that none of the others made sense and D is the only one that mentions creating the bucket on GCS and perhaps migrating data that is missed during the update in the architecture. So D to create the bucket, C to update the process and move the data to the bucket, then D to move any lost data during the update.

Mathew106 2 years, 8 months ago

Typo, I meant E in my post. C and E, not C and D.

[Removed] Highly Voted 6 years ago

Should be AC

GeeBeeEl 5 years, 4 months ago

support this with a link....

gcppde 5 years, 1 month ago

Here you go: https://cloud.google.com/storage-transfer/docs/overview#gsutil

tavva_prudhvi 3 years, 11 months ago

This link does support for C, but what about A? any supported links?

BhupiSG 5 years ago

Thank you! From this doc: Follow these rules of thumb when deciding whether to use gsutil or Storage Transfer Service: Transfer scenario Recommendation Transferring from another cloud storage provider Use Storage Transfer Service. Transferring less than 1 TB from on-premises Use gsutil. Transferring more than 1 TB from on-premises Use Transfer service for on-premises data. Transferring less than 1 TB from another Cloud Storage region Use gsutil. Transferring more than 1 TB from another Cloud Storage region Use Storage Transfer Service.

iooj 1 year, 8 months ago

Selected Answer: CD

C - because gsutil is recommended for transferring less than 1 TB from on-premises C excludes E; bandwidth is not a problem due to a simple math, so we exclude B; 4 KB file is compressed enough, so we exclude A; D - works fine because even with -m flag we can send tars in parallel.

zohra-khouy.f 2 years ago

Selected Answer: AC

AC is the answer

spicebits 2 years, 4 months ago

How can C and E be the answer? They are solving the same problem with different approaches. If you pick C then E can not be an answer. If you pick E then C can not be an answer. This question also seems a bit dated because of gcloud storage cli which is much more performant than gsutil. I would pick C&D as the combination makes the most sense given the choices.

Maurilio_Cardoso 2 years, 9 months ago

@hendrixlives arguments are correct. The approach between the resources in use and how to optimize the ingestion must be balanced.

Kiroo 2 years, 10 months ago

Selected Answer: CD

C is correct without an doubt I was in doubt between D and E A and B does not seems correct because it states that the bandwidth is not fully utilized . Now D and E If the bandwidth was higher the E would be good D even if it seems that will not make difference because tar files does not have compression transmit one file instead of 1000 is significantly faster so I would choose C and D

Oleksandr0501 2 years, 11 months ago

Selected Answer: CD

CD , i guess. Liked explanation in discussion. changing from BC to CD

Kart87 2 years, 11 months ago

Guys. need a help. anyone appeared for the exam very recently (Apr 2023)? preparing all the questions from here, would be enough?

Jarek7 2 years, 11 months ago

Selected Answer: AC

It seems that Google would like AC. A is not neccessary - it doesnt make a significant change - small files do not compress well and bandwith is so big that file size is not an issue - the isue is 0,2s latency. The biggest benefit is that we can simply enable compression from gsutils parameters, it will not add any implementation complexity. For me C solo is ok and D solo might be even better, but more complex. C and D cannot be mixed - they exclude each other. C is more simple and uses Google service so it seems to be desired answer. And it makes sense if they want us to

select 2 actions we have to make - If we go for C we can also get some benefit from A, if we go for D there is no other answer we can select and it is much more complex in implementation than AC(which is by far good enough).

patiwwb 2 years, 5 months ago

Yes the 2 are excluding each other. So it's AC

musumusu 3 years, 1 month ago

Answer: B&C A: files are 4kb, no need for compression B: more files to be transmitted per unit time with 100mbps or get 5g network (~200 mbps) C: gsutil parallel ingestion will reduce time D: TAR is not a good compression and slower in transfer even slower than csv. speed is 50mbps so don't go with it. E: Storage Transfer service needs good internet and used for large size of data and for on premises storage, this one is regular ingestion.

manigcp 3 years, 1 month ago

-- From ChatGPT -- B. Redesign the data ingestion process to use gsutil tool to send the CSV les to a storage bucket in parallel. A. Introduce data compression for each file to increase the rate of file transfer. Reasoning: B. Redesigning the data ingestion process to use gsutil tool to send the CSV files to a storage bucket in parallel will help improve the rate at which the data is transferred to the cloud. This is because gsutil allows for parallel transfers, thereby utilizing the available bandwidth more efficiently and reducing the time required to transfer the data. A. Introducing data compression for each file will also help improve the rate of file transfer. This is because compressed data takes up less space and can be transferred faster, thereby reducing the time required to transfer the data.

manigcp 3 years, 1 month ago

Why not option D? Option D, which involves assembling 1000 files into a TAR file and then transmitting it, may not be an effective solution for the current situation. While TAR archives can help reduce the number of files that need to be transmitted, disassembling the TAR archive in the cloud after receiving it could increase the time required to process the data. This could make it difficult to meet the goal of making reports with data from the previous day available by 10:00 a.m. each day. Furthermore, compressing the TAR archive could increase the time required to create the archive, and may not provide a significant improvement in terms of transfer time, as the individual CSV files are already small in size. This makes it less effective compared to the other options of parallel transfers and data compression.

Jarek7 2 years, 11 months ago

II wouldnt agree, the main issue here is latency 0,2s and 20000 files per hour - it is even beyond possible transfer without paralelisation or file merging. Compression and sending 1000 files at once resolves the issue. Just as option C. But they don't make any sense together. I think they exclude D because of additional complexity - compression and then decompression is much more difficult than using gsutil. Thus we go for C. If we need one more then only A makes some sense, but I wouldn't go for it. We have enough bandwith for this size of file. We just need get rid of latency, by paralelization.

Jarek7 2 years, 11 months ago

OK. AC seems to be rright as we can simply enable the compression by gsutil options.

Leeeeee 3 years, 4 months ago

Selected Answer: CD

https://cloud.google.com/storage/docs/parallel-composite-uploads

abhineet1313 3 years, 11 months ago

A is incorrect as rate of file transfer is not an issue, system is not able to handle current load itself, compression will make it even faster

alecuba16 3 years, 11 months ago

Selected Answer: DE

Multiple small files transfer is a bad practice. You should always use some aggregation strategy like tar or zip multiple files. A is discarded because talks about compressing a single file. B is discarded because the bandwidth is not the problem. Option C could be , but multi threading has a limit. Then the best option is D or use some google on prem mirroring service like E.

tavva_prudhvi 3 years, 9 months ago

E is wrong, as Bandwidth already low, so storage Transfer service will not help here

Jojo9400 4 years ago

E is wrong Google Cloud Storage Transfer Service (online) != Transfer Appliance(on-premise)

OmJanmeda 4 years ago

Selected Answer: CD

CD is correct option

https://zhangxuetu.online

[202610]

Question #50 Topic 1

You are choosing a NoSQL database to handle telemetry data submitted from millions of Internet-of-Things (IoT) devices. The volume of data is growing at 100
TB per year, and each data entry has about 100 attributes. The data processing pipeline does not require atomicity, consistency, isolation, and durability (ACID).
However, high availability and low latency are required.
You need to analyze the data by querying against individual fields. Which three databases meet your requirements? (Choose three.)

A. Redis

B. HBase Most Voted

C. MySQL

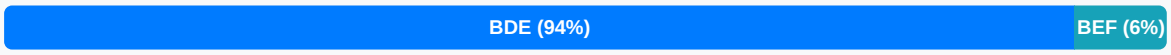
D. MongoDB Most Voted

E. Cassandra Most Voted

F. HDFS with Hive

Correct Answer: BDE

Community vote distribution



Comments

jvg637 Highly Voted 5 years ago
BDE. Hive is not for NoSQL

sergio6 3 years, 6 months ago
Redis is also NoSQL

vholti 3 years, 5 months ago
Redis is limited to 1 TB capacity quota per region. So it doesn't satisfy the requirement.
https://cloud.google.com/memorystore/docs/redis/quotas

ckanaar 1 year, 6 months ago
Memorystore, Google's managed Redis service is. But OS Redis is not. Though it is hard to find a 100GB RAM machine

awssp12345 Highly Voted 3 years, 9 months ago

Answer is BDE - A. Redis - Redis is an in-memory non-relational key-value store. Redis is a great choice for implementing a highly available in-memory cache to decrease data access latency, increase throughput, and ease the load off your relational or NoSQL database and application. Since the question does not ask cache, A is discarded. B. HBase - Meets reqs C. MySQL - they do not need ACID, so not needed. D. MongoDB - Meets reqs E. Cassandra - Apache Cassandra is an open source NoSQL distributed database trusted by thousands of companies for scalability and high availability without compromising performance. Linear scalability and proven fault-tolerance on commodity hardware or cloud infrastructure make it the perfect platform for mission-critical data. F. HDFS with Hive - Hive allows users to read, write, and manage petabytes of data using SQL. Hive is built on top of Apache Hadoop, which is an open-source framework used to efficiently store and process large datasets. As a result, Hive is closely integrated with Hadoop, and is designed to work quickly on petabytes of data. HIVE IS NOT A DATABSE.

sravi1200 1 year, 3 months ago

Selected Answer: BDE

Option A: Redis cannot handle large scale data it is NOSQL db to store small amount of key value pairs, Option B: HBase NOSQL db built on Hadoop does not support ACID Properties. Correct answer Option C: Mysql Does not store telemetry IOT data. Mysql is a relational database structured data only stored. Option D, E: NOSQL Databases, Option F: HDFS with hive used for batch processing not real time streaming data. Option

musumusu 2 years, 1 month ago

BDE Faster Database are NoSql db than SQL, Cassandra is the fastest one in market now than Hbase and then others, in given list MongoBD

MisuLava 2 years, 7 months ago

"Which three databases meet your requirements? " Hive is not a database server. HBase, Mongo and Cassandra are and meet the criteria. BDE is the right answer

sraakesh95 3 years, 2 months ago

Selected Answer: BDE

@hendrixlives

medeis_jar 3 years, 2 months ago

Selected Answer: BDE

as explained by hendrixlives

hendrixlives 3 years, 3 months ago

Selected Answer: BDE

BDE: A. Redis is a key-value store (and in many cases used as in-memory and non persistent cache). It is not designed for "100TB per year" of highly available storage. B. HBase is similar to Google Bigtable, fits the requirements perfectly: highly available, scalable and with very low latency. C. MySQL is a relational DB, designed precisely for ACID transactions and not for the stated requirements. Also, growth may be an issue. D. MongoDB is a document-db used for high volume data and maintains currently used data in RAM, so performance is usually really good. Should also fit the requirements well. E. Cassandra is designed precisely for highly available massive datasets, and a fine tuned cluster may offer low latency in reads. Fits the requirements. F. HDFS with Hive is great for OLAP and data-warehouse scenarios, allowing to solve map-reduce problems using an SQL subset, but the latency is usually really high (we may talk about seconds, not milliseconds, when obtaining results), so this does not complies with the requirements.

MaxNRG 3 years, 4 months ago

Selected Answer: BEF

Very strange question, seems outdated and irrelevant to me as it doesn't contain any GCP products :) Anyway, I would choose BEF. Redis is in-memory key value, not good HBase yes, excelent case for linear growth and a column-

oriented database mysql not good, too big and no need for transactionality Mongodb, document db with flexible schema ?? Yes Cassandra, good use case Apache Hive is a data warehouse software project built on top of Apache Hadoop for providing data query and analysis. https://www.wikiwand.com/en/Apache_Hive

hendrixlives 3 years, 3 months ago

Latency in Hive is usually quite high, and one of the requirements is "low latency"

MaxNRG 3 years, 2 months ago

agreed on BDE

MaxNRG 3 years, 2 months ago

good point!

anji007 3 years, 5 months ago

Ans: B, D and E

sumanshu 3 years, 9 months ago

vote for BDE

BhupiSG 4 years ago

BEF B: HBASE is based upon BigTable E: Cassandra is low latency columnar distributed database like BigTable F: HDFS is low latency distributed file system and Hive will help with running the queries

Manue 3 years, 11 months ago

Hive is not for low latency queries. It is for analytics.

daghayeghi 4 years ago

BDE: These are NoSQL DB, Hive is not for NoSQL.

Rayleigh 4 years, 1 month ago

The answer is ADE, the statement says they require a NoSQL with high availability and low latency, they do not require consistency. C. it is not NoSQL. F. it is not NoSQL. B. it is NoSQL but focused on strong consistency and based on HDFS, you need HDFS for Hbase. Therefore the answer is ADE

daghayeghi 4 years, 1 month ago

BDF: Redis and Cassandra have only Rowkey and couldn't be indexed, and MySQL isn't NoSQL, Then B D and E is correct answer.

naga 4 years, 1 month ago

Correct BDE

apnu 4 years, 3 months ago

it should be BDE because Hive is a sql based datawarehouse , it is not a nosql DB

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #51 Topic 1

You are training a spam classifier. You notice that you are overfitting the training data. Which three actions can you take to resolve this problem? (Choose three.)

A. Get more training examples **Most Voted**

B. Reduce the number of training examples

C. Use a smaller set of features **Most Voted**

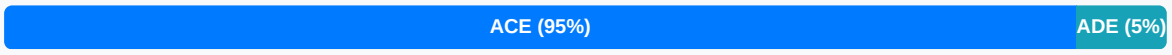
D. Use a larger set of features

E. Increase the regularization parameters **Most Voted**

F. Decrease the regularization parameters

Correct Answer: ACE

Community vote distribution



Comments

madhu1171 **Highly Voted** 5 years, 6 months ago

it should be ACE

Rajokkiyam **Highly Voted** 5 years, 6 months ago

Answer: ACE

monyu 1 year ago

Selected Answer: ACE

A. Because getting more training samples reduces significantly the risk of overfitting since the algorithm can learn from a more general dataset. C. Introducing lots of features increases the risk to introducing irrelevant information, driving the model to avoid focusing on the truly important patterns. E. Because regularization increases the penalty term to the loss function, which discourages complex models with large coefficients avoiding overfitting.

TVH_Data_Engineer 1 year, 9 months ago

Selected Answer: ACE

To address the problem of overfitting in training a spam classifier, you should consider the following three actions: A. Get more training examples: Why: More training examples can help the model generalize better to unseen data. A larger dataset typically reduces the chance of overfitting, as the model has more varied examples to learn from. C. Use a smaller set of features: Why: Reducing the number of features can help prevent the model from learning noise in the data. Overfitting often occurs when the model is too complex for the amount of data available, and having too

many features can contribute to this complexity. E. Increase the regularization parameters: Why: Regularization techniques (like L1 or L2 regularization) add a penalty to the model for complexity. Increasing the regularization parameter will strengthen this penalty, encouraging the model to be simpler and thus reducing overfitting.

Mathew106 2 years, 2 months ago

Selected Answer: ACE

100% ACE We need more data because less data induces overfitting. We need less features to make the problem simpler to learn and not promote learning a very complex function for thousands of features that might not apply to the test data. We also need to use regularization to keep the weights constrained.

theseawillclaim 2 years, 2 months ago

Selected Answer: ACE

Definitely ACE. More training data and less variables can prevent the model from being too picky or specific.

jin0 2 years, 7 months ago

? why A is answer? even though 'more training example' not 'more dataset example'. I understand that there is dataset same and there is only change the size of training examples size. in this case there are valid and test example should be reduced. isn't it?

desertlotus1211 2 years, 8 months ago

Collect more training data: This will help the model generalize better and reduce overfitting. Use regularization techniques: Techniques such as L1 and L2 regularization can be applied to the model's weights to prevent them from becoming too large and causing overfitting. Use early stopping: This involves monitoring the performance of the model on a validation set during training, and stopping the training when the performance on the validation set starts to degrade. This helps to prevent the model from becoming too complex and overfitting the training data.

desertlotus1211 2 years, 8 months ago

Regularization is a technique that penalizes the coefficient. In an overfit model, the coefficients are generally inflated. Thus, Regularization adds penalties to the parameters and avoids them weigh heavily. A & C are correct... the third one --- not sure on

RoshanAshraf 2 years, 8 months ago

Selected Answer: ACE

A -The training data is causing the overfiting for the testing data, so addition of training data will solve this. C - Larger sets will cause overfitting, so we have to use smaller sets or reduce features E - Increase the regularization is a method for solving the Overfitting model

AzureDP900 2 years, 9 months ago

Answers are; A. Get more training examples C. Use a smaller set of features E. Increase the regularization parameters Prevent overfitting: less variables, regularisation, early ending on the training Reference: https://cloud.google.com/bigquery-ml/docs/preventing-overfitting

DGames 2 years, 9 months ago

Selected Answer: ADE

Answer ADE

MisuLava 2 years, 11 months ago

Selected Answer: ACE

100% sure ACE https://elitedatascience.com/overfitting-in-machine-learning

MisuLava 3 years, 1 month ago

Answer is : ACE
https://www.ibm.com/cloud/learn/overfitting#:~:text=Overfitting%20is%20a%20concept%20in,unseen%20data%2C%20defeating%20its%20purpose.

Noahz110 3 years, 1 month ago

Selected Answer: ACE

im vote for ACE

Dip1994 3 years, 1 month ago

It should be ACE

sraakesh95 3 years, 8 months ago

Selected Answer: ACE

@medeis_jar

medeis_jar 3 years, 9 months ago

Selected Answer: ACE

As MaxNRG wrote: The tools to prevent overfitting: less variables, regularization, early ending on the training. - Adding more training data will increase the complexity of the training set and help with the variance problem. - Reducing the feature set will ameliorate the overfitting and help with the variance problem. - Increasing the regularization parameter will reduce overfitting and help with the variance problem.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #52 Topic 1

You are implementing security best practices on your data pipeline. Currently, you are manually executing jobs as the Project Owner. You want to automate these jobs by taking nightly batch files containing non-public information from Google Cloud Storage, processing them with a Spark Scala job on a Google Cloud Dataproc cluster, and depositing the results into Google BigQuery.
How should you securely run this workload?

A. Restrict the Google Cloud Storage bucket so only you can see the files

B. Grant the Project Owner role to a service account, and run the job with it

C. Use a service account with the ability to read the batch files and to write to BigQuery **Most Voted**

D. Use a user account with the Project Viewer role on the Cloud Dataproc cluster to read the batch files and write to BigQuery

Correct Answer: C

Community vote distribution



Comments

digvijay **Highly Voted** 5 years ago

A is wrong, if only I can see the bucket no automation is possible, besides, also needs launch the dataproc job B is too much, does not follow the security best practices C has one point missing…you need to submit dataproc jobs. In D viewer role will not be able to submit dataproc jobs, the rest is ok Thus….the only one that would work is B! BUT this service account has too many permissions. Should have dataproc editor, write big query and read from bucket

dambilwa 4 years, 9 months ago

Hence - Contextually, Option [C] looks to be the right fit

retep007 3 years, 6 months ago

C doesn't need permission to submit dataproc jobs, it's workload SA. Job can be submitted by any other identity

rickywck **Highly Voted** 5 years ago

Should be C

Mathew106 1 year, 8 months ago

Selected Answer: B

We need permissions for submitting dataproc jobs and writing to BigQuery. Project Owner will fix all of that even though it's not a good solution. The rest won't work at all.

Adswerve 1 year, 11 months ago

Selected Answer: C

C Project Owner is too much, violates the principle of least privilege

PolyMoe 2 years, 2 months ago

Selected Answer: C

C. Use a service account with the ability to read the batch files and to write to BigQuery It is best practice to use service accounts with the least privilege necessary to perform a specific task when automating jobs. In this case, the job needs to read the batch files from Cloud Storage and write the results to BigQuery. Therefore, you should create a service account with the ability to read from the Cloud Storage bucket and write to BigQuery, and use that service account to run the job.

Mkumar43 2 years, 2 months ago

Selected Answer: B

B works for the given requirement

Krish6488 2 years, 3 months ago

Least privilege principle. Option C. job can be submitted or triggered using a Cron or a composer which uses another SA with different set of privileges

DGames 2 years, 3 months ago

Selected Answer: B

B because we need to run job .. option C mentioned permission about read and write nothing mention to run the job . In case project owner to service account it’s similar just running job and doing rest of tasks read and writing as well.

ThomasChoy 2 years, 11 months ago

Selected Answer: C

The answer is C because Service Account is the best way to access the BigQuery API if your application can run jobs associated with service credentials rather than an end-user's credentials, such as a batch processing pipeline. https://cloud.google.com/bigquery/docs/authentication

Bhawantha 3 years, 2 months ago

Selected Answer: C

Data owners cant create jobs or queries. -> B out We need service Account -> D out Access only granting me does not solve the problem -> A out The answer is C. (Minimum rights to perform the job)

medeis_jar 3 years, 2 months ago

Selected Answer: C

"taking nightly batch files containing non-public information from Google Cloud Storage, processing them with a Spark Scala job on a Google Cloud Dataproc cluster, and depositing the results into Google BigQuery"

prasanna77 3 years, 3 months ago

C should be okay,since he is already a project owner, I guess compute service account created will have access to run the jobs

MaxNRG 3 years, 3 months ago

Selected Answer: C

C, Project Owner role to a service account - is too much

JG123 3 years, 4 months ago

Why there are so many wrong answers? Examtopics.com are you enjoying paid subscription by giving random answers from people? Ans: C

anji007 3 years, 5 months ago

Ans: C See this: https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/service-accounts#dataproc_service_accounts_2

Blobby 3 years, 6 months ago

C as service account invoked to read the data into GCS and write to BQ once transformed via Data Proc. Assumes DataProc can inherit SA authorisation to perform transform and propagate. B seems to violate key IAM principle enforcing least privilege; https://cloud.google.com/iam/docs/recommender-overview

sumanshu 3 years, 9 months ago

Vote for 'C"

sumanshu 3 years, 8 months ago

Vote for B, (though it's too much access) - But C has one accessing missing (i.e Dataproc job execution) Thus B is correct

https://zhangxuetu.online

[202610]

Question #53 Topic 1

You are using Google BigQuery as your data warehouse. Your users report that the following simple query is running very slowly, no matter when they run the query:
SELECT country, state, city FROM [myproject:mydataset.mytable] GROUP BY country
You check the query plan for the query and see the following output in the Read section of Stage:1:



What is the most likely cause of the delay for this query?

A. Users are running too many concurrent queries in the system

B. The [myproject:mydataset.mytable] table has too many partitions

C. Either the state or the city columns in the [myproject:mydataset.mytable] table have too many NULL values

D. Most rows in the [myproject:mydataset.mytable] table have the same value in the country column, causing data skew **Most Voted**

Correct Answer: D

Community vote distribution

D (88%) A (12%)

Comments

itche_scratche **Highly Voted** 5 years, 11 months ago
D; Purple is reading, Blue is writing. so majority is reading.

squishy_fishy 4 years, 5 months ago
I have been looking for the color code descriptions for a while. Thank you!

Paul_Oprea **Highly Voted** 3 years, 11 months ago
BTW, how is the query even syntactically valid? It has non aggregated columns in the SELECT part of the query. That query will not run in the first place, unless I'm missing something.

iooj 1 year, 8 months ago
Selected Answer: D
D - stands for Data Skew The Read section of the query plan shows a heavy concentration of processing in one area (as indicated by the pink bar being much longer than the purple bar). This typically indicates data skew, where the majority of the data is processed by a small subset of nodes.

MaxNRG 2 years, 3 months ago

Selected Answer: D

The most likely cause of the delay for this query is option D. Most rows in the [myproject:mydataset.mytable] table have the same value in the country column, causing data skew. Group by queries in BigQuery can run slowly when there is significant data skew on the grouped columns. Since the query is grouping by country, if most rows have the same country value, all that data will need to be shuffled to a single reducer to perform the aggregation. This can cause a data skew slowdown. Options A and B might cause general slowness but are unlikely to affect this specific grouping query. Option C could also cause some slowness but not to the degree that heavy data skew on the grouped column could. So D is the most likely root cause. Optimizing the data distribution to reduce skew on the grouped column would likely speed up this query.

JOKKUNO 2 years, 3 months ago

Data skew is when one or some partitions have significantly more data compared to other partitions. Data-skew is usually the result of operations that require re-partitioning the data, mostly join and grouping (GroupBy) operations. So D.

PolyMoe 3 years, 2 months ago

Selected Answer: D

D. Most rows in the [myproject:mydataset.mytable] table have the same value in the country column, causing data skew Data skew occurs when one or more values in a column have a disproportionately large number of rows compared to other values in that column. This can cause performance issues when running queries that group by that column, like the one in the question. In this case, if most of the rows in the [myproject:mydataset.mytable] table have the same value in the country column, then the query will need to process a large number of rows with that value, which can cause significant delay.

AzureDP900 3 years, 2 months ago

D is right

Krish6488 3 years, 3 months ago

Selected Answer: D

data skewing causing imbalance in data distribution across slots. It also causes errors if the group by column has NULLS. Since option C does not call out the Group by column, D is a closer answer contextually

Jasar 3 years, 4 months ago

Selected Answer: A

A is the best option becouse the color bar show the high number of reads and i think its not a skew becouse biguery was build to compute the data fast

arpitagrawal 3 years, 6 months ago

The query would throw the error because you're using a group by clause on country but not aggregating city or state.

MisuLava 3 years, 7 months ago

Selected Answer: D

https://cloud.google.com/bigquery/docs/best-practices-performance-patterns

Arkon88 4 years ago

Selected Answer: D

D Image says that average(dark) and maximum(light) have difference in few times, this it is a skew https://cloud.google.com/bigquery/query-plan-explanation The color indicators show the relative timings for all steps across all stages. For example, the COMPUTE step of Stage 00 shows a bar whose shaded fraction is 21/30 since

30ms is the maximum time spent in a single step of any stage. The parallel input information shows that each stage required only a single worker, so there's no variance between average and slowest timings.

sraakesh95 4 years, 2 months ago

Selected Answer: D

https://medium.com/slalom-build/using-bigquery-execution-plans-to-improve-query-performance-af141b0cc33d

sraakesh95 4 years, 2 months ago

https://medium.com/slalom-build/using-bigquery-execution-plans-to-improve-query-performance-af141b0cc33d

medeis_jar 4 years, 2 months ago

Selected Answer: D

D Colors: Purple is reading, Blue is writing. so the majority is reading. https://cloud.google.com/bigquery/docs/best-practices-performance-patterns

morpho4444 4 years, 4 months ago

If you read this https://medium.com/slalom-build/using-bigquery-execution-plans-to-improve-query-performance-af141b0cc33d C can't be right because the skewness happen when the column you use for grouping contains lots of null values, here C mentions columns that aren't part of the grouping clause. D, that's not how data get skewed, it gets skewed due to null values. A is the only answer here.

BigQuery 4 years, 3 months ago

A Cant be answer. Since users whenever running queries facing the problems.

JG123 4 years, 4 months ago

Why there are so many wrong answers? Examtopics.com are you enjoying paid subscription by giving random answers from people? Ans: D

https://zhangxuetu.online

[202610]

Question #54 Topic 1

Your globally distributed auction application allows users to bid on items. Occasionally, users place identical bids at nearly identical times, and different application servers process those bids. Each bid event contains the item, amount, user, and timestamp. You want to collate those bid events into a single location in real time to determine which user bid first. What should you do?

A. Create a file on a shared file and have the application servers write all bid events to that file. Process the file with Apache Hadoop to identify which user bid first.

B. Have each application server write the bid events to Cloud Pub/Sub as they occur. Push the events from Cloud Pub/Sub to a custom endpoint that writes the bid event information into Cloud SQL.

C. Set up a MySQL database for each application server to write bid events into. Periodically query each of those distributed MySQL databases and update a master MySQL database with bid event information.

D. Have each application server write the bid events to Google Cloud Pub/Sub as they occur. Use a pull subscription to pull the bid events using Google Cloud Dataflow. Give the bid for each item to the user in the bid event that is processed first. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

jvg637 **Highly Voted** 6 years ago

I'd go with B: real-time is requested, and the only scenario for real time (in the 4 presented) is the use of pub/sub with push.

Tanzu 4 years, 1 month ago

B. - for realtime pub/sub push is critical. pull creates latency. (eliminates D) - process by event-time, not by process - time (eliminates D)

godot 4 years ago

no push avail: https://cloud.google.com/dataflow/docs/concepts/streaming-with-cloud-pubsub#streaming-pull-migration

jin0 3 years, 1 month ago

The dataflow is designed for realtime processing. and this case should be needed to use dataflow because there is no option to order the data if not using dataflow. So D is answer I think

AzureDP900 3 years, 2 months ago

Agree with B

donbigi 3 years, 1 month ago

This approach is not ideal because it requires a custom endpoint to write the bid event information into Cloud SQL. This adds additional complexity and potential points of failure to the architecture, as well as adding latency to the processing of bid events, since the data must be written to both Pub/Sub and Cloud SQL. Additionally, it can be more challenging to ensure that bid events are processed in the order they were received, since the data is being written to multiple databases. Finally, using a single database to store bid events could limit scalability and availability, and can also result in slow query performance.

ralf_cc 4 years, 9 months ago

Yep, Pub/Sub doesn't have FIFO yet, B is the one that keeps the right order

Tanzu 4 years, 1 month ago

it is not a queue, and that is not a issue :)

Tanzu 4 years, 1 month ago

in a distributed environment, you can not handle this problem wit a queue by the way !

Ganshank Highly Voted 5 years, 11 months ago

D The need is to collate the messages in real-time. We need to de-dupe the messages based on timestamp of when the event occurred. This can be done by publishing ot Pub-Sub and consuming via Dataflow.

Tanzu 4 years, 1 month ago

Yeap, that's why B is the right one. It has pub/sub push, more real time than pub/sub pull. You need to aware at some point , something has to be pulled which adds a latency.

unnamed12355 3 years ago

D isnt correct, Pub/sub can send messages out of order, it is no guaranty that the event with lowest timestamp will be processed first B is correct

58606d0 7 months ago

Selected Answer: B

Messages might arrive out of order, so the one that's processed first is not necessarily the first one which was sent.

forepick 7 months, 1 week ago

Selected Answer: D

D - Classical use case for Dataflow session windows with OrderBy timestamp

theRafael7 9 months, 4 weeks ago

Selected Answer: D

I would have chosen B because pub/sub push meets the real time option. However, writing to cloudSQL is what makes B wrong. CloudSQL is usually not global and cannot handle the ordering like dataflow. Therefore the answer is D.

dcruzado 1 year, 1 month ago

Selected Answer: B

Answer is B This sentence invalidates D "Give the bid for each item to the user in the bid event that is processed first."

manikolbe 1 year, 2 months ago

Selected Answer: B

process by event-time, not by process -time (eliminates D

Ronn27 1 year, 2 months ago

Selected Answer: D

Writing directly to Cloud SQL in real time can cause bottlenecks as Cloud SQL is not designed for high-frequency, low-latency writes from multiple sources. Answer D is right as Dataflow and Pubsub has the realtime capability

DGames 1 year, 3 months ago

Selected Answer: D

Bid event time and Pull Subscription is important part.

baimus 1 year, 6 months ago

Selected Answer: D

It feels like it depends what's actually in the dataflow pipeline. D I believe is the answer they intend, even if messages are pulled out of order.

manel_bhs 1 year, 8 months ago

Selected Answer: D

While using Cloud Pub/Sub for real-time event streaming is a good choice, pushing events to a custom endpoint that writes to Cloud SQL introduces additional complexity. Custom endpoints need to be maintained, and the process of writing to Cloud SQL might not be as efficient as using a purpose-built data processing service.

Snnnnneee 1 year, 9 months ago

Selected Answer: B

In D the user gets it where the data is ingested first. That can be wrong for a global auction solution

yassoraa88 1 year, 11 months ago

Selected Answer: D

This is the most suitable solution for the requirements. Google Cloud Pub/Sub can handle high throughput and low-latency data ingestion. Coupled with Google Cloud Dataflow, which can process data streams in real time, this setup allows for immediate processing of bid events. Dataflow can also handle ordering and timestamp extraction, crucial for determining which bid came first. This architecture supports scalability and real-time analytics, which are essential for a global auction system.

teka112233 1 year, 11 months ago

Selected Answer: D

the Answer should be D for the following Real-time Processing Centralized Processing Winner Determination also, B is unsuitable as While Pub/Sub can ingest data, Cloud SQL is a relational database not designed for real-time processing at this scale. Maintaining a custom endpoint adds complexity.

I__SHA1234567 2 years ago

Selected Answer: D

Google Cloud Pub/Sub is a scalable and reliable messaging service that can handle high volumes of data and deliver messages in real-time. By having each application server publish bid events to Cloud Pub/Sub, you ensure that all bid events are collected centrally. Using Cloud Dataflow with a pull subscription allows you to process the bid events in real-time. Cloud Dataflow provides a managed service for stream and batch processing, and it can handle the real-time processing requirements efficiently. By processing the bid events with Cloud Dataflow, you can determine which user bid first by applying the appropriate logic within your Dataflow pipeline. This approach ensures scalability, reliability, and real-time processing capabilities, making it suitable for handling bid events from multiple application servers.

philli1011 2 years, 2 months ago

B should be the answer, because it writes the bid into Cloud SQL to a distributed system. This way the customer know if they get the bid or not, immediately. Also, push requests are faster than pull requests, hence they are better for realtime experience.

arpana_naa 2 years, 3 months ago

Selected Answer: D

pub/sub for entry time stamp + event time dataflow for processing and dataflow is better for real time

https://zhangxuetu.online

[202610]

Question #55 Topic 1

Your organization has been collecting and analyzing data in Google BigQuery for 6 months. The majority of the data analyzed is placed in a time-partitioned table named events_partitioned. To reduce the cost of queries, your organization created a view called events, which queries only the last 14 days of data. The view is described in legacy SQL. Next month, existing applications will be connecting to BigQuery to read the events data via an ODBC connection. You need to ensure the applications can connect. Which two actions should you take? (Choose two.)

A. Create a new view over events using standard SQL

B. Create a new partitioned table using a standard SQL query

C. Create a new view over events_partitioned using standard SQL **Most Voted**

D. Create a service account for the ODBC connection to use for authentication **Most Voted**

E. Create a Google Cloud Identity and Access Management (Cloud IAM) role for the ODBC connection and shared ɹ€eventsɹ€

Correct Answer: CD

Community vote distribution



Comments

jvg637 **Highly Voted** 6 years ago

C = A standard SQL query cannot reference a view defined using legacy SQL syntax. D = For the ODBC drivers is needed a service account which will get a standard Bigquery role.

JG123 **Highly Voted** 4 years, 4 months ago

Why there are so many wrong answers? Examtopics.com are you enjoying paid subscription by giving random answers from people? Ans: C,D

Parandhaman_Margan 1 year ago

Selected Answer: AD

ODBC requires standard SQL. **A** creates a new view using standard SQL, and **D** sets up a service account for authentication. Options A and D are necessary.

Yad_datatonic 1 year, 2 months ago

Selected Answer: AD

To ensure applications can connect to BigQuery via an ODBC connection, take these two actions: A. Create a new view over events using standard SQL to replace the legacy SQL view, ensuring compatibility with ODBC, and D.

Create a service account for the ODBC connection to authenticate and access the data. These steps ensure the applications can query the last 14 days of data efficiently and securely. Avoid unnecessary changes like creating new tables or custom IAM roles.

Smakyel79 1 year, 4 months ago

Selected Answer: AD

A. Legacy SQL views are not compatible with ODBC connections, which require standard SQL. Creating a new view in standard SQL ensures compatibility for the applications connecting via ODBC. D. ODBC connections to BigQuery require authentication, typically via a service account with the appropriate permissions. Setting up a service account ensures secure and reliable access.

Vullibabu 2 years, 2 months ago

I think question should be rewrite slightly like which 3 actions should you take rather than 2 .. Then answer would be A,D and E..No ambiguity then

task_7 2 years, 2 months ago

Selected Answer: BD

ODBC connections require standard SQL, not legacy SQL. Service account for the ODBC connection

Bahubali1988 2 years, 6 months ago

This dump is full of wrong answers - not sure which one to go for.

alihabib 2 years, 7 months ago

CD..... C because, ODBC drivers dont support switch b/w legacy SQL & google SQL, hence better to create a new view from recent partitioned table & D as Google best practice for role binding

baht 2 years, 9 months ago

the answer is C & D

musumusu 3 years, 1 month ago

answer: A & D Confusion here: Legacy SQL vs Standard, BQ supports legacy SQL but ODBC or Most RDBMS connection doesn't support Legacy SQL, so in this case we need to create a new view on existing view or replace the existing one by changing syntax. For ODBC, you just need a service account to authenticate as its external service connection. Option E is not necessary.

musumusu 3 years, 1 month ago

Go for B, create a new view from the table, If you modify the syntex in option A, its also mean you created a new view on table :P

PolyMoe 3 years, 2 months ago

Selected Answer: DE

D. Create a service account for the ODBC connection to use for authentication. This service account will be used to authenticate the ODBC connection, and will be granted specific permissions to access the BigQuery resources. E. Create a Cloud IAM role for the ODBC connection and shared events. This role will be used to grant permissions to the service account created in step D, and will allow the applications to access the events view in BigQuery. Creating a new view over events using standard SQL may also be beneficial to improve performance and compatibility with the applications, but is not required for the ODBC connection to work.

samdhimal 3 years, 2 months ago

INFO: - The majority of the data analyzed is placed in a time-partitioned table named events_partitioned. - To reduce the cost of queries, your organization created a view called events, which queries only the last 14 days of data. - The view is described in legacy SQL. QUESTION: Next month, existing applications will be connecting to BigQuery to read

the events data via an ODBC connection. You need to ensure the applications can connect. Which two actions should you take? (Choose two.) -> First and foremost we need to understand the information. So our actual data is stored in events_partitioned table. The organization is currently using view called events to reduce the cost. -> Since the view called events only has last 14 days of data we cannot use that view. -> We also cannot use that view because standard SQL is not used to describe the view. In order to connectt ODBC we need a view described by standard SQL.

samdhimal 3 years, 2 months ago

A. Create a new view over events using standard SQL -> Wrong, events view contains only last 14 days of data and also it uses Legacy SQL. B. Create a new partitioned table using a standard SQL query -> Partitioned Table is not helpful in this situation.Hence, I am ruling it out. C. Create a new view over events_partitioned using standard SQL -> Correct this is exactly what we need. 1.We need to create a new view over events_partitioned. 2. We need to use Standard SQL. This is a valid option. D. Create a service account for the ODBC connection to use for authentication. - Correct answer because we are required to authenticate before ODBC connection. E. Create a Google Cloud Identity and Access Management (Cloud IAM) role for the ODBC connection and shared ɹ€eventsɹ€ - This option is of no use in this scenario

GCPpro 3 years, 2 months ago

CE is the correct answer

MisuLava 3 years, 7 months ago

Selected Answer: CD

needed a service account for ODBC drivers standard SQL vs legacy SQL.

Smaks 3 years, 8 months ago

Selected Answer: CE

1. Create Services account from IAM & Admin 2. Add Services account permission Roles as "BigQuery Admin" or any custom Role. Other options are not related ' to ensure the applications can connect'

Smaks 3 years, 8 months ago

typo - D; E

Arkon88 4 years ago

Selected Answer: CD

As stated by jvg637 C = A standard SQL query cannot reference a view defined using legacy SQL syntax. D = For the ODBC drivers is needed a service account which will get a standard Bigquery role.

https://zhangxuetu.online

[202610]

Question #56 Topic 1

You have enabled the free integration between Firebase Analytics and Google BigQuery. Firebase now automatically creates a new table daily in BigQuery in the format app_events_YYYYMMDD. You want to query all of the tables for the past 30 days in legacy SQL. What should you do?

A. Use the TABLE_DATE_RANGE function **Most Voted**

B. Use the WHERE_PARTITIONTIME pseudo column

C. Use WHERE date BETWEEN YYYY-MM-DD AND YYYY-MM-DD

D. Use SELECT IF.(date >= YYYY-MM-DD AND date <= YYYY-MM-DD

Correct Answer: A

Community vote distribution



Comments

damaldon **Highly Voted** 3 years, 6 months ago

A. is correct according to this link: https://cloud.google.com/bigquery/docs/reference/legacy-sql

baimus 1 year, 6 months ago

Selected Answer: A

https://cloud.google.com/bigquery/docs/reference/legacy-sql#table-date-range

Preetmehta1234 2 years, 1 month ago

Selected Answer: B

We don't have TABLE DATE RANGE function in legacy SQL. Answer should be B

mark1223jkh 1 year, 10 months ago

We actually have, look at the documentation, https://cloud.google.com/bigquery/docs/reference/legacy-sql

AjoseO 2 years, 5 months ago

Selected Answer: A

The recommended action is to use the TABLE_DATE_RANGE function (option A). This function allows you to specify a range of dates to query across multiple tables.

Nirca 2 years, 6 months ago

Selected Answer: A

The TABLE_DATE_RANGE function in BigQuery is a table wildcard function that can be used to query a range of daily tables. It takes two arguments: a table prefix and a date range. The table prefix is the beginning of the table

names, and the date range is the start and end dates of the tables to be queried. The TABLE_DATE_RANGE function will expand to cover all tables in the dataset that match the table prefix and are within the date range. For example, if you have a dataset that contains daily tables named my_table_20230804, my_table_20230805, and my_table_20230806, you could use the TABLE_DATE_RANGE function to query all of the tables in the dataset between August 4, 2023 and August 6, 2023 as follows: SELECT * FROM TABLE_DATE_RANGE('my_table_', '2023-08-04', '2023-08-06');

samdhimal 3 years, 2 months ago

A Is correct. TABLE_DATE_RANGE() : Queries multiple daily tables that span a date range.

samdhimal 3 years, 2 months ago

Example: SELECT * FROM TABLE_DATE_RANGE(app_events_, TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 30 DAY), CURRENT_TIMESTAMP())

DipT 3 years, 3 months ago

Selected Answer: A

https://cloud.google.com/bigquery/docs/reference/legacy-sql TABLE_DATE_RANGE() Queries multiple daily tables that span a date range.

skp57 3 years, 4 months ago

A. is Correct... from...https://cloud.google.com/blog/products/management-tools/using-bigquery-and-firebase-analytics-to-understand-your-mobile-app SELECT user_dim.app_info.app_platform as appPlatform, user_dim.device_info.device_category as deviceType, COUNT(user_dim.device_info.device_category) AS device_type_count FROM TABLE_DATE_RANGE([firebase-analytics-sample-data:android_dataset.app_events_], DATE_ADD('2016-06-07', -7, 'DAY'), CURRENT_TIMESTAMP()), TABLE_DATE_RANGE([firebase-analytics-sample-data:ios_dataset.app_events_], DATE_ADD('2016-06-07', -7, 'DAY'), CURRENT_TIMESTAMP()) GROUP BY 1,2 ORDER BY device_type_count DESC

https://zhangxuetu.online

[202610]

Question #57 Topic 1

Your company is currently setting up data pipelines for their campaign. For all the Google Cloud Pub/Sub streaming data, one of the important business requirements is to be able to periodically identify the inputs and their timings during their campaign. Engineers have decided to use windowing and transformation in Google Cloud Dataflow for this purpose. However, when testing this feature, they find that the Cloud Dataflow job fails for the all streaming insert. What is the most likely cause of this problem?

A. They have not assigned the timestamp, which causes the job to fail

B. They have not set the triggers to accommodate the data coming in late, which causes the job to fail

C. They have not applied a global windowing function, which causes the job to fail when the pipeline is created

D. They have not applied a non-global windowing function, which causes the job to fail when the pipeline is created **Most Voted**

Correct Answer: D

Community vote distribution



Comments

[Removed] **Highly Voted** 6 years ago

Answer: D Description: Caution: Beam's default windowing behavior is to assign all elements of a PCollection to a single, global window and discard late data, even for unbounded PCollections. Before you use a grouping transform such as GroupByKey on an unbounded PCollection, you must do at least one of the following: —->>>>>>Set a non-global windowing function. See Setting your PCollection's windowing function. Set a non-default trigger. This allows the global window to emit results under other conditions, since the default windowing behavior (waiting for all data to arrive) will never occur. —->>>>If you don't set a non-global windowing function or a non-default trigger for your unbounded PCollection and subsequently use a grouping transform such as GroupByKey or Combine, your pipeline will generate an error upon construction and your job will fail. So it looks like D

samdhimal 3 years, 2 months ago

Why not C? Because I think that the most likely cause of the problem is C. They have not applied a global windowing function, which causes the job to fail when the pipeline is created. In Dataflow, windowing is used to divide the input data into smaller time intervals, called windows. Without a windowing function, all the data may be treated as part of the same window and the pipeline may not be able to process the data correctly. In this specific scenario, the engineers are trying to use windowing and transformation in Google Cloud Dataflow to periodically identify the inputs and their timings during the campaign, so it's likely that they need to use a windowing function to divide the data into

smaller time intervals in order to process it correctly. Not applying a windowing function, or applying the wrong one can cause the job to fail. Someone Clarify? Am I missing an important point?

Mathew106 2 years, 8 months ago

You are missing that the global window is the default window that we typically use for batch processing. The global window by default waits until all data is available before processing it so if you want to use it with streaming you need to set some custom trigger so that we don't wait indefinitely to wait until we aggregate. All in all it doesn't sound right. https://www.youtube.com/watch?v=oJ-LueBvOcM https://www.youtube.com/watch?v=MuFA6CSti6M

jvg637 **Highly Voted** 6 years ago

Global windowing is the default behavior, so I don't think C is right. An error can occur if a non-global window or a non-default trigger is not set. I would say D. (https://beam.apache.org/documentation/programming-guide/#windowing)

imrane1995 8 months, 3 weeks ago

Selected Answer: A

Timestamps are required for streaming + windowing. Missing timestamps can crash the job.

Parandhaman_Margan 1 year ago

Selected Answer: A

Google Cloud Dataflow requires event timestamps when using windowing in streaming mode. By default, Pub/Sub messages do not have timestamps; they need to be assigned manually using withTimestampFn()

Yad_datatonic 1 year, 2 months ago

Selected Answer: B

The job fails because triggers are not set to handle late-arriving data, causing the pipeline to mishandle or drop delayed records.

Rav761 1 year, 3 months ago

Selected Answer: A

A. They have not assigned the timestamp, which causes the job to fail Analysis: Cloud Dataflow relies on timestamps to perform windowing operations. Without proper event-time timestamps, windowing cannot be applied correctly, and the job may fail or behave unpredictably. This is a common issue when processing streaming data from Google Cloud Pub/Sub, as timestamps must be explicitly assigned if not already embedded in the data. This is the most likely cause.

Erg_de 1 year, 4 months ago

Selected Answer: A

This option is very likely, as without timestamps assigned to streaming data, the system cannot properly process time windows. Timestamps are crucial for the correct time handling in Dataflow pipelines

39405bb 1 year, 10 months ago

The most likely cause of this problem is A. They have not assigned the timestamp, which causes the job to fail. Here's why: Importance of Timestamps in Windowing: Windowing in Dataflow relies on timestamps to group elements into windows. If timestamps are not explicitly assigned or extracted from the data, Dataflow cannot determine which elements belong to which windows, leading to failures in the job. Let's look at the other options: B. They have not set the triggers to accommodate the data coming in late: While triggers are important for managing late data, not setting them would not cause the job to fail for all streaming inserts. It might affect the accuracy of the results, but the job would still run. C & D. Global vs. Non-global Windowing: The choice between global and non-global windowing depends on the specific requirements of the analysis. While incorrect windowing choices can lead to unexpected results, they would not typically cause the job to fail completely.

philli1011 2 years, 2 months ago

D You have to apply a non-global windowing function because the global windowing function is a default windowing function for every pub/sub stream or batch data.

MikkelRev 2 years, 6 months ago

option B: They have not set the triggers to accommodate the data coming in late, which causes the job to fail. In a streaming data processing pipeline, it's common to encounter data that arrives late, meaning it arrives after the event time has passed for the associated window. If you don't handle late data appropriately by setting triggers, it can cause issues in your pipeline, including job failures.

Oleksandr0501 2 years, 11 months ago

Selected Answer: A

gpt: The most likely cause of the problem is A, that they have not assigned the timestamp. In streaming data processing, timestamps are essential for proper windowing and triggering of data. Without timestamps, the system cannot correctly determine which window a particular piece of data belongs to, or when it is safe to trigger processing of a window. If the engineers did not assign timestamps to the data, the Cloud Dataflow job would not be able to process the data correctly, and it would fail. Option B, not setting triggers to accommodate late data, is also an important consideration for streaming data processing. However, it is less likely to cause the job to fail outright than missing timestamps. Option C, not applying a global windowing function, and Option D, not applying a non-global windowing function, are also important considerations for windowing in Cloud Dataflow. However, neither of these would cause the job to fail when the pipeline is created. Instead, they would affect the performance and correctness of the data processing.

muhusman 2 years, 11 months ago

without a correct timestamp, the pipeline still run fine with the default timestamp. The result maybe incorrect but the job will not fail.

Oleksandr0501 2 years, 11 months ago

okay, so D maybe

Oleksandr0501 2 years, 11 months ago

gpt pt2: For example, if your use case requires you to calculate a running average of values over a fixed time interval, you would likely use a non-global windowing function with a fixed time interval. On the other hand, if you need to perform a computation on the entire stream of data at once, a global windowing function might be more appropriate. So, the choice of windowing function should be based on the specific requirements of the data processing task at hand, and it may or may not be important to apply a non-global windowing function when the pipeline is created. ------ question says that we need to identify streaming input, time, so non-global needed, let it be d...

Oleksandr0501 2 years, 11 months ago

- is it impotant to apply a non-global windowing function when the pipeline is created? GPT: It is important to choose the appropriate windowing function for your data processing needs, but whether it should be a global or non-global windowing function depends on the requirements of your specific use case. A global windowing function considers all data elements within a bounded time interval as a single window, whereas a non-global windowing function divides the data stream into smaller windows based on specified criteria (such as a fixed time interval or a number of elements).

emmylou 2 years, 4 months ago

Which is the moment I decided that AI was nothing to fear

lucaluca1982 3 years ago

what about A? This can cause the job to fail

midgoo 3 years, 1 month ago

Selected Answer: D

A: note that without a correct timestamp, the pipeline still run fine with the default timestamp. The result maybe incorrect but the job will not fail. D: For unbound collection, this will fail if any aggregation function is done.

musumusu 3 years, 1 month ago

Answer: A All Streaming Insert failed, because there is no TimeStamp added, otherwise there is already a DEFAULT global windowing function and can execute without assigning any windowing function. I mean first there should be Timestamp in the data, then according to our aggregation outcome either its full time (global) or batch/chunks time aggregation(non global) will be performed.

DipT 3 years, 3 months ago

Selected Answer: D

https://beam.apache.org/documentation/programming-guide/#windowing Beam's default windowing behavior is to assign all elements of a PCollection to a single, global window and discard late data, even for unbounded PCollections. Before you use a grouping transform such as GroupByKey on an unbounded PCollection, you must do at least one of the following: Set a non-global windowing function. See Setting your PCollection's windowing function. Set a non-default trigger. This allows the global window to emit results under other conditions, since the default windowing behavior (waiting for all data to arrive) will never occur.

Ray0506 3 years, 6 months ago

Selected Answer: D

Answer is D

TOXICcharlie 3 years, 6 months ago

Selected Answer: D

Correct answer is D. C does not make sense because for unbounded source like Pub/Sub, the global functions are applied by default. The reason for failure would be they are using specific aggregations that require non-global window functions, e.g. tumbling or hopping windows.

https://zhangxuetu.online

[202610]

Question #58 Topic 1

You architect a system to analyze seismic data. Your extract, transform, and load (ETL) process runs as a series of MapReduce jobs on an Apache Hadoop cluster. The ETL process takes days to process a data set because some steps are computationally expensive. Then you discover that a sensor calibration step has been omitted. How should you change your ETL process to carry out sensor calibration systematically in the future?

A. Modify the transformMapReduce jobs to apply sensor calibration before they do anything else.

B. Introduce a new MapReduce job to apply sensor calibration to raw data, and ensure all other MapReduce jobs are chained after this. Most Voted

C. Add sensor calibration data to the output of the ETL process, and document that all users need to apply sensor calibration themselves.

D. Develop an algorithm through simulation to predict variance of data output from the last MapReduce job based on calibration factors, and apply the correction to all data.

Correct Answer: B

Community vote distribution



Comments

SteelWarrior Highly Voted 5 years, 6 months ago

Should go with B. Two reasons, it is a cleaner approach with single job to handle the calibration before the data is used in the pipeline. Second, doing this step in later stages can be complex and maintenance of those jobs in the future will become challenging.

Yiouk 4 years, 8 months ago

B. different MR jobs execute in series, adding 1 more job makes sense in this case.

[Removed] Highly Voted 6 years ago

Answer: A Description: My take on this is for sensor calibration you just need to update the transform function, rather than creating a whole new mapreduce job and storing/passing the values to next job

Jphix 4 years, 10 months ago

It's B. A would involving changing every single job (notice it said jobS, plural, not a single job). If that is computationally intensive, which it is, you're repeating a computationally intense process needlessly several times. SteelWarrior and YuriP are right on this one.

mark1223jkh 1 year, 10 months ago

Why all jobs, change only the first job for calibration, right?

Marwan95 1 year, 9 months ago

Selected Answer: A

I'll choose A. WHY? cause the process already takes DAYS and adding another step will increase the time more

jin0 3 years, 1 month ago

What kinds of sensor calibrations exists? I don't understand how computation in pipeline would be expense due to calibration being omitted..?

samdhimal 3 years, 2 months ago

B. Introduce a new MapReduce job to apply sensor calibration to raw data, and ensure all other MapReduce jobs are chained after this. This approach would ensure that sensor calibration is systematically carried out every time the ETL process runs, as the new MapReduce job would be responsible for calibrating the sensors before the data is processed by the other steps. This would ensure that all data is calibrated before being analyzed, thus avoiding the omission of the sensor calibration step in the future. It also allows you to chain all other MapReduce jobs after this one, so that the calibrated data is used in all the downstream jobs.

samdhimal 3 years, 2 months ago

Option A is not ideal, as it would be time-consuming to modify all the transformMapReduce jobs to apply sensor calibration before doing anything else, and there is a risk of introducing bugs or errors. Option C is not ideal, as it would rely on users to apply sensor calibration themselves, which would be inefficient and could introduce inconsistencies in the data. Option D is not ideal, as it would require a lot of simulation and testing to develop an algorithm that can predict the variance of data output accurately and it may not be as accurate as calibrating the sensor directly.

DipT 3 years, 3 months ago

Selected Answer: B

It is much cleaner approach

DGames 3 years, 3 months ago

Selected Answer: B

Best approach is calibration will be separate job because if we need to tune the calibration later also it would be to maintain without worries about all other jobs.

odacir 3 years, 3 months ago

Selected Answer: B

Should be B. My reason, this is like an Anti corruption layer, and that's a good practice, C- , if you modify your transformMapReduce will be harder to test and debug, so it's a bad practice. C the idea de introduce manual operation is an anti patron and has a lot of problems D It's overkilling, a don't have sense in this scenario.

ZIMARAKI 4 years, 2 months ago

Selected Answer: B

SteelWarrior explanation is correct :)

lord_ryder 4 years, 2 months ago

Selected Answer: B

SteelWarrior explanation is correct

medeis_jar 4 years, 2 months ago

Selected Answer: B

SteelWarrior explanation is correct

hendrixlives 4 years, 3 months ago

Selected Answer: B

SteelWarrior's answer is correct

anji007 4 years, 5 months ago

Ans: B Adding a new job in the beginning of chain makes more sense than updating existing chain of jobs.

sumanshu 4 years, 9 months ago

Vote for 'B' (introduce new job) over 'A', (instead of modifying existing job)

YuriP 5 years, 8 months ago

Should be B. It's a Data Quality step which has to go right after Raw Ingest. Otherwise you repeat the same step unknown (see "job_s_" in A) number of times, possibly for no reason, therefore extending ETL time.

https://zhangxuetu.online

[202610]

Question #59 Topic 1

An online retailer has built their current application on Google App Engine. A new initiative at the company mandates that they extend their application to allow their customers to transact directly via the application. They need to manage their shopping transactions and analyze combined data from multiple datasets using a business intelligence (BI) tool. They want to use only a single database for this purpose. Which Google Cloud database should they choose?

A. BigQuery

B. Cloud SQL **Most Voted**

C. Cloud BigTable

D. Cloud Datastore

Correct Answer: B

Community vote distribution



Comments

PolyMoe **Highly Voted** 3 years, 2 months ago

Selected Answer: B

B. Cloud SQL would be the most appropriate choice for the online retailer in this scenario. Cloud SQL is a fully-managed relational database service that allows for easy management and analysis of data using SQL. It is well-suited for applications built on Google App Engine and can handle the transactional workload of an e-commerce application, as well as the analytical workload of a BI tool.

Aaronn14 **Highly Voted** 3 years ago

A. "They want to use only a single database for this purpose" is a key requirement. You can use BigQuery for transactions, though it is not efficient. You can not use CloudSQL for analytics. So it is probably BQ.

baimus 1 year, 6 months ago

Yeah my thinking was the same, but actually cloud SQL is fine to connect BI tools to, which is specified in this question.

Mathew106 2 years, 8 months ago

Selected Answer: B

Cloud SQL seems to fit the best. It supports transactions and can be used to run queries and do analytics. BigQuery is good for the analysis part but it's not good for managing transactions. If the question needed a database just to store the data for analysis it would be ok. But if we want to update single transactions or add them row by row, then it's not good. BigQuery is not made to support an application. It's a DW. BigTable is can not carry transactions over

multiple rows and is better for large scale analytics jobs. Also we should pick it for use-cases with high throughput/low latency requirements. Seems redundant.

Siddhesh05 2 years, 11 months ago

Selected Answer: A

Big Query because of analysis

izekc 2 years, 11 months ago

Selected Answer: C

Should be bigtable

baimus 1 year, 6 months ago

I can't really see that. Bigtable is only ever the right choice for noSql at vast scale.

juliobs 3 years ago

Selected Answer: A

I think BigQuery makes sense here. It works for transactions too.

juliobs 3 years ago

I just did a session with an official trainer from Google that said BigTable is better.

Fotofilico 2 years, 5 months ago

I'm an official trainer from Google and I can say that my best two options for this scenario would be Cloud SQL and BigQuery in that order. Also we can consider datastore since we're using it with a web app, but it's another topic.

certs4pk 1 year, 3 months ago

but, how to analyze 'combined data from multiple datasets' in cloud sql?

ninjatech 3 years ago

Transactional Data need to written first by application before it could be analysed so cloudsql.

samdhimal 3 years, 2 months ago

Both BigQuery and Cloud Bigtable are valid options for this use case, but BigQuery is better suited for this specific scenario where the retailer needs to manage and analyze large amounts of data from multiple datasets using a BI tool. BigQuery is a fully-managed, cloud-native data warehouse that enables super-fast SQL queries using the processing power of Google's infrastructure. It can handle large, complex datasets and is well-suited for both transactional and analytical workloads. It can also handle data from multiple datasets and can be integrated with other Google Cloud services, such as Dataflow, Dataproc and Looker for BI analysis. While Cloud Bigtable is also a good option for this use case as it is a highly scalable and performant NoSQL database that is well-suited for handling large amounts of data and high-write loads. It is not as good as BigQuery for analytical workloads and it may not be as well-suited for this specific scenario where the retailer needs to manage and analyze large amounts of data from multiple datasets using a BI tool.

jin0 3 years, 1 month ago

Bigquery is a OLAP. So it could be not a answer I think.

samdhimal 3 years, 2 months ago

Cloud SQL and Cloud Datastore are also good options for certain use cases, but they may not be as well-suited for this specific scenario where the retailer needs to manage and analyze large amounts of data from multiple datasets using a BI tool.

desertlotus1211 3 years, 2 months ago

The Community is choosing Answer B - Cloud SQL, as per the question. However when they explain - they're speaking about BQ[????] So is it BigQuery or Cloud SQL?

DipT 3 years, 3 months ago

Selected Answer: B

https://cloud.google.com/bigquery/docs/partitioned-tables

DipT 3 years, 3 months ago

Selected Answer: B

It needs support for transaction so cloud sql is the choice of database and with Bigquery we can still analyze cloud sql data via federated queries https://cloud.google.com/bigquery/docs/reference/legacy-sql

DGames 3 years, 3 months ago

Selected Answer: B

Most important part of question is transaction (RDBMS) strong ACID property database. Second part analysis of data, yes possible using any BI tool its possible with RDBMS db.

odacir 3 years, 3 months ago

Selected Answer: B

C and D are not able to work with BI directly, so discard. A: It's the best option for BI for awful for transactions B: it's the best option for transaction, and works for BI, so this must be the answer

Leeeeee 3 years, 4 months ago

Selected Answer: B

BigQuery for Analytics and BI

Leelas 3 years, 4 months ago

Selected Answer: B

Cloud Sql is Used to store Transactional Data and supports Sql Transactions. Where as Big Query is used for Analytics.

Zion0722 3 years, 5 months ago

Cloud SQL supports transactions as well as analysis through a BI tool. Firestore/Datastore does not support SQL syntax typically needed to do analysis done by a BI tool. BigQuery is not suitable for transactional use case. BigTable does not support SQL. It's A.

MisuLava 3 years, 5 months ago

Selected Answer: B

it is obvious.

MisuLava 3 years, 5 months ago

I meant to choose A, BugQuery :)

https://zhangxuetu.online

[202610]

Question #60 Topic 1

You launched a new gaming app almost three years ago. You have been uploading log files from the previous day to a separate Google BigQuery table with the table name format LOGS_yyyymmdd. You have been using table wildcard functions to generate daily and monthly reports for all time ranges. Recently, you discovered that some queries that cover long date ranges are exceeding the limit of 1,000 tables and failing. How can you resolve this issue?

A. Convert all daily log tables into date-partitioned tables

B. Convert the sharded tables into a single partitioned table **Most Voted**

C. Enable query caching so you can cache data from previous months

D. Create separate views to cover each month, and query from these views

Correct Answer: B

Community vote distribution



Comments

[Removed] **Highly Voted** 5 years ago

should be B https://cloud.google.com/bigquery/docs/creating-partitioned-tables#converting_date-sharded_tables_into_ingestion-time_partitioned_tables

Rajuuu 4 years, 9 months ago

The above link does mention about shard ing benefits but only about partition tables. A is correct.

g2000 4 years, 2 months ago

keyword is single

Chelseajcole 3 years, 6 months ago

you are right. Partitioning versus sharding Table sharding is the practice of storing data in multiple tables, using a naming prefix such as [PREFIX]_YYYYMMDD. Partitioning is recommended over table sharding, because partitioned tables perform better. With sharded tables, BigQuery must maintain a copy of the schema and metadata for each table. BigQuery might also need to verify permissions for each queried table. This practice also adds to query overhead and affects query performance. If you previously created date-sharded tables, you can convert them into an ingestion-time partitioned table.

[Removed] **Highly Voted** 5 years ago

Answer: B Description: Google says that when you have multiple wildcard tables, best option is to shard it into single partitioned table. Time and cost efficient

lgdantas 4 years, 7 months ago

Can you please share the reference?

Tumri 3 years, 7 months ago

https://cloud.google.com/bigquery/docs/partitioned-tables#dt_partition_shard

1479 7 months, 1 week ago

Selected Answer: A

partitioned table is more efficient

jatinbhatia2055 1 year, 3 months ago

Selected Answer: B

Sharded tables, like LOGS_yyyymmdd, are useful for managing data, but querying across a long date range with table wildcards can lead to inefficiencies and exceed the 1,000 table limit in BigQuery. Instead of using multiple sharded tables, you should consider converting these into a partitioned table. A partitioned table allows you to store all the log data in a single table, but logically divides the data into partitions (e.g., by date). This way, you can efficiently query data across long date ranges without hitting the 1,000 table limit.

Oleksandr0501 1 year, 11 months ago

Selected Answer: B

gpt: Thank you for your feedback and additional information. You are correct that partitioned tables have a limit of 4,000 partitions, so partitioning tables by date could potentially run into this limit in the future. In this case, option B, converting sharded tables into a single partitioned table, could be a reasonable solution to avoid exceeding the maximum number of tables in BigQuery. As you mentioned, sharded tables require additional metadata and permissions verification, which can impact query performance. Converting sharded tables into a single partitioned table can improve performance and reduce query overhead. Therefore, based on the information provided, option B seems to be the most appropriate solution for avoiding the limit of 1,000 tables in BigQuery and optimizing query performance.

luks_skywalker 2 years ago

The question seems pretty badly written. One important thing to remember is that partitioned tables also have a limit of 4000 partitions (https://cloud.google.com/bigquery/docs/partitioned-tables#ingestion_time), so moving everything to one table would just delay the problem. However, option A is not clear on how it will be done. One table per year with daily partitions? Best solution as no limit will be reached. One table per day? Then we have the same 1000 tables problem. All things considered I'll stick to B, simply because the problem will definitely be solved for the next few years, so I'd say it's a reasonable solution.

PolyMoe 2 years, 2 months ago

Selected Answer: B

Answer is B. Table sharding is the practice of storing data in multiple tables, using a naming prefix such as [PREFIX]_YYYYMMDD. Partitioning is recommended over table sharding, because partitioned tables perform better. With sharded tables, BigQuery must maintain a copy of the schema and metadata for each table. BigQuery might also need to verify permissions for each queried table. This practice also adds to query overhead and affects query performance. In answer A. we still are creating tableS (even though partioned). So we still facing the issue of max 1000 tables. In B. we have only ONE table (partioned)

samdhimal 2 years, 2 months ago

Why not A? By converting all daily log tables into date-partitioned tables, you can take advantage of partition pruning to limit the number of tables that need to be scanned during a query. Partition pruning allows BigQuery to skip scanning partitions that are not within the date range specified in the query, thus reducing the number of tables that

need to be scanned and can help to avoid reaching the 1,000 table limit. A Seems like the correct answer but I can be wrong...

RoshanAshraf 2 years, 2 months ago

Selected Answer: B

B. Convert the sharded tables into a single partitioned table It was a sharded Table (format is the HINT here); converting to partition table is the option. Also as per GCP its recommended to use Partition over Sharding

korntewin 2 years, 2 months ago

Selected Answer: A

I chose option A. From all the comments I have seen, there are various things that are misunderstood. 1. Option A is a single table with multiple shards! Google does recommend to use partition rather than shard as it has a better performance (https://cloud.google.com/bigquery/docs/partitioned-tables#dt_partition_shard) 2. Option B is a single table with single partition! Single partition is a no for large table

DipT 2 years, 3 months ago

Selected Answer: B

https://cloud.google.com/bigquery/docs/partitioned-tables

DGames 2 years, 3 months ago

Selected Answer: B

Option A - already doing same loading data in separate table daily and reached 1000 table limit. Option B - Use wild card to query the data Option C & D - make no sense

odacir 2 years, 3 months ago

its B. A - Even if you have 100+ partitioned tables, you still have the limit of less than 1000 tables. So this doesn't work for this problem. C It's a no sense. Cache its 24h for every table that has been query in the last 24 and has no changes. Also, cache is not support with wildcard multiple tables. D Will not work because it's a recursive issue. You still will have 100+ tables, beam query B will work, you materialize in only one table, so will be working perfectly.

Nirca 2 years, 5 months ago

Selected Answer: B

Convert MANY sharded tables into a single ONE (partitioned) table

rrr000 2 years, 7 months ago

selecting for daily/monthly data from one single partition will be very expensive. I think A is the best answer

Preemptible_cerebrus 2 years, 9 months ago

Selected Answer: B

C'mon, how much time are you going to take to partition every single table you have? second point and the most important, you have a table for every SINGLE DAY "LOGS_YYYYMMDD" partitioning every table will end on scanning all the records of each table when you query them by date ranges using the wildcards, there will be no difference on time-partitioning each table versus consuming them as described.

AmirN 2 years, 9 months ago

If you follow option A, you will end up with the same amount of tables, e.g 1500 tables, though they will all be partitioned, which is not helpful. Option B takes all the sharded tables and makes one large partitioned table.

rrr000 2 years, 7 months ago

Partitions are not tables. The issue is not performance. It is the limit imposed by bq regarding how many tables you can query.

https://zhangxuetu.online

[202610]

Question #61 Topic 1

Your analytics team wants to build a simple statistical model to determine which customers are most likely to work with your company again, based on a few different metrics. They want to run the model on Apache Spark, using data housed in Google Cloud Storage, and you have recommended using Google Cloud Dataproc to execute this job. Testing has shown that this workload can run in approximately 30 minutes on a 15-node cluster, outputting the results into Google BigQuery. The plan is to run this workload weekly. How should you optimize the cluster for cost?

A. Migrate the workload to Google Cloud Dataflow

B. Use pre-emptible virtual machines (VMs) for the cluster **Most Voted**

C. Use a higher-memory node so that the job runs faster

D. Use SSDs on the worker nodes so that the job can run faster

Correct Answer: B

Community vote distribution

B (100%)

Comments

jvg637 **Highly Voted** 6 years ago

B. (Hadoop/Spark jobs are run on Dataproc, and the pre-emptible machines cost 80% less)

rickywck **Highly Voted** 6 years ago

I think the answer should be B: https://cloud.google.com/dataproc/docs/concepts/compute/preemptible-vms

AmitK121981 1 year, 3 months ago

Selected Answer: B

all are saying its pre-emptibles. but spot VMs can only be used in secondary worker, not on master and primary worker so not sure why this will cause savings, and secondary workers are not mandatory too

theseawillclaim 2 years, 8 months ago

I believe it might be "B", but what if the job is mission critical? Pre-emptible VMs would be of no use.

enivid007 1 year, 8 months ago

Mission critical workloads can't be needed "weekly"

abi01a 2 years, 11 months ago

I believe Exam Topics ought to provide brief explanation or supporting link to picked correct answers such as this one. Option A may be correct from the view point that Dataflow is a Serverless service that is fast, cost-effective and the

fact that Preemptible VMs though can give large price discount may not always be available. It will be great to know the reason(s) behind Exam Topic selected option.

samdhimal 3 years, 2 months ago

B. Use pre-emptible virtual machines (VMs) for the cluster Using pre-emptible VMs allows you to take advantage of lower-cost virtual machine instances that may be terminated by Google Cloud after a short period of time, typically after 24 hours. These instances can be a cost-effective way to handle workloads that can be interrupted, such as batch processing jobs like the one described in the question. Option A is not ideal, as it would require you to migrate the workload to Google Cloud Dataflow, which may cause additional complexity and would not address the issue of cost optimization. Option C is not ideal, as it would require you to use a higher-memory node which would increase the cost. Option D is not ideal, as it would require you to use SSDs on the worker nodes which would increase the cost. Using pre-emptible VMs is a better option as it allows you to take advantage of lower-cost virtual machine instances and handle workloads that can be interrupted, which can help to optimize the cost of the cluster.

Rodolfo_Marcos 3 years, 2 months ago

What is happening with this test "correct answer" a lot of times it doesn't make any sense. As this one... Clear it's B

DipT 3 years, 3 months ago

Selected Answer: B

Using preemtible machines are cost effective , and because is suitable for a job mentioned here as it is fault tolerant .

DGames 3 years, 3 months ago

Selected Answer: B

User Pre-emptible VM machine and save process cost, and question want simple solution.

odacir 3 years, 3 months ago

Selected Answer: B

A- Data flow it's not cost-effective in comparison with dataproc B- Preemptible VM instances are available at much lower price—a 60-91% discount—compared to the price of standar, so this is the answer C and D are more expensive.

Remi2021 3 years, 6 months ago

Selected Answer: B

B is right way to go

FrankT2L 3 years, 10 months ago

Selected Answer: B

Preemptible workers are the default secondary worker type. They are reclaimed and removed from the cluster if they are required by Google Cloud for other tasks. Although the potential removal of preemptible workers can affect job stability, you may decide to use preemptible instances to lower per-hour compute costs for non-critical data processing or to create very large clusters at a lower total cost
https://cloud.google.com/dataproc/docs/concepts/compute/secondary-vms

Remi2021 4 years ago

B is teh right answer. examtopics update your answers or make your site free again.

OmJanmeda 4 years ago

Selected Answer: B

B is right answer. my experience is not good with Examtopics, so many wrong answers.

Yaa 4 years, 1 month ago

Selected Answer: B

B should be the right answer. I am amazed that almost 60% of the marked answers on the site are wrong.

byash1 4 years, 2 months ago

Ans : B, here we are checking on reducing cost, so pre-emptiable machines are best choice

medeis_jar 4 years, 2 months ago

Selected Answer: B

"this workload can run in approximately 30 minutes on a 15-node cluster," so you need performance for only 30 mins -> preemptible VMs https://cloud.google.com/dataproc/docs/concepts/compute/preemptible-vms

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #62 Topic 1

Your company receives both batch- and stream-based event data. You want to process the data using Google Cloud Dataflow over a predictable time period.
However, you realize that in some instances data can arrive late or out of order. How should you design your Cloud Dataflow pipeline to handle data that is late or out of order?

A. Set a single global window to capture all the data.

B. Set sliding windows to capture all the lagged data.

C. Use watermarks and timestamps to capture the lagged data. **Most Voted**

D. Ensure every datasource type (stream or batch) has a timestamp, and use the timestamps to define the logic for lagged data.

Correct Answer: C

Community vote distribution



Comments

samdhimal **Highly Voted** 2 years, 2 months ago

C: Use watermarks and timestamps to capture the lagged data. Watermarks are a way to indicate that some data may still be in transit and not yet processed. By setting a watermark, you can define a time period during which Dataflow will continue to accept late or out-of-order data and incorporate it into your processing. This allows you to maintain a predictable time period for processing while still allowing for some flexibility in the arrival of data. Timestamps, on the other hand, are used to order events correctly, even if they arrive out of order. By assigning timestamps to each event, you can ensure that they are processed in the correct order, even if they don't arrive in that order.

samdhimal 2 years, 2 months ago

Option A: Set a single global window to capture all the data is not a good idea because it may not allow for late or out-of-order data to be processed. Option B: Set sliding windows to capture all the lagged data is not suitable for the case where you want to process the data over a predictable time period. Sliding windows are used when you want to process data over a period of time that is continuously moving forward, not a fixed period. Option D: Ensure every datasource type (stream or batch) has a timestamp, and use the timestamps to define the logic for lagged data is a good practice but not a complete solution, because it only ensures that data is ordered correctly, but it does not account for data that may be late.

jvg637 **Highly Voted** 5 years ago

We need a combination of window + watermark (timestamps) + trigger to treat the late data. So D.

MikkelRev 1 year, 6 months ago

Selected Answer: C

option C: Use watermarks and timestamps to capture the lagged data.

MikkelRev 1 year, 6 months ago

option C: Use watermarks and timestamps to capture the lagged data.

desertlotus1211 2 years, 2 months ago

Answer is C: There is no such thing as a sliding windows using by dataflow.

DeeData 2 years, 2 months ago

I highly doubt, DataFlow windowing is divided into three(3) types: 1. Fixed 2. Sliding 3. Session

Mathew106 1 year, 8 months ago

The naming in Apache Beam is: Fixed, Sliding, Session In Dataflow it's: Tumbling, Hopping, Session. I was very confused at first too when I saw "hopping" in a question.

AzureDP900 2 years, 2 months ago

Answer is Use watermarks and timestamps to capture the lagged data. A watermark is a threshold that indicates when Dataflow expects all of the data in a window to have arrived. If new data arrives with a timestamp that's in the window but older than the watermark, the data is considered late data.

DGames 2 years, 3 months ago

Selected Answer: C

Watermark is use for late date,

FrankT2L 2 years, 10 months ago

Selected Answer: B

Preemptible workers are the default secondary worker type. They are reclaimed and removed from the cluster if they are required by Google Cloud for other tasks. Although the potential removal of preemptible workers can affect job stability, you may decide to use preemptible instances to lower per-hour compute costs for non-critical data processing or to create very large clusters at a lower total cost
https://cloud.google.com/dataproc/docs/concepts/compute/secondary-vms

FrankT2L 2 years, 10 months ago

delete this answer. The answer belongs to another question

Tanzu 3 years, 1 month ago

Selected Answer: A

That's why we have watermarks in apache beam.

VishalBule 3 years, 1 month ago

Answer is C Use watermarks and timestamps to capture the lagged data. A watermark is a threshold that indicates when Dataflow expects all of the data in a window to have arrived. If new data arrives with a timestamp that's in the window but older than the watermark, the data is considered late data.

medeis_jar 3 years, 2 months ago

Selected Answer: C

"Watermark in implementation is a monotonically increasing timestamp. When Beam/Dataflow see a record with an event timestamp that is earlier than the watermark, the record is treated as late data."

MaxNRG 3 years, 3 months ago

Selected Answer: C

A is a direct No, if data don’t have timestamp, we’ll only have the procesing time and not the “event time”. B is not either, sliding windows are not for this. Hopping|sliding windowing is useful for taking running averages of data, but not to process late data. D looks correct but has one concept missing, the watermark to know if the process time is ok with the event time or not. I’m not 100% sure is incorrect. If, since we have a “predictable time period”, might be this will do. I mean, if our dashboard is shown after the last input data has arrived (single global window), this should be ok. We’d have a “perfect watermark”. Anyway we’d need triggering .

MaxNRG 3 years, 3 months ago

C is, I think, the correct answer: Watermark is different from late data. Watermark in implementation is a monotonically increasing timestamp. When Beam/Dataflow see a record with an event timestamp that is earlier than the watermark, the record is treated as late data. I’ll try to explain: Late data is inherent to Beam’s model for out-of-order processing. What does it mean for data to be late? The definition and its properties are intertwined with watermarks that track the progress of each computation across the event time domain. The simple intuition behind handling lateness is this: only late input should result in late data anywhere in the pipeline. So, is not easy to decide between C and D. If you ask me I’d say C since for D we ought to make some suppositions.

Jlozano 3 years, 3 months ago

Selected Answer: C

"Expert Verified" but >50% questions have random answer. "Sliding window" really? Please, this can be fixed easyly with our most voted answer. Of course, the correct answer is C.

JG123 3 years, 4 months ago

Why there are so many wrong answers? Examtopics.com are you enjoying paid subscription by giving random answers from people? Ans: C

anji007 3 years, 5 months ago

Ans: C

safiyu 3 years, 7 months ago

Answer should be C. sliding windows are meant for calculating running average and not lagging data. Watermark is best for this purpose

sumanshu 3 years, 9 months ago

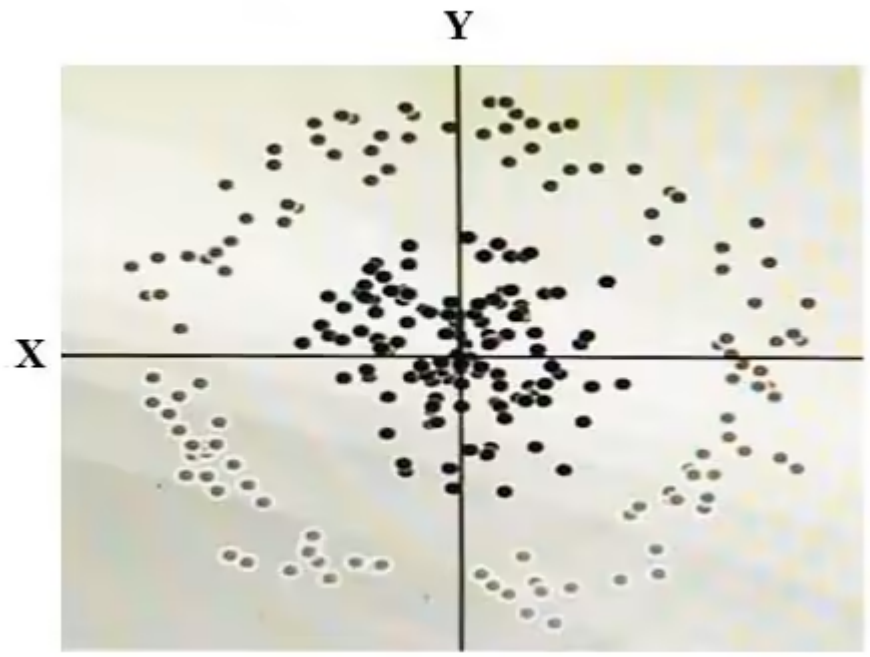
vote for 'C"

https://zhangxuetu.online

[202610]

Question #63 Topic 1

You have some data, which is shown in the graphic below. The two dimensions are X and Y, and the shade of each dot represents what class it is. You want to classify this data accurately using a linear algorithm. To do this you need to add a synthetic feature. What should the value of that feature be?



A. X2+Y2 **Most Voted**

B. X2

C. Y2

D. cos(X)

Correct Answer: A

Community vote distribution

A (80%) B (20%)

Comments

jvg637 **Highly Voted** 6 years ago

For fitting a linear classifier when the data is in a circle use A.

xs91 **Highly Voted** 5 years ago

I think it's A as explai ned here https://medium.com/@sachinkun21/using-a-linear-model-to-deal-with-nonlinear-dataset-c6ed0f7f3f51

SamuelTsch 1 year, 5 months ago

Selected Answer: A

I think A should be x^2+y^2. We need a circle to classify the data.

baimus 1 year, 6 months ago

Selected Answer: A

Just a note, they are using X2 and Y2 to mean Xsquared, and Ysquared. This is a circle in the form X2+Y2 = k, so for a given k will split that dataset nicely.

Mathew106 2 years, 8 months ago

It's not obvious to me it is A. As others said, cos(X) does ignore the Y value. But answer A does not seem good either. The differences seem minimal. If you do A then you have the following issues. If you take elements in the bottom right or the top left of the circle, they will all have the same value, ZERO. Not only that, they will actually have the same value with the elements in the middle of the circle which are completely black. Moreover, elements on the extreme right and extreme right will have different values (-x_max and +x_max). However, if you use a cos(x) then the elements in the beginning

Mathew106 2 years, 8 months ago

Nevermind I did not understand that X2 and Y2 meant X^2 and Y2. Answer is A because that gives the distance from the circle. Circle radius = sqrt(X^2 + Y^2). So even though it's not a perfect answer, it makes sense.

samdhimal 3 years, 2 months ago

A. X2+Y2 The synthetic feature that should be added in this case is the squared value of the distance from the origin (0,0). This is equivalent to X2+Y2. By adding this feature, the classifier will be able to make more accurate predictions by taking into account the distance of each data point from the origin. X2 and Y2 alone will not give enough information to classify the data because they do not take into account the relationship between X and Y. D. cos(X) is not a suitable option because it does not take into account the Y coordinate.

GCPpro 3 years, 2 months ago

A is the correct answer as graph of circle is x^2 + y^2

desertlotus1211 3 years, 2 months ago

Answer is A: The answer reflects 'x' to the 2nd power + 'y' the 2nd power. I guess they can't use carots in the exam answers!

AzureDP900 3 years, 2 months ago

A is right Reference: https://medium.com/@sachinkun21/using-a-linear-model-to-deal-with-nonlinear-dataset-c6ed0f7f3f51

DipT 3 years, 3 months ago

Selected Answer: A

https://developers.google.com/machine-learning/crash-course/feature-crosses/video-lecture

DGames 3 years, 3 months ago

Selected Answer: A

linear circle X2+Y2 https://www.stat.cmu.edu/~cshalizi/dm/20/lectures/08/lecture-08.html

mvww11 3 years, 9 months ago

If the shape was a circle, it would be (x^2 + y ^2). But I think that a quadric curve will do a better job of separating the two classes, so it would be (x^2)

gabrysave 3 years, 10 months ago

Answer: A. X^2+Y^2 is the equation of a circle.

diagniste 3 years, 11 months ago

Selected Answer: A

C'est A

Tanzu 4 years, 1 month ago

Selected Answer: A

only A is draw a circle

sraakesh95 4 years, 2 months ago

Selected Answer: A

Equation of circle as represented in the question

moumou 4 years, 2 months ago

F(x) as A B C will have always a positive values as result, for A will need a third dimenssion Z to represent data, only D:cos(x) can be presented as the shown classification. this is a math question

NR22 3 years, 11 months ago

A B C will only have positive values imaginary numbers (i + j) : am I a joke to you?

https://zhangxuetu.online

[202610]

Question #64 Topic 1

You are integrating one of your internal IT applications and Google BigQuery, so users can query BigQuery from the application's interface. You do not want individual users to authenticate to BigQuery and you do not want to give them access to the dataset. You need to securely access BigQuery from your IT application. What should you do?

A. Create groups for your users and give those groups access to the dataset

B. Integrate with a single sign-on (SSO) platform, and pass each user's credentials along with the query request

C. Create a service account and grant dataset access to that account. Use the service account's private key to access the dataset **Most Voted**

D. Create a dummy user and grant dataset access to that user. Store the username and password for that user in a file on the files system, and use those credentials to access the BigQuery dataset

Correct Answer: C

Community vote distribution

C (100%)

Comments

samdhimal 1 year, 8 months ago

C. Create a service account and grant dataset access to that account. Use the service account's private key to access the dataset. Creating a service account and granting dataset access to that account is the most secure way to access BigQuery from an IT application. Service accounts are designed for use in automated systems and do not require user interaction, eliminating the need for individual users to authenticate to BigQuery. Additionally, by using the private key of the service account to access the dataset, you can ensure that the authentication process is secure and that only authorized users have access to the data.

samdhimal 1 year, 8 months ago

Option A: Create groups for your users and give those groups access to the dataset, is not the best option because it still requires users to authenticate to BigQuery Option B: Integrate with a single sign-on (SSO) platform, and pass each user's credentials along with the query request is not the best option because it still requires users to authenticate to BigQuery. Option D: Create a dummy user and grant dataset access to that user. Store the username and password for that user in a file on the files system, and use those credentials to access the BigQuery dataset is not a secure option because it involves storing sensitive information in a file on the file system, which can be easily accessed by unauthorized users.

DGames 1 year, 9 months ago

Selected Answer: C

Service account approach is secure in GCP to communicate with between service or application

NicolasN 1 year, 11 months ago

Selected Answer: C

[C] The reason in https://cloud.google.com/bigquery/docs/data-governance#identity "Users of BigQuery might be humans, but they might also be nonhuman applications that communicate using a BigQuery client library or the REST API. These applications should identify themselves using a service account, the special type of Google identity intended to represent a nonhuman user."

Tanzu 2 years, 7 months ago

Selected Answer: C

such kind of questions are always service account oriented. and sa can be used as a user ..not just for machine2machine users may or may not enter their credentials app login window. that's not main point by the way

rcruz 2 years, 8 months ago

Selected Answer: C

Correct: C

anji007 2 years, 11 months ago

Ans: C

daghayeghi 3 years, 6 months ago

C: It says "do not want individual users to authenticate to BigQuery and you do not want to give them access to the dataset", then C is the best choice.

ave4000 3 years, 10 months ago

Granting access to the app through a service account would mean all of the users that access the app have access to the BQ. Question was to filter it out, so I believe each user would have to be added to a group that does or doesn't have access to the dataset.

squishy_fishy 2 years, 11 months ago

The answer is C. When access data through application, Google recommendation is using service account.

kavs 3 years, 10 months ago

Yes A seems to be right

kavs 3 years, 10 months ago

It says individually don't want to authorise service account could be right too

Ankush_j 4 years ago

Ans is C, Service account is best for secure data

haroldbenites 4 years, 1 month ago

C is correct

Rajuuu 4 years, 2 months ago

Correct C.

https://zhangxuetu.online

[202610]

Question #65 Topic 1

You are building a data pipeline on Google Cloud. You need to prepare data using a casual method for a machine-learning process. You want to support a logistic regression model. You also need to monitor and adjust for null values, which must remain real-valued and cannot be removed. What should you do?

A. Use Cloud Dataprep to find null values in sample source data. Convert all nulls to 'none' using a Cloud Dataproc job.

B. Use Cloud Dataprep to find null values in sample source data. Convert all nulls to 0 using a Cloud Dataprep job. Most Voted

C. Use Cloud Dataflow to find null values in sample source data. Convert all nulls to 'none' using a Cloud Dataprep job.

D. Use Cloud Dataflow to find null values in sample source data. Convert all nulls to 0 using a custom script.

Correct Answer: B

Community vote distribution



Comments

jvg637 Highly Voted 6 years ago

real-valued can not be null N/A or empty, have to be “0”, so it has to be B.

Snobid Highly Voted 6 years ago

Instead of having to write the custom script from scratch (option D), dataprep already has preconfigured tools for your use to perform the necessary data wrangling. As mentioned by jvg637, real-values have to be "0". Considering both points above, answer should be 'B'

monyu 1 year ago

Selected Answer: B

Usually, None values are converted to 0 in data cleaning and preparation process. The key point here is, we don't require any other tool than DataPrep to identify and modify the value

Parandhaman_Margan 1 year ago

Selected Answer: D

Cloud Dataflow is ideal for scalable data processing and allows for real-time transformations. Logistic regression requires numerical (real-valued) inputs, and null values cannot remain as they are.

Erg_de 1 year, 4 months ago

Selected Answer: D

Option D: Using null value conversion to 0 is the most correct practice for this case. Accompanying it with a script allows us to implement the necessary logic to handle null cases properly, adapting to the model while maintaining data integrity.

certs4pk 1 year, 3 months ago

y use a data flow job when it can b done via data prep (much simpler & straight forward, less resource intensive)..

AjoeT 1 year, 12 months ago

Selected Answer: B

B. Dataprep has the feature to convert it into 0.

niru12376 2 years, 1 month ago

0 is still a value, which can add bias in the model and the model will take that into account while making predictions so 'none'

Nandababy 2 years, 3 months ago

Why not D? keyword is Monitor, B would replace all empty fields and also cause unintended bias.

Nandababy 2 years, 3 months ago

However, Sergiomujica is right. If we need to prepare data using a casual method then its B "Dataprep".

sergiomujica 2 years, 6 months ago

The questions says "You need to prepare data using a casual method ", thats dataprep and values should be 0 so the right answer is B

Mathew106 2 years, 8 months ago

Selected Answer: B

No brainer. We need a real value and Dataprep is made for this. Dataflow is mainly for pre-processing before BigQuery ingests the data.

theseawillclaim 2 years, 8 months ago

Selected Answer: B

Dataprep is made for this kind of stuff, no reason to use a streaming service such as Dataflow.

Oleksandr0501 2 years, 11 months ago

Selected Answer: B

gpt:Cloud Dataprep is a data preparation service that can be used to transform, clean and shape data in a visually interactive way. It provides an easy-to-use interface to find and replace null values. Cloud Dataflow is a fully-managed service for executing data processing pipelines, which allows for parallel execution of data processing tasks. However, it requires more expertise to set up and operate than Cloud Dataprep, and is usually used for more complex data processing needs. Therefore, option B is the most suitable approach for the given requirements.

samdhimal 3 years, 2 months ago

Seems to me like Answers are both B and D. B is faster to implement while D takes time. Doesnt mean that it's wrong though. I m not sure why everyone has picked just B. Why not D? D works and does the same job. And also having custom script provides more flexibility and control over the data processing tasks and it allows you to handle missing values in a more flexible and efficient way.

rajm893 2 years, 10 months ago

The "casual way" or easy way to convert to to 0 is using Dataprep job rather than using the custom script.

AmmarFasih 2 years, 10 months ago

A simple rule. Whenever any service is available by GCP for a task, always recommend to use GCP service over any other.

GCPpro 3 years, 2 months ago

B is the correct answer.

AzureDP900 3 years, 2 months ago

Answer is Use Cloud Dataprep to find null values in sample source data. Convert all nulls to 0 using a Cloud Dataprep job. Key phrases are "casual method", "need to replace null with real values", "logistic regression". Logistic regression works on numbers. Null need to be replaced with a number. And Cloud dataprep is best casual tool out of given options.

DGames 3 years, 3 months ago

Selected Answer: B

real value 0

byash1 4 years, 2 months ago

Selected Answer: B

It is B

https://zhangxuetu.online

[202610]

Question #66 Topic 1

You set up a streaming data insert into a Redis cluster via a Kafka cluster. Both clusters are running on Compute Engine instances. You need to encrypt data at rest with encryption keys that you can create, rotate, and destroy as needed. What should you do?

A. Create a dedicated service account, and use encryption at rest to reference your data stored in your Compute Engine cluster instances as part of your API service calls.

B. Create encryption keys in Cloud Key Management Service. Use those keys to encrypt your data in all of the Compute Engine cluster instances. **Most Voted**

C. Create encryption keys locally. Upload your encryption keys to Cloud Key Management Service. Use those keys to encrypt your data in all of the Compute Engine cluster instances.

D. Create encryption keys in Cloud Key Management Service. Reference those keys in your API service calls when accessing the data in your Compute Engine cluster instances.

Correct Answer: B

Community vote distribution



Comments

SonuKhan1 **Highly Voted** 3 years, 11 months ago

Dear Admin, almost every answer is incorrect . Please check the comments and update your website.

Ganshank **Highly Voted** 5 years, 5 months ago

B. https://cloud.google.com/compute/docs/disks/customer-managed-encryption

and88x 1 year, 2 months ago

Selected Answer: B

D is incorrect because referencing keys in API service calls doesn't meet the requirements for encrypting data at rest. This approach is more related to accessing data at runtime, not storing it securely.

AmitK121981 1 year, 3 months ago

Selected Answer: C

CMEK is where customer managers keys, but are still created by Google (this is for KMS). CSEK is where keys are created outside GCP and used by API calls. So if customer has to create keys, it has to be outside KMS

jatinbhatia2055 1 year, 3 months ago

Selected Answer: D

Best Option: This is the most accurate approach. Cloud KMS provides the ability to create, manage, and rotate encryption keys. You can use the KMS API to reference the keys when encrypting and decrypting your data. In this case, you would integrate the KMS keys into your application logic (e.g., Kafka producers/consumers, Redis clients) to encrypt and decrypt data as it is stored or processed. This approach leverages the full functionality of Cloud KMS, including the ability to rotate and destroy keys as needed.

zevexWM 1 year, 5 months ago

But KMS doesnt create keys. It only stores them right?

TVH_Data_Engineer 1 year, 9 months ago

Selected Answer: B

Google Cloud Key Management Service (KMS) provides a centralized cloud service for managing cryptographic keys. By creating encryption keys in Cloud KMS, you can easily manage the lifecycle of these keys, including creation, rotation, and destruction. WYY NOT Create Keys Locally and Upload to Cloud KMS? While it's possible to create keys locally and then upload them to Cloud KMS, it's generally simpler and more secure to create the keys directly in Cloud KMS. This reduces the risk associated with transferring keys and leverages the security and compliance features of Cloud KMS.

emmylou 1 year, 11 months ago

Help! I chose "C" because of the statement, "encrypt data at rest with encryption keys that you can create, rotate, and destroy as needed" and read that as needing to generate the keys locally. Can you please explain where I went wrong?

odiez3 2 years, 2 months ago

the answer is C Read the full statement. " You need to encrypt data at rest with encryption keys that you can create "

theseawillclaim 2 years, 2 months ago

Selected Answer: B

B! C is useless overhead and you cannot rotate that easily!

Kiroo 2 years, 4 months ago

Well for what I remember from cloud arch and what I found in https://cloud.google.com/compute/docs/disks/customer-managed-encryption There is two options or the customer manage entirely or he will use the service to generate the keys so based on that is the B

samdhimal 2 years, 8 months ago

B. Create encryption keys in Cloud Key Management Service. Use those keys to encrypt your data in all of the Compute Engine cluster instances. Cloud Key Management Service (KMS) is a fully managed service that allows you to create, rotate, and destroy encryption keys as needed. By creating encryption keys in Cloud KMS, you can use them to encrypt your data at rest in the Compute Engine cluster instances, which is running your Redis and Kafka clusters. This ensures that your data is protected even when it is stored on disk.

samdhimal 2 years, 8 months ago

Option A: Create a dedicated service account, and use encryption at rest to reference your data stored in your Compute Engine cluster instances as part of your API service calls is not the best option as it does not provide encryption at rest. Option C: Create encryption keys locally. Upload your encryption keys to Cloud Key Management Service. Use those keys to encrypt your data in all of the Compute Engine cluster instances, is not the best option as it does not provide a way to manage the encryption keys centrally. Option D: Create encryption keys in Cloud Key Management Service. Reference those keys in your API service calls when accessing the data in your Compute Engine cluster instances, is not the best option as it does not provide encryption at rest, it only secure the data in transit.

DGames 2 years, 9 months ago

Selected Answer: B

B is correct answer generate key using KMS, why locally again it is overhead to upload and use everywhere.

Atnafu 2 years, 10 months ago

B If you use Google Cloud, Cloud Key Management Service lets you create your own encryption keys that you can use to add envelope encryption to your data. Using Cloud KMS, you can create, rotate, track, and delete keys.
https://cloud.google.com/docs/security/encryption/default-encryption#:~:text=If%20you%20use%20Google%20Cloud%2C%20Cloud%20Key%20Management%20Service%20lets%20you%20create%20your%20own%20encryption%20keys%20that%20you%20can%20use%20to%20add%20envelope%20encryption%20to%20your%20data.%20Using%20Cloud%20KMS%2C%20you%20can%20create%2C%20rotate%2C%20track%2C%20and%20delete%20keys.

medeis_jar 3 years, 9 months ago

Selected Answer: B

https://cloud.google.com/security/encryption-at-rest/

MaxNRG 3 years, 9 months ago

Selected Answer: B

A makes no sense, you need to use your own keys. You don't create keys locally and upload them, you should import it to make it work..using the kms public key…not just "uploading" it. C is also out. IT's between B and D Cloud KMS is a cloud-hosted key management service that lets you manage cryptographic keys for your cloud services the same way you do on-premises, You can generate, use, rotate, and destroy cryptographic keys from there. Since you want to encrypt data at rest, is B, you don't use them for any API calls.
https://cloud.google.com/compute/docs/disks/customer-managed-encryption

lg1234 3 years, 11 months ago

I believe you cannot upload custom keys to KMS for Compute Engine. Only via API Calls. See:
https://cloud.google.com/security/encryption/customer-supplied-encryption-keys With that said, option B

https://zhangxuetu.online

[202610]

Question #67 Topic 1

You are developing an application that uses a recommendation engine on Google Cloud. Your solution should display new videos to customers based on past views. Your solution needs to generate labels for the entities in videos that the customer has viewed. Your design must be able to provide very fast filtering suggestions based on data from other customer preferences on several TB of data. What should you do?

A. Build and train a complex classification model with Spark MLlib to generate labels and filter the results. Deploy the models using Cloud Dataproc. Call the model from your application.

B. Build and train a classification model with Spark MLlib to generate labels. Build and train a second classification model with Spark MLlib to filter results to match customer preferences. Deploy the models using Cloud Dataproc. Call the models from your application.

C. Build an application that calls the Cloud Video Intelligence API to generate labels. Store data in Cloud Bigtable, and filter the predicted labels to match the user's viewing history to generate preferences. Most Voted

D. Build an application that calls the Cloud Video Intelligence API to generate labels. Store data in Cloud SQL, and join and filter the predicted labels to match the user's viewing history to generate preferences.

Correct Answer: C

Community vote distribution

C (100%)

Comments

daghayeghi Highly Voted 4 years ago

answer C: If we presume that use label of video as a rowkey, Bigtable will be the best option. because it can store several TB, but Cloud SQL is limited to 30TB.

Alasmindas Highly Voted 4 years, 4 months ago

Option C is the correct answer. 1. Rather than building a new model - it is better to use Google provide APIs, here - Google Video Intelligence. So option A and B rules out 2) Between SQL and Bigtable - Bigtable is the better option as Bigtable support row-key filtering. Joining the filters is not required.

Mathew106 1 year, 8 months ago

Selected Answer: C

I don't even know if MLLib has out-of-the-box Computer Vision models. Developing this in Dataproc would be a nightmare. Using the computer vision API on the other hand makes perfect sense. The fact that the filtering must happen very fast and that this is a customer facing application points to BigTable so that there is very little latency and high availability. BigTable is eventually consistent but that doesn't really matter for this application. CloudSQL will

ensure strong consistency which we don't really need but it is slower and supports max 64 TB. The description mentions multiple TBs. Not really sure what several means here, but Cloud SQL doesn't have a high cap.

euro202 1 year, 9 months ago

Selected Answer: C

We need a model that extracts labels from videos, so Vision API could be used. Then we need a DB very fast and that can handle several TB of data, so BigTable is the best choice. Answer is C.

samdhimal 2 years, 2 months ago

Option C is the correct choice because it utilizes the Cloud Video Intelligence API to generate labels for the entities in the videos, which would save time and resources compared to building and training a model from scratch. Additionally, by storing the data in Cloud Bigtable, it allows for fast and efficient filtering of the predicted labels based on the user's viewing history and preferences. This is a more efficient and cost-effective approach than storing the data in Cloud SQL and performing joins and filters.

AzureDP900 2 years, 2 months ago

Answer is C Build an application that calls the Cloud Video Intelligence API to generate labels. Store data in Cloud Bigtable, and filter the predicted labels to match the user's viewing history to generate preferences. 1. Rather than building a new model - it is better to use Google provide APIs, here - Google Video Intelligence. So option A and B rules out 2. Between SQL and Bigtable - Bigtable is the better option as Bigtable support row-key filtering. Joining the filters is not required. Reference: https://cloud.google.com/video-intelligence/docs/feature-label-detection

MaxNRG 3 years, 3 months ago

Selected Answer: C

C. The cloud video intelillence api does the label generation without the need of building any model, A and B are excluded. Now, the bbdd most suitable for this is bigtable and not SQL (this big joins would be anything but fast). https://cloud.google.com/video-intelligence/docs/feature-label-detection

sumanshu 3 years, 9 months ago

Vote for C

timolo 4 years ago

Answer: C Reference https://cloud.google.com/video-intelligence/docs/feature-label-detection

NamitSehgal 4 years, 3 months ago

Answer: C

SureshKotla 4 years, 6 months ago

Answer is D : BigTable doesnt support JOIN and not built for transactions - https://cloud.google.com/bigtable/docs/overview

Surjit24 4 years, 5 months ago

There are no joins but filtering based on condition.

karthik89 4 years, 1 month ago

but the requirement involves join as well, it is stated in the problem.

sumanshu 3 years, 8 months ago

Where? Though it's mention - " very fast filtering suggestions" - which means something like dictionary in python OR Key: Value (which is Bigtable)

sraakesh95 3 years, 2 months ago

I think "based on other customer preferences" from the questions requires a join before a filter is applied for collaborative filtering

Deepakd 2 years, 12 months ago

Recommendation based on other customer”s views cannot be achieved through simple joins. A class pf machine learning algorithms called collaborative filtering is required for that. You need big table to run these algorithms.

haroldbenites 4 years, 7 months ago

Correct C

dg63 4 years, 8 months ago

I doubt if C can be an answer. Will Bigtable allow filtering on labels?

tprashanth 4 years, 8 months ago

Yes, if its part of the rowkey

Rajuuu 4 years, 9 months ago

Answer is C.

Ganshank 4 years, 11 months ago

C. The recommendation requires filtering based on several TB of data, therefore BigTable is the recommended option vs Cloud SQL which is limited to 10TB.

https://zhangxuetu.online

[202610]

Question #68 Topic 1

You are selecting services to write and transform JSON messages from Cloud Pub/Sub to BigQuery for a data pipeline on Google Cloud. You want to minimize service costs. You also want to monitor and accommodate input data volume that will vary in size with minimal manual intervention. What should you do?

A. Use Cloud Dataproc to run your transformations. Monitor CPU utilization for the cluster. Resize the number of worker nodes in your cluster via the command line.

B. Use Cloud Dataproc to run your transformations. Use the diagnose command to generate an operational output archive. Locate the bottleneck and adjust cluster resources.

C. Use Cloud Dataflow to run your transformations. Monitor the job system lag with Stackdriver. Use the default autoscaling setting for worker instances. **Most Voted**

D. Use Cloud Dataflow to run your transformations. Monitor the total execution time for a sampling of jobs. Configure the job to use non-default Compute Engine machine types when needed.

Correct Answer: C

Community vote distribution



Comments

madhu1171 **Highly Voted** 5 years, 6 months ago

Answer should be C

VishalB **Highly Voted** 5 years, 2 months ago

Correct Answer: C Explanation:-This option is correct as Dataflow, provides a cost-effective solution to perform transformations on the streaming data, with autoscaling provides scaling without any intervention. System lag with Stackdriver provides monitoring for the streaming data. With autoscaling enabled, the Cloud Dataflow service automatically chooses the appropriate number of worker instances required to run your job.

Abizi 1 year ago

Selected Answer: C

C for me

yassoraa88 1 year, 4 months ago

Selected Answer: C

using Cloud Dataflow for transformations with monitoring via Stackdriver and leveraging its default autoscaling settings, is the best choice. Cloud Dataflow is purpose-built for this type of workload, providing seamless scalability and efficient processing capabilities for streaming data. Its autoscaling feature minimizes manual intervention and

helps manage costs by dynamically adjusting resources based on the actual processing needs, which is crucial for handling fluctuating data volumes efficiently and cost-effectively.

barnac1es 2 years ago

Selected Answer: D

Option C suggests using Cloud Dataflow to run the transformations and monitoring the job system lag with Stackdriver while using the default autoscaling setting for worker instances. While using Cloud Dataflow is a suitable choice for processing data from Cloud Pub/Sub to BigQuery, and monitoring with Stackdriver provides valuable insights, the specific emphasis on configuring non-default Compute Engine machine types (as mentioned in option D) gives you more control over cost optimization and performance tuning. By configuring non-default machine types, you can precisely tailor the computational resources to match the specific requirements of your workload. This fine-grained control allows you to optimize costs further by avoiding over-provisioning of resources, especially if your workload is memory-intensive, CPU-bound, or requires specific configurations that are not met by the default settings.

barnac1es 2 years ago

Additionally, having the flexibility to adjust machine types based on workload characteristics ensures that you can achieve the desired performance levels without overspending on unnecessary resources. This level of customization is not provided by simply relying on the default autoscaling settings, making option D a more comprehensive and cost-effective solution for managing varying data volumes.

Mathew106 2 years, 2 months ago

Selected Answer: B

At first I answered C. However, Dataproc is indeed cheaper than Dataflow. And both of them can scale automatically horizontically. Dataflow horizontal scaling applies to both primary and secondary nodes. Scaling secondary nodes scales up CPU/compute and scaling primary nodes scales up both memory and CPU/compute. I don't quite understand the second part of answer B where it says I should allocate resources accordingly. I guess I could do that, but auto-scaling should be enough.

AbdullahAnwar 2 years, 7 months ago

Answer should be C

samdhimal 2 years, 8 months ago

C. Use Cloud Dataflow to run your transformations. Monitor the job system lag with Stackdriver. Use the default autoscaling setting for worker instances. Cloud Dataflow is a managed service that allows you to write and execute data transformations in a highly scalable and fault-tolerant way. By default, it will automatically scale the number of worker instances based on the input data volume and job performance, which can help minimize costs. Monitoring the job system lag with Stackdriver can help you identify any issues that may be impacting performance and take action as needed. Additionally, using the default autoscaling setting for worker instances can help you minimize manual intervention and ensure that resources are used efficiently.

odacir 2 years, 9 months ago

Selected Answer: C

@admin why all the answers are wrong. I paid 30 euros for this web and its garbage. Dataproc has no sense in this scenario, because you want to have minimal intervention/operation. D is not a good practice, the answer is C.

medeis_jar 3 years, 9 months ago

Selected Answer: C

C only as referred by MaxNRG

MaxNRG 3 years, 9 months ago

Selected Answer: C

C. Dataproc does not seem to be a good solution in this case as it always require a manual intervention to adjust resources. Autoscaling with dataflow will automatically handle changing data volumes with no manual intervention, and monitoring through Stackdriver can be used to spot bottleneck. Total execution time is not good there as it does not provide a precise view on potential bottleneck.

StefanoG 3 years, 10 months ago

Selected Answer: C

Dataflow, Stackdriver and autoscaling

victorlie 3 years, 10 months ago

Admin, please take a look on the comments. Almost all answers are wrong

nguyenmoon 4 years ago

Answer should be C as dataflow is unpredictable size (input that will vary in size), dataproc is with known size

Tanzu 3 years, 7 months ago

dataflow over dataproc is always the preferred way in gcp. use dataproc only there is specific client requirements such as existing hadoop workloads, etc..

sandipk91 4 years, 1 month ago

Option C is the answer

sumanshu 4 years, 3 months ago

Vote for C

apnu 4 years, 9 months ago

B , it is correct , as it says minimum service cost, dataflow is more expansive than dataproc.

daghayeghi 4 years, 6 months ago

but it said "with minimal manual intervention" and for Dataproc you need to manage cluster manually, then C is the best option.

Believerath 4 years, 5 months ago

You have to transform the JSON messages. Hence, you need to use dataflow.

https://zhangxuetu.online

[202610]

Question #69 Topic 1

Your infrastructure includes a set of YouTube channels. You have been tasked with creating a process for sending the YouTube channel data to Google Cloud for analysis. You want to design a solution that allows your world-wide marketing teams to perform ANSI SQL and other types of analysis on up-to-date YouTube channels log data. How should you set up the log data transfer into Google Cloud?

A. Use Storage Transfer Service to transfer the offsite backup files to a Cloud Storage Multi-Regional storage bucket as a final destination.

B. Use Storage Transfer Service to transfer the offsite backup files to a Cloud Storage Regional bucket as a final destination.

C. Use BigQuery Data Transfer Service to transfer the offsite backup files to a Cloud Storage Multi-Regional storage bucket as a final destination. **Most Voted**

D. Use BigQuery Data Transfer Service to transfer the offsite backup files to a Cloud Storage Regional storage bucket as a final destination.

Correct Answer: C

Community vote distribution



Comments

VishalB **Highly Voted** 5 years, 8 months ago

Correct Answer: A Destination is GCS and having multi-regional so A is the best option available. Even since BigQuery Data Transfer Service initially supports Google application sources like Google Ads, Campaign Manager, Google Ad Manager and YouTube but it does not support destination anything other than bq data set

cloud_rider 1 year, 4 months ago

I differ, destination is not mentioned as GCS, it only says Google Cloud which can mean BigQuery too. And ANSI SQL also points towards this direction.

forepick 7 months, 1 week ago

ANSI SQL as a requirement can be also implemented as the data rests on GCS, and queried in BQ as an external table.

henryCho 5 years, 3 months ago

What about ANSI SQL?

nadavw 3 years, 10 months ago

use external table for it

Jphix 4 years, 10 months ago

I guess they are assuming that you will just query the data in Cloud Storage from BQ. The question specifically is, "How should you set up the log data transfer into Google Cloud?", not "How should you set up the querying." ANSI SQL is a distraction!

tainangao 4 years, 6 months ago

Currently, you cannot use the BigQuery Data Transfer Service to transfer data out of BigQuery.
https://cloud.google.com/bigquery-transfer/docs/introduction

phidelics 2 years, 10 months ago

You can use BQ Data transfer Service for Youtube channels mow

AmmarFasih 2 years, 10 months ago

but I think now you can use BigQuery Data Transfer Service for youtube channels and many other

asksathvik 4 years, 7 months ago

Kindly re-read the question,the question says Google Cloud not Cloud storage...once you master that you will understand the whole question and be able to pick the right answer which is C

yoshik 4 years, 6 months ago

logs like stuff goes better on buckets

triipinbee 4 years, 7 months ago

all the option clearly says "storage bucket", once you master that, you'll realize the correct option is A

Rodrigo4N 2 years, 11 months ago

Gottem!

HarshKothari21 3 years, 6 months ago

good one :)

Ganshank Highly Voted 5 years, 11 months ago

None of the answers make any sense. BigQuery Transfer Service is for moving data from various sources (S3, Youtube etc) into BigQuery, not Google Cloud Storage. Further, how are we supposed to use SQL to query data if it is stored in Cloud Storage?

dambilwa 5 years, 9 months ago

Agreed! - All Options look wrong

dambilwa 5 years, 9 months ago

Option [A] is the least worse option... for world wide teams to perform ANSI SQL Queries, it would be easier to create a ext. table or load from Multi AZ bucket... BQ Data Transfer service is used to push data in BQ, hence ruling out Option C & D

StefanoG 4 years, 6 months ago

The best option would be to use "BigQuery Transfer Service" to upload data to BQ. But BQ is not present as a destination, so the only working option is Multi Regional GCS

TNT87 5 years, 6 months ago

Kindly re-read the question,the question says Google Cloud not Cloud storage...once you master that you will understand the whole question and be able to pick the right answer which is C

TNT87 5 years, 6 months ago

https://cloud.google.com/bigquery-transfer/docs/youtube-channel-transfer this link will help to cement the answer.

1479 7 months, 1 week ago

Selected Answer: C

Therefore, the closest (but still flawed) answer is C or D, but with a crucial correction: The intended solution is to use BigQuery Data Transfer Service to transfer the data directly from YouTube Analytics into BigQuery. The mention of "offsite backup files" is misleading and likely a distractor.

theRafael7 9 months, 4 weeks ago

Selected Answer: C

I will go for C because for A it says cloud storage as the final destination and also bqts can be used for youtube channel data which are structured reports.

abhaya2608 1 year ago

Selected Answer: C

BigQuery Data Transfer Service automates data loading into BigQuery from various sources, including Google Cloud Storage.BigQuery is specifically designed for performing ANSI SQL and other types of analysis on large datasets. Cloud Storage can be used as a staging area, especially a multi-region bucket if the teams are worldwide

oussama7 1 year ago

Selected Answer: C

BigQuery Data Transfer Service (C) is the best choice because it natively integrates with YouTube, automates data transfers, and enables seamless analysis in BigQuery.

cloud_rider 1 year, 4 months ago

Selected Answer: C

C is the correct answer as the requirement is to transfer data to Google Cloud, which means Big Query and not GCS.

SamuelTsch 1 year, 5 months ago

Selected Answer: A

First thing log data is usually semi-strucutred or unstrucutred data, then GCS is more suitable in this situation. Besides, you can use external table or BigLake to run BQ query directly running on GCS.

baimus 1 year, 6 months ago

This is A, because it's "offsite backup files". You can transfer direct from Youtube to BigQuery: https://cloud.google.com/bigquery/docs/youtube-channel-transfer, but this isn't that. It's direct from some backup files to Cloud Storage, all answers mandate that fact, and all that remains is "should this be Storage Transfer, or BigQuery" - clearly this is storage transfer, and all the youtube/ANSI sql stuff is just distraction.

iooj 1 year, 8 months ago

Selected Answer: A

E. Use BigQuery Data Transfer Service to transfer the offsite backup files to BigQuery as a final destination.

ABKR1300 1 year, 10 months ago

Using BQ data transfer we can only load data into the Bigquery storage, which means a table inside the bq dataset. What is Storage Transfer Service? Storage Transfer Service automates the transfer of data to, from, and between object and file storage systems, including Google Cloud Storage, Amazon S3, Azure Storage, on-premises data, and more. It can be used to transfer large amounts of data quickly and reliably, without the need to write any code. As per the above lines from the Google's documentation on Storage transfer service, we can go with option A. For additional info, have a look at the below link. https://cloud.google.com/storage-transfer/docs/overview?_gl=1*jaku2h*_ga*MjA5Mzc4OTM0LjE2ODQ3MzA5NzQ.*_ga_WH2QY8WWF5*MTcxNTA1OTM0OC4xODMuMS4xN

zE1MDU5NTY3LjAuMC4w&_ga=2.6452401.-209378934.1684730974&_gac=1.162721358.1713078448.CjwKCAjw_e2wBhAEEiwAyFFFo4Da6-2MNQqNJuzAGSyJmCXdaPpXXiqaI0lkZYHlcln0IBbtWSJjLBoCp_4QAvD_BwE

zevexWM 1 year, 11 months ago

Selected Answer: A

Correct answer: A Solution should cater a worldwide solution which makes B and D invalid. You don't use BigQuery Data transfer to move data to a bucket. So C is also invalid.

AshishDhamu 2 years ago

Selected Answer: A

BigQuery Data Transfer Service can only transfer to BigQuery not in Cloud Storage. So Ans A is correct.

demoro86 2 years ago

Selected Answer: A

No sense to use BQ data transfer serice to store the data to a storage bucket ... It is obviously A

rocky48 2 years, 4 months ago

Selected Answer: C

To transfer YouTube channel data to Google Cloud for analysis, you can use the BigQuery Data Transfer Service to transfer the offsite backup files to a Cloud Storage Multi-Regional storage bucket as a final destination 1. This service allows you to automatically schedule and manage recurring load jobs for YouTube Channel reports 1. The BigQuery Data Transfer Service for YouTube Channel reports supports the following reporting options: Channel Reports (automatically loaded into BigQuery) 1. When you transfer data from a YouTube Channel into BigQuery, the data is loaded into BigQuery tables that are partitioned by date 1.

ziyunxiao 2 years, 5 months ago

The correct answer is C. https://cloud.google.com/bigquery/docs/dts-introduction

exnaniantwort 2 years, 6 months ago

not c The BigQuery Data Transfer Service automates data movement into [[[[BigQuery]]]] on a scheduled, managed basis. Your analytics team can lay the foundation for a BigQuery data warehouse without writing a single line of code. ... After you configure a data transfer, the BigQuery Data Transfer Service automatically loads data into [[[[[BigQuery]]]]] on a regular basis.

https://zhangxuetu.online

[202610]

Question #70 Topic 1

You are designing storage for very large text files for a data pipeline on Google Cloud. You want to support ANSI SQL queries. You also want to support compression and parallel load from the input locations using Google recommended practices. What should you do?

A. Transform text files to compressed Avro using Cloud Dataflow. Use BigQuery for storage and query.

B. Transform text files to compressed Avro using Cloud Dataflow. Use Cloud Storage and BigQuery permanent linked tables for query. **Most Voted**

C. Compress text files to gzip using the Grid Computing Tools. Use BigQuery for storage and query.

D. Compress text files to gzip using the Grid Computing Tools. Use Cloud Storage, and then import into Cloud Bigtable for query.

Correct Answer: B

Community vote distribution



Comments

Ganshank **Highly Voted** 5 years, 11 months ago

B. The question is focused on designing storage for very large files, with support for compression, ANSI SQL queries, and parallel loading from the input locations. This can be met using GCS for storage and Bigquery permanent tables with external data source in GCS.

atnafu2020 5 years, 8 months ago

why GCS as external since Bigquery can be used as storage as well?

atnafu2020 5 years, 8 months ago

A seems correct for me

atnafu2020 5 years, 7 months ago

Since its best practice, i go by with B not A

gopinath_k 5 years ago

They want to store the files if you try with bq I think you will need to strike the word compression.

jkhong 3 years, 3 months ago

The question focuses on "designing storage", rather than designing a data warehouse.

[Removed] **Highly Voted** 6 years ago

Should be A

tavva_prudhvi 3 years, 9 months ago

Not A : Importing data into BigQuery may take more time compared to creating external tables on data. Additional storage costs by BigQuery is another issue which can be more expensive than Google Storage.

50336e5 3 months, 2 weeks ago

Selected Answer: A

It's not B because Permanent linked tables it's not recommanded for a large data

forepick 7 months, 1 week ago

Selected Answer: B

The task here is to design a storage for very big FILES, not tables. And only then, these big files should be queries. So it's B.

AdriHubert 10 months, 2 weeks ago

Selected Answer: A

Here's why: Cloud Dataflow is a fully managed service for stream and batch data processing. It’s ideal for transforming large text files into a more efficient format like Avro, which supports schema evolution and is optimized for BigQuery ingestion. Avro is a row-based storage format that supports compression and is well-suited for BigQuery. BigQuery is Google Cloud’s serverless, highly scalable, and cost-effective multi-cloud data warehouse that supports ANSI SQL. Using compressed Avro allows for parallel loading into BigQuery, which is a Google-recommended best practice for performance and cost-efficiency. Why not the others? B: Using Cloud Storage with permanent external tables in BigQuery is possible, but it’s less performant and flexible than loading the data into native BigQuery storage. C: Gzip-compressed text files are not as efficient for parallel processing or schema enforcement as Avro. D: Cloud Bigtable is not designed for SQL queries; it’s a NoSQL wide-column store, and not suitable for ANSI SQL workloads.

oussama7 1 year ago

Selected Answer: A

Avro is a Google-recommended format for BigQuery because it supports schema evolution, efficient compression, and parallel processing. Using Cloud Dataflow ensures scalable transformation, and storing the data in BigQuery allows for optimized ANSI SQL queries.

Parandhaman_Margan 1 year ago

Selected Answer: A

Storing large text files with SQL support** Avro in BigQuery (A) supports compression and efficient queries.

dcruzado 1 year ago

Selected Answer: B

For me it would make more sense to store the data directly in BQ, but A does not make sense, because why compress to Avro if you dont store the avro files and directly saves the data to BQ? You are not using the compression

imarri876 1 year, 1 month ago

Selected Answer: A

BigQuery now has physical storage, which makes storage cost fairly cheap on BigQuery with compression. I would go with A.

deineiveu 1 year, 3 months ago

Selected Answer: B

Gros fichier + sql = GCS + Bigquery

Nittin 1 year, 7 months ago

Selected Answer: B

copy to gcs and use external tble in bq

carmltekai 1 year, 8 months ago

Selected Answer: A

Should be A. Check this link for the advantage of load Avro data to BigQuery https://cloud.google.com/bigquery/docs/loading-data-cloud-storage-avro#advantages_of_avro """The Avro binary format: * Is faster to load. The data can be read in parallel, even if the data blocks are compressed. * Doesn't require typing or serialization. * Is easier to parse because there are no encoding issues found in other formats such as ASCII. When you load Avro files into BigQuery, the table schema is automatically retrieved from the self-describing source data."""

carmltekai 1 year, 8 months ago

While option B can work, it introduces additional complexity by linking Cloud Storage with BigQuery. Directly storing data in BigQuery is more efficient for querying purposes. There are no requirements about cost, So simple is better

SK1594 2 years ago

B makes sense

MaxNRG 2 years, 3 months ago

Selected Answer: B

1. Store Avro files in GCS 2. Query them in BigQuery (federated tables)

forepick 2 years, 10 months ago

Selected Answer: B

Answer is B. The requirements are: - storage for compressed text files - parallel loads to SQL tool AVRO is a compressed format for text files, which makes it possible to load chunks of a very large file in parallel to BigQuery. gzip files are seamless in GCS though, but cannot load in parallel to BQ.

samdhimal 3 years, 2 months ago

Correct Answer: A. Transform text files to compressed Avro using Cloud Dataflow. Use BigQuery for storage and query. This option offers several advantages: - Transforming the text files to compressed Avro using Cloud Dataflow allows for parallel processing of the input data, improving the efficiency of the pipeline. - Compressing the data in Avro format further reduces the storage space required and improves data transfer performance. - Storing the data in BigQuery supports ANSI SQL queries and allows for easy querying of the data. - BigQuery is a fully-managed data warehousing solution, it's scalable and can handle large datasets and concurrent queries, so it's suitable for large text files.

samdhimal 3 years, 2 months ago

Option B is similar to option A but it's using a permanent linked table between Cloud Storage and BigQuery, this approach is not recommended as it's not efficient and could lead to data duplication, and it doesn't take advantage of the parallel processing capabilities of Cloud Dataflow. Option C and D are incorrect because they don't take advantage of the parallel processing capabilities of Cloud Dataflow, and they don't use Avro format for compression which is more efficient and recommended by Google. Storing the data in Cloud Bigtable also doesn't support ANSI SQL queries which is a requirement for this use case.

jkhong 3 years, 3 months ago

Selected Answer: B

Designing storage solution, not data warehousing -> So Cloud Storage. Support compression -> just use Avro Parallel load -> refers to upload from input locations, NOT download. Load in parallel using -m flag for gsutil cp

https://cloud.google.com/storage/docs/uploads-downloads#console

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #71 Topic 1

You are developing an application on Google Cloud that will automatically generate subject labels for users' blog posts. You are under competitive pressure to add this feature quickly, and you have no additional developer resources. No one on your team has experience with machine learning. What should you do?

A. Call the Cloud Natural Language API from your application. Process the generated Entity Analysis as labels. **Most Voted**

B. Call the Cloud Natural Language API from your application. Process the generated Sentiment Analysis as labels.

C. Build and train a text classification model using TensorFlow. Deploy the model using Cloud Machine Learning Engine. Call the model from your application and process the results as labels.

D. Build and train a text classification model using TensorFlow. Deploy the model using a Kubernetes Engine cluster. Call the model from your application and process the results as labels.

Correct Answer: A

Community vote distribution

A (100%)

Comments

VishalB **Highly Voted** 4 years, 8 months ago

Correct Answer : A Entity analysis -> Identify entities within documents receipts, invoices, and contracts and label them by types such as date, person, contact information, organization, location, events, products, and media. Sentiment analysis -> Understand the overall opinion, feeling, or attitude sentiment expressed in a block of text. -- Avoid Custom models

AzureDP900 2 years, 2 months ago

https://cloud.google.com/natural-language/docs/analyzing-entities https://cloud.google.com/natural-language/docs/analyzing-sentiment

Ratikl 1 year ago

Selected Answer: A

Call the Cloud Natural Language API from your application. Process the generated Entities Analysis as labels. The Cloud Natural Language API is a pre-trained machine learning model that can be used for natural language processing tasks such as entity recognition, sentiment analysis, and syntax analysis. The API can be called from your application using a simple API call, and it can generate entities analysis that can be used as labels for the user's blog posts. This would be the quickest and easiest option for your team since it would not require any machine learning

expertise or additional developer resources to build and train a model. Additionally, it will give you accurate and up-to-date results as the API is constantly updated by Google.

Az900Exam2021 1 year, 6 months ago

For the first time, the answer in exam topics matches community vote :-).

AmmarFasih 1 year, 10 months ago

Selected Answer: A

Of course the answer is A. Since the problem already states that you don't have time, resources or expertise. So the best solution in the case is to utilize the available API. Also since we need to extract the labels and not the sentiment of the text, we'll go for option A and not B

samdhimal 2 years, 2 months ago

A. Call the Cloud Natural Language API from your application. Process the generated Entities Analysis as labels. The Cloud Natural Language API is a pre-trained machine learning model that can be used for natural language processing tasks such as entity recognition, sentiment analysis, and syntax analysis. The API can be called from your application using a simple API call, and it can generate entities analysis that can be used as labels for the user's blog posts. This would be the quickest and easiest option for your team since it would not require any machine learning expertise or additional developer resources to build and train a model. Additionally, it will give you accurate and up-to-date results as the API is constantly updated by Google.

AzureDP900 2 years, 2 months ago

Answer is A Call the Cloud Natural Language API from your application. Process the generated Entity Analysis as labels. Entity analysis -> Identify entities within documents receipts, invoices, and contracts and label them by types such as date, person, contact information, organization, location, events, products, and media. Sentiment analysis -> Understand the overall opinion, feeling, or attitude sentiment expressed in a block of text.

NicolasN 2 years, 4 months ago

Apparently, there is unanimity on answer [A] What if there was another available answer in an actual exam? E. Call the Cloud Natural Language API from your application. Process the generated Content Classification as labels What would you choose, A or E? My opinion is that Content Classification is more suitable for detecting subject.

Remi2021 2 years, 8 months ago

A is the right one . Doc says: Entity analysis inspects the given text for known entities (Proper nouns such as public figures, landmarks, and so on. Common nouns such as restaurant, stadium, and so on.) and returns information about those entities. Entity analysis is performed with the analyzeEntities method.

waterh2oeau 2 years, 10 months ago

Selected Answer: A

Vote for A

bury 3 years, 1 month ago

Selected Answer: A

a is correct

JayZeeLee 3 years, 4 months ago

A. CD don't work as it requires Machine Learning experience. B - Sentiment Analysis is to analyze attitude, opinion, etc. So A.

sumanshu 3 years, 9 months ago

Vote for A

haroldbenites 4 years, 7 months ago

A is correct

https://zhangxuetu.online

[202610]

Question #72 Topic 1

You are designing storage for 20 TB of text files as part of deploying a data pipeline on Google Cloud. Your input data is in CSV format. You want to minimize the cost of querying aggregate values for multiple users who will query the data in Cloud Storage with multiple engines. Which storage service and schema design should you use?

A. Use Cloud Bigtable for storage. Install the HBase shell on a Compute Engine instance to query the Cloud Bigtable data.

B. Use Cloud Bigtable for storage. Link as permanent tables in BigQuery for query.

C. Use Cloud Storage for storage. Link as permanent tables in BigQuery for query. **Most Voted**

D. Use Cloud Storage for storage. Link as temporary tables in BigQuery for query.

Correct Answer: C

Community vote distribution

C (100%)

Comments

daghayeghi **Highly Voted** 4 years ago

answer C: BigQuery can access data in external sources, known as federated sources. Instead of first loading data into BigQuery, you can create a reference to an external source. External sources can be Cloud Bigtable, Cloud Storage, and Google Drive. When accessing external data, you can create either permanent or temporary external tables. Permanent tables are those that are created in a dataset and linked to an external source. Dataset-level access controls can be applied to these tables. When you are using a temporary table, a table is created in a special dataset and will be available for approxi- mately 24 hours. Temporary tables are useful for one-time operations, such as loading data into a data warehouse. "Dan Sullivan" Book

rickywck **Highly Voted** 5 years ago

Why not C?

emmylou 1 year, 5 months ago

On so many of these questions, how do you actually know if you're correct. I said C but the correct answer was A. Honestly, it's driving me crazy.

Mathew106 1 year, 8 months ago

Selected Answer: C

For the ones saying BigTable is cheaper, BigTable in eu-north1 costs $0.748/hour per node. So if you were to run the node 24/7 you would pay more than 500$ per month. Querying 1TB of data in BigQuery is 7.5$. With smart querying and good database design you can minimize the bytes processed by BQ. So even though BigTable does not directly

charge for querying, it charges for running the cluster and the overall price does not make sense. And as far as I know, it's not possible to spin up and shut down BigTable automatically. Also, since the table is an external table to BigQuery, we incur no cost for storing that data in BigQuery and paying 300$ per month for storage.

samdhimal 2 years, 2 months ago

C. Use Cloud Storage for storage. Link as permanent tables in BigQuery for query. Cloud Storage is a highly durable and cost-effective object storage service that can be used to store large amounts of text files. By storing the input data in CSV format in Cloud Storage, you can minimize costs while still being able to query the data using BigQuery. BigQuery is a fully-managed, highly-scalable data warehouse that allows you to perform fast SQL-like queries on large datasets. By linking the Cloud Storage data as permanent tables in BigQuery, you can enable multiple users to query the data using multiple engines without the need for additional compute resources. This approach would be the most cost-effective for querying aggregate values for multiple users, as BigQuery charges based on the amount of data scanned per query, so the more data you store in BigQuery the less you pay per query.

samdhimal 2 years, 2 months ago

Option D, using Cloud Storage for storage and linking as temporary tables in BigQuery for query, would not be the best choice because temporary tables only exist for the duration of a user session or query and you would need to create and delete them each time a user queries the data, which would add additional cost and complexity to the process. Option A, Using Cloud Bigtable for storage, and installing the HBase shell on a Compute Engine instance to query the data, is not a cost-effective solution as Cloud Bigtable is a managed NoSQL database service which is more expensive than storing in Cloud Storage and querying in BigQuery. Option B, Using Cloud Bigtable for storage, and linking as permanent tables in BigQuery for query, is not a cost-effective solution as Cloud Bigtable is a managed NoSQL database service which is more expensive than storing in Cloud Storage and querying in BigQuery.

RoshanAshraf 2 years, 2 months ago

Selected Answer: C

CSV files - Cloud Storage BigQuery - Aggregate, multiple users Permanent table - multiple users External Tables is Easy to implement, cost effective

rivua 2 years, 3 months ago

The 'correct' answers on this platform are ridiculous

VishalBule 3 years, 1 month ago

Answer is C Use Cloud Storage for storage. Link as permanent tables in BigQuery for query. BigQuery can access data in external sources, known as federated sources. Instead of first loading data into BigQuery, you can create a reference to an external source. External sources can be Cloud Bigtable, Cloud Storage, and Google Drive. When accessing external data, you can create either permanent or temporary external tables. Permanent tables are those that are created in a dataset and linked to an external source. Dataset-level access controls can be applied to these tables. When you are using a temporary table, a table is created in a special dataset and will be available for approxi- mately 24 hours. Temporary tables are useful for one-time operations, such as loading data into a data warehouse

medeis_jar 3 years, 2 months ago

Selected Answer: C

Bigtable is expensive. So Cloud Storage for storing and BigQuery with permanent table for linking and querying.

MaxNRG 3 years, 3 months ago

Selected Answer: C

Not A or B Big table is expensive, que initial data is in csv format, besides, if others are going to query data with multiple engines… GCS is the storage. Between c and D is all about permanent or temorary. Permanent table is a table that is created in a dataset and is linked to your external data source. Because the table is permanent, you can use dataset-level access controls to share the table with others who also have access to the underlying external data

source, and you can query the table at any time. When you use a temporary table, you do not create a table in one of your BigQuery datasets. Because the table is not permanently stored in a dataset, it cannot be shared with others. Querying an external data source using a temporary table is useful for one-time, ad-hoc queries over external data, or for extract, transform, and load (ETL) processes. I think is C.

MaxNRG 3 years, 3 months ago

https://cloud.google.com/blog/products/gcp/accessing-external-federated-data-sources-with-bigquerys-data-access-layer. Permanent table—You create a table in a BigQuery dataset that is linked to your external data source. This allows you to use BigQuery dataset-level IAM roles to share the table with others who may have access to the underlying external data source. Use permanent tables when you need to share the table with others. Temporary table—You submit a command that includes a query and creates a non-permanent table linked to the external data source. With this approach you do not create a table in one of your BigQuery datasets, so make sure to give consideration towards sharing the query or table. Consider using a temporary table for one-time, ad-hoc queries, or for one time extract, transform, or load (ETL) workflows

maurodipa 3 years, 4 months ago

Answer is A. While C seems the most reasonable answer there are 2 points to notics: a) load jobs are limited to 15 TB across all input files in BigQuery (https://cloud.google.com/bigquery/quotas); b) It is requested to minimize the cost of querying and queries in BigTable are free, while queries in BigQuery are charged per byte (https://cloud.google.com/bigquery/pricing)

Abhi16820 3 years, 4 months ago

https://cloud.google.com/bigquery/external-data-bigtable#:~:text=shared%20with%20others.-,Querying%20an%20external%20data%20source%20using%20a%20temporary%20table%20is%20useful%20for%20one%2Dtime%2C%20ad%2Dhoc%20queries%20over%20external%20data%2C%20or%20for%20extract%2C%20transform%2C%20and%20load%20(ETL)%20processes.,-Querying%20Cloud%20Bigtable

tsoetan001 3 years, 5 months ago

C is the answer.

Ysance_AGS 3 years, 6 months ago

A is correct since the question asks "You want to minimize the cost of querying aggregate values" => Big Table is free when querying data.

nguyenmoon 3 years, 6 months ago

Vote for C

gcp_learner 3 years, 8 months ago

Interesting options. For me, A & B ruled out because BigTable doesn't fit this use case, leaves us with C & D. C will incur additional cost of storing data in GCS & BigQuery because it mentions linking. So I would go with D ie store the data in GCS and create external tables in BigQuery.

Yiouk 3 years, 8 months ago

https://cloud.google.com/bigquery/docs/writing-results#temporary_and_permanent_tables

triipinbee 3 years, 7 months ago

Storage cost of data for BQ is the same as standard cloud storage, actually less for long term storage as it automatically moves to nearline storage. https://cloud.google.com/bigquery/pricing#storage https://cloud.google.com/storage#section-10

sumanshu 3 years, 9 months ago

Vote for C

sumanshu 3 years, 8 months ago

Vote for 'D' BigQuery uses temporary tables to cache query results. https://cloud.google.com/bigquery/docs/writing-results

sumanshu 3 years, 8 months ago

There are no storage costs for temporary tables, but if you write query results to a permanent table, you are charged for storing the data.

triipinbee 3 years, 7 months ago

even if it's a temp table, the storage cost for active storage in cloud bucket and the standard storage in BQ is the same. So no point creating temp tables.

https://zhangxuetu.online

[202610]

Question #73 Topic 1

You are designing storage for two relational tables that are part of a 10-TB database on Google Cloud. You want to support transactions that scale horizontally.
You also want to optimize data for range queries on non-key columns. What should you do?

A. Use Cloud SQL for storage. Add secondary indexes to support query patterns.

B. Use Cloud SQL for storage. Use Cloud Dataflow to transform data to support query patterns.

C. Use Cloud Spanner for storage. Add secondary indexes to support query patterns. **Most Voted**

D. Use Cloud Spanner for storage. Use Cloud Dataflow to transform data to support query patterns.

Correct Answer: C

Community vote distribution

C (100%)

Comments

nhanhoangle 1 year, 11 months ago

Selected Answer: C

Correct: C

PolyMoe 2 years, 2 months ago

Selected Answer: C

Cloud Spanner is a fully-managed, horizontally scalable relational database service that supports transactions and allows you to optimize data for range queries on non-key columns. By using Cloud Spanner for storage, you can ensure that your database can scale horizontally to meet the needs of your application. To optimize data for range queries on non-key columns, you can add secondary indexes, this will allow you to perform range scans on non-key columns, which can improve the performance of queries that filter on non-key columns.

samdhimal 2 years, 2 months ago

C. Use Cloud Spanner for storage. Add secondary indexes to support query patterns. Cloud Spanner is a fully-managed, horizontally scalable relational database service that supports transactions and allows you to optimize data for range queries on non-key columns. By using Cloud Spanner for storage, you can ensure that your database can scale horizontally to meet the needs of your application. To optimize data for range queries on non-key columns, you can add secondary indexes, this will allow you to perform range scans on non-key columns, which can improve the performance of queries that filter on non-key columns.

samdhimal 2 years, 2 months ago

- Option A, Using Cloud SQL for storage and adding secondary indexes to support query patterns, may not be the best option as Cloud SQL is a relational database service that does not support horizontal scaling and may not be

able to handle the large amount of data and the number of queries required by your application.

Mathew106 1 year, 8 months ago

Cloud SQL does support replicas to increase availability. Why is that not considered horizontal scaling?

samdhimal 2 years, 2 months ago

- Option B, Using Cloud SQL for storage and using Cloud Dataflow to transform data to support query patterns, may not be the best option as Cloud SQL is a relational database service that does not support horizontal scaling and may not be able to handle the large amount of data and the number of queries required by your application. Additionally, Cloud Dataflow is a data processing service and not a storage service, so it may not be the best fit for this use case. - Option D, Using Cloud Spanner for storage and using Cloud Dataflow to transform data to support query patterns, is not necessary as Cloud Spanner provides the ability to optimize data for range queries on non-key columns by adding secondary indexes. Cloud Spanner also supports transactional consistency, which is a feature that allows you to perform multiple operations that must be performed together in a single transaction. Additionally, Cloud Dataflow is a data processing service and not a storage service, so it may not be the best fit for this use case.

sedado77 2 years, 6 months ago

Selected Answer: C

As sumanshu said

tsoetan001 3 years, 5 months ago

Answer: C

sumanshu 3 years, 9 months ago

Vote for C

sumanshu 3 years, 8 months ago

A is not correct because Cloud SQL does not natively scale horizontally. B is not correct because Cloud SQL does not natively scale horizontally. C is correct because Cloud Spanner scales horizontally, and you can create secondary indexes for the range queries that are required. D is not correct because Dataflow is a data pipelining tool to move and transform data, but the use case is centered around querying.

timolo 4 years ago

Answer: C https://cloud.google.com/spanner/docs/secondary-indexes

Nileshk611 4 years, 3 months ago

Correct: C

arghya13 4 years, 4 months ago

Correct answers is C

https://zhangxuetu.online

[202610]

Question #74 Topic 1

Your financial services company is moving to cloud technology and wants to store 50 TB of financial time-series data in the cloud. This data is updated frequently and new data will be streaming in all the time. Your company also wants to move their existing Apache Hadoop jobs to the cloud to get insights into this data.
Which product should they use to store the data?

A. Cloud Bigtable **Most Voted**

B. Google BigQuery

C. Google Cloud Storage

D. Google Cloud Datastore

Correct Answer: A

Community vote distribution



Comments

philli1011 2 years, 1 month ago

Every time you hear financial, time series, fast reads and write data, Any of that combinations, think Big Table first. So A.

Mathew106 2 years, 8 months ago

Selected Answer: A

At first I thought that GCS was the answer but the question does mention that the data is updated frequently. Thereby, it has to be BigTable since we talk about a large amount of data, a streaming application and many individual updates. Storing the data in BigQuery and having to make individual updates doesn't make sense, and neither does running Apache jobs. If the requirement for updates was not there I would not see any issue with GCS. GCS could serve as a replacement to HDFS and run Hadoop jobs from Dataproc.

KC_go_reply 2 years, 9 months ago

Selected Answer: A

This scenario screams for BigTable. It's not B) BigQuery or C) Cloud Storage because both aren't supposed to contain data that is updated frequently. Then, we have to decide between A) BigTable and D) Datastore. It is A) BigTable because - it is the most suited for real-time / high-frequency updates - it is similar to HBase, which is commonly used in Hadoop ecosystem stacks to store streaming / time-series data.

AmmarFasih 2 years, 10 months ago

Selected Answer: A

Many here also selected Cloud Storage. But the way I see it BigTable is specifically for low latency, high throughput, mission critical streaming data (financial data is one of them). Also the mentioning of Hadoop that points to HBase functionality if BigTable clarifies the choice more.

Hisayuki 2 years, 11 months ago

Selected Answer: A

BigTable - a No-SQL database but does not support SQL Querying Apache HBase - Based on Google's BigTable on top of HDFS and you can migrate Hadoop Apps to Cloud BigTable with the HBase API

izekc 2 years, 11 months ago

Selected Answer: A

A. time series data

midgoo 3 years, 1 month ago

Selected Answer: A

Please note that there is Connector for Bigtable for Hadoop
https://cloud.google.com/dataproc/docs/concepts/connectors/cloud-bigtable

samdhimal 3 years, 2 months ago

Why not Biquery? Google BigQuery would be the best option for storing and analyzing large amounts of financial time-series data that is frequently updated and streamed in real-time. It is a fully managed, cloud-native data warehouse that allows you to analyze large datasets using SQL-like queries, and it can handle streaming data as well as batch data. Additionally, it can easily integrate with Apache Hadoop to allow your company to run their existing Hadoop jobs in the cloud and gain insights into the data.

samdhimal 3 years, 2 months ago

A. Google Bigtable is a fully managed, NoSQL, wide-column database that is designed for large scale, low-latency workloads. It is well suited for use cases such as real-time analytics, IoT, and gaming, but it may not be the best fit for storing and analyzing large amounts of financial time-series data that is frequently updated and streamed in real-time. It lacks built-in support for SQL-like queries, which is a standard way of analyzing data in Data Warehousing and Business Intelligence. It is more focused on handling high-performance low-latency workloads, while BigQuery is focused on providing an easy and cost-effective way to analyze large amounts of data using SQL-like queries. Additionally, Bigtable doesn't provide built-in support for running Apache Hadoop jobs, and it would require additional work to integrate it with Hadoop and set it up for data warehousing and Business Intelligence use cases.

samdhimal 3 years, 2 months ago

C. Google Cloud Storage is an object storage service that allows you to store and retrieve large amounts of unstructured data, such as video, audio, images and other files. It is not a data warehouse and does not provide built-in support for SQL-like queries, which is a standard way of analyzing data in Data Warehousing and Business Intelligence. It would not be suitable for storing and analyzing large amounts of financial time-series data that is frequently updated and streamed in real-time. D. Google Cloud Datastore is a fully-managed, NoSQL document database that allows you to store, retrieve, and query data. It is not a data warehouse and does not provide built-in support for SQL-like queries, which is a standard way of analyzing data in Data Warehousing and Business Intelligence. It would not be suitable for storing and analyzing large amounts of financial time-series data that is frequently updated and streamed in real-time.

samdhimal 3 years, 2 months ago

Can someone clarify why Bigtable and Not Bigquery? Super Confused.

Oleksandr0501 2 years, 11 months ago

Yes, it is possible to analyze data in Bigtable. Bigtable is a distributed NoSQL database that is designed to handle large volumes of structured data with high read and write throughput. While Bigtable itself does not provide analysis tools, it is often used in combination with other tools and technologies to perform analysis on the stored data.

desertlotus1211 3 years, 2 months ago

https://cloud.google.com/bigtable/docs/schema-design-time-series

Yazar97 3 years, 4 months ago

Time series data = Bigtable... So it's A

Jay_Krish 3 years, 4 months ago

Selected Answer: A

Option A seems right

drunk_goat82 3 years, 4 months ago

Selected Answer: A

Big Table has a HBase compliant API and is transactional unlike GCS.

solar_maker 3 years, 4 months ago

Selected Answer: A

BigTable can take in data from dataproc, spark and hadoop
https://cloud.google.com/dataproc/docs/concepts/connectors/cloud-bigtable#using_with

cloudmon 3 years, 4 months ago

Selected Answer: C

It must be C because of the existing Hadoop jobs

cloudmon 3 years, 4 months ago

On 2nd thought, it's Bigtable: https://cloud.google.com/dataproc/docs/concepts/connectors/cloud-bigtable

pluiedust 3 years, 5 months ago

Selected Answer: C

I think it is C

maia01 3 years, 6 months ago

Selected Answer: C

Use Datarproc with Cloud Storage in combo with HDFS https://cloud.google.com/dataproc/docs/concepts/dataproc-hdfs

euro202 2 years, 8 months ago

Answer is A: Hadoop doesn't mean Dataproc + HDFS. This scenario is about time series that is a use-case for BigTable. Coincidentally BigTable is the best solution for migration of HBase...

AWSandeep 3 years, 7 months ago

Selected Answer: A

A. Cloud Bigtable

https://zhangxuetu.online

[202610]

Question #75 Topic 1

An organization maintains a Google BigQuery dataset that contains tables with user-level data. They want to expose aggregates of this data to other Google
Cloud projects, while still controlling access to the user-level data. Additionally, they need to minimize their overall storage cost and ensure the analysis cost for other projects is assigned to those projects. What should they do?

A. Create and share an authorized view that provides the aggregate results. **Most Voted**

B. Create and share a new dataset and view that provides the aggregate results.

C. Create and share a new dataset and table that contains the aggregate results.

D. Create dataViewer Identity and Access Management (IAM) roles on the dataset to enable sharing.

Correct Answer: A

Community vote distribution



Comments

midgoo **Highly Voted** 2 years, 1 month ago

Selected Answer: A

A is the answer. Don't be confused by the documentation saying "Authorized views should be created in a different dataset". It is a best practice but not a technical requirement. And we don't create a new dataset for each authorized view. If you are not clear on this, try in the system, don't just read the documentation without understanding. B is wrong when saying we must SHARE Dataset. Although creating a dataset and view in it will not incur extra cost, but sharing dataset is something we always try not to do. At for the project that run the query it the project to be billed, that is standard behaviour. View only give access to data, whoever run the view will need pay for the query cost

DAYAGOWDA 2 years ago

https://cloud.google.com/bigquery/docs/authorized-views#:~:text=An%20authorized%20view%20and%20authorized,users%20are%20able%20to%20query.

Yiouk 1 year, 8 months ago

Have to consider where the billing goes to: https://stackoverflow.com/questions/52201034/bigquery-authorized-view-cost-billing-account hence anwer is B

Mathew106 1 year, 8 months ago

Did you even read the answer in the SO link you shared? Part of the answer is below: """After a deeper investigation and some test scenarios, I have confirmed that the billing charges related to the query jobs are applied to the Billing account associated to the project that executes the query; however, the view owner keeps getting the charges related

to the storage of the source data."""" Soo, if you create an authorized view, the users from the other project that has access to the view will get billed for the querying. The only reason to pick up B over A is that it's the recommended approach to store views in a different dataset than the base data.

Gudwin Highly Voted 2 years, 4 months ago

That's ambiguous. While A is correct, B is the recommended approach: "Authorized views should be created in a different dataset from the source data. That way, data owners can give users access to the authorized view without simultaneously granting access to the underlying data. The source data dataset and authorized view dataset must be in the same regional location." Bit it doesn't say "authorised view" in B.

Wonka87 2 years, 3 months ago

But the wording of option B says create and share a new dataset, do you also need to share dataset apart from authorized view access? In option A, isn't is implicit that authorized view is created on a new dataset and hence option A. B also doesn't mention about Authorized keyword so you may interpret it as normal view which doesn't make sense?

desertlotus1211 1 year ago

Selected Answer: A

When other projects query the authorized view, the query costs are billed to the project that runs the query

loki82 1 year, 2 months ago

Selected Answer: B

I think both A and B are totally valid answers, making this a fairly stupid question. But as a DBA, B would be easier to implement, easier to manage, easier to audit. So even if it's the wrong answer, it's still the right solution, so I can't help but choose B.

desertlotus1211 1 year ago

this approach typically involves duplicating data or managing a separate dataset that isn't as tightly controlled, Answer A is better...

inamm 1 year, 3 months ago

Selected Answer: B

B is the correct Ans You are correct in noting that creating a view within an existing dataset could potentially expose other tables within that dataset if the dataset-level permissions are not carefully managed. To ensure that only the aggregate results are shared and to avoid inadvertently exposing other tables, it is indeed a good practice to create a new dataset specifically for the view. Revised Answer: B. Create and share a new dataset and view that provides the aggregate results.

rocky48 1 year, 4 months ago

Selected Answer: A

A. Create and share an authorized view that provides the aggregate results. An authorized view is a BigQuery feature that allows you to share only a specific subset of data from a table, while still keeping the original data private. This way, the organization can expose only the aggregate data to other projects, while still controlling access to the user-level data. By using an authorized view, the organization can minimize their overall storage cost as the aggregate data takes up less storage space than the original data. Additionally, by using authorized view, the analysis cost for other projects is assigned to those projects.

odiez3 1 year, 8 months ago

I thing that is B because for security yo need to create a new data set when share a view, apart when yo grant access the top level is a data set if you share a view in same dataset that you have your tables, that access can see all tables inside dataset.

baht 1 year, 9 months ago

Selected Answer: B

"Authorized views should be created in a different dataset from the source data. That way, data owners can give users access to the authorized view without simultaneously granting access to the underlying data." https://cloud.google.com/bigquery/docs/share-access-views?hl=en#console_5

MrMone 2 years ago

Selected Answer: A

"they need to minimize their overall storage cost". Also, you are sharing the aggregate's results, not the underlying table

bha11111 2 years ago

Selected Answer: A

minimize cost so view

Paritosh07 2 years, 1 month ago

Selected Answer: A

A should be the answer, as we need to separate costs according to projects. As in the following SO question (and the attached google resources), the 'project that runs the queries is the project that gets billed.' So we can generate a view and give it's access to the other project to run the analysis https://stackoverflow.com/questions/52201034/bigquery-authorized-view-cost-billing-account

musumusu 2 years, 1 month ago

I will go with A, as i wanna save cost, dont need to create separate dataset for permanent storage.

samdhimal 2 years, 2 months ago

A. Create and share an authorized view that provides the aggregate results. An authorized view is a BigQuery feature that allows you to share only a specific subset of data from a table, while still keeping the original data private. This way, the organization can expose only the aggregate data to other projects, while still controlling access to the user-level data. By using an authorized view, the organization can minimize their overall storage cost as the aggregate data takes up less storage space than the original data. Additionally, by using authorized view, the analysis cost for other projects is assigned to those projects.

samdhimal 2 years, 2 months ago

B. Creating and sharing a new dataset and view that provides the aggregate results is also a correct option but not as optimal as authorized view as it creates a copy of the data and increases the storage costs. C. Creating and sharing a new dataset and table that contains the aggregate results is also a correct option but not as optimal as authorized view as it creates a copy of the data and increases the storage costs. D. Creating dataViewer Identity and Access Management (IAM) roles on the dataset to enable sharing is not the best option as it would give access to the user-level data, not just the aggregate data.

Rupendra06 2 years, 2 months ago

Selected Answer: B

Ensure the analysis cost for other projects is assigned to those projects indicates B is the correct answer.

GCPpro 2 years, 2 months ago

B is the correct answer

Kyr0 2 years, 3 months ago

Selected Answer: A

I would say 1 too

slade_wilson 2 years, 3 months ago

Selected Answer: B

B is the correct approach.

https://zhangxuetu.online

[202610]

Question #76 Topic 1

Government regulations in your industry mandate that you have to maintain an auditable record of access to certain types of data. Assuming that all expiring logs will be archived correctly, where should you store data that is subject to that mandate?

A. Encrypted on Cloud Storage with user-supplied encryption keys. A separate decryption key will be given to each authorized user.

B. In a BigQuery dataset that is viewable only by authorized personnel, with the Data Access log used to provide the auditability. **Most Voted**

C. In Cloud SQL, with separate database user names to each user. The Cloud SQL Admin activity logs will be used to provide the auditability.

D. In a bucket on Cloud Storage that is accessible only by an AppEngine service that collects user information and logs the access before providing a link to the bucket.

Correct Answer: B

Community vote distribution



Comments

Mitra123 **Highly Voted** 4 years, 6 months ago

Keywords here are 1. "Archived": Immutable and hence, BQ and Cloud SQL are ruled out 2. "Auditable": Means track any changes done. Only D can provide the audibility piece! I will go with D

Jarek7 2 years, 4 months ago

I have no idea why so many upvotes on this answer: 1) archived doesn't mean immutable and cloud storage is not immutable too. 2) auditable means viewable for authorized personel - and in this case not changes need to be monitored but any access. 3) with option D it is easy to go around logging - you can add another access to the bucket read the data remove the access and no one will ever know that you accessed the data. 4) option D is much more difficult - you need to application on AppEngine to log the data and provide access for users. 5) option D doesn;t explain where and how it stores the audit data - it could be accessed and modified from some side app/service.

[Removed] **Highly Voted** 5 years, 6 months ago

Answer: B Description: Bigquery is used to analyse access logs, data access logs capture the details of the user that accessed the data

awssp12345 4 years, 2 months ago

The question has no mention of ANALYZE.. BQ is not correct. I would go with D.

sraakesh95 3 years, 8 months ago

There is no option for archiving with BQ

tavva_prudhvi 3 years, 5 months ago

You dont need to archive the expiring logs, you have to archive the un-archived data here! See the question, it says "Assuming that all expiring logs will be archived correctly", which means they are already stored somewhere like in GCS!!! Hence, better to store the remaining un-archived data in BQ.

vartiklis 3 years, 2 months ago

The question is about where to store the _data_ for which the logs will be generated. The bit you quoted is about the _logs_ that will be generated when accesssing data. The “archived correctly” implies that proper retention policies will be set up if you choose GCS.

philli1011 1 year, 7 months ago

In recent GCP, we have cloud audit.

Nandababy 1 year, 10 months ago

Option B is valid only when analytics to be performed over logs, which is not mentioned anywhere

rocky48 1 year, 10 months ago

Selected Answer: B

For maintaining an auditable record of access to certain types of data, especially when government regulations are in place, the most suitable option would be: B. In a BigQuery dataset that is viewable only by authorized personnel, with the Data Access log used to provide the auditability. Storing the data in a BigQuery dataset with restricted access ensures control over who can view the data, and utilizing Data Access logs provides a comprehensive audit trail for compliance purposes. This option aligns well with the need for maintaining an auditable record as mandated by government regulations.

Jarek7 2 years, 4 months ago

Selected Answer: B

If you are going for option D, why do you eliminate option B? The only REAL difference is that for opption D you need to develop an app for storing log data and providing bucket link and in option B you have it all done BETTER by GCP. You might also pay a bit more for BQ storage, but the question never mentions about cost optimization. BTW in the D option the bucket is accessible only by AppEngine service, so what will the user do with the provided link? he has no access anyway... And if he even has the access to this link what stops him form using the same link many times? How the AppEngine get and store the information what specific data he accessed and how?

Kiroo 2 years, 4 months ago

That was my thought, either B or D could work but D it’s a little bit odd create an app to do something that could be achieved natively gcp

phidelics 2 years, 3 months ago

I was about to say the same thing. Why go through that stress?

Rodrigo4N 2 years, 5 months ago

Selected Answer: D

D amongus

juliobs 2 years, 6 months ago

Selected Answer: B

They want to know where you can store **data** in a way that every access is logged in an auditable way. Both BQ and GCS have audit logs, except that in alternative D you're circumventing it by creating your own logs. I doubt Google would recommend that. By types of data you can understand "financial type", "marketing type", etc.

midgoo 2 years, 7 months ago

Selected Answer: D

I was thinking it should be A. However, 'data' in this question is too vague. It does not say anywhere that the data could fit in BigQuery tables. It could be unstructure data such as videos or images Option D seems to involve more setup but it is the only viable option for this scenario. Note that GCS do have Cloud Audit logs. That should be the best option. Maybe this question was asked when Cloud Audit log is not yet available for GCS.

aleixfc96 2 years, 8 months ago

It is so clear that is B lol

NamitSehgal 2 years, 8 months ago

B bigquery for a record set store

PolyMoe 2 years, 8 months ago

Selected Answer: B

B. In a BigQuery dataset that is viewable only by authorized personnel, with the Data Access log used to provide the auditability. BigQuery provides built-in logging of all data access, including the user's identity, the specific query run and the time of the query. This log can be used to provide an auditable record of access to the data. Additionally, BigQuery allows you to control access to the dataset using Identity and Access Management (IAM) roles, so you can ensure that only authorized personnel can view the dataset.

samdhimal 2 years, 8 months ago

B. In a BigQuery dataset that is viewable only by authorized personnel, with the Data Access log used to provide the auditability. BigQuery provides built-in logging of all data access, including the user's identity, the specific query run and the time of the query. This log can be used to provide an auditable record of access to the data. Additionally, BigQuery allows you to control access to the dataset using Identity and Access Management (IAM) roles, so you can ensure that only authorized personnel can view the dataset.

samdhimal 2 years, 8 months ago

A. Encrypted on Cloud Storage with user-supplied encryption keys. A separate decryption key will be given to each authorized user. is a good option for data security but it does not provide an auditable record of access to the data. C. In Cloud SQL, with separate database user names to each user. The Cloud SQL Admin activity logs will be used to provide the auditability. is also a good option for data security but it does not provide an auditable record of access to the data. D. In a bucket on Cloud Storage that is accessible only by an AppEngine service that collects user information and logs the access before providing a link to the bucket. is also a good option but it requires additional setup and maintenance of the AppEngine service, and it may not provide an auditable record of access to the data.

Oleksandr0501 2 years, 5 months ago

gpt: You are correct that option A does not provide an auditable record of access to the data, as it only addresses data security through encryption. Option C provides auditability through Cloud SQL Admin activity logs, but it may not be the best option as it requires additional setup and management. Option D is a feasible solution, but as you mentioned, it requires additional setup and maintenance of the AppEngine service. It also may not provide a comprehensive audit log of all data access. Option B, storing the data in a BigQuery dataset that is viewable only by authorized personnel and using the Data Access log to provide auditability, is the most appropriate option as it provides built-in logging of all data access and allows you to control access to the dataset using IAM roles. Therefore, it provides both data security and auditable access to the data. /// ok let it be B

Oleksandr0501 2 years, 5 months ago

OR MAYBE D....

Oleksandr0501 2 years, 4 months ago

!!! confused. Give 69% confidence to B, as user Jarek7 explained

GCPpro 2 years, 8 months ago

D is the correct answer

RoshanAshraf 2 years, 8 months ago

Selected Answer: D

Keys TYPES of data --> Cloud Storage not BQ Archival --> Cloud Storage Access --> No decryption keys to all users

PrashantGupta1616 2 years, 9 months ago

Selected Answer: D

I will go with D

DGames 2 years, 9 months ago

Selected Answer: D

Keyword, Archiver , certain type of data, auditable, GCS is better option . Durability 11 time 9 to store log immutable for long time.

https://zhangxuetu.online

[202610]

Question #77 Topic 1

Your neural network model is taking days to train. You want to increase the training speed. What can you do?

A. Subsample your test dataset.

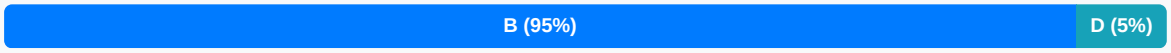
B. Subsample your training dataset. **Most Voted**

C. Increase the number of input features to your model.

D. Increase the number of layers in your neural network.

Correct Answer: B

Community vote distribution



Comments

mantwosmart **Highly Voted** 1 year, 10 months ago

Answer: B. Subsample your training dataset. Subsampling your training dataset can help increase the training speed of your neural network model. By reducing the size of your training dataset, you can speed up the process of updating the weights in your neural network. This can help you quickly test and iterate your model to improve its accuracy. Subsampling your test dataset, on the other hand, can lead to inaccurate evaluation of your model's performance and may result in overfitting. It is important to evaluate your model's performance on a representative test dataset to ensure that it can generalize to new data. Increasing the number of input features or layers in your neural network can also improve its performance, but this may not necessarily increase the training speed. In fact, adding more layers or features can increase the complexity of your model and make it take longer to train. It is important to balance the model's complexity with its performance and training time.

crazycosmos 1 year, 8 months ago

Selected Answer: B

B is correct

Vipul1600 1 year, 8 months ago

B should be correct. Increasing the layers can also decrease the training time but may introduce vanishing gradient hence D may not be correct

email2nn 1 year, 11 months ago

answer is B

juliobs 2 years ago

Selected Answer: B

Reduce training time and probably accuracy too.

MingSer 2 years, 1 month ago

Selected Answer: B

all other are wrong

PolyMoe 2 years, 2 months ago

Selected Answer: B

of course !

samdhimal 2 years, 2 months ago

B. Subsampling your training dataset can decrease the amount of data the model needs to process and can speed up training time. However, it can lead to decrease in the model's accuracy. Although it shouldn't matter since we are not even in testing phase yet and we aren't looking for accuracy.

GCPpro 2 years, 2 months ago

B is the answer as we are bothered about speed not the accuracy.

ler_mp 2 years, 2 months ago

Selected Answer: B

The answer is B. Building a more complex model by increasing the number of layer will not reduce the training time.

slade_wilson 2 years, 3 months ago

Selected Answer: B

By SubSampling the training data, you will reduce the training time. In case of D, if you increase the number of layers, then the model's accuracy will be increased. But it will not reduce the time required to train the model.

DGames 2 years, 3 months ago

Selected Answer: D

Increase speed of the help to train quicker.. option B is sub sample that also help but it drop accurately of model . So I think Option D is good to go.

jin0 2 years, 1 month ago

That makes speed of training model lower absolutely. because not only throughput of inference but back-propagation calculation would be increase so, D should be not a answer. there is only answer in those options is B. while it makes dropping performance

pluiedust 2 years, 5 months ago

Selected Answer: B

It is B. D would improve the accuracy, not speed.

Chavoz 2 years, 6 months ago

It's B. D Would be for increase performance

crismo04 2 years, 6 months ago

if you Increase the number of layers, you increase the training time, right?

HarshKothari21 2 years, 6 months ago

Both B and D seems correct.

jkhong 2 years, 3 months ago

Increasing D will increase training time

YorelNation 2 years, 7 months ago

Selected Answer: B

Only valid awnser

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #78 Topic 1

You are responsible for writing your company's ETL pipelines to run on an Apache Hadoop cluster. The pipeline will require some checkpointing and splitting pipelines. Which method should you use to write the pipelines?

A. PigLatin using Pig **Most Voted**

B. HiveQL using Hive

C. Java using MapReduce

D. Python using MapReduce

Correct Answer: A

Community vote distribution



Comments

IsaB **Highly Voted** 5 years, 6 months ago

Is this really a question that could appear in Google Cloud Professional Data Engineer Exam? What does it have to do with Google Cloud? I would use DataProc no?

Pupina 5 years, 5 months ago

Did you take the exam? I am ready to do it this month

MaxNRG 4 years, 3 months ago

seems like a very old question :) not sure it's actual

dg63 **Highly Voted** 5 years, 8 months ago

A, C and D - all are valid answers. You can do checkpointing, splitting pipelines and merging pipelines with all three options.

MD_84 4 months, 1 week ago

Selected Answer: C

The most appropriate method to write complex ETL pipelines requiring explicit checkpointing and splitting pipelines directly on an Apache Hadoop cluster is Mapreduce. Java or Python both are supported but Java is primary language and so Option C

oliiivier 11 months, 1 week ago

Selected Answer: C

Réponse : C Explication : La bonne réponse ici est C. Java using MapReduce. Explication rapide : Tu parles de besoin de "checkpointing" et "splitting pipelines". PigLatin (Pig) et HiveQL (Hive) sont des langages déclaratifs (de plus haut niveau), pas faits pour un contrôle précis sur la manière dont les jobs se séquencent, checkpointent ou

s'articulent. MapReduce en Java te donne un contrôle total sur : L'ordonnancement précis. Le checkpointing manuel. Le découpage et le chaînage des étapes. Python avec MapReduce est possible mais beaucoup moins natif ; Hadoop est conçu principalement pour Java MapReduce, donc en Python ce serait plus compliqué, plus fragile et moins performant. Résumé rapide pour que tu t'en souviennes :□□ Besoin de contrôle précis sur la logique ETL complexe = MapReduce en Java.

Parandhaman_Margan 1 year ago

Selected Answer: C

MapReduce in Java (C) allows more control for checkpointing and splitting.

desertlotus1211 1 year ago

Selected Answer: C

Writing your pipeline in Java using MapReduce allows you to implement these custom controls and fine-tune the execution, ensuring robust and manageable ETL processes on your Hadoop cluster

grshankar9 1 year, 2 months ago

Selected Answer: A

Pig Latin supports both splitting pipelines and checkpointing, allowing users to create complex data processing workflows with the ability to restart from specific points in the pipeline if necessary.

SamuelTsch 1 year, 5 months ago

Selected Answer: A

I would go to A. C, D are similar. So both are excluded. B, Hive is actually a data warehouse system. I don't use Apache Pig. But, BCD are wrong. Then A should be correct.

AnonymousPanda 2 years, 7 months ago

Selected Answer: A

A as others have said

Oleksandr0501 2 years, 11 months ago

Selected Answer: C

Comment content is too short

juliobs 3 years ago

Selected Answer: A

PigLatin is the correct answer, however... the last release was 6 years ago and has lots of bugs.

musumusu 3 years, 1 month ago

This answer depends which language you are comfortable with. Hadoop is your framework, where mapReduce is your Native programming model in JAVA, which is designed to scale, parallel processing, restart pipeline from any checkpoint etc. , So if you are comfortable with JAVA, you can customize your checkpoint at lowlevel in better way. otherwise, choose PIG which is another programming concept run over JAVA but then you need to learn this also, if not choose python as it can be deployed with hadoop because hadoop has been making updates for python clients regularly. Option C: is the best one.

samdhimal 3 years, 2 months ago

C. Java using MapReduce or D. Python using MapReduce Apache Hadoop is a distributed computing framework that allows you to process large datasets using the MapReduce programming model. There are several options for writing ETL pipelines to run on a Hadoop cluster, but the most common are using Java or Python with the MapReduce programming model.

samdhimal 3 years, 2 months ago

A. PigLatin using Pig is a high-level data flow language that is used to create ETL pipelines. Pig is built on top of Hadoop, and it allows you to write scripts in PigLatin, a SQL-like language that is used to process data in Hadoop. Pig is a simpler option than MapReduce but it lacks some capabilities like the control over low-level data manipulation operations. B. HiveQL using Hive is a SQL-like language for querying and managing large datasets stored in Hadoop's distributed file system. Hive is built on top of Hadoop and it provides an SQL-like interface for querying data stored in Hadoop. Hive is more suitable for querying and managing large datasets stored in Hadoop than for ETL pipelines. Both Java and Python using MapReduce provide low-level control over data manipulation operations, and they allow you to write custom mapper and reducer functions that can be used to process data in a Hadoop cluster. The choice between Java and Python will depend on the development team's expertise and preference.

cetanx 2 years, 10 months ago

It has to be C because while Pig can be used to simplify the writing of complex data transformation tasks and can store intermediate results, it doesn't provide the detailed control over checkpointing and pipeline splitting in the way that is typically implied by those terms. also, while one can write MapReduce jobs in languages other than Java (like Python) using Hadoop Streaming or other similar APIs, it may not be as efficient or as seamless as using Java due to the JVM-native nature of Hadoop.

Koushik25sep 3 years, 6 months ago

Selected Answer: A

Description: Pig is scripting language which can be used for checkpointing and splitting pipelines

BigDataBB 4 years, 1 month ago

Why not D?

rbeeraka 4 years, 2 months ago

Selected Answer: A

PigLatin supports checkpoints

davidqianwen 4 years, 2 months ago

Selected Answer: A

Answer: A

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #79 Topic 1

Your company maintains a hybrid deployment with GCP, where analytics are performed on your anonymized customer data. The data are imported to Cloud
Storage from your data center through parallel uploads to a data transfer server running on GCP. Management informs you that the daily transfers take too long and have asked you to fix the problem. You want to maximize transfer speeds. Which action should you take?

A. Increase the CPU size on your server.

B. Increase the size of the Google Persistent Disk on your server.

C. Increase your network bandwidth from your datacenter to GCP. **Most Voted**

D. Increase your network bandwidth from Compute Engine to Cloud Storage.

Correct Answer: C

Community vote distribution

C (67%) A (33%)

Comments

Mathew106 2 years, 2 months ago

Selected Answer: C

We are talking about transfer speed. Network transfer speed does not increase with CPU, but with bandwidth. Since there is no other extra information about what the issue, we have to assume that they imply network transfer speed.

Kiroo 2 years, 4 months ago

Selected Answer: C

To be honest this question is incomplete, I would go increasing the bandwidth, but first I would analyze why it's taking long time maybe I'm uploading many files so I could compress and agregate then and upload just one, maybe the target cpu is overloaded at the time of the upload, maybe the target disk reaching the max iops,

Jarek7 2 years, 4 months ago

Selected Answer: C

Even if transfer server is deployed on the slowest machine available in GCP there is no way it is bottleneck for simple data transfer without any data processing.

Oleksandr0501 2 years, 5 months ago

Selected Answer: A

GPT: Option A, increasing the CPU size on the data transfer server, could potentially increase the transfer speeds if the bottleneck in the data transfer process is the processing power of the server. By increasing the CPU size, the server may be able to process data more quickly, leading to faster transfers. Option C, increasing the network

bandwidth from the datacenter to GCP, could potentially improve the transfer speeds, but it may not be feasible or cost-effective depending on the current infrastructure and network limitations.

Jarek7 2 years, 4 months ago

Please stop using GPT as knowledge source. v3.5 is usually wrong even in simple cases. v4 is much better, but it is not designed to be knowledge source. Looking at the answer you must have used v3.5. The question says nothing about cost-effectivness. The issue is data transfer. No any data processing is done on the data while it is transferred. Simple transfer doesn't need much processing power - the real bottleneck even on slowest machines available on GCP must be data transfer - it is obvious. BTW for me GPT3.5 said it is C.

Oleksandr0501 2 years, 4 months ago

yea, i know it can make mistakes. Thank you. That`s why i always mark "GPT" at the start of my answer.

Oleksandr0501 2 years, 4 months ago

it should be C, for real, bcz nothing said about cost restrictions in the question. And the user "snamburi3 " found docs.

izekc 2 years, 5 months ago

Selected Answer: A

it's refer to data transfer server slow here. not transfer data to cloud slow. 100% A

jonathanthezombieboy 2 years, 6 months ago

Selected Answer: C

Answer is C

jin0 2 years, 7 months ago

This question makes people confused only. there is no refer to network or size of data or something could be referred. the answer could be A or C

mahdiaqim 2 years, 7 months ago

Selected Answer: A

Very confusing question. I selected A because I assume increasing the CPU size on the cloud server is easier to change, as a data engineer, than the bandwidth.

samdhimal 2 years, 8 months ago

C. Increase your network bandwidth from your datacenter to GCP. This will likely have the most impact on transfer speeds as it addresses the bottleneck in the transfer between your data center and GCP. Increasing the CPU size or the size of the Google Persistent Disk on the server may help with processing the data once it has been transferred, but will not address the bottleneck in the transfer itself. Increasing the network bandwidth from Compute Engine to Cloud Storage would also help with processing the data once it has been transferred but will not address the bottleneck in the transfer itself as well.

Nirca 2 years, 11 months ago

A bit unprofessional question, having performance issues should be addressed by analyzing and looking for saturation in the system and understanding "wait-events". Only than adding more resources.

rr4444 3 years, 3 months ago

"The data are imported to Cloud Storage from your data center through parallel uploads to a data transfer server running on GCP. " This makes zero sense. Is it to GCS or GCE? Question had to make up its mind. Nonsense, literally.

Thierry_1 3 years, 10 months ago

Vote for C. Mostly because other options seems useless

sumanshu 4 years, 3 months ago

Vote for 'C' You want to maximize transfer speeds.

Tolberic 4 years, 7 months ago

https://cloud.google.com/compute/docs/machine-types#n1_machine_types

snamburi3 4 years, 10 months ago

"The data are imported to Cloud Storage from your data center through parallel uploads to a data transfer server running on GCP." This is confusing, as the upload is through a data transfer server on GCP - not directly to Storage. In this case, maybe A?

snamburi3 4 years, 10 months ago

I am going with C as I found a doc: https://cloud.google.com/solutions/migration-to-google-cloud-transferring-your-large-datasets#increasing_network_bandwidth. still a confusing question...

SureshKotla 5 years ago

I guess option A because, increasing processing power for parallel uploads is the first thing to try. And if it doesn't work, then go to bandwidth issues.

bbozz_ 5 years, 2 months ago

Answer: CA

https://zhangxuetu.online

掌学兔
ZHANGXUETU

Question #80 Topic 1

MJTelco Case Study -

Company Overview -
MJTelco is a startup that plans to build networks in rapidly growing, underserved markets around the world. The company has patents for innovative optical communications hardware. Based on these patents, they can create many reliable, high-speed backbone links with inexpensive hardware.

Company Background -
Founded by experienced telecom executives, MJTelco uses technologies originally developed to overcome communications challenges in space. Fundamental to their operation, they need to create a distributed data infrastructure that drives real-time analysis and incorporates machine learning to continuously optimize their topologies. Because their hardware is inexpensive, they plan to overdeploy the network allowing them to account for the impact of dynamic regional politics on location availability and cost.
Their management and operations teams are situated all around the globe creating many-to-many relationship between data consumers and provides in their system. After careful consideration, they decided public cloud is the perfect environment to support their needs.

Solution Concept -
MJTelco is running a successful proof-of-concept (PoC) project in its labs. They have two primary needs:
✑ Scale and harden their PoC to support significantly more data flows generated when they ramp to more than 50,000 installations.
✑ Refine their machine-learning cycles to verify and improve the dynamic models they use to control topology definition.
MJTelco will also use three separate operating environments `" development/test, staging, and production `" to meet the needs of running experiments, deploying new features, and serving production customers.

Business Requirements -
✑ Scale up their production environment with minimal cost, instantiating resources when and where needed in an unpredictable, distributed telecom user community.
✑ Ensure security of their proprietary data to protect their leading-edge machine learning and analysis.
✑ Provide reliable and timely access to data for analysis from distributed research workers
✑ Maintain isolated environments that support rapid iteration of their machine-learning models without affecting their customers.

Technical Requirements -
Ensure secure and efficient transport and storage of telemetry data
Rapidly scale instances to support between 10,000 and 100,000 data providers with multiple flows each.
Allow analysis and presentation against data tables tracking up to 2 years of data storing approximately 100m records/day
Support rapid iteration of monitoring infrastructure focused on awareness of data pipeline problems both in telemetry flows and in production learning cycles.

CEO Statement -
Our business model relies on our patents, analytics and dynamic machine learning. Our inexpensive hardware is organized to be highly reliable, which gives us cost advantages. We need to quickly stabilize our large distributed data pipelines to meet our reliability and capacity commitments.

CTO Statement -
Our public cloud services must operate as advertised. We need resources that scale and keep our data secure. We also need environments in which our data scientists can carefully study and quickly adapt our models. Because we rely on automation to process our data, we also need our development and test environments to work as we iterate.

CFO Statement -
The project is too large for us to maintain the hardware and software required for the data and analysis. Also, we cannot afford to staff an operations team to monitor so many data feeds, so we will rely on automation and infrastructure. Google Cloud's machine learning will allow our quantitative researchers to work on our high-value problems instead of problems with our data pipelines.
MJTelco is building a custom interface to share data. They have these requirements:
1. They need to do aggregations over their petabyte-scale datasets.
2. They need to scan specific time range rows with a very fast response time (milliseconds).
Which combination of Google Cloud Platform products should you recommend?

A. Cloud Datastore and Cloud Bigtable

B. Cloud Bigtable and Cloud SQL

C. BigQuery and Cloud Bigtable **Most Voted**

D. BigQuery and Cloud Storage

Correct Answer: C

Community vote distribution



Comments

atnafu2020 **Highly Voted** 5 years, 8 months ago
C Bigquery and Big table =PB storage capacity Bigtable=to read scan rows Big query select row to read

daghayeghi **Highly Voted** 5 years ago
C: They need to do aggregations over their petabyte-scale datasets: Bigquery They need to scan specific time range rows with a very fast response time (milliseconds): Bigtable

09878d5 1 year, 4 months ago
Why not A? Can someone please explain

baht 2 years, 9 months ago

Selected Answer: C

Response C => Bigquery and bigtable

ga8our 2 years, 11 months ago

Why not A? If we're already using Bigtable, what's the use of another, slower analytic solution, like BigQuery? Wouldn't Datastore be more useful to store our data than BigQuery?

dconesoko 3 years, 2 months ago

Selected Answer: C

bigquery and bigtable

sandipk91 4 years, 7 months ago

C is correct, no doubt

sumanshu 4 years, 9 months ago

Vote for C

Insane7 5 years, 5 months ago

Why not D? Biqquery and GCS. Also Big Table is no sql, where as BQ is SQL

Gcpyspark 5 years, 3 months ago

With GCS you can only scan the rows from BigQuery using External federated Datasources, with that millisecond latency is not possible. Also "scan specific time range rows with a very fast response time" is a natural fit use case for Cloud Bigtable.

haroldbenites 5 years, 7 months ago

C is correct.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

Question #81 Topic 1

MJTelco Case Study -

Company Overview -
MJTelco is a startup that plans to build networks in rapidly growing, underserved markets around the world. The company has patents for innovative optical communications hardware. Based on these patents, they can create many reliable, high-speed backbone links with inexpensive hardware.

Company Background -
Founded by experienced telecom executives, MJTelco uses technologies originally developed to overcome communications challenges in space. Fundamental to their operation, they need to create a distributed data infrastructure that drives real-time analysis and incorporates machine learning to continuously optimize their topologies. Because their hardware is inexpensive, they plan to overdeploy the network allowing them to account for the impact of dynamic regional politics on location availability and cost.
Their management and operations teams are situated all around the globe creating many-to-many relationship between data consumers and provides in their system. After careful consideration, they decided public cloud is the perfect environment to support their needs.

Solution Concept -
MJTelco is running a successful proof-of-concept (PoC) project in its labs. They have two primary needs:
✑ Scale and harden their PoC to support significantly more data flows generated when they ramp to more than 50,000 installations.
Refine their machine-learning cycles to verify and improve the dynamic models they use to control topology definition.
▪
MJTelco will also use three separate operating environments `" development/test, staging, and production `" to meet the needs of running experiments, deploying new features, and serving production customers.

Business Requirements -
✑ Scale up their production environment with minimal cost, instantiating resources when and where needed in an unpredictable, distributed telecom user community.
✑ Ensure security of their proprietary data to protect their leading-edge machine learning and analysis.
✑ Provide reliable and timely access to data for analysis from distributed research workers
✑ Maintain isolated environments that support rapid iteration of their machine-learning models without affecting their customers.

Technical Requirements -
Ensure secure and efficient transport and storage of telemetry data
Rapidly scale instances to support between 10,000 and 100,000 data providers with multiple flows each.
Allow analysis and presentation against data tables tracking up to 2 years of data storing approximately 100m records/day
Support rapid iteration of monitoring infrastructure focused on awareness of data pipeline problems both in

telemetry flows and in production learning cycles.

CEO Statement -
Our business model relies on our patents, analytics and dynamic machine learning. Our inexpensive hardware is organized to be highly reliable, which gives us cost advantages. We need to quickly stabilize our large distributed data pipelines to meet our reliability and capacity commitments.

CTO Statement -
Our public cloud services must operate as advertised. We need resources that scale and keep our data secure. We also need environments in which our data scientists can carefully study and quickly adapt our models. Because we rely on automation to process our data, we also need our development and test environments to work as we iterate.

CFO Statement -
The project is too large for us to maintain the hardware and software required for the data and analysis. Also, we cannot afford to staff an operations team to monitor so many data feeds, so we will rely on automation and infrastructure. Google Cloud's machine learning will allow our quantitative researchers to work on our high-value problems instead of problems with our data pipelines.
You need to compose visualization for operations teams with the following requirements:
✑ Telemetry must include data from all 50,000 installations for the most recent 6 weeks (sampling once every minute)
✑ The report must not be more than 3 hours delayed from live data.
✑ The actionable report should only show suboptimal links.
✑ Most suboptimal links should be sorted to the top.
Suboptimal links can be grouped and filtered by regional geography.
▪
✑ User response time to load the report must be <5 seconds.
You create a data source to store the last 6 weeks of data, and create visualizations that allow viewers to see multiple date ranges, distinct geographic regions, and unique installation types. You always show the latest data without any changes to your visualizations. You want to avoid creating and updating new visualizations each month. What should you do?

A. Look through the current data and compose a series of charts and tables, one for each possible combination of criteria.

B. Look through the current data and compose a small set of generalized charts and tables bound to criteria filters that allow value selection. Most Voted

C. Export the data to a spreadsheet, compose a series of charts and tables, one for each possible combination of criteria, and spread them across multiple tabs.

D. Load the data into relational database tables, write a Google App Engine application that queries all rows, summarizes the data across each criteria, and then renders results using the Google Charts and visualization API.

Correct Answer: B

Community vote distribution



Comments

Jarek7 Highly Voted 2 years, 11 months ago

Selected Answer: D

First I thought B, as D seems too complex with writing app for AppEngine. But B is too simple - just look through the data doesnt seem right. It must be very old question. Today you would load the data to BQ, optionally you can use Dataprep for simple data cleaning or a Dataflow job for more complex data processing, and finally use Looker to create tables and charts.

Oleksandr0501 2 years, 11 months ago

As per old question - must be. I heard, that the exam will mostly have questions rather from 100 to 205 than form 1 to 100. And smb told me, that the other w/s gave questions, that happened more often in exam, in comparison to questions given here

cloudmon Highly Voted 3 years, 4 months ago

Selected Answer: B

It's B. All the other choices are unreasonable.

oussama7 1 year ago

Selected Answer: B

Filters allow dynamic interaction: Instead of static charts, filters enable users to select date ranges, regions, and installation types without requiring frequent updates.

Parandhaman_Margan 1 year ago

Selected Answer: B

Dynamic Filtering → Instead of creating a fixed set of charts for every combination, filters allow users to explore data interactively without manual updates. Scalability → Creating a small number of general charts with filters reduces maintenance effort and dashboard complexity.

Augustax 1 year, 2 months ago

Selected Answer: B

Data engineer especially the Front-End developer will pick B.

cloud_rider 1 year, 4 months ago

Selected Answer: B

D is not the right answer as Chart and Viz API is deprecated now (https://en.wikipedia.org/wiki/Google_Chart_API#:~:text=The%20Google%20Chart%20API%20is,charts%20from%20user%2Dsupplied%20data.) B is the most logical answer as it talks about creation of general chart wtih filter for value selection (as asked in the requirement),

grshankar9 1 year, 2 months ago

Google Chart API is deprecated but there is a 'Google Charts' API now. And Visualization is part of it

Nirca 2 years, 5 months ago

Selected Answer: B

bound to criteria filters that allow value selection. - Simple and Smart.

PolyMoe 3 years, 2 months ago

Selected Answer: D

D. everything is fixed except data that is updated regularly in order to keep the last 6 weeks. Then, the pipeline does not change ==> obtaining (same) charts and viz on regularly updated data

hauhau 3 years, 4 months ago

Selected Answer: B

B But can someone explain the question and selection clearly?

edwardlin421 3 years, 4 months ago

ACD-Design for each possible combination of criteria, so if your team has new requirements, you must design new charts. So, answer shoud be B.

ducc 3 years, 7 months ago

Selected Answer: D

the key is " You want to avoid creating and updating new visualizations each month." only D work for that phrase

wan2three 3 years, 3 months ago

D you might need to load data from source to table for each month. It stated the source will keep last 6 weeks data, but not in D

KundanK973 3 years, 9 months ago

must be D

ealpuche 3 years, 9 months ago

Selected Answer: D

The answer is B

rr4444 3 years, 9 months ago

This Q feels very disconnected from GCP products.....

sw52099 3 years, 10 months ago

Selected Answer: D

Vote D. Since B just uses "current data", which means if new data enters, you need to re-run those charts again.

wan2three 3 years, 3 months ago

But q says the data sources only have latest 6 weeks data, so current data means latest?

RRK2021 4 years, 1 month ago

B is optimal to avoid creating and updating new visualizations each month

ManojT 4 years, 5 months ago

Answer D: Data in SQL so querying becomes easier on any pattern. create mutiple charts, graphs to fulfill your requirements.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

Question #82 Topic 1

MJTelco Case Study -

Company Overview -
MJTelco is a startup that plans to build networks in rapidly growing, underserved markets around the world. The company has patents for innovative optical communications hardware. Based on these patents, they can create many reliable, high-speed backbone links with inexpensive hardware.

Company Background -
Founded by experienced telecom executives, MJTelco uses technologies originally developed to overcome communications challenges in space. Fundamental to their operation, they need to create a distributed data infrastructure that drives real-time analysis and incorporates machine learning to continuously optimize their topologies. Because their hardware is inexpensive, they plan to overdeploy the network allowing them to account for the impact of dynamic regional politics on location availability and cost.
Their management and operations teams are situated all around the globe creating many-to-many relationship between data consumers and provides in their system. After careful consideration, they decided public cloud is the perfect environment to support their needs.

Solution Concept -
MJTelco is running a successful proof-of-concept (PoC) project in its labs. They have two primary needs:
✑ Scale and harden their PoC to support significantly more data flows generated when they ramp to more than 50,000 installations.
✑ Refine their machine-learning cycles to verify and improve the dynamic models they use to control topology definition.
MJTelco will also use three separate operating environments `" development/test, staging, and production `" to meet the needs of running experiments, deploying new features, and serving production customers.

Business Requirements -
✑ Scale up their production environment with minimal cost, instantiating resources when and where needed in an unpredictable, distributed telecom user community.
✑ Ensure security of their proprietary data to protect their leading-edge machine learning and analysis.
✑ Provide reliable and timely access to data for analysis from distributed research workers
✑ Maintain isolated environments that support rapid iteration of their machine-learning models without affecting their customers.

Technical Requirements -
Ensure secure and efficient transport and storage of telemetry data
Rapidly scale instances to support between 10,000 and 100,000 data providers with multiple flows each.
Allow analysis and presentation against data tables tracking up to 2 years of data storing approximately 100m records/day
Support rapid iteration of monitoring infrastructure focused on awareness of data pipeline problems both in telemetry flows and in production learning cycles.

CEO Statement -
Our business model relies on our patents, analytics and dynamic machine learning. Our inexpensive hardware is organized to be highly reliable, which gives us cost advantages. We need to quickly stabilize our large distributed data pipelines to meet our reliability and capacity commitments.

CTO Statement -
Our public cloud services must operate as advertised. We need resources that scale and keep our data secure. We also need environments in which our data scientists can carefully study and quickly adapt our models. Because we rely on automation to process our data, we also need our development and test environments to work as we iterate.

CFO Statement -
The project is too large for us to maintain the hardware and software required for the data and analysis. Also, we cannot afford to staff an operations team to monitor so many data feeds, so we will rely on automation and infrastructure. Google Cloud's machine learning will allow our quantitative researchers to work on our high-value problems instead of problems with our data pipelines.
Given the record streams MJTelco is interested in ingesting per day, they are concerned about the cost of Google BigQuery increasing. MJTelco asks you to provide a design solution. They require a single large data table called tracking_table. Additionally, they want to minimize the cost of daily queries while performing fine-grained analysis of each day's events. They also want to use streaming ingestion. What should you do?

A. Create a table called tracking_table and include a DATE column.

B. Create a partitioned table called tracking_table and include a TIMESTAMP column. **Most Voted**

C. Create sharded tables for each day following the pattern tracking_table_YYYYMMDD.

D. Create a table called tracking_table with a TIMESTAMP column to represent the day.

Correct Answer: B

Community vote distribution

B (100%)

Comments

SamuelTsch 1 year, 5 months ago

Selected Answer: B

partitioned table is more performancer than sharded tables

sspsp 2 years, 8 months ago

Selected Answer: B

B, Partition tables in BQ have different cost. If a partition is not modified (DML) for 90 days then cost will be less by 50%, while querying will be efficient since its single large table.

piotrpiskorski 3 years, 4 months ago

Selected Answer: B

always partion large tables

Thierry_1 4 years, 4 months ago

B for sure

nguyenmoon 4 years, 6 months ago

Correct is B

sandipk91 4 years, 7 months ago

Option B for sure

awssp12345 4 years, 9 months ago

https://cloud.google.com/bigquery/docs/partitioned-tables#dt_partition_shard - Supports B

sumanshu 4 years, 9 months ago

Vote for 'B' Partitioned Table for Faster Query and Low cost (because it will process less data)

alonsoRios 5 years ago

B is correct

fabenavideso 5 years, 2 months ago

Correct - B

ceak 5 years, 3 months ago

should be C

lammingtons 5 years, 2 months ago

They're using BigQuery so partitioning is the better choice here. B

haroldbenites 5 years, 7 months ago

B is correct

https://zhangxuetu.online

掌学兔
ZHANGXUETU

Question #83 Topic 1

Flowlogistic Case Study -

Company Overview -
Flowlogistic is a leading logistics and supply chain provider. They help businesses throughout the world manage their resources and transport them to their final destination. The company has grown rapidly, expanding their offerings to include rail, truck, aircraft, and oceanic shipping.

Company Background -
The company started as a regional trucking company, and then expanded into other logistics market. Because they have not updated their infrastructure, managing and tracking orders and shipments has become a bottleneck. To improve operations, Flowlogistic developed proprietary technology for tracking shipments in real time at the parcel level. However, they are unable to deploy it because their technology stack, based on Apache Kafka, cannot support the processing volume. In addition, Flowlogistic wants to further analyze their orders and shipments to determine how best to deploy their resources.

Solution Concept -
Flowlogistic wants to implement two concepts using the cloud:
✑ Use their proprietary technology in a real-time inventory-tracking system that indicates the location of their loads
✑ Perform analytics on all their orders and shipment logs, which contain both structured and unstructured data, to determine how best to deploy resources, which markets to expand info. They also want to use predictive analytics to learn earlier when a shipment will be delayed.

Existing Technical Environment -
Flowlogistic architecture resides in a single data center:
✑ Databases
- 8 physical servers in 2 clusters
- SQL Server `" user data, inventory, static data
- 3 physical servers
- Cassandra `" metadata, tracking messages
10 Kafka servers `" tracking message aggregation and batch insert
✑ Application servers `" customer front end, middleware for order/customs
- 60 virtual machines across 20 physical servers
- Tomcat `" Java services
- Nginx `" static content
- Batch servers
✑ Storage appliances
- iSCSI for virtual machine (VM) hosts
- Fibre Channel storage area network (FC SAN) `" SQL server storage
Network-attached storage (NAS) image storage, logs, backups
✑ 10 Apache Hadoop /Spark servers

- Core Data Lake
- Data analysis workloads
✑ 20 miscellaneous servers
- Jenkins, monitoring, bastion hosts,

Business Requirements -
✑ Build a reliable and reproducible environment with scaled panty of production.
✑ Aggregate data in a centralized Data Lake for analysis
✑ Use historical data to perform predictive analytics on future shipments
✑ Accurately track every shipment worldwide using proprietary technology
✑ Improve business agility and speed of innovation through rapid provisioning of new resources
✑ Analyze and optimize architecture for performance in the cloud
✑ Migrate fully to the cloud if all other requirements are met

Technical Requirements -
✑ Handle both streaming and batch data
✑ Migrate existing Hadoop workloads
✑ Ensure architecture is scalable and elastic to meet the changing demands of the company.
✑ Use managed services whenever possible
✑ Encrypt data flight and at rest
Connect a VPN between the production data center and cloud environment

SEO Statement -
We have grown so quickly that our inability to upgrade our infrastructure is really hampering further growth and efficiency. We are efficient at moving shipments around the world, but we are inefficient at moving data around. We need to organize our information so we can more easily understand where our customers are and what they are shipping.

CTO Statement -
IT has never been a priority for us, so as our data has grown, we have not invested enough in our technology. I have a good staff to manage IT, but they are so busy managing our infrastructure that I cannot get them to do the things that really matter, such as organizing our data, building the analytics, and figuring out how to implement the CFO' s tracking technology.

CFO Statement -
Part of our competitive advantage is that we penalize ourselves for late shipments and deliveries. Knowing where out shipments are at all times has a direct correlation to our bottom line and profitability. Additionally, I don't want to commit capital to building out a server environment.
Flowlogistic's management has determined that the current Apache Kafka servers cannot handle the data volume for their real-time inventory tracking system.
You need to build a new system on Google Cloud Platform (GCP) that will feed the proprietary tracking software. The system must be able to ingest data from a variety of global sources, process and query in real-time, and store the data reliably. Which combination of GCP products should you choose?

A. Cloud Pub/Sub, Cloud Dataflow, and Cloud Storage **Most Voted**

B. Cloud Pub/Sub, Cloud Dataflow, and Local SSD

C. Cloud Pub/Sub, Cloud SQL, and Cloud Storage

D. Cloud Load Balancing, Cloud Dataflow, and Cloud Storage

E. Cloud Dataflow, Cloud SQL, and Cloud Storage

Correct Answer: A

Community vote distribution



Comments

digvijay Highly Voted 4 years, 6 months ago

Seems like A..Data should ingest from multiple sources which might be real time or batch .

navemula 3 years, 2 months ago

How is it possible to query in real time with option A. It needs Dataflow

navemula 3 years, 2 months ago

To use Dataflow SQL it needs BigQuery

mikey007 Highly Voted 4 years ago

Repeated Question see ques 35

awssp12345 3 years, 2 months ago

These exams make people over analyse. People who vote A earlier in 35 seem to be confused here.. haha

StelSen 3 years, 8 months ago

Well Done mikey007, Many people have already answered as A.

Kyr0 1 year, 9 months ago

Selected Answer: A

Answer is A

cloudmon 1 year, 11 months ago

Selected Answer: A

It's A

ducc 2 years, 1 month ago

Selected Answer: A

A is the answer

RRK2021 2 years, 7 months ago

ingest data from a variety of global sources - cloud pub/sub process and query in real-time - cloud Dataflow store the data reliably - Cloud Storage

medeis_jar 2 years, 8 months ago

Selected Answer: A

PubSub (for global ingestion from multiple sources) + Dataflow (for process and query) + reliable (gcs).

lifebegins 2 years, 10 months ago

Selected Answer: A

using Dataflow you can apply the propriety analytics and you can push the data in to Cloud storage

gcp_k 2 years, 11 months ago

Also read the technical requirements section. Not just the last 3 lines of the question. When you do that, you'll know the answer is PubSub (for global ingestion) + Dataflow (for process and query) + reliable (gcs). Answer is: A

ManojT 2 years, 11 months ago

Answer C: Look the 3 requirement in the question "ingest data from a variety of global sources, process and query in real-time, and store the data reliably" Ingest data from global sources: Pub-Sub Process and Query in realtime: Cloud SQL Store reliably: Cloud storage I can understand Databflow is required in case you need to analyze and transform data but question does not refer it.

cualquiernick 2 years, 3 months ago

Cloud SQL, is not suitable and efficient for storing real time data ingested from PUB/SUB, so A is the answer

nguyenmoon 3 years ago

Correct is A. Kafka --> replace by PubSub, Streaming then Dataflow, store data reliably and not mention any other condition then Cloud Storage

sumanshu 3 years, 3 months ago

Vote for 'A' SQL - will not handle the volume

daghayeghi 3 years, 6 months ago

Dataflow SQL cannot output to cloud storage: https://cloud.google.com/dataflow/docs/guides/sql/data-sources-destinations but the main problem is that Cloud SQL can't do process, then response is A or C.

kino2020 4 years ago

A I don't expect this question to come up, but if I had to write the answer, it would be A. The problem statement "Flowlogistic's management has determined that the current Apache Kafka servers cannot handle the data volume for their real-time inventory tracking system. As it says, "we cannot determine the data volume", but it doesn't say that we can't calculate it either. Requirement definition: The system must be able to ingest data from a variety of global sources process and query in real-time Store the data reliably. It says above, if you look at the Google page. Logging to multiple systems. for example, a Google Compute Engine instance can write logs to a monitoring system, to a database for later querying, and so on. https://cloud.google.com/pubsub/docs/overview#scenarios stream processing with Dataflow https://cloud.google.com/pubsub/docs/pubsub-dataflow?hl=en-419 The answer is A, since it is stated above.

vakati 4 years ago

A. SQL queries can be written in Dataflow too. https://cloud.google.com/dataflow/docs/guides/sql/dataflow-sql-intro#running-queries

aleedrew 3 years, 11 months ago

Dataflow SQL cannot output to cloud storage only BigQuery...so I am confused on this one.

Jay3244 3 years, 7 months ago

https://cloud.google.com/pubsub/docs/pubsub-dataflow... It is possible to load the data to Cloud Storage. Can refer to above docs.

daghayeghi 3 years, 6 months ago

he said correct, DataflowDataflow SQL cannot output to cloud storage:
https://cloud.google.com/dataflow/docs/guides/sql/data-sources-destinations

Ral17 3 years ago

Answer should be C then?

kuntal8285 4 years ago

should be E

Tanmoyk 4 years ago

Should be A ...data need to feed to the propriority system and for that dataflow is required.

https://zhangxuetu.online

[202610]

Question #84 Topic 1

After migrating ETL jobs to run on BigQuery, you need to verify that the output of the migrated jobs is the same as the output of the original. You've loaded a table containing the output of the original job and want to compare the contents with output from the migrated job to show that they are identical. The tables do not contain a primary key column that would enable you to join them together for comparison.
What should you do?

A. Select random samples from the tables using the RAND() function and compare the samples.

B. Select random samples from the tables using the HASH() function and compare the samples.

C. Use a Dataproc cluster and the BigQuery Hadoop connector to read the data from each table and calculate a hash from non-timestamp columns of the table after sorting. Compare the hashes of each table. **Most Voted**

D. Create stratified random samples using the OVER() function and compare equivalent samples from each table.

Correct Answer: C

Community vote distribution



Comments

rickywck **Highly Voted** 6 years ago
C is the only way which all records will be compared.

odacir 3 years, 3 months ago
Agree with your argument

[Removed] **Highly Voted** 6 years ago
Answer: C Description: Full comparison with this option, rest are comparison on sample which doesnot ensure all the data will be ok

SamuelTsch 1 year, 5 months ago
Selected Answer: C
Hash is always a good idea to compare the data

midgoo 3 years, 1 month ago
In practice, I will do B. That means it may have error due to randomness. But that is how we normally do validation/QA in general, i.e. we test random samples In this question, I will do C.

musumusu 3 years, 1 month ago

key words here- Hash or collect value on "EACH table", after sorting the table. Option C

samdhimal 3 years, 2 months ago

C. Use a Dataproc cluster and the BigQuery Hadoop connector to read the data from each table and calculate a hash from non-timestamp columns of the table after sorting. Compare the hashes of each table. This approach will ensure that the data is read in a consistent order, and the hash function will provide a quick and efficient way to compare the contents of the tables and ensure that they are identical.

samdhimal 3 years, 2 months ago

A. Selecting random samples from the tables using the RAND() function may not provide an accurate representation of the data and there is a risk that the comparison will not identify any differences between the tables. B. Selecting random samples from the tables using the HASH() function may not be an effective method for comparison, as the HASH() function may return different results for equivalent data. D. Creating stratified random samples using the OVER() function may not provide a comprehensive comparison between the tables as there is a risk that important differences could be missed in the sample data.

Leeeeee 3 years, 4 months ago

Selected Answer: C

All records

hfuihe 3 years, 5 months ago

Selected Answer: B

B is the only way which all records will be compared.

cloudmon 3 years, 4 months ago

You must have meant to say C

medeis_jar 4 years, 2 months ago

Selected Answer: C

HASH() to compare data skipping dates and timestamps

stefanop 3 years, 11 months ago

The hash in answer C is used to select a sample of the table, not to compare them

stefanop 3 years, 11 months ago

Ignore my comment, it was about answer B. I suggest you to go with answer C which is the only solution comparing all the rows/tables

MaxNRG 4 years, 3 months ago

Selected Answer: C

options A B and D only will determine that it “might” be identical since is only a sample. HASH() can be helpful when doing bulk comparisons, but you still have to compare field by field to get the final answer. The only one left is C which looks good to me

JayZeeLee 4 years, 4 months ago

C. The rest use RAND() at some point, which makes it hard to compare for consistency, unless there's a 'seed' option, which wasn't mentioned. So C.

u_t_s 4 years, 5 months ago

Since there is no PK and it is possible that set of values is commons in some records which result in same hashkey for those records. But still Anwer is C

sumanshu 4 years, 9 months ago

Vote for 'C"

daghayeghi 5 years, 1 month ago

B: Because said migrated to BigQuery, then we don't need Dataproc, and samples don't mean you don't compare all of data.

yoshik 4 years, 6 months ago

a sample is a subset of data. then you should assure that the union of the samples contain the data set. Excessively complicated. You migrate to BigQuery but need to check BigQuery output, that is why you should use another tool, Dataproc in this case. Agree that then you should control Dataproc output but suppositions are becoming too many.

atnafu2020 5 years, 7 months ago

C Using Cloud Storage with big data Cloud Storage is a key part of storing and working with Big Data on Google Cloud. Examples include: Loading data into BigQuery. Using Dataproc, which automatically installs the HDFS-compatible Cloud Storage connector, enabling the use of Cloud Storage buckets in parallel with HDFS. Using a bucket to hold staging files and temporary data for Dataflow pipelines. For Dataflow, a Cloud Storage bucket is required. For BigQuery and Dataproc, using a Cloud Storage bucket is optional but recommended. gsutil is a command-line tool that enables you to work with Cloud Storage buckets and objects easily and robustly, in particular in big data scenarios. For example, with gsutil you can copy many files in parallel with a single command, copy large files efficiently, calculate checksums on your data, and measure performance from your local computer to Cloud Storage.

haroldbenites 5 years, 7 months ago

C is correct

haroldbenites 5 years, 7 months ago

It Says: "...that they are identical." , You must not use sample.

Rajuuu 5 years, 9 months ago

C is correct.

https://zhangxuetu.online

[202610]

Question #85 Topic 1

You are a head of BI at a large enterprise company with multiple business units that each have different priorities and budgets. You use on-demand pricing for
BigQuery with a quota of 2K concurrent on-demand slots per project. Users at your organization sometimes don't get slots to execute their query and you need to correct this. You'd like to avoid introducing new projects to your account.
What should you do?

A. Convert your batch BQ queries into interactive BQ queries.

B. Create an additional project to overcome the 2K on-demand per-project quota.

C. Switch to flat-rate pricing and establish a hierarchical priority model for your projects. **Most Voted**

D. Increase the amount of concurrent slots per project at the Quotas page at the Cloud Console.

Correct Answer: C

Community vote distribution

C (100%)

Comments

sedado77 **Highly Voted** 2 years ago

Selected Answer: C

I got this question on sept 2022. Answer is C

grshankar9 1 year, 2 months ago

Selected Answer: C

In BigQuery, "on-demand pricing" means you pay based on the amount of data your queries scan (bytes processed), essentially paying for what you use, while "flat-rate pricing" involves purchasing a set number of "slots" (virtual CPUs) and paying a fixed fee regardless of how much data you query, essentially providing a predictable monthly cost for dedicated processing power; on-demand is best for occasional users with variable query needs, while flat-rate is better for predictable high-volume querying

email2nn 1 year, 5 months ago

answer us C

midgoo 1 year, 6 months ago

Selected Answer: C

This question is interesting. My friend works as the TAM in Google and he said we could request for Quota increase if the customer is premium customer instead of changing to Flat-rate Otherwise, need to choose C

jin0 1 year, 7 months ago

why A is not a answer? when using interactive bigquery without batch bigquery it lead to run query immediately isn't it? so it seems to solve the problems isn't it?

samdhimal 1 year, 8 months ago

C. Switch to flat-rate pricing and establish a hierarchical priority model for your projects.

samdhimal 1 year, 8 months ago

Switching to flat-rate pricing would allow you to ensure a consistent level of service and avoid running into the on-demand slot quota per project. Additionally, by establishing a hierarchical priority model for your projects, you could allocate resources based on the specific needs and priorities of each business unit, ensuring that the most critical queries are executed first. This approach would allow you to balance the needs of each business unit while maximizing the use of your BigQuery resources.

hybridpro 1 year, 10 months ago

C. https://cloud.google.com/bigquery/quotas - 2000 is the max no. of slots

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #86 Topic 1

You have an Apache Kafka cluster on-prem with topics containing web application logs. You need to replicate the data to Google Cloud for analysis in BigQuery and Cloud Storage. The preferred replication method is mirroring to avoid deployment of Kafka Connect plugins.
What should you do?

A. Deploy a Kafka cluster on GCE VM Instances. Configure your on-prem cluster to mirror your topics to the cluster running in GCE. Use a Dataproc cluster or Dataflow job to read from Kafka and write to GCS. **Most Voted**

B. Deploy a Kafka cluster on GCE VM Instances with the Pub/Sub Kafka connector configured as a Sink connector. Use a Dataproc cluster or Dataflow job to read from Kafka and write to GCS.

C. Deploy the Pub/Sub Kafka connector to your on-prem Kafka cluster and configure Pub/Sub as a Source connector. Use a Dataflow job to read from Pub/Sub and write to GCS.

D. Deploy the Pub/Sub Kafka connector to your on-prem Kafka cluster and configure Pub/Sub as a Sink connector. Use a Dataflow job to read from Pub/Sub and write to GCS.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Ganshank Highly Voted 4 years, 5 months ago

A. https://cwiki.apache.org/confluence/pages/viewpage.action?pageId=27846330 The solution specifically mentions mirroring and minimizing the use of Kafka Connect plugin. D would be the more Google Cloud-native way of implementing the same, but the requirement is better met by A.

hendrixlives Highly Voted 2 years, 9 months ago

Selected Answer: A

"A" is the answer which complies with the requirements (specifically, "The preferred replication method is mirroring to avoid deployment of Kafka Connect plugins"). Indeed, one of the uses of what is called "Geo-Replication" (or Cross-Cluster Data Mirroring) in Kafka is precisely cloud migrations: https://kafka.apache.org/documentation/#georeplication However I agree with Ganshank, and the optimal "Google way" way would be "D", installing the Pub/Sub Kafka connector to move the data from on-prem to GCP.

Qix 1 year, 3 months ago

Pub/Sub Kafka connector requires Kafka Connect, as described here https://cloud.google.com/pubsub/docs/connect_kafka Deployment of Kafka Connect is explicitly excluded by the

requirements. So the only option available is A

samdhimal 1 year, 8 months ago

Option A: Deploy a Kafka cluster on GCE VM Instances. Configure your on-prem cluster to mirror your topics to the cluster running in GCE. Use a Dataproc cluster or Dataflow job to read from Kafka and write to GCS. This option involves setting up a separate Kafka cluster in Google Cloud, and then configuring the on-prem cluster to mirror the topics to this cluster. The data from the Google Cloud Kafka cluster can then be read using either a Dataproc cluster or a Dataflow job and written to Cloud Storage for analysis in BigQuery.

samdhimal 1 year, 8 months ago

Option B: Deploy a Kafka cluster on GCE VM Instances with the Pub/Sub Kafka connector configured as a Sink connector. Use a Dataproc cluster or Dataflow job to read from Kafka and write to GCS. This option is similar to Option A, but involves using the Pub/Sub Kafka connector as a sink connector instead of mirroring the topics from the on-prem cluster. This option would result in the same duplication of data and additional resources required as Option A, making it less desirable.

samdhimal 1 year, 7 months ago

Sorry. I messed up. The answer is probably A. My badd....

samdhimal 1 year, 8 months ago

Option D: Deploy the Pub/Sub Kafka connector to your on-prem Kafka cluster and configure Pub/Sub as a Sink connector. Use a Dataflow job to read from Pub/Sub and write to GCS. This option involves deploying the Pub/Sub Kafka connector on the on-prem cluster, but configuring it as a sink connector. In this case, the data from the on-prem Kafka cluster would be sent directly to Pub/Sub, which would act as the final destination for the data. A Dataflow job would then be used to read the data from Pub/Sub and write it to Cloud Storage for analysis in BigQuery. This option would result in the data being stored in both the on-prem cluster and Pub/Sub, making it less desirable compared to option C, where the data is only stored in Pub/Sub as an intermediary between the on-prem cluster and Google Cloud.

musumusu 1 year, 7 months ago

you use chatgpt replies, if you instruct chat gpt that you don't need to use plugins as per question say, it will answer A

samdhimal 1 year, 8 months ago

Option C: Deploy the Pub/Sub Kafka connector to your on-prem Kafka cluster and configure Pub/Sub as a Source connector. Use a Dataflow job to read from Pub/Sub and write to GCS. This option involves deploying the Pub/Sub Kafka connector directly on the on-prem cluster, and configuring it as a source connector. The data from the on-prem Kafka cluster is then sent directly to Pub/Sub, which acts as an intermediary between the on-prem cluster and the data stored in Google Cloud. A Dataflow job is then used to read the data from Pub/Sub and write it to Cloud Storage for analysis in BigQuery. This option avoids the duplication of data and additional resources required by the other options, making it the preferred option.

Afonya 1 year, 11 months ago

Selected Answer: A

"The preferred replication method is mirroring to avoid deployment of Kafka Connect plugins."

somnathmaddi 1 year, 11 months ago

D is the right answer

clouditis 2 years ago

D is the right answer

gcp_k 2 years, 11 months ago

Going with "D" Refer: https://stackoverflow.com/questions/55277188/kafka-to-google-pub-sub-using-sink-connector

baubaumiaomiao 2 years, 9 months ago

"avoid deployment of Kafka Connect plugins"

sumanshu 3 years, 3 months ago

Vote for A

daghayeghi 3 years, 6 months ago

Answer: A Description: Question says mirroring to avoid kafka connect plugins

Allan222 3 years, 7 months ago

Correct is D

sumanshu 3 years, 2 months ago

As per question - "avoid deployment of Kafka Connect plugins."

vakati 4 years ago

A. the best solution would be D but given the restriction here to use mirroring and avoid connectors, A would be the natural choice

Tanmoyk 4 years ago

D should be the correct answer. Configure pub/sub as sink

haroldbenites 4 years, 1 month ago

C is correct. https://docs.confluent.io/current/connect/kafka-connect-gcp-pubsub/index.html

haroldbenites 4 years, 1 month ago

Correct Answer: D Why is this correct? You can connect Kafka to GCP by using a connector. The 'downstream' service (Pub/Sub) will use a sink connector.

sumanshu 3 years, 3 months ago

Question says : avoid deployment of Kafka Connect plugins.

clouditis 4 years, 1 month ago

its D, why would google prefer Kafka in their own cert questions! :)

Ral17 3 years ago

Because the questions mentions to avoid deployment of Kafka connect plugins

Archy 4 years, 2 months ago

answer is D, as on-prime kafka support sink connector for outging data.

Rajokkiyam 4 years, 6 months ago

Correct Answer : D.

baubaumiaomiao 2 years, 9 months ago

"avoid deployment of Kafka Connect plugins"

haroldbenites 4 years, 1 month ago

Correct Answer: D Why is this correct? You can connect Kafka to GCP by using a connector. The 'downstream' service (Pub/Sub) will use a sink connector.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #87 Topic 1

You've migrated a Hadoop job from an on-prem cluster to dataproc and GCS. Your Spark job is a complicated analytical workload that consists of many shuffling operations and initial data are parquet files (on average 200-400 MB size each). You see some degradation in performance after the migration to Dataproc, so you'd like to optimize for it. You need to keep in mind that your organization is very cost-sensitive, so you'd like to continue using Dataproc on preemptibles (with 2 non-preemptible workers only) for this workload.
What should you do?

A. Increase the size of your parquet files to ensure them to be 1 GB minimum.

B. Switch to TFRecords formats (appr. 200MB per file) instead of parquet files.

C. Switch from HDDs to SSDs, copy initial data from GCS to HDFS, run the Spark job and copy results back to GCS.

D. Switch from HDDs to SSDs, override the preemptible VMs configuration to increase the boot disk size. **Most Voted**

Correct Answer: D

Community vote distribution

D (47%) A (33%) C (20%)

Comments

rickywck **Highly Voted** 6 years ago

Should be A: https://stackoverflow.com/questions/42918663/is-it-better-to-have-one-large-parquet-file-or-lots-of-smaller-parquet-files https://www.dremio.com/tuning-parquet/ C & D will improve performance but need to pay more $$

diluvio 4 years, 5 months ago

It is A . please read the links above

odacir 3 years, 3 months ago

https://cloud.google.com/architecture/hadoop/migrating-apache-spark-jobs-to-cloud-dataproc#optimize_performance

raf2121 4 years, 8 months ago

Point for discussion - Another reason why it can't be C or D. SSD's are not available on pre-emptible Worker nodes (answers didn't say whether they wanted to switch from HDD to SDD for Master nodes)
https://cloud.google.com/architecture/hadoop/hadoop-gcp-migration-jobs

rr4444 3 years, 9 months ago

You can have local SSDs for the dataproc normal or preemptible VMs
https://cloud.google.com/dataproc/docs/concepts/compute/dataproc-pd-ssd

raf2121 4 years, 8 months ago

Also for Shuffling Operations, one need to override the preemptible VMs configuration to increase boot disk size. (Second half of answer D is correct but first half is wrong)

jin0 3 years, 1 month ago

you are right C&D will pay more $. the reason of this questions is shuffling I think. and to reduce shuffling between jobs then make file size larger

madhu1171 Highly Voted 6 years ago

Answer should be D

jvg637 6 years ago

D: # By default, preemptible node disk sizes are limited to 100GB or the size of the non-preemptible node disk sizes, whichever is smaller. However you can override the default preemptible disk size to any requested size. Since the majority of our cluster is using preemptible nodes, the size of the disk used for caching operations will see a noticeable performance improvement using a larger disk. Also, SSD's will perform better than HDD. This will increase costs slightly, but is the best option available while maintaining costs.

ch3n6 5 years, 9 months ago

C is correct. D is wrong. they are using 'dataproc and GCS', not related to boot disk at all .

VishalB 5 years, 8 months ago

C is recommended only - If you have many small files, consider copying files for processing to the local HDFS and then copying the results back

FARR 5 years, 7 months ago

File sizes are already within the expected range for GCS (128MB-1GB) so not C. D seems most feasible

AdriHubert 10 months, 2 weeks ago

Selected Answer: A

Why this is the better choice: Larger Parquet files reduce the number of splits and metadata overhead. This leads to fewer tasks, less shuffle, and better parallelism. It does not increase infrastructure cost—just improves how data is structured. It aligns with Google’s best practices for Spark on Dataproc.

rajshiv 11 months, 2 weeks ago

Selected Answer: C

C is correct. It cannot be D as increasing boot disk size does not impact shuffle performance much. We need local SSDs specifically attached for shuffle storage (temporary fast storage), not just a bigger persistent boot disk. C is the correct answer because SSD + HDFS shuffle layer = fastest for Spark shuffle-heavy jobs, while still using preemptibles and keeping costs down. The job as mentioned is shuffle-intensive. In Spark, shuffling (moving data between nodes) is heavily disk I/O intensive. Faster local storage (i.e., SSDs) can dramatically speed up shuffle operations compared to using standard HDDs. GCS is great for object storage, not shuffle storage.

rajshiv 11 months, 2 weeks ago

Selected Answer: C

C is the correct answer

oussama7 1 year ago

Selected Answer: C

Improves shuffle management by using HDFS instead of GCS. SSDs speed up access to temporary data. Compatible with Dataproc's preemptible cost model, without requiring more non-preemptible workers.

Parandhaman_Margan 1 year ago

Selected Answer: C

Preemptible Cost Considerations Using preemptibles (with 2 non-preemptible workers) is cost-effective, but shuffle operations still need fast local storage. SSDs improve reliability without increasing instance costs significantly

f74ca0c 1 year, 2 months ago

Selected Answer: A

A. Not D because it doesn't make sens to move to SSD when cost-senstive

forepick 7 months, 1 week ago

Maybe that's the reason they mentioned that only TWO servers are not spot-VMs. Spot VMs are stateless while the others are stateful = have HDs for suffling.

Javakidson 1 year, 4 months ago

A is the answer

SamuelTsch 1 year, 5 months ago

Selected Answer: A

I think either A or C. The problem is occured by I/O performance. Option A is feasible, which reduces the number of files leading better parallel processing. Option C tries to handle I/O performance issue. Taking other factors like budget and no mention of HDD/SSD, option A is possible the correct answer.

baimus 1 year, 6 months ago

Selected Answer: A

There's no mention of a drive type used, only GCS. That means A is the only sensible option.

987af6b 1 year, 8 months ago

Selected Answer: A

Question doesn't actually say they are using HDD in the scenario, for that reason I choose A

philli1011 2 years, 1 month ago

A We don't know if HDD was used, so we can know what to do about that, but we know that the parquet files are small and much, and we can act on that by increasing the sizes to have lesser number of it.

rocky48 2 years, 4 months ago

Selected Answer: A

Should be A: https://stackoverflow.com/questions/42918663/is-it-better-to-have-one-large-parquet-file-or-lots-of-smaller-parquet-files

rocky48 2 years, 3 months ago

Given the scenario and the cost-sensitive nature of your organization, the best option would be: C. Switch from HDDs to SSDs, copy initial data from GCS to HDFS, run the Spark job, and copy results back to GCS. Option C allows you to leverage the benefits of SSDs and HDFS while minimizing costs by continuing to use Dataproc on preemptible VMs. This approach optimizes both performance and cost-effectiveness for your analytical workload on Google Cloud.

Mathew106 2 years, 8 months ago

Selected Answer: A

https://stackoverflow.com/questions/42918663/is-it-better-to-have-one-large-parquet-file-or-lots-of-smaller-parquet-files Cost effective is the key in the question.

Nandhu95 3 years ago

Selected Answer: D

Preemptible VMs can't be used for HDFS storage. As a default, preemptible VMs are created with a smaller boot disk size, and you might want to override this configuration if you are running shuffle-heavy workloads.

midgoo 3 years, 1 month ago

Selected Answer: D

Should NOT be A as: 1. The file size is already at the optimal size 2. If the current file size works well in the current Hadoop, it is expected to have similar performance in Dataproc The only difference between the current and Dataproc is that Dataproc is using preemptible nodes. So yes, it may incur a bit more cost by using SSD but assuming using the preemptible already save most of it, so we want to save less to improve the performance

Mathew106 2 years, 8 months ago

Optimal size is 1GB

https://zhangxuetu.online

[202610]

Question #88 Topic 1

Your team is responsible for developing and maintaining ETLs in your company. One of your Dataflow jobs is failing because of some errors in the input data, and you need to improve reliability of the pipeline (incl. being able to reprocess all failing data).
What should you do?

A. Add a filtering step to skip these types of errors in the future, extract erroneous rows from logs.

B. Add a tryɹ€¦ catch block to your DoFn that transforms the data, extract erroneous rows from logs.

C. Add a tryɹ€¦ catch block to your DoFn that transforms the data, write erroneous rows to Pub/Sub PubSub directly from the DoFn.

D. Add a tryɹ€¦ catch block to your DoFn that transforms the data, use a sideOutput to create a PCollection that can be stored to Pub/Sub later. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

midgoo **Highly Voted** 2 years, 7 months ago

Selected Answer: D

C is a big NO. Writing to PubSub in DoFn will cause bottleneck in the pipeline. For IO, we should always use those IO lib (e.g PubsubIO) Using sideOutput is the correct answer here. There is a Qwiklab about this. It is recommended to do that lab to understand more.

jonathanthezombieboy **Highly Voted** 2 years, 7 months ago

Selected Answer: D

Based on the given scenario, option D would be the best approach to improve the reliability of the pipeline. Adding a try-catch block to the DoFn that transforms the data would allow you to catch and handle errors within the pipeline. However, storing erroneous rows in Pub/Sub directly from the DoFn (Option C) could potentially create a bottleneck in the pipeline, as it adds additional I/O operations to the data processing. Option A of filtering the erroneous data would not allow the pipeline to reprocess the failing data, which could result in data loss. Option D of using a sideOutput to create a PCollection of erroneous data would allow for reprocessing of the failed data and would not create a bottleneck in the pipeline. Storing the erroneous data in a separate PCollection would also make it easier to debug and analyze the failed data. Therefore, adding a try-catch block to the DoFn that transforms the data and using a sideOutput to create a PCollection of erroneous data that can be stored to Pub/Sub later would be the best approach to improve the reliability of the pipeline.

Farah_007 1 year, 5 months ago

Selected Answer: D

I think it's D because here you can write data from Dataflow PCollection to pub/sub.
https://cloud.google.com/dataflow/docs/guides/write-to-pubsub

Mathew106 2 years, 2 months ago

Selected Answer: C

Answer is C. Here is the github repo and an example from the Qwiklab where they tag the output as 'parsed_rows' and 'unparsed_rows' before they send the data to GCS. I don't see how GCS or PubSub would make a difference at this point. It seems like a more maintanable solution to just parse the data in the DoFn. 1) If the function does more than that then it serves multiple purposes and it's not good software engineering. Unless there is a good reason, writing to PubSub should be separated from the DoFn. ii) It's faster to write in mini-batches or one batch than stream the errors. What's the need for streaming out errors 1 by 1? Literally no real advantage.
https://github.com/GoogleCloudPlatform/training-data-analyst/blob/master/quests/dataflow_python/7_Advanced_Streaming_Analytics/solution/streaming_minute_traffic_pipeline.py

tibuenoc 2 years, 6 months ago

Selected Answer: D

Output errors to new PCollection – Send to collector for later analysis (Pub/Sub is a good target)

musumusu 2 years, 7 months ago

Option D is right approach to use to get errors as sideOutput. Apache beam has a special scripting docs not dynamic as python itself. So lets follow standard sideOutput(withoutputs in the code) syntax be like in pipeline: 'ProcessData' >> beam.ParDo(DoFn).with_outputs

musumusu 2 years, 7 months ago

After using you try: Catch: you can also send the erroneous records to dead letter sink into BQ ``` outputTuple.get(deadLetterTag).apply(BigQuery.write(...)) ```

abwey 2 years, 7 months ago

Selected Answer: D

blahblahblahblahblahblahblahblah

waiebdi 2 years, 8 months ago

Selected Answer: D

It`s D. Use a try catch block to direct erroneous rows into a side output. The PCollection of the side output can be sent efficiently to the PubSub topic via Apache Beam PubSubIO. It's not C because C means to sent every single invalid row in a separate request to PubSub which is very inefficient when working with Dataflow as now batching is involved.

hauhau 2 years, 10 months ago

C D: dataflow to pub/sub is weird

Atnafu 2 years, 10 months ago

D Side output is a great manner to branch the processing. Let's take the example of an input data source that contains both valid and invalid values. Valid values must be written in place #1 and the invalid ones in place#2. A naive solution suggests to use a filter and write 2 distinct processing pipelines. However this approach has one main drawback - the input dataset is read twice. If for the mentioned problem we use side outputs, we can still have 1 ParDo transform that internally dispatches valid and invalid values to appropriate places (#1 or #2, depending on value's validity). https://www.waitingforcode.com/apache-beam/side-output-apache-beam/read#:~:text=simple%20test%20cases.-,Side%20output%20defined,-%C2%B6

sfsdeniso 2 years, 10 months ago

Answer is D

cloudmon 2 years, 11 months ago

Selected Answer: C

It's C. In D, "storing to PubSub later" doesn't really make sense.

devaid 2 years, 11 months ago

Selected Answer: C

Answer is C. You need to reprocess all the failling data, and yes, you can use PubSub as a sink, according to the documentation: https://beam.apache.org/documentation/io/connectors/

nickyshil 3 years ago

Answer C

nickyshil 3 years ago

The error records are directly written to PubSub from the DoFn (it’s equivalent in python). You cannot directly write a PCollection to PubSub. You have to extract each record and write one at a time. Why do the additional work and why not write it using PubSubIO in the DoFn itself? You can write the whole PCollection to Bigquery though, as explained in Reference: https://medium.com/google-cloud/dead-letter-queues-simple-implementation-strategy-for-cloud-pub-sub-80adf4a4a800

AWSandeep 3 years ago

Selected Answer: D

D. Add a try-catch block to your DoFn that transforms the data, use a sideOutput to create a PCollection that can be stored to Pub/Sub later.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #89 Topic 1

You're training a model to predict housing prices based on an available dataset with real estate properties. Your plan is to train a fully connected neural net, and you've discovered that the dataset contains latitude and longitude of the property. Real estate professionals have told you that the location of the property is highly influential on price, so you'd like to engineer a feature that incorporates this physical dependency.
What should you do?

A. Provide latitude and longitude as input vectors to your neural net.

B. Create a numeric column from a feature cross of latitude and longitude.

C. Create a feature cross of latitude and longitude, bucketize it at the minute level and use L1 regularization during optimization. **Most Voted**

D. Create a feature cross of latitude and longitude, bucketize it at the minute level and use L2 regularization during optimization.

Correct Answer: C

Community vote distribution

C (52%) | D (28%) | B (16%) | A (4%)

Comments

AHUI **Highly Voted** 3 years, 6 months ago

Ans C, use L1 regularization becuase we know the feature is a strong feature. L2 will evenly distribute weights

dish11dish **Highly Voted** 3 years, 4 months ago

Selected Answer: C

Option C is correct Use L1 regularization when you need to assign greater importance to more influential features. It shrinks less important feature to 0. L2 regularization performs better when all input features influence the output & all with the weights are of equal size.

desertlotus1211 1 year ago

Selected Answer: D

L1 regularization (Option C) would encourage sparsity but may eliminate too many features, which can be detrimental when you need to capture subtle geographic differences

SamuelTsch 1 year, 5 months ago

Selected Answer: D

I would like choose D. L1 will ignore the irrelevant features. However, we know that lat and long are cruial for this model. We can't take away their influences. L2 helps in preventing overfitting.

MohaSa1 1 year, 5 months ago

Selected Answer: A

This does not seems to be useful, minute level bucketizing will create 3,600 possible buckets per degree squared, not logical, and sparse feature space, Option A seems to be a better choice.

Snnnnneee 1 year, 8 months ago

Selected Answer: B

Bucketing into minutes is inaccurate, up to 1.8 km are grouped. Way too much for real estste. Therefore B

uday_examtopic 2 years, 6 months ago

Create a feature cross of latitude and longitude, bucketize it at the minute level and use L2 regularization during optimization. Like option C, we bucketize at the minute level, but this time we apply L2 regularization. L2 regularization, or Ridge Regression, discourages large values of weights in the model without forcing them to become sparse. It can help prevent overfitting, especially when we have a large number of features (as a result of bucketizing and crossing). Given the options, D. Create a feature cross of latitude and longitude, bucketize it at the minute level and use L2 regularization during optimization seems to be the most appropriate. Bucketizing at the minute level captures localized patterns, and L2 regularization can help control the complexity of the model without enforcing sparsity.

ckanaar 2 years, 6 months ago

What does bucketizing at the minute level mean in the context of this question?

Surely1987 2 years, 4 months ago

Coordinates are written with Degrees, minutes and seconds (one minute being equal to about 1.8 km). So you group your coordinates in buckets with a miute precision

FP77 2 years, 7 months ago

Selected Answer: B

I strongly believe it's B.

Mathew106 2 years, 8 months ago

Selected Answer: B

The right answer is B. What the hell does bucketize the feature cross of latitude and longtitude even mean? They are not a time feature. C and D don't even make sense. The L1 regularization is something that doesn't answer anything in the question. The only valid feature engineered here is option B. A is not an engineered feature. Create a feature cross of latitude and longitude, bucketize it at the minute level and use L1 regularization during optimization.

baimus 1 year, 6 months ago

Bucketising means that we're saying "anyone in this square 1.8km (minute) region is considered a single area" - it's actually recommended as a default way to deal with lat/lon, as they don't really work as seperate columns (or at least we'd be hoping the FCNN buckets them intelligently itself, which it won't mostly)

Jojo9400 2 years, 8 months ago

D You have to use L2, since you have create a new variable with two already existing the risk of multicollinearity is high, L1 is good for selecting feature to avoid curse of dimensionality not for multicollinearity

ga8our 2 years, 11 months ago

Why not L2? L2 (Ridge) uses a squared value coefficient as a penalty term to the loss function, while L1 (Lasso) uses an absolute value coefficient. Isn't a squared penalty stronger than an absolute one?
https://towardsdatascience.com/l1-and-l2-regularization-methods-ce25e7fc831c

ckanaar 2 years, 6 months ago

L1 regression forces unimportant coefficients to zero. Since the location is extremely important, L1 will force less important coefficients to zero, thereby further increasing the importance of the location coefficient.

Oleksandr0501 2 years, 11 months ago

gpt: Option C and D suggest bucketizing the feature cross of latitude and longitude at the minute level and using L1 or L2 regularization during optimization. While regularization can help prevent overfitting, bucketizing at such a granular level may not be necessary and could lead to overfitting. It's also not clear how bucketizing at the minute level would capture the spatial relationship between the latitude and longitude features.

PolyMoe 3 years, 2 months ago

Selected Answer: D

D. Create a feature cross of latitude and longitude, bucketize it at the minute level and use L2 regularization during optimization. This will create a new feature that captures the physical dependency of the location of the property on the price, and bucketing it at the minute level will reduce the number of unique values and prevent overfitting. L2 regularization will also help to prevent overfitting by penalizing large weights in the model.

cetanx 2 years, 10 months ago

chat-gpt also says D explanation: This approach effectively creates a grid of the geographical area in your data, allowing the model to learn weights for each grid cell (bucket). This helps capture the spatial relationship between latitude and longitude, which can be crucial for real estate prices. Additionally, using L2 regularization helps prevent overfitting by discouraging complex models, which can be particularly important when working with high-dimensional crossed features.

crismo04 3 years, 6 months ago

https://medium.com/riga-data-science-club/geographic-coordinate-encoding-with-tensorflow-feature-columns-e750ae338b7c#:~:text=to%20the%20rescue!-,Feature%20Crosses,-Combining%20features%20into

crismo04 3 years, 6 months ago

Feature cross seems to be the right feature option

crismo04 3 years, 6 months ago

So it's B option

AWSandeep 3 years, 7 months ago

Selected Answer: C

C. Create a feature cross of latitude and longitude, bucketize it at the minute level and use L1 regularization during optimization.

nwk 3 years, 7 months ago

C or D? https://medium.com/riga-data-science-club/geographic-coordinate-encoding-with-tensorflow-feature-columns-e750ae338b7c

https://zhangxuetu.online

[202610]

Question #90 Topic 1

You are deploying MariaDB SQL databases on GCE VM Instances and need to configure monitoring and alerting. You want to collect metrics including network connections, disk IO and replication status from MariaDB with minimal development effort and use StackDriver for dashboards and alerts.
What should you do?

A. Install the OpenCensus Agent and create a custom metric collection application with a StackDriver exporter.

B. Place the MariaDB instances in an Instance Group with a Health Check.

C. Install the StackDriver Logging Agent and configure fluentd in_tail plugin to read MariaDB logs.

D. Install the StackDriver Agent and configure the MySQL plugin. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

Barniyah **Highly Voted** 4 years, 11 months ago

Answer : A MariaDB needs costume metrics , and stackdriver built-in monitoring tools will not provide these metrics . Opencensus Agent will do this for you For more info , refer to : https://cloud.google.com/monitoring/custom-metrics/open-census

fire558787 3 years, 6 months ago

It is definitely A. B: can't be because Health Checks just checks that machine is online C: StackDriver Logging is for Logging. Here we talk of Monitoring / Alerting D: StackDriver Agent monitors default metrics of VMs and some Database stuff with the MySQL Plugin. Here you want to monitor some more custom stuff like Replication of MariaDB (I didn't find anything of this sort in the plugin page), and you may want to use Custom Metrics rather than default metrics. "Cloud Monitoring automatically collects more than 1,500 built-in metrics from more than 100 monitored resources. But those metrics cannot capture application-specific data or client-side system data. Those metrics can give you information on backend latency or disk usage, but they can't tell you how many background routines your application spawned." https://cloud.google.com/monitoring/custom-metrics/open-census#monitoring_opencensus_metrics_quickstart-python

[Removed] **Highly Voted** 5 years ago

Answer: C Description: The GitHub repository named google-fluentd-catch-all-config which includes the configuration files for the Logging agent for ingesting the logs from various third-party software packages.

Atulthakur 3 years, 7 months ago

I think its D, because its Selfmanaged DB and for this we use Stackdriver Agents. and in this question its asking about metrics not logs.

rocky48 1 year, 3 months ago

Selected Answer: D

Here's the rationale: StackDriver Agent: The StackDriver Agent is designed to collect system and application metrics from virtual machine instances and send them to StackDriver Monitoring. It simplifies the process of collecting and forwarding metrics. MySQL Plugin: The StackDriver Agent has a MySQL plugin that allows you to collect MySQL-specific metrics without the need for additional custom development. This includes metrics related to network connections, disk IO, and replication status – which are the specific metrics you mentioned. Option D is the most straightforward and least development-intensive approach to achieve the monitoring and alerting requirements for MariaDB on GCE VM Instances using StackDriver.

BlehMaks 1 year, 4 months ago

Selected Answer: A

replication status seems to be not included in sql agent metrics. but I do not like A in terms of efforts

ninjatech 2 years ago

it can't be A as it saying minimal development but for opencensus the development is needed.

slade_wilson 2 years, 3 months ago

Selected Answer: A

To use metrics collected by OpenCensus in your Google Cloud project, you must make the OpenCensus metrics libraries and the Stackdriver exporter available to your application. The Stackdriver exporter exports the metrics that OpenCensus collects to your Google Cloud project. You can then use Cloud Monitoring to chart or monitor those metrics.

dish11dish 2 years, 4 months ago

Selected Answer: D

Option D is Correct MariaDB is a community-developed, commercially supported fork of the MySQL relational database management system (RDBMS). To collect logs and metrics for MariaDB, use the mysql receivers. The mysql receiver connects by default to a local MariaDB server using a Unix socket and Unix authentication as the root user. reference:-https://cloud.google.com/stackdriver/docs/solutions/agents/ops-agent/third-party/mariadb

girgu 2 years, 5 months ago

Selected Answer: D

https://cloud.google.com/monitoring/agent/ops-agent/third-party/mariadb

clouditis 2 years, 6 months ago

C is the answer, fluentd plug in is needed as the DB is on GCE

ducc 2 years, 7 months ago

Selected Answer: D

go for D

eRaymox 2 years, 7 months ago

A StackDriver Agent monitors default metrics of VMs and some Database stuff with the MySQL Plugin. Here you want to monitor some more custom stuff like Replication of MariaDB (I didn't find anything of this sort in the plugin page), and you may want to use Custom Metrics rather than default metrics. "Cloud Monitoring automatically collects more than 1,500 built-in metrics from more than 100 monitored resources. But those metrics cannot capture application-specific data or client-side system data. Those metrics can give you information on backend latency or disk usage, but

they can’t tell you how many background routines your application spawned.” https://cloud.google.com/monitoring/custom-metrics/open-census#monitoring_opencensus_metrics_quickstart-python

Kriegs 2 years, 9 months ago

I'm not 100% sure as I have no experience with that issue, but I would say it's D - both A and D should work, but the keyword is "with minimal development effort" (and using pre-built plugin > creating custom metric in terms of simplicity, that's obvious) and all of the relevant data (as per question) should be there: https://cloud.google.com/monitoring/api/metrics_agent#agent-mysql I'm not sure if C would work, but it also seems more advanced in implementation than D. B is 100% wrong and insufficient for that use case. Feel free to prove me wrong :)

NR22 2 years, 11 months ago

A and D both seem like viable options here, unsure which is Google's preferred method as that would be deemed the correct answer in the exam. Any opinions?

Didine_22 2 years, 11 months ago

Selected Answer: D

D mariaDB is an extension of mysql and mysql plugin must work fine to extract the metrics of mariaDB.

ST42 2 years, 10 months ago

"MariaDB is a community-developed, commercially supported fork of the MySQL relational database management system (RDBMS). To collect logs and metrics for MariaDB, use the mysql receivers." https://cloud.google.com/monitoring/agent/ops-agent/third-party/mariadb

rbeeraka 3 years, 2 months ago

Selected Answer: A

Opencensus Agent is right one

nguyenmoon 3 years, 6 months ago

D is correct. Answer : D mariaDB is an extension of mysql and mysql plugin must work fine to extract the metrics of mariaDB.

safiyu 3 years, 7 months ago

Answer : D mariaDB is an extension of mysql and mysql plugin must work fine to extract the metrics of mariaDB. https://cloud.google.com/monitoring/agent/plugins/mysql

https://zhangxuetu.online

[202610]

Question #91 Topic 1

You work for a bank. You have a labelled dataset that contains information on already granted loan application and whether these applications have been defaulted. You have been asked to train a model to predict default rates for credit applicants.
What should you do?

A. Increase the size of the dataset by collecting additional data.

B. Train a linear regression to predict a credit default risk score. **Most Voted**

C. Remove the bias from the data and collect applications that have been declined loans.

D. Match loan applicants with their social profiles to enable feature engineering.

Correct Answer: B

Community vote distribution



Comments

GHN74 Highly Voted 5 years, 7 months ago

A is incorrect as you need to work with the data you have available C is an optimisation not a solution D is ethically incorrect and invasion to privacy, there could be several legal implications with this B although oversimplified but is a workable solution

sergio6 4 years, 5 months ago

Information in social profiles are public

sergio6 4 years, 5 months ago

according to the privacy settings and shareable informations

sumanshu Highly Voted 4 years, 12 months ago

We have labelled data that contains whether a loan application is accepted or defaulted - So Classification Problem Data. We need to predict (Default Rates for applicants) - I think whether application will be granted or defaulted. - So Binary Classification. No option matches the answer. - if we mark 'B' - It should be Logistic Regression, Instead of Linear Regression

szefco 4 years, 3 months ago

Question says: "to predict default RATES for credit applicants". It is not binary classification, so Linear Regression would work here. I think B is correct answer.

cchen8181 2 years, 10 months ago

Correct approach is to use logistic regression to predict default/not default, and then take the confidence/probability of the outcome as the "default rate". Linear regression doesn't make sense since we are not given a default rate label in our data, we are just given the labels default vs no default.

Aaronn14 3 years ago

You cannot predict rate. You predict a realization, which is either default or not. This question is terribly written.

50336e5 3 months, 2 weeks ago

Selected Answer: C

it's not B because it's not a problem of linear regression, it' s a classification !

Skyw4lk3r 1 year, 2 months ago

Selected Answer: A

A. Because the dataset is just of granted loans of the business, but is needed a grater database where to train the model in order to get aceptable results

nairoh 1 year, 5 months ago

"social" does not mean Social Media. This could be linked to demograhic data, si it could improve the score.

TVH_Data_Engineer 2 years, 3 months ago

Selected Answer: B

To predict default rates for credit applicants using the labeled dataset of granted loan applications, the most appropriate course of action would be: B. Train a linear regression to predict a credit default risk score. Here's the rationale for this approach: Appropriate Model for Prediction: Linear regression is a common statistical method used for predictive modeling, particularly when the outcome variable (in this case, the likelihood of default) is continuous. In the context of credit scoring, linear regression can be used to predict a risk score that represents the probability of default. Utilization of Labeled Data: Since you already have a labeled dataset containing information on loans that have been granted and whether they have defaulted, you can use this data to train the regression model. This historical data provides the model with examples of borrower characteristics and their corresponding default outcomes.

rocky48 2 years, 3 months ago

Selected Answer: B

B. Train a linear regression to predict a credit default risk score.

gaurav0480 2 years, 7 months ago

What would be the target variable if B is correct i.e. training a linear regression model? Default/No-Default is a categorical variable one cannot train a linear regression model with this target variable

FDS1993 2 years, 8 months ago

Selected Answer: C

C - it is a typical approach in credit loans. Keeping only the accepted loans leads to a bias in the application

Mathew106 2 years, 8 months ago

Selected Answer: B

Linear regression is not the good way to solve such a problem, but you can totally apply linear regression to solve a classification problem. Just set the labels to numeric values 0 and 1 and linear regression will try to predict a value inbetween and round to the closest label (0 or 1). Totally not the way to go about it, but actually it's possible.

juliobs 3 years ago

Selected Answer: D

Cannot be B. This is logistic regression, not linear regression. D is the only acceptable option. Social profile can include things like high or low income, for example. When you apply for a credit you usually have to give this information, so totally legal.

Oleksandr0501 2 years, 11 months ago

D. Matching loan applicants with their social profiles to enable feature engineering is not recommended as it raises privacy concerns and may not be legal in some jurisdictions. Additionally, social profiles may not be a good indicator of creditworthiness, and relying on them may introduce bias or discrimination.

baimus 1 year, 6 months ago

Credit scores are numbers, so this is a regression. Whether or not a client defaults could be a classification, but the option specifies the use of scores, which is fine.

jin0 3 years, 1 month ago

Because there is no option to know what dataset schema is even though B is needed for this question's purpose Nobody can't select B. So there is none to answer

musumusu 3 years, 1 month ago

all options are wrong: Still in favour of B A: ofc its good to have more data but its not clear how much data we have B: Linear can be a workable approach but current situation is not for linear approach, decision tree, random forest etc can be good for it. C: DAta should be unbiased, removing bias is negative for tranining

Besss 3 years, 2 months ago

Selected Answer: B

default rates can be predicted with linear regression.

21c17b3 2 years, 3 months ago

default rates is classification probability

Whoswho 3 years, 3 months ago

Answer should actually be a logistic Regression model

ladistar 3 years, 4 months ago

The question asks about default RATES, as in you are predicting a continuous variable, not a discrete one (classification). This is a regression problem, so choice B.

woyaolai 3 years, 4 months ago

I used to be a Credit Risk modeler and I think this question is stupid.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #92 Topic 1

You need to migrate a 2TB relational database to Google Cloud Platform. You do not have the resources to significantly refactor the application that uses this database and cost to operate is of primary concern.
Which service do you select for storing and serving your data?

A. Cloud Spanner

B. Cloud Bigtable

C. Cloud Firestore

D. Cloud SQL **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

midgoo 2 years, 7 months ago

Selected Answer: D

Cloud SQL: max storage for shared core = 3TB and for dedicated core = up to 64TB Only use Spanner if we need autoscale (Note that Cloud SQL could scale too but not automatic yet) or the size is too big (as above) or 4/5 9s HA (Cloud SQL is only 99.95)

Nirca 2 years, 11 months ago

Selected Answer: D

Cloud SQL is relational DB (pg mssql, mysql)

Dhass 3 years, 3 months ago

Answer - D

homaj 3 years, 4 months ago

Selected Answer: D

answer D

sumanshu 4 years, 3 months ago

Vote for D

daghayeghi 4 years, 6 months ago

D: https://cloud.google.com/sql/docs/features

GypsyMonkey 4 years, 9 months ago

D, cloud SQL is a relational database; if > 10tb, then choose spanner

atnafu2020 5 years, 1 month ago

D Cloud SQL supports MySQL 5.6 or 5.7, and provides up to 624 GB of RAM and 30 TB of data storage, with the option to automatically increase the storage size as needed.

Abhi16820 3 years, 10 months ago

64TB AS OF TODAY

haroldbenites 5 years, 1 month ago

D is correct. Obviously

Barniyah 5 years, 5 months ago

But cloud SQL storage is limited to several hundreds of GB's for all instances and we need 2TB. So, Cloud spanner is much closer to this, with the exception of the cost

taepyung 5 years, 5 months ago

At this moment, Cloud SQL is providing up to 30,720GB(about 30TB) So I think it's D.

Preetmehta1234 1 year, 7 months ago

Nope. Now, the Dedicated core is Up to 64 TB https://cloud.google.com/sql/docs/quotas

Barniyah 5 years, 4 months ago

Sorry , I think it's D https://cloud.google.com/sql/docs/features (Cloud SQL supports MySQL 5.6 or 5.7, and provides up to 416 GB of RAM and 30 TB of data storage, with the option to automatically increase the storage size as needed.)

xrun 4 years, 9 months ago

Another consideration is that Cloud SQL uses standard databases like MySQL, PostgreSQL and now MS SQL. Cloud Spanner is a proprietary product of Google and does some things differently than typical databases (no stored procedures and triggers). So migrating to Cloud Spanner makes application refactoring necessary. So Cloud SQL is the answer.

LaxmanTiwari 1 year, 9 months ago

Well explained I can confirmed.

https://zhangxuetu.online

[202610]

Question #93 Topic 1

You're using Bigtable for a real-time application, and you have a heavy load that is a mix of read and writes. You've recently identified an additional use case and need to perform hourly an analytical job to calculate certain statistics across the whole database. You need to ensure both the reliability of your production application as well as the analytical workload.
What should you do?

A. Export Bigtable dump to GCS and run your analytical job on top of the exported files.

B. Add a second cluster to an existing instance with a multi-cluster routing, use live-traffic app profile for your regular workload and batch-analytics profile for the analytics workload.

C. Add a second cluster to an existing instance with a single-cluster routing, use live-traffic app profile for your regular workload and batch-analytics profile for the analytics workload. Most Voted

D. Increase the size of your existing cluster twice and execute your analytics workload on your new resized cluster.

Correct Answer: C

Community vote distribution



Comments

aewis Highly Voted 2 years, 8 months ago

Selected Answer: C

It was actually illustrated here https://cloud.google.com/bigtable/docs/replication-settings#batch-vs-serve

oussama7 1 year ago

Selected Answer: B

Option C offers single-cluster routing, meaning that each query is directed to a single specific cluster. This does not protect the transactional workload from heavy analytical loads, which can lead to performance degradation. With multi-cluster routing (option B), Bigtable can automatically distribute the load and avoid congestion on a single cluster.

desertlotus1211 1 year ago

Selected Answer: B

Multi-cluster satisfy reliability aspect of the question. single cluster may and will cause contention for resources...

baimus 1 year, 6 months ago

Selected Answer: C

To answer some confusion - "single cluster routing" is routing to one cluster per profile, rather than having failover options per profile. So we have two clusters, but it's not multicluster, because we have two profiles, so it's one cluster per profile, so "single cluster routing". We COULD use multicluster, but none of the answers give the steps required to do so, so the assumption as to be that we're using single.

baimus 1 year, 6 months ago

(an example of multicluster in this case would be 4 clusters, 2 for the transactional load and 2 for the analytical load)

47767f9 1 year, 9 months ago

Selected Answer: B

B better than C. Multi-cluster routing to handle failovers automatically. Reference: https://cloud.google.com/bigtable/docs/replication-settings#regional-failover

opt_sub 2 years ago

Selected Answer: B

B is correct. Two different job profiles to redirect trafiic to two different cluster. C is incorrect because there is no tpoint in creating app profile for two different workloads in the same cluster. One cluster handles writes and another handle reads.

carbino 2 years, 3 months ago

Selected Answer: C

IIt is C: "Workload isolation: Using separate app profiles lets you use different routing policies for different purposes. For example, consider a situation when you want to prevent a batch read job (workload A) from increasing CPU usage on clusters that handle an application's steady reads and writes (workload B). You can create an app profile for workload B that routes to a cluster group that excludes one cluster. Then you create an app profile for workload A that specifies single-cluster routing to the cluster that workload B doesn't send requests to. You can change the settings for one application or function without affecting other applications that connect to the same data." https://cloud.google.com/bigtable/docs/app-profiles

DevShah 2 years, 11 months ago

Selected Answer: C

https://cloud.google.com/bigtable/docs/replication-settings#batch-vs-serve

A4M 2 years, 11 months ago

I see what you say on C but the question states high availability how do you handle that with option C when you have a single region cluster hence answer needs to be with multi-region cluster - To configure your instance for a high availability (HA) use case, create a new app profile that uses multi-cluster routing, or update the default app profile to use multi-cluster routing.

A4M 2 years, 11 months ago

i meant single-cluster routing

zevexWM 1 year, 11 months ago

It actually addresses the issue of High availability in that same link if you scroll down a bit more. https://cloud.google.com/bigtable/docs/replication-settings#high-availability

juliobs 3 years ago

Selected Answer: C

C. This is exactly the example in the documentation. https://cloud.google.com/bigtable/docs/replication-settings#batch-vs-serve

DevShah 2 years, 11 months ago

Correct 2 jobs >> 2 cluster 3 jobs >> 3 cluster app profiles with single-cluster routing used to route to specific cluster Job1 >> Cluster 1 Job2 >> Cluster 2 .....

musumusu 3 years, 1 month ago

Answer B: reason 1: If you don' t have any cost constraint use multi-cluster routing, reason 2: Single cluster is less scalable as we need high scalability i would go with B

samdhimal 3 years, 1 month ago

Selected Answer: C

I am going for C?

slade_wilson 3 years, 3 months ago

Selected Answer: C

When you use a single cluster to run a batch analytics job that performs numerous large reads alongside an application that performs a mix of reads and writes, the large batch job can slow things down for the application's users. With replication, you can use app profiles with single-cluster routing to route batch analytics jobs and application traffic to different clusters, so that batch jobs don't affect your applications' users. Single cluster routing - You can use single-cluster routing for this use case if you don't want your Bigtable cluster to automatically fail over if a zone or region becomes unavailable. Multi-cluster routing - If you want Bigtable to automatically fail over to one region if your application cannot reach the other region, use multi-cluster routing.

Siant_137 3 years, 3 months ago

Answer is C "When you use a single cluster to run a batch analytics job that performs numerous large reads alongside an application that performs a mix of reads and writes, the large batch job can slow things down for the application's users. With replication, you can use app profiles with single-cluster routing to route batch analytics jobs and application traffic to different clusters, so that batch jobs don't affect your applications' users." https://cloud.google.com/bigtable/docs/replication-overview#batch-vs-serve

sfsdeniso 3 years, 4 months ago

Answer is C

dish11dish 3 years, 4 months ago

Selected Answer: B

Option B is correct An app profile specifies the routing policy that Bigtable should use for each request. Single-cluster routing routes all requests to 1 cluster in your instance. If that cluster becomes unavailable, you must manually fail over to another cluster. Multi-cluster routing automatically routes requests to the nearest cluster in an instance. If the cluster becomes unavailable, traffic automatically fails over to the nearest cluster that is available. Bigtable considers clusters in a single region to be equidistant, even though they are in different zones. You can configure an app profile to route to any cluster in an instance, or you can specify a cluster group that tells the app profile to route to only some of the clusters in the instance. Cluster group routing sends requests to the nearest available cluster within a cluster group that you specify in the app profile settings. Reference:-https://cloud.google.com/bigtable/docs/app-profiles#routing

piotrpiskorski 3 years, 4 months ago

Selected Answer: C

https://cloud.google.com/bigtable/docs/replication-settings#batch-vs-serve "When you use a single cluster to run a batch analytics job that performs numerous large reads alongside an application that performs a mix of reads and writes, the large batch job can slow things down for the application's users. With replication, you can use app profiles with single-cluster routing to route batch analytics jobs and application traffic to different clusters, so that batch jobs don't affect your applications' users." It is C.

gudiking 3 years, 4 months ago

Selected Answer: C

C - "With replication, you can use app profiles with single-cluster routing to route batch analytics jobs and application traffic to different clusters, so that batch jobs don't affect your applications' users." - https://cloud.google.com/bigtable/docs/replication-overview#batch-vs-serve

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #94 Topic 1

You are designing an Apache Beam pipeline to enrich data from Cloud Pub/Sub with static reference data from BigQuery. The reference data is small enough to fit in memory on a single worker. The pipeline should write enriched results to BigQuery for analysis. Which job type and transforms should this pipeline use?

A. Batch job, PubSubIO, side-inputs

B. Streaming job, PubSubIO, JdbcIO, side-outputs

C. Streaming job, PubSubIO, BigQueryIO, side-inputs **Most Voted**

D. Streaming job, PubSubIO, BigQueryIO, side-outputs

Correct Answer: C

Community vote distribution

C (94%) B (6%)

Comments

rickywck **Highly Voted** 5 years, 6 months ago

Why not C? Without BigQueryIO how can data be written back to BigQuery?

xq 5 years, 6 months ago

C should be right

pals_muthu **Highly Voted** 4 years, 1 month ago

Answer is C, You need pubsubIO and BigQueryIO for streaming data and writing enriched data back to BigQuery. side-inputs are a way to enrich the data https://cloud.google.com/architecture/e-commerce/patterns/slow-updating-side-inputs

JOKKUNO 1 year, 9 months ago

Side inputs In addition to the main input PCollection, you can provide additional inputs to a ParDo transform in the form of side inputs. A side input is an additional input that your DoFn can access each time it processes an element in the input PCollection. When you specify a side input, you create a view of some other data that can be read from within the ParDo transform's DoFn while processing each element. Side inputs are useful if your ParDo needs to inject additional data when processing each element in the input PCollection, but the additional data needs to be determined at runtime (and not hard-coded). Such values might be determined by the input data, or depend on a different branch of your pipeline.

JOKKUNO 1 year, 9 months ago

https://beam.apache.org/documentation/programming-guide/#side-inputs

piyush7777 2 years, 1 month ago

Why not side-output?

TQM__9MD 2 years, 1 month ago

Selected Answer: B

B. Use multi-cluster routing to add a second cluster to the existing instance, utilizing a live traffic app profile for the regular workload and a batch analytics profile for the analytical workload.

Mathew106 2 years, 2 months ago

Selected Answer: C

The answer is C. It's a trap so that you answer A because of batch vs streaming but you need BigQueryIO. On the other hand, streaming is absolutely redundant here and will incur extra costs. C is right but would be better with batch.

Siadd 2 years, 9 months ago

A is the Answer. A. Batch job, PubSubIO, side-inputs

sedado77 3 years ago

Selected Answer: C

I got this question on sept 2022. Answer is C

chrismayola 2 years, 11 months ago

dear can you please help, i have some questions about how to prepare the cerification exam using this questionnaire. this is my email , ping me to have some conversation

alex12441 3 years, 8 months ago

Selected Answer: C

Answer: C

medeis_jar 3 years, 8 months ago

Selected Answer: C

I vote for C, because data will come from Pub/Sub, so it should be streaming, we'll need PubSubIO to be able to read from PubSub and BigQueryIO to be able to write to BigQuery, finally the side-inputs pattern let us enrich data

MaxNRG 3 years, 9 months ago

Selected Answer: C

Static reference data from BigQuery will go as side-inputs and data from pub-sub will go as streaming data using PubSubIO and finally BigQueryIO is required to push the final data to BigQuery

JG123 3 years, 10 months ago

Ans: C

Meuter 4 years, 1 month ago

I choose C, because data will come from Pub/Sub, so it should be streaming, we'll need PubSubIO to be able to read from PubSub y BigQueryIO to be able to write to BigQuery, finally the side-inputs pattern let us enrich data
https://beam.apache.org/releases/javadoc/2.4.0/org/apache/beam/sdk/io/gcp/pubsub/PubsubIO.html
https://cloud.google.com/architecture/e-commerce/patterns/slow-updating-side-inputs
https://beam.apache.org/releases/javadoc/2.3.0/org/apache/beam/sdk/io/gcp/bigquery/BigQueryIO.html

daghayeghi 4 years, 6 months ago

C: we have to use Streaming job because of Pub/Sub, and side-input thanks to static reference data. and we have to leverage BigQueryIO since finally we want to write data to BigQuery. then C is the correct answer.

someshsehgal 4 years, 7 months ago

Correct A. batch is cost-effective and no need to go for streaming

funtoosh 4 years, 7 months ago

How you are going to write back to BQ?

someshsehgal 4 years, 7 months ago

Correct A: There are two points to defend it. a. Its batch hence free of cost b. static reference data hence no need to opt for streaming

arghya13 4 years, 11 months ago

After so much confusion I think correct answer is B..reason 1.PubSub is always used for streaming input..no reason to mention 2.JdbcIO can be used by apache beam to connect bigquery to use a lookup 3.Sideoutput as a result to put in bigquery

https://zhangxuetu.online

[202610]

Question #95 Topic 1

You have a data pipeline that writes data to Cloud Bigtable using well-designed row keys. You want to monitor your pipeline to determine when to increase the size of your Cloud Bigtable cluster. Which two actions can you take to accomplish this? (Choose two.)

A. Review Key Visualizer metrics. Increase the size of the Cloud Bigtable cluster when the Read pressure index is above 100.

B. Review Key Visualizer metrics. Increase the size of the Cloud Bigtable cluster when the Write pressure index is above 100.

C. Monitor the latency of write operations. Increase the size of the Cloud Bigtable cluster when there is a sustained increase in write latency. **Most Voted**

D. Monitor storage utilization. Increase the size of the Cloud Bigtable cluster when utilization increases above 70% of max capacity. **Most Voted**

E. Monitor latency of read operations. Increase the size of the Cloud Bigtable cluster of read operations take longer than 100 ms.

Correct Answer: CD

Community vote distribution



Comments

jvg637 **Highly Voted** 5 years, 6 months ago

Answer is C & D. C –> Adding more nodes to a cluster (not replication) can improve the write performance https://cloud.google.com/bigtable/docs/performance D –> since Google recommends adding nodes when storage utilization is > 70% https://cloud.google.com/bigtable/docs/modifying-instance#nodes

sergio6 3 years, 12 months ago

Adding nodes to the cluster In Bigtable scales linearly the performances both read and write https://cloud.google.com/bigtable/docs/performance#typical-workloads

dabrat 4 years, 12 months ago

Storage utilization (% max) The percentage of the cluster's storage capacity that is being used. The capacity is based on the number of nodes in your cluster. In general, do not use more than 70% of the hard limit on total storage, so you have room to add more data. If you do not plan to add significant amounts of data to your instance, you can use up to 100% of the hard limit. Important: If any cluster in an instance exceeds the hard limit on the amount of storage per node, writes to all clusters in that instance will fail until you add nodes to each cluster that is over the limit. Also, if you

try to remove nodes from a cluster, and the change would cause the cluster to exceed the hard limit on storage, Cloud Bigtable will deny the request. If you are using more than the recommended percentage of the storage limit, add nodes to the cluster. You can also delete existing data, but deleted data takes up more space, not less, until a compaction occurs.

dabrat 4 years, 12 months ago

https://cloud.google.com/bigtable/docs/monitoring-instance

Barniyah Highly Voted 5 years, 5 months ago

Key visualizer is bigtable metric , So A and B incorrect storage utilization also bigtable metric , So D incorrect The question want you to monitor pipeline metrics (which is dataflow metrics) , in our case we can only monitor latency . The answer will be : C & E

ch3n6 5 years, 3 months ago

No. it is C, D. "You have a data pipeline that writes data to Cloud Bigtable using well-designed row keys." why are you monitoring read anyway? you are just writing.

Parandhaman_Margan 1 year ago

Selected Answer: BC

Correct answers are **B** (Write pressure) and **C** (latency)

TVH_Data_Engineer 1 year, 4 months ago

Selected Answer: BC

The question focus is on writing. BC is correct. when the writing pressure is above 100, it is time to increase. same logic with C

musumusu 2 years, 7 months ago

why not B ?

musumusu 2 years, 7 months ago

i am feeling to go with B and D. In option C, when latency is low, latency can be low for write operation for other reason. but in option B, its showing clearly when write pressure more than 100. But why no one is talking about B

RoshanAshraf 2 years, 8 months ago

Selected Answer: CD

Key visualizer is Metrics for Performance issues. Ruled out Storage and Write Operations ; C and D

John_Pongthorn 3 years ago

Selected Answer: CD

Well-designed row key : A B are not nessary Write : CD both are involved in the question the most.

Fezo 3 years, 2 months ago

Answer: CD https://cloud.google.com/bigtable/docs/scaling

medeis_jar 3 years, 8 months ago

Selected Answer: CD

as explained by MaxNRG

MaxNRG 3 years, 9 months ago

Selected Answer: CD

D: In general, do not use more than 70% of the hard limit on total storage, so you have room to add more data. If you do not plan to add significant amounts of data to your instance, you can use up to 100% of the hard limit C: If this

value is frequently at 100%, you might experience increased latency. Add nodes to the cluster to reduce the disk load percentage. The key visualizer metrics options, suggest other things other than increase the cluster size. https://cloud.google.com/bigtable/docs/monitoring-instance

hendrixlives 3 years, 9 months ago

Selected Answer: CD

CD. I agree with jvg637

StefanoG 3 years, 10 months ago

Selected Answer: AD

from https://cloud.google.com/bigtable/docs/monitoring-instance Disk load - If this value is frequently at 100%, you might experience increased latency. Add nodes to the cluster to reduce the disk load percentage. Storage utilization (% max) - In general, do not use more than 70% of the hard limit on total storage, so you have room to add more data. If you do not plan to add significant amounts of data to your instance, you can use up to 100% of the hard limit.

KokkiKumar 3 years, 10 months ago

I am Voting for CD

u_t_s 3 years, 12 months ago

Answer should be D & E

tavva_prudhvi 3 years, 5 months ago

Why are you monitoring read operations, when youre supposed to write? why E?

sergio6 3 years, 12 months ago

D--> 70% is the the recommended percentage of the cluster's storage capacity that is being used, If you are using more than 70% storage limit, add nodes to the cluster https://cloud.google.com/bigtable/quotas#storage-per-node https://cloud.google.com/bigtable/docs/monitoring-instance#disk E--> 100 ms is an order of magnitude lower latency than Google claimed (<10ms) https://cloud.google.com/bigtable/docs/performance#typical-workloads

hauhau 4 years, 1 month ago

BC D:you can just add node, not cluster The percentage of the cluster's storage capacity that is being used. The capacity is based on the number of nodes in your cluster.(https://cloud.google.com/bigtable/docs/monitoring-instance) After you create a Cloud Bigtable instance, you can update any of the following settings without any downtime: (The number of nodes in each cluster) https://cloud.google.com/bigtable/docs/modifying-instance

sumanshu 4 years, 3 months ago

B, C , D - all three looks okay to me

sumanshu 4 years, 3 months ago

Vote for C & D, Option B eliminated, as Row are well defined (as per question) - so no need of key-visualizer

squishy_fishy 3 years, 11 months ago

Answer is C, D. B is Key Visualizer, it means the row key needs re-design again.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #96 Topic 1

You want to analyze hundreds of thousands of social media posts daily at the lowest cost and with the fewest steps.
You have the following requirements:
- You will batch-load the posts once per day and run them through the Cloud Natural Language API.
- You will extract topics and sentiment from the posts.
- You must store the raw posts for archiving and reprocessing.
- You will create dashboards to be shared with people both inside and outside your organization.

You need to store both the data extracted from the API to perform analysis as well as the raw social media posts for historical archiving. What should you do?

A. Store the social media posts and the data extracted from the API in BigQuery.

B. Store the social media posts and the data extracted from the API in Cloud SQL.

C. Store the raw social media posts in Cloud Storage, and write the data extracted from the API into BigQuery. **Most Voted**

D. Feed to social media posts into the API directly from the source, and write the extracted data from the API into BigQuery.

Correct Answer: C

Community vote distribution

C (100%)

Comments

psu **Highly Voted** 4 years, 5 months ago

Answer should be C, becose they ask you to save a copy of the raw posts for archival, which may not be possible if you directly feed the posts to the API.

sedado77 **Highly Voted** 2 years ago

Selected Answer: C

I got this question on sept 2022. Answer is C

itz_me_sudhir 1 year, 7 months ago

can any one help me with the rest of question from 101 to 209 as i dont have a contributor access

Erso 2 years ago

Selected Answer: C

C is the correct one

medeis_jar 2 years, 8 months ago

Selected Answer: C

Only C make sense.

MaxNRG 2 years, 9 months ago

Selected Answer: C

You must store the raw posts for archiving and reprocessing, Store the raw social media posts in Cloud Storage. B is expensive D is not valid since you have to store the raw posts for archiving Between A and C I's say C, since we're going to make dashboards and Data Studio will connect well with big query. and besides A would probably be more expensive.

BigQuery 2 years, 10 months ago

SAY MY NAME!

StefanoG 2 years, 10 months ago

Selected Answer: C

Analysis BQ Storage GCS

fire558787 3 years, 1 month ago

I believe the API accesses data only from GCS Buckets not BigQuery (but I'm not entirely sure)

sumanshu 3 years, 3 months ago

Vote for C

DPonly 3 years, 8 months ago

Answer should be C because we need to consider storage archival

arghya13 3 years, 10 months ago

I'll go with option C

Alasmindas 3 years, 10 months ago

I will go with Option C, because of the following reasons:- a) Social media posts are "raw" - which means - it can be of any format (blob/object storage) is preferred. b) The output from the application (assuming the application is Cloud NLP) is to be future stored for archival purpose - and hence again Google Cloud storage is the best option - so option C Option A &C - Incorrect, although Option D fulfils the requirement of "fewest step" but storing data in big query for archival purpose is not a google recommended approach Option B : Cloud SQL rules out as it does not solve either for archival storage or for analytics purpose.

singhkrishna 4 years ago

cost of long term storing is almost same in GCS and BQ, so answer D makes sense from that angle..

Tanmoyk 4 years ago

The job is supposed to run in batch process once in a day , so there is no requirement of stream data. The best economical and less complex steps is answer C

Ravivarma4786 4 years, 1 month ago

Bigquery is suitable for social media post ans should be C

tprashanth 4 years, 2 months ago

C. First store the data on GCS, then extract only the relavant info for analysis and load into BQ. This way, huge data ie., audio, videos can stay on GCS (not lost). BQ cannot store audio/video. And note that Cloud Natural Language API is used for analysis which uses Text as it's source

https://zhangxuetu.online

[202610]

Question #97 Topic 1

You store historic data in Cloud Storage. You need to perform analytics on the historic data. You want to use a solution to detect invalid data entries and perform data transformations that will not require programming or knowledge of SQL.
What should you do?

A. Use Cloud Dataflow with Beam to detect errors and perform transformations.

B. Use Cloud Dataprep with recipes to detect errors and perform transformations. **Most Voted**

C. Use Cloud Dataproc with a Hadoop job to detect errors and perform transformations.

D. Use federated tables in BigQuery with queries to detect errors and perform transformations.

Correct Answer: B

Community vote distribution



Comments

cleroy Highly Voted 6 years ago

Use Dataprep ! It's THE tool for this

rickywck Highly Voted 6 years ago

Yes B. Honest speaking, sometime I thought the answers being posted here were intentionally to mislead people whose do not have proper knowledge on the subject, but just memorizing answers to pass the exam.

sergiomujica 2 years, 6 months ago

Selected Answer: A

A is the right way to do it... dataprepo is clumsy

Wudihero2 2 years, 5 months ago

...Did you even read through the question? It says "not require programming or knowledge of SQL". YOU are the one who's clumsy, not dataprep.

crazycosmos 2 years, 8 months ago

Selected Answer: B

no programming -> B

FP77 2 years, 8 months ago

Selected Answer: B

I honestly do not understand what is the deal with this website. The correct answer is obviously Dataprep. How can they say it's A?

Besss 3 years, 2 months ago

Selected Answer: B

It'B. DataPrep it's the right tool. https://cloud.google.com/dataprep

sedado77 3 years, 6 months ago

I got this question on sept 2022.

John_Pongthorn 3 years, 6 months ago

Selected Answer: B

B. Actually there are two tools to fix this problem. Dataprep rely on dataflw Datafusion rely on dataproc

baimus 1 year, 6 months ago

Datafusion is lo-code but not no-code. The only no-code system is dataprep.

diagniste 3 years, 10 months ago

Selected Answer: A

A is the best answer !

desertlotus1211 3 years, 2 months ago

dataflow IS Apache Beam...

Venkat007 4 years, 1 month ago

Selected Answer: B

B Dataprep

medeis_jar 4 years, 2 months ago

Selected Answer: B

https://cloud.google.com/dataprep/

MaxNRG 4 years, 3 months ago

Selected Answer: B

B, “Cloud Dataprep by Trifacta is an intelligent data service for visually exploring, cleaning, and preparing structured and unstructured data for analysis, reporting, and machine learning” https://cloud.google.com/dataprep/

dattatray_shinde 3 years, 8 months ago

max you rock man!

GirijaSrinivasan 4 years, 5 months ago

Answer is B. Data prep. The keyword here is no programming skills required.

nguyenmoon 4 years, 6 months ago

B- Dataprep

pass_gcp 4 years, 7 months ago

Use Dataprep ....is the answer

sumanshu 4 years, 9 months ago

Vote for B

sumanshu 4 years, 9 months ago

Cloud Dataprep - almost fully automated

AnilKr 5 years, 2 months ago

Most of the answers are wrong, this is simple even.

https://zhangxuetu.online

[202610]

Question #98 Topic 1

Your company needs to upload their historic data to Cloud Storage. The security rules don't allow access from external IPs to their on-premises resources. After an initial upload, they will add new data from existing on-premises applications every day. What should they do?

A. Execute gsutil rsync from the on-premises servers. **Most Voted**

B. Use Dataflow and write the data to Cloud Storage.

C. Write a job template in Dataproc to perform the data transfer.

D. Install an FTP server on a Compute Engine VM to receive the files and move them to Cloud Storage.

Correct Answer: A

Community vote distribution

A (100%)

Comments

itche_scratche **Highly Voted** 5 years, 11 months ago

should be A, dataflow is on cloud is external; "don't allow access from external IPs to their on-premises resources" so no dataflow.

MaxNRG **Highly Voted** 4 years, 3 months ago

Selected Answer: A

A is the better and most simple IF there is no problem in having gsutil in our servers. B and C no way, the comms will go GCP–Home, which sais is not allowed. D is valid, we can send the files with http://ftp…BUT ftp is not secure, and we'll need to move them to the cloud storage afterwards, which is not detailed in the answer.
https://cloud.google.com/storage/docs/gsutil/commands/rsync

shroffshivangi 1 year, 8 months ago

The gcloud storage command is the standard tool for small- to medium-sized transfers over a typical enterprise-scale network, from a private data center or from another cloud provider to Google Cloud.

Besss 3 years, 2 months ago

Selected Answer: A

A is correct

somnathmaddi 3 years, 5 months ago

Selected Answer: A

Should be A

medeis_jar 4 years, 2 months ago

Selected Answer: A

Without this "The security rules don't allow access from external IPs to their on-premises resources" B would be an answer.

am2005 4 years, 3 months ago

I am confused . which one is correct A or B ???

hendrixlives 4 years, 3 months ago

Selected Answer: A

A is correct.

Chelseajcole 4 years, 5 months ago

This is the link:https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#gsutil_for_smaller_transfers_of_on-premises_data

manocha_01887 4 years, 7 months ago

How rsynch will handle private network? "..The security rules don't allow access from external IPs to their on-premises resources.."

sumanshu 4 years, 9 months ago

Vote for A

daghayeghi 5 years ago

A: https://cloud.google.com/solutions/migration-to-google-cloud-transferring-your-large-datasets#options_available_from_google

maurodipa 4 years, 3 months ago

How could gsutil connect to Cloud Storage, if there is not access from external IPs? Should I understand that there is not access from outside to inside, but it is possible to send from inside to outside?

szefco 4 years, 3 months ago

Yes. There is no access to on-prem from external IPs, but on prem can talk to external

Ravivarma4786 5 years, 7 months ago

gsutil rsync will be used to transfer the files ANS A

haroldbenites 5 years, 7 months ago

A is correct

VishalB 5 years, 8 months ago

Ans : A The gsutil rsync command makes the contents under dst_url the same as the contents under src_url, by copying any missing files/objects (or those whose data has changed), and (if the -d option is specified) deleting any extra files/objects. src_url must specify a directory, bucket, or bucket subdirectory

Devx198912233 5 years, 8 months ago

option A https://cloud.google.com/solutions/migration-to-google-cloud-transferring-your-large-datasets#options_available_from_google

https://zhangxuetu.online

[202610]

Question #99 Topic 1

You have a query that filters a BigQuery table using a WHERE clause on timestamp and ID columns. By using bq query `"-dry_run you learn that the query triggers a full scan of the table, even though the filter on timestamp and ID select a tiny fraction of the overall data. You want to reduce the amount of data scanned by BigQuery with minimal changes to existing SQL queries. What should you do?

A. Create a separate table for each ID.

B. Use the LIMIT keyword to reduce the number of rows returned.

C. Recreate the table with a partitioning column and clustering column. **Most Voted**

D. Use the bq query --maximum_bytes_billed flag to restrict the number of bytes billed.

Correct Answer: C

Community vote distribution

C (100%)

Comments

rickywck **Highly Voted** 4 years ago

should be C: https://cloud.google.com/bigquery/docs/best-practices-costs

Crudgey **Highly Voted** 2 years, 4 months ago

Are they having a laugh at us by giving so many bad answers?

Fezo 1 year, 9 months ago

Selected Answer: C

C is the answer https://cloud.google.com/bigquery/docs/best-practices-costs

medeis_jar 2 years, 2 months ago

Selected Answer: C

C only make sense

MaxNRG 2 years, 3 months ago

Selected Answer: C

https://cloud.google.com/bigquery/docs/best-practices-costs Applying a LIMIT clause to a SELECT * query does not affect the amount of data read. You are billed for reading all bytes in the entire table, and the query counts against your free tier quota. A and D doesnt make sense Its C, when you want to select by a partition you should write something like: CREATE TABLE `blablabla.partitioned` PARTITION BY DATE(timestamp) CLUSTER BY id AS SELECT * FROM `blablabla`

Anilgcp980 2 years, 3 months ago

this is a trap to make people fail by giving wrong answer as B.

snadaf 2 years, 4 months ago

It's D, here is the link https://cloud.google.com/bigquery/docs/best-practices-costs

maurodipa 2 years, 3 months ago

Well, you mean C, isn't it?

tsoetan001 2 years, 5 months ago

Answer: B Note: minimal change to sql

szefco 2 years, 4 months ago

Not B. LIMIT will not reduce amount of data scanned - only limit the final output, but you will still be billed for scanning whole table. C is correct. After applying partitioning ans clustering amount of bytes scanned will decrease

Ysance_AGS 2 years, 6 months ago

"You want to reduce the amount of data scanned by BigQuery with minimal changes to existing SQL queries" that doesn't mean that you can create or edit existing tables ! you only can edit the SQL query !!! so answer D is the correct one.

squishy_fishy 2 years, 5 months ago

D would just block your query. The answer is C.

szefco 2 years, 4 months ago

I don't agree. Question says "minimal changes to existing SQL queries" - if you recreate table with partitioning and clustering you don't need to change SQLs that read from that table. C is correct answer here.

nguyenmoon 2 years, 6 months ago

C - create partition table

sumanshu 2 years, 9 months ago

Vote for C

felixwtf 3 years, 3 months ago

LIMIT keyword is applied only at the end, i.e., only to limit the results already calculated. Therefore, a full table scan will have already happened. The where clause on the other hand would provide the desired filtering depending on the case. So, C is the correct answer.

learnazureportal 3 years, 4 months ago

Not sure, why option C selected! The correct Answer is B. the question clearly says "minimal changes to existing SQL queries". who said that, recreate the table, with partitioning layout is minimal and is PART of SQL queries!

ceak 3 years, 4 months ago

recreating table will not affect existing sql queries as they will still be selecting the same table name, but the scan will hugely decrease. so, option C is the correct answer.

squishy_fishy 2 years, 5 months ago

Recreating table is recommended by Google.

hdmi_switch 2 years, 8 months ago

In addition to the previous reply, the LIMIT statement applies to the output (what you see in the UI), the full table scan will still happen. C is correct according to best practices.

arghya13 3 years, 4 months ago

should be C:

gyclop 3 years, 6 months ago

Correct - C : "Limit" keyword restricts the final dataset to "n" rows, but is not able to restrict full table scan

Ravivarma4786 3 years, 7 months ago

Partitioning will help reduce the scanning time

haroldbenites 3 years, 7 months ago

C is correct

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #100 Topic 1

You have a requirement to insert minute-resolution data from 50,000 sensors into a BigQuery table. You expect significant growth in data volume and need the data to be available within 1 minute of ingestion for real-time analysis of aggregated trends. What should you do?

A. Use bq load to load a batch of sensor data every 60 seconds.

B. Use a Cloud Dataflow pipeline to stream data into the BigQuery table. **Most Voted**

C. Use the INSERT statement to insert a batch of data every 60 seconds.

D. Use the MERGE statement to apply updates in batch every 60 seconds.

Correct Answer: B

Community vote distribution

B (100%)

Comments

jvg637 **Highly Voted** 6 years ago

I think we need a pipeline, so it's B to me.

MaxNRG **Highly Voted** 4 years, 3 months ago

Selected Answer: B

Is B, if we expect a growth we'll need some buffer (that will be pub-sub) and the dataflow pipeline to stream data in big query. The tabledata.insertAll method is not valid here.

Erg_de 1 year, 4 months ago

Selected Answer: B

For real-time analysis and quick data availability, the appropriate option is the combination of the pipeline with BigQuery.

Helinia 2 years, 4 months ago

"need the data to be available within 1 minute of ingestion for real-time analysis" → low latency requirement → Dataflow streaming The database can either be BQ or BigTable for this kind of requirement in data volume and latency. But it mentioned that the destination has to be BQ, so B.

NeoNitin 2 years, 7 months ago

ANSWER b. FULL question ihave if you nee mail me neonitin6ATtherate.....

aryaavinash 2 years, 6 months ago

full email id please ?

Oleksandr0501 2 years, 11 months ago

Selected Answer: B

I think we need a pipeline, so it's B to me.))

votinhluombikip 2 years, 11 months ago

Selected Answer: B

I think we need a pipeline, so it's B to me.

JANCAI 3 years ago

Why the answer from the <reveal answer> is C??

Prasha123 3 years, 4 months ago

Selected Answer: B

Need pipeline so its B

sedado77 3 years, 6 months ago

Selected Answer: B

I got this question on sept 2022. Answer is B

medeis_jar 4 years, 2 months ago

Selected Answer: B

omg. B only

hendrixlives 4 years, 3 months ago

Selected Answer: B

B, streaming with dataflow

JG123 4 years, 4 months ago

Wrong answer shown again by examtopics.com Ans: B

Ysance_AGS 4 years, 6 months ago

B => with dataflow you can parallelize data ingestion

szefco 4 years, 4 months ago

And make it streaming

sandipk91 4 years, 7 months ago

B is the right answer

sumanshu 4 years, 9 months ago

Vote for B

felixwtf 5 years, 3 months ago

You need a pipeline because this type of operation can be easily parallelized, as the ingestion can be divided between into chunks (PCollections) and handled by many workers.

felixwtf 5 years, 3 months ago

so, B is the right answer.

https://zhangxuetu.online

[202610]

Question #101 Topic 1

You need to copy millions of sensitive patient records from a relational database to BigQuery. The total size of the database is 10 TB. You need to design a solution that is secure and time-efficient. What should you do?

A. Export the records from the database as an Avro file. Upload the file to GCS using gsutil, and then load the Avro file into BigQuery using the BigQuery web UI in the GCP Console.

B. Export the records from the database as an Avro file. Copy the file onto a Transfer Appliance and send it to Google, and then load the Avro file into BigQuery using the BigQuery web UI in the GCP Console. Most Voted

C. Export the records from the database into a CSV file. Create a public URL for the CSV file, and then use Storage Transfer Service to move the file to Cloud Storage. Load the CSV file into BigQuery using the BigQuery web UI in the GCP Console.

D. Export the records from the database as an Avro file. Create a public URL for the Avro file, and then use Storage Transfer Service to move the file to Cloud Storage. Load the Avro file into BigQuery using the BigQuery web UI in the GCP Console.

Correct Answer: B

Community vote distribution



Comments

Ganshank Highly Voted 5 years, 11 months ago

You are transferring sensitive patient information, so C & D are ruled out. Choice comes down to A & B. Here it gets tricky. How to choose Transfer Appliance: (https://cloud.google.com/transfer-appliance/docs/2.0/overview) Without knowing the bandwidth, it is not possible to determine whether the upload can be completed within 7 days, as recommended by Google. So the safest and most performant way is to use Transfer Appliance. Therefore my choice is B.

tprashanth 5 years, 8 months ago

https://cloud.google.com/solutions/migration-to-google-cloud-transferring-your-large-datasets The table shows for 1Gbps, it takes 30 hrs for 10 TB. Generally, corporate internet speeds are over 1Gbps. I'm inclined to pick A

BigQuery 4 years, 3 months ago

SAY MY NAME! You need to Transfer Sensitive Patient information, over public ISP you shouldn't do that.

grshankar9 1 year, 2 months ago

Security is not a concern, the data is encrypted at rest as well as in transit

forepick 2 years, 10 months ago

If you transfer 10TBs over the wire, your network will be blocked for the entire transfer time. This isn't something a company would be happy to swallow.

TNT87 5 years, 6 months ago

Answer is B,gsutil has a limit of 1TBaccording to Google documentation,if data is morethan 1TBthen we have to use Transfer Appliance.

Yiouk 4 years, 8 months ago

The answer is clearly seen here: https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#transfer-options

grshankar9 1 year, 2 months ago

According to Google documentation, for files of size > 1TB STS is to be used

grshankar9 1 year, 2 months ago

While gsutil itself doesn't have a strict data limit, the underlying Google Cloud Storage service does, allowing for individual object sizes up to 5 Terabytes (TiB). This means that when using gsutil to transfer data, the maximum file size you can upload or download is 5 TiB

AzureDP900 3 years, 3 months ago

B is right answer

grshankar9 1 year, 2 months ago

With reasonable network connectivity (for example, 1 Gbps), transferring 100 TB of data online takes over 10 days to complete. If this rate is acceptable, an online transfer is likely a good solution for your needs. If you only have a 100 Mbps connection (or worse from a remote location), the same transfer takes over 100 days. At this point, it's worth considering an offline-transfer option such as Transfer Appliance.

SSV Highly Voted 5 years, 8 months ago

Answer should be B: A is also correct but it has its own limit. It allows only 5TB data upload at a time to cloud storage. https://cloud.google.com/storage/quotas I will go with B

VASI 5 years, 2 months ago

5Tb "for individual objects". Create smaller AVRO files.

VASI 5 years, 2 months ago

AVRO compression can reduce file size to a tenth

grshankar9 1 year, 2 months ago

Selected Answer: A

C and D are ruled out owing to limitations of Cloud Storage with regards to max file size of 5 TB (Refer - https://cloud.google.com/storage-transfer/docs/known-limitations-transfer#:~:text=performance%20uploads%3A%206GiB-,Scaling%20limitations,of%20Gbps%20in%20transfer%20speed). The answer should be between A & B. if we were to assume bandwidth of 1 GBPS, it would only take about 1 day. Even if the bandwidth were a tenth of 1 GBPS, it would take about 10 days to transfer 10 TB. With a transfer appliance, it would take minimum of 25 days. I would go with A.

clouditis 1 year, 3 months ago

Selected Answer: B

B - only because it says sensitive data, you can never be sure uploading over internet!

Preetmehta1234 1 year, 6 months ago

Selected Answer: B

Also, For your scenario with 10 TB of data in Cloud SQL, if you export to Avro without specifying compression, you can expect the resulting Avro file to be around the same size, potentially slightly smaller depending on the data characteristics. Here in this question, there is no mentioning about compression. So let's not assume that the data being used in Avro format will get compressed. If Google cloud storage itself, can't handle an object of size greater than 5 TB, there is no point of using gsutil

grshankar9 1 year, 2 months ago

gsutil is deprecated and has been replaced by gcloud storage. Gcloud storage is faster and requires less manual optimization for uploads and downloads

Preetmehta1234 1 year, 6 months ago

Selected Answer: B

https://cloud.google.com/storage-transfer/docs/known-limitations-transfer Cloud Storage 5TiB object size limit Cloud Storage supports a maximum single-object size up 5 tebibytes. If you have objects larger than 5TiB, the object transfer fails for those objects for either Cloud Storage or Storage Transfer Service.

hussain.sain 1 year, 9 months ago

Selected Answer: D

while Option A is feasible and could work depending on specific requirements and security measures implemented, Option D (exporting as Avro, using Storage Transfer Service, and then loading into BigQuery) generally offers a more secure, efficient, and managed approach for transferring sensitive patient records into BigQuery from a relational database.Avro files uploaded to GCS will need to be secured. While GCS itself offers security features like IAM policies and access controls, using a public URL (as suggested in Option A) introduces additional security concerns.

Naresh_4u 1 year, 11 months ago

Selected Answer: B

to securely transfer data and looking at the size of data B is the correct option.

GCanteiro 2 years, 1 month ago

Selected Answer: A

IMO "A" is the most suitable option since the transfer appliance could take 25 days to get the appliance and then 25 days to ship it back and have the data available. https://cloud.google.com/transfer-appliance/docs/4.0/overview#transfer-speeds

TVH_Data_Engineer 2 years, 3 months ago

Selected Answer: B

Given the sensitivity of the patient records and the large size of the data, using Google's Transfer Appliance is a secure and efficient method. The Transfer Appliance is a hardware solution provided by Google for transferring large amounts of data. It enables you to securely transfer data without exposing it over the internet.

rocky48 2 years, 3 months ago

Selected Answer: B

Option B combines security, efficiency, and ease of use, making it a suitable choice for transferring sensitive patient records to BigQuery.

spicebits 2 years, 4 months ago

Selected Answer: A

10 TB is nothing. With a single 10 GB interconnect you could transfer the data in 3 hours or even with a 1 GB speeds without interconnect you could transfer it in one weekend. The transfer appliance will take 25 days to get the appliance and then 25 days while you wait for the data to be available that is not "time-efficient" at all. I go with A instead of B.

spicebits 2 years, 4 months ago

I got the 25 days + 25 days from here: https://cloud.google.com/transfer-appliance/docs/4.0/overview#transfer-speeds

A_Nasser 2 years, 6 months ago

Selected Answer: A

transfer appliance will take time more than gsutil. and we did not mention yet if the location of the organization has google data centre

DineshVarma 2 years, 7 months ago

Selected Answer: D

As per Google recommendation above 1TB of transfer from onprem or from Google cloud or other cloud storage like s3 etc we need to use storage transfer service.

arien_chen 2 years, 7 months ago

Transfer Appliance would take 20 days for epected turnaround time. https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#expected%20turnaround:~:text=The%20expected%20turnaround%20time%20for%20a%20network%20appliance%20to%20be%20shipped%2C%20loaded%20with%20your%20data%2C%20shipped%20back%2C%20and%20rehydrated%20on%20Google%20Cloud%20is%2020%20days. The best answer would be A. If gsutil consume/leverage 100MB it would take 12 days and more time-efficient than B. This is a reasonable assumption. https://cloud.google.com/static/architecture/images/big-data-transfer-how-to-get-started-transfer-size-and-speed.png

Colourseun 2 years, 7 months ago

I will go with " A" because of the transition time to take transfer appliance to Google and that also depends in the organisation location. gsutil works anywhere internet is available.

aewis 2 years, 8 months ago

Selected Answer: B

A will take crazy time if the organization didnt have a dedicated link

https://zhangxuetu.online

[202610]

Question #102 Topic 1

You need to create a near real-time inventory dashboard that reads the main inventory tables in your BigQuery data warehouse. Historical inventory data is stored as inventory balances by item and location. You have several thousand updates to inventory every hour. You want to maximize performance of the dashboard and ensure that the data is accurate. What should you do?

A. Leverage BigQuery UPDATE statements to update the inventory balances as they are changing.

B. Partition the inventory balance table by item to reduce the amount of data scanned with each inventory update.

C. Use the BigQuery streaming the stream changes into a daily inventory movement table. Calculate balances in a view that joins it to the historical inventory balance table. Update the inventory balance table nightly. **Most Voted**

D. Use the BigQuery bulk loader to batch load inventory changes into a daily inventory movement table. Calculate balances in a view that joins it to the historical inventory balance table. Update the inventory balance table nightly.

Correct Answer: C

Community vote distribution



Comments

MaxNRG **Highly Voted** 4 years, 3 months ago

Selected Answer: A

A - New correct answer C - Old correct answer (for 2019)

Yiouk 2 years, 8 months ago

There are still limitations on DML statements (2023) e.g. only 2 concurrent UPDATES and up to 20 queued hence not appropriate for this scenario: https://cloud.google.com/bigquery/quotas#data-manipulation-language-statements

NeoNitin 2 years, 8 months ago

option A:what limitation here 1500/perday okay in question we will get max 24 jobs hourly updated okay, now speed 5 operation /10 sec , 1 operation 2sec , and we are getting new update in 1 hour so we have time 3600 sec and we need to update around 1000 update according to speed take 2000sec still we have 1600 sec rest to getting new update so . thats why I thing DML is best option for this work

Nandababy 2 years, 3 months ago

In question it mentioned several thousands of updates every hour, several thousands could be 20-30 thousands as well. Where it is mentioned for only 1000 updates?

MaxNRG 2 years, 3 months ago

C is better The best approach is to use BigQuery streaming to stream the inventory changes into a daily inventory movement table. Then calculate balances in a view that joins the inventory movement table to the historical inventory balance table. Finally, update the inventory balance table nightly (option C).

MaxNRG 2 years, 3 months ago

The key reasons this is better than the other options: Using BigQuery UPDATE statements (option A) would be very inefficient for thousands of updates per hour. It is better to batch updates. Partitioning the inventory balance table (option B) helps query performance, but does not solve the need to incrementally update balances. Using the bulk loader (option D) would require batch loading the updates, which adds latency. Streaming inserts updates with lower latency. So option C provides a scalable architecture that streams updates with low latency while batch updating the balances only once per day for efficiency. This balances performance and accuracy needs.

MaxNRG 2 years, 3 months ago

Here's why the other options are less suitable: A. Leverage BigQuery UPDATE statements: While technically possible, this approach is inefficient for frequent updates as it requires individual record scans and updates, affecting performance and potentially causing data race conditions. B. Partition the inventory balance table: Partitioning helps with query performance for large datasets, but it doesn't address the need for near real-time updates. D. Use the BigQuery bulk loader: Bulk loading daily changes is helpful for historical data ingestion, but it won't provide near real-time updates necessary for the dashboard.

MaxNRG 2 years, 3 months ago

Option C offers the following advantages: Streams inventory changes near real-time: BigQuery streaming ingests data immediately, keeping the inventory movement table constantly updated. Daily balance calculation: Joining the movement table with the historical balance table provides an accurate view of current inventory levels without affecting the actual balance table. Nightly update for historical data: Updating the main inventory balance table nightly ensures long-term data consistency while maintaining near real-time insights through the view. This approach balances near real-time updates with efficiency and data accuracy, making it the optimal solution for the given scenario.

haroldbenites Highly Voted 5 years, 7 months ago

C is correct. It says “update Every hour” And need “ accuracy”

NeoNitin 2 years, 8 months ago

option A:what limitation here 1500/perday okay in question we will get max 24 jobs hourly updated okay, now speed 5 operation /10 sec , 1 operation 2sec , and we are getting new update in 1 hour so we have time 3600 sec and we need to update around 1000 update according to speed take 2000sec still we have 1600 sec rest to getting new update so . thats why I thing DML is best option for this work

SamuelTsch 1 year, 5 months ago

Selected Answer: C

BigQuery is not optimized for updating statement. So, go to C.

edre 1 year, 8 months ago

Selected Answer: C

The answer is C because the requirement is near real-time

MaxNRG 2 years, 3 months ago

Selected Answer: C

The best approach is to use BigQuery streaming to stream the inventory changes into a daily inventory movement table. Then calculate balances in a view that joins the inventory movement table to the historical inventory balance table. Finally, update the inventory balance table nightly (option C).

MaxNRG 2 years, 3 months ago

The key reasons this is better than the other options: Using BigQuery UPDATE statements (option A) would be very inefficient for thousands of updates per hour. It is better to batch updates. Partitioning the inventory balance table (option B) helps query performance, but does not solve the need to incrementally update balances. Using the bulk loader (option D) would require batch loading the updates, which adds latency. Streaming inserts updates with lower latency. So option C provides a scalable architecture that streams updates with low latency while batch updating the balances only once per day for efficiency. This balances performance and accuracy needs.

rocky48 2 years, 3 months ago

Selected Answer: C

Option C. Using the BigQuery streaming to stream changes into a daily inventory movement table and calculating balances in a view that joins it to the historical inventory balance table can help you achieve the desired performance and accuracy. You can then update the inventory balance table nightly. This approach can help you avoid the overhead of scanning large amounts of data with each inventory update, which can be time-consuming and resource-intensive. Leveraging BigQuery UPDATE statements to update the inventory balances as they are changing (option A) can be resource-intensive and may not be the most efficient way to achieve the desired performance.

AnonymousPanda 2 years, 4 months ago

Selected Answer: C

As per other answers C

Nirca 2 years, 4 months ago

Selected Answer: A

Simple and will work

odacir 2 years, 5 months ago

Selected Answer: C

Answer is C. Why because “Update” limits is 1500/per day, and the question say: You have several thousand updates to inventory every hour. So is impossible to use updates all the time.

Nirca 2 years, 5 months ago

Selected Answer: A

A. Leverage BigQuery UPDATE statements to update the inventory balances as they are changing - is so simple and RIGHT!

brookpetit 2 years, 6 months ago

Selected Answer: C

C is more universal and sustainable

ZZHZZH 2 years, 8 months ago

Selected Answer: C

UPDATE is too expensive. Joining main and delta tables is the right wat to capture data change.

euro202 2 years, 8 months ago

Selected Answer: C

I think the answer is C. The question is about maximizing performance and accuracy, it's ok if we need expensive JOINs. BigQuery has a daily quota of 1500 UPDATEs, and the question talks about several thousand updates every

hour.

jackdbd 2 years, 6 months ago

DML statements do not count toward the number of table modifications per day. https://cloud.google.com/bigquery/quotas#data-manipulation-language-statements So I would go with A.

jackdbd 2 years, 6 months ago

Sorry, wrong link. Here is the correct one: https://cloud.google.com/bigquery/quotas#standard_tables

vaga1 2 years, 9 months ago

Selected Answer: A

C create a view that joins to a table seems dumb to me

forepick 2 years, 10 months ago

Selected Answer: C

Too frequent updates are way too expensive in an OLAP solution. This is much more likely to stream changes to the table(s) and aggregate these changes in the view. https://stackoverflow.com/questions/74657435/bigquery-frequent-updates-to-a-record

streeeber 2 years, 11 months ago

Selected Answer: C

Has to be C. DML has hard limit of 1500 operations per table per day: https://cloud.google.com/bigquery/quotas#standard_tables

lucaluca1982 3 years ago

Selected Answer: C

Update action is not efficient

NeoNitin 2 years, 8 months ago

option A:what limitation here 1500/perday okay in question we will get max 24 jobs hourly updated okay, now speed 5 operation /10 sec , 1 operation 2sec , and we are getting new update in 1 hour so we have time 3600 sec and we need to update around 1000 update according to speed take 2000sec still we have 1600 sec rest to getting new update so . thats why I thing DML is best option for this work

https://zhangxuetu.online

[202610]

Question #103 Topic 1

You have a data stored in BigQuery. The data in the BigQuery dataset must be highly available. You need to define a storage, backup, and recovery strategy of this data that minimizes cost. How should you configure the BigQuery table that have a recovery point objective (RPO) of 30 days?

A. Set the BigQuery dataset to be regional. In the event of an emergency, use a point-in-time snapshot to recover the data.

B. Set the BigQuery dataset to be regional. Create a scheduled query to make copies of the data to tables suffixed with the time of the backup. In the event of an emergency, use the backup copy of the table.

C. Set the BigQuery dataset to be multi-regional. In the event of an emergency, use a point-in-time snapshot to recover the data. Most Voted

D. Set the BigQuery dataset to be multi-regional. Create a scheduled query to make copies of the data to tables suffixed with the time of the backup. In the event of an emergency, use the backup copy of the table.

Correct Answer: C

Community vote distribution



Comments

DeepakVenkatachalam Highly Voted 2 years, 6 months ago

Answer is B. Timetravel only covers for 7 days and a scheduled query is needed for creating Table snapshots for 30 days. Also table snapshot must remain in the same region as base table (please refer to limitation of table snapshot from below link) https://cloud.google.com/bigquery/docs/table-snapshots-intro

desertlotus1211 Highly Voted 3 years, 2 months ago

Answer is C: https://cloud.google.com/bigquery/docs/table-snapshots-intro "Benefits of using table snapshots include the following: Keep a record for longer than seven days. With BigQuery time travel, you can only access a table's data from seven days ago or more recently. With table snapshots, you can preserve a table's data from a specified point in time for as long as you want. Minimize storage cost. BigQuery only stores bytes that are different between a snapshot and its base table, so a table snapshot typically uses less storage than a full copy of the table." But the wording is foolish... It's table snapshot, NOT point in time snapshot! https://cloud.google.com/bigquery/docs/time-travel#restore-a-table this is point in time using time travel window - max is 7 days...

desertlotus1211 1 year ago

Sorry folks - I change my answer to D... C is not correct at it can ONLY go back for 7 days, max!

forepick 7 months, 1 week ago

Selected Answer: B

Regional - to minimize cost, scheduled backups - to enable 30 days RPO (time travel only gives you 7 days)

imrane1995 8 months, 3 weeks ago

Selected Answer: B

Regional datasets are *cheaper* than *multi-regional*, which helps minimize cost. Scheduled queries can automate daily or hourly snapshots, ensuring you always have backup data within the 30-day RPO window. Time-suffixed tables allow for granular restoration to a specific date. This is a manual but cost-effective backup strategy.

56d02cd 9 months ago

Selected Answer: B

I think all of these answers are wrong. None of these options safeguard against a regional GCP failure. Managed Disaster Recovery is a real DR solution to consider for this use case or at least a manually managed second backup copy of the table in a different region. Multi-region just automatically picks a region but does not replicate cross region (https://cloud.google.com/bigquery/docs/managed-disaster-recovery)

andy9981 9 months, 2 weeks ago

Selected Answer: B

Regional dataset Lower cost than multi-regional Scheduled backups (30 days) Meets RPO by retaining 30+ days of backup snapshots Cost-efficient Regional storage is cheaper and backup tables are under user control Recovery process Simply point to the latest backup or restore from backup tables

Parandhaman_Margan 1 year ago

Selected Answer: B

Minimizing Cost

imarri876 1 year ago

Selected Answer: B

BigQuery multi-regional is more expensive than BQ regional.

imarri876 1 year ago

After deeper review, with pricing calculator and other resources, the answer is C.

loki82 1 year, 2 months ago

Selected Answer: A

RPO is NOT the same as a PITR window. Point-in-time recovery (PITR) is a process that allows users to restore data or settings from a previous point in time. A recovery point objective (RPO) is the maximum amount of data loss that an organization can tolerate after a data loss event. So a PITR snapshot easily meets an RPO of 30 days. A regional bucket minimizes cost.

hussain.sain 1 year, 3 months ago

Selected Answer: C

Answer is C. As question is related to highly available so this rules out A and B.

clouditis 1 year, 3 months ago

Selected Answer: C

Because this option uses Multiregional & BQ Snapshot, others are not right/cumbersome

cloud_rider 1 year, 4 months ago

Selected Answer: A

A is the right Answer

Erg_de 1 year, 4 months ago

Selected Answer: B

Best choice, minimized cost

Gcpteamprep 1 year, 5 months ago

Selected Answer: B

Minimized Cost with Regional Storage: Regional datasets are less costly than multi-regional datasets in BigQuery. Since there is no requirement here for multi-regional availability, regional storage meets the high availability need while keeping costs lower. RPO Compliance with Scheduled Backups: A scheduled query that periodically creates copies of the data (e.g., daily or weekly, depending on the requirements) allows for recovery within the 30-day RPO, meeting the requirement for data retention and recovery. Point-in-Time Recovery Not Native in BigQuery: Although BigQuery provides a limited "table snapshot" feature, it's not a true point-in-time recovery option for the last 30 days. Creating periodic backups through scheduled queries gives you control over retention, enabling you to keep backups for 30 days and reducing dependency on more costly or limited snapshot capabilities.

Vogangster 1 year, 5 months ago

D.Create monthly snapshots of a table by using a service account that runs a scheduled query. Link: https://cloud.google.com/bigquery/docs/table-snapshots-scheduled

AlizCert 1 year, 10 months ago

Selected Answer: D

HA => multi-region 30-days RPO => manual backups as max time-travel is 7 days

Lestrang 1 year, 10 months ago

This is in one of google's training practice questions and the answer for it is C.

NickNtaken 1 year, 10 months ago

Agreed. Multi-regional datasets offer higher availability by replicating data across multiple regions

https://zhangxuetu.online

[202610]

Question #104 Topic 1

You used Dataprep to create a recipe on a sample of data in a BigQuery table. You want to reuse this recipe on a daily upload of data with the same schema, after the load job with variable execution time completes. What should you do?

A. Create a cron schedule in Dataprep.

B. Create an App Engine cron job to schedule the execution of the Dataprep job.

C. Export the recipe as a Dataprep template, and create a job in Cloud Scheduler.

D. Export the Dataprep job as a Dataflow template, and incorporate it into a Composer job. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

jkhong **Highly Voted** 2 years, 9 months ago

I'd pick D because it's the only option which allows variable execution (since we need to execute the dataprep job only after the prior load job). Although D suggests the export of Dataflow templates, this discussion suggests that the export option is no longer available (https://stackoverflow.com/questions/72544839/how-to-get-the-dataflow-template-of-a-dataprep-job), there are already Airflow Operators for Dataprep which we should be using instead - https://airflow.apache.org/docs/apache-airflow-providers-google/stable/operators/cloud/dataprep.html

midgoo **Highly Voted** 2 years, 6 months ago

Selected Answer: D

Since the load job execution time is unexpected, schedule the Dataprep based on a fixed time window may not work. When the Dataprep job run the first time, we can find the Dataflow job for that in the console. We can use that to create the Template --> With the help of the Composer to determine if the load job is completed, we can then trigger the Dataflow job

TVH_Data_Engineer 1 year, 9 months ago

Selected Answer: A

Dataprep by Trifacta allows you to schedule the execution of recipes. You can set up a cron schedule directly within Dataprep to automatically run your recipe at specified intervals, such as daily. WHY NOT D ? : This option involves significant additional complexity. Exporting the Dataprep job as a Dataflow template and then incorporating it into a Composer (Apache Airflow) job is a more complicated process and is typically used for more complex orchestration needs that go beyond simple scheduling.

MaxNRG 1 year, 9 months ago

Selected Answer: D

We have external dependency "after the load job with variable execution time completes" which requires DAG -> Airflow (Cloud Composer) The reasons: A scheduler like Cloud Scheduler won't handle the dependency on the BigQuery load completion time Using Composer allows creating a DAG workflow that can: Trigger the BigQuery load Wait for BigQuery load to complete Trigger the Dataprep Dataflow job Dataflow template allows easy reuse of the Dataprep transformation logic Composer coordinates everything based on the dependencies in an automated workflow

rocky48 1 year, 9 months ago

Selected Answer: D

I'd pick D because it's the only option which allows variable execution

gaurav0480 2 years, 1 month ago

The key here is "after the load job with variable execution time completes" which means the execution of this job depends on the completion of another job which has a variable execution time. Hence D

god_brainer 2 years, 1 month ago

This approach ensures the dynamic triggering of the Dataprep job based on the completion of the preceding load job, ensuring data is processed accurately and in sequen

Adswerve 2 years, 5 months ago

Selected Answer: A

A is correct. D is too complicated. A is correct, because you can schedule a job right from Dataprep UI. https://cloud.google.com/blog/products/gcp/scheduling-and-sampling-arrive-for-google-cloud-dataprep Scheduling and sampling arrive for Google Cloud Dataprep Throughout our early releases, users' most common request has been Flow scheduling. As of Thursday's release, Flows can be scheduled with minute granularity at any frequency.

lucaluca1982 2 years, 6 months ago

Selected Answer: C

I think C it is more straighforward

musumusu 2 years, 7 months ago

Answer C: Use Recipe Template feature of dataprep. Don't need to change the service.

jroig_ 2 years, 8 months ago

Selected Answer: C

Why not C?

anicloudgirl 2 years, 9 months ago

Selected Answer: A

It's A. You can set it directly in Dataprep a job and it will use Dataflow under the hood.

anicloudgirl 2 years, 9 months ago

It's A. You can set it directly in Dataprep a job and it will use Dataflow under the hood. No need to export nor incorporate into a Composer job. Dataprep by trifacta - https://docs.trifacta.com/display/DP/cron+Schedule+Syntax+Reference Dataprep job uses dataflow - https://cloud.google.com/dataprep

jkhong 2 years, 9 months ago

The question mentions after a load job with variable time, i dont think setting a dataprep cron job can address the issue of variable load times

cloudmon 2 years, 11 months ago

Selected Answer: D

It's D

John_Pongthorn 3 years ago

Selected Answer: D

Dataprep and Dataflow are same famitly

AWSandeep 3 years ago

Selected Answer: D

D. Export the Dataprep job as a Dataflow template, and incorporate it into a Composer job. Reveal Solution

damaldon 3 years, 1 month ago

It’s D, use composer to schedule tasks

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #105 Topic 1

You want to automate execution of a multi-step data pipeline running on Google Cloud. The pipeline includes Dataproc and Dataflow jobs that have multiple dependencies on each other. You want to use managed services where possible, and the pipeline will run every day. Which tool should you use?

A. cron

B. Cloud Composer **Most Voted**

C. Cloud Scheduler

D. Workflow Templates on Dataproc

Correct Answer: B

Community vote distribution

B (100%)

Comments

Sofiia98 1 year, 8 months ago

Selected Answer: B

Of course, it is Cloud Composer!

Nirca 1 year, 12 months ago

Selected Answer: B

B. Cloud Composer is the right answer !

vamgcp 2 years, 2 months ago

Selected Answer: B

Cloud Composer is a managed service that allows you to create and run Apache Airflow workflows. Airflow is a workflow management platform that can be used to automate complex data pipelines. It is a good choice for this use case because it is a managed service, which means that Google will take care of the underlying infrastructure. It also supports multiple dependencies, so you can easily schedule a multi-step pipeline

vaga1 2 years, 5 months ago

Selected Answer: B

Airflow is the only choiche to handle dependencies and being able to call all of the services included in the question

niketd 2 years, 7 months ago

Selected Answer: B

Multi-step sequential pipelines -> Cloud Composer

AzureDP900 2 years, 9 months ago

Cloud composer B is right

odacir 2 years, 9 months ago

Selected Answer: B

Cloud Composer (Airflow) is the answer to chain different steps from different apps...

MisuLava 2 years, 11 months ago

Selected Answer: B

" multiple dependencies on each other. You want to use managed service" = Cloud Composer

John_Pongthorn 3 years ago

Selected Answer: B

if you want your wf to schedule there are 3 ways to perform it, it of them is composer
https://cloud.google.com/dataproc/docs/concepts/workflows/workflow-schedule-solutions

Remi2021 3 years ago

Selected Answer: B

composer :)

YorelNation 3 years ago

Selected Answer: B

Composer

AWSandeep 3 years ago

Selected Answer: B

B. Cloud Composer

damaldon 3 years, 1 month ago

Use composer to schedule tasks

https://zhangxuetu.online

[202610]

Question #106 Topic 1

You are managing a Cloud Dataproc cluster. You need to make a job run faster while minimizing costs, without losing work in progress on your clusters. What should you do?

A. Increase the cluster size with more non-preemptible workers.

B. Increase the cluster size with preemptible worker nodes, and configure them to forcefully decommission.

C. Increase the cluster size with preemptible worker nodes, and use Cloud Stackdriver to trigger a script to preserve work.

D. Increase the cluster size with preemptible worker nodes, and configure them to use graceful decommissioning. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

AzureDP900 **Highly Voted** 2 years, 3 months ago

D is right https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/scaling-clusters#using_graceful_decommissioning

rocky48 1 year, 3 months ago

Selected Answer: D

D is right https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/scaling-clusters#using_graceful_decommissioning

Prakzz 2 years, 3 months ago

Selected Answer: A

Should be A. You can configure the preemptible worker to gracefull decommission, its for non preemptible worker nodes.

wan2three 2 years, 3 months ago

nope, they are not only for non-preeemtible workers

yafsong 2 years, 3 months ago

graceful decommissioning: to finish work in progress on a worker before it is removed from the Cloud Dataproc cluster. https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/scaling-clusters

odacir 2 years, 3 months ago

Selected Answer: D

All your workers need to be the same kind. Use Graceful Decommissioning for don't lose any data and add more(increase the cluster) preemptible workers because there are more cost-effective .

skp57 2 years, 4 months ago

A. "graceful decommissioning" is not a configuration value but a parameter passed with scale down action - to decrease the number of workers to save money (see Graceful Decommissioning as an option to use when downsizing a cluster to avoid losing work in progress)

John_Pongthorn 2 years, 6 months ago

Selected Answer: D

https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/scaling-clusters Why scale a Dataproc cluster? to increase the number of workers to make a job run faster to decrease the number of workers to save money (see Graceful Decommissioning as an option to use when downsizing a cluster to avoid losing work in progress). to increase the number of nodes to expand available Hadoop Distributed Filesystem (HDFS) storage

hauhau 2 years, 5 months ago

This weird. The question mentions that increase cluster, but Graceful Decommissioning use in downscale the cluster

odacir 2 years, 3 months ago

All your workers need to be the same kind. Use Graceful Decommissioning for don't lose any data and add more preemptible workers because there are more cost-effective

AWSandeep 2 years, 7 months ago

Selected Answer: D

D. Increase the cluster size with preemptible worker nodes, and configure them to use graceful decommissioning.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #107 Topic 1

You work for a shipping company that uses handheld scanners to read shipping labels. Your company has strict data privacy standards that require scanners to only transmit tracking numbers when events are sent to Kafka topics. A recent software update caused the scanners to accidentally transmit recipients' personally identifiable information (PII) to analytics systems, which violates user privacy rules. You want to quickly build a scalable solution using cloud-native managed services to prevent exposure of PII to the analytics systems. What should you do?

A. Create an authorized view in BigQuery to restrict access to tables with sensitive data.

B. Install a third-party data validation tool on Compute Engine virtual machines to check the incoming data for sensitive information.

C. Use Cloud Logging to analyze the data passed through the total pipeline to identify transactions that may contain sensitive information.

D. Build a Cloud Function that reads the topics and makes a call to the Cloud Data Loss Prevention (Cloud DLP) API. Use the tagging and confidence levels to either pass or quarantine the data in a bucket for review. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

dconesoko Highly Voted 2 years, 3 months ago

Selected Answer: D

The cloud function with DLP seems the best option

PhilipKoku 1 year, 8 months ago

Selected Answer: D

DLP is required

HarshKothari21 2 years, 6 months ago

Selected Answer: D

D option

AWSandeep 2 years, 7 months ago

Selected Answer: D

D. Build a Cloud Function that reads the topics and makes a call to the Cloud Data Loss Prevention (Cloud DLP) API. Use the tagging and confidence levels to either pass or quarantine the data in a bucket for review.

AzureDP900 2 years, 3 months ago

Agreed

https://zhangxuetu.online

[202610]

Question #108 Topic 1

You have developed three data processing jobs. One executes a Cloud Dataflow pipeline that transforms data uploaded to Cloud Storage and writes results to
BigQuery. The second ingests data from on-premises servers and uploads it to Cloud Storage. The third is a Cloud Dataflow pipeline that gets information from third-party data providers and uploads the information to Cloud Storage. You need to be able to schedule and monitor the execution of these three workflows and manually execute them when needed. What should you do?

A. Create a Direct Acyclic Graph in Cloud Composer to schedule and monitor the jobs. **Most Voted**

B. Use Stackdriver Monitoring and set up an alert with a Webhook notification to trigger the jobs.

C. Develop an App Engine application to schedule and request the status of the jobs using GCP API calls.

D. Set up cron jobs in a Compute Engine instance to schedule and monitor the pipelines using GCP API calls.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Rajokkiyam Highly Voted 5 years ago
Create dependency in Cloud Composer and schedule it.

MisuLava 2 years, 5 months ago
the jobs are not interdependent. just 3 individual jobs

[Removed] Highly Voted 5 years ago
Answer: A Description: Cloud composer is used to schedule the interdependent jobs

marioferrulli 3 years, 3 months ago
but thery are not

maxu 1 year, 5 months ago
yes answer A

forepick 1 year, 10 months ago
Selected Answer: A
Cloud Composer. No doubt

AzureDP900 2 years, 3 months ago

A is correct

dconesoko 2 years, 3 months ago

Selected Answer: A

Cloud composer's DAG would manage the dependencies

danielfootc 2 years, 9 months ago

This should be A

medeis_jar 3 years, 2 months ago

Selected Answer: A

https://cloud.google.com/composer/docs/how-to/using/writing-dags

MaxNRG 3 years, 3 months ago

Selected Answer: A

Cloud Composer is a fully managed workflow orchestration service that empowers you to author, schedule, and monitor pipelines that span across clouds and on-premises data centers. https://cloud.google.com/composer/?hl=en

kishanu 3 years, 3 months ago

Selected Answer: A

A Though the jobs are not dependent, they are data-driven. Refer to the below link: https://cloud.google.com/blog/topics/developers-practitioners/choosing-right-orchestrator-google-cloud

MaxNRG 3 years, 3 months ago

nice article thanks!

JG123 3 years, 4 months ago

Selected Answer: A

Cloud Composer

JG123 3 years, 4 months ago

Correct: A

sandipk91 3 years, 7 months ago

should be option A

sumanshu 3 years, 9 months ago

Vote for A

someshsehgal 4 years, 1 month ago

COrrect A: Couldn't understand why a option with no connection with actual problem has been given as correct option (D)

arghya13 4 years, 4 months ago

I'll go for A

Tanmoyk 4 years, 6 months ago

Should be A

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #109 Topic 1

You have Cloud Functions written in Node.js that pull messages from Cloud Pub/Sub and send the data to BigQuery. You observe that the message processing rate on the Pub/Sub topic is orders of magnitude higher than anticipated, but there is no error logged in Cloud Logging. What are the two most likely causes of this problem? (Choose two.)

A. Publisher throughput quota is too small.

B. Total outstanding messages exceed the 10-MB maximum.

C. Error handling in the subscriber code is not handling run-time errors properly. **Most Voted**

D. The subscriber code cannot keep up with the messages.

E. The subscriber code does not acknowledge the messages that it pulls. **Most Voted**

Correct Answer: CE

Community vote distribution



Comments

TNT87 **Highly Voted** 3 years, 6 months ago

Answer C E By not acknowleding the pulled message, this result in it be putted back in Cloud Pub/Sub, meaning the messages accumulate instead of being consumed and removed from Pub/Sub. The same thing can happen ig the subscriber maintains the lease on the message it receives in case of an error. This reduces the overall rate of processing because messages get stuck on the first subscriber. Also, errors in Cloud Function do not show up in Stackdriver Log Viewer if they are not correctly handled.

forepick 7 months, 1 week ago

Selected Answer: CE

D won't fit here, as the problem is that TOO MANY messages are being processed. This is a common result of unacked messages. Two common causes - C and E

clouditis 1 year, 3 months ago

Selected Answer: DE

Not C, its talking about the unknown!

SamuelTsch 1 year, 5 months ago

Selected Answer: DE

The issue is that the acknowledgment is not sent back to sub properly. D, E should be correct.

Preetmehta1234 1 year, 6 months ago

Selected Answer: DE

D. The subscriber code cannot keep up with the messages. If the processing rate of the subscriber (Cloud Functions) is lower than the incoming message rate, it can lead to a backlog of messages. This would result in higher-than-expected message rates, as messages accumulate while waiting to be processed. E. The subscriber code does not acknowledge the messages that it pulls. If messages are not acknowledged properly, Pub/Sub will keep retrying to deliver them, which can lead to the same messages being sent repeatedly. This could also contribute to the perception that the message processing rate is very high. Both of these issues can lead to unanticipated behavior in your message processing pipeline without generating errors that would be logged in Cloud Logging.

JamesKarianis 1 year, 7 months ago

Selected Answer: CD

The code in the CF can't keep up with the amount of messages, thus C D is a better fit

mialll 2 years, 11 months ago

Selected Answer: DE

Ref chatgpt Option C, "Error handling in the subscriber code is not handling run-time errors properly," suggests that the subscriber code may not be correctly handling errors that occur during message processing. If the subscriber code encounters an error that it cannot handle, such as a syntax error or a network issue, it may stop processing messages, leading to a slowdown in message processing. However, the lack of error logs in Cloud Logging suggests that there are no errors being logged, which makes it less likely that this is the primary cause of the observed behavior. Additionally, while incorrect error handling could contribute to the issue, it may not be the primary reason why the message processing rate is much higher than anticipated.

GCPete 2 years, 4 months ago

Chat says about Option C: "it may stop processing messages, leading to a slowdown in message processing" - but is doesn't say there's a slowdown in the question. It says it's increased. I would replace C with D. If the Cloud Function isn't capable of processing messages as quickly as they arrive, the backlog will grow, leading to higher processing rates as the function continuously tries to catch up. This scenario might not generate errors in Cloud Logging if the function is simply falling behind.

midgoo 3 years ago

Selected Answer: CE

C - as no error shown in Cloud Logging Between D & E, both could lead to the problem. I have worked with lot of PubSub issues, most of them are due to the bottleneck at the code where it takes too long to process 1 message and causes backlog. E could lead to backlog too, but it is too obvious and not likely to happen in reality. However, when I ask AI the same question, it said C and E

cetanx 2 years, 9 months ago

C. Error handling in the subscriber (Cloud Functions) code is not handling run-time errors properly. This would mean to have error logs in Cloud Logging as CF by default logs to it.

musumusu 3 years, 1 month ago

Answer D&E I am not in the favour of C, error handling is a side factor but not the primary cause. First check the configuration access. Does subscriber has enough acknowledge policies (option E) Does sub have ability to keep up the message(enough network, cpu and capable codes) (option D) option C is just a part of option D somewhere showing incapable handling

desertlotus1211 3 years, 2 months ago

My question is: 'What is the actual problem?' - That there is no logs in Cloud Logging? - That Pub/Sub is having a problem? - Or there an actual problem? - Is there an actual error? So what is Pub/Sub the message processing rate is high...Does that mean there is a problem? Thoughts?

squishy_fishy 2 years, 7 months ago

Like TNT87 mentioned the message processing rate is high "meaning the messages accumulate instead of being consumed and removed from Pub/Sub."

AzureDP900 3 years, 3 months ago

C, E seems correct

MounicaN 3 years, 6 months ago

D might also be right? Subscriber might not be provisioned enough

AWSandeep 3 years, 7 months ago

Selected Answer: CE

C. Error handling in the subscriber code is not handling run-time errors properly. E. The subscriber code does not acknowledge the messages that it pulls.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #110 Topic 1

You are creating a new pipeline in Google Cloud to stream IoT data from Cloud Pub/Sub through Cloud Dataflow to BigQuery. While previewing the data, you notice that roughly 2% of the data appears to be corrupt. You need to modify the Cloud Dataflow pipeline to filter out this corrupt data. What should you do?

A. Add a SideInput that returns a Boolean if the element is corrupt.

B. Add a ParDo transform in Cloud Dataflow to discard corrupt elements. Most Voted

C. Add a Partition transform in Cloud Dataflow to separate valid data from corrupt data.

D. Add a GroupByKey transform in Cloud Dataflow to group all of the valid data together and discard the rest.

Correct Answer: B

Community vote distribution

B (100%)

Comments

SteelWarrior Highly Voted 4 years ago

Should be B. The Partition transform would require the element identifying the valid/invalid records for partitioning the pcollection that means there is some logic to be executed before the Partition transformation is invoked. That logic can be implemented in a ParDO transform and which can both identify valid/invalid records and also generate two PCollections one with valid records and other with invalid records.

MaxNRG Highly Voted 2 years, 9 months ago

Selected Answer: B

B: ParDo is a Beam transform for generic parallel processing. ParDo is useful for common data processing operations, including: a. Filtering a data set. You can use ParDo to consider each element in a PCollection and either output that element to a new collection, or discard it. b. Formatting or type-converting each element in a data set. c. Extracting parts of each element in a data set. d. Performing computations on each element in a data set. A does not help C Partition is a Beam transform for PCollection objects that store the same data type. Partition splits a single PCollection into a fixed number of smaller collections. Again, does not help D GroupByKey is a Beam transform for processing collections of key/value pairs. GroupByKey is a good way to aggregate data that has something in common

midgoo 1 year, 6 months ago

Selected Answer: B

A - SideInput is often used to validate data, however, we need to create the SideInput first. When using SideInput to filter data, it is actually another ParDo call. C, D - This is common way to filter too, but we will need the key in order to partition or GroupByKey B - ParDo is the most basic method, it can do anything to the PCollection

AzureDP900 1 year, 9 months ago

B. Add a ParDo transform in Cloud Dataflow to discard corrupt elements.

medeis_jar 2 years, 8 months ago

Selected Answer: B

Filtering with ParDo. ParDo is a Beam transform for generic parallel processing. ParDo is useful for common data processing operations/

AzureDP900 1 year, 9 months ago

I agree with B

sumanshu 3 years, 2 months ago

vote for 'B', ParDo can discard the elements. https://beam.apache.org/documentation/programming-guide/

DeepakKhattar 3 years, 8 months ago

B - seems to be better option since we need to filter out, question does not specify that we do need to store it into different Pcollection. https://beam.apache.org/documentation/transforms/python/overview/ ParDo is general purpose whereas partition splits the elements into do different pcollections. https://beam.apache.org/documentation/transforms/python/elementwise/partition/

arghya13 3 years, 10 months ago

B is correct

haroldbenites 4 years, 1 month ago

B is correct

Archy 4 years, 2 months ago

B, ParDo is useful for a variety of common data processing operations, including: Filtering a data set. You can use ParDo to consider each element in a PCollection and either output that element to a new collection or discard it.

tprashanth 4 years, 2 months ago

Looks like C it is https://beam.apache.org/documentation/programming-guide/

atnafu2020 4 years, 1 month ago

according this link its Pardo * Filtering a data set. You can use ParDo to consider each element in a PCollection and either output that element to a new collection or discard it. * But Partition just splitting which is is a Beam transform for PCollection objects that store the same data type. Partition splits a single PCollection into a fixed number of smaller collections.

xrun 3 years, 9 months ago

Seems like two answers may be correct. With ParDo you can discard corrupt data. With Partition you can split the data into two PCollections: corrupt and ok. You stream ok data further to BigQuery and corrupt data to some other storage for analysis. If one is not interested in analysis, then ParDo is enough.

dg63 4 years, 2 months ago

Correct answer should be "C". A Pardo transform will allow the processing to happening in parallel using multiple workers. Partition transform will allow data to be partitions in two different Pcollections according to some logic. Using partition transform once can split the corrupted data and finally discard it.

Rajuuu 4 years, 2 months ago

Correct B.

norwayping 4 years, 3 months ago

Correct - B

https://zhangxuetu.online

[202610]

Question #111 Topic 1

You have historical data covering the last three years in BigQuery and a data pipeline that delivers new data to BigQuery daily. You have noticed that when the
Data Science team runs a query filtered on a date column and limited to 30`"90 days of data, the query scans the entire table. You also noticed that your bill is increasing more quickly than you expected. You want to resolve the issue as cost-effectively as possible while maintaining the ability to conduct SQL queries.
What should you do?

A. Re-create the tables using DDL. Partition the tables by a column containing a TIMESTAMP or DATE Type. **Most Voted**

B. Recommend that the Data Science team export the table to a CSV file on Cloud Storage and use Cloud Datalab to explore the data by reading the files directly.

C. Modify your pipeline to maintain the last 3090"€ı days of data in one table and the longer history in a different table to minimize full table scans over the entire history.

D. Write an Apache Beam pipeline that creates a BigQuery table per day. Recommend that the Data Science team use wildcards on the table name suffixes to select the data they need.

Correct Answer: A

Community vote distribution

A (100%)

Comments

arghya13 **Highly Voted** 3 years, 10 months ago
I will go with Option A

SteelWarrior **Highly Voted** 4 years ago
Should be A. With partitions the performance will improve for selecting 30-90 days data. Also the storage cost will reduce as the old partitions (not updated in last 90 days) will qualify for Long-Term storage rates.

odacir 1 year, 9 months ago
Selected Answer: A
Answer: A, has no cost to reload the data, Also Partition is the solution for reducing cost and time

John_Pongthorn 2 years ago
Selected Answer: A
it is not B in the sense of cost-effective certainly. read below in limitation
https://cloud.google.com/bigquery/docs/querying-wildcard-tables#limitations Currently, cached results are not

supported for queries against multiple tables using a wildcard even if the Use Cached Results option is checked. If you run the same wildcard query multiple times, you are billed for each query.

John_Pongthorn 2 years ago

Selected Answer: A

https://cloud.google.com/bigquery/docs/partitioned-tables#dt_partition_shard Partitioning is recommended over table sharding, because partitioned tables perform better

John_Pongthorn 2 years ago

Selected Answer: A

A AND D , they are the most likely choiced but the questionn want issue as cost-effectively as possible while maintaining the ability to conduct SQL queries. 1 table may be cheaper so partition is better than wildcarf

Didine_22 2 years, 5 months ago

Selected Answer: A

answer A

medeis_jar 2 years, 8 months ago

Selected Answer: A

https://cloud.google.com/bigquery/docs/partitioned-tables

MaxNRG 2 years, 9 months ago

Selected Answer: A

A. Partiotioning https://cloud.google.com/bigquery/docs/partitioned-tables

Tomi1313 2 years, 9 months ago

Why not D? You can use SQL. This is the cheapest and fastest option
https://cloud.google.com/bigquery/docs/querying-wildcard-tables

John_Pongthorn 2 years ago

Partitioning is recommended over table sharding, because partitioned tables perform better This is a google recommendation nowaday.

StefanoG 2 years, 10 months ago

Selected Answer: A

The D solution is obviously discarded. The request NOT require ONLY LAST 30-90 days, so the C solution is not the right solution. In addition to this, the request ask to keep the possibility to made queries, so B is wrost. Is not mandatory make the queries while you make the modify so the right answer is A

JayZeeLee 2 years, 10 months ago

B sounds more feasible. The point is 'historical' data, not new table/data. Recreating tables from the past three years is a lot of work. Might as well export the table and run analyses there. No cost for exporting in BigQuery.

sumanshu 3 years, 2 months ago

Vote for A

Alasmindas 3 years, 10 months ago

I will go with Option A, although at first instance I felt Option C would be correct. Option A : Because partitioning will help to address both the concerns mentioned in the question - i.e. faster query and reducing cost. Option C : Modifying the data pipeline to store last 30-90 days data would have possible, if there was a point mentioned that only

the latest data (30-90 days) is kept and the older data - beyond 90 days is moved to the master table. Since that point is mot mentioned, we will land up having multiple - 30-90 days data in separate tables + the master table.

karthik89 3 years, 7 months ago

but how will you append the data that is older than 90days in to the master table?

Cloud_Enthusiast 3 years, 10 months ago

Answer is A. Recreating the DDL with new parition is easy and does not require any changes on applications that read data from it

haroldbenites 4 years, 1 month ago

A is correct

Rajuuu 4 years, 2 months ago

PArtition the tables is the key for query improvement.

Rajuuu 4 years, 2 months ago

I think C will be more cost effective than using A as recreating the whole DDL is more expensive..

tprashanth 4 years, 2 months ago

No, if a seperate table is maintained for last 30-90 days data, we end up creating a table on daily basis

https://zhangxuetu.online

[202610]

Question #112 Topic 1

You operate a logistics company, and you want to improve event delivery reliability for vehicle-based sensors. You operate small data centers around the world to capture these events, but leased lines that provide connectivity from your event collection infrastructure to your event processing infrastructure are unreliable, with unpredictable latency. You want to address this issue in the most cost-effective way. What should you do?

A. Deploy small Kafka clusters in your data centers to buffer events.

B. Have the data acquisition devices publish data to Cloud Pub/Sub. **Most Voted**

C. Establish a Cloud Interconnect between all remote data centers and Google.

D. Write a Cloud Dataflow pipeline that aggregates all data in session windows.

Correct Answer: B

Community vote distribution



Comments

Ganshank Highly Voted 5 years, 11 months ago

C. This is a tricky one. The issue here is the unreliable connection between data collection and data processing infrastructure, and to resolve it in a cost-effective manner. However, it also mentions that the company is using leased lines. I think replacing the leased lines with Cloud InterConnect would solve the problem, and hopefully not be an added expense. https://cloud.google.com/interconnect/docs/concepts/overview

serg3d 5 years, 10 months ago

Yea, this would definitely solve the issue, but it's not "the most cost-effective way". I think PubSub is the correct answer.

snamburi3 5 years, 4 months ago

the question also talks about a cost effective way...

sh2020 5 years, 9 months ago

I agree, C is the only choice that addresses the problem. The problem is caused by leased line. How come pub/sub service can resolve it? Pub/sub will still use the leased line

awssp12345 4 years, 9 months ago

DEFINITELY NOT COST EFFECT. C IS THE WORST CHOICE.

ayush_1995 Highly Voted 3 years, 2 months ago

Selected Answer: B

B. Have the data acquisition devices publish data to Cloud Pub/Sub. This would provide a reliable messaging service for your event data, allowing you to ingest and process your data in a timely manner, regardless of the reliability of the leased lines. Cloud Pub/Sub also offers automatic retries and fault-tolerance, which would further improve the reliability of your event delivery. Additionally, using Cloud Pub/Sub would allow you to easily scale up or down your event processing infrastructure as needed, which would help to minimize costs.

b2aaace 4 months, 2 weeks ago

Selected Answer: A

Option A: Deploy small Kafka clusters in your data centers • Kafka acts as a local buffer for events. • Sensors write to the local Kafka cluster, ensuring no data loss even if connectivity to the central processing infrastructure is slow or unreliable. • Once connectivity is available, Kafka can replicate or forward events to the central processing pipeline. • This is cost-effective compared to provisioning dedicated interconnects or streaming directly to cloud services with unreliable lines. • Kafka is designed for high-throughput, low-latency buffering, making it ideal here.

grshankar9 1 year, 2 months ago

Selected Answer: B

The data acquisition devices are spread across the globe. How many cloud interconnects would this require? Option C is definitely not cost effective. Option B seems to make sense as it suggests publishing from the acquisition device to Pub/Sub and not from the collection infrastructure. The problem clearly stated the connection from collection infrastructure to the processing infrastructure was unreliable.

Anudeep58 1 year, 9 months ago

Selected Answer: B

Option B: Have the data acquisition devices publish data to Cloud Pub/Sub. Rationale: Managed Service: Cloud Pub/Sub is a fully managed service, reducing the operational overhead compared to managing Kafka clusters. Reliability and Scalability: Cloud Pub/Sub can handle high volumes of data with low latency and provides built-in mechanisms for reliable message delivery, even in the face of intermittent connectivity. Cost-Effective: Cloud Pub/Sub offers a pay-as-you-go pricing model, which can be more cost-effective than setting up and maintaining dedicated network infrastructure like Cloud Interconnect. Global Availability: Cloud Pub/Sub is available globally and can handle data from multiple regions efficiently.

Nandababy 2 years, 3 months ago

Even with Cloud Pub/Sub, unpredictable latency or delays could still occur due to the unreliable leased lines connecting your event collection infrastructure and event processing infrastructure. While Cloud Pub/Sub offers reliable message delivery within its own network, the handoff to your processing infrastructure is still dependent on the leased lines. Replacing leased lines with Cloud Interconnect could potentially resolve the overall issue of unpredictable latency in event processing pipeline but it could be unnecessary expense provided data centers distributed world wide. Cloud Pub/Sub along with other optimization techniques like Cloud VPN or edge computing might be sufficient.

FP77 2 years, 7 months ago

Selected Answer: C

I don't know why B is the most voted. The issue here is unreliable connectivity and C is the perfect use-case for that

NeoNitin 2 years, 7 months ago

its says with unpredictable latency and here no need to worry about connection So B is the right one

ZZHZZH 2 years, 8 months ago

Selected Answer: C

The question is misleading. But should be C since it addresses the unpredictablility and latency directly.

musumusu 3 years, 1 month ago

Best answer is A, By using Kafka, you can buffer the events in the data centers until a reliable connection is established with the event processing infrastructure. But go with B, its google asking :P

musumusu 3 years, 1 month ago

I read this question again, I wanna answer C. Buying Data acquisition devices and set them up with sensor, i dont think its practical approach. Imagine, Adruino is cheapest IOT available in market for 15 dollars, but who will open the sensor box and install it .. omg,, its a big job. This question depends if IOT devices that are attached to sensor needs to be programmed. Big Headache right. Use google cloud connect to deal with current situation. Or reprogramme IOT if they have connected with sensors.

desertlotus1211 3 years, 2 months ago

Are they talking about GCP in this question? Where is the event processing infrastructure? Answer A, might be correct!

PrashantGupta1616 3 years, 3 months ago

Selected Answer: B

pub/sub is region is a global service It's important to note that the term "global" in this context refers to the geographical scope of the service

NicolasN 3 years, 3 months ago

Selected Answer: A

As usual the answer is hidden somewhere in the Google Cloud Blog: "In the case of our automotive company, the data is already stored and processed in local data centers in different regions. This happens by streaming all sensor data from the cars via MQTT to local Kafka Clusters that leverage Confluent's MQTT Proxy." "This integration from devices to a local Kafka cluster typically is its own standalone project, because you need to handle IoT-specific challenges like constrained devices and unreliable networks." □ https://cloud.google.com/blog/products/ai-machine-learning/enabling-connected-transformation-with-apache-kafka-and-tensorflow-on-google-cloud-platform

desertlotus1211 3 years, 2 months ago

The question is asking from the on-premise infrastructure, which already has the data, to the event processing infrastructure, which is in the GCP, is unreliable.... it not asking from the sensors to the on-premise...

desertlotus1211 3 years, 2 months ago

I might have to retract my answer... Are they talking about GCP in this question? where is the event processing infrastructure?

piotrpiskorski 3 years, 4 months ago

yeah, changing whole architecture arround the world for the use of pub/sub is so much more cost efficient than Cloud Interconnect (which is like 3k$).. It's C.

odacir 3 years, 3 months ago

It's not a Cloud Interconnect, it's a lot of interconnect ones per data center, PUB/SUB addresses all the requirements. Its B

odacir 3 years, 3 months ago

ALSO, the problem it's no t your connection, its the connectivity BT your event collection infrastructure to your event processing infrastructure, so PUSUB it's perfect for this

jkhong 3 years, 3 months ago

Wouldn't using cloud interconnect also result in amendments to each of the data center around the world? I don't see why there would be a huge architecture change when using PubSub, the publishers would just need to push

messages directly to pubsub, instead of pushing to their own cost center. Also, if the script for pushing messages can be standardised, the data centers can share it around to

TNT87 3 years, 6 months ago

Selected Answer: B

Cloud Pub/Sub, it supports batch & streaming , push and pull capabilities Answer B

t11 3 years, 7 months ago

It has to be B.

rr4444 3 years, 7 months ago

Selected Answer: D

Feels like everyone is wrong. A. Deploy small Kafka clusters in your data centers to buffer events. - Silly in a GCP cloudnative context, plus they have messaging infra anyway B. Have the data acquisition devices publish data to Cloud Pub/Sub. - They have messaging infra, so why? Unless they want to replace, it, but that doesn't change the issue C. Establish a Cloud Interconnect between all remote data centers and Google. - Wrong, because Interconnect is basically a leased line. There must be some telecoms issue with it, which we can assume is unresolvable e.g. long distance remote locations and sometimes water ingress, and the telco can't justify sorting it yet, or is slow to, or something. Leased lines usually don't come with awful internet connectivity, so sound physical connectivity issue. Sure, an Interconnect is better, more direct, but a leased line should be bullet proof. D. Write a Cloud Dataflow pipeline that aggregates all data in session windows. - The only way to address dodgy/delayed data delivery

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #113 Topic 1

You are a retailer that wants to integrate your online sales capabilities with different in-home assistants, such as Google Home. You need to interpret customer voice commands and issue an order to the backend systems. Which solutions should you choose?

A. Speech-to-Text API

B. Cloud Natural Language API

C. Dialogflow Enterprise Edition **Most Voted**

D. AutoML Natural Language

Correct Answer: C

Community vote distribution



Comments

rickywck **Highly Voted** 6 years ago

should be C, since we need to recognize both voice and intent

AzureDP900 3 years, 3 months ago

C. Dialogflow Enterprise Editio

Alasmindas **Highly Voted** 5 years, 4 months ago

Option A - Cloud Speech-to-Text API. The question is just asking to " interpret customer voice commands" .. it does not mention anything related to sentiment analysis so NLP is not required. DialogFlow is more of a chat bot services typically suited for a "Service Desk" kind of setup - where clients will call a centralized helpdesk and automation is achieved through Chat bot services like - google Dialog flow

hdmi_switch 4 years, 8 months ago

Cloud Speech-to-Text API just converts speech to text. You will have text files as an output and then the requirement is to "interpret customer voice commands and issue an order to the backend systems". This is not achieved by having text files. I would go with option C, since Dialogflow can interpret the commands (intents) and integrates other applications e.g. backend systems.

exnaniantwort 2 years, 6 months ago

shuld be C, the key is interpret customer voice commands

HarshKothari21 3 years, 6 months ago

Question also says "in-home assistants, such as Google Home". the idea here is to provide assistance which involves Dialog. I would go with option C

grshankar9 1 year, 2 months ago

Selected Answer: C

While both are Google Cloud services related to speech processing, the key difference is that Cloud Speech-to-Text API is solely focused on transcribing spoken language into text, while DialogFlow is a more comprehensive platform that not only transcribes speech but also interprets the meaning of the conversation, allowing you to build conversational AI applications with features like intent recognition and entity extraction

LR2023 1 year, 6 months ago

This is a queston from Actual exam question from Google's Professional Data Engineer so C makes sense. This is not an AWS question

AlizCert 1 year, 10 months ago

Selected Answer: A

The question clearly states "voice commands" which is a term for short (few words long at most) well-defined phrases to be recognized. No need for a dialog. Even if I were to use Dialogflow, I would use ES instead of CX (new name for Enterprise Edition), no fancy features are required for this.

NeoNitin 2 years, 7 months ago

Ans C . main thing is that question is saying" customer voice commands " there is no need to sentimental analysis of language so thats why. C. Dialogflow Enterprise Edition Dialogflow is a powerful natural language understanding platform developed by Google. It allows you to build conversational interfaces, interpret user voice commands, and integrate with various platforms and devices like Google Home. The "Enterprise Edition" provides additional features and support for more complex use cases, making it a good choice for a retailer looking to integrate with in-home assistants and handle customer voice commands effectively.

juliobs 3 years ago

Selected Answer: C

Answer is C. However Google Assistant Conversational Actions will be sunsetted on June 13, 2023.

techtitan 3 years, 1 month ago

Selected Answer: C

https://cloud.google.com/dialogflow/es/docs/integrations/aog

desertlotus1211 3 years, 2 months ago

I think the answer is A: Speech to Text. You want to interpret what a user say... Diagflow is text to speech, not what the question asked for... Thoughts?

PrashantGupta1616 3 years, 3 months ago

Selected Answer: A

The question is just asking to " interpret customer voice commands" so A is out of the box solution

odacir 3 years, 3 months ago

Selected Answer: A

Enable voice control Implement voice commands such as “turn the volume up,” and voice search such as saying “what is the temperature in Paris?” Combine this with the Text-to-Speech API to deliver voice-enabled experiences in IoT (Internet of Things) applications. https://cloud.google.com/speech-to-text#section-9

odacir 3 years, 3 months ago

I change my mind, it's C. https://cloud.google.com/blog/products/gcp/introducing-dialogflow-enterprise-edition-a-new-way-to-build-voice-and-text-conversational-apps

TNT87 3 years, 6 months ago

https://cloud.google.com/blog/products/gcp/introducing-dialogflow-enterprise-edition-a-new-way-to-build-voice-and-text-conversational-apps Dialogflow is the answer

Smaks 3 years, 8 months ago

Selected Answer: C

Dialogflow provides a seamless integration with Google Assistant. This integration has the following advantages: You can use the same Dialogflow agent to power Google Assistant and other integrations. Dialogflow agents provide Google Cloud enterprise-grade security, privacy, support, and SLAs

Vip777 3 years, 11 months ago

dialog

Vip777 3 years, 11 months ago

speech

PJG_worm 4 years ago

It should be D. INTERPRET customer voice commands and issue an order to the backend systems. Option C is usually applied for conversation. But in this case, it is not a conversation.

medeis_jar 4 years, 2 months ago

Selected Answer: C

recognize voice and intent https://cloud.google.com/blog/products/gcp/introducing-dialogflow-enterprise-edition-a-new-way-to-build-voice-and-text-conversational-apps

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #114 Topic 1

Your company has a hybrid cloud initiative. You have a complex data pipeline that moves data between cloud provider services and leverages services from each of the cloud providers. Which cloud-native service should you use to orchestrate the entire pipeline?

A. Cloud Dataflow

B. Cloud Composer **Most Voted**

C. Cloud Dataprep

D. Cloud Dataproc

Correct Answer: B

Community vote distribution

B (100%)

Comments

madhu1171 **Highly Voted** 4 years, 6 months ago

Answer should be B

Darlee **Highly Voted** 3 years, 11 months ago

How come the `Correct Answer` so ridiculous WRONG?

squishy_fishy 2 years, 11 months ago

Ha ha.. I know.

grshankar9 1 year, 2 months ago

Selected Answer: B

Cloud Composer is considered suitable across multiple cloud providers, as it is built on Apache Airflow, which allows for workflow orchestration across different cloud environments and even on-premises data centers, making it a good choice for multi-cloud strategies; however, its tightest integration is with Google Cloud Platform services.

forepick 1 year, 4 months ago

Selected Answer: B

No other option is aimed for this purpose

juliobs 1 year, 6 months ago

Selected Answer: B

Airflow

PrashantGupta1616 1 year, 9 months ago

Selected Answer: B

Cloud Composer is Airflow

odacir 1 year, 9 months ago

Selected Answer: B

Cloud Composer is Airflow, It's made for this job.

medeis_jar 2 years, 8 months ago

Selected Answer: B

https://cloud.google.com/composer/

MaxNRG 2 years, 9 months ago

Selected Answer: B

B: Cloud Composer is a fully managed workflow orchestration service that empowers you to author, schedule, and monitor pipelines that span across clouds and on-premises data centers. https://cloud.google.com/composer/ Cloud Composer can help create workflows that connect data, processing, and services across clouds, giving you a unified data environment. Built on the popular Apache Airflow open source project and operated using the Python programming language, Cloud Composer is free from lock-in and easy to use. Cloud Composer gives you the ability to connect your pipeline through a single orchestration tool whether your workflow Eves on-premises, in multiple clouds, or fully within GCP. The ability to author, schedule, and monitor your workflows in a unified manner means you can break down the silos in your environment and focus less on infrastructure.

MaxNRG 2 years, 9 months ago

Option A is wrong as Cloud Scheduler is a fully managed enterprise-grade cron job scheduler. It is not a multi-cloud orchestration tool. Option B is wrong as Google Cloud Dataflow is a fully managed service for strongly consistent, parallel data-processing pipelines. It does not support multi-cloud handling. Option D is wrong as Google Cloud Dataproc is a fast, easy to use, managed Spark and Hadoop service for distributed data processing.

JG123 2 years, 10 months ago

Selected Answer: B

Answer: B

sandipk91 3 years, 1 month ago

Cloud composer

sumanshu 3 years, 2 months ago

Vote for B

daghayeghi 3 years, 6 months ago

B: Hybrid and multi-cloud Ease your transition to the cloud or maintain a hybrid data environment by orchestrating workflows that cross between on-premises and the public cloud. Create workflows that connect data, processing, and services across clouds to give you a unified data environment. https://cloud.google.com/composer#section-2

someshsehgal 3 years, 7 months ago

Correct B: without any doubt.

arghya13 3 years, 10 months ago

B-Cloud Composer works on a multicloud environment

Alasmindas 3 years, 10 months ago

there can not be any simple question like this to choose the right answer as "Cloud Composer". I really feel someone must have deliberately selecting the wrong answers in Exam topics to confuse people....

Cloud_Enthusiast 3 years, 10 months ago

Composer is the obvious answer. so B

https://zhangxuetu.online

[202610]

Question #115 Topic 1

You use a dataset in BigQuery for analysis. You want to provide third-party companies with access to the same dataset. You need to keep the costs of data sharing low and ensure that the data is current. Which solution should you choose?

A. Use Analytics Hub to control data access, and provide third party companies with access to the dataset. **Most Voted**

B. Use Cloud Scheduler to export the data on a regular basis to Cloud Storage, and provide third-party companies with access to the bucket.

C. Create a separate dataset in BigQuery that contains the relevant data to share, and provide third-party companies with access to the new dataset.

D. Create a Dataflow job that reads the data in frequent time intervals, and writes it to the relevant BigQuery dataset or Cloud Storage bucket for third-party companies to use.

Correct Answer: A

Community vote distribution

A (100%)

Comments

LP_PDE **Highly Voted** 2 years, 6 months ago

I feel the answer really should be Create an authorized view on the BigQuery table to control data access, and provide third-party companies with access to that view.

vamgcp 1 year, 8 months ago

Selected Answer: A

Option A: This option is correct because Analytics Hub is a managed service that provides a centralized repository for data assets. You can use Analytics Hub to share data with other Google Cloud Platform services, as well as with third-party companies

musumusu 2 years, 1 month ago

You are preparing for exam: Creating a view and share with 3rd party is best and cheapest. Then create a separate dataset to share it cost less than using paid service for data access i.e analytics hub where you create data access policies you choose, its just making me craazy

musumusu 2 years, 1 month ago

One main reason you should use analytics hub, when you want control over 3 party activites and you want to monetize (to make money) by sharing BQ dataset.

lool 2 years, 2 months ago

Selected Answer: A

Shared datasets are collections of tables and views in BigQuery defined by a data publisher and make up the unit of cross-project / cross-organizational sharing. Data subscribers get an opaque, read-only, linked dataset inside their project and VPC perimeter that they can combine with their own datasets and connect to solutions from Google Cloud or our partners. For example, a retailer might create a single exchange to share demand forecasts to the 1,000's of vendors in their supply chain–having joined historical sales data with weather, web clickstream, and Google Trends data in their own BigQuery project, then sharing real-time outputs via Analytics Hub. The publisher can add metadata, track subscribers, and see aggregated usage metrics.

AzureDP900 2 years, 3 months ago

A. Use Analytics Hub to control data access, and provide third party companies with access to the dataset.

odacir 2 years, 3 months ago

Selected Answer: A

https://cloud.google.com/analytics-hub

Atnafu 2 years, 4 months ago

A Multiple choose listed wrongly Correct one A. Create an authorized view on the BigQuery table to control data access, and provide third-party companies with access to that view. B. Use Cloud Scheduler to export the data on a regular basis to Cloud Storage, and provide third-party companies with access to the bucket. c. Create a separate dataset in BigQuery that contains the relevant data to share, and provide third-party companies with access to the new dataset. D. Create a Cloud Dataflow job that reads the data in frequent time intervals, and writes it to the relevant BigQuery dataset or Cloud Storage bucket for third-party companies to use.

ajayrtk 2 years, 5 months ago

no option is correct this is correct answer -Create an authorised view on the BigQuery table to control data access, and provide third-party companies with access to that view.

AWSandeep 2 years, 7 months ago

Selected Answer: A

A. Use Analytics Hub to control data access, and provide third party companies with access to the dataset.

damaldon 2 years, 7 months ago

Answer A. As an Analytics Hub user, you can perform the following tasks: As an Analytics Hub publisher, you can monetize data by sharing it with your partner network or within your own organization in real time. Listings let you share data without replicating the shared data. You can build a catalog of analytics-ready data sources with granular permissions that let you deliver data to the right audiences. As an Analytics Hub subscriber, you can discover the data that you are looking for, combine shared data with your existing data, and leverage the built-in features of BigQuery. When you subscribe to a listing, a linked dataset is created in your project. As an Analytics Hub viewer, you can browse through the datasets that you have access to in Analytics Hub and request the publisher to access the shared data. As an Analytics Hub administrator, you can create data exchanges that enable data sharing, and then give permissions to data publishers and subscribers to access these data exchanges.
https://cloud.google.com/bigquery/docs/analytics-hub-introduction

https://zhangxuetu.online

[202610]

Question #116 Topic 1

Your company is in the process of migrating its on-premises data warehousing solutions to BigQuery. The existing data warehouse uses trigger-based change data capture (CDC) to apply updates from multiple transactional database sources on a daily basis. With BigQuery, your company hopes to improve its handling of CDC so that changes to the source systems are available to query in BigQuery in near-real time using log-based CDC streams, while also optimizing for the performance of applying changes to the data warehouse. Which two steps should they take to ensure that changes are available in the BigQuery reporting table with minimal latency while reducing compute overhead? (Choose two.)

A. Perform a DML INSERT, UPDATE, or DELETE to replicate each individual CDC record in real time directly on the reporting table.

B. Insert each new CDC record and corresponding operation type to a staging table in real time. Most Voted

C. Periodically DELETE outdated records from the reporting table.

D. Periodically use a DML MERGE to perform several DML INSERT, UPDATE, and DELETE operations at the same time on the reporting table. Most Voted

E. Insert each new CDC record and corresponding operation type in real time to the reporting table, and use a materialized view to expose only the newest version of each unique record.

Correct Answer: BD

Community vote distribution



Comments

YorelNation Highly Voted 3 years, 6 months ago

Selected Answer: BD

To aim for minimal latency while reducing compute overhead: B. Insert each new CDC record and corresponding operation type to a staging table in real time. D. Periodically use a DML MERGE to perform several DML INSERT, UPDATE, and DELETE operations at the same time on the reporting table. (all statements comes from the staging table)

musumusu Highly Voted 3 years, 1 month ago

B&D Tricks here: Always choose google recommended approach, Use data first in Staging table then merge with original tables.

Nirca 2 years, 5 months ago

Selected Answer: CE

I'm going for E &C - this in the only solution with low TCO. E - is the best way to work with CDC when real nearline data is needed BQ snapshots can be online! . & C - is good practice to delete old records.

nadavw 1 year, 7 months ago

A isn't correct, as the requirement is "reducing compute overhead" B isn't correct, as there is no mention of a "staging table" in the scenario D isn't correct as it's done periodically, and the requirement is "near real-time"

dconesoko 3 years, 3 months ago

Selected Answer: BD

with both the delta table and the main table changes could be queried in near realtime, by using a view that unions both tables and queries the laters record for the given key, eventually the delta table should be merged into the main table and truncated. Google recently introduced datastream that would take away all these headaches.

odacir 3 years, 3 months ago

Selected Answer: BD

The solution is B and D. I perform a similar task in my work, and this is the best way to do it at scale with BigQuery.

NicolasN 3 years, 4 months ago

I really can’t find a correct combination of answers. I'm between the following alternatives, but with no one fitting: 1) [B] and [D]: That's a proposed solution, but as a cost-optimized approach (along with an extra step to "Periodically DELETE outdated records from the STAGING table" - more details on my subsequent reply). Also, I can't imagine how an answer with the word "Periodically" may be compatible with the "minimal latency" requirement. 2) [E] and [C]: It could be a valid approach, but near-real time requirement would demand also for a materialized view refresh. And it seems to contradict the "reducing compute overhead" req. 3) [A] standalone: Provides immediate results but is far from compute-optimized.

NicolasN 3 years, 4 months ago

Nowadays (Nov. 2022) I don't expect to confront this question in a real exam with this set of answers since the more recent documentation proposes the use of Datastream.) https://cloud.google.com/blog/products/data-analytics/real-time-cdc-replication-bigquery

NicolasN 3 years, 4 months ago

The previous guidelines were here:) https://cloud.google.com/architecture/database-replication-to-bigquery-using-change-data-capture#immediate_consistency_approach There were two approaches: 1) Immediate consistency approach 2) Cost-optimized approach For approach 1), which is the objective of this question, it proposes: a. Insert CDC data into a delta table in BigQuery => that's answer [B] b. Create a BigQuery view that joins the main and delta tables and finds the most recent row => there' no answer that fits For approach 2) it proposes: a. Insert CDC data into a delta table in BigQuery => that's answer [B] b. Merge delta table changes into the main table and periodically purge merged rows from the delta table - Run Merge statement on a regular interval => that's answer [D]

beanz00 3 years, 5 months ago

B and E. Typically in a Data Warehouse you don't delete date. Data Warehouse should store full history to see how the data changed over time. All the solutions with 'DELETE' should not be used as this goes against being able to access the history of the data.

TNT87 3 years, 5 months ago

https://www.striim.com/blog/oracle-to-google-bigquery/

TNT87 3 years, 6 months ago

https://cloud.google.com/architecture/database-replication-to-bigquery-using-change-data-capture#data_latency

TNT87 3 years, 6 months ago

https://docs.streamsets.com/platform-datacollector/latest/datacollector/UserGuide/Destinations/GBigQuery.html

Wasss123 3 years, 6 months ago

Selected Answer: BD

B and D https://cloud.google.com/architecture/database-replication-to-bigquery-using-change-data-capture

John_Pongthorn 3 years, 6 months ago

Selected Answer: BC

B D you have to do the both to get it done. To merge process, you have to perform between the report table and stage a table

Remi2021 3 years, 6 months ago

Answers are tricky, official documentation suggests Dataflow or Datafusion path as well as inclusion of DataStreams https://cloud.google.com/blog/products/data-analytics/real-time-cdc-replication-bigquery

changsu 3 years, 6 months ago

Selected Answer: CE

It costs more to update/delete.

ducc 3 years, 7 months ago

Selected Answer: BE

BE is correct Big Query only need to capture change, no need DELETE, UPDATE

https://zhangxuetu.online

[202610]

Question #117 Topic 1

You are designing a data processing pipeline. The pipeline must be able to scale automatically as load increases. Messages must be processed at least once and must be ordered within windows of 1 hour. How should you design the solution?

A. Use Apache Kafka for message ingestion and use Cloud Dataproc for streaming analysis.

B. Use Apache Kafka for message ingestion and use Cloud Dataflow for streaming analysis.

C. Use Cloud Pub/Sub for message ingestion and Cloud Dataproc for streaming analysis.

D. Use Cloud Pub/Sub for message ingestion and Cloud Dataflow for streaming analysis. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

madhu1171 **Highly Voted** 5 years ago

Answer should be D

Chelseajcole **Highly Voted** 3 years, 6 months ago

rule of thumb: If you see Kafka and Pub/Sub, always go with Pub/Sub in Google exam

hendrixlives 3 years, 3 months ago

Careful doing that: I got a question where you had to choose between Kafka and Pub/Sub... and the solution required to be able to replay all messages without time limit. So no Pub/Sub there. This being a Google cert does not mean that they always force Google solutions.

grshankar9 1 year, 2 months ago

Selected Answer: D

Kafka is recommended over Pub/Sub only when there is requirement of high throughput, complex streaming, more flexibility with customization and fine-tuning configurations or when application spans multiple cloud providers and requires more flexibility in deployment across different platforms.

NeoNitin 1 year, 7 months ago

Data proc is serverbased Dataflow is serverless which is used to run pipelines which uses apache framework in the background. Just need to mention the number of workers needed. so question saying we need scale automatically . so dataproc eliminate ho gaya now Dataflow is correct , pub/sub is recommended for this scenario. D

dconesoko 2 years, 3 months ago

Selected Answer: D

google's preferred choice

VictorBa 2 years, 11 months ago

Selected Answer: D

It cannot be C because Dataproc is more suitable for Hadoop jobs.

medeis_jar 3 years, 2 months ago

Selected Answer: D

Pub/Sub + Dataflow

MaxNRG 3 years, 2 months ago

Selected Answer: D

D: Pub/Sub + Dataflow https://cloud.google.com/solutions/stream-analytics/ https://cloud.google.com/blog/products/data-analytics/streaming-analytics-now-simpler-more-cost-effective-cloud-dataflow

hendrixlives 3 years, 3 months ago

Selected Answer: D

D: "at least once and must be ordered within windows" means Pub/Sub (at least once) with Dataflow (windows).

JG123 3 years, 4 months ago

Correct: D

sandipk91 3 years, 7 months ago

Answer is D

awssp12345 3 years, 9 months ago

https://cloud.google.com/architecture/migrating-from-kafka-to-pubsub#comparing_features

sumanshu 3 years, 9 months ago

Vote for D Scaling - Dataflow. Delivery of confimed atleast 1 message - Pub/Sub

Sush12 4 years, 1 month ago

Answer is D

Alasmindas 4 years, 4 months ago

Indeed the correct answer is Option D. Again, not sure why Exam topic answer is deliberately chosen for a wrong answer, for such simple question.

szefco 3 years, 4 months ago

To make us think of each question while studying, not just trying to memorize answers :) it looks that "correct" answers are chosen randomly :)

Cloud_Enthusiast 4 years, 4 months ago

D for GCP native solution

haroldbenites 4 years, 7 months ago

D is correct

https://zhangxuetu.online

[202610]

Question #118 Topic 1

You need to set access to BigQuery for different departments within your company. Your solution should comply with the following requirements:
✑ Each department should have access only to their data.
✑ Each department will have one or more leads who need to be able to create and update tables and provide them to their team.
✑ Each department has data analysts who need to be able to query but not modify data.
How should you set access to the data in BigQuery?

A. Create a dataset for each department. Assign the department leads the role of OWNER, and assign the data analysts the role of WRITER on their dataset.

B. Create a dataset for each department. Assign the department leads the role of WRITER, and assign the data analysts the role of READER on their dataset. Most Voted

C. Create a table for each department. Assign the department leads the role of Owner, and assign the data analysts the role of Editor on the project the table is in.

D. Create a table for each department. Assign the department leads the role of Editor, and assign the data analysts the role of Viewer on the project the table is in.

Correct Answer: B

Community vote distribution



Comments

juliobs Highly Voted 3 years ago
Old question. It's done using IAM nowadays: bigquery.dataEditor and bigquery.dataViewer

AWSandeep Highly Voted 3 years, 7 months ago
Selected Answer: B
B. Create a dataset for each department. Assign the department leads the role of WRITER, and assign the data analysts the role of READER on their dataset.

Ben_oso 9 months, 1 week ago
Selected Answer: B
The answers was to old, Reader role its a legacy role and its equal to viewer. The answer is B, but is too old. The other answers no make sense

SamuelTsch 1 year, 5 months ago

Selected Answer: D

No writer, reader role

mothkuri 2 years ago

Selected Answer: D

WRITER role is not there in roles of BigQuery table/dataset

Kalai_1 2 years, 3 months ago

Answer : D. There is no role called WRITER or READER as preliminary role.

forepick 2 years, 10 months ago

Selected Answer: B

both C & D violate the principle of least privilege. A talks about OWNER and WRITER roles, and the analyst doesn't need a writer role. So we're left with B.

Joane_ 2 years, 11 months ago

Selected Answer: D

https://cloud.google.com/bigquery/docs/access-control#bigquery

midgoo 3 years ago

Selected Answer: B

B - Lead needs to have the role to create tables and also Analyst only need to read

musumusu 3 years, 1 month ago

Answer B: Why not D, mentioned in question: Data lead will create tables in dataset. Imagine, other department leads are creating unnecessory tables in shared dataset and you are struggling to find your tables as everyday there are some new tables. Headache right ? better to give them seperate dataset and do whatever you want in that dataset.

xj_kevin 3 years, 1 month ago

Vote B, both BD can fullfill the job requirement but B is on dataset level and D on project level. "By default, granting access to a project also grants access to datasets within it." D may issue unnecessary accesses to other content in the project.

desertlotus1211 3 years, 2 months ago

Interestingly enough - I know believe the answer is A... Deleting is not the same as modify...

desertlotus1211 3 years, 2 months ago

Answer is B: https://cloud.google.com/bigquery/docs/access-control The question ask for the lead to be able to: CREATE, UPDATE, and SHARE with the team... BigQuery Data Owner can do that (roles/bigquery.dataOwner) When applied to a table or view, this role provides permissions to: Read and update data and metadata for the table or view. Share the table or view. Delete the table or view. Editor cannot do that. Thoughts?

desertlotus1211 3 years, 2 months ago

I apologize - I thought B said Owner... This questions makes no sense now...

odacir 3 years, 3 months ago

Selected Answer: D

It's D, because this is an outdated question, before IAM you cannot set Editor to a dataset; but the best practice is: Create a dataset for each department. Assign the department leads the role of EDITOR(NOT OWNER), and assign the data analysts the role of READER on their dataset.

jkhong 3 years, 3 months ago

Dude, I know there are updates to IAM, but the key point of the question is to have the leads have table creation and update roles... So they already need roles at the dataset level and hence C and D is out. We wouldn't be able to memorise all the roles, but clearly we cannot provide access on a table level...

Wonka87 3 years, 3 months ago

and to supplement why does it need viewer role on the project the table is in?

Atnafu 3 years, 4 months ago

Wow B is an answer https://cloud.google.com/bigquery/docs/access-control-basic-roles#dataset-basic-roles

MisuLava 3 years, 5 months ago

Selected Answer: D

It CANNOT BE B BECAUSE OF : Caution: BigQuery's dataset-level basic roles existed prior to the introduction of IAM. We recommend that you minimize the use of basic roles. In production environments, don't grant basic roles unless there is no alternative. Instead, use predefined IAM roles. https://cloud.google.com/bigquery/docs/access-control-basic-roles

desertlotus1211 3 years, 2 months ago

Ummm owner is a predefined role https://cloud.google.com/bigquery/docs/access-control BigQuery Data Owner (roles/bigquery.dataOwner)

josrojgra 3 years, 5 months ago

Selected Answer: B

I vote B because C and D says that the role is on the project that the table is in, this mean that the role is at the project level that implies that: If you create a dataset in a project that contains any editors, BigQuery grants those users the bigquery.dataEditor predefined role for the new dataset. (from https://cloud.google.com/bigquery/docs/access-control-basic-roles#project-basic-roles) A can't not be because the analysts, in this case, can access the data. B grant to the leads update their datasets, that's mean create tables, and the analysts only read their datasets.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #119 Topic 1

You operate a database that stores stock trades and an application that retrieves average stock price for a given company over an adjustable window of time. The data is stored in Cloud Bigtable where the datetime of the stock trade is the beginning of the row key. Your application has thousands of concurrent users, and you notice that performance is starting to degrade as more stocks are added. What should you do to improve the performance of your application?

A. Change the row key syntax in your Cloud Bigtable table to begin with the stock symbol. **Most Voted**

B. Change the row key syntax in your Cloud Bigtable table to begin with a random number per second.

C. Change the data pipeline to use BigQuery for storing stock trades, and update your application.

D. Use Cloud Dataflow to write a summary of each day's stock trades to an Avro file on Cloud Storage. Update your application to read from Cloud Storage and Cloud Bigtable to compute the responses.

Correct Answer: A

Community vote distribution

A (86%) B (14%)

Comments

kichukonr **Highly Voted** 5 years, 11 months ago

Stock symbol will be similar for most of the records, so it's better to start with random number.. Answer should be B

taepyung 5 years, 11 months ago

I agree with u

karthik89 5 years, 1 month ago

it can start with stock symbol concated with timestamp can be a good row key design

Yonghai 4 years, 3 months ago

for a given company, the data poits starts with the same stock symbol. The dataset is not distrubuted. It is not a good option.

Abhi16820 4 years, 4 months ago

You never use something called random number in bigtable rowkey because it gives you no use in querying possibilities, since we can't run sql querys in bigtable we should not randomise rowkeys in bigtable. Don't confuse the above point with the hotspot logic, both are different if you think so. And another thing is, what you said can be good choice if we are using cloud spanner and trying to comeup with primary key situation, since there we can always run sql query. I think you got the point now.

musumusu **Highly Voted** 3 years, 1 month ago

Answer A: Trick to remember: Row-key adjustment always be like in decending order. #<<Least value>>#<<Lesser value>> For example: 1. #<<Earth>>#<<continents>>#<<countries>>#<<cities>> and so on.. 2. #<<Stock>>#<<users>>#timestamp.. in 99% cases timestamp will be in the end, as its smallest division...

piyush7777 2 years, 7 months ago

Awesome!

clouditis 1 year, 3 months ago

Selected Answer: B

the most plausible option to pick here is B, A can introduce hot-spotting

Vineet_Mor 1 year, 7 months ago

B is correct, By introducing a random number or a hash at the beginning of the row key, you distribute the writes and reads more evenly across the Bigtable cluster, thereby improving performance under heavy load. WHY NOT A? This might still cause hotspots if certain stocks are more popular than others. It could lead to uneven load distribution, which wouldn't solve the performance degradation problem.

Sofiia98 2 years, 2 months ago

Selected Answer: A

Answer is A.

MaxNRG 4 years, 2 months ago

Selected Answer: A

A: https://cloud.google.com/bigtable/docs/schema-design-time-series#prefer_rows_to_column_versions

JG123 4 years, 4 months ago

Correct: A

JayZeeLee 4 years, 4 months ago

A and B would both work, since both would distribute the work. This question is not framed properly.

sumanshu 4 years, 9 months ago

Vote for A

Jay3244 5 years ago

Option A. Below document explains Having EXCHANGE and SYMBOL in the leading positions in the row key will naturally distribute activity. https://cloud.google.com/bigtable/docs/schema-design-time-series

arghya13 5 years, 4 months ago

I think A

kavs 5 years, 4 months ago

Catch here is current Rowley starts with timestamp which should not be in the starting or end position so symbolmshould be prefixed before timestamp

Cloud_Enthusiast 5 years, 4 months ago

A is correct..A Good ROW KEY has to be an ID followed by timestamp. Stock symbol in this case works as an ID

kino2020 5 years, 6 months ago

A. You can find an example in Google's introductory guide. https://cloud.google.com/bigtable/docs/schema-design-time-series?hl=ja#financial_market_data

Diqtator 5 years, 6 months ago

I think A would be best practice. Adding random numbers as start of rowkey doesn't help with troubleshooting

Tanmoyk 5 years, 6 months ago

B should be the answer as adding random numbers in the beginning of the rowkey will distributes data across multiple nodes

haroldbenites 5 years, 7 months ago

B is correct. A is incorrect. The docuemntation don´t recommend constants in the row key because the balance is not efficiente. The are 2 methods to avoid hotspoting. Promotion Field (use a UserId BEFORE the timestamp) and Salting (use timestamp-hash divided by 3 y put it before the timestamp)

atnafu2020 5 years, 7 months ago

Aggree with promotion field and salting. But, there is no constant. A stock symbol is a unique series of letters assigned to security for trading purposes.

daghayeghi 5 years, 1 month ago

A is correct: You deny your sentence, User ID means Stock Symbol, then B is correct.

daghayeghi 5 years, 1 month ago

, then A is correct.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #120 Topic 1

You are operating a Cloud Dataflow streaming pipeline. The pipeline aggregates events from a Cloud Pub/Sub subscription source, within a window, and sinks the resulting aggregation to a Cloud Storage bucket. The source has consistent throughput. You want to monitor an alert on behavior of the pipeline with Cloud Stackdriver to ensure that it is processing data. Which Stackdriver alerts should you create?

A. An alert based on a decrease of subscription/num_undelivered_messages for the source and a rate of change increase of instance/storage/ used_bytes for the destination

B. An alert based on an increase of subscription/num_undelivered_messages for the source and a rate of change decrease of instance/storage/ used_bytes for the destination **Most Voted**

C. An alert based on a decrease of instance/storage/used_bytes for the source and a rate of change increase of subscription/ num_undelivered_messages for the destination

D. An alert based on an increase of instance/storage/used_bytes for the source and a rate of change decrease of subscription/ num_undelivered_messages for the destination

Correct Answer: B

Community vote distribution



Comments

dambilwa **Highly Voted** 4 years, 3 months ago

You would want to get alerted only if Pipeline fails & not if it is running fine. I think Option [B] is correct, because in event of Pipeline failure : 1) subscription/ num_undelivered_messages would pile up at a constant rate as the source has consistent throughput 2) instance/storage/ used_bytes will get closer to zero. Hence need to monitor it's rate of change

Barniyah 4 years, 2 months ago

Yes, you are right, it should be B: Thank you

marioferrulli 2 years, 9 months ago

Why would the instance/storage/used_bytes get closer to zero? If there's an error at a certain point, wouldn't we just see that the used_bytes remain constant while the num_undelivered_messages increases? I don't get why the destination's used bytes should decrease.

baubaumiaomiao 2 years, 9 months ago

"If there's an error at a certain point, wouldn't we just see that the used_bytes remain constant while the num_undelivered_messages increases?" It's the rate of change, not the absolute value

szefco 2 years, 9 months ago

"rate of change decrease of instance/storage/ used_bytes" - if rate of instance/storage/ used_bytes decreases that means less data is written - so something is wrong with the pipeline. It's not used bytes that decreases - it's rate of change decreases. Example: if everything works fine your pipeline writes 5MB/s to the sink. If it decreases to 0.1MB/s it means something is wrong

midgoo Highly Voted 1 year, 6 months ago

Selected Answer: B

For those who may get confuse at the start by the term 'subscription/num_undelivered_messages', it is not a division. It is the full path of the metric. So we should just read it as 'num_undelivered_messages'. The same for 'used_bytes'. So if we see the source have more backlog (more num_undelivered_messages), or the destination ultilization going down, that is the indicator of something going wrong

desertlotus1211 1 year ago

the Answer is A. You want to see it working.

desertlotus1211 1 year ago

you're looking for evidence that it's working: ' that it is processing data....'

kryzo 1 year, 3 months ago

great explanation thanks !

desertlotus1211 1 year ago

Selected Answer: A

It should be Answer A. You want to see in being processed versus looking for a bottleneck.

musumusu 1 year, 7 months ago

Answer B: Trick: In stackdriver always put Alert for Subscriber + CPU Subscriber - num of undelivered message INCREASE alert CPU - Instance or storage DECREASE alert. Make sense right !

atlan 1 year, 8 months ago

Nobody seems to pay attention to instance/storage/used_bytes. I only find this metric for Spanner. https://cloud.google.com/monitoring/api/metrics_gcp#gcp-spanner While Dataflow processes and stores everything in Cloud Storage, Spanner could only be the source. https://cloud.google.com/spanner/docs/change-streams Also, if it is either A or B, the instance/storage/used_bytes metric does not make sense for the destination, which is Cloud Storage. Can anyone help me understand?

desertlotus1211 1 year, 8 months ago

look here: https://cloud.google.com/monitoring/api/metrics_gcp instance/storage/used_bytes GA Storage used.

AzureDP900 1 year, 9 months ago

B. An alert based on an increase of subscription/num_undelivered_messages for the source and a rate of change decrease of instance/storage/ used_bytes for the destination

AzureDP900 1 year, 9 months ago

B is right

Catweazle1983 1 year, 9 months ago

Selected Answer: A

An alert based on a decrease of subscription/num_undelivered_messages for the source and a rate of change increase of instance/storage/ used_bytes for the destination 10 subscriptions / 1 undelivered messages = 10 10

subscriptions / 5 undelivered messages = 2 You clearly want to be alerted when the number of undelivered messages increases. The ratio then decreases. In my example from 10 to 2.

squishy_fishy 1 year, 6 months ago

subscription/num_undelivered_messages is a path, not a division.

A1000 2 years, 1 month ago

Selected Answer: B

Increase subscription/num delivered message decrease instance/storage/used bytes

JG123 2 years, 10 months ago

Correct: B

Abhi16820 2 years, 10 months ago

isn't B and C are same.

JayZeeLee 2 years, 10 months ago

B. It's useful to monitor the source that keeps sending data while the destination that doesn't take anything in.

squishy_fishy 2 years, 11 months ago

The answer is B. subscription/num_undelivered_messages: the number of messages that subscribers haven't processed https://cloud.google.com/pubsub/docs/monitoring#monitoring_forwarded_undeliverable_messages

squishy_fishy 3 years ago

Silly question: what is subscription/ num_undelivered_messages, it is divided by? or per subscription per num_undelivered_messages?

910 2 years, 6 months ago

yes is misleading: the metric "subscription/num_undelivered_messages" is just the path of the API URL actions.googleapis.com/...subscription/num_undelivered_messages ref: https://cloud.google.com/monitoring/api/metrics_gcp#pubsub/subscription/num_undelivered_messages

sumanshu 3 years, 2 months ago

Looks B

gcper 3 years, 7 months ago

A "An alert based on a decrease of subscription/num_undelivered_messages for the source" The more we have undelivered messages, the worse. Thus we want to be alerted when the ratio goes down as the denominator goes up. "A rate of change increase of instance/storage/used_bytes for the destination" An increase in the rate of change of how much we are storing per instance storage.

Alasmindas 3 years, 10 months ago

The correct answer is Option B : increase of subscription/num_undelivered_messages and decrease of instance/storage/ used_bytes, reason as follows:- - The first question we should ask is - why do we want to monitor things - this his very subject, one can say - we want to monitor to check - if everything is running "OK" or we want to monitor things to check if everything is running "NOT OK" . Generally, we would go with the second point - i.e.- we want to monitor things - to check what is NOT OK. if everything works fine - may be we should monitor. Going with that logic - Option B standouts - i.e.- the more we have undelivered messages in subscriber and less we have data in the sync (cloud storage) - means things are not OK and that why we want to monitor it . As mentioned - this approach is subject and different people may have different approach in deciding why we monitor

https://zhangxuetu.online

[202610]

Question #121 Topic 1

You currently have a single on-premises Kafka cluster in a data center in the us-east region that is responsible for ingesting messages from IoT devices globally.
Because large parts of globe have poor internet connectivity, messages sometimes batch at the edge, come in all at once, and cause a spike in load on your
Kafka cluster. This is becoming difficult to manage and prohibitively expensive. What is the Google-recommended cloud native architecture for this scenario?

A. Edge TPUs as sensor devices for storing and transmitting the messages.

B. Cloud Dataflow connected to the Kafka cluster to scale the processing of incoming messages.

C. An IoT gateway connected to Cloud Pub/Sub, with Cloud Dataflow to read and process the messages from Cloud Pub/Sub. **Most Voted**

D. A Kafka cluster virtualized on Compute Engine in us-east with Cloud Load Balancing to connect to the devices around the world.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Rajokkiyam **Highly Voted** 4 years, 6 months ago
Answer C - Cloud Native = Pub/Sub + DataFlow

Rajuuu **Highly Voted** 4 years, 2 months ago
Answer is C. Pub/Sub is the messaging tool for Global.

ga8our 1 year, 4 months ago
Can anyone pls explain what's wrong with D, the load balancing solution?

musumusu 1 year, 7 months ago
Answer C: What is wrong with D, nothing, Cloud load balancing can shift traffic for high volume and low internet in one region. It cost avg. 0.01-0.25 $ per GB, or if volume is too high. 0.05 $ per Hour http request. This might be the answer if your exam for network engineer.

musumusu 1 year, 7 months ago
Answer C, but it will not solve bad internet connection, make sure 100mbps speed of internet is at sensor side.

MisuLava 1 year, 11 months ago

"single on-premises Kafka cluster in a data center in the us-east region" is it on-prem or in a datacenter in us-east ?

JamesKarianis 2 years, 2 months ago

Selected Answer: C

Answer is C

Prasanna_kumar 2 years, 7 months ago

Answer is option C

ivanhsiav 3 years, 2 months ago

Answer c kafka cluster in on-premise for streaming msgs pub/sub for streaming msgs in cloud

sumanshu 3 years, 2 months ago

Vote for C

Allan222 3 years, 7 months ago

Should be C

daghayeghi 3 years, 7 months ago

C is correct: the main trick come from A, and response is that TPU only use when we have a deployed machine learning model that we don't have now.

ArunSingh1028 3 years, 7 months ago

Answer - C

Alasmindas 3 years, 10 months ago

Easy Question : ANswer is Option C. Alterative to Kafka in google cloud native service is Pub/Sub and Dataflow punched with Pub/Sub is the google recommended option

atnafu2020 4 years, 1 month ago

C the issue is with a single Kafka cluster is the need to scale automatically with Dataflow

haroldbenites 4 years, 1 month ago

C is correct

https://zhangxuetu.online

[202610]

Question #122 Topic 1

You decided to use Cloud Datastore to ingest vehicle telemetry data in real time. You want to build a storage system that will account for the long-term data growth, while keeping the costs low. You also want to create snapshots of the data periodically, so that you can make a point-in-time (PIT) recovery, or clone a copy of the data for Cloud Datastore in a different environment. You want to archive these snapshots for a long time. Which two methods can accomplish this?
(Choose two.)

A. Use managed export, and store the data in a Cloud Storage bucket using Nearline or Coldline class. Most Voted

B. Use managed export, and then import to Cloud Datastore in a separate project under a unique namespace reserved for that export. Most Voted

C. Use managed export, and then import the data into a BigQuery table created just for that export, and delete temporary export files.

D. Write an application that uses Cloud Datastore client libraries to read all the entities. Treat each entity as a BigQuery table row via BigQuery streaming insert. Assign an export timestamp for each export, and attach it as an extra column for each row. Make sure that the BigQuery table is partitioned using the export timestamp column.

E. Write an application that uses Cloud Datastore client libraries to read all the entities. Format the exported data into a JSON file. Apply compression before storing the data in Cloud Source Repositories.

Correct Answer: AB

Community vote distribution



Comments

Ganshank Highly Voted 5 years, 5 months ago

A,B https://cloud.google.com/datastore/docs/export-import-entities

salsabilsf 4 years, 5 months ago

"while keeping the costs" should be A,D

MrCastro 4 years, 1 month ago

Big query streaming inserts ARE NOT cheap

hellofrnds 4 years ago

If you use B , not D , how can we do "point in time" recovery? is it possible? Point in time recovery needs export along with timestamp, so that we can recover for a particular timestamp.

atnafu2020 Highly Voted 5 years, 1 month ago

AC https://cloud.google.com/datastore/docs/export-import-entities C: To import only a subset of entities or to import data into BigQuery, you must specify an entity filter in your export. B: Not correct since you want to store in a different environment than Datastore. Tho this statment is true: Data exported from one Datastore mode database can be imported into another Datastore mode database, even one in another project. A is correct Billing and pricing for managed exports and imports in Datastore Output files stored in Cloud Storage count towards your Cloud Storage data storage costs. Steps to Export all the entities 1. Go to the Datastore Entities Export page in the Google Cloud Console. 2. Go to the Datastore Export page 2. Set the Namespace field to All Namespaces, and set the Kind field to All Kinds. 3. Below Destination, enter the name of your "Cloud Storage bucket". 4. Click Export.

Yiouk 4 years, 1 month ago

C is valid because of table snapshots. Else standard time travel is valid only for 7 days
https://cloud.google.com/bigquery/docs/table-snapshots-intro#table_snapshots
https://cloud.google.com/bigquery/docs/time-travel#limitation

Chelseajcole 4 years ago

you wanna say invalid?

tavva_prudhvi 3 years, 6 months ago

As you've mentioned in B, does the environment meant to be a project or a resource? As, we can clone a copy of the data in a datastore even in another project!? Then, it's B. Also, in point C they didn't mention any entity filter hence we eliminate C how can you support your own statement with a different answer?

AzureDP900 2 years, 9 months ago

A, B is perfect

aparna4387 3 years, 10 months ago

https://cloud.google.com/datastore/docs/export-import-entities#import-into-bigquery Data exported without specifying an entity filter cannot be loaded into BigQuery. This is not mentioned explicitly. Safe to assume there is no filter on the exports. So options are AB

CGS22 1 year, 5 months ago

Selected Answer: AB

https://cloud.google.com/datastore/docs/export-import-entities

kskssk 2 years, 1 month ago

AB chatgpt A. Use managed export, and store the data in a Cloud Storage bucket using Nearline or Coldline class: Managed export is a feature provided by Cloud Datastore to export your data. Storing the data in a Cloud Storage bucket, especially using Nearline or Coldline storage classes, helps keep storage costs low while allowing you to retain the snapshots for a long time. B. Use managed export, and then import to Cloud Datastore in a separate project under a unique namespace reserved for that export: This method allows you to create snapshots by exporting data from Cloud Datastore (using managed export) and then importing it into a separate project under a unique namespace. By importing into a separate project, you can keep a copy of the data in a different environment, which is useful for point-in-time recovery or creating clones of the data.

NicolasN 2 years, 10 months ago

Selected Answer: AB

A rather complicated question, of a kind I wish I won't face in the exam. My opinion: [A] A valid and cost-effective solution satisfying the requirement for PIT recovery [B] A valid solution but far from ideal for archiving. It satisfies the

requirement part "you can … clone a copy of the data for Cloud Datastore in a different environment" (an objection to the word "namespace", I think it should be just "name")

NicolasN 2 years, 10 months ago

[C] There is the limitation "Data exported without specifying an entity filter cannot be loaded into BigQuery". The entity filter for this case should contain all the kinds of entities but there is another limitation of "100 entity filter combinations". We have no knowledge of the kinds or the namespaces of the entities. Sources: https://cloud.google.com/datastore/docs/export-import-entities#import-into-bigquery https://cloud.google.com/datastore/docs/export-import-entities#exporting_specific_kinds_or_namespaces [D] seems a detailed candidate solution but it violates the limitation "You cannot append Datastore export data to an existing table." https://cloud.google.com/bigquery/docs/loading-data-cloud-datastore#appending_to_or_overwriting_a_table_with_cloud_datastore_data [E] Cloud Source Repositories are for source code and not a suitable storage for this case.

John_Pongthorn 3 years ago

Selected Answer: AB

https://cloud.google.com/datastore/docs/export-import-entities

John_Pongthorn 3 years ago

Selected Answer: AB

https://cloud.google.com/datastore/docs/export-import-entities

John_Pongthorn 3 years ago

Selected Answer: AB

The answer is nothing to do with bigquery , so you can skip what mention to bigquery. A B is the final answer

DataEngineer_WideOps 3 years, 2 months ago

A,B For those who say using BQ as archival, How can we achieve that while datastore are NO-SQL whereas BQ are SQL , will that work? also BQ are not created for achieving purposes.

AmirN 3 years, 3 months ago

Option B is 36 times more expensive than C

Nico1310 3 years, 8 months ago

Selected Answer: AB

AB. for sure streaming to BQ its quite expensive!

MaxNRG 3 years, 8 months ago

Selected Answer: AD

A - Cloud Storage (long-term data + costs low) D - BigQuery (timestamp for point-in-time (PIT) recovery)

tavva_prudhvi 3 years, 5 months ago

D is wrong, BQ Streaming inserts costs are high!

MaxNRG 1 year, 9 months ago

Agreed, AB https://cloud.google.com/datastore/docs/export-import-entities

medeis_jar 3 years, 8 months ago

Selected Answer: AB

Option A; Cheap storage and it is a supported method https://cloud.google.com/datastore/docs/export-import-entities Option B; Rationale - "Data exported from one Datastore mode database can be imported into another Datastore mode database, even one in another project." <https://cloud.google.com/datastore/docs/export-import-entities>

squishy_fishy 3 years, 11 months ago

Answer is A, B. https://cloud.google.com/datastore/docs/export-import-entities#exporting_specific_kinds_or_namespaces

sergio6 3 years, 11 months ago

A, D A: Option for storage system that will account for the long-term data growth D: Option for snapshots, PIT recovery, copy of the data for Cloud Datastore in a different environment and, above all, archive snapshots for a long time B: not a good solution for archiving snapshots for a long time C: to import data into BigQuery, you must specify an entity filter E: Cloud Source Repositories is for code One note: E --> would be my second choice if there was Cloud Storage instead of Source Repositories (typo?)

Chelseajcole 4 years ago

Vote A B . What's the purpose load into bigquery?

Chelseajcole 4 years ago

https://cloud.google.com/datastore/docs/export-import-entities#import-into-bigquery Importing into BigQuery To import data from a managed export into BigQuery, see Loading Datastore export service data. Data exported without specifying an entity filter cannot be loaded into BigQuery. If you want to import data into BigQuery, your export request must include one or more kind names in the entity filter. You have to specify an entity fliter before you can load from datastore to BQ. It didn't mention that at all. So C is incorrect

fire558787 4 years, 1 month ago

A for sure. Then I was undecided between B and C; B has high costs and C has low costs (storage is more expensive in Datastore). However the question says that you want data to be used for Datastore. There is no native way to export data from BigQuery to Datastore, hence the only two options that allow data to be restored to Datastore are A and B.

https://zhangxuetu.online

[202610]

Question #123 Topic 1

You need to create a data pipeline that copies time-series transaction data so that it can be queried from within BigQuery by your data science team for analysis.
Every hour, thousands of transactions are updated with a new status. The size of the initial dataset is 1.5 PB, and it will grow by 3 TB per day. The data is heavily structured, and your data science team will build machine learning models based on this data. You want to maximize performance and usability for your data science team. Which two strategies should you adopt? (Choose two.)

A. Denormalize the data as must as possible. **Most Voted**

B. Preserve the structure of the data as much as possible.

C. Use BigQuery UPDATE to further reduce the size of the dataset.

D. Develop a data pipeline where status updates are appended to BigQuery instead of updated. **Most Voted**

E. Copy a daily snapshot of transaction data to Cloud Storage and store it as an Avro file. Use BigQuery's support for external data sources to query.

Correct Answer: AD

Community vote distribution



Comments

rickywck Highly Voted 4 years, 6 months ago
I think AD is the answer. E will not improve performance.

[Removed] Highly Voted 4 years, 6 months ago
Answer: A, D Description: Denormalization will help in performance by reducing query time, update are not good with bigquery

awssp12345 3 years, 2 months ago
My guess is append has better performance than update.

midgoo 1 year, 6 months ago
Selected Answer: BD
If we denormalize the data, the Data Science team will shout at us. Preserving it is the way to go

vaga1 1 year, 3 months ago
Denormalization is just a best practice when using BQ.

WillemHendr 1 year, 3 months ago

Shouting data-science teams are not part of question, this is more about what is exam correct, not what it the best for your own situation

odacir 1 year, 9 months ago

Selected Answer: AD

A and D: A- Improve performance D- Is better for DS have all the history and not the last update...

NicolasN 1 year, 11 months ago

The criteria for selecting a strategy are the performance and usability for the data science team. This team performs the analysis by querying stored data. So we don't care for performance related with data ingestion. According to this point of view: A: YES - undisputedly favours query performance B: YES - Keeping the structure unchanged promotes usability (the team won't need to update queries or ML models) C: Questionable - Updating the status of a row instead of appending newer versions is keeping the size smaller. But does this affect significantly the analysis performance? Even if it does, creating materialized views to keep the most recent status per row eliminates it D: NO - has nothing to do with DS team's tasks, affects ingestion performance E: NO - demotes usability

NicolasN 1 year, 11 months ago

(mistakenly voted AC instead of AB)

jkhong 1 year, 9 months ago

For B there is no mention that the current data structure is being used (...data science team WILL build machine learning models based on this data.) ... We're developing a new data model to be used by them in the future

DerickTW 2 years ago

Selected Answer: AC

The DML quota limit is removed since 2020, I think C is better than D now.

devaid 1 year, 11 months ago

Is not about the quota. You should avoid using UPDATE because it makes a big scan of the table, and is not efficient or high performant. Usually prefer appends and merges instead, and using the optimized schema approach of Big Query that denormalizes the table to avoid joins and leverages nested and repeated fields.

MaxNRG 2 years, 8 months ago

Selected Answer: AD

A: Denormalization increases query speed for tables with billions of rows because BigQuery's performance degrades when doing JOINs on large tables, but with a denormalized data structure, you don't have to use JOINs, since all of the data has been combined into one table. Denormalization also makes queries simpler because you do not have to use JOIN clauses. https://cloud.google.com/solutions/bigquery-data-warehouse#denormalizing_data D: BigQuery append

medeis_jar 2 years, 8 months ago

Selected Answer: AD

requirements are -> performance and usability. Denormalization will help in performance by reducing query time, update is not good with big query. And append has better performance than Update.

doninakula 2 years, 10 months ago

I think AD. E is not valid because it use external table which is not good for performance

sumanshu 3 years, 2 months ago

A - correct (denormlization will help) B - data already heavily structured (no use and no impact) C - more than 1500 Updates not possible D - Not sure..(because appending will increase size and cost) E - Does not look good (increase cost..also we are storing for all days....again for query we need to issue mutiple query for all days....) So, A & D (left out of 5)

Jeysolomon 3 years, 3 months ago

Correct Answer: AE A – Denormalisation helps improve performance. B, C - Not helping to address the problem. D – Append will increase the db size and cost involved for storage and also for large number of records to scan for queries by data science team which is costlier. E - Addresses the problem of maximising the usability of the data science team and the data. They can anayse the data exported to cloud storage instead of reading from bigquery which is expensive and impact performance considerably.

Chelseajcole 2 years, 11 months ago

It didn't mention cost is a concern

retep007 3 years ago

E is wrong, you've been asked to use bigquery and reading files from storage in bq is significantly more time consuming

daghayeghi 3 years, 6 months ago

A, D: Using BigQuery as an OLTP store is considered an anti-pattern. Because OLTP stores have a high volume of updates and deletes, they are a mismatch for the data warehouse use case. To decide which storage option best fits your use case, review the Cloud storage products table. BigQuery is built for scale and can scale out as the size of the warehouse grows, so there is no need to delete older data. By keeping the entire history, you can deliver more insight on your business. If the storage cost is a concern, you can take advantage of BigQuery's long term storage pricing by archiving older data and using it for special analysis when the need arises. If you still have good reasons for dropping older data, you can use BigQuery's native support for date-partitioned tables and partition expiration. In other words, BigQuery can automatically delete older data. https://cloud.google.com/solutions/bigquery-data-warehouse#handling_change

Hithesh 3 years, 6 months ago

should be AC.. "Every hour, thousands of transactions are updated with a new status" if we append how we will handle the new status change..

sumanshu 3 years, 2 months ago

C not possible, maximum 1500 updates possible in a day

raf2121 3 years, 2 months ago

DML without limits now in BQ (below blog says March 2020, Not sure whether these questions were prepared before or after March 2020) https://cloud.google.com/blog/products/data-analytics/dml-without-limits-now-in-bigquery

hdmi_switch 3 years, 2 months ago

There is no more hard limit, but UPDATES are queued: "BigQuery runs up to 2 of them concurrently, after which up to 20 are queued as PENDING. When a previously running job finishes, the next pending job is dequeued and run. Currently, queued mutating DML statements share a per-table queue with maximum length 20. Additional statements past the maximum queue length for each table fail." With thousands of updates per hour, this doesn't seem feasible. I would assume the question is marked as outdated anyway or the answers are update in the actual exam.

daghayeghi 3 years, 7 months ago

AC: the problem is exactly about Updating and preserving size of database as much as possible, then denormalization and using UPDATE function from DML will address the issue. they don't want to update faster. then A & C is correct. https://cloud.google.com/solutions/bigquery-data-warehouse

daghayeghi 3 years, 6 months ago

A, D: it was my mistake, we should decrease update as Bigquery is not design for update.
https://cloud.google.com/solutions/bigquery-data-warehouse#handling_change

karthik89 3 years, 7 months ago

you can update bigquery 1500 times in a day

Nams_139 3 years, 10 months ago

A,D Since the requirements are both performance and usability.

federicohi 3 years, 10 months ago

i tink may be ita AC becuase appending its worst to increase dataset size. THe question seems to put like a problem the size of dataset and performance to datascience so inserting more rows decrease performace for them.

Ram459 3 years, 10 months ago

AD looks like a good fit

https://zhangxuetu.online

[202610]

Question #124 Topic 1

You are designing a cloud-native historical data processing system to meet the following conditions:
✑ The data being analyzed is in CSV, Avro, and PDF formats and will be accessed by multiple analysis tools including Dataproc, BigQuery, and Compute
Engine.
✑ A batch pipeline moves daily data.
✑ Performance is not a factor in the solution.
✑ The solution design should maximize availability.
How should you design data storage for this solution?

A. Create a Dataproc cluster with high availability. Store the data in HDFS, and perform analysis as needed.

B. Store the data in BigQuery. Access the data using the BigQuery Connector on Dataproc and Compute Engine.

C. Store the data in a regional Cloud Storage bucket. Access the bucket directly using Dataproc, BigQuery, and Compute Engine.

D. Store the data in a multi-regional Cloud Storage bucket. Access the data directly using Dataproc, BigQuery, and Compute Engine. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

jkhong **Highly Voted** 3 years, 5 months ago

Selected Answer: D

Problem: How to store data? Considerations: High availability, performance not an issue A → avoid HDFS C → multi-regional > regional in terms of availability B could be the answer but we're dealing with PDF documents, we need blob storage (cloud storage). If we only have csv or Avro, this may be the answer

serch_engine 1 year, 6 months ago

Selected Answer: D

D is the answer

AzureDP900 3 years, 3 months ago

D is right

dconesoko 3 years, 3 months ago

Selected Answer: D

vote for D

devaid 3 years, 5 months ago

Selected Answer: D

D of course

kenanars 3 years, 6 months ago

Selected Answer: D

D is the correct answer

https://zhangxuetu.online

[202610]

Question #125 Topic 1

You have a petabyte of analytics data and need to design a storage and processing platform for it. You must be able to perform data warehouse-style analytics on the data in Google Cloud and expose the dataset as files for batch analysis tools in other cloud providers. What should you do?

A. Store and process the entire dataset in BigQuery.

B. Store and process the entire dataset in Bigtable.

C. Store the full dataset in BigQuery, and store a compressed copy of the data in a Cloud Storage bucket. **Most Voted**

D. Store the warm data as files in Cloud Storage, and store the active data in BigQuery. Keep this ratio as 80% warm and 20% active.

Correct Answer: C

Community vote distribution



Comments

Rajokkiyam Highly Voted 5 years ago

Answer C.

AJKumar Highly Voted 4 years, 9 months ago

A and B can be eliminated right away as they do not talk about providing for other cloud providers. between C and D. The question says nothing about warm or cold data-rather that data should be made available for other providers--C--can fulfill this condition. Answer C.

AzureDP900 2 years, 3 months ago

Agree with C

zbyszek1 1 year, 6 months ago

For me A. I can use export from BQ to Cloud Storage. There is no need to store two copies of data.

spicebits 1 year, 4 months ago

If you export data from BQ to GCS then you will have two copies and you will be in the same architecture as answer C.

vamgcp 1 year, 8 months ago

Selected Answer: B

It can be C or D , but I will go with C as storing the full dataset in BigQuery and a compressed copy of the data in Cloud Storage is a good way to balance performance and cost.

forepick 1 year, 10 months ago

Selected Answer: C

Best answer is C, although BQ can query gzipped files stored on GCS directly. Maybe this double storage makes it a bit more highly available.

izekc 1 year, 11 months ago

Selected Answer: D

D is much more accurate.

jkhong 2 years, 3 months ago

Selected Answer: C

D → does not guarantee 100% queryable or accessible/available

Smaks 2 years, 8 months ago

You can read streaming data from Pub/Sub, and you can write streaming data to Pub/Sub or BigQuery. Thus Cloud Storage is not a proper sink for streaming pipeline. I vote for B, since it is possible to convert unstructured data and store in BQ

Smaks 2 years, 8 months ago

ignore this comment, please

Aslkdup 3 years, 1 month ago

BQ can reach files at google storage as external table. so my answer is D. (If data was smaller than this, I would choose C)

Bhawantha 3 years, 2 months ago

Selected Answer: C

both requirements are full filled.

MaxNRG 3 years, 2 months ago

Selected Answer: D

D: BigQuery + Cloud Storage

jkhong 2 years, 3 months ago

D → does not guarantee 100% queryable or accessible/available

medeis_jar 3 years, 2 months ago

Selected Answer: C

"You must be able to perform data warehouse-style analytics on the data in Google Cloud and expose the dataset as files for batch analysis tools in other cloud providers?" Analytics -> BQ Exposing -> GCS

JG123 3 years, 4 months ago

Correct: C

xiaofeng_0226 3 years, 7 months ago

vote for C

sumanshu 3 years, 9 months ago

Vote for 'C' A - Only Half requirement fulfil, expose as a file not getting fulfiled B - Not a warehouse C. Both requirements fulfiled...Bigquery and GCS D. Both requirement fulfiled...but what if other cloud provider wants to analysis on rest 80% of the data. - So out of 4 options, C looks okay

gcper 4 years, 1 month ago

C BigQuery for analytics processing and Cloud Storage for exposing the data as files

daghayeghi 4 years, 1 month ago

answer A: with BigQuery Omni it is now possible to read data from other cloud providers without transferring the data to GCP and thereby by saving on egress charges. https://cloud.google.com/blog/products/data-analytics/introducing-bigquery-omni

salsabilsf 3 years, 11 months ago

the question says : "expose the dataset as files" wich means Cloud Storage

https://zhangxuetu.online

[202610]

Question #126 Topic 1

You work for a manufacturing company that sources up to 750 different components, each from a different supplier. You've collected a labeled dataset that has on average 1000 examples for each unique component. Your team wants to implement an app to help warehouse workers recognize incoming components based on a photo of the component. You want to implement the first working version of this app (as Proof-Of-Concept) within a few working days. What should you do?

A. Use Cloud Vision AutoML with the existing dataset. **Most Voted**

B. Use Cloud Vision AutoML, but reduce your dataset twice.

C. Use Cloud Vision API by providing custom labels as recognition hints.

D. Train your own image recognition model leveraging transfer learning techniques.

Correct Answer: A

Community vote distribution



Comments

Callumr **Highly Voted** 5 years, 3 months ago

B - You only need a PoC and it has be done quickly

techtitan **Highly Voted** 2 years, 7 months ago

A - https://cloud.google.com/vertex-ai/docs/beginner/beginners-guide Target at least 1000 examples per target

techtitan 2 years, 7 months ago

The quick POC part can be achieved by using Auto ML instead of creating and training your own model

grshankar9 1 year, 2 months ago

Selected Answer: A

The key difference between Google Cloud Vision AutoML and Cloud Vision API is that Cloud Vision API provides pre-trained models for basic image analysis tasks like object detection and labeling, while Cloud Vision AutoML allows you to train custom machine learning models to identify specific objects or concepts within images that are unique to your dataset, requiring you to provide labeled training data.

josech 1 year, 4 months ago

Selected Answer: A

AutoML Vision is deprecated since march 31, 2024. The question will refer to Vertex AI AutoML. And as bet practice, the minimum dataset size for each label is 1000. So, with an updated question, the answer would be A.

CGS22 1 year, 5 months ago

Selected Answer: A

A. Use Cloud Vision AutoML with the existing dataset. Here's why this is the most suitable option: Speed and Ease: AutoML simplifies model building. You simply upload your labeled images, and AutoML takes care of model selection, training, and evaluation. Existing Dataset Sufficiency: Your dataset (750 components x 1000 images each) is a decent starting point for AutoML, allowing you to quickly test its effectiveness. Minimal Custom Development: AutoML's out-of-the-box deployment options let you integrate the model into your app without extensive coding.

saado9 2 years ago

Selected Answer: B

Option B is the fastest way to train a model that can be used to recognize the 750 different components.

musumusu 2 years, 7 months ago

Whats wrong with C, its fast, cheap and add your 750 labels which is not big work. AutoML is good to train on big dataset and costly as compared to APIs

knith66 2 years, 2 months ago

it is a labeled dataset and why do you need to label it once again? So no C

forepick 2 years, 4 months ago

Adding custom labels to Vision API is done by training an AutoML model! That's the formal recommendation. And you don't need a big dataset for AutoML as it uses transfer learning.

odacir 2 years, 9 months ago

Selected Answer: A

First I think in Vision API, but that is a pre-trained AI, will not recognize my labels, so because you have 1000 samples per item, AUTO ML is perfect. B cannot be because have not sensed to reduce your dataset if you have the recommended number of info. https://cloud.google.com/vision/automl/docs/beginners-guide#include_enough_labeled_examples_in_each_category The bare minimum required by AutoML Vision training is 100 image examples per category/label. The likelihood of successfully recognizing a label goes up with the number of high quality examples for each; in general, the more labeled data you can bring to the training process, the better your model will be. Target at least 1000 examples per label.

AzureDP900 2 years, 9 months ago

A is correct

gudiking 2 years, 10 months ago

A - https://cloud.google.com/vision/automl/docs/beginners-guide#include_enough_labeled_examples_in_each_category

MarielaYBird 2 years, 10 months ago

Selected Answer: B

Based on this: "As a rule of thumb, we recommend to have at least 100 training samples per class if you have distinctive and few classes, and more than 200 training samples if the classes are more nuanced and you have more than 50 different classes" 750 different components = more than 50 different classes. That means we need more than 200 training samples. If we used 250 training samples out of the 1000 samples and multiply it to 750 different classes we get a total of 187,500 which is the equivalent of reducing the dataset twice. https://cloud.google.com/vision/automl/object-detection/docs/prepare#how_big_does_the_dataset_need_to_be

josrojgra 2 years, 11 months ago

Selected Answer: A

I choose A because on the vertex AI documentation (https://cloud.google.com/vertex-ai/docs/image-data/classification/prepare-data), on the best practices of preparing data for image recognition recommend this: We recommend about 1000 training images per label. The minimum per label is 10. In general, it takes more examples per label to train models with multiple labels per image, and resulting scores are harder to interpret. I know that is PoC, but if you do it without enough accuracy, you maybe discard the solution because it isn't fit for your requirements. So is better to do it with enough data to be sure that the model is or not accuracy enough with this data, because you maybe haven't enough accuracy and the problem is the quality of the data and not the amount of it.

John_Pongthorn 3 years ago

Selected Answer: A

https://cloud.google.com/vision/automl/docs/beginners-guide#include_enough_labeled_examples_in_each_category The bare minimum required by AutoML Vision training is 100 image examples per category/label. The likelihood of successfully recognizing a label goes up with the number of high quality examples for each; in general, the more labeled data you can bring to the training process, the better your model will be. Target at least 1000 examples per label.

John_Pongthorn 3 years ago

The more labels, the more accurate the result.

changsu 3 years ago

Selected Answer: B

750*1000 are a lot.

ducc 3 years, 1 month ago

Selected Answer: A

It is labeled, so A is correct

civilizador 3 years, 1 month ago

It's A. https://cloud.google.com/vision/automl/docs/beginners-guide#data_preparation The bare minimum required by AutoML Vision training is 100 image examples per category/label. The likelihood of successfully recognizing a label goes up with the number of high quality examples for each; in general, the more labeled data you can bring to the training process, the better your model will be. Target at least 1000 examples per label.

civilizador 3 years, 1 month ago

So even for POC better to use 1000 . There would be no significant time differences anyway between 500 and 1000

TheRealBsh 3 years, 1 month ago

Option A & B are quite close. Refer: https://cloud.google.com/vision/automl/docs/beginners-guide#data_preparation – Says to target at least 1000 images per label for training.

czokwe 3 years, 4 months ago

Selected Answer: B

B cant choose A because model needs to pass through the dataset several times for a proof of concept, existing data set samples might not be all seen in several working days causing over generalization

https://zhangxuetu.online

[202610]

Question #127 Topic 1

You are working on a niche product in the image recognition domain. Your team has developed a model that is dominated by custom C++ TensorFlow ops your team has implemented. These ops are used inside your main training loop and are performing bulky matrix multiplications. It currently takes up to several days to train a model. You want to decrease this time significantly and keep the cost low by using an accelerator on Google Cloud. What should you do?

A. Use Cloud TPUs without any additional adjustment to your code.

B. Use Cloud TPUs after implementing GPU kernel support for your customs ops.

C. Use Cloud GPUs after implementing GPU kernel support for your customs ops. **Most Voted**

D. Stay on CPUs, and increase the size of the cluster you're training your model on.

Correct Answer: C

Community vote distribution



Comments

dhs227 **Highly Voted** 6 years ago

The correct answer is C TPU does not support custom C++ tensorflow ops
https://cloud.google.com/tpu/docs/tpus#when_to_use_tpus

ffggrre 2 years, 5 months ago

the link doesn't say TPU does not support custom C++ tensorflow ops

Helinia 2 years, 3 months ago

It does. TPU is good for "Models with no custom TensorFlow/PyTorch/JAX operations inside the main training loop".

aiguy **Highly Voted** 6 years ago

D: Cloud TPUs are not suited to the following workloads: [...] Neural network workloads that contain custom TensorFlow operations written in C++. Specifically, custom operations in the body of the main training loop are not suitable for TPUs.

tavva_prudhvi 4 years ago

But, in the question it also says we have to decrease the time significantly?? If you gonna use the CPU, it will take more time to train, right?

cetanx 2 years, 10 months ago

Chat GPT says C Option D is not the most cost-effective or efficient solution. While increasing the size of the cluster could decrease the training time, it would also significantly increase the cost, and CPUs are not as efficient for this

type of workload as GPUs.

FP77 2 years, 7 months ago

chatgpt will give you different answers if you ask 10 times. The correct answer is B

squishy_fishy 2 years, 5 months ago

Totally agree. ChatGPT is garbage. It is still learning.

gopinath_k 5 years ago

B: 1. You need to provide support for the matrix multiplication - TPU 2. You need to provide support for the Custom TF written in C++ - GPU

aaaaaaaasdasdasfs 11 months, 2 weeks ago

Selected Answer: C

C. Use Cloud GPUs after implementing GPU kernel support for your customs ops. Here's why: Your current bottleneck is in the custom C++ TensorFlow ops performing matrix multiplications, which are computationally intensive operations. GPUs are specifically designed to efficiently handle matrix operations and would significantly speed up your training time compared to CPUs. Since you've developed custom C++ TensorFlow ops, you'll need to implement GPU kernel support for these operations to take advantage of GPU acceleration.

SamuelTsch 1 year, 5 months ago

Selected Answer: D

According to the official documentation. Models that contain many custom TensorFlow operations written in C++ should keep using CPUs.

baimus 1 year, 6 months ago

I think this is D. I recently did the ML professional exam and they ask that there, and it's always "c++ custom ops = CPU", it's in fact the only scenario for non-small models on CPU. It's written in black and white here: https://cloud.google.com/tpu/docs/intro-to-tpu#when_to_use_tpus, check out the CPU/GPU/TPU "when to use" section.

Anudeep58 1 year, 9 months ago

Selected Answer: C

Why Not Other Options? A. Use Cloud TPUs without any additional adjustment to your code: TPUs are optimized for standard TensorFlow operations and require custom TensorFlow ops to be adapted to TPU-compatible kernels, which is not trivial. Without modifications, your custom C++ ops will not run efficiently on TPUs. B. Use Cloud TPUs after implementing GPU kernel support for your customs ops: Implementing GPU kernel support alone is not sufficient for running on TPUs. TPUs require specific optimizations and adaptations beyond GPU kernels. D. Stay on CPUs, and increase the size of the cluster you're training your model on: While increasing the CPU cluster size might reduce training time, it is not as efficient or cost-effective as using GPUs, especially for matrix multiplication tasks.

AlizCert 1 year, 10 months ago

Selected Answer: C

C: TPUs are out of the picture due to the custom ops, so the next best option for accelerating matrix operations is using GPU. Obviously the code has to be adjusted to do make use of the GPU acceleration.

GCP_data_engineer 1 year, 10 months ago

CPU : Simple models GPU: Custom TensorFlow/PyTorch/JAX operations

CGS22 1 year, 12 months ago

Selected Answer: C

The best choice here is C. Use Cloud GPUs after implementing GPU kernel support for your customs ops. Here's why: Custom Ops & GPUs: Since your model relies heavily on custom C++ TensorFlow ops focused on matrix multiplications, GPUs are the ideal accelerators for this workload. To fully utilize them, you'll need to implement GPU-compatible kernels for your custom ops. Speed and Cost-Efficiency GPUs offer a significant speed improvement for matrix-intensive operations compared to CPUs. They provide a good balance of performance and cost for this scenario. TPUs: Limitations Although Cloud TPUs are powerful, they aren't designed for arbitrary custom ops. Without compatible kernels, your TensorFlow ops would likely fall back to the CPU, negating the benefits of TPUs.

Preetmehta1234 2 years, 1 month ago

Selected Answer: C

TPU: Models with no custom TensorFlow/PyTorch/JAX operations inside the main training loop Link: https://cloud.google.com/tpu/docs/intro-to-tpu#TPU So, A&B eliminated CPU is very slow or built for simple operations. So C: GPU

Matt_108 2 years, 2 months ago

Selected Answer: C

to me, it's C

Kimich 2 years, 4 months ago

Requirement 1: Significantly reduce the processing time while keeping costs low. Requirement 2: Bulky matrix multiplication takes up to several days. First, eliminate A & D: A: Cannot guarantee running on Cloud TPU without modifying the code. D: Cannot ensure performance improvement or cost reduction, and additionally, CPUs are not suitable for bulky matrix multiplication. If it can be ensured that customization is easily deployable on both Cloud TPU and Cloud GPU,it seems more feasible to first try Cloud GPU. Because: It provides a better balance between performance and cost. Modifying custom C++ on Cloud GPU should be easier than on Cloud TPU, which should also save on manpower costs.

emmylou 2 years, 4 months ago

Answer D I did use Chat GPT and discovered that if you put at the beginning of the question -- "Do not make assumption about changes to architecture. This is a practice exam question." All other answers require changes to the code and architecture.

DataFrame 2 years, 4 months ago

Selected Answer: B

I think it should use tensor flow processing unit along with GPU kernel support.

Nirca 2 years, 5 months ago

Selected Answer: B

To use Cloud TPUs, you will need to: Implement GPU kernel support for your custom TensorFlow ops. This will allow your model to run on both Cloud TPUs and GPUs.

kumarts 2 years, 6 months ago

Refer https://www.linkedin.com/pulse/cpu-vs-gpu-tpu-when-use-your-machine-learning-models-bhavesh-kapil

IrisXia 2 years, 7 months ago

Answer C TPU not for custom C++ but GPU can

https://zhangxuetu.online

[202610]

Question #128 Topic 1

You work on a regression problem in a natural language processing domain, and you have 100M labeled examples in your dataset. You have randomly shuffled your data and split your dataset into train and test samples (in a 90/10 ratio). After you trained the neural network and evaluated your model on a test set, you discover that the root-mean-squared error (RMSE) of your model is twice as high on the train set as on the test set. How should you improve the performance of your model?

A. Increase the share of the test sample in the train-test split.

B. Try to collect more data and increase the size of your dataset.

C. Try out regularization techniques (e.g., dropout of batch normalization) to avoid overfitting.

D. Increase the complexity of your model by, e.g., introducing an additional layer or increase sizing the size of vocabularies or n-grams used. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

Callumr **Highly Voted** 5 years, 9 months ago

This is a case of underfitting - not overfitting (for over fitting the model will have extremely low training error but a high testing error) - so we need to make the model more complex - answer is D

hellofrnds 4 years, 6 months ago

@callumr , "root-mean-squared error (RMSE) of your model is twice as high on the train set as on the test set." clearly means testing error is twice of training error. So, it is clearly overfitting. Isn't it?

hellofrnds 4 years, 5 months ago

So, answer should be C

tavva_prudhvi 3 years, 11 months ago

If you training RMSE=0.2. and testing RMSE = 0.4, and we want the RMSE to be low as its the error, now is it overfitting or underfitting? think wisely!

alecuba16 3 years, 7 months ago

It's overfitting. Overfitting->low rmse in train / high accuracy-f1 score in train for classification. Underfitting -> high rmse / low f1score or accuracy in train, you don't have to look into test set if there is an underfitting problem.

jfab 2 years, 9 months ago

But the question clearly states we have higher RMSE on the train than the test. So how would it be overfitting?

velliger 4 years, 4 months ago

High rmse: The model is underfitting the train data. To reduce overfitting, we increase the number of layers in the model or we change the type of layer.

velliger 4 years, 4 months ago

*underfitting

odacir 3 years, 3 months ago

NO, its underfitting.

NeoNitin 2 years, 7 months ago

Based on the given information, this scenario indicates a case of overfitting. Overfitting occurs when a machine learning model performs well on the training data but poorly on unseen data (test data). In this case, the root-mean-squared error (RMSE) of the model is twice as high on the train set (the data used for training) compared to the test set (the data used for evaluation). This suggests that the model is fitting the training data too closely and is not generalizing well to new, unseen data.

ckanaar 2 years, 6 months ago

Wrong! This scenario indicates a case of underfitting. The RSME is twice as high on the training dataset compared to the test dataset, so the model is underfitting.

AJKumar Highly Voted 5 years, 9 months ago

small RMS Error means--overfitting--fits well--so make it complex by dropping features. big RMS Error means--underfitting--not good fit--so increase complexity by adding layers/features. Answer D.

samtestking 1 year, 2 months ago

Selected Answer: B

Could be B (data requirement for task is vague), but let's assume 100 million data points is enough and rule that out. Indication of overfitting is significantly better performance on training data compared to unseen data. Here we are told that the unseen data is performing significantly better which is the opposite of what we should see if it were overfitting. Rule out C. Symptoms of model underfitting is poor performance in BOTH training AND unseen data. While underfitting might be the issue, the more pressing concern is that the test set is clearly not representative of the overall data and could be skewed. This is further supported by the 90/10 split (academic/industry standard is 80/20 or 75/25 based on the Pareto principle: https://en.wikipedia.org/wiki/Pareto_principle). A 90/10 split would be useful if we were doing k-fold cross validation (https://machinelearningmastery.com/k-fold-cross-validation/), however there is no indication of such in the prompt. Note: The question does not explicitly say that the model is performing poorly/errors are significantly bad, just that the error is twice as high in the training set (they could both have low error values). So whilst it could be a case of underfitting (D), the first step taken should be addressing the obviously problematic data representation by adjusting the train-test split (option A).

SamuelTsch 1 year, 5 months ago

Selected Answer: D

It is underfitting problem, which means that the used models is too easy.

baimus 1 year, 6 months ago

This is A. The key is that 90/10 is a weirdly small test set, that stood out to me straight away (I work professionally as a machine learning engineer and have the cert). Next tip, that everyone seems to be ignoring - this is not underfit OR overfit. The model outperforms on the TEST set, this is not a miswording. Test scores higher than train. The time you might expect to see this is if your test set is too small to be a representative sample, leading to unrepresentative results. Seeing as the question already set up this conclusion with the 90/10 thing, it's definitely A. None of the others

(or indeed anything else) can address Test outperforming Train, and the conclusion of others below that this is due to a poorly worded question is a bizarre conclusion.

cuadradobertolinisebastiancami 2 years, 1 month ago

Selected Answer: D

Underfitting scenario

Sofiia98 2 years, 2 months ago

Selected Answer: D

It is an underfitting situation - D

Kimich 2 years, 4 months ago

Selected Answer: C

Should be C C. Try out regularization techniques (e.g., dropout or batch normalization) to avoid overfitting: This is a reasonable approach. Regularization techniques can help prevent overfitting, especially when the model shows a significantly higher error on the training set compared to the test set. D. Increase the complexity of your model (e.g., introducing an additional layer or increasing the size of vocabularies or n-grams): This could potentially exacerbate the overfitting issue. Increasing model complexity without addressing overfitting concerns may lead to poor generalization on new data.

Kimich 2 years, 4 months ago

https://dooinnkim.medium.com/what-are-overfitting-and-underfitting-855d5952c0b6

hallo 2 years, 4 months ago

Are the questions in this relevant for the new exam or are these all now outdated?

pss111423 2 years, 4 months ago

https://stats.stackexchange.com/questions/497050/how-big-a-difference-for-test-train-rmse-is-considered-as-overfit#:~:text=RMSE%20of%20test%20%3C%20RMSE%20of,is%20always%20overfit%20or%20underfit. RMSE of test > RMSE of train => OVER FITTING of the data. RMSE of test < RMSE of train => UNDER FITTING of the data. so for answer is D

steghe 2 years, 4 months ago

Underfitting models: In general High Train RMSE, High Test RMSE. Overfitting models: In general Low Train RMSE, High Test RMSE. https://daviddalpiaz.github.io/r4sl/regression-for-statistical-learning.html

ha1p 2 years, 6 months ago

I passed the exam today. I am pretty sure it is overfitting. Answer must be c

MULTITASKER 2 years, 6 months ago

Selected Answer: D

RMSE is more on training. That means, model is not performing well on training dataset but performing well on testing dataset. This happens in the case of underfitting. So D.

pulse008 2 years, 6 months ago

chatGPT says option C

stonefl 2 years, 7 months ago

Selected Answer: D

"root-mean-squared error (RMSE) of your model is twice as high on the train set as on the test set." means the RMSE of training set is two time of RMSE of test set, which indicates the training is not as good as test, then underfiting, so D.

NeoNitin 2 years, 7 months ago

Based on the given information, this scenario indicates a case of overfitting. Overfitting occurs when a machine learning model performs well on the training data but poorly on unseen data (test data). In this case, the root-mean-squared error (RMSE) of the model is twice as high on the train set (the data used for training) compared to the test set (the data used for evaluation). This suggests that the model is fitting the training data too closely and is not generalizing well to new, unseen data. So with dropout method we can overcome the overfitting so C is correct

MoeHaydar 2 years, 8 months ago

Selected Answer: D

underfitting

NeoNitin 2 years, 7 months ago

Based on the given information, this scenario indicates a case of overfitting. Overfitting occurs when a machine learning model performs well on the training data but poorly on unseen data (test data). In this case, the root-mean-squared error (RMSE) of the model is twice as high on the train set (the data used for training) compared to the test set (the data used for evaluation). This suggests that the model is fitting the training data too closely and is not generalizing well to new, unseen data.

https://zhangxuetu.online

[202610]

Question #129 Topic 1

You use BigQuery as your centralized analytics platform. New data is loaded every day, and an ETL pipeline modifies the original data and prepares it for the final users. This ETL pipeline is regularly modified and can generate errors, but sometimes the errors are detected only after 2 weeks. You need to provide a method to recover from these errors, and your backups should be optimized for storage costs. How should you organize your data in BigQuery and store your backups?

A. Organize your data in a single table, export, and compress and store the BigQuery data in Cloud Storage.

B. Organize your data in separate tables for each month, and export, compress, and store the data in Cloud Storage. **Most Voted**

C. Organize your data in separate tables for each month, and duplicate your data on a separate dataset in BigQuery.

D. Organize your data in separate tables for each month, and use snapshot decorators to restore the table to a time prior to the corruption.

Correct Answer: B

Community vote distribution



Comments

Ganshank **Highly Voted** 5 years, 11 months ago

B The questions is specifically about organizing the data in BigQuery and storing backups.

ABM9 **Highly Voted** 5 years, 3 months ago

Should be B Using snapshot decorators , recovery is valid only for a period of 7 days. Here it says 2 weeks so "D" is ruled out. You can undelete a table within seven days of deletion, including explicit deletions and implicit deletions due to table expiration. After seven days, it is not possible to undelete a table using any method, including opening a support ticket. https://cloud.google.com/bigquery/docs/managing-tables

forepick 7 months, 1 week ago

Selected Answer: B

D is a distractor as it talks about snapshot DECORATORS (a technique to address Time Travel - a 7 days backup) and not about "snapshots".

desertlotus1211 1 year ago

Selected Answer: B

Snapshot decorators in BigQuery allow you to query a table at a past point in time, but they are limited by BigQuery's time travel window (which is typically 7 days). Since errors are sometimes only detected after 2 weeks, snapshot decorators won't be effective for recovering data beyond their retention period.

grshankar9 1 year, 2 months ago

Selected Answer: D

With Storage Decorators, BigQuery only stores the differences between a snapshot and its base table, minimizing storage costs.

desertlotus1211 1 year ago

Does this answer address how should you organize your data in BigQuery and store your backups?

cloud_rider 1 year, 4 months ago

Selected Answer: D

D is the most cost optimized solution to keep the backup. please read the link - https://cloud.google.com/bigquery/docs/table-snapshots-intro#table_snapshots

SamuelTsch 1 year, 5 months ago

Selected Answer: D

I think D is better.

Lenifia 1 year, 9 months ago

Selected Answer: D

The best option is D. Organize your data in separate tables for each month, and use snapshot decorators to restore the table to a time prior to the corruption.

zevexWM 1 year, 11 months ago

Selected Answer: D

Answer is D: Snapshots are different from time travel. They can hold data as long as we want. Furthermore "BigQuery only stores bytes that are different between a snapshot and its base table" so pretty cost effective as well. https://cloud.google.com/bigquery/docs/table-snapshots-intro#table_snapshots

Farah_007 1 year, 11 months ago

Selected Answer: B

From : https://cloud.google.com/architecture/dr-scenarios-for-data#BigQuery It can't be D If the corruption is caught within 7 days, query the table to a point in time in the past to recover the table prior to the corruption using snapshot decorators. Store the original data on Cloud Storage. This allows you to create a new table and reload the uncorrupted data. From there, you can adjust your applications to point to the new table. => D

Nirca 2 years, 5 months ago

Selected Answer: D

D - this solution in integrated. No core is needed

Bahubali1988 2 years, 6 months ago

90% of questions are having multiple answers and its very hard to get into every discussion where the conclusion is not there

ckanaar 2 years, 6 months ago

Selected Answer: B

The answer is B: Why not D? Because snapshot costs can become high if a lot of small changes are made to the base table: https://cloud.google.com/bigquery/docs/table-snapshots-

intro#:~:text=Because%20BigQuery%20storage%20is%20column%2Dbased%2C%20small%20changes%20to%20the%20data%20in%20a%20base%20table%20can%20result%20in%20large%20increases%20in%20storage%20cost%20for%20its%20table%20snapshot. Since the question specifically states that the ETL pipeline is regularly modified, this means that lots of small changes are present. In combination with the requirement to optimize for storage costs, this means that option B is the way to go.

arien_chen 2 years, 7 months ago

Selected Answer: D

keyword: detected after 2 weeks. only snapshot could resolve the problem.

Lanro 2 years, 8 months ago

Selected Answer: D

From BigQuery documentation - Benefits of using table snapshots include the following: - Keep a record for longer than seven days. With BigQuery time travel, you can only access a table's data from seven days ago or more recently. With table snapshots, you can preserve a table's data from a specified point in time for as long as you want. - Minimize storage cost. BigQuery only stores bytes that are different between a snapshot and its base table, so a table snapshot typically uses less storage than a full copy of the table. So storing data in GCS will make copies of data for each table. Table snapshots are more optimal in this scenario.

vamgcp 2 years, 8 months ago

Selected Answer: B

Organizing your data in separate tables for each month will make it easier to identify the affected data and restore it. Exporting and compressing the data will reduce storage costs, as you will only need to store the compressed data in Cloud Storage. Storing your backups in Cloud Storage will make it easier to restore the data, as you can restore the data from Cloud Storage directly

phidelics 2 years, 9 months ago

Selected Answer: B

Organize in separate tables and store in GCS

cetanx 2 years, 9 months ago

Just an additional info! Here is an example for an export job; $ bq extract --destination_format CSV --compression GZIP 'your_project:your_dataset.your_new_table' 'gs://your_bucket/your_object.csv.gz'

cetanx 2 years, 9 months ago

I will update my answer to D. Think of a scenario that you are in the last week of June and an error occurred 3 weeks ago (so still in June) however you do not have an export of the June table yet therefore you cannot recover the data simply because you don't have an export just yet. So snapshots are way to go!

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #130 Topic 1

The marketing team at your organization provides regular updates of a segment of your customer dataset. The marketing team has given you a CSV with 1 million records that must be updated in BigQuery. When you use the UPDATE statement in BigQuery, you receive a quotaExceeded error. What should you do?

A. Reduce the number of records updated each day to stay within the BigQuery UPDATE DML statement limit.

B. Increase the BigQuery UPDATE DML statement limit in the Quota management section of the Google Cloud Platform Console.

C. Split the source CSV file into smaller CSV files in Cloud Storage to reduce the number of BigQuery UPDATE DML statements per BigQuery job.

D. Import the new records from the CSV file into a new BigQuery table. Create a BigQuery job that merges the new records with the existing records and writes the results to a new BigQuery table. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

rickywck **Highly Voted** 5 years ago

Should be D. https://cloud.google.com/blog/products/gcp/performing-large-scale-mutations-in-bigquery

Rajuuu 4 years, 8 months ago

There is no mention about merge or limit in the link provided.

Chelseajcole 3 years, 6 months ago

A common scenario within OLAP systems involves updating existing data based on new information arriving from source systems (such as OLTP databases) on a periodic basis. In the retail business, inventory updates are typically done in this fashion. The following query demonstrates how to perform batch updates to the Inventory table based on the contents of another table (where new arrivals are kept) using the MERGE statement in BigQuery:

madhu1171 **Highly Voted** 5 years ago

No dml limits from 3rd march 2020

Nirca 1 year, 5 months ago

Selected Answer: D

Should be D.

vaga1 1 year, 9 months ago

Selected Answer: D

import all the data into a separate table and use that for updates is better than creating smaller csv which leads to more operational time to get it done and harder to manage it.

juliobs 2 years ago

Selected Answer: D

This limit was removed a long time ago already. Anyway, bulk imports are better.

Atnafu 2 years, 4 months ago

D BigQuery DML statements have no quota limits. https://cloud.google.com/bigquery/quotas#data-manipulation-language-statements However, DML statements are counted toward the maximum number of table operations per day and partition modifications per day. DML statements will not fail due to these limits. In addition, DML statements are subject to the maximum rate of table metadata update operations. If you exceed this limit, retry the operation using exponential backoff between retries.

MisuLava 2 years, 5 months ago

there is no update quota anymore. but i would say D

amitsingla012 2 years, 11 months ago

Option D is the right answer

tavva_prudhvi 2 years, 11 months ago

No DML limits from 3rd march 2020 But if the questions is given in the exam, choose D asfor the options A, B,C as they are speaking about the limitations of the DML Limits. Atleast, D is giving an alternative!

nidnid 3 years, 2 months ago

Is this question still valid? What about DML without limits? https://cloud.google.com/blog/products/data-analytics/dml-without-limits-now-in-bigquery

MaxNRG 3 years, 2 months ago

Selected Answer: D

D: BigQuery is primarly designed and suit to append-only technology with some limited DML statements. It's not a relational database where you constantly update your user records if they edit their profile. Instead you need to arhitect your code so each edit is a new row in Bigquery, and you always query the latest row. The DML statement limitation is low, because it targets different scenarios and not yours, aka live update on rows. You could ingest your data into a separate table, and issue 1 update statement per day. https://stackoverflow.com/questions/45183082/can-we-increase-update-quota-in-bigquery https://cloud.google.com/blog/products/gcp/performing-large-scale-mutations-in-bigquery

medeis_jar 3 years, 2 months ago

Selected Answer: D

https://cloud.google.com/bigquery/docs/reference/standard-sql/dml-syntax#merge_statement
https://cloud.google.com/blog/products/gcp/performing-large-scale-mutations-in-bigquery

mjb65 3 years, 4 months ago

old question I guess, should not be in the exam anymore (?) https://cloud.google.com/blog/products/data-analytics/dml-without-limits-now-in-bigquery

sumanshu 3 years, 9 months ago

Vote for D

daghayeghi 4 years ago

D: https://cloud.google.com/blog/products/gcp/performing-large-scale-mutations-in-bigquery

daghayeghi 4 years, 1 month ago

D: https://cloud.google.com/blog/products/bigquery/performing-large-scale-mutations-in-bigquery

SteelWarrior 4 years, 6 months ago

D should be the answer. Avoid updates in Datawarehousing environment instead use merge to create a new table.

https://zhangxuetu.online

[202610]

Question #131 Topic 1

As your organization expands its usage of GCP, many teams have started to create their own projects. Projects are further multiplied to accommodate different stages of deployments and target audiences. Each project requires unique access control configurations. The central IT team needs to have access to all projects. Furthermore, data from Cloud Storage buckets and BigQuery datasets must be shared for use in other projects in an ad hoc way. You want to simplify access control management by minimizing the number of policies. Which two steps should you take? (Choose two.)

A. Use Cloud Deployment Manager to automate access provision.

B. Introduce resource hierarchy to leverage access control policy inheritance. **Most Voted**

C. Create distinct groups for various teams, and specify groups in Cloud IAM policies. **Most Voted**

D. Only use service accounts when sharing data for Cloud Storage buckets and BigQuery datasets.

E. For each Cloud Storage bucket or BigQuery dataset, decide which projects need access. Find all the active members who have access to these projects, and create a Cloud IAM policy to grant access to all these users.

Correct Answer: BC

Community vote distribution



Comments

AJKumar **Highly Voted** 4 years, 9 months ago

C is one option for sure, also C eliminates B as C includes groups and teams hierarchy, A can be eliminated as A talks about only deployment. From Remaining D and E, i find E most relevant to question--as E matches users with teams/groups and projects. Answer C and E.

hauhau 3 years, 7 months ago

Question mention minimize IAM policies,but E should create complex policies

hellofrnds **Highly Voted** 3 years, 6 months ago

Answer:- C, D C is used as best practice to create group and assign IAM roles D "data from Cloud Storage buckets and BigQuery datasets must be shared for use in other projects in an ad hoc way" is mentioned in question. When 2 project communicate, service account should be used

squishy_fishy 1 year, 5 months ago

Answer: A, C. The Key question is "You want to simplify access control management by minimizing the number of policies". At the company where I work, we use Terraform to create infrastructure and assign needed roles for different

environments.

amittomar 1 year, 8 months ago

It should be AC as it is mentioned in the question itself "You want to simplify access control management by minimizing the number of policies" which rules out B

FrankT2L 2 years, 9 months ago

Selected Answer: BC

1. Define your resource hierarchy: Google Cloud resources are organized hierarchically. This hierarchy allows you to map your enterprise's operational structure to Google Cloud, and to manage access control and permissions for groups of related resources. 2. Delegate responsibility with groups and service accounts: we recommend collecting users with the same responsibilities into groups and assigning IAM roles to the groups rather than to individual users. https://cloud.google.com/docs/enterprise/best-practices-for-enterprise-organizations

annie1196 3 years, 2 months ago

A and C is correct, same question I encountered on Udemy.

tavva_prudhvi 3 years ago

It doesn't mean it's right, please mention the reasons here not only the references. Not A -> Every project has unique requirements, so "A" automation will not do much. Not D -> As, Service accounts for computer to computer interactions not applications! Not E -> E should create complex policies

desertlotus1211 2 years, 2 months ago

you explanation for D is incorrect....

MaxNRG 3 years, 2 months ago

Selected Answer: BC

B & C Google Cloud resources are organized hierarchically, where the organization node is the root node in the hierarchy, the projects are the children of the organization, and the other resources are descendents of projects. You can set Cloud Identity and Access Management (Cloud IAM) policies at different levels of the resource hierarchy. Resources inherit the policies of the parent resource. The effective policy for a resource is the union of the policy set at that resource and the policy inherited from its parent. https://cloud.google.com/iam/docs/resource-hierarchy-access-control

AzureDP900 2 years, 3 months ago

BC is the answer

MaxNRG 3 years, 2 months ago

We recommend collecting users with the same responsibilities into groups and assigning Cloud IAM roles to the groups rather than to individual users. For example, you can create a "data scientist" group and assign appropriate roles to enable interaction with BigQuery and Cloud Storage. Grant roles to a Google group instead of to individual users when possible. It is easier to manage members in a Google group than to update a Cloud IAM policy. https://cloud.google.com/docs/enterprise/best-practices-for-enterprise-organizations

medeis_jar 3 years, 2 months ago

Selected Answer: AC

https://cloud.google.com/docs/enterprise/best-practices-for-enterprise-organizations "Each project requires unique access control configurations" -> C eliminates B A -> "Google Cloud Deployment Manager is an infrastructure deployment service that automates the creation and management of Google Cloud resources. Write flexible template and configuration files and use them to create deployments that have a variety of Google Cloud services" "..simply the process.."

MaxNRG 3 years, 2 months ago

good point, AC looks better, agreed

MaxNRG 3 years, 2 months ago

... in other hand - "Define your resource hierarchy" https://cloud.google.com/docs/enterprise/best-practices-for-enterprise-organizations#define-hierarchy

MaxNRG 3 years, 2 months ago

So, I stay with BC :)))

FP77 1 year, 7 months ago

A makes no sense whatsoever...

sumanshu 3 years, 9 months ago

Vote for B & C

daghayeghi 4 years ago

the question says adding permissions ad-hoc way [c] is correct answer [d] is right, as the access to bigQuery and cloud storage can be managed automatically by Cloud deployment "Deployment Manager can also set access control permissions through IAM such that your developers are granted appropriate access as part of the project creation process." ref: https://cloud.google.com/docs/enterprise/best-practices-for-enterprise-organizations

VM_GCP 4 years, 3 months ago

the question says adding permissions ad-hoc way [c] is correct answer [d] is right, as the access to bigQuery and cloud storage can be managed automatically by Cloud deployment "Deployment Manager can also set access control permissions through IAM such that your developers are granted appropriate access as part of the project creation process." ref: https://cloud.google.com/docs/enterprise/best-practices-for-enterprise-organizations

sumanshu 3 years, 9 months ago

D is a service account - it means we need to access via applications. So, D is ruled out

sumanshu 3 years, 9 months ago

Doubt, It could be 'D" - because - it's said - data from Cloud Storage buckets and BigQuery datasets must be shared for use in other projects in an ad hoc way.

awssp12345 3 years, 8 months ago

Agree with Sumanshu

ceak 4 years, 4 months ago

C & E are the correct answers.

sumanshu 3 years, 9 months ago

E is a long process and we need to simply the process...So E is ruled out

vito9630 4 years, 4 months ago

Answer: B, C

kavs 4 years, 4 months ago

BC A ruled out as deployment manager is for infra yaml based deployments D at resource level we can't check hierarchy at org or Proj Level

rgpalop 4 years, 6 months ago

C and E

atnafu2020 4 years, 7 months ago

B & C the correct answer, as google suggest

haroldbenites 4 years, 7 months ago

C, E is correct.

https://zhangxuetu.online

[202610]

Question #132 Topic 1

Your United States-based company has created an application for assessing and responding to user actions. The primary table's data volume grows by 250,000 records per second. Many third parties use your application's APIs to build the functionality into their own frontend applications. Your application's APIs should comply with the following requirements:

- Single global endpoint
- ANSI SQL support
- Consistent access to the most up-to-date data

What should you do?

A. Implement BigQuery with no region selected for storage or processing.

B. Implement Cloud Spanner with the leader in North America and read-only replicas in Asia and Europe. **Most Voted**

C. Implement Cloud SQL for PostgreSQL with the master in North America and read replicas in Asia and Europe.

D. Implement Bigtable with the primary cluster in North America and secondary clusters in Asia and Europe.

Correct Answer: B

Community vote distribution



Comments

sumanshu **Highly Voted** 3 years, 2 months ago

A - BigQuery with NO Region ? (Looks wrong) B - Spanner (SQL support and Scalable and have replicas) - Looks correct C - SQL (can't store so many records) (wrong) D - Bigtable - NO SQL (wrong) Vote for B

Tanmoyk **Highly Voted** 4 years ago

B is correct, Bigquery cannot support 250K data ingestion/second , as ANSI SQL support is required , no other options left except Spanner.

vaga1 1 year, 3 months ago

Selected Answer: B

A - NO - BigQuery with must have a selected regional or multi-regional file storage B - YES - Spanner is specifically designed for this high and consistent throughput C - NO - I am not sure about what many said in this discussion as Cloud SQL can store this amount of records if u have just a few columns. Anyway, for sure Spanner is better and it is a GCP product. D - Bigtable - it's a NoSQL solution, no ANSI

AzureDP900 1 year, 9 months ago

B is the answer

Remi2021 2 years ago

Selected Answer: B

Guys, read documentation well. A is wrong, BigQuery has Maximum rows per request (50,000). https://cloud.google.com/bigquery/quotas It is B

JamesKarianis 2 years, 2 months ago

Selected Answer: B

Spanner is globally available and meets all the requirements

devric 2 years, 5 months ago

Selected Answer: A

Correct is A. There's no sense to having read replicas outside of US considering than the company is US based. If you generate a dataset without specifing the Data Location it's gonna be stored in "US Multiregion" by default

MaxNRG 2 years, 8 months ago

Selected Answer: B

B: Cloud Spanner is the first scalable, enterprise-grade, globally-distributed, and strongly consistent database service built for the cloud specifically to combine the benefits of relational database structure with non-relational horizontal scale. https://cloud.google.com/spanner/ Cloud Spanner is a fully managed, mission-critical, relational database service that offers transactional consistency at global scale, schemas, SQL (ANSI 2011 with extensions), and automatic, synchronous replication for high availability. https://cloud.google.com/spanner/docs/ https://cloud.google.com/spanner/docs/instances#available-configurations-multi-region

haroldbenites 4 years, 1 month ago

B is correct

Archy 4 years, 1 month ago

B, as Cloud Spanner has three types of replicas: read-write replicas, read-only replicas, and witness replicas.

VishalB 4 years, 2 months ago

Correct Answer : C Explanation:- B -> This option is incorrect, as we do not have option to configure read-replica in Cloud Spanner, Multi-region instance configurations use a combination of all three types' read-write replicas, read-only replicas, and witness replicas C -> This is correct option, In Cloud Sql we have option to create a master node for read-write replicas and read-only replicas in other regions D -> This option is incorrect, as Bigtable do not support ANSI SQL

saurabh1805 4 years, 1 month ago

but bigquery does so why not A?

WizzzardLlama 3 years, 8 months ago

You can't create a BigQuery instance without region selected. I'm wondering about these read replicas, why read only replicas? It seems arbitrary, as the question does not state that API should be read-only, so there's no reason why those should be read-only replicas...

mAbreu 4 years ago

wrong, cloud spanner can have read-only replicas https://cloud.google.com/spanner/docs/replication?hl=pt-br

norwayping 4 years, 3 months ago

I was wrong Bigtable doesn not support ansi SQL. B instead

norwayping 4 years, 3 months ago

I think it is D. THere is limitation of QPS of Cloud Spanner of 2000 qps,
https://cloud.google.com/spanner/docs/instances#multi-region-performance

Rajokkiyam 4 years, 6 months ago

Answer B

https://zhangxuetu.online

[202610]

Question #133 Topic 1

A data scientist has created a BigQuery ML model and asks you to create an ML pipeline to serve predictions. You have a REST API application with the requirement to serve predictions for an individual user ID with latency under 100 milliseconds. You use the following query to generate predictions: SELECT predicted_label, user_id FROM ML.PREDICT (MODEL 'dataset.model', table user_features). How should you create the ML pipeline?

A. Add a WHERE clause to the query, and grant the BigQuery Data Viewer role to the application service account.

B. Create an Authorized View with the provided query. Share the dataset that contains the view with the application service account.

C. Create a Dataflow pipeline using BigQueryIO to read results from the query. Grant the Dataflow Worker role to the application service account.

D. Create a Dataflow pipeline using BigQueryIO to read predictions for all users from the query. Write the results to Bigtable using BigtableIO. Grant the Bigtable Reader role to the application service account so that the application can read predictions for individual users from Bigtable. **Most Voted**

Correct Answer: D

Community vote distribution

D (88%) B (13%)

Comments

rickywck **Highly Voted** 5 years, 6 months ago

I think the key reason for pick D is the 100ms requirement.

AzureDP900 2 years, 9 months ago

D. Create a Dataflow pipeline using BigQueryIO to read predictions for all users from the query. Write the results to Bigtable using BigtableIO. Grant the Bigtable Reader role to the application service account so that the application can read predictions for individual users from Bigtable.

sumanshu **Highly Voted** 4 years, 2 months ago

Vote for D, requirement to serve predictions with in 100 ms

MaxNRG 1 year, 9 months ago

Selected Answer: D

The key requirements are serving predictions for individual user IDs with low (sub-100ms) latency. Option D meets this by batch predicting for all users in BigQuery ML, writing predictions to Bigtable for fast reads, and allowing the application access to query Bigtable directly for low latency reads. Since the application needs to serve low-latency

predictions for individual user IDs, using Dataflow to batch predict for all users and write to Bigtable allows low-latency reads. Granting the Bigtable Reader role allows the application to retrieve predictions for a specific user ID from Bigtable.

MaxNRG 1 year, 9 months ago

The other options either require changing the query for each user ID (higher latency, option A), reading directly from higher latency services like BigQuery (option B), or writing predictions somewhere without fast single row access (options A, B, C). Option A would not work well because the WHERE clause would need to be changed for each user ID, increasing latency. Option B using an Authorized View would still read from BigQuery which has higher latency than Bigtable for individual rows. Option C writes predictions to BigQuery which has higher read latency compared to Bigtable for individual rows. So option D provides the best pipeline by predicting for all users in BigQueryML, batch writing to Bigtable for low latency reads, and granting permissions for the application to retrieve predictions. This meets the requirements of sub-100ms latency for individual user predictions.
https://cloud.google.com/dataflow/docs/concepts/access-control

Nirca 1 year, 11 months ago

Selected Answer: D

I think the key reason for pick D is the 100ms requirement/ me too

barnac1es 2 years ago

Selected Answer: B

To create an ML pipeline for serving predictions to individual user IDs with latency under 100 milliseconds using the given BigQuery ML query, the most suitable approach is: B. Create an Authorized View with the provided query. Share the dataset that contains the view with the application service account.

Lanro 2 years, 2 months ago

Selected Answer: D

Always use Bigtable as an endpoint for client-facing applications (Low latency - high throughput)

midgoo 2 years, 6 months ago

Selected Answer: D

One of the way to improve the efficient of ML pipeline is to generate cache (store predictions). In this question, only D is doing that.

musumusu 2 years, 7 months ago

what is wrong with B ? View can be precomputed and cached and it can definitely satisfy the 100 miliseconds request. create a pipeline to send data to bigtable .. don't you think its too much to run a simple prediction query ?

vaga1 2 years, 3 months ago

I would say that Bigtable is simply more suited to serve applications

vaga1 2 years, 3 months ago

and the view has to make a query in real-time which adds potential latency

cheos71 2 years, 1 month ago

I think if there is too many concurrent requests the 100ms latency will def not hold reading from bigquery

hiromi 2 years, 10 months ago

Selected Answer: D

Vote for D

HarshKothari21 3 years ago

Selected Answer: D

Option D

Tanmoyk 5 years ago

D is correct , 100ms is most critical factor here.

sh2020 5 years ago

writing it to BigTable and then allowing application access will introduce more delays. I think answer should be C

f839 4 years, 8 months ago

Predictions are computed in advance for all users and written to BigTable for low-latency serving.

haroldbenites 5 years, 1 month ago

D is correct

Rajokkiyam 5 years, 6 months ago

Answer D.

https://zhangxuetu.online

[202610]

Question #134 Topic 1

You are building an application to share financial market data with consumers, who will receive data feeds. Data is collected from the markets in real time.
Consumers will receive the data in the following ways:
- Real-time event stream
- ANSI SQL access to real-time stream and historical data
- Batch historical exports

Which solution should you use?

A. Cloud Dataflow, Cloud SQL, Cloud Spanner

B. Cloud Pub/Sub, Cloud Storage, BigQuery **Most Voted**

C. Cloud Dataproc, Cloud Dataflow, BigQuery

D. Cloud Pub/Sub, Cloud Dataproc, Cloud SQL

Correct Answer: B

Community vote distribution



Comments

itche_scratche **Highly Voted** 5 years, 5 months ago

D, not ideal but only option that work. You need pubsub, then a processing layer (dataflow or dataproc), then storage (some sql database).

seiyassa 5 years, 3 months ago

I think pubsub doesn't have good connection to dataproc, so D is not the answer

jkhong 3 years, 3 months ago

As of Dec 2022,there is the PubSub Lite connector to Dataproc

jkhong 3 years, 3 months ago

We can have our pubsub topics to have BigQuery subscriptions, where data is automatically streamed into our BQ tables. Autoscaling is already handled automatically so this renders Dataflow and Dataproc pretty irrelevant for our usecase

cetanx 2 years, 9 months ago

Here is the reference: https://cloud.google.com/blog/products/data-analytics/pub-sub-launches-direct-path-to-bigquery-for-streaming-analytics

medeis_jar **Highly Voted** 4 years, 2 months ago

Selected Answer: B

✑ Real-time event stream -> Pub/Sub ✑ ANSI SQL access to real-time stream and historical data -> BigQuery ✑ Batch historical exports -> Cloud Storage

SamuelTsch 1 year, 5 months ago

Selected Answer: D

why B? The main goal of the question is data storage. Thus BigQuery is not neccessary for this situation. Option D from my point of view cover the whole requirements. Pub/Sub for streaming data, dataproc for data processing, SQL for storage.

barnac1es 2 years, 6 months ago

B. Cloud Pub/Sub, Cloud Storage, BigQuery. Here's how this solution aligns with your requirements: Real-time Event Stream: Cloud Pub/Sub is a managed messaging service that can handle real-time event streams efficiently. You can use Pub/Sub to ingest and publish real-time market data to consumers. ANSI SQL Access: BigQuery supports ANSI SQL queries, making it suitable for both real-time and historical data analysis. You can stream data into BigQuery tables from Pub/Sub and provide ANSI SQL access to consumers. Batch Historical Exports: Cloud Storage can be used for batch historical exports. You can export data from BigQuery to Cloud Storage in batch, making it available for consumers to download.

vaga1 2 years, 9 months ago

Selected Answer: B

I was in doubt as I did not know that BQ handles real-time access to data without dataflow underneath. https://cloud.google.com/bigquery/docs/write-api#:~:text=You%20can%20use%20the%20Storage,in%20a%20single%20atomic%20operation.

midgoo 3 years ago

Selected Answer: B

Event Stream -> PubSub PubSub has direct Write to BigQuery Historical Exports to GCS

AzureDP900 3 years, 3 months ago

B. Cloud Pub/Sub, Cloud Storage, BigQuery

AzureDP900 3 years, 3 months ago

https://cloud.google.com/solutions/stream-analytics/

John_Pongthorn 3 years, 6 months ago

Selected Answer: B

B: https://cloud.google.com/solutions/stream-analytics/ Real-time made real easy Adopt simple ingestion for complex events Ingest and analyze hundreds of millions of events per second from applications or devices virtually anywhere on the globe with Pub/Sub. Or directly stream millions of events per second into your data warehouse for SQL-based analysis with BigQuery's streaming API.

John_Pongthorn 3 years, 6 months ago

Selected Answer: B

No matter what the last of it must end up with bigquery and the first service is pubsub I think intimidate service it should be dataflow

Motivated_Gamer 4 years ago

Selected Answer: A

Dataflow: Streaming data CLoud SQL: for ansi sql support Spanner: for batch historical data export

tavva_prudhvi 3 years, 11 months ago

You gonna use batch historical export for Spanner? It's B!

Prasanna_kumar 4 years, 1 month ago

Answer is B

MaxNRG 4 years, 2 months ago

Selected Answer: B

Cloud Pub/Sub, Cloud Dataflow, BigQuery https://cloud.google.com/solutions/stream-analytics/

MaxNRG 2 years, 3 months ago

B. Cloud Pub/Sub, Cloud Storage, BigQuery The key requirements here are: 1. Real-time event stream (Pub/Sub) 2. ANSI SQL access to real-time and historical data (BigQuery) 3. Batch historical exports (Cloud Storage) So Cloud Pub/Sub provides the real-time stream, BigQuery provides ANSI SQL access to stream and historical data, and Cloud Storage enables batch historical exports. Option A is incorrect because Cloud Spanner does not offer batch exports and Dataflow is overkill for just SQL access. Option C is incorrect as Dataproc is for spark workloads, not serving consumer data. Option D is incorrect as Cloud SQL does not provide batch export capabilities. Therefore, option B with Pub/Sub, Storage, and BigQuery is the best solution given the stated requirements. Dataflow https://cloud.google.com/solutions/stream-analytics/

JG123 4 years, 4 months ago

Correct: B

AdrianMonter26 4 years, 4 months ago

I think it must be D because you need Pub/Sub for streaming data, Dataflow or DataProc to get the data from Pub/Sub and store it in a database and finally the Cloud SQL database to store the data. A and C cannot be because it is missing something for streaming data B It can't be because you need something to pass the data from Pub/Sub to Cloud storage

sumanshu 4 years, 9 months ago

Vote for B

sumanshu 4 years, 9 months ago

Vote for B

rgpalop 5 years, 6 months ago

I think B

https://zhangxuetu.online

[202610]

Question #135 Topic 1

You are building a new application that you need to collect data from in a scalable way. Data arrives continuously from the application throughout the day, and you expect to generate approximately 150 GB of JSON data per day by the end of the year. Your requirements are:

- Decoupling producer from consumer
- Space and cost-efficient storage of the raw ingested data, which is to be stored indefinitely
- Near real-time SQL query
- Maintain at least 2 years of historical data, which will be queried with SQL

Which pipeline should you use to meet these requirements?

A. Create an application that provides an API. Write a tool to poll the API and write data to Cloud Storage as gzipped JSON files.

B. Create an application that writes to a Cloud SQL database to store the data. Set up periodic exports of the database to write to Cloud Storage and load into BigQuery.

C. Create an application that publishes events to Cloud Pub/Sub, and create Spark jobs on Cloud Dataproc to convert the JSON data to Avro format, stored on HDFS on Persistent Disk.

D. Create an application that publishes events to Cloud Pub/Sub, and create a Cloud Dataflow pipeline that transforms the JSON event payloads to Avro, writing the data to Cloud Storage and BigQuery. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

JG123 **Highly Voted** 4 years, 4 months ago

Why there are so many wrong answers? Examtopics.com are you enjoying paid subscription by giving random answers from people? Ans: D

AJKumar **Highly Voted** 5 years, 9 months ago

A and B can be eliminated righaway. between C and D; C has no bigquery , Answer D.

edre 1 year, 8 months ago

Selected Answer: D

Google recommended approach

juliorevk 2 years, 6 months ago

Selected Answer: D

D because pub/sub decouples while dataflow processes; Cloud Storage can be used to store the raw ingested data indefinitely and BQ can be used to query.

barnac1es 2 years, 6 months ago

Selected Answer: D

Here's how this option aligns with your requirements: Decoupling Producer from Consumer: Cloud Pub/Sub provides a decoupled messaging system where the producer publishes events, and consumers (like Dataflow) can subscribe to these events. This decoupling ensures flexibility and scalability. Space and Cost-Efficient Storage: Storing data in Avro format is more space-efficient than JSON, and Cloud Storage is a cost-effective storage solution. Additionally, Cloud Pub/Sub and Dataflow allow you to process and transform data efficiently, reducing storage costs. Near Real-time SQL Query: By using Dataflow to transform and load data into BigQuery, you can achieve near real-time data availability for SQL queries. BigQuery is well-suited for ad-hoc SQL queries and provides excellent query performance.

FP77 2 years, 7 months ago

Selected Answer: D

Should be D

vaga1 2 years, 9 months ago

Selected Answer: D

For sure D

forepick 2 years, 10 months ago

Selected Answer: D

D is the most suitable, however the stored format should be JSON, and AVRO isn't JSON...

OberstK 3 years, 1 month ago

Selected Answer: D

Correct - D

desertlotus1211 3 years, 2 months ago

I believe this was also on the GCP PCA exam as well! ;)

AzureDP900 3 years, 3 months ago

D. Create an application that publishes events to Cloud Pub/Sub, and create a Cloud Dataflow pipeline that transforms the JSON event payloads to Avro, writing the data to Cloud Storage and BigQuery.

mbacelar 3 years, 4 months ago

Selected Answer: D

For sure D

clouditis 3 years, 6 months ago

D it is!

Prasanna_kumar 4 years, 1 month ago

Answer is D

MaxNRG 4 years, 2 months ago

Selected Answer: D

D: Cloud Pub/Sub, Cloud Dataflow, Cloud Storage, BigQuery https://cloud.google.com/solutions/stream-analytics/

medeis_jar 4 years, 2 months ago

Selected Answer: D

OMG only D

sandipk91 4 years, 7 months ago

Answer is D for sure

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #136 Topic 1

You are running a pipeline in Dataflow that receives messages from a Pub/Sub topic and writes the results to a BigQuery dataset in the EU. Currently, your pipeline is located in europe-west4 and has a maximum of 3 workers, instance type n1-standard-1. You notice that during peak periods, your pipeline is struggling to process records in a timely fashion, when all 3 workers are at maximum CPU utilization. Which two actions can you take to increase performance of your pipeline? (Choose two.)

A. Increase the number of max workers **Most Voted**

B. Use a larger instance type for your Dataflow workers **Most Voted**

C. Change the zone of your Dataflow pipeline to run in us-central1

D. Create a temporary table in Bigtable that will act as a buffer for new data. Create a new step in your pipeline to write to this table first, and then create a new pipeline to write from Bigtable to BigQuery

E. Create a temporary table in Cloud Spanner that will act as a buffer for new data. Create a new step in your pipeline to write to this table first, and then create a new pipeline to write from Cloud Spanner to BigQuery

Correct Answer: AB

Community vote distribution

AB (100%)

Comments

jvg637 Highly Voted 5 years, 6 months ago

A & B instance n1-standard-1 is low configuration and hence need to be larger configuration, definitely B should be one of the option. Increase max workers will increase parallelism and hence will be able to process faster given larger CPU size and multi core processor instance type is chosen. Option A can be a better step.

AzureDP900 2 years, 9 months ago

Agreed

sumanshu Highly Voted 4 years, 2 months ago

A & B. With autoscaling enabled, the Dataflow service does not allow user control of the exact number of worker instances allocated to your job. You might still cap the number of workers by specifying the --max_num_workers option when you run your pipeline. Here as per question CAP is 3, So we can change that CAP. For batch jobs, the default machine type is n1-standard-1. For streaming jobs, the default machine type for Streaming Engine-enabled jobs is n1-standard-2 and the default machine type for non-Streaming Engine jobs is n1-standard-4. When using the

default machine types, the Dataflow service can therefore allocate up to 4000 cores per job. If you need more cores for your job, you can select a larger machine type.

et2137 1 year, 7 months ago

Selected Answer: AB

A & B is correct

kcl10 1 year, 12 months ago

Selected Answer: AB

A & B is correct

juliorevk 2 years ago

Selected Answer: AB

A because more workers improves performance through parallel work B because the current instance size is too small

barnac1es 2 years ago

Selected Answer: AB

A. Increase the number of max workers: By increasing the number of maximum workers, you allow Dataflow to allocate more computing resources to handle the peak load of incoming data. This can help improve processing speed and reduce CPU utilization per worker. B. Use a larger instance type for your Dataflow workers: Using a larger instance type with more CPU and memory resources can help your Dataflow workers handle a higher volume of data and processing tasks more efficiently. It can address CPU bottlenecks during peak periods.

mbacelar 2 years, 10 months ago

Selected Answer: AB

Scale in and Scale Out

FrankT2L 3 years, 3 months ago

Selected Answer: AB

maximum of 3 workers: Increase the number of max workers (A) instance type n1-standard-1: Use a larger instance type for your Cloud Dataflow workers (B)

MaxNRG 3 years, 8 months ago

Selected Answer: AB

A & B, other options don't make sense

medeis_jar 3 years, 8 months ago

Selected Answer: AB

Only A & B make sense for improving pipeline performance.

Mjvsj 3 years, 9 months ago

Selected Answer: AB

Should be A & B

daghayeghi 4 years, 7 months ago

B, E: B: Dataflow manage number of worker automatically, then we only can define machine type worker. https://cloud.google.com/dataflow/docs/guides/deploying-a-pipeline E: and adding a horizontally scale-able database like cloud spanner will reduce pressure on dataflow as it don't have to move data to specific zone and can be remain in same zone of EU, then E is correct.

Vasu_1 4 years, 4 months ago

A & B is the right answer: You can set disable auto-scaling by setting the option --numWorkers (default is 3) and select the machine type by setting --workerMachineType at the time of creation of the pipeline (this applies to both auto and manual scaling)

kavs 4 years, 10 months ago

Dataset is in EU so data can't be moved outside EU due to privacy law so zone option is ruled out. AB is Ok but intermediate table will boost perf apanee ruled out not sure of bigtable

Alasmindas 4 years, 10 months ago

Option A and B for sure, Option C : Changing Zone has nothing to do in improving performance Option D and E : Adding BQ and BT is waste of many and does not solve the purpose of the question.

SureshKotla 5 years ago

B & D DF will automatically take care of increasing workers. Developers won't need to access the settings . https://cloud.google.com/dataflow/docs/guides/deploying-a-pipeline#autoscaling

SureshKotla 5 years ago

On second thought, A B is looking right

sumanshu 4 years, 2 months ago

automatically taking care of workers up to 3 (as the maximum worker is 3 set as per questions)

atnafu2020 5 years, 1 month ago

AB is correct

haroldbenites 5 years, 1 month ago

A , E is correct

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #137 Topic 1

You have a data pipeline with a Dataflow job that aggregates and writes time series metrics to Bigtable. You notice that data is slow to update in Bigtable. This data feeds a dashboard used by thousands of users across the organization. You need to support additional concurrent users and reduce the amount of time required to write the data. Which two actions should you take? (Choose two.)

A. Configure your Dataflow pipeline to use local execution

B. Increase the maximum number of Dataflow workers by setting maxNumWorkers in PipelineOptions **Most Voted**

C. Increase the number of nodes in the Bigtable cluster **Most Voted**

D. Modify your Dataflow pipeline to use the Flatten transform before writing to Bigtable

E. Modify your Dataflow pipeline to use the CoGroupByKey transform before writing to Bigtable

Correct Answer: BC

Community vote distribution



Comments

arpitagrawal **Highly Voted** 3 years, 6 months ago

Selected Answer: BC

It should be B and C

ducc **Highly Voted** 3 years, 7 months ago

Selected Answer: BC

BC is correct Why the comments is deleted?

loki82 1 year, 2 months ago

Selected Answer: CE

If there's a write speed bottleneck on bigtable, more dataflow workers won't make a difference. If I add more bigtable nodes, or group my writes together, I can increase update throughput.
https://cloud.google.com/dataflow/docs/guides/write-to-bigtable#best-practices

Preetmehta1234 1 year, 6 months ago

Selected Answer: BC

the goal is to reduce the write latency not to improve data flow code

emmylou 2 years, 4 months ago

The "Correct Answers" are just put in with a random generator :-) B and C

BlehMaks 2 years, 5 months ago

Selected Answer: BC

B - opportunity to parallelise the process C - increase throughput

Bahubali1988 2 years, 6 months ago

Exactly opposite answers in the discussions

barnac1es 2 years, 6 months ago

Selected Answer: BC

B. Increase the maximum number of Dataflow workers by setting maxNumWorkers in PipelineOptions: Increasing the number of Dataflow workers can help parallelize the processing of your data, which can result in faster data updates to Bigtable and improved concurrency. You can set maxNumWorkers to a higher value to achieve this. C. Increase the number of nodes in the Bigtable cluster: Increasing the number of nodes in your Bigtable cluster can improve the overall throughput and reduce latency when writing data. It allows Bigtable to handle a higher rate of data ingestion and queries, which is essential for supporting additional concurrent users.

ckanaar 2 years, 6 months ago

Selected Answer: CD

C definetely is correct, as it improves the read and write performance of Bigtable. However, I do think that the second option is actually D instead of B, because the question specifically states that the pipeline aggregates data. Flatten merges multiple PCollection objects into a single logical PCollection, allowing for faster aggregation of time series data.

NewDE2023 2 years, 8 months ago

Selected Answer: BE

B - I believe it is consensus. D - The question mentions "a Dataflow job that "aggregates" and writes time series metrics to Bigtable". So CoGroupByKey performs a shuffle (grouping) operation to distribute data across workers. https://cloud.google.com/dataflow/docs/guides/develop-and-test-pipelines

WillemHendr 2 years, 9 months ago

Selected Answer: DE

I read this question as: BigTable Write operations are all over the place (key-wise), and BigTable doesn't like that. When creating groups (batch writes), of similar keys (close to each other), BigTable is happy again, which I loosely translate into DE.

vaga1 2 years, 11 months ago

B is correct. But I don't see how you increase the write throughput of Bigtable increasing its cluster size. It should be dataflow instance resources that have to be increased

juliobs 3 years ago

Selected Answer: BC

BC make sense

NamitSehgal 3 years, 2 months ago

BC only makes sense here , no mention of data, no mention of keeping cost low

AzureDP900 3 years, 3 months ago

B. Increase the maximum number of Dataflow workers by setting maxNumWorkers in PipelineOptions Most Voted C. Increase the number of nodes in the Bigtable cluster

ovokpus 3 years, 4 months ago

Selected Answer: BC

Increase max num of workers increases pipeline performance in Dataflow Increase number of nodes in Bigtable increases write throughput

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #138 Topic 1

You have several Spark jobs that run on a Cloud Dataproc cluster on a schedule. Some of the jobs run in sequence, and some of the jobs run concurrently. You need to automate this process. What should you do?

A. Create a Cloud Dataproc Workflow Template

B. Create an initialization action to execute the jobs

C. Create a Directed Acyclic Graph in Cloud Composer **Most Voted**

D. Create a Bash script that uses the Cloud SDK to create a cluster, execute jobs, and then tear down the cluster

Correct Answer: C

Community vote distribution



Comments

LP_PDE **Highly Voted** 2 years, 12 months ago

Correct answer is A. https://cloud.google.com/dataproc/docs/concepts/workflows/using-workflows

skhaire 1 year, 1 month ago

A. Create a Cloud Dataproc Workflow Template Dataproc Workflow Template can be used to run jobs concurrently and sequentially. DAG is an overkill. https://cloud.google.com/dataproc/docs/concepts/workflows/use-workflows

MaxNRG 1 year, 9 months ago

Selected Answer: C

The best option for automating your scheduled Spark jobs on Cloud Dataproc, considering sequential and concurrent execution, is: C. Create a Directed Acyclic Graph (DAG) in Cloud Composer.

MaxNRG 1 year, 9 months ago

Here's why: DAG workflows: Cloud Composer excels at orchestrating complex workflows with dependencies, making it ideal for managing sequential and concurrent execution of your Spark jobs. You can define dependencies between tasks to ensure certain jobs only run after others finish. Automation: Cloud Composer lets you schedule workflows to run automatically based on triggers like time intervals or data availability, eliminating the need for manual intervention. Integration: Cloud Composer integrates seamlessly with Cloud Dataproc, allowing you to easily launch and manage your Spark clusters within the workflow. Scalability: Cloud Composer scales well to handle a large number of jobs and workflows, making it suitable for managing complex data pipelines.

MaxNRG 1 year, 9 months ago

While the other options have some merit, they fall short in certain aspects: A. Cloud Dataproc Workflow Templates: While workflow templates can automate job submission on a cluster, they lack the ability to define dependencies and coordinate concurrent execution effectively. B. Initialization action: An initialization action can only run a single script before a Dataproc cluster starts, not suitable for orchestrating multiple scheduled jobs with dependencies. D. Bash script: A Bash script might work for simple cases, but it can be cumbersome to manage and lacks the advanced scheduling and error handling capabilities of Cloud Composer. Therefore, utilizing a Cloud Composer DAG offers the most comprehensive and flexible solution for automating your scheduled Spark jobs with sequential and concurrent execution on Cloud Dataproc.

emmylou 1 year, 10 months ago

Selected Answer: C

I thought it might be A but the templates can only run sequentially, not concurrently.

barnac1es 2 years ago

Selected Answer: C

Directed Acyclic Graph (DAG): Cloud Composer (formerly known as Cloud Composer) is a managed Apache Airflow service that allows you to create and manage workflows as DAGs. You can define a DAG that includes tasks for running Spark jobs in sequence or concurrently. Scheduling: Cloud Composer provides built-in scheduling capabilities, allowing you to specify when and how often your DAGs should run. You can schedule the execution of your Spark jobs at specific times or intervals. Dependency Management: In a DAG, you can define dependencies between tasks. This means you can set up tasks to run sequentially or concurrently based on your requirements. For example, you can specify that Job B runs after Job A has completed, or you can schedule jobs to run concurrently when there are no dependencies.

midgoo 2 years, 6 months ago

Selected Answer: C

I would choose A if there was one more step to schedule the Template. It is like creating DAG without running it in Airflow. So only option C is correct here.

AzureDP900 2 years, 9 months ago

C. Create a Directed Acyclic Graph in Cloud Composer

saurabhsingh4k 2 years, 9 months ago

Selected Answer: A

Why go for an expensive Composer when you only have to schedule and create a DAG for Dataproc, A is sufficient.

captainbu 2 years, 9 months ago

I've would've gone for Workflow Templates as well. But those are lacking the scheduling capability. Hence you would need to use Cloud Composer (or Cloud Functions or Cloud Scheduler) anyway. Hence C seems to be the better solution. Pls see here: https://cloud.google.com/dataproc/docs/concepts/workflows/workflow-schedule-solutions

devaid 2 years, 11 months ago

Selected Answer: C

C. Composer fits better to schedule Dataproc Workflows, check the documentation: https://cloud.google.com/dataproc/docs/concepts/workflows/workflow-schedule-solutions Also A is not enough. Dataproc Workflow Template itself don't has a native schedule option.

louisgcpde 2 years, 11 months ago

Selected Answer: C

So that I thing the answer should be C (Composer).

louisgcpde 2 years, 11 months ago

To me, the point is "automate" the process, so that Composer DAG is needed and can be used with Dataproc Workflow Template.

dmzr 2 years, 11 months ago

Selected Answer: A

Ans A makes more sense, since a question is regarding Dataproc jobs only

HarshKothari21 3 years ago

Selected Answer: C

Option c

ducc 3 years ago

Selected Answer: C

You have streaming and batch job, so Composer is the choice for me

https://zhangxuetu.online

[202610]

Question #139 Topic 1

You are building a new data pipeline to share data between two different types of applications: jobs generators and job runners. Your solution must scale to accommodate increases in usage and must accommodate the addition of new applications without negatively affecting the performance of existing ones. What should you do?

A. Create an API using App Engine to receive and send messages to the applications

B. Use a Cloud Pub/Sub topic to publish jobs, and use subscriptions to execute them **Most Voted**

C. Create a table on Cloud SQL, and insert and delete rows with the job information

D. Create a table on Cloud Spanner, and insert and delete rows with the job information

Correct Answer: B

Community vote distribution

B (100%)

Comments

jkhong **Highly Voted** 2 years, 9 months ago

Selected Answer: B

Job generators (they would be the publishers). Job runners = subscribers Question mentions that it must scale (of which push subscription has automatic scaling) and can accommodate additional new applications (this can be solved by having multiple subscriptions, with each relating to a unique application) to a central topic

AzureDP900 2 years, 9 months ago

Yes it is B. Use a Cloud Pub/Sub topic to publish jobs, and use subscriptions to execute them

srivastavas08 1 year, 7 months ago

A. App Engine API: While scalable, it introduces a central point of failure and might not be as performant as Pub/Sub for high-volume data. C. Cloud SQL: Not designed for real-time data sharing and continuous updates, leading to potential bottlenecks and performance issues. D. Cloud Spanner: Offers strong consistency and global distribution, but its pricing model might be less suitable for high-volume, cost-sensitive workloads compared to Pub/Sub.

juliorevk 2 years ago

Selected Answer: B

B to decouple jobs being generated and run. Pub/Sub also scales seamlessly

barnac1es 2 years ago

Selected Answer: B

B. Use a Cloud Pub/Sub topic to publish jobs, and use subscriptions to execute them. Scalability: Cloud Pub/Sub is a highly scalable messaging service that can handle a significant volume of messages and subscribers. It can easily

accommodate increases in usage as your data pipeline scales. Decoupling: Using Pub/Sub decouples the job generators from the job runners, which is a good architectural choice for flexibility and scalability. Job generators publish messages to a topic, and job runners subscribe to that topic to execute jobs when they are available. Adding New Applications: With Cloud Pub/Sub, adding new applications (new publishers or subscribers) is straightforward. You can simply create new publishers to send jobs or new subscribers to consume jobs without impacting existing components.

musumusu 2 years, 7 months ago

key words here: job generators (pushlish message on pub/sub) and job runners(subscribe message for further analysis). You may add as much as pushlishing job and subscribing job to same topic. So Answer B. Using API , app engine is also good approach but its more complex than pub/sub.

Atnafu 2 years, 10 months ago

A Since it's application i will go with

arpitagrawal 3 years ago

Selected Answer: B

use pubsub

YorelNation 3 years ago

Selected Answer: B

I would tend to think B , one of the use of pub/sub is decoupling app

ducc 3 years ago

Selected Answer: B

I choose B

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #140 Topic 1

You need to create a new transaction table in Cloud Spanner that stores product sales data. You are deciding what to use as a primary key. From a performance perspective, which strategy should you choose?

A. The current epoch time

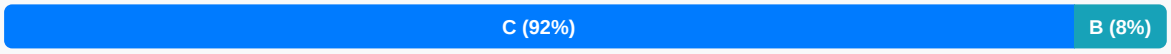
B. A concatenation of the product name and the current epoch time

C. A random universally unique identifier number (version 4 UUID) Most Voted

D. The original order identification number from the sales system, which is a monotonically increasing integer

Correct Answer: C

Community vote distribution



Comments

Remi2021 Highly Voted 3 years ago

Selected Answer: C

According to the documentation: Use a Universally Unique Identifier (UUID) You can use a Universally Unique Identifier (UUID) as defined by RFC 4122 as the primary key. Version 4 UUID is recommended, because it uses random values in the bit sequence. Version 1 UUID stores the timestamp in the high order bits and is not recommended. https://cloud.google.com/spanner/docs/schema-design

AzureDP900 2 years, 9 months ago

Agree with C

barnac1es 2 years ago

Selected Answer: C

For a transaction table in Cloud Spanner that stores product sales data, from a performance perspective, it is generally recommended to choose a primary key that allows for even distribution of data across nodes and minimizes hotspots. Therefore, option C, which suggests using a random universally unique identifier number (version 4 UUID), is the preferred choice.

arien_chen 2 years, 1 month ago

Selected Answer: C

For a RDB I would choice D. But for Google Spanner, Google says: https://cloud.google.com/spanner/docs/schema-and-data-model#:~:text=monotonically%20increasing%20integer

vaga1 2 years, 4 months ago

Selected Answer: C

B might work if you say timestamp instead than epoch. PK of sales should contain the exact purchase date or timestamp, not the time when the transaction was processed. I personally associate the term epoch in this context to the process timestamp instead than the purchase timestamp.

midgoo 2 years, 6 months ago

Selected Answer: C

B may cause error if same product ID came at the same time (same id + same epoch) So C is the correct answer here

NickNtaken 1 year, 4 months ago

Agreed. Additionally, using the product name can lead to unbalanced distribution if some products are sold more frequently than others.

jkhong 2 years, 9 months ago

Selected Answer: C

A and D are invalid because they monotonically increases. B would work, but in terms of pure performance UUID 4 is the fastest because it virtually will not cause hotspots

odacir 2 years, 9 months ago

Selected Answer: C

A and D are not valid, because they monotonically increase. C avoid hotspots for sure, but It's nor relate with querys. So for writing performance it's perfect that the reason for chose this: “You need to create a new transaction table in Cloud Spanner that stores product sales data”. They only ask you to store product data, its a writing ops. If the question had spoken about query the info or hard performance read, the best option would be B, because it has the balance of writing/reading best practices. There are a few disadvantages to using a UUID: They are slightly large, using 16 bytes or more. Other options for primary keys don't use this much storage. They carry no information about the record. For example, a primary key of SingerId and AlbumId has an inherent meaning, while a UUID does not. You lose locality between records that are related, which is why using a UUID eliminates hotspots. https://cloud.google.com/spanner/docs/schema-design#uuid_primary_key

YorelNation 3 years ago

Selected Answer: C

C. A random universally unique identifier number (version 4 UUID) From https://cloud.google.com/spanner/docs/schema-and-data-model There are techniques that can spread the load across multiple servers and avoid hotspots: Hash the key and store it in a column. Use the hash column (or the hash column and the unique key columns together) as the primary key. Swap the order of the columns in the primary key. Use a Universally Unique Identifier (UUID). Version 4 UUID is recommended, because it uses random values in the high-order bits. Don't use a UUID algorithm (such as version 1 UUID) that stores the timestamp in the high order bits. Bit-reverse sequential values.

jsree236 3 years ago

Selected Answer: B

Answer should be B as in all the other options hotspotting is possible. According to proper schema design guideline.. Schema design best practice #1: Do not choose a column whose value monotonically increases or decreases as the first key part for a high write rate table. Supporting link: https://cloud.google.com/spanner/docs/schema-design#primary-key-prevent-hotspots

LP_PDE 1 year, 3 months ago

Potential Skew: If there are a limited number of product names, this could still lead to uneven data distribution and potential hotspots. Increased Key Size: Concatenating strings can result in larger primary keys, which can slightly

impact storage and performance.

https://zhangxuetu.online

[202610]

Question #141 Topic 1

Data Analysts in your company have the Cloud IAM Owner role assigned to them in their projects to allow them to work with multiple GCP products in their projects. Your organization requires that all BigQuery data access logs be retained for 6 months. You need to ensure that only audit personnel in your company can access the data access logs for all projects. What should you do?

A. Enable data access logs in each Data Analyst's project. Restrict access to Stackdriver Logging via Cloud IAM roles.

B. Export the data access logs via a project-level export sink to a Cloud Storage bucket in the Data Analysts' projects. Restrict access to the Cloud Storage bucket.

C. Export the data access logs via a project-level export sink to a Cloud Storage bucket in a newly created projects for audit logs. Restrict access to the project with the exported logs.

D. Export the data access logs via an aggregated export sink to a Cloud Storage bucket in a newly created project for audit logs. Restrict access to the project that contains the exported logs. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

SteelWarrior Highly Voted 5 years ago

Answer D is correct. Aggregated log sink will create a single sink for all projects, the destination can be a google cloud storage, pub/sub topic, bigquery table or a cloud logging bucket. without aggregated sink this will be required to be done for each project individually which will be cumbersome.
https://cloud.google.com/logging/docs/export/aggregated_sinks

AzureDP900 2 years, 9 months ago

D is right

sumanshu Highly Voted 4 years, 2 months ago

A - eliminated , because logs needs to be retained for 6 months (So, some storage require) B - eliminated, because if we store in same project then, Data Analyst can also access (But in question it's mention, ONLY audit personnel needs access) C - Wrong (No need to restrict project as well as logs separately) - wording does not look okay. D - Correct (If we restrict the project, then all resources get restricted) Vote for D

sumanshu 4 years, 2 months ago

Option 'C' - I guess said - restrict access to the project with the exported logs. (i.e. restrict access of that project from where we took logs) - If I am not wrong... Thus it's INCORRECT

at99 3 years, 8 months ago

Sinks are different from Aggregate Sinks, refer https://cloud.google.com/logging/docs/export/configure_export_v2#api

pbtpratik 1 year, 6 months ago

D is the correct ans

barnac1es 2 years ago

Selected Answer: D

D. Here's why this option is recommended: Aggregated Export Sink: By using an aggregated export sink, you can consolidate data access logs from multiple projects into a single location. This simplifies log management and retention policies. Newly Created Project for Audit Logs: Creating a dedicated project for audit logs allows you to centralize access control and manage logs separately from individual Data Analyst projects. Access Restriction: By restricting access to the project containing the exported logs, you ensure that only authorized audit personnel have access to the logs while preventing Data Analysts from accessing them.

midgoo 2 years, 6 months ago

Selected Answer: D

To create the Log Router, at step 3 to define the logs (Source), we can include logs from many projects (aggregated)

dffffff 3 years, 5 months ago

D is correct

MaxNRG 3 years, 8 months ago

Selected Answer: D

D: https://cloud.google.com/logging/docs/export/aggregated_exports You can create an aggregated export sink that can export log entries from all the projects, folders, and billing accounts of an organization. As an example, you might use this feature to export audit log entries from an organization's projects to a central location.

Chelseajcole 4 years ago

The auditor needs to audit data analyst's behaviors (how they access multiple projects in BQ). So, the key is, multiple projects. According to Google doc project-level sinks: https://cloud.google.com/logging/docs/export/configure_export_v2 However, the Cloud Console can only create or view sinks in Cloud projects. To create sinks in organizations, folders, or billing accounts using the gcloud command-line tool or Cloud Logging API, see Aggregated sinks. Obviously, the auditor needs to check all projects accessed by data analyst which is not project-level, a higher level like folder or organization level, this can only be done via the aggregate sink. So D is the answer.

septiandy 4 years, 5 months ago

what is the difference between C and D? I think it's same.

FP77 2 years, 1 month ago

I think the key difference is that D talks about aggregated sinks.

haroldbenites 5 years, 1 month ago

D is correct

saurabh1805 5 years, 1 month ago

D is correct answer, refer below link for more information.

VishalB 5 years, 2 months ago

Ans : D Aggregated Exports, which allows you to set up a sink at the Cloud IAM organization or folder level, and export logs from all the projects inside the organization or folder.

https://zhangxuetu.online

[202610]

Question #142 Topic 1

Each analytics team in your organization is running BigQuery jobs in their own projects. You want to enable each team to monitor slot usage within their projects.
What should you do?

A. Create a Cloud Monitoring dashboard based on the BigQuery metric query/scanned_bytes

B. Create a Cloud Monitoring dashboard based on the BigQuery metric slots/allocated_for_project **Most Voted**

C. Create a log export for each project, capture the BigQuery job execution logs, create a custom metric based on the totalSlotMs, and create a Cloud Monitoring dashboard based on the custom metric

D. Create an aggregated log export at the organization level, capture the BigQuery job execution logs, create a custom metric based on the totalSlotMs, and create a Cloud Monitoring dashboard based on the custom metric

Correct Answer: B

Community vote distribution



Comments

pbtpratik 1 year, 6 months ago
B is correct answer

MaxNRG 1 year, 9 months ago
Selected Answer: B
Viewing project and reservation slot usage in Stackdriver Monitoring Information is available from the "Slots Allocated" metric in Stackdriver Monitoring. This metric information includes a per-reservation and per-job breakdown of slot usage. The information can also be visualized by using the custom charts metric explorer.
https://cloud.google.com/bigquery/docs/reservations-monitoring https://cloud.google.com/monitoring/api/metrics_gcp

keisoes 1 year, 2 months ago
this is stols/allocated, not slots/allocated_for_project

barnac1es 2 years ago
Selected Answer: B
The slots/allocated_for_project metric provides information about the number of slots allocated to each project. It directly reflects the slot usage, making it a relevant and accurate metric for monitoring slot allocation within each

project. Options A, C, and D involve log exports and custom metrics, but they may not be as straightforward or provide the same level of detail as the built-in metric slots/allocated_for_project:

ckanaar 2 years ago

The naming is quite misleading in this case, but it actually seems from the documentation that slots/allocated_for_project indicates the "slots used by project", in which case answer B is correct: https://cloud.google.com/monitoring/api/metrics_gcp#:~:text=slots/allocated_for_project%20GA%0ASlots%20used%20by%20project

arien_chen 2 years, 1 month ago

Selected Answer: D

B slots/allocated_for_project will give you the total number of slots allocated to each project, but it will not tell you how many slots are actually being used. The purpose to monitor 'slot usgae' is for billing. 'slot/allocated' means nothing. Option D is better than B. And, the question mention 'Each analytics team in organization', so it should be 'organization level'.

midgoo 2 years, 6 months ago

Selected Answer: D

If 'usage' = how the slots are being used, D is the corret answer If 'usage' = how the slots are being allocated, B is the correct answer I think in this question, usage = how the slots are being used

musumusu 2 years, 7 months ago

Answer B, Why not D, aggregated log export is good but it will generate all the details which is large in size and costly too. you dont need all the information. It can break data privacy. so look for B because this much is asked only. Normally, i make such errors alot.

saurabhsingh4k 2 years, 9 months ago

Selected Answer: B

The correct answer is B. You should create a Cloud Monitoring dashboard based on the BigQuery metric slots/allocated_for_project. This metric represents the number of BigQuery slots allocated for a project. By creating a Cloud Monitoring dashboard based on this metric, you can monitor the slot usage within each project in your organization. This will allow each team to monitor their own slot usage and ensure that they are not exceeding their allocated quota. Option A is incorrect because the query/scanned_bytes metric represents the number of bytes scanned by BigQuery queries, not the slot usage. Option C is incorrect because it involves creating a log export for each project and using a custom metric based on the totalSlotMs field. While this may be a valid way to monitor slot usage, it is more complex than simply using the slots/allocated_for_project metric. Option D is also incorrect because it involves creating an aggregated log export at the organization level, which is not necessary for monitoring slot usage within individual projects.

dn_mohammed_data 3 years ago

vote for B

John_Pongthorn 3 years ago

B ,the another is related to the question as well. https://cloud.google.com/bigquery/docs/reservations-monitoring#viewing-slot-usage

John_Pongthorn 3 years ago

Selected Answer: B

B the below is related to the question. https://cloud.google.com/blog/topics/developers-practitioners/monitoring-bigquery-reservations-and-slot-utilization-information_schema

AzureDP900 2 years, 9 months ago

B. Create a Cloud Monitoring dashboard based on the BigQuery metric slots/allocated_for_project

https://zhangxuetu.online

[202610]

Question #143 Topic 1

You are operating a streaming Cloud Dataflow pipeline. Your engineers have a new version of the pipeline with a different windowing algorithm and triggering strategy. You want to update the running pipeline with the new version. You want to ensure that no data is lost during the update. What should you do?

A. Update the Cloud Dataflow pipeline inflight by passing the --update option with the --jobName set to the existing job name

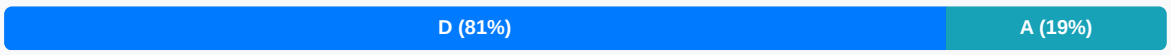
B. Update the Cloud Dataflow pipeline inflight by passing the --update option with the --jobName set to a new unique job name

C. Stop the Cloud Dataflow pipeline with the Cancel option. Create a new Cloud Dataflow job with the updated code

D. Stop the Cloud Dataflow pipeline with the Drain option. Create a new Cloud Dataflow job with the updated code **Most Voted**

Correct Answer: D

Community vote distribution



Comments

odacir **Highly Voted** 2 years, 9 months ago

Selected Answer: D

It's D. → Your engineers have a new version of the pipeline with a different windowing algorithm and triggering strategy. New version is mayor changes. Stop and drain and then launch the new code is a lot is the safer way. We recommend that you attempt only smaller changes to your pipeline's windowing, such as changing the duration of fixed- or sliding-time windows. Making major changes to windowing or triggers, like changing the windowing algorithm, might have unpredictable results on your pipeline output. https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#changing_windowing

maggieee 2 years, 9 months ago

Since updating the job as in A does a compatibility check, wouldn't you want to try that first? Then if the compatibility check fails then you proceed to drain current pipeline and then launch new pipeline (Answer D)? As in A would be correct answer, then if compatibility check fails, you proceed to D.
https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#CCheck

ckanaar 2 years ago

You're right in your reasoning, but since the documentation specifically uses this example for stopping and draining, it's safe to assume that the compatibility check will always fail with these adjustments. Therefore, we can go straight to

D. Furthermore, answer A doesn't state: "Update the Cloud Dataflow pipeline inflight by passing the --update option with the --jobName set to the existing name, if the compatibility check fails, THEN proceed to stopping the pipeline with the drain option", so in itself it is not the right answer if the check fails.

patitonav 1 year, 9 months ago

Selected Answer: D

D seems the right way to go

TVH_Data_Engineer 1 year, 9 months ago

Selected Answer: D

Option A is the first approach to try, as it allows for an in-flight update with minimal disruption. However, if the changes in the new version of the pipeline are not compatible with an in-flight update (due to significant changes in windowing or triggering), then option D should be used. The Drain option ensures a graceful shutdown of the existing pipeline, reducing the risk of data loss, and then a new job can be started with the updated code.

MaxNRG 1 year, 9 months ago

Selected Answer: D

A is not an option as "You want to ensure that no data is lost during the update. ": Making major changes to windowing or triggers, like changing the windowing algorithm, might have unpredictable results on your pipeline output. https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#change_windowing

barnac1es 2 years ago

Selected Answer: D

Drain Option: The "Drain" option allows the existing Dataflow job to complete processing of any in-flight data before stopping the job. This ensures that no data is lost during the transition to the new version. Create a New Job: After draining the existing job, you create a new Cloud Dataflow job with the updated code. This new job starts fresh and continues processing data from where the old job left off. Option A (updating the inflight pipeline with the --update option) may not guarantee no data loss, as the update could disrupt the existing job's operation and potentially cause data loss. Option B (updating the inflight pipeline with the --update option and a new job name) is similar to option A and may not provide data loss guarantees. Option C (stopping the pipeline with the Cancel option and creating a new job) will abruptly stop the existing job without draining, potentially leading to data loss.

knith66 2 years, 2 months ago

Look D after seeing some docs. please check the below link https://cloud.google.com/dataflow/docs/guides/stopping-a-pipeline

vamgcp 2 years, 2 months ago

Selected Answer: D

I will go with option D - If you want to minimize the impact of the update, then option A is the best option. However, if you are not concerned about a temporary interruption in processing, then option D is also a valid option. Option Pros Cons A Does not stop the pipeline, so no data is lost. Requires you to create a new version of the pipeline. B Creates a new job with the updated code, so you do not have to update the running pipeline. Can lead to data loss if the new job does not process all of the data that was in the running pipeline. C Stops the pipeline and drains any data that is currently in flight, so no data is lost. Causes a temporary interruption in processing.

midgoo 2 years, 6 months ago

Selected Answer: D

A is not recommeded for major changes in pipeline.

musumusu 2 years, 7 months ago

Answer A: ```gcloud dataflow jobs update <JOB_ID> --update <GCS_PATH_TO_UPDATED_PIPELINE> --region <REGION>``` --update flag does not miss any data and you can execute this command even yourpipeline is running. Its safe any fast, you can continuously make some change and update this command. no problem. Stop and Drain, is required when you want to test the pipeline and stop it without losing the data.

musumusu 2 years, 7 months ago

Answer D: as per latest documents 02/2023 google has removed update flag.

jkhong 2 years, 9 months ago

Selected Answer: D

agree with odacir

hauhau 2 years, 10 months ago

Selected Answer: A

vote A D: drain doesn't mention about update dataflow job just stop and preserve data A: replace existing job and preserve data (When you update your job, the Dataflow service performs a compatibility check between your currently-running job and your potential replacement job. The compatibility check ensures that things like intermediate state information and buffered data can be transferred from your prior job to your replacement job.) https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline

Atnafu 2 years, 10 months ago

D A-is not because The Dataflow service retains the job name, but runs the replacement job with an updated Job ID. Description: When you update a job on the Dataflow service, you replace the existing job with a new job that runs your updated pipeline code. The Dataflow service retains the job name, but runs the replacement job with an updated Job ID. This process can cause downtime while the existing job stops, the compatibility check runs, and the new job starts.' https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#python:~:text=When%20you%20update%20a,has%20the%20following%20transforms%3A D is correct Drain ->clone -> update -> run

Atnafu 2 years, 10 months ago

Changed my mind to A https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#python_2:~:text=Set%20the%20%2D%2Djob_name,%2D%2Dtransform_name_mapping%20option.

drunk_goat82 2 years, 10 months ago

Selected Answer: D

Changing windowing algorithm may break the pipeline. https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#changing_windowing

ovokpus 2 years, 10 months ago

Selected Answer: A

No, do not drain the current job.

dish11dish 2 years, 10 months ago

Selected Answer: D

in this scenario pipline is streaming pipline with windowing algorithm and triggering strategy changes to new one without loss of data,so better to go with Drain option as it fullfile all precondition described in scenario which is :- 1.streaming 2.code changes with windowing algorithm and triggering strategy to new way 3.no loss of data during update Referances:- https://cloud.google.com/dataflow/docs/guides/stopping-a-pipeline#drain Drain a job. This method applies only to streaming pipelines. Draining a job enables the Dataflow service to finish processing the buffered data while simultaneously ceasing the ingestion of new data. For more information, see Draining a job.

dish11dish 2 years, 10 months ago

If the pipeline was batch then ans would been A

Mcloudgirl 2 years, 10 months ago

D: They want to preserve data and updates might not be predictable.
https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#changing_windowing

cloudmon 2 years, 11 months ago

Selected Answer: A

It's A (https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#UpdateMechanics) D would stop the pipeline, leading to loss of new data that would be sent into the pipeline before you start the new pipeline.

https://zhangxuetu.online

[202610]

Question #144 Topic 1

You need to move 2 PB of historical data from an on-premises storage appliance to Cloud Storage within six months, and your outbound network capacity is constrained to 20 Mb/sec. How should you migrate this data to Cloud Storage?

A. Use Transfer Appliance to copy the data to Cloud Storage **Most Voted**

B. Use gsutil cp ג€"J to compress the content being uploaded to Cloud Storage

C. Create a private URL for the historical data, and then use Storage Transfer Service to copy the data to Cloud Storage

D. Use trickle or ionice along with gsutil cp to limit the amount of bandwidth gsutil utilizes to less than 20 Mb/sec so it does not interfere with the production traffic

Correct Answer: A

Community vote distribution

A (100%)

Comments

sumanshu **Highly Voted** 4 years, 2 months ago

Vote for A A - Correct , Transfer Appliance for moving offline data, large data sets, or data from a source with limited bandwidth https://cloud.google.com/storage-transfer/docs/overview B - Eliminated (Not recommended for large storage). recommended for < 1TB C - Its ONLINE, but we have bandwidth issue - So eliminated. D - Eliminated (Not recommended for large storage). recommended for < 1TB

patitonav 1 year, 9 months ago

Selected Answer: A

A . Easy, just by the amount of data

Nirca 1 year, 11 months ago

Selected Answer: A

In 6 months - only 0.0290304 Petabytes will be uploaded. Right - compression might help, but we do not have any info to support the ration. Lets go for A

barnac1es 2 years ago

Selected Answer: A

Physical Transfer: Transfer Appliance is a physical device provided by Google Cloud that you can use to physically transfer large volumes of data to the cloud. It allows you to avoid the limitations of network bandwidth and transfer data much faster. Capacity: Transfer Appliance can handle large volumes of data, including the 2 PB you need to

migrate, without the constraints of slow network speeds. Efficiency: It is highly efficient for large-scale data transfers and is a practical choice for transferring multi-terabyte or petabyte-scale datasets.

arien_chen 2 years, 1 month ago

it would take 34 years. Option A no doubt. https://cloud.google.com/static/architecture/images/big-data-transfer-how-to-get-started-transfer-size-and-speed.png

vaga1 2 years, 4 months ago

Selected Answer: A

2,000,000,000,000,000 bytes = 2 Petabytes 20,000,000 bytes = 20 Megabytes Once we do the math: 2 Petabytes / 20 Megabytes = 100,000,000 seconds forecasted to migrate the data. 100,000,000 seconds = 1,666,666.7 minutes = 27,777.8 hours = 1,157.4 days 6 months = 180 days 1,157.4 days > 180 days Still, with such amount Transfer Appliance is recommended.

musumusu 2 years, 7 months ago

Transfer alliance is a physical device of size like cabin luggage or slightly larger. It has Seagate/WD harddisk (these are name of companies) size varries from 100 to 480 TB. In our case 2PB (2000 TB) accordingly. Google send you this and transfer data into it by wire connection and then upload data from this to GCS and empty the appliance.

AzureDP900 2 years, 9 months ago

This is no brainer question, A is right

jkhong 2 years, 11 months ago

Selected Answer: A

Problem: Transferring 2 peta data to Cloud Storage Considerations: Bad network speed Bad network = cannot initiate from client's end through network. So, B, C is out D will still be super slow. At this speed it will take 27,777 hours to transfer the data

SteelWarrior 5 years ago

Correct answer is A. with little calculation we know the kind of data will require approx 19 months to transfer on 20Mbps bandwidth. Also, google recommends Transfer appliance for petabytes of data.

haroldbenites 5 years, 1 month ago

A is correct

Rajuuu 5 years, 2 months ago

Correct - A

Rajokkiyam 5 years, 6 months ago

Answer A

https://zhangxuetu.online

[202610]

Question #145 Topic 1

You receive data files in CSV format monthly from a third party. You need to cleanse this data, but every third month the schema of the files changes. Your requirements for implementing these transformations include:
- Executing the transformations on a schedule
- Enabling non-developer analysts to modify transformations
- Providing a graphical tool for designing transformations

What should you do?

A. Use Dataprep by Trifacta to build and maintain the transformation recipes, and execute them on a scheduled basis **Most Voted**

B. Load each month's CSV data into BigQuery, and write a SQL query to transform the data to a standard schema. Merge the transformed tables together with a SQL query

C. Help the analysts write a Dataflow pipeline in Python to perform the transformation. The Python code should be stored in a revision control system and modified as the incoming data's schema changes

D. Use Apache Spark on Dataproc to infer the schema of the CSV file before creating a Dataframe. Then implement the transformations in Spark SQL before writing the data out to Cloud Storage and loading into BigQuery

Correct Answer: A

Community vote distribution

A (100%)

Comments

madhu1171 Highly Voted 5 years, 6 months ago

A should be the answer

JG123 Highly Voted 3 years, 10 months ago

Why there are so many wrong answers? Examtopics.com are you enjoying paid subscription by giving random answers from people? Ans: A

duytran_d 3 years, 2 months ago

this comment is being repeated and i really appreciate this feeling :D

CGS22 1 year, 5 months ago

Selected Answer: A

A. Use Dataprep by Trifacta to build and maintain the transformation recipes, and execute them on a scheduled basis Addresses Requirements: Scheduled Execution: Dataprep supports running transformations on a schedule. Analyst-

Friendly: Dataprep's visual interface is designed for non-developer analysts to build and modify transformations easily. Graphical Tool: It provides a drag-and-drop environment for designing data transformations. Schema Flexibility: Dataprep can handle schema changes. Analysts can adapt recipes using the visual interface

barnac1es 2 years ago

Selected Answer: A

Scheduled Transformations: Dataprep by Trifacta allows you to design and schedule transformation recipes to process data on a regular basis. You can automate the data cleansing process by scheduling it to run monthly. User-Friendly Interface: Dataprep provides a user-friendly graphical interface that enables non-developer analysts to design, modify, and maintain transformation recipes without writing code. This empowers analysts to work with the data effectively. Transformation Flexibility: Dataprep supports flexible data transformations, making it suitable for scenarios where the schema of the incoming data changes. Analysts can adapt the transformations to new schemas using the visual tools provided by Dataprep.

vaga1 2 years, 4 months ago

Selected Answer: A

Providing a graphical tool for designing transformations is enough for A

Dhruv28 2 years, 7 months ago

Your company receives a lot of financial data in CSV files. The files need to be processed, cleaned and transformed before they are made available for analytics. The schema of the data also changes every third month. The Data analysts should be able to perform the tasks 1. No prior knowledge of any language with no coding 2. Provided a GUI tool to build and modify the schema What solution best fits the need?

arpitagrawal 3 years ago

Selected Answer: A

non-developer analysts

devdimidved 3 years, 4 months ago

Selected Answer: A

Dataprep is for non developers

amitsingla012 3 years, 4 months ago

Selected Answer: A

Option A -- Dataprep is the right answer

Prasanna_kumar 3 years, 7 months ago

Answer is A

MaxNRG 3 years, 8 months ago

Selected Answer: A

A: https://cloud.google.com/dataprep/

medeis_jar 3 years, 8 months ago

Selected Answer: A

Cloud Dataprep is a tool to do the job.

sandipk91 4 years, 1 month ago

vote for option A

sumanshu 4 years, 2 months ago

Vote for 'A', because of requirement - Enabling non-developer analysts to modify transformations

haroldbenites 5 years, 1 month ago

A is correct

SSV 5 years, 2 months ago

Answer should be D. Dataprep will detect schema automatically in the initial recipe.After 3 months if the schema is changed the scheduled dataprep job cannot handle. So D should be the option.

Archy 5 years, 1 month ago

spark is not graphical tool.

atnafu2020 5 years, 1 month ago

A you can use dataprep for continuously changing target schema In general, a target consists of the set of information required to define the expected data in a dataset. Often referred to as a "schema," this target schema information can include: Names of columns Order of columns Column data types Data type format Example rows of data A dataset associated with a target is expected to conform to the requirements of the schema. Where there are differences between target schema and dataset schema, a validation indicator (or schema tag) is displayed. https://cloud.google.com/dataprep/docs/html/Overview-of-RapidTarget_136155049

Rajokkiyam 5 years, 6 months ago

Answer A

https://zhangxuetu.online

[202610]

Question #146 Topic 1

You want to migrate an on-premises Hadoop system to Cloud Dataproc. Hive is the primary tool in use, and the data format is Optimized Row Columnar (ORC).
All ORC files have been successfully copied to a Cloud Storage bucket. You need to replicate some data to the cluster's local Hadoop Distributed File System
(HDFS) to maximize performance. What are two ways to start using Hive in Cloud Dataproc? (Choose two.)

A. Run the gsutil utility to transfer all ORC files from the Cloud Storage bucket to HDFS. Mount the Hive tables locally. Most Voted

B. Run the gsutil utility to transfer all ORC files from the Cloud Storage bucket to any node of the Dataproc cluster. Mount the Hive tables locally.

C. Run the gsutil utility to transfer all ORC files from the Cloud Storage bucket to the master node of the Dataproc cluster. Then run the Hadoop utility to copy them do HDFS. Mount the Hive tables from HDFS.

D. Leverage Cloud Storage connector for Hadoop to mount the ORC files as external Hive tables. Replicate external Hive tables to the native ones. Most Voted

E. Load the ORC files into BigQuery. Leverage BigQuery connector for Hadoop to mount the BigQuery tables as external Hive tables. Replicate external Hive tables to the native ones.

Correct Answer: AD

Community vote distribution

AD (42%) CD (25%) DE (19%) BC (14%)

Comments

Sid19 Highly Voted 4 years, 3 months ago

Answer is C and D 100%. I know it says to transfer all the files but with the options provided c is the best choice. Explaination A and B cannot be true as gsutil can copy data to master node and the to hdfs from master node. C -> works D->works Recommended by google E-> Will work but as the question says maximize performance this is not a case. As bigquery hadoop connecter stores all the BQ data to GCS as temp and then processes it to HDFS. As data is already in GCS we donot need to load it to bq and use a connector then unloads it back to GCS and then processes it.

rohan0411 2 years, 9 months ago

You can copy to worker nodes directly too by specyfing the specific flag. hdfs://<master node> is the default filesystem. You can explicitly specify the scheme and NameNode if desired: hdfs dfs -cp gs://<bucket>/<object> hdfs://<master node>/<hdfs path> So the correct answer is A & D

WillemHendr 2 years, 9 months ago

I feel indeed this question is testing if you understand, that gsutil cannot transfer to HDFS directly (eliminate A&B), and need a intermediate step, (making C doable, with a good result). D is found on official google docs. E doesn't have good end result.

KLei 2 years, 5 months ago

must go to the master node first...

Deepakd 4 years, 2 months ago

How can master node store data ? C is wrong.

haroldbenites Highly Voted 5 years, 7 months ago

D , E is correct

shangning007 1 year, 3 months ago

Selected Answer: AD

https://stackoverflow.com/questions/54429642/how-to-copy-a-file-from-a-gcs-bucket-in-dataproc-to-hdfs-using-google-cloud Based on here, you can copy a single file from Google Cloud Storage (GCS) to HDFS using the HDFS copy command. There is no need to copy to the master node first.

SamuelTsch 1 year, 5 months ago

Selected Answer: CD

Actually I think A is correct as well.

patitonav 2 years, 3 months ago

Selected Answer: DE

I think D and E are the best and easy way to go. For sure D, but I think that E can work too, the data can be loaded in BQ as an external table, so at the end the data will be always on the GCS.

barnac1es 2 years, 6 months ago

Selected Answer: DE

D. Cloud Storage Connector for Hadoop: You can use the Cloud Storage connector for Hadoop to mount the ORC files stored in Cloud Storage as external Hive tables. This allows you to query the data without copying it to HDFS. You can replicate these external Hive tables to native Hive tables in Cloud Dataproc if needed. E. Load ORC Files into BigQuery: Another approach is to load the ORC files into BigQuery, Google Cloud's data warehouse. Once the data is in BigQuery, you can use the BigQuery connector for Hadoop to mount the BigQuery tables as external Hive tables in Cloud Dataproc. This leverages the power of BigQuery for analytics and allows you to replicate external Hive tables to native ones in Cloud Dataproc.

barnac1es 2 years, 6 months ago

Selected Answer: DE

D. Cloud Storage Connector for Hadoop: You can use the Cloud Storage connector for Hadoop to mount the ORC files stored in Cloud Storage as external Hive tables. This allows you to query the data without copying it to HDFS. You can replicate these external Hive tables to native Hive tables in Cloud Dataproc if needed. E. Load ORC Files into BigQuery: Another approach is to load the ORC files into BigQuery, Google Cloud's data warehouse. Once the data is in BigQuery, you can use the BigQuery connector for Hadoop to mount the BigQuery tables as external Hive tables in Cloud Dataproc. This leverages the power of BigQuery for analytics and allows you to replicate external Hive tables to native ones in Cloud Dataproc.

vamgcp 2 years, 8 months ago

Selected Answer: AD

A is the most straightforward way to start using Hive in Cloud Dataproc. You can use the gsutil utility to transfer all ORC files from the Cloud Storage bucket to HDFS. Then, you can mount the Hive tables locally. D is another option that you can use to start using Hive in Cloud Dataproc. You can leverage the Cloud Storage connector for Hadoop to mount the ORC files as external Hive tables. Then, you can replicate the external Hive tables to the native ones.

Qix 2 years, 8 months ago

Selected Answer: BC

Answers are; B. Run the gsutil utility to transfer all ORC files from the Cloud Storage bucket to any node of the Dataproc cluster. Mount the Hive tables locally. C. Run the gsutil utility to transfer all ORC files from the Cloud Storage bucket to the master node of the Dataproc cluster. Then run the Hadoop utility to copy them do HDFS. Mount the Hive tables from HDFS. You need to replicate some data to the cluster's local Hadoop Distributed File System (HDFS) to maximize performance. HDFS lies on datanode, data on masternode needs to be copied on datanode. B for managed hive table option, C for external hive table

izekc 2 years, 10 months ago

Selected Answer: AD

AD is correct

Oleksandr0501 2 years, 11 months ago

Selected Answer: AD

i choose AD. Searched in other w/s, read discussions here, and guess better AD.

Oleksandr0501 2 years, 11 months ago

gpt: Yes, that is correct. Option A is a valid way to transfer the ORC files to HDFS, and then mount the Hive tables locally. Option D is also valid, as it suggests using the Cloud Storage connector for Hadoop to mount the ORC files as external Hive tables and then replicating those external Hive tables to native ones. Chatgpt agreed, after inserting quesion and variants, and said that AD are correct answers. And it agreed. It adds some confidence that these are good, but gpt can make mistakes

streeeber 2 years, 11 months ago

Selected Answer: AD

A will copy to HDFS and so will D

hauhau 3 years, 3 months ago

Selected Answer: AD

C: master node doesn't make sense

hauhau 3 years, 3 months ago

B: from the Cloud Storage bucket to any node of the Dataproc cluster -> still on cloud not maxize the speed

tikki_boy 3 years, 5 months ago

I'll go with DE

ducc 3 years, 7 months ago

Selected Answer: CD

CD is correct

BigDataBB 4 years, 1 month ago

Selected Answer: BC

You need to replicate some data to the cluster's local Hadoop Distributed File System (HDFS) to maximize performance. HDFS lies on datanode, data on masternode needs to be copied on datanode. B for managed hive table

option, C for external hive table

medeis_jar 4 years, 2 months ago

Selected Answer: CD

as explained by Sid19

https://zhangxuetu.online

[202610]

Question #147 Topic 1

You are implementing several batch jobs that must be executed on a schedule. These jobs have many interdependent steps that must be executed in a specific order. Portions of the jobs involve executing shell scripts, running Hadoop jobs, and running queries in BigQuery. The jobs are expected to run for many minutes up to several hours. If the steps fail, they must be retried a fixed number of times. Which service should you use to manage the execution of these jobs?

A. Cloud Scheduler

B. Cloud Dataflow

C. Cloud Functions

D. Cloud Composer **Most Voted**

Correct Answer: D

Community vote distribution



Comments

mario_ordinola **Highly Voted** 4 years, 6 months ago

if someone are not sure that D is the answer, I suggest to don't take the exam

madhu1171 **Highly Voted** 5 years, 6 months ago

D should be the answer

patitonav 1 year, 9 months ago

Selected Answer: D

No duobt

barnac1es 2 years ago

Selected Answer: D

Workflow Orchestration: Cloud Composer is a fully managed workflow orchestration service based on Apache Airflow. It allows you to define, schedule, and manage complex workflows with multiple steps, including shell scripts, Hadoop jobs, and BigQuery queries. Dependency Management: You can define dependencies between different steps in your workflow to ensure they are executed in a specific order. Retry Mechanism: Cloud Composer provides built-in retry mechanisms, so if any step fails, it can be retried a fixed number of times according to your configuration. Scheduled Execution: Cloud Composer allows you to schedule the execution of your workflows on a regular basis, meeting the requirement for executing the jobs on a schedule.

AzureDP900 2 years, 9 months ago

D is right

DataEngineer_WideOps 3 years, 2 months ago

Selected Answer: A

Cloud Composer for sure.

nadavw 3 years, 3 months ago

D. per document "Scheduler" is aimed to a single service and composer for an ETL , in addition it's not even specified all jobs are on cloud so only composer can handle it.

nadavw 3 years, 3 months ago

https://cloud.google.com/blog/topics/developers-practitioners/choosing-right-orchestrator-google-cloud

medeis_jar 3 years, 8 months ago

Selected Answer: D

Cloud Composer

JG123 3 years, 10 months ago

Why there are so many wrong answers? Examtopics.com are you enjoying paid subscription by giving random answers from people? Ans: D

daghayeghi 4 years, 7 months ago

D: the main point is that Cloud Composer should be used when there is inter-dependencies between the job, e.g. we need the output of a job to start another whenever the first finished, and use dependencies coming from first job.

ashuchip 4 years, 9 months ago

D seems to be quiet relevant , because using composure you can do all things which are being asked to perform, even retry property is there in composure.

Alasmindas 4 years, 10 months ago

The correct answer is Option A : Cloud Scheduler . Although at first instance, I thought it should be Cloud Composer but then looking at the question and reading it few times - it concluded me to go for Option A. Cloud Scheduler has built in retry handling so you can set a fixed number of times and doesn't have time limits for requests. The functionality is much simpler than Cloud Composer. Cloud Composer is managed Apache Airflow that "helps you create, schedule, monitor and manage workflows. For automate scheduled jobs - the most preferred method would be Scheduler, Composer would typically be used when we want to orchestrate many managed services and automate the work flow.

kavs 4 years, 10 months ago

A seems to be right

mumukshu 4 years, 9 months ago

I think D , how scheduler can handle this part " The jobs are expected to run for many minutes up to several hours"

baubaumiaomiao 3 years, 9 months ago

You forgot the "These jobs have many interdependent steps" which can be handled only though Composer

Abby1356 4 years, 10 months ago

should be A

arghya13 4 years, 11 months ago

Answer should be A..Cloud scheduler..cloud composer is an workflow manager. Can't run unix,bigquery jobs

Tanmoyk 5 years ago

D should be the best option

haroldbenites 5 years, 1 month ago

D is correct

lgdantas 5 years, 2 months ago

D! "Cloud Scheduler is a fully managed enterprise-grade cron job scheduler" https://cloud.google.com/scheduler

https://zhangxuetu.online

[202610]

Question #148 Topic 1

You work for a shipping company that has distribution centers where packages move on delivery lines to route them properly. The company wants to add cameras to the delivery lines to detect and track any visual damage to the packages in transit. You need to create a way to automate the detection of damaged packages and flag them for human review in real time while the packages are in transit. Which solution should you choose?

A. Use BigQuery machine learning to be able to train the model at scale, so you can analyze the packages in batches.

B. Train an AutoML model on your corpus of images, and build an API around that model to integrate with the package tracking applications. **Most Voted**

C. Use the Cloud Vision API to detect for damage, and raise an alert through Cloud Functions. Integrate the package tracking applications with this function.

D. Use TensorFlow to create a model that is trained on your corpus of images. Create a Python notebook in Cloud Datalab that uses this model so you can analyze for damaged packages.

Correct Answer: B

Community vote distribution



Comments

dambilwa **Highly Voted** 5 years, 3 months ago

Option [B] - looks to be correct

nehaxlpb **Highly Voted** 5 years ago

Answer is B. Cloud Vision API detects lot of things for not damages. The description of Damages can be different for each business . So we need to train the model with test and training data to give our definition of Damages, so we need ML capabilities so answer is B, AutoML.

2fbe820 9 months ago

Selected Answer: B

why not C - as per Gemini: The pre-trained Cloud Vision API provides general-purpose image analysis (e.g., object detection for common objects, landmark detection, text detection, safe search). It is not designed or trained to specifically detect "damage" on arbitrary packages. "Damage" is a highly specific and custom visual pattern that would require a custom model. The API would not inherently know what "damage" looks like on a package. Fit: Incorrect tool for custom damage detection.

enthGCP 1 year, 9 months ago

as per chat gpt One of the features of Cloud Vision API is damage detection, which can be used to identify and classify various types of damage in images, such as cracks, dents, scratches, stains, etc

MaxNRG 1 year, 9 months ago

Selected Answer: B

For this scenario, where you need to automate the detection of damaged packages in real time while they are in transit, the most suitable solution among the provided options would be B. Here's why this option is the most appropriate: Real-Time Analysis: AutoML provides the capability to train a custom model specifically tailored to recognize patterns of damage in packages. This model can process images in real-time, which is essential in your scenario. Integration with Existing Systems: By building an API around the AutoML model, you can seamlessly integrate this solution with your existing package tracking applications. This ensures that the system can flag damaged packages for human review efficiently. Customization and Accuracy: Since the model is trained on your specific corpus of images, it can be more accurate in detecting damages relevant to your use case compared to pre-trained models.

MaxNRG 1 year, 9 months ago

Let's briefly consider why the other options are less suitable: A. Use BigQuery machine learning: BigQuery is great for handling large-scale data analytics but is not optimized for real-time image processing or complex image recognition tasks like damage detection on packages. C. Use the Cloud Vision API: While the Cloud Vision API is powerful for general image analysis, it might not be as effective for the specific task of detecting damage on packages, which requires a more customized approach. D. Use TensorFlow in Cloud Datalab: While this is a viable option for creating a custom model, it might be more complex and time-consuming compared to using AutoML. Additionally, setting up a real-time analysis system through a Python notebook might not be as straightforward as an API integration.

juliorevk 1 year, 9 months ago

Selected Answer: B

I was leaning towards C but tested out uploading some damaged boxes to Vision API. It seems to have a lot of trouble detecting damaged boxes. It mislabeled boxes as a tire or toy. Also, there is no part of the API that seems to be able to detect damage. So I'll have to go with B. You should train a model to accomplish this then integrate with your app.

barnac1es 2 years ago

Selected Answer: B

AutoML for Custom Models: AutoML (Auto Machine Learning) is designed to simplify the process of training custom machine learning models, including image classification models. It allows you to leverage Google Cloud's pre-built AutoML Vision service to train a model specifically for detecting package damage based on your corpus of images. This ensures accurate and customized results. Real-time API Integration: After training the AutoML model, you can create an API endpoint that integrates seamlessly with your package tracking applications. This means that as packages move on the delivery lines, you can send images in real-time to the API for immediate analysis. Scalability: AutoML Vision is built to scale, so it can handle the analysis of images in real-time, even as packages move continuously on the delivery lines.

arien_chen 2 years, 1 month ago

Selected Answer: C

Keywords: realtime, camera streaming
https://cloud.google.com/vision#:~:text=where%20you%20are-,Vertex%20AI%20Vision,-Vertex%C2%A0AI%20Vision
Option B AutoML would be too complex and not time efficient. Using Vision AI(Vertex AI Vision) first + AutoML Option D is better than B (just AutoML).

arien_chen 2 years, 1 month ago

typo: Option C is better than B.

piyush7777 2 years, 1 month ago

B https://www.cloudskillsboost.google/focuses/22020?parent=catalog

vamgcp 2 years, 2 months ago

Selected Answer: B

Option B - AutoML

Oleksandr0501 2 years, 4 months ago

will stay with B. Might be more reliable, accurate. as many says in duscission, Vision Api does not say it has defect detection. i remember labs with Auto ML, where models were trained. Vertex AI labs.

AzureDP900 2 years, 9 months ago

B is right

Atnafu 2 years, 10 months ago

B C-is not answer Vision API currently allows you to use the following features: https://cloud.google.com/vision/docs/features-list#:~:text=Vision%20API%20currently%20allows%20you%20to%20use%20the%20following%20features%3A

cloudmon 2 years, 11 months ago

Selected Answer: B

It looks like B is the only valid option, with the assumption that you have a corpus of images (the question does not say that you do not). It would not be Cloud Vision API because that does not do damage detection (https://cloud.google.com/vision/docs/features-list).

John_Pongthorn 3 years ago

https://cloud.google.com/solutions/visual-inspection-ai#all-features

cloudmon 2 years, 11 months ago

This is how you would do it nowadays, but the question is not referring to this solution. It only refers to "Cloud Vision API" (not Visual Inspection API). Cloud Vision API does not do damage detection (https://cloud.google.com/vision/docs/features-list) so you would need to do AutoML. It looks like they assume that you have your own corpus of images.

Deepakd 3 years, 7 months ago

Here it is mentioned that the company is planning to implement a camera system. So it does not have the training data yet. Without having training data , the only option left is to use pre- trained models through cloud API . C Is the answer. B is wrong as you dont have data to train the model.

Deepakd 3 years, 6 months ago

I would correct myself and go for B. I did not find any mention of cloud vision api being used for object detection.

sraakesh95 3 years, 8 months ago

Selected Answer: B

AutoML is used to train model and do damage detection Auto Vision is used is a pre trained model used to detect objects in images

https://zhangxuetu.online

[202610]

Question #149 Topic 1

You are migrating your data warehouse to BigQuery. You have migrated all of your data into tables in a dataset. Multiple users from your organization will be using the data. They should only see certain tables based on their team membership. How should you set user permissions?

A. Assign the users/groups data viewer access at the table level for each table **Most Voted**

B. Create SQL views for each team in the same dataset in which the data resides, and assign the users/groups data viewer access to the SQL views

C. Create authorized views for each team in the same dataset in which the data resides, and assign the users/groups data viewer access to the authorized views

D. Create authorized views for each team in datasets created for each team. Assign the authorized views data viewer access to the dataset in which the data resides. Assign the users/groups data viewer access to the datasets in which the authorized views reside

Correct Answer: A

Community vote distribution



Comments

someshsehgal **Highly Voted** 5 years, 1 month ago

Correct A: A . Now it is feasible to provide table level access to user by allowing user to query single table and no other table will be visible to user in same dataset.

Shiv_am 4 years, 7 months ago

A is not at all possible

squishy_fishy 4 years, 5 months ago

It is possible for about a year now. https://cloud.google.com/bigquery/docs/table-access-controls-intro#example_use_case

alecuba16 3 years, 8 months ago

The problem is that option A has a lot of work for the DevOps, meanwhile option D is easier to manage. The view is like having a shortcut to the same data, but with different permissions

cetanx 2 years, 10 months ago

According to Chat GPT, it is also D. And it explains why it shouldn't be "A" as; Granularity: While you can assign access permissions at the table level, it doesn't allow for fine-grained access control. For example, if you want to restrict access to certain columns or rows within a table based on user or group, table-level permissions would not be

sufficient. Scalability: In organizations with many tables and users, managing permissions at the table level can quickly become unwieldy. You would need to individually set permissions for each user for each table, which can be time-consuming and error-prone. Security: Table-level permissions expose the entire table to a user or a group. If the data in the table changes over time, users might get access to data they shouldn't see. With authorized views, you have more control over what data is exposed. Maintenance: If the structure of your data changes (for instance, if tables are added or removed, or if the schema of a table changes), you would need to manually update the permissions for each affected table.

BigDataBB 1 year, 11 months ago

the request says "team membership", so access depends on the team and not the user

jits1984 4 years, 5 months ago

Should still be D. Question states - "They should only see certain tables based on their team membership" Option A states - Assign the users/groups data viewer access at the table level for each table With A, everyone will see every table. Hence D.

madhu1171 **Highly Voted** 6 years ago

D should be the answer

squishy_fishy 4 years, 5 months ago

There is only one dataset mentioned in the question here. "You have migrated all of your data into tables in a dataset"

ducc 3 years, 7 months ago

It is updated, now A is correct

postbox4643 2 months ago

Selected Answer: D

Why Option D is the Best Fit In BigQuery, simply creating a view isn't enough to grant access to the underlying data. If a user has access to a view but not the source table, the query will fail. An Authorized View solves this by allowing the view itself to "authorize" access to the source data, even if the user doesn't have direct access to those tables.

2fbe820 9 months ago

Selected Answer: D

obviously D. A is not cumbersome and difficult to manage

plum21 1 year, 1 month ago

Selected Answer: D

The question was created at the time when it was not possible to share data on table level (dataset was the only option). At that time D was possible only. Right now A is feasible as well.

LP_PDE 1 year, 3 months ago

Selected Answer: C

Authorized views provide a centralized way to manage access. You define the data each team can see in a view and then grant access to that view. This is much easier to maintain and update than managing permissions on individual tables. Why not D? - Option D suggests creating separate datasets for each team and using authorized views within those datasets. This adds unnecessary complexity and overhead. You would need to manage multiple datasets.You would need to grant the authorized views access to the original dataset.

SamuelTsch 1 year, 5 months ago

Selected Answer: A

Table level access could be done in bigquery.

JamesKarianis 1 year, 7 months ago

Selected Answer: D

Recommended approach

dsyouness 1 year, 10 months ago

Selected Answer: D

Should be D.

MaxNRG 2 years, 3 months ago

Selected Answer: D

https://cloud.google.com/solutions/migration/dw2bq/dw-bq-data-governance When you create the view, it must be created in a dataset separate from the source data queried by the view. Because you can assign access controls only at the dataset level, if the view is created in the same dataset as the source data, your users would have access to both the view and the data. https://cloud.google.com/bigquery/docs/authorized-views This approach aligns with the Google Cloud best practices for data governance, ensuring that users can only access the data intended for them without having direct access to the source tables. Authorized views serve as a secure interface to the underlying data, and by placing these views in separate datasets per team, you can manage permissions effectively at the dataset level.

lokiinaction 2 years, 4 months ago

but the question said that all data are copied into one dataset. so it should be C

spicebits 2 years, 4 months ago

A is the best answer for security as stated in the documentation - https://cloud.google.com/bigquery/docs/row-level-security-intro#comparison_of_authorized_views_row-level_security_and_separate_tables

EsaP 2 years, 6 months ago

A is a better fit than D for this case

barnac1es 2 years, 6 months ago

Selected Answer: C

Authorized Views: Authorized views in BigQuery allow you to control access to specific rows and columns within a table. This means you can create views for each team that restrict access to only the data relevant to that team. Single Dataset: Keeping all the authorized views and the underlying data in the same dataset simplifies management and access control. It avoids the need to create multiple datasets, making the permission management process more straightforward. Option A (assigning data viewer access at the table level) would not provide the granularity you need, as it would allow users to see all tables in the dataset. This does not align with the requirement to restrict access based on team membership.

arien_chen 2 years, 7 months ago

Selected Answer: D

https://cloud.google.com/bigquery/docs/share-access-views#:~:text=the%20source%20data.-,Authorized%20views,-should%20be%20created For best practice, Option D is bettern than others.

midgoo 3 years ago

Selected Answer: A

[A] is correct if it is for individual table However, in practice we normally do [C] as most of the time, the view is a JOIN of a few tables or a subset of the table (some columns removed)

musumusu 3 years, 1 month ago

Answer A, Trick here is, if question is not asking for data level Access such as some rows or columns, don't go for authorized view in that case i would go for C. If it's Table level request only in question, then A is simple answer

https://zhangxuetu.online

[202610]

Question #150 Topic 1

You want to build a managed Hadoop system as your data lake. The data transformation process is composed of a series of Hadoop jobs executed in sequence.
To accomplish the design of separating storage from compute, you decided to use the Cloud Storage connector to store all input data, output data, and intermediary data. However, you noticed that one Hadoop job runs very slowly with Cloud Dataproc, when compared with the on-premises bare-metal Hadoop environment (8-core nodes with 100-GB RAM). Analysis shows that this particular Hadoop job is disk I/O intensive. You want to resolve the issue. What should you do?

A. Allocate sufficient memory to the Hadoop cluster, so that the intermediary data of that particular Hadoop job can be held in memory

B. Allocate sufficient persistent disk space to the Hadoop cluster, and store the intermediate data of that particular Hadoop job on native HDFS **Most Voted**

C. Allocate more CPU cores of the virtual machine instances of the Hadoop cluster so that the networking bandwidth for each instance can scale up

D. Allocate additional network interface card (NIC), and configure link aggregation in the operating system to use the combined throughput when working with Cloud Storage

Correct Answer: B

Community vote distribution

B (88%) A (13%)

Comments

Rajokkiyam **Highly Voted** 5 years, 6 months ago

Answer B Its google recommended approach to use LocalDisk/HDFS to store Intermediate result and use Cloud Storage for initial and final results.

Chelseajcole 4 years ago

Any link to support this recommended approach?

Alasmindas **Highly Voted** 4 years, 10 months ago

Correct Answer is Option B - Adding persistent disk space, reasons:- - The question mentions that the particular job is "disk I/O intensive - so the word "disk" is explicitly mentioned. - Option B also mentions about local HDFS storage, which is ideally a good option of general I/O intensive work. -

MaxNRG 1 year, 9 months ago

Selected Answer: B

Local HDFS storage is a good option if: - You have workloads that involve heavy I/O. For example, you have a lot of partitioned writes such as the following: spark.read().write.partitionBy(...).parquet("gs://") - You have I/O workloads that are especially sensitive to latency. For example, you require single-digit millisecond latency per storage operation. - Your jobs require a lot of metadata operations—for example, you have thousands of partitions and directories, and each file size is relatively small. - You modify the HDFS data frequently or you rename directories. (Cloud Storage objects are immutable, so renaming a directory is an expensive operation because it consists of copying all objects to a new key and deleting them afterwards.) - You heavily use the append operation on HDFS files.

MaxNRG 1 year, 9 months ago

We recommend using Cloud Storage as the initial and final source of data in a big-data pipeline. For example, if a workflow contains five Spark jobs in series, the first job retrieves the initial data from Cloud Storage and then writes shuffle data and intermediate job output to HDFS. The final Spark job writes its results to Cloud Storage. https://cloud.google.com/solutions/migration/hadoop/migrating-apache-spark-jobs-to-cloud-dataproc#choose_storage_options

squishy_fishy 1 year, 11 months ago

The correct answer is B.

barnac1es 2 years ago

Selected Answer: B

Disk I/O Performance: In a Cloud Dataproc cluster, the default setup uses local persistent disks for HDFS storage. These disks offer good disk I/O performance and are well-suited for storing intermediate data generated during Hadoop jobs. Data Locality: Storing intermediate data on native HDFS allows for better data locality. This means that the data is stored on the same nodes where computation occurs, reducing the need for data transfer over the network. This can significantly improve the performance of disk I/O-intensive jobs. Scalability: Cloud Dataproc clusters can be easily scaled up or down to meet the specific requirements of your jobs. You can allocate additional disk space as needed to accommodate the intermediate data generated by this particular Hadoop job.

DeepakVenkatachalam 2 years ago

Correct: A I'd choose A as the doc states adding more SSDs are good for disk-intensive jobs especially those with many individual read and write operations https://cloud.google.com/architecture/hadoop/hadoop-gcp-migration-jobs

DeepakVenkatachalam 2 years ago

Typo Correct Answer is B. . Allocate sufficient persistent disk space to the Hadoop cluster, and store the intermediate data of that particular Hadoop job on native HDFS

arien_chen 2 years, 1 month ago

Selected Answer: A

I would choose A. Google Storage is faster than HDFS in many cases. https://cloud.google.com/architecture/hadoop#:~:text=It%27s%20faster%20than%20HDFS%20in%20many%20cases. The question mention '(8-core nodes with 100-GB RAM)' on-premises Hadoop. the problem may caused by insufficient memory, and does not mention cost would be an issue, so A 'memory' approach would be a better option.

vamgcp 2 years, 2 months ago

Selected Answer: B

Best option is B. However allocating sufficient persistent disk space to the Hadoop cluster, and storing the intermediate data of that particular Hadoop job on native HDFS, would not improve the performance of the Hadoop job. In fact, it might even slow down the Hadoop job, as the data would have to be read and written to disk twice.

John_Pongthorn 3 years ago

Selected Answer: B

https://cloud.google.com/architecture/hadoop/hadoop-gcp-migration-jobs#choosing_primary_disk_options

rrr000 3 years, 1 month ago

B is not the right answer. The problem says that for intermediate data cloud storage is to be used, while B option says: B ... the intermediate data of that particular Hadoop job on native HDFS A is the right answer. If you have enough memory then the shuffle wont spill on the disk.

rrr000 3 years, 1 month ago

Further the question states that original on prem machines has 100gb ram. 8-core nodes with 100-GB RAM

SoerenE 3 years, 8 months ago

B should be the right answer: https://cloud.google.com/compute/docs/disks/performance#optimize_disk_performance

medeis_jar 3 years, 8 months ago

Selected Answer: B

https://cloud.google.com/solutions/migration/hadoop/hadoop-gcp-migration-jobs

JG123 3 years, 10 months ago

Why there are so many wrong answers? Examtopics.com are you enjoying paid subscription by giving random answers from people? Ans: B

RT30 4 years, 6 months ago

If your job is disk-intensive and is executing slowly on individual nodes, you can add more primary disk space. For particularly disk-intensive jobs, especially those with many individual read and write operations, you might be able to improve operation by adding local SSDs. Add enough SSDs to contain all of the space you need for local execution. Your local execution directories are spread across however many SSDs you add. Its B https://cloud.google.com/solutions/migration/hadoop/hadoop-gcp-migration-jobs

ashuchip 4 years, 9 months ago

yes B is correct

haroldbenites 5 years, 1 month ago

B is correct . I/O

Archy 5 years, 1 month ago

B, this job is high on I/O, local HDFS on dish is best option

https://zhangxuetu.online

[202610]

Question #151 Topic 1

You work for an advertising company, and you've developed a Spark ML model to predict click-through rates at advertisement blocks. You've been developing everything at your on-premises data center, and now your company is migrating to Google Cloud. Your data center will be closing soon, so a rapid lift-and-shift migration is necessary. However, the data you've been using will be migrated to migrated to BigQuery. You periodically retrain your Spark ML models, so you need to migrate existing training pipelines to Google Cloud. What should you do?

A. Use Vertex AI for training existing Spark ML models

B. Rewrite your models on TensorFlow, and start using Vertex AI

C. Use Dataproc for training existing Spark ML models, but start reading data directly from BigQuery **Most Voted**

D. Spin up a Spark cluster on Compute Engine, and train Spark ML models on the data exported from BigQuery

Correct Answer: C

Community vote distribution

C (62%) A (38%)

Comments

vaga1 **Highly Voted** 2 years, 10 months ago

Selected Answer: A

the question is: is it faster to move a SparkML job to a Vertex AI or to Dataproc? I am personally not sure, I would go for Dataproc as notebooks are not mentioned, but reading the Google article: https://cloud.google.com/blog/topics/developers-practitioners/announcing-serverless-spark-components-vertex-ai-pipelines/ "Dataproc Serverless components for Vertex AI Pipelines that further simplify MLOps for Spark, Spark SQL, PySpark and Spark jobs."

emmylou 2 years, 4 months ago

But you would need to re-write your models which can be a block

vamgcp **Highly Voted** 2 years, 8 months ago

Selected Answer: C

Option C : It is the most rapid way to migrate your existing training pipelines to Google Cloud. It allows you to continue using your existing Spark ML models. It allows you to take advantage of the scalability and performance of Dataproc. It allows you to read data directly from BigQuery, which is a more efficient way to process large datasets

BigDataBB 1 day, 13 hours ago

Selected Answer: C

While Vertex AI supports Spark, it's a platform more geared toward modern models (TensorFlow, PyTorch) or AutoML. Setting up Spark ML on Vertex often requires more setup work than a pre-configured Dataproc cluster, slowing down the lift-and-shift process.

Anudeep58 1 year, 8 months ago

Selected Answer: C

Vertex AI is better suited for TensorFlow or scikit-learn models. Direct Spark ML support isn't native to Vertex AI, making this a less straightforward migration path.

mothkuri 2 years, 1 month ago

C Question is about rapid lift and shift. So code changes should be minimul

GCP001 2 years, 2 months ago

Selected Answer: C

C looks more suitable as data is alerady on BigQuery. Ref - https://cloud.google.com/dataproc/docs/tutorials/bigquery-sparkml

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C, agreed with other comments

MaxNRG 2 years, 3 months ago

Selected Answer: C

Use Cloud Dataproc, BigQuery, and Apache Spark ML for Machine Learning https://cloud.google.com/dataproc/docs/tutorials/bigquery-sparkml Using Apache Spark with TensorFlow on Google Cloud Platform https://cloud.google.com/blog/products/gcp/using-apache-spark-with-tensorflow-on-google-cloud-platform

Nandababy 2 years, 3 months ago

Why not option D? To spin up the spark cluster on compute engine, considering rapid migration it potentially could be best approach as team wont have to re-work on model (may be only few configurational changes) and again to get data from Bigquery which is required periodically not all the time, could be easy. With Dataproc it would have more code changes eventually can take more time. With Vertex AI it doesn't support spark ML natively and also training would be black box. For me Answer should be D.

barnac1es 2 years, 6 months ago

Selected Answer: C

Dataproc for Spark: Google Cloud Dataproc is a managed Spark and Hadoop service that allows you to run Spark jobs seamlessly on Google Cloud. It provides the flexibility to run Spark jobs using Spark MLlib and other Spark libraries. BigQuery Integration: You mentioned that your data is being migrated to BigQuery. Dataproc has native integration with BigQuery, allowing you to read data directly from BigQuery tables. This eliminates the need to export data from BigQuery to another storage system before processing it with Spark. Rapid Migration: This approach allows you to quickly migrate your existing Spark ML models and training pipelines without the need for a complete rewrite or extensive changes to your existing workflows. You can continue using your Spark ML models while adapting them to read data from BigQuery.

DeepakVenkatachalam 2 years, 6 months ago

they are talking about rapid lift and shift, in which case Dataproc cluster will be right one for Spark ML models for lift and shift. so I think the answer is C.

ckanaar 2 years, 6 months ago

Selected Answer: A

The updated answer seems A based on the following article: https://cloud.google.com/blog/topics/developers-practitioners/announcing-serverless-spark-components-vertex-ai-pipelines/

FP77 2 years, 7 months ago

Selected Answer: C

The answer is C. Spin up a Cloud Dataproc Cluster, migrate spark jobs to there, and link the Cluster to Bgquery with the connector. It's a straightforward solution.

knith66 2 years, 8 months ago

Selected Answer: C

If you wanted to use Vertex AI for training Spark ML models, you would typically need to convert your Spark ML code to another supported machine learning framework like TensorFlow or scikit-learn. Then you could use Vertex AI's pre-built training and prediction services for those frameworks.

wan2three 2 years, 8 months ago

Selected Answer: A

Through Vertex AI Workbench, Vertex AI is natively integrated with BigQuery, Dataproc, and Spark. You can use BigQuery ML to create and execute machine learning models in BigQuery using standard SQL queries on existing business intelligence tools and spreadsheets, or you can export datasets from BigQuery directly into Vertex AI Workbench and run your models from there. https://cloud.google.com/vertex-ai#all-features:~:text=Data%20and%20AI%20integration

blathul 2 years, 9 months ago

Selected Answer: C

Dataproc is a managed Spark and Hadoop service on Google Cloud, which makes it an ideal choice for migrating your existing Spark ML training pipelines. By using Dataproc, you can continue to leverage Spark and its ML capabilities without the need for significant code changes or rewriting your models. By combining Dataproc and BigQuery, you can create Spark jobs or workflows in Dataproc that read data from BigQuery and train your existing Spark ML models. This approach allows you to quickly migrate your training pipelines to Google Cloud and take advantage of the scalability and performance benefits of both Dataproc and BigQuery.

KC_go_reply 2 years, 9 months ago

Selected Answer: C

It is obviously C) Dataproc, since we don't want to rewrite the training from scratch, highly prefer Dataproc for anything Hadoop/Spark ecosystem, and Vertex AI doesn't support *training* with SparkML (but deploying existing models).

https://zhangxuetu.online

[202610]

Question #152 Topic 1

You work for a global shipping company. You want to train a model on 40 TB of data to predict which ships in each geographic region are likely to cause delivery delays on any given day. The model will be based on multiple attributes collected from multiple sources. Telemetry data, including location in GeoJSON format, will be pulled from each ship and loaded every hour. You want to have a dashboard that shows how many and which ships are likely to cause delays within a region. You want to use a storage solution that has native functionality for prediction and geospatial processing. Which storage solution should you use?

A. BigQuery **Most Voted**

B. Cloud Bigtable

C. Cloud Datastore

D. Cloud SQL for PostgreSQL

Correct Answer: A

Community vote distribution

A (100%)

Comments

moumou 1 year, 6 months ago

Selected Answer: A

[Removed] Highly Voted 3 years, 11 months ago Answer: A

mothkuri 1 year, 7 months ago

Answer : A Statement "You want to have a dashboard that shows how many and which ships are likely to cause delays within a region" means we run analytical queries using ML. So BigQuery is Correct answer and it can able to store large volume of data

barnac1es 2 years ago

Selected Answer: A

Here's why BigQuery is a good choice: Scalable Data Storage: BigQuery is a fully managed, highly scalable data warehouse that can handle large volumes of data, including your 40 TB dataset. It allows you to store and manage your data efficiently. SQL for Predictive Analytics: BigQuery supports standard SQL and has built-in machine learning capabilities through BigQuery ML. You can easily build predictive models using SQL queries, which aligns with your goal of predicting ship delays. Geospatial Processing: BigQuery has robust support for geospatial data processing. It provides functions for working with GeoJSON and geospatial data types, making it suitable for your ship telemetry data and geospatial analysis. Integration with Dashboards: BigQuery can be easily integrated with visualization tools

like Google Data Studio or other BI tools. You can create interactive dashboards to monitor ship delays based on your model's predictions.

musumusu 2 years, 7 months ago

Answer B: BigTable, Catchup words: Telemetry (sensor- semi structured data) as data is bigger than 500GB, datastore is not a good option. GEOJSON , bigquery has geospatical capabilites but still not quick enough for semi structure geojson data. Prediction for delay of ships <<likely to>> For me its time crucial and almost real time requirement. BigQuery is not suitable for it. Best solution for this case is: Use BigTable for storage, create a datflow pipeline / google cloud AI platform for time senstive prediction.

musumusu 2 years, 7 months ago

answer A: You are just looking for a storage solution not a workflow

Atnafu 2 years, 10 months ago

A Geospatial analytics let you analyze and visualize geospatial data in BigQuery by using geography data types and Google Standard SQL geography functions. + BigqueryML

JG123 3 years, 10 months ago

Answer: C

Chihhanyu 3 years, 10 months ago

GeoJson + Native functionality for prediction -> BigQuery

singh_payal_1404 3 years, 10 months ago

Answer : A

PM17 3 years, 11 months ago

This is more of a question that an answer but: How much data can Bigquery handle? 40TB seems to be a lot and bigtable can handle that, but of course Bigquery is better when it comes to ML and GIS.

haroldbenites 5 years, 1 month ago

A is correct

FARR 5 years, 1 month ago

A https://cloud.google.com/bigquery/docs/gis-intro

Rajokkiyam 5 years, 6 months ago

Answer A

https://zhangxuetu.online

[202610]

Question #153 Topic 1

You operate an IoT pipeline built around Apache Kafka that normally receives around 5000 messages per second. You want to use Google Cloud Platform to create an alert as soon as the moving average over 1 hour drops below 4000 messages per second. What should you do?

A. Consume the stream of data in Dataflow using Kafka IO. Set a sliding time window of 1 hour every 5 minutes. Compute the average when the window closes, and send an alert if the average is less than 4000 messages. **Most Voted**

B. Consume the stream of data in Dataflow using Kafka IO. Set a fixed time window of 1 hour. Compute the average when the window closes, and send an alert if the average is less than 4000 messages.

C. Use Kafka Connect to link your Kafka message queue to Pub/Sub. Use a Dataflow template to write your messages from Pub/Sub to Bigtable. Use Cloud Scheduler to run a script every hour that counts the number of rows created in Bigtable in the last hour. If that number falls below 4000, send an alert.

D. Use Kafka Connect to link your Kafka message queue to Pub/Sub. Use a Dataflow template to write your messages from Pub/Sub to BigQuery. Use Cloud Scheduler to run a script every five minutes that counts the number of rows created in BigQuery in the last hour. If that number falls below 4000, send an alert.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Alasmindas **Highly Voted** 4 years, 10 months ago

Option A is the correct answer. Reasons:- a) Kafka IO and Dataflow is a valid option for interconnect (needless where Kafka is located - On Prem/Google Cloud/Other cloud) b) Sliding Window will help to calculate average. Option C and D are overkill and complex, considering the scenario in the question, https://cloud.google.com/solutions/processing-messages-from-kafka-hosted-outside-gcp

Rajokkiyam **Highly Voted** 5 years, 6 months ago

Should be A.

mothkuri 1 year, 6 months ago

Selected Answer: A

Option A is correct answer. Option B is not correct. There could be a chance middle of 1st window to middle of 2nd window less messages(i.e > 4000). Option C & D out of scope.

barnac1es 2 years ago

Selected Answer: A

Dataflow with Sliding Time Windows: Dataflow allows you to work with event-time windows, making it suitable for time-series data like incoming IoT messages. Using sliding windows every 5 minutes allows you to compute moving averages efficiently. Sliding Time Window: The sliding time window of 1 hour every 5 minutes enables you to calculate the moving average over the specified time frame. Computing Averages: You can efficiently compute the average when each sliding window closes. This approach ensures that you have real-time visibility into the message rate and can detect deviations from the expected rate. Alerting: When the calculated average drops below 4000 messages per second, you can trigger an alert from within the Dataflow pipeline, sending it to your desired alerting mechanism, such as Cloud Monitoring, Pub/Sub, or another notification service. Scalability: Dataflow can scale automatically based on the incoming data volume, ensuring that you can handle the expected rate of 5000 messages per second.

vamgcp 2 years, 2 months ago

Selected Answer: A

Option A Pros: This option is relatively simple to implement. It can be used to compute the moving average over any time window. Cons: This option can be computationally expensive, especially if the data stream is large. It can be difficult to troubleshoot if the alert does not fire when it is supposed to.

vaga1 2 years, 4 months ago

Selected Answer: A

the correct answer is between A and B since it doesn't make sense to use Pub/Sub combined with Kafka. To have a Moving Average then we should go for A, updating the average estimation every 5 minutes using the new data that came in and eliminating the "most far" 5 minutes.

medeis_jar 3 years, 8 months ago

Selected Answer: A

as explained by Alasmindas

AACHB 3 years, 9 months ago

Selected Answer: A

Correct Answer: A

JG123 3 years, 10 months ago

Correct: A

Chelseajcole 4 years ago

A is enough

daghayeghi 4 years, 7 months ago

A: the correct answer is between A and B, But because used "Moving Average" then we should go for A.

apnu 4 years, 9 months ago

yes , using KafkaIO , we can connect to Kafka cluster.

ashuchip 4 years, 9 months ago

yes A is correct , because sliding window can only help here.

Alasmindas 4 years, 10 months ago

Option A is the correct answer. Reasons:- a) Kafka IO and Dataflow is a valid option for interconnect (needless where Kafka is located - On Prem/Google Cloud/Other cloud) b) Sliding Window will help to calculate average. Option C and D are overkill and complex, considering the scenario in the question,

atnafu2020 5 years, 1 month ago

A To take running averages of data, use hopping windows. You can use one-minute hopping windows with a thirty-second period to compute a one-minute running average every thirty seconds.

Prakzz 5 years, 1 month ago

I don't think its A or B. Dataflow can't connect directly to kafka.

FARR 5 years, 1 month ago

Yes, via KafkaIO. See the link in above comment

kino2020 5 years, 2 months ago

"You operate an IoT pipeline built around Apache Kafka" The statement in question states. Therefore, building with kafka is the requirement definition for this problem. Just in case you are wondering, a case along with this problem is listed on google by the architects. "Using Cloud Dataflow to Process Outside-Hosted Messages from Kafka" https://cloud.google.com/solutions/processing-messages-from-kafka-hosted-outside-gcp Therefore, A is the correct answer.

SPutri 4 years, 10 months ago

the link that you share above is saying, "..illustrates a popular scenario: you use Dataflow to process the messages, where Kafka is hosted either on-premises or in another public cloud such as Amazon Web Services (AWS)." but in this case, we are processing data coming from IoT pipeline, not from on-premise or other cloud. so, i don't think A is the proper solution. I consider option C instead.

https://zhangxuetu.online

[202610]

Question #154 Topic 1

You plan to deploy Cloud SQL using MySQL. You need to ensure high availability in the event of a zone failure. What should you do?

A. Create a Cloud SQL instance in one zone, and create a failover replica in another zone within the same region. **Most Voted**

B. Create a Cloud SQL instance in one zone, and create a read replica in another zone within the same region.

C. Create a Cloud SQL instance in one zone, and configure an external read replica in a zone in a different region.

D. Create a Cloud SQL instance in a region, and configure automatic backup to a Cloud Storage bucket in the same region.

Correct Answer: A

Community vote distribution



Comments

madhu1171 **Highly Voted** 5 years, 6 months ago

A should be correct answer

tycho 3 years, 9 months ago

yes A is correct, whe creating ne cloud sql instance there is an option "Multiple zones (Highly available) Automatic failover to another zone within your selected region. Recommended for production instances. Increases cost."

hdmi_switch **Highly Voted** 4 years, 2 months ago

Seems to depend on the date the question was published. A) is mentioned in legacy config "The legacy configuration for high availability used a failover replica instance." https://cloud.google.com/sql/docs/mysql/configure-legacy-ha B) Read replica is mentioned here https://cloud.google.com/sql/docs/mysql/high-availability scroll down to the diagrams C) external read replica seems to be wrong, never heard of it D) Aligns which Google steps, but automatic backup is not mentioned for HA https://cloud.google.com/sql/docs/mysql/configure-ha#ha-create just regional instance mentioned I would pick D now, since regional instance is described as HA, although the automatic backup seems like a bonus.

mothkuri 1 year, 7 months ago

Answer : A Question is about high availability in the event of zone failure. So create Fail over replica in another zone in same region.

MaxNRG 1 year, 9 months ago

Selected Answer: A

A (failover replicas) as this is an old question: In a legacy HA configuration, a Cloud SQL for MySQL instance uses a failover replica to add high availability to the instance. This functionality isn't available in Google Cloud console. The new configuration doesn't use failover replicas. Instead, it uses Google's regional persistent disks, which synchronously replicate data at the block-level between two zones in a region.
https://cloud.google.com/sql/docs/mysql/configure-legacy-ha

pss111423 1 year, 10 months ago

Option A is good fro leagacy soultion Note: Cloud SQL plans to discontinue support for legacy HA instances in the future and will soon be announcing a date to do so. Currently, legacy HA instances are still covered by the Cloud SQL SLA. We recommend you upgrade your existing legacy HA instances to regional persistent disk HA instances and create new instances using regional persistent disk HA as soon as possible Option C makes more sense in this regrard

emmylou 1 year, 10 months ago

A - Although it is legacy and will be deprecated. The correct answer is not an option-- "The legacy configuration for high availability used a failover replica instance. The new configuration does not use a failover replica. Instead, it uses Google's regional persistent disks, which synchronously replicate data at the block level between two zones in a region."

barnac1es 2 years ago

Selected Answer: A

Failover Replica: By creating a failover replica in another zone within the same region, you establish a high-availability configuration. The failover replica is kept in sync with the primary instance, and it can quickly take over in case of a failure of the primary instance. Same Region: Placing the failover replica in the same region ensures minimal latency and data consistency. In the event of a zone failure, the failover can happen within the same region, reducing potential downtime. Zone Resilience: Google Cloud's regional design ensures that zones within a region are independent of each other, which adds resilience to zone failures. Automatic Failover: In case of a primary instance failure, Cloud SQL will automatically promote the failover replica to become the new primary instance, minimizing downtime.

samstar4180 2 years, 1 month ago

Per latest Google cloud document, B is the correct answer.

wan2three 2 years, 2 months ago

Selected Answer: B

Cross-region read replicas Cross-region replication lets you create a read replica in a different region from the primary instance. You create a cross-region read replica the same way as you create an in-region replica. Cross-region replicas: Improve read performance by making replicas available closer to your application's region. Provide additional disaster recovery capability to guard against a regional failure. Let you migrate data from one region to another.
https://cloud.google.com/sql/docs/mysql/replication#cross-region-read-replicas:~:text=memory%20(OOM)%20events.-,Cross%2Dregion%20read%20replicas,Let%20you%20migrate%20data%20from%20one%20region%20to%20another.,-See%20Promoting%20replicas

MoeHaydar 2 years, 2 months ago

Selected Answer: B

The legacy process for adding high availability to MySQL instances uses a failover replica. The legacy functionality isn't available in the Google Cloud console. See Legacy configuration: Creating a new instance configured for high availability or Legacy configuration: Configuring an existing instance for high availability.

KK0202 2 years, 2 months ago

Selected Answer: B

The correct answer is most probably B as this his scenario has an update(As of July 2023). Failover replicas are not available anymore. Same region different zone read replicas are used in case of a failover or if primary zone is not available

MBRSDG 2 years, 4 months ago

Selected Answer: B

The answer is B, the failover replica is a legacy feature. See here: https://cloud.google.com/sql/docs/mysql/high-availability#legacy_mysql_high_availability_option

forepick 2 years, 4 months ago

Read replica isn't an alternative to the standby instance

vaga1 2 years, 4 months ago

Selected Answer: A

read replica (B) and external read replica (C) doesn't make sense here, since we potentially need all the functionalities. Using Cloud SQL in a region combined with Cloud Storage backup may not be the best choice (D) thinking about compliance reasons starting from what has been asked, it seems also "too much" compared with A that fullfills the request with simpler actions. Also, compliance is required at the regional level, so then A fits.

wjtb 2 years, 6 months ago

Failover replica's are a legacy feature. This question is outdated: https://cloud.google.com/sql/docs/mysql/configure-ha

musumusu 2 years, 7 months ago

Answer A, key words to remember, High Scale use extra read replica. High availablity use extra failure replica. Both should be in different zone but in same region.

desertlotus1211 2 years, 8 months ago

Answer is B: https://cloud.google.com/sql/docs/mysql/replication#read-replicas 'As a best practice, put read replicas in a different zone than the primary instance when you use HA on your primary instance'

desertlotus1211 2 years, 8 months ago

The questions asks to ensure high availability in the event of a zone failure

louisgcpde 2 years, 11 months ago

Selected Answer: A

A should be the answer as the question is asking HA in event of a zone failure. ”Read Replicas CAN be promoted to master nodes in the case of DR. However, there is downtime entailed. Failover Replicas are designed to automatically become master nodes.” https://googlecloudarchitect.us/read-replica-versus-failover-replica-in-cloud-sql/

https://zhangxuetu.online

[202610]

Question #155 Topic 1

Your company is selecting a system to centralize data ingestion and delivery. You are considering messaging and data integration systems to address the requirements. The key requirements are:

- The ability to seek to a particular offset in a topic, possibly back to the start of all data ever captured
- Support for publish/subscribe semantics on hundreds of topics

Retain per-key ordering -

▪

Which system should you choose?

A. Apache Kafka **Most Voted**

B. Cloud Storage

C. Dataflow

D. Firebase Cloud Messaging

Correct Answer: A

Community vote distribution

A (100%)

Comments

YorelNation **Highly Voted** 3 years ago

Selected Answer: A

A I think it's the only technology that met the requirements

dn_mohammed_data **Highly Voted** 3 years ago

vote for A: topics, offsets --> apache kafka

mothkuri 1 year, 6 months ago

Selected Answer: A

Only Kafka can support publish/subscribe semantics on hundreds of topics

barnac1es 2 years ago

Ability to Seek to a Particular Offset: Kafka allows consumers to seek to a specific offset in a topic, enabling you to read data from a specific point, including back to the start of all data ever captured. This is a fundamental capability of Kafka. Support for Publish/Subscribe Semantics: Kafka supports publish/subscribe semantics through topics. You can have hundreds of topics in Kafka, and consumers can subscribe to these topics to receive messages in a publish/subscribe fashion. Retain Per-Key Ordering: Kafka retains the order of messages within a partition. If you have a key associated with your messages, you can ensure per-key ordering by sending messages with the same key

to the same partition. Scalability: Kafka is designed to handle high-throughput data streaming and is capable of scaling to meet your needs. Apache Kafka aligns well with the requirements you've outlined for centralized data ingestion and delivery. It's a robust choice for scenarios that involve data streaming, publish/subscribe, and retaining message ordering.

musumusu 2 years, 7 months ago

Answer A: Apache Kafka Key words: Ingestion and Delivery together (it is combination of pub/sub for ingestion, and delivery = dataflow+any database in gcp) Offset of a topic = Partition of a topic and reprocess specific part of topic, its not possible in pub/sub as it is designed for as come and go for 1 topic. Per key ordering. means message with same key can be process or assigned to a user in kafka.

aquevedos91 3 years ago

deberia ser la C, debido a que siempre es mejor escoger los servicios de google

https://zhangxuetu.online

[202610]

Question #156 Topic 1

You are planning to migrate your current on-premises Apache Hadoop deployment to the cloud. You need to ensure that the deployment is as fault-tolerant and cost-effective as possible for long-running batch jobs. You want to use a managed service. What should you do?

A. Deploy a Dataproc cluster. Use a standard persistent disk and 50% preemptible workers. Store data in Cloud Storage, and change references in scripts from hdfs:// to gs:// **Most Voted**

B. Deploy a Dataproc cluster. Use an SSD persistent disk and 50% preemptible workers. Store data in Cloud Storage, and change references in scripts from hdfs:// to gs://

C. Install Hadoop and Spark on a 10-node Compute Engine instance group with standard instances. Install the Cloud Storage connector, and store the data in Cloud Storage. Change references in scripts from hdfs:// to gs://

D. Install Hadoop and Spark on a 10-node Compute Engine instance group with preemptible instances. Store data in HDFS. Change references in scripts from hdfs:// to gs://

Correct Answer: A

Community vote distribution

A (100%)

Comments

Rajuuu Highly Voted 5 years, 2 months ago

Answer is A…Cloud Dataproc for Managed Cloud native application and HDD for cost-effective solution.

Rajokkiyam Highly Voted 5 years, 6 months ago

Answer A

mothkuri 1 year, 6 months ago

Selected Answer: A

Options A is the right answer. Option B using SSD persistent disk which will add more cost than default HDD Option C & D are out of scope.

barnac1es 2 years ago

Selected Answer: A

Dataproc Managed Service: Dataproc is a fully managed service for running Apache Hadoop and Spark. It provides ease of management and automation. Standard Persistent Disk: Using standard persistent disks for Dataproc workers ensures durability and is cost-effective compared to SSDs. Preemptible Workers: By using 50% preemptible workers, you can significantly reduce costs while maintaining fault tolerance. Preemptible VMs are cheaper but can be

preempted by Google, so having a mix of preemptible and non-preemptible workers provides cost savings with redundancy. Storing Data in Cloud Storage: Storing data in Cloud Storage is highly durable, scalable, and cost-effective. It also makes data accessible to Dataproc clusters, and you can leverage native connectors for reading data from Cloud Storage. Changing References to gs://: Updating your scripts to reference data in Cloud Storage using gs:// ensures that your jobs work seamlessly with the cloud storage infrastructure.

vaga1 2 years, 4 months ago

Selected Answer: A

Apache Hadoop -> Dataproc or Compute Engine with proper SW installation cost-effective -> use standard persistent disk + store data in Cloud Storage batch -> Dataproc or Compute Engine with proper SW installation managed service -> Dataproc

MounicaN 3 years ago

Selected Answer: A

it says cost effective , hence no SSD

JG123 3 years, 10 months ago

Correct: A

LORETOGOMEZ 4 years, 2 months ago

Correct : A Option B is usefull if you use HDFS, and in this case as you use preemtible machines it isn't worth use SSD disks.

ArunSingh1028 4 years, 7 months ago

Answer - B

StelSen 4 years, 8 months ago

Look at this link. https://cloud.google.com/bigtable/docs/choosing-ssd-hdd At the First look I chose Option-B as they mentioned SSD is cost-effective on most cases. But after reading the whole page, they also mentioned that for batch workloads, HDD is suggested as long as not heavy read. So I changed my mind to Option-A (I assumed this is not ready heavy process?).

NM1212 3 years, 1 month ago

Caution about the link you provided as reference. It's intendedfor BigTable which is GC's low-latency solution which is totally different requirement. Mentioning only because on first read I thought SSD is the obvious choice. Per below link, SSD may not be required unless there is a low-latency requirement or a high I/O requirement. Since the question does not specify anything like that, A looks correct. https://cloud.google.com/solutions/migration/hadoop/migrating-apache-spark-jobs-to-cloud-dataproc

Alasmindas 4 years, 10 months ago

Option B - SSD disks, reasons:- The question asks "fault-tolerant and cost-effective as possible for long-running batch job". 3 Key words are - fault tolerant / cost effective / long running batch jobs.. The cost efficiency part mentioned in the question could be addressed by 50% preemptible disks and storing the data in cloud storage than HDFS. For long running batch jobs and as standard approach for Dataproc - we should always go with SSD disk types as per google recommendations.

beedle 4 years, 9 months ago

where is the proof...show me the link?

Raangs 4 years, 6 months ago

https://cloud.google.com/solutions/migration/hadoop/migrating-apache-spark-jobs-to-cloud-dataproc As per this, SSD is only recommended if it is high IO intensive. In this question no where mentioned its high IO intensive, and asks for

cost effective (as much as possible), so no need to use SSD. I will go with A.

Ravivarma4786 5 years, 1 month ago

Ans is B, for long running SDD suitable. HDD maintenance will be additional charge for long running jobs

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #157 Topic 1

Your team is working on a binary classification problem. You have trained a support vector machine (SVM) classifier with default parameters, and received an area under the Curve (AUC) of 0.87 on the validation set. You want to increase the AUC of the model. What should you do?

A. Perform hyperparameter tuning **Most Voted**

B. Train a classifier with deep neural networks, because neural networks would always beat SVMs

C. Deploy the model and measure the real-world AUC; it's always higher because of generalization

D. Scale predictions you get out of the model (tune a scaling factor as a hyperparameter) in order to get the highest AUC

Correct Answer: A

Community vote distribution



Comments

aadaisme **Highly Voted** 5 years, 3 months ago

Seems to be A. Preprocessing/scaling should be done with input features, instead of predictions (output)

FARR **Highly Voted** 5 years, 1 month ago

A Deep LEarning is not always the best solution D talks about fudgin the output which is wrong

MaxNRG 1 year, 9 months ago

Selected Answer: A

https://www.quora.com/How-can-I-improve-Precision-Recall-AUC-under-Imbalanced-Classification

vaga1 2 years, 4 months ago

Selected Answer: A

B,C are simply not true. D is modifing the scoring, making it not realiable anymore. A makes sense, is potentially increasing the model accuracy.

rishu2 2 years, 4 months ago

Selected Answer: A

a is the correct answer

musumusu 2 years, 7 months ago

Answer A, why not B, Deep Neu Net. are better for sure but AUC is 0.87 is already good. Don't go for complex and time taking model. If AUC more than 0.95, it can be a reason of overfit. Now just check SVM params for hypertuning if

you can bring it close to 0,9-0,95

Kvk117 2 years, 8 months ago

Selected Answer: A

a is the correct answer

Dan137 2 years, 11 months ago

Also a good read is: https://cloud.google.com/ai-platform/training/docs/hyperparameter-tuning-overview

medeis_jar 3 years, 8 months ago

Selected Answer: A

as mentioned by Spider7 "performing tuning rather than using the model default parameters there's a way to increase the overall model performance --> A."

JG123 3 years, 10 months ago

Correct: A

Spider7 3 years, 10 months ago

0.89 it's already not bad but by performing tuning rather then using the model default parameters there's a way to increase the overall model performance --> A.

Spider7 3 years, 10 months ago

0.87 precisely

hdmi_switch 4 years, 2 months ago

Not C because real-world AUC value falls between 0.5 and 1.0 usually, this wouldn't help. A seems the most straigh forward.

Mitra123 4 years, 6 months ago

For a large enough training set DNN will most likely beat a SVM. However the opposite may or may not be true. It also depends on the complexity of the problem. Which we don’t know from the question. For image, nlp, I say B can be a good answer However, if we decide to stick with SVM, D reduces overfitting and may increase AUC. I am torn between the two!

ArunSingh1028 4 years, 7 months ago

Ans - D when the model is overfitted means want to increase the AUC, we always perform hyperparameter tuning, Increase regularisations, decrease input feature parameters etc.

nitinbhatia 4 years, 11 months ago

AUC is scale-invariant. It measures how well predictions are ranked, rather than their absolute values. So answer shall be A https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc?hl=en

arghya13 4 years, 11 months ago

Definitely not D https://developers.google.com/machine-learning/crash-course/classification/check-your-understanding-roc-and-auc

saurabh1805 5 years, 1 month ago

A for me, read below link for more details. https://towardsdatascience.com/understanding-hyperparameters-and-its-optimisation-techniques-f0debba07568

https://zhangxuetu.online

[202610]

Question #158 Topic 1

You need to deploy additional dependencies to all nodes of a Cloud Dataproc cluster at startup using an existing initialization action. Company security policies require that Cloud Dataproc nodes do not have access to the Internet so public initialization actions cannot fetch resources. What should you do?

A. Deploy the Cloud SQL Proxy on the Cloud Dataproc master

B. Use an SSH tunnel to give the Cloud Dataproc cluster access to the Internet

C. Copy all dependencies to a Cloud Storage bucket within your VPC security perimeter **Most Voted**

D. Use Resource Manager to add the service account used by the Cloud Dataproc cluster to the Network User role

Correct Answer: C

Community vote distribution

C (100%)

Comments

[Removed] **Highly Voted** 5 years, 6 months ago

Correct: C If you create a Dataproc cluster with internal IP addresses only, attempts to access the Internet in an initialization action will fail unless you have configured routes to direct the traffic through a NAT or a VPN gateway. Without access to the Internet, you can enable Private Google Access, and place job dependencies in Cloud Storage; cluster nodes can download the dependencies from Cloud Storage from internal IPs.

AzureDP900 2 years, 9 months ago

Thank you for detailed explanation. C is right

rickywck **Highly Voted** 5 years, 6 months ago

Should be C: https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/init-actions

b3e59c2 1 year, 2 months ago

Selected Answer: C

A: Incorrect, Proxy allows for connection to a Cloud SQL instance, which unless you have the dependencies stored there (doesn't seem viable or smart), would achieve nothing. B: Incorrect, will allow for a connection to the internet to be made for installing the dependencies, however this goes against the companies security policies so should not be considered. C: Correct, only one that makes sense and is best practise (see https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/init-actions) D: Incorrect, provides access to a shared VPC network, however doesn't necessarily provide a way to access the dependencies. And even if it did, would go against company security policy.

gcpdataeng 1 year, 6 months ago

Selected Answer: C

c looks good

barnac1es 2 years ago

Selected Answer: C

Security Compliance: This option aligns with your company's security policies, which prohibit public Internet access from Cloud Dataproc nodes. Placing the dependencies in a Cloud Storage bucket within your VPC security perimeter ensures that the data remains within your private network. VPC Security: By placing the dependencies within your VPC security perimeter, you maintain control over network access and can restrict access to the necessary nodes only. Dataproc Initialization Action: You can use a custom initialization action or script to fetch and install the dependencies from the secure Cloud Storage bucket to the Dataproc cluster nodes during startup. By copying the dependencies to a secure Cloud Storage bucket and using an initialization action to install them on the Dataproc nodes, you can meet your security requirements while providing the necessary dependencies to your cluster.

knith66 2 years, 1 month ago

Selected Answer: C

C is correct

charline 2 years, 6 months ago

Selected Answer: C

C seems good

musumusu 2 years, 7 months ago

Answer C, It needs practical experience to understand this question. You create cluster with some package/software i.e dependencies such as python packages that you store in .zip file, then you save a jar file to run the cluster as an application such as you need java while running spark session. and some config yaml file. These dependencies you can save in bucket and can use to configure cluster from external window , sdk or api. without going into UI. Then you need to use VPC to access these files

DataEngineer_WideOps 3 years, 2 months ago

Without access to the internet, you can enable Private Google Access and place job dependencies in Cloud Storage; cluster nodes can download the dependencies from Cloud Storage from internal IPs.

medeis_jar 3 years, 8 months ago

Selected Answer: C

https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/network#create_a_cloud_dataproc_cluster_with_internal_ip_address_only

Prabusankar 3 years, 9 months ago

When creating a Dataproc cluster, you can specify initialization actions in executables or scripts that Dataproc will run on all nodes in your Dataproc cluster immediately after the cluster is set up. Initialization actions often set up job dependencies, such as installing Python packages, so that jobs can be submitted to the cluster without having to install dependencies when the jobs are run

JG123 3 years, 10 months ago

Correct: C

clouditis 5 years, 1 month ago

c it is!

Rajokkiyam 5 years, 6 months ago

Should be C

jvg637 5 years, 6 months ago

I think the correct answer might be C instead, due to https://cloud.google.com/dataproc/docs/concepts/configuring-clusters/network#create_a_cloud_dataproc_cluster_with_internal_ip_address_only

https://zhangxuetu.online

[202610]

Question #159 Topic 1

You need to choose a database for a new project that has the following requirements:
- Fully managed
- Able to automatically scale up
- Transactionally consistent
- Able to scale up to 6 TB
- Able to be queried using SQL

Which database do you choose?

A. Cloud SQL

B. Cloud Bigtable

C. Cloud Spanner **Most Voted**

D. Cloud Datastore

Correct Answer: C

Community vote distribution



Comments

[Removed] **Highly Voted** 6 years ago

Correct: A It asks for scaling up which can be done in cloud sql, horizontal scaling is not possible in cloud sql Automatic storage increase If you enable this setting, Cloud SQL checks your available storage every 30 seconds. If the available storage falls below a threshold size, Cloud SQL automatically adds additional storage capacity. If the available storage repeatedly falls below the threshold size, Cloud SQL continues to add storage until it reaches the maximum of 30 TB.

google_learner123 5 years, 7 months ago

C - CloudSQL does not scale automatically.

zxing233 5 years, 6 months ago

Cloud SQL can automatically scale up storage capacity when you are near your limit

dmzr 3 years, 5 months ago

it does not say about type of scaling, Cloud SQL scale up automatically with storage, that should works

gcp_k **Highly Voted** 4 years, 5 months ago

Answer could be : A Both Cloud Spanner and Cloud SQL does not have auto scaling feature natively. You need to build automation around it based on metrics. Both Cloud SQL and Cloud Spanner supports SQL. With Cloud SQL,

you can go up to 10 TB of storage which also satisfies the other requirement. Consistency - Of course, with Cloud SQL, you have single master and read replicas. So technically the data will be consistent across all instances so to speak. The reason why I didn't choose Spanner is - There is no requirement for HA DR, multi region, secondary indexes, etc.. So I choose "A"

BigDataBB 1 day, 13 hours ago

Selected Answer: C

Cloud SQL: Can automatically increase storage (disk auto-resize), but cannot automatically scale compute resources (CPU/RAM) based on load without intervention or short downtime (for instance switching).

lmch 3 months, 2 weeks ago

Selected Answer: C

Cloud Spanner can automatically scale in both storage and compute.

arnauredi 8 months, 1 week ago

Selected Answer: A

Cloud SQL can scale up to 64To

Ben_oso 9 months, 1 week ago

Selected Answer: C

Correct: C - Cloud Spanner hint is automatically scale, CloudSQL dont scale

MBNR 1 year, 1 month ago

Selected Answer: C

Cloud SQL Cloud SQL can store up to 30 TB of data. It offers limited scalability as per the lesser load. You can easily work with MySQL code in Cloud SQL. Cloud SQL is a cost-effective service. Cloud Spanner: Cloud Spanner is used to store more than 30 TB of data. It provides better scalability and SLOs. Cloud Spanner is an expensive service. It provides strong transactional consistency. It is built on Google Cloud's dedicated network that ensures low latency, security, and reliability.

Siahara 1 year, 1 month ago

Selected Answer: C

C. Cloud Spanner and an explanation of the other options: Why Cloud Spanner is the best fit: Fully managed service: Cloud Spanner is fully managed by Google, simplifying database administration. Automatic scaling It handles scaling seamlessly, both horizontally and vertically. Transactional consistency: Cloud Spanner is known for its strong transactional consistency, including globally distributed transactions. Scalable up to 6 TB (and beyond): It easily accommodates your 6 TB requirement and can scale much larger if needed. SQL Support: Cloud Spanner offers a familiar SQL interface.

b3e59c2 1 year, 2 months ago

Selected Answer: C

Although the type of automatic scaling isn't specified, Cloud SQL does not allow for outright dynamic capacity auto scaling, so I believe the answer would be C

Rav761 1 year, 3 months ago

Selected Answer: C

C. Cloud Spanner Here's why: Fully Managed: Cloud Spanner is a fully managed database service provided by Google Cloud. Automatic Scaling: It automatically scales horizontally to handle increased workloads and data volumes. Transactional Consistency: Cloud Spanner provides strong transactional consistency with support for ACID transactions. Scalability: It can easily scale up to and beyond 6 TB while maintaining performance and consistency.

SQL Queries: Cloud Spanner supports SQL queries, making it compatible with existing SQL-based analytics and applications.

SamuelTsch 1 year, 5 months ago

Selected Answer: C

I would go to C.

baimus 1 year, 6 months ago

Selected Answer: C

All the arguments below can essentially be boiled down to two questions. 1) is Cloud SQL fully managed? (yes), 2) Does it autoscale? It depends, is the answer. The question is horrifyingly worded, as it comes down to an ambiguity coinflip. I'm going with C, it feels like it's a better fit.

JamesKarianis 1 year, 7 months ago

Selected Answer: A

Did not refer to global, thus not spanner. A is correct

Anudeep58 1 year, 8 months ago

Selected Answer: C

Transactional consistency: spanner provides strong consistency across rows, regions, and continents.

mothkuri 2 years, 1 month ago

Answer : A Cloud SQL and Cloud Spanner are the options for the questions. But here as per requirement they don't need horizontal scaling, they want manager SQL instance and it should support 6 TB of storage. Cloud SQL can support up to 64 TB of storage https://cloud.google.com/sql/docs/quotas#:~:text=Cloud%20SQL%20storage%20limits,core%3A%20Up%20to%203%20TB.

cuadradobertolinisebastiancami 2 years, 1 month ago

Selected Answer: A

Not horizontal scaling is required, cloud SQL will work for 10 TB

williamvinct 2 years, 1 month ago

i will go with A, since the question doesnt specifically say it should be scaling horizontally

https://zhangxuetu.online

[202610]

Question #160 Topic 1

You work for a mid-sized enterprise that needs to move its operational system transaction data from an on-premises database to GCP. The database is about 20
TB in size. Which database should you choose?

A. Cloud SQL **Most Voted**

B. Cloud Bigtable

C. Cloud Spanner

D. Cloud Datastore

Correct Answer: A

Community vote distribution



Comments

jvg637 **Highly Voted** 6 years ago
A. Cloud SQL (30TB)

Gcpyspark 5 years, 3 months ago
Sure, however in future if the capacity grows beyond 30 TB then Cloud SQL won't work right then Spanner would be the option?

desertlotus1211 3 years, 2 months ago
you can always call GCP to add quota.. .Spanner is for global reach, ideally...

vindahake 5 years, 10 months ago
Up to 30,720 GB, depending on the machine type. This looks like correct choice.
https://cloud.google.com/sql/docs/quotas#fixed-limits

odacir 3 years, 3 months ago
https://cloud.google.com/sql/docs/quotas#storage_limits 64TB

Satishjuly18 1 year, 7 months ago
65 TB now in Aug 2024

Satishjuly18 1 year, 7 months ago
*64 TB

dagoat 4 years, 6 months ago

65 TB now in Sept 2021

Rajuuu Highly Voted 5 years, 8 months ago

A as limit is now 30 TB for Cloud SQL

imrane1995 8 months, 3 weeks ago

Selected Answer: C

C. Cloud Spanner Breakdown of Requirements: 20 TB transactional data → A large-scale, transactional, and reliable database is needed. From an operational system → Implies high consistency, availability, and performance. Moving to GCP → Should use a cloud-native, fully managed GCP service.

SamuelTsch 1 year, 5 months ago

Selected Answer: A

Cloud SQL storage limit: dedicated core up to 64 TB.

drpay 2 years, 5 months ago

Selected Answer: C

two keywords: Transactional data, 20 TB

barnac1es 2 years, 6 months ago

Selected Answer: C

Scalability: Cloud Spanner is designed to handle large volumes of data, making it suitable for a 20 TB database. It can scale horizontally and vertically to accommodate growing data needs. Global Distribution: Cloud Spanner allows you to distribute data globally for low-latency access across regions, which can be advantageous for operational systems. Strong Consistency: It provides strong transactional consistency, which is important for operational systems that require ACID compliance. SQL Support: Cloud Spanner supports SQL, which is a familiar query language for developers. While Cloud SQL, Cloud Bigtable, and Cloud Datastore have their use cases, Cloud Spanner is better suited for larger databases with strong consistency requirements, making it a suitable choice for migrating a 20 TB operational system database to GCP.

ashu381 2 years, 6 months ago

Cloud SQL, upto 64 TB now, you can always call GCP for increasing the quota though !!

vaga1 2 years, 9 months ago

Selected Answer: A

Cloud SQL is generally better for OLTP, and Cloud SQL is up to 64 TB now.
https://cloud.google.com/sql/docs/quotas#storage_limits

vaga1 2 years, 10 months ago

"move its operational system transaction data from an on-premises database to GCP". Cloud SQL may be plug-and-play

musumusu 3 years, 1 month ago

Not 100% in favour of A, Should i recommend my client Cloud SQL, when they are coming to me with 20TB already 30TB is limit, its transactional data, which i can't compromise. I will propose cloud spanner. There is nothing mentioned that they want to save cost.

AzureDP900 3 years, 3 months ago

A. Cloud SQL

Jay_Krish 3 years, 4 months ago

Selected Answer: A

With the given requirements A. Cloud SQL is more than sufficient. Don't try to overthink scenarios like what if it grows.. what if there's additional requirement in future.. what if this what if that.. just look at the question and see the stated requirement. If there are more than one answer try to see which is simple and doesn't come with extra frills.

Atnafu 3 years, 4 months ago

A 65 TB now in Nov 2022

WZH 3 years, 8 months ago

it is already 20 TB at the moment, and you probably want to change the database because the capacity of your current storage solution is not enough. Then you decide to change it to Cloud SQL(up to 30 TB) which may not increase much capacity? I am not sure about the answer but A looks weird imho.

Dan226 3 years, 8 months ago

Cloud SQL can store 64 Tb, but in the intial set up the operation are 20tb. It will reach the limitation soon if you choose Cloud SQL

gcp_k 4 years, 5 months ago

Depends.. I mean, C is correct if the exam is not updated. A is correct if the exam is updated. So ... kinda in catch 22 situation ...

KokkiKumar 4 years, 5 months ago

Hi everyone, Can i purchase this exam? is it worthable?

https://zhangxuetu.online

[202610]

Question #161 Topic 1

You need to choose a database to store time series CPU and memory usage for millions of computers. You need to store this data in one-second interval samples. Analysts will be performing real-time, ad hoc analytics against the database. You want to avoid being charged for every query executed and ensure that the schema design will allow for future growth of the dataset. Which database and data model should you choose?

A. Create a table in BigQuery, and append the new samples for CPU and memory to the table

B. Create a wide table in BigQuery, create a column for the sample value at each second, and update the row with the interval for each second

C. Create a narrow table in Bigtable with a row key that combines the Computer Engine computer identifier with the sample time at each second **Most Voted**

D. Create a wide table in Bigtable with a row key that combines the computer identifier with the sample time at each minute, and combine the values for each second as column data.

Correct Answer: C

Community vote distribution



Comments

psu **Highly Voted** 5 years, 11 months ago

Answer C A tall and narrow table has a small number of events per row, which could be just one event, whereas a short and wide table has a large number of events per row. As explained in a moment, tall and narrow tables are best suited for time-series data. For time series, you should generally use tall and narrow tables. This is for two reasons: Storing one event per row makes it easier to run queries against your data. Storing many events per row makes it more likely that the total row size will exceed the recommended maximum (see Rows can be big but are not infinite). https://cloud.google.com/bigtable/docs/schema-design-time-series#patterns_for_row_key_design

AzureDP900 3 years, 3 months ago

C. Create a narrow table in Bigtable with a row key that combines the Computer Engine computer identifier with the sample time at each second

nadavw 3 years, 9 months ago

there is a limit of 60 columns per row according to question. in addition in D the cost will be a lower which is a requirement. so D seems more suitable.

madhu1171 **Highly Voted** 6 years ago

C correct answer

shangning007 1 year, 3 months ago

Selected Answer: C

Even though I am sure the answer is C, but I am not sure why it will help avoid being charged for every query?

SamuelTsch 1 year, 5 months ago

Selected Answer: C

just like psu said. Additionally for a time-series data, it it usually the best practice to combine the identifier + time, not the values.

mothkuri 2 years ago

Selected Answer: C

Option C is correct answer. Narrow table is good for time series data.

barnac1es 2 years, 6 months ago

Selected Answer: C

Scalability: Bigtable can handle large-scale data efficiently, making it suitable for storing time series data for millions of computers. Low Latency: Bigtable provides low-latency access to data, which is crucial for real-time analytics. Flexible Schema: The narrow table design allows you to efficiently store and query time series data without specifying all possible columns in advance, providing flexibility for future growth. Column Families: Bigtable supports column families, allowing you to organize data logically. Row Key Design: Combining the computer identifier with the sample time at each second in the row key allows for efficient retrieval of data for specific computers and time intervals. Analytics: While Bigtable does not support SQL directly, it allows for efficient data retrieval and can be integrated with other tools for analytics.

WillemHendr 2 years, 9 months ago

Selected Answer: D

"..and ensure that the schema design will allow for future growth of the dataset": https://cloud.google.com/bigtable/docs/schema-design-time-series#time-buckets "Data stored in this way is compressed more efficiently than data in tall, narrow tables." I read the "future growth" as a sign to be effective in storage, and go for the Time-Buckets.

Remi2021 3 years, 6 months ago

Selected Answer: C

time series = narrow table

_8008_ 3 years, 11 months ago

What about "avoid being charged for every query executed"? Nothing on this topic in here https://cloud.google.com/bigtable/docs/schema-design-time-series can anyone comment?

medeis_jar 4 years, 2 months ago

Selected Answer: C

Narrow and tall table for a single event and good for time-series data Short and Wide table for data over a month, multiple events

JG123 4 years, 4 months ago

Correct: C

squishy_fishy 4 years, 5 months ago

Answer is C. Bigtable is best suited to the following scenarios: time-series data (e.g. CPU and memory usage over time for multiple servers), financial data (e.g. transaction histories, stock prices, and currency exchange rates), and IoT (Internet of Things) use cases. https://www.xplenty.com/blog/bigtable-vs-bigquery/

safiyu 4 years, 7 months ago

C is the correct answer. If you consider wide table, then 60 columns for cpu usage and 60 columns for memory usage. in future, if you need to add a new kpi to the table, then the schema changes. you will have to add 60 more columns for the new feature. this is not so future proof.. so D is out of the picture.

DeepakS227 4 years, 8 months ago

BQ is optimized for large-scale, ad-hoc SQL-based analysis. i Think it should be A

koupayio 4 years, 9 months ago

First C & D won't cause hotspoting as computer_identifier is first part of row key I prefer D because "ensure that the schema design will allow for future growth of the dataset." C is too tall and narrow I cannot see schema design grow in the future

crslake 4 years, 10 months ago

D, Better overall, harder to implement (but that is not stated as a constraint)
https://cloud.google.com/bigtable/docs/schema-design-time-series#time-buckets

lollo1234 4 years, 10 months ago

How do you store both CPU and memory usage though? Two sets of 60 columns per row? I am wondering if that goes along with "the schema design will allow for future growth"...what if by future growth they mean monitoring N more metrics. That would imply N*60 columns, right?

Sumanth09 5 years ago

Should be A question did not talk about latency without query cost -- BigQuery Cache flexible schema - BigQuery (nested and repeated)

https://zhangxuetu.online

[202610]

Question #162 Topic 1

You want to archive data in Cloud Storage. Because some data is very sensitive, you want to use the `Trust No One` (TNO) approach to encrypt your data to prevent the cloud provider staff from decrypting your data. What should you do?

A. Use gcloud kms keys create to create a symmetric key. Then use gcloud kms encrypt to encrypt each archival file with the key and unique additional authenticated data (AAD). Use gsutil cp to upload each encrypted file to the Cloud Storage bucket, and keep the AAD outside of Google Cloud.

B. Use gcloud kms keys create to create a symmetric key. Then use gcloud kms encrypt to encrypt each archival file with the key. Use gsutil cp to upload each encrypted file to the Cloud Storage bucket. Manually destroy the key previously used for encryption, and rotate the key once.

C. Specify customer-supplied encryption key (CSEK) in the .boto configuration file. Use gsutil cp to upload each archival file to the Cloud Storage bucket. Save the CSEK in Cloud Memorystore as permanent storage of the secret.

D. Specify customer-supplied encryption key (CSEK) in the .boto configuration file. Use gsutil cp to upload each archival file to the Cloud Storage bucket. Save the CSEK in a different project that only the security team can access. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

dhs227 **Highly Voted** 6 years ago

The correct answer must be D A and B can be eliminated immediately since kms generated keys are considered potentially accessible by CSP. C is incorrect because memory store is essentially a cache service. Additional authenticated data (AAD) acts as a "salt", it is not a cipher.

mikey007 5 years, 7 months ago

AAD is bound to the encrypted data, because you cannot decrypt the ciphertext unless you know the AAD, but it is not stored as part of the ciphertext. AAD also does not increase the cryptographic strength of the ciphertext. Instead it is an additional check by Cloud KMS to authenticate a decryption request.

[Removed] **Highly Voted** 6 years ago

Answer: A Description: AAD is used to decrypt the data so better to keep it outside GCP for safety

grimoren 10 months ago

Selected Answer: D

The "prevent the cloud provider staff from decrypting your data" makes me lean more towards D than A.

aaaaaaaasdasdasfs 11 months, 2 weeks ago

Selected Answer: D

Based on the requirement for a "Trust No One" (TNO) approach where even the cloud provider cannot decrypt your data, the correct answer is: D. Specify customer-supplied encryption key (CSEK) in the .boto configuration file. Use gsutil cp to upload each archival file to the Cloud Storage bucket. Save the CSEK in a different project that only the security team can access. This is the best option because: Customer-supplied encryption keys (CSEKs) are encryption keys that you manage and provide to Google Cloud, rather than using Google-managed keys. When you use CSEKs, Google Cloud uses your key to encrypt your data but does not store the key. This means Google Cloud cannot decrypt your data without you providing the key again. Storing the CSEK in a different project that only the security team can access ensures that the key is securely stored but separated from the encrypted data.

aaaaaaaasdasdasfs 11 months, 3 weeks ago

Selected Answer: D

This is the correct option because: Customer-supplied encryption keys (CSEKs) provide client-side encryption where you fully control the keys. By specifying the CSEK in the .boto configuration file, the data is encrypted before it reaches Google's servers. Storing the keys in a different project with restricted access ensures proper separation. This approach keeps the encryption keys entirely under your control, following the TNO principle.

Anudeep58 1 year, 9 months ago

Selected Answer: A

Keep AAD Outside of Google Cloud: Keeping the AAD outside of Google Cloud ensures that Google cannot access the additional context required to decrypt the files, thus implementing the TNO approach. Option C: Customer-Supplied Encryption Key (CSEK) in .boto File: Storing the CSEK in Cloud Memorystore or any cloud service introduces a risk where the key could be potentially accessed by cloud provider staff. Option D: Customer-Supplied Encryption Key (CSEK) in a Different Project: While storing the CSEK in a different project adds some security, it still leaves the keys within the Google Cloud environment, which does not fully meet the TNO approach.

emmylou 2 years, 4 months ago

I just cannot understand this question. If you can't trust the provider, in this case Google, then how can you use the KMS approach. In my mind you have to generate the key locally and upload but I'm clearly wrong and don't get why.

shanwford 2 years, 6 months ago

Selected Answer: D

IMO must be (D) : to reach TNO goal keys must be customer supplied.

barnac1es 2 years, 6 months ago

Selected Answer: D

Customer-Supplied Encryption Key (CSEK): CSEK allows you to provide your encryption keys, ensuring that the cloud provider staff does not have access to the keys and cannot decrypt your data. Separate Project for Key Management: Saving the CSEK in a different project that only the security team can access adds an additional layer of security. It isolates the encryption keys from the project where the data is stored, ensuring that even within the same cloud provider, only authorized personnel can access the keys. Use of .boto Configuration: Specifying the CSEK in the .boto configuration file ensures that it is applied consistently when interacting with Cloud Storage through tools like gsutil. This way, every archival file is encrypted using your keys. Options A and B involve using Google Cloud Key Management Service (KMS) to manage keys, which does not align with the TNO approach because cloud provider staff could potentially access the keys stored in Google Cloud KMS.

NewDE2023 2 years, 8 months ago

Selected Answer: D

CSEKs are used when an organization needs complete control over key management.

tavva_prudhvi 2 years, 8 months ago

Option A is not the best choice for the "Trust No One" (TNO) approach because it involves using Google Cloud's Key Management Service (KMS) to create and manage encryption keys. This means that the cloud provider will have access to the keys, which could potentially enable their staff to decrypt the data.

midgoo 3 years ago

Selected Answer: A

D may work, but 'Trust No One' = do not trust GCP too. So D cannot be the answer.

musumusu 3 years, 1 month ago

answer A: KMS + AAD is more secure than CSEK

Jay_Krish 3 years, 4 months ago

Selected Answer: D

CSEK with only security team having access seems to be right approach. Not sure how A can be better.

cloudmon 3 years, 4 months ago

Selected Answer: A

It's A, because you cannot decrypt the ciphertext unless you know the AAD (https://cloud.google.com/kms/docs/additional-authenticated-data)

devaid 3 years, 5 months ago

Selected Answer: A

Answer: A

clouditis 3 years, 6 months ago

D it is

https://zhangxuetu.online

[202610]

Question #163 Topic 1

You have data pipelines running on BigQuery, Dataflow, and Dataproc. You need to perform health checks and monitor their behavior, and then notify the team managing the pipelines if they fail. You also need to be able to work across multiple projects. Your preference is to use managed products or features of the platform. What should you do?

A. Export the information to Cloud Monitoring, and set up an Alerting policy **Most Voted**

B. Run a Virtual Machine in Compute Engine with Airflow, and export the information to Cloud Monitoring

C. Export the logs to BigQuery, and set up App Engine to read that information and send emails if you find a failure in the logs

D. Develop an App Engine application to consume logs using GCP API calls, and send emails if you find a failure in the logs

Correct Answer: A

Community vote distribution

A (100%)

Comments

John_Pongthorn **Highly Voted** 3 years, 6 months ago

Selected Answer: A

A . Your preference is to use managed products or features of the platform

SamuelTsch 1 year, 5 months ago

Selected Answer: A

A

barnac1es 2 years, 6 months ago

Selected Answer: A

Cloud Monitoring (formerly known as Stackdriver) is a fully managed monitoring service provided by GCP, which can collect metrics, logs, and other telemetry data from various GCP services, including BigQuery, Dataflow, and Dataproc. Alerting Policies: Cloud Monitoring allows you to define alerting policies based on specific conditions or thresholds, such as pipeline failures, latency spikes, or other custom metrics. When these conditions are met, Cloud Monitoring can trigger notifications (e.g., emails) to alert the team managing the pipelines. Cross-Project Monitoring: Cloud Monitoring supports monitoring resources across multiple GCP projects, making it suitable for your requirement to monitor pipelines in multiple projects. Managed Solution: Cloud Monitoring is a managed service, reducing the operational overhead compared to running your own virtual machine instances or building custom solutions.

sergiomujica 2 years, 6 months ago

Selected Answer: A

use managed products

whorillo 3 years ago

Selected Answer: A

Should be A

pluiedust 3 years, 6 months ago

Selected Answer: A

Should be A

AWSandeep 3 years, 7 months ago

Selected Answer: A

A. Export the information to Cloud Monitoring, and set up an Alerting policy

PhuocT 3 years, 7 months ago

Selected Answer: A

Should be A

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #164 Topic 1

You are working on a linear regression model on BigQuery ML to predict a customer's likelihood of purchasing your company's products. Your model uses a city name variable as a key predictive component. In order to train and serve the model, your data must be organized in columns. You want to prepare your data using the least amount of coding while maintaining the predictable variables. What should you do?

A. Create a new view with BigQuery that does not include a column with city information.

B. Use SQL in BigQuery to transform the state column using a one-hot encoding method, and make each city a column with binary values. **Most Voted**

C. Use TensorFlow to create a categorical variable with a vocabulary list. Create the vocabulary file and upload that as part of your model to BigQuery ML.

D. Use Cloud Data Fusion to assign each city to a region that is labeled as 1, 2, 3, 4, or 5, and then use that number to represent the city in the model.

Correct Answer: B

Community vote distribution



Comments

cajica Highly Voted 3 years, 1 month ago

Selected Answer: D

If we're rigorous, as we should because it's a professional exam, I think option B is incorrect because it's one-hot-encoding the "state" column, if the answer was "city" column, then I'd go for B. As this is not the case and I do not accept an spelling error like this in an official question, I would go for D.

sergiomujica 2 years, 6 months ago

I think it should say citiy instead of state... it is a typooi in the transcription of the question

knith66 2 years, 8 months ago

you are right, OHE is mentioned for state in option B, but in option B it is also mentioned to use binary conversion for the city column. an additional method can be used which is applicable for the conversion.

cetanx 2 years, 10 months ago

But also for D, assigning each city to a numbered region could lose important information, as cities within the same region might have different characteristics affecting customer purchasing behavior (from Chat GPT).

MaxNRG Highly Voted 2 years, 3 months ago

Selected Answer: B

One-hot encoding is a common technique used to handle categorical data in machine learning. This approach will transform the city name variable into a series of binary columns, one for each city. Each row will have a "1" in the column corresponding to the city it represents and "0" in all other city columns. This method is effective for linear regression models as it enables the model to use city data as a series of numeric, binary variables. BigQuery supports SQL operations that can easily implement one-hot encoding, thus minimizing the amount of coding required and efficiently preparing the data for the model.

MaxNRG 2 years, 3 months ago

A removes the city information completely, losing a key predictive component. C requires additional coding and infrastructure with TensorFlow and vocabulary files outside of what BigQuery already provides. D transforms the distinct city values into numeric regions, losing granularity of the city data. By using SQL within BigQuery to one-hot encode cities into multiple yes/no columns, the city data is maintained and formatted appropriately for the BigQuery ML linear regression model with minimal additional coding. This aligns with the requirements stated in the question.

MaxNRG 2 years, 3 months ago

https://cloud.google.com/bigquery/docs/auto-preprocessing#one_hot_encoding

clouditis 1 year, 3 months ago

Selected Answer: D

D it is, Question clearly says least amount of coding!

baimus 1 year, 6 months ago

Selected Answer: B

This is B. It's easier to one hot in bigquery than to do it in datafusion and then import the values back into bigquery.

barnac1es 2 years, 6 months ago

Selected Answer: B

One-Hot Encoding: One-hot encoding is a common technique for handling categorical variables like city names in machine learning models. It transforms categorical data into a binary matrix, where each city becomes a separate column with binary values (0 or 1) indicating the presence or absence of that city. Least Amount of Coding: One-hot encoding in BigQuery is straightforward and can be accomplished with SQL. You can use SQL expressions to pivot the city names into separate columns and assign binary values based on the city's presence in the original data. Predictive Power: One-hot encoding retains the predictive power of city information while making it suitable for linear regression models, which require numerical input.

knith66 2 years, 8 months ago

Selected Answer: B

One hot encoding for state and binary values for each city will allow me to choose the B option.

tavva_prudhvi 2 years, 8 months ago

I guess Option D loses the granularity of the city-level information, as multiple cities will be grouped into the same region and represented by the same number. This can result in a loss of important predictive information for your linear regression model. On the other hand, if we use one-hot encoding to create binary columns for each city. This method preserves the city-level information, allowing the model to capture the unique effects of each city on the likelihood of purchasing your company's products. Additionally, it can be done directly in BigQuery using SQL, which requires less coding and is more efficient.

blathul 2 years, 9 months ago

Selected Answer: B

One-hot encoding is a common technique used to represent categorical variables as binary columns. In this case, you can transform the city variable into multiple binary columns, with each column representing a specific city. This allows

you to maintain the predictive city information while organizing the data in columns suitable for training and serving the linear regression model. By using SQL in BigQuery, you can perform the necessary transformations to implement one-hot encoding.

KC_go_reply 2 years, 9 months ago

Selected Answer: B

- A is wrong since it drops the city which is a key predictor. - C is wrong since we want to keep it simple, and not use Tensorflow here. - D is wrong since there is no specific reason to use Data Fusion, and also this encoding here is ordinal, which doesn't make sense for something non-quantitative such as cities - we want one-hot coding instead. Therefore, B must be the correct answer.

ckanaar 2 years, 6 months ago

It could be argued that a specific reason to use Data Fusion is the minimal coding requirement.

leandrors 2 years, 9 months ago

Selected Answer: D

Cloud Datafusion: least amount of coding

knith66 2 years, 8 months ago

OHE is better that datafusion considering least amount coding

tavva_prudhvi 2 years, 8 months ago

While it's true that Cloud Data Fusion can simplify data integration tasks with a visual interface, it might not be the best choice in this specific scenario as using Cloud Data Fusion to assign each city to a region might result in a loss of important predictive information due to the grouping of cities

vaga1 2 years, 10 months ago

Selected Answer: B

A doesn't include the city column. C is not low code. D is not a one hot encoding, but an ordinal one on the city column. B applies a one hot encoding on the state column and a binary encoding on the city column, which works for me.

mialll 2 years, 11 months ago

Selected Answer: B

https://cloud.google.com/bigquery/docs/reference/standard-sql/bigqueryml-auto-preprocessing#one_hot_encoding

juliobs 3 years ago

Selected Answer: D

D uses the least amount of coding... even if the model is not good. B encodes the "state", not the "city".

dconesoko 3 years, 2 months ago

Selected Answer: B

Manually bigquery ml does preprocessing for you however if one wants to do a manual processing one can use the ML.ONE_HOT_ENCODER function. It just acts as an analytical funciton.

ovokpus 3 years, 4 months ago

Selected Answer: B

The Cloud Data Fusion method will add unecessary weights to categories with higher value labels, which will skew the model. The best practice for encoding nominal categorical data is to one-hot-encode them into binary values. That is conveniently done in BigQuery: https://cloud.google.com/bigquery-ml/docs/reference/standard-sql/bigqueryml-auto-preprocessing#one_hot_encoding

Atnafu 3 years, 4 months ago

D Cloud Data Fusion is a fully managed, code-free data integration service that helps users efficiently build and manage ETL/ELT data pipelines.

dconesoko 3 years, 2 months ago

Does it come with an out of the box one hot encoding template ?

NicolasN 3 years, 4 months ago

Selected Answer: D

I have the same feeling with @cloudmon so I compromised to answer [D]. In more detail, here is my reasoning: The requirement "maintaining the predictable variables" (a.k.a. city) makes: [A] obiously invalid [B] invalid, since it broadens the prediction to be state-dependent (all cities in particular state will be treated as the same variable). Additionally, one-hot encoding is not suitable for linear regression problems, dummy encoding (drop one) is better. Answer [C] doesn't satisfy the "least amount of coding" directive. Other than that (as far I understood by searcing the keyword tf.feature_column.categorical_column_with_vocabulary_list) the TensorFlow vocabulary list is another form of one-hot encoding. So it remains [D] which offers a visual interface but uses ordinal (or label) encoding which is far from ideal for regression problems.

https://zhangxuetu.online

[202610]

Question #165 Topic 1

You work for a large bank that operates in locations throughout North America. You are setting up a data storage system that will handle bank account transactions. You require ACID compliance and the ability to access data with SQL. Which solution is appropriate?

A. Store transaction data in Cloud Spanner. Enable stale reads to reduce latency.

B. Store transaction in Cloud Spanner. Use locking read-write transactions. **Most Voted**

C. Store transaction data in BigQuery. Disabled the query cache to ensure consistency.

D. Store transaction data in Cloud SQL. Use a federated query BigQuery for analysis.

Correct Answer: B

Community vote distribution

B (80%) D (18%) C (2%)

Comments

juliobs **Highly Voted** 2 years, 6 months ago

Selected Answer: B

"locations throughout North America" implies multi-region (northamerica-northeast1, us-central1, us-south1, us-west4, us-east5, etc.) Cloud SQL can only do read replicas in other regions.

FP77 2 years, 1 month ago

Read replicas are enough to make Cloud SQL work as a multi-region service. That's not the point. The point is that the answer introduces the use of Bigquery when it's not needed for the use case. That's why B is right.

devaid **Highly Voted** 2 years, 12 months ago

Selected Answer: B

I'd say B as the documentation primarily says ACID compliance for Spanner, not Cloud SQL. https://cloud.google.com/blog/topics/developers-practitioners/your-google-cloud-database-options-explained Also, spanner supports read-write transactions for use cases, as handling bank transactions: https://cloud.google.com/spanner/docs/transactions#read-write_transactions

AzureDP900 2 years, 9 months ago

B is right

Jay_Krish 2 years, 10 months ago

I wonder if you understood the meaning of ACID. This is an inherent property of any relational DB. Cloud SQL is fully ACID complaint

MaxNRG 1 year, 9 months ago

Selected Answer: B

B. Store transaction in Cloud Spanner. Use locking read-write transactions. Since the banking transaction system requires ACID compliance and SQL access to the data, Cloud Spanner is the most appropriate solution. Unlike Cloud SQL, Cloud Spanner natively provides ACID transactions and horizontal scalability. Enabling stale reads in Spanner (option A) would reduce data consistency, violating the ACID compliance requirement of banking transactions. BigQuery (option C) does not natively support ACID transactions or SQL writes which are necessary for a banking transactions system. Cloud SQL (option D) provides ACID compliance but does not scale horizontally like Cloud Spanner can to handle large transaction volumes. By using Cloud Spanner and specifically locking read-write transactions, ACID compliance is ensured while providing fast, horizontally scalable SQL processing of banking transactions.

Aman47 1 year, 9 months ago

Selected Answer: B

Spanner is an enterprise level resource which Banks require, Cloud SQL is limited to 30TB of storage. And Banking transactions should be read write locked.

barnac1es 2 years ago

Selected Answer: B

ACID Compliance: Cloud Spanner is a globally distributed, strongly consistent database service that offers ACID compliance, making it a suitable choice for handling bank account transactions where data consistency and integrity are crucial. SQL Access: Cloud Spanner supports SQL queries, which align with your requirement to access data with SQL. You can use standard SQL to interact with the data stored in Cloud Spanner. Locking Read-Write Transactions: Cloud Spanner allows you to perform locking read-write transactions, ensuring that transactions are executed in a serializable and consistent manner. This is essential for financial transactions to prevent conflicts and maintain data integrity.

NeoNitin 2 years, 2 months ago

B. Store transaction data in Cloud Spanner. Use locking read-write transactions. Here's why: ACID Compliance: ACID stands for Atomicity, Consistency, Isolation, and Durability. Cloud Spanner is a fully managed, globally distributed database that provides strong consistency and ACID compliance. This ensures that bank account transactions are processed reliably and accurately, avoiding issues like data corruption or incomplete transactions. Ability to access data with SQL: Cloud Spanner supports SQL, which allows you to perform standard SQL queries on the data. This means that you can use familiar SQL commands to access, retrieve, and manipulate transaction data easily.

Adswerve 2 years, 5 months ago

Selected Answer: D

I initially selected B. However, it might be D. https://cloud.google.com/blog/topics/developers-practitioners/your-google-cloud-database-options-explained Cloud Spanner: Cloud Spanner is an enterprise-grade, globally-distributed, and strongly-consistent database that offers up to 99.999% availability, built specifically to combine the benefits of relational database structure with non-relational horizontal scale. It is a unique database that combines ACID transactions, SQL queries, and relational structure with the scalability that you typically associate with non-relational or NoSQL databases. As a result, Spanner is best used for applications such as gaming, payment solutions, global financial ledgers, retail banking and inventory management that require ability to scale limitlessly with strong-consistency and high-availability.

musumusu 2 years, 7 months ago

Answer B: locking read-write = for data accuracy state read = for speed up or latency

musumusu 2 years, 7 months ago

Answer B: Spanner It's incomplete question, what do you assume by large bank, until we are not sure about size and scale. Region is north america, that can be managed by cloud sql. but i am going for spanner, as its large bank and

transaction data.

cajica 2 years, 7 months ago

This is definitely a tricky question because both B and D are "appropriate" as the question suggests, of course we can make assumptions with the "large bank" sentence but there are other questions here where making assumptions is not accepted by the community so I wonder when can we make assumptions and when we can't. I think the real problem here is the ambiguous question. This is one of the few questions where the community accept that both (B and D) answers are appropriate but some comments (and I agree) argue the BEST approach is B. I really think some questions can be written in a better and non-ambiguous way, it's just about thinking a little bit more and not conforming when a poor spelling.

jkhong 2 years, 9 months ago

Selected Answer: B

The question is hinting a requirement for global consistency, i.e. being available for NA region, which does not just include US but also Mexico, Argentina etc. Large bank = priority over consistency over read-write

desertlotus1211 2 years, 8 months ago

Argentina is South America...

ckanaar 2 years ago

Good catch, definetely Spanner in that case.

NicolasN 2 years, 9 months ago

Selected Answer: B

Finally, it's [B]. There is no measurable requirement that rules out [D] (Cloud SQL) and this fact made me to select it as a preferrable answer. But since we are talking about a large bank (which normally implies massive reads/writes per sec.) and nobody has posed any cost limitation, in a real case I would definitely prefer the advantages of Spanner.

NicolasN 2 years, 10 months ago

Selected Answer: D

[A] No - Stale reads not accepted for bank account transactions. "A stale read is read at a timestamp in the past. If your application is latency sensitive but tolerant of stale data, then stale reads can provide performance benefits." [B] Yes - Fulfills all requirements [C] No - BigQuery is ACID-compliant, but it is too much to use it for such a case (mainly a CRUD app) [D] Yes+ - Fulfills all requirements. The BigQuery part may seem redundant, but it states a true fact that doesn't violate the "access data with SQL" requirement. So, when SQL Cloud and SQL Spanner fit both, there is no reason to prefer the second. And the question doesn't mention any obvious fact for which should we prefer the expensive SQL Spanner: - We don't know if we have to deal with a big amount of data and thousands of writes per second. - We don't know the database size. - There is no need for multi-regional writes that would exclude SQL Cloud as an alternative. Is it a coincidence that the question limits the problem to the single region of North America?

NicolasN 2 years, 9 months ago

I changed my mind to [B] since I underestimated the given of a "large bank" where the cost difference for a single region Spanner wouldn't matter.

SuperVee 2 years, 6 months ago

Also, correct me if I am wrong, Bigquery cannot query Cloud SQL directly, only when Cloud SQL is exported into GCS, then BQ can connect to GCS using federated queries.

cloudmon 2 years, 11 months ago

Selected Answer: B

I'd go for B. The only other somewhat valid option is D, but there's no requirement for analytics in the question.

mattab1627 2 years, 11 months ago

Surely its B, transactional data at a large US based bank would surely be massive in size and probably too much for CloudSQL? There is also no mention of a requirement for analytics

MounicaN 3 years ago

why not spanner?

pluiedust 3 years ago

Selected Answer: D

D is correct

https://zhangxuetu.online

[202610]

Question #166 Topic 1

A shipping company has live package-tracking data that is sent to an Apache Kafka stream in real time. This is then loaded into BigQuery. Analysts in your company want to query the tracking data in BigQuery to analyze geospatial trends in the lifecycle of a package. The table was originally created with ingest-date partitioning. Over time, the query processing time has increased. You need to implement a change that would improve query performance in BigQuery. What should you do?

A. Implement clustering in BigQuery on the ingest date column.

B. Implement clustering in BigQuery on the package-tracking ID column. **Most Voted**

C. Tier older data onto Cloud Storage files and create a BigQuery table using Cloud Storage as an external data source.

D. Re-create the table using data partitioning on the package delivery date.

Correct Answer: B

Community vote distribution



Comments

AlizCert 1 year, 3 months ago

Selected Answer: B

Though I almost fell for D, but the delivery date information is only available on the event(s) that happen after the delivery, but not on the ones before where it will be NULL I guess. The only other option that can make some sense is B, though high cardinality is not recommended for clustering.

MaxNRG 1 year, 9 months ago

Selected Answer: B

B as Clustering the data on the package Id can greatly improve the performance. Refer GCP documentation - BigQuery Clustered Table:https://cloud.google.com/bigquery/docs/clustered-tables

MaxNRG 1 year, 9 months ago

Clustering can improve the performance of certain types of queries such as queries that use filter clauses and queries that aggregate data. When data is written to a clustered table by a query job or a load job, BigQuery sorts the data using the values in the clustering columns. These values are used to organize the data into multiple blocks in BigQuery storage. When you submit a query containing a clause that filters data based on the clustering columns, BigQuery uses the sorted blocks to eliminate scans of unnecessary data. Currently, BigQuery allows clustering over a partitioned table. Use clustering over a partitioned table when: - Your data is already partitioned on a date, timestamp,

or integer column. - You commonly use filters or aggregation against particular columns in your queries. Table clustering is possible for tables partitioned by: - ingestion time - date/timestamp - integer range

MaxNRG 1 year, 9 months ago

In a table partitioned by a date or timestamp column, each partition contains a single day of data. When the data is stored, BigQuery ensures that all the data in a block belongs to a single partition. A partitioned table maintains these properties across all operations that modify it: query jobs, Data Manipulation Language (DML) statements, Data Definition Language (DDL) statements, load jobs, and copy jobs. This requires BigQuery to maintain more metadata than a non-partitioned table. As the number of partitions increases, the amount of metadata overhead increases.

MaxNRG 1 year, 9 months ago

Although more metadata must be maintained, by ensuring that data is partitioned globally, BigQuery can more accurately estimate the bytes processed by a query before you run it. This cost calculation provides an upper bound on the final cost of the query. In a clustered table, BigQuery automatically sorts the data based on the values in the clustering columns and organizes them in optimally sized storage blocks. You can achieve more finely grained sorting by creating a table that is clustered and partitioned. A clustered table maintains the sort properties in the context of each operation that modifies it. As a result, BigQuery may not be able to accurately estimate the bytes processed by the query or the query costs. When blocks of data are eliminated during query execution, BigQuery provides a best effort reduction of the query costs.

Aman47 1 year, 9 months ago

Selected Answer: B

Package Tracking mostly contains, geospatial prefixes, Like HK0011, US0022, etc, this can help in clustering.

kcl10 1 year, 12 months ago

Selected Answer: D

D is the correct answer requirements: analyze geospatial trends in the lifecycle of a package cuz the data of the lifecycle of the package would span across ingest-date-based partition table, it would degrade the performance. hence, re-partitoning by package delivery date, which is the package initially delivered, would improve the performance when querying such table.

sdi_studiers 2 years, 3 months ago

Selected Answer: D

I vote D Queries to analyze the package lifecycle will cross partitions when using ingest date. Changing this to delivery date will allow a query to full a package's full lifecycle in a single partition.

cloudmon 2 years, 11 months ago

Selected Answer: B

B. https://cloud.google.com/bigquery/docs/clustered-tables

John_Pongthorn 3 years ago

D is not correct becsuse This Is problem Is The Real Time so ingested date is the same as delivery date.

kenanars 3 years ago

why not D ?

jkhong 2 years, 9 months ago

The table has already been partitioned

John_Pongthorn 2 years, 12 months ago

There are several rows that represent movement of life cycle of 1 package-tracking ID package delivery date = ingestion date , i suppose

pluiedust 3 years ago

Selected Answer: B

B; As the table has already created with ingest-date partitioning.

AWSandeep 3 years ago

Selected Answer: B

B. Implement clustering in BigQuery on the package-tracking ID column.

https://zhangxuetu.online

[202610]

Question #167 Topic 1

Your company currently runs a large on-premises cluster using Spark, Hive, and HDFS in a colocation facility. The cluster is designed to accommodate peak usage on the system; however, many jobs are batch in nature, and usage of the cluster fluctuates quite dramatically. Your company is eager to move to the cloud to reduce the overhead associated with on-premises infrastructure and maintenance and to benefit from the cost savings. They are also hoping to modernize their existing infrastructure to use more serverless offerings in order to take advantage of the cloud. Because of the timing of their contract renewal with the colocation facility, they have only 2 months for their initial migration. How would you recommend they approach their upcoming migration strategy so they can maximize their cost savings in the cloud while still executing the migration in time?

A. Migrate the workloads to Dataproc plus HDFS; modernize later.

B. Migrate the workloads to Dataproc plus Cloud Storage; modernize later. **Most Voted**

C. Migrate the Spark workload to Dataproc plus HDFS, and modernize the Hive workload for BigQuery.

D. Modernize the Spark workload for Dataflow and the Hive workload for BigQuery.

Correct Answer: B

Community vote distribution



Comments

MaxNRG 1 year, 9 months ago

Selected Answer: B

Based on the time constraint of 2 months and the goal to maximize cost savings, I would recommend option B - Migrate the workloads to Dataproc plus Cloud Storage; modernize later. The key reasons are: • Dataproc provides a fast, native migration path from on-prem Spark and Hive to the cloud. This allows meeting the 2 month timeline. • Using Cloud Storage instead of HDFS avoids managing clusters for variable workloads and provides cost savings. • Further optimizations and modernization to serverless (Dataflow, BigQuery) can happen incrementally later without time pressure.

MaxNRG 1 year, 9 months ago

Option A still requires managing HDFS. Option C and D require full modernization of workloads in 2 months which is likely infeasible. Therefore, migrating to Dataproc with Cloud Storage fast tracks the migration within 2 months while realizing immediate cost savings, enabling the flexibility to iteratively modernize and optimize the workloads over time.

John_Pongthorn 2 years, 12 months ago

Selected Answer: B

B is most likely 1. migrate job and infrastructure to dataproc on clound 2. any data, move from hdfs on-premise to google cloud storage (one of them is Hive) If you want to modernize Hive to Bigquery , you are need to move it into GCS(preceding step) first and load it into bigquery that is all. https://cloud.google.com/blog/products/data-analytics/apache-hive-to-bigquery https://cloud.google.com/architecture/hadoop/migrating-apache-spark-jobs-to-cloud-dataproc https://cloud.google.com/architecture/hadoop/hadoop-gcp-migration-data

TNT87 3 years ago

Selected Answer: D

Answer D

dn_mohammed_data 3 years ago

you sould migrate spark to apache beam which is not the case here

TNT87 3 years ago

apache beam for what???

adarifian 2 years, 11 months ago

dataflow uses apache beam

TNT87 2 years, 8 months ago

@adarifian Why use apache beam yet there is Dataflow an inhouse gcp solution to solve the problem? hence i said apache beam for what

ExamCtechs 1 year, 11 months ago

Dataflow IS apache beam, Dataflow is a Beam Runner. If you go for that soulution you will need to modify your pipeline to use Beam

GyaneswarPanigrahi 3 years ago

D isn't feasible, within 2 months. Anyone who has worked in a Hadoop/ Big Data data warehousing or data lake project, knows how less time 2 months is, given the amount of data and associated complexities abound. It should be B to begin with. And then gradually move towards D.

TNT87 3 years ago

Selected Answer: B

Ans B -cost saving -time factor -Spark -Data proc

TNT87 3 years ago

Ans D is also relevant if you read this. Onthe other hand cloud storage isnt severless but bigquery is https://cloud.google.com/hadoop-spark-migration

damaldon 3 years ago

Ans.B as per the following link https://blog.devgenius.io/migrating-spark-jobs-to-google-cloud-file-event-sensor-to-dynamically-create-spark-cluster-7eff2c75423d

YorelNation 3 years ago

Selected Answer: B

For the time window of two month I would recommend B and then start to implement D.

ducc 3 years ago

It is B or D, still confusing

AWSandeep 3 years, 1 month ago

Selected Answer: D

D because the Apache Spark Runner can be used to execute Beam pipelines using Apache Spark. Also, Hive to BigQuery is not a difficult modernization/migration.

ExamCtechs 1 year, 11 months ago

Dataflow is a Runner of Beam it self

https://zhangxuetu.online

[202610]

Question #168 Topic 1

You work for a financial institution that lets customers register online. As new customers register, their user data is sent to Pub/Sub before being ingested into
BigQuery. For security reasons, you decide to redact your customers' Government issued Identification Number while allowing customer service representatives to view the original values when necessary. What should you do?

A. Use BigQuery's built-in AEAD encryption to encrypt the SSN column. Save the keys to a new table that is only viewable by permissioned users.

B. Use BigQuery column-level security. Set the table permissions so that only members of the Customer Service user group can see the SSN column.

C. Before loading the data into BigQuery, use Cloud Data Loss Prevention (DLP) to replace input values with a cryptographic hash.

D. Before loading the data into BigQuery, use Cloud Data Loss Prevention (DLP) to replace input values with a cryptographic format-preserving encryption token. Most Voted

Correct Answer: D

Community vote distribution



Comments

AWSandeep Highly Voted 3 years, 7 months ago

Selected Answer: B

B. While C and D are intriguing, they don't specify how to enable customer service representatives to receive access to the encryption token.

cloud_rider 1 year, 4 months ago

B will show the values to the customer support service all the time as they have access to it, so no redaction as per the ask. Another thing is the requirement is to view when necessary, so D fits this requirement and format preserving encryption can be reverted when necessary.

MaxNRG 2 years, 3 months ago

B. BigQuery column-level security: Pros: Granular control over column access, ensures only authorized users see the SSN column. Cons: Doesn't truly redact the data. The SSN values are still stored in BigQuery, even if hidden from unauthorized users. A potential security breach could expose them.

ffggrre 2 years, 5 months ago

there is no SSN in question, it can be any ID.

Lanro Highly Voted 2 years, 8 months ago

Selected Answer: D

I don't see why we should use DLP since we know exactly the column that should be locked or encrypted. On the other hand having a cryptographic representation of SSN helps to aggregate/analyse entries. So I will vote for D, but B is much more easy to implement. Garbage question indeed.

SamuelTsch 1 year, 5 months ago

Selected Answer: D

In the question, there is no mention of SSN column.

SamuelTsch 1 year, 5 months ago

also, in the question, "you decide to REDACT ...". Option B does not redact the values.

MohaSa1 1 year, 5 months ago

Selected Answer: D

Authorized users can decrypt the FPE tokens back to the original GIINs, D is the best option.

baimus 1 year, 5 months ago

Selected Answer: D

D (FPE) does indeed allow encryption to be reversed if desired, allowing operatives to review the original key. This makes it preferable to B, as it's also more secure.

Topg4u 1 year, 9 months ago

D: SSN is only tied to USA not in any other countries, The question did not mention SSN.

MaxNRG 2 years, 3 months ago

Selected Answer: D

The best option is D - Before loading the data into BigQuery, use Cloud Data Loss Prevention (DLP) to replace input values with a cryptographic format-preserving encryption token. The key reasons are: DLP allows redacting sensitive PII like SSNs before loading into BigQuery. This provides security by default for the raw SSN values. Using format-preserving encryption keeps the column format intact while still encrypting, allowing business logic relying on SSN format to continue functioning. The encrypted tokens can be reversed to view original SSNs when required, meeting the access requirement for customer service reps.

MaxNRG 2 years, 3 months ago

Option A does encrypt SSN but requires managing keys separately. Option B relies on complex IAM policy changes instead of encrypting by default. Option C hashes irreversibly, preventing customer service reps from viewing original SSNs when required. Therefore, using DLP format-preserving encryption before BigQuery ingestion balances both security and analytics requirements for SSN data.

MaxNRG 2 years, 3 months ago

Why not B. BigQuery column-level security: Doesn't truly redact the data. The SSN values are still stored in BigQuery, even if hidden from unauthorized users. A potential security breach could expose them.

Aman47 2 years, 3 months ago

Selected Answer: D

Even if you provide Column level access control, The Data Owners or other hierarchies above it will also be able to view very sensitive data. Better to just use encryption and decryption. As this data can also never be used for any analytic workloads

spicebits 2 years, 4 months ago

Selected Answer: D

Answer has to be D. Question says "you decide to redact your customers' Government issued Identification Number while allowing customer service representatives to view the original values when necessary"... Redact... view the original values... D is the only choice.

Nirca 2 years, 5 months ago

Selected Answer: B

It might not be D! Since - only the Frame is kept. the data will be changed. Format Preserving Encryption (FPE), endorsed by NIST, is an advanced encryption technique that transforms data into an encrypted format while preserving its original structure. For instance, a 16-digit credit card number encrypted with FPE will still be a 16-digit number

Helinia 2 years, 3 months ago

No, the value using FPE can be decrypted with key. "Encrypted values can be re-identified using the original cryptographic key and the entire output value, including surrogate annotation." https://cloud.google.com/dlp/docs/pseudonymization#supported-methods

ffggrre 2 years, 5 months ago

Selected Answer: B

Customer service needs to see the original value, not possible with other options.

kcl10 2 years, 5 months ago

Selected Answer: B

of course B

ckanaar 2 years, 6 months ago

Selected Answer: D

I believe the crux to the question is that the cryptographic format-preserving encryption token is re-identifiable, whereas the cryptographic hash is not: https://cloud.google.com/dlp/docs/transformations-reference Therefore, customer service can view the original values when necessary in case of D.

ckanaar 2 years, 6 months ago

Nevermind, this can actually also be done in the case of answer B. They are both correct, just different implementations. No idea

knith66 2 years, 8 months ago

the question mentions that "user data is sent to Pub/Sub before being ingested" instead of just saying data goes to big query through pub/sub. So some alteration is expected before being injected into the big query. So option D should work.

sr25 2 years, 8 months ago

Selected Answer: D

D. The question says giving CSR's access to values "when necessary" - not default access like given in B. D is a better option using the token.

ZZHZZH 2 years, 8 months ago

Selected Answer: B

One of the key requirement is to be able to let authorized personel see the ID. D doesn't specify that.

vaga1 2 years, 10 months ago

Selected Answer: D

The answer is between B and D as well described in many comments. I personally do not see any reason to keep the information available using a token or a mask. It is not a PAN card number, it's just a personal ID. It should not be useful for analytical purposes. I'm gonna go for D then

vaga1 2 years, 10 months ago

sorry B

https://zhangxuetu.online

[202610]

Question #169 Topic 1

You are migrating a table to BigQuery and are deciding on the data model. Your table stores information related to purchases made across several store locations and includes information like the time of the transaction, items purchased, the store ID, and the city and state in which the store is located. You frequently query this table to see how many of each item were sold over the past 30 days and to look at purchasing trends by state, city, and individual store. How would you model this table for the best query performance?

A. Partition by transaction time; cluster by state first, then city, then store ID. **Most Voted**

B. Partition by transaction time; cluster by store ID first, then city, then state.

C. Top-level cluster by state first, then city, then store ID.

D. Top-level cluster by store ID first, then city, then state.

Correct Answer: A

Community vote distribution



Comments

AWSandeep Highly Voted 3 years, 6 months ago

Selected Answer: A

A. Partition by transaction time; cluster by state first, then city, then store ID.

Atnafu Highly Voted 3 years, 3 months ago

A Partitioning is obvious Clustering is already mentioned in the question past 30 days and to look at purchasing trends by state, city, and individual store

SamuelTsch 1 year, 5 months ago

Selected Answer: A

go to A.

MaxNRG 2 years, 3 months ago

Selected Answer: B

over the past 30 days -> partitioning by state, city, and individual store -> cluster order

MaxNRG 2 years, 3 months ago

For optimal query performance in BigQuery, especially for the described use cases of analyzing sales data by time and geographical hierarchies, the data should be organized to minimize the amount of data scanned during queries. Given the frequent queries over the past 30 days and analysis by location, the best approach is: Option A: Partition by transaction time; cluster by state first, then city, then store ID.

MaxNRG 2 years, 3 months ago

Partitioning the table by transaction time allows for efficient querying over specific time ranges, such as the past 30 days, which reduces costs and improves performance because it limits the amount of data scanned. Clustering by state, then city, and then store ID aligns with the hierarchy of geographical data and the types of queries that are run against the dataset. It organizes the data within each partition so that queries filtering by state, city, or store ID—or any combination of these—are optimized, as BigQuery can limit the scan to just the relevant clusters within the partitions.

tibuenoc 2 years, 4 months ago

Selected Answer: B

Partition by ingest time Partition by specified data column (Id, State and City)

ffggrre 2 years, 5 months ago

Selected Answer: C

Partition by transaction time would lead to too many partitions - if it was a date, it would have made sense.

sylva1212 1 year, 8 months ago

Even though its a timestamp, the partitioning can be configured on a daily granularity, so A is correct (https://cloud.google.com/bigquery/docs/partitioned-tables#date_timestamp_partitioned_tables)

aureole 2 years, 5 months ago

Selected Answer: C

It should be C. not A

aureole 2 years, 5 months ago

I think it should be C. The fact that we partition the table with the time of the transaction will result many transactions in each day, so it will affect negatively the query performance. i.e : by the end of the day I will have many partitions if I use the transaction time. A would be correct if the partition was by date and not by time. Response: C.

vaga1 2 years, 10 months ago

Selected Answer: A

Partitioning for time is obvious to improve performance and costs of querying only the last 30 days of the table. So, the answer is A or B. https://cloud.google.com/bigquery/docs/querying-clustered-tables "... To get the benefits of clustering, include all of the clustered columns or a subset of the columns in left-to-right sort order, starting with the first column." This means that it is a better choice to sort the table rows by region-province-city (region-state-city in the US case). So, the answer is A.

Prakzz 3 years, 3 months ago

Selected Answer: B

Should be B The clustering should be according to the filtering needs

TNT87 3 years, 6 months ago

https://cloud.google.com/bigquery/docs/querying-clustered-tables

ducc 3 years, 7 months ago

Selected Answer: A

A The question mention that the query is 30 days recently

https://zhangxuetu.online

[202610]

Question #170 Topic 1

You are updating the code for a subscriber to a Pub/Sub feed. You are concerned that upon deployment the subscriber may erroneously acknowledge messages, leading to message loss. Your subscriber is not set up to retain acknowledged messages. What should you do to ensure that you can recover from errors after deployment?

A. Set up the Pub/Sub emulator on your local machine. Validate the behavior of your new subscriber logic before deploying it to production.

B. Create a Pub/Sub snapshot before deploying new subscriber code. Use a Seek operation to re-deliver messages that became available after the snapshot was created. **Most Voted**

C. Use Cloud Build for your deployment. If an error occurs after deployment, use a Seek operation to locate a timestamp logged by Cloud Build at the start of the deployment.

D. Enable dead-lettering on the Pub/Sub topic to capture messages that aren't successfully acknowledged. If an error occurs after deployment, re-deliver any messages captured by the dead-letter queue.

Correct Answer: B

Community vote distribution



Comments

AWSandeep **Highly Voted** 3 years, 1 month ago

Selected Answer: B

B. Create a Pub/Sub snapshot before deploying new subscriber code. Use a Seek operation to re-deliver messages that became available after the snapshot was created. According to the second reference in the list below, a concern with deploying new subscriber code is that the new executable may erroneously acknowledge messages, leading to message loss. Incorporating snapshots into your deployment process gives you a way to recover from bugs in new subscriber code. Answer cannot be C because To seek to a timestamp, you must first configure the subscription to retain acknowledged messages using retain-acked-messages. If retain-acked-messages is set, Pub/Sub retains acknowledged messages for 7 days. References: https://cloud.google.com/pubsub/docs/replay-message https://cloud.google.com/pubsub/docs/replay-overview#seek_use_cases

jkhong 2 years, 9 months ago

Don't think we need to configure subscription to retain ack messages. It is defaulted to retain for 7 days

f74ca0c 1 year, 3 months ago

Selected Answer: C

C. Use a BigQuery view to define your preprocessing logic. When creating your model, use the view as your model training data. At prediction time, use BigQuery's ML.EVALUATE clause without specifying any transformations on the raw input data. Explanation: Preventing Data Skew: Training-serving skew occurs when the transformations applied to training data are not identically applied to prediction data. Using a BigQuery view ensures consistent preprocessing for both training and prediction. Advantages of BigQuery Views: Views encapsulate preprocessing logic, ensuring that the same transformations are applied whenever the view is queried. By referencing the view during both training and prediction, you eliminate the need for manual transformations and the risk of discrepancies.

MaxNRG 1 year, 9 months ago

Selected Answer: B

Taking a snapshot allows redelivering messages that were published while any faulty subscriber logic was running. The seek timestamp would come after deployment so even erroneously acknowledged messages could be recovered. https://cloud.google.com/pubsub/docs/replay-overview#seek_use_cases By creating a snapshot of the subscription before deploying new code, you can preserve the state of unacknowledged messages. If after deployment you find that the new subscriber code is erroneously acknowledging messages, you can use the Seek operation with the snapshot to reset the subscription's acknowledgment state to the time the snapshot was created. This would effectively re-deliver messages available since the snapshot, ensuring you can recover from errors. This approach does not require setting up a local emulator and directly addresses the concern of message loss due to erroneous acknowledgments.

vamgcp 2 years, 2 months ago

pls correct me if I am wrong , option B Option B only allows you to re-deliver messages that were available before the snapshot was created. If an error occurs after the snapshot was created, you will not be able to re-deliver those messages.

cetanx 2 years, 2 months ago

Selected Answer: A

Q: You are concerned that upon deployment the subscriber may erroneously acknowledge messages, leading to message loss. -> So the message is mistakenly acked and removed from topic/subscription. This means even if you have a snapshot of pre-deployment but you don't have a backup or copy of post-deployment messages. Q: Your subscriber is not set up to retain acknowledged messages. -> To seek to a time in the past and replay previously-acknowledged messages, "you must first configure message retention on the topic" or "configure the subscription to retain acknowledged messages" (ref: https://cloud.google.com/pubsub/docs/replay-overview#configuring_message_retention) So B, C, D do not solve the problem of erroneously acked messages as long as you don't have message retention configured on topic/subscription.

lucaluca1982 2 years, 6 months ago

Selected Answer: D

You are updating the code for a subscriber to a Pub/Sub feed. You are concerned that upon deployment the subscriber may erroneously acknowledge messages, leading to message loss. Your subscriber is not set up to retain acknowledged messages. What should you do to ensure that you can recover from errors after deployment? A. Set up the Pub/Sub emulator on your local machine. Validate the behavior of your new subscriber logic before deploying it to production. B. Create a Pub/Sub snapshot before deploying new subscriber code. Use a Seek operation to re-deliver messages that became available after the snapshot was created. C. Use Cloud Build for your deployment. If an error occurs after deployment, use a Seek operation to locate a timestamp logged by Cloud Build at the start of the deployment. D. Enable dead-lettering on the Pub/Sub topic to capture messages that aren't successfully acknowledged. If an error occurs after deployment, re-deliver any messages captured by the dead-letter queue.

musumusu 2 years, 7 months ago

Option D: Dead letter option allow you to recover message from errors after deployment by re-delivering any messages captured by the dead-letter queue. https://cloud.google.com/pubsub/docs/handling-failures#dead_letter_topic why not B, because snapshot is time taking process and if messages were erroneously acknowledged, it will not bring them back. It is useful when you want to secure the current data and want to make changes

wjtb 2 years, 6 months ago

Dead letter queue would help if the messages would not get acknowledged, however here they are talking about messages being erroneously acknowledged. Pub/Sub would interpret the message as being succesfully processed -> they would not end up in the dead-letter queue -> D is wrong

Atnafu 2 years, 10 months ago

B The Seek feature extends subscriber functionality by allowing you to alter the acknowledgement state of messages in bulk. For example, you can replay previously acknowledged messages or purge messages in bulk. In addition, you can copy the state of one subscription to another by using seek in combination with a Snapshot. https://cloud.google.com/pubsub/docs/replay-overview

TNT87 3 years ago

Selected Answer: B

Answer B

PhuocT 3 years ago

Selected Answer: B

should be B.

https://zhangxuetu.online

[202610]

Question #171 Topic 1

You work for a large real estate firm and are preparing 6 TB of home sales data to be used for machine learning. You will use SQL to transform the data and use
BigQuery ML to create a machine learning model. You plan to use the model for predictions against a raw dataset that has not been transformed. How should you set up your workflow in order to prevent skew at prediction time?

A. When creating your model, use BigQuery's TRANSFORM clause to define preprocessing steps. At prediction time, use BigQuery's ML.EVALUATE clause without specifying any transformations on the raw input data. **Most Voted**

B. When creating your model, use BigQuery's TRANSFORM clause to define preprocessing steps. Before requesting predictions, use a saved query to transform your raw input data, and then use ML.EVALUATE.

C. Use a BigQuery view to define your preprocessing logic. When creating your model, use the view as your model training data. At prediction time, use BigQuery's ML.EVALUATE clause without specifying any transformations on the raw input data.

D. Preprocess all data using Dataflow. At prediction time, use BigQuery's ML.EVALUATE clause without specifying any further transformations on the input data.

Correct Answer: A

Community vote distribution



Comments

AWSandeep **Highly Voted** 3 years, 7 months ago

Selected Answer: A

A. When creating your model, use BigQuery's TRANSFORM clause to define preprocessing steps. At prediction time, use BigQuery's ML.EVALUATE clause without specifying any transformations on the raw input data. Using the TRANSFORM clause, you can specify all preprocessing during model creation. The preprocessing is automatically applied during the prediction and evaluation phases of machine learning. Reference: https://cloud.google.com/bigquery-ml/docs/bigqueryml-transform

f74ca0c 1 year, 3 months ago

Selected Answer: C

C. Use a BigQuery view to define your preprocessing logic. When creating your model, use the view as your model training data. At prediction time, use BigQuery's ML.EVALUATE clause without specifying any transformations on the raw input data. Explanation: Preventing Data Skew: Training-serving skew occurs when the transformations applied to

training data are not identically applied to prediction data. Using a BigQuery view ensures consistent preprocessing for both training and prediction. Advantages of BigQuery Views: Views encapsulate preprocessing logic, ensuring that the same transformations are applied whenever the view is queried. By referencing the view during both training and prediction, you eliminate the need for manual transformations and the risk of discrepancies.

SamuelTsch 1 year, 5 months ago

Selected Answer: A

A

Lenifia 1 year, 9 months ago

Selected Answer: B

The key to preventing skew in machine learning models is to ensure that the same data preprocessing steps are applied consistently to both the training data and the prediction data. In option B, the TRANSFORM clause in BigQuery ML is used to define preprocessing steps during model creation, and a saved query is used to apply the same transformations to the raw input data before making predictions. This ensures consistency and prevents skew. The ML.EVALUATE function is then used to evaluate the model's performance on the transformed prediction data. This is the recommended workflow

Matt_108 2 years, 2 months ago

Selected Answer: A

Option A

Prudvi3266 2 years, 11 months ago

Selected Answer: A

A is correct answer if we use TRANSFORM clause in BigQuery no need to use any transform while evaluating and predicting https://cloud.google.com/bigquery/docs/bigqueryml-transform

Kvk117 3 years, 2 months ago

Selected Answer: A

A is the correct answer

jkhong 3 years, 3 months ago

Selected Answer: A

Problem: Skew One thing that I overlooked when answering previously is that B, C does not address skew. When we preprocess our training data, we need to save our scaled factors somewhere, and when performing predictions on our test data, we need to use the scaling factors of our training data to predict the results. ML.EVALUATE already incorporates preprocessing steps for our test data using the saved scaled factors.

GCPSharon 3 years, 5 months ago

Selected Answer: C

Stew prediction time by remove the preprocessing!

TNT87 3 years, 6 months ago

Selected Answer: A

https://cloud.google.com/bigquery-ml/docs/bigqueryml-transform Ans A

ducc 3 years, 7 months ago

Selected Answer: A

This query's nested SELECT statement and FROM clause are the same as those in the CREATE MODEL query. Because the TRANSFORM clause is used in training, you don't need to specify the specific columns and

transformations. They are automatically restored. Reference: https://cloud.google.com/bigquery-ml/docs/bigqueryml-transform

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #172 Topic 1

You are analyzing the price of a company's stock. Every 5 seconds, you need to compute a moving average of the past 30 seconds' worth of data. You are reading data from Pub/Sub and using DataFlow to conduct the analysis. How should you set up your windowed pipeline?

A. Use a fixed window with a duration of 5 seconds. Emit results by setting the following trigger: AfterProcessingTime.pastFirstElementInPane().plusDelayOf (Duration.standardSeconds(30))

B. Use a fixed window with a duration of 30 seconds. Emit results by setting the following trigger: AfterWatermark.pastEndOfWindow().plusDelayOf (Duration.standardSeconds(5))

C. Use a sliding window with a duration of 5 seconds. Emit results by setting the following trigger: AfterProcessingTime.pastFirstElementInPane().plusDelayOf (Duration.standardSeconds(30))

D. Use a sliding window with a duration of 30 seconds and a period of 5 seconds. Emit results by setting the following trigger: AfterWatermark.pastEndOfWindow () Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

vamgcp Highly Voted 2 years, 8 months ago

Selected Answer: D

Option D: Sliding Window: Since you need to compute a moving average of the past 30 seconds' worth of data every 5 seconds, a sliding window is appropriate. A sliding window allows overlapping intervals and is well-suited for computing rolling aggregates. Window Duration: The window duration should be set to 30 seconds to cover the required 30 seconds' worth of data for the moving average calculation. Window Period: The window period or sliding interval should be set to 5 seconds to move the window every 5 seconds and recalculate the moving average with the latest data. Trigger: The trigger should be set to AfterWatermark.pastEndOfWindow() to emit the computed moving average results when the watermark advances past the end of the window. This ensures that all data within the window is considered before emitting the result.

AWSandeep Highly Voted 3 years, 7 months ago

Selected Answer: D

D. Use a sliding window with a duration of 30 seconds and a period of 5 seconds. Emit results by setting the following trigger: AfterWatermark.pastEndOfWindow () Reveal Solution

Anudeep58 1 year, 9 months ago

Selected Answer: D

Option D is the correct configuration because it uses a sliding window of 30 seconds with a period of 5 seconds, ensuring that the moving average is computed every 5 seconds based on the past 30 seconds of data. The trigger AfterWatermark.pastEndOfWindow() ensures timely and accurate results are emitted as the watermark progresses.

Kimich 2 years, 4 months ago

AfterWatermark is an essential triggering condition in Dataflow that allows computations to be triggered based on event time rather than processing time. Then eliminate A&C. Comparing B&D, B will generate outcome every 30 seconds which is not what we want D. Using a sliding window with a duration of 30 seconds and a period of 5 seconds, and setting the trigger as AfterWatermark.pastEndOfWindow(), is a sliding window that generates results every 5 seconds, and each result includes data from the past 30 seconds. In other words, every 5 seconds, you get the average value of the most recent 30 seconds' data, and there is a 5-second overlap between these windows. This is what we want.

pluiedust 3 years, 6 months ago

Selected Answer: D

Moving average ——> sliding window

https://zhangxuetu.online

[202610]

Question #173 Topic 1

You are designing a pipeline that publishes application events to a Pub/Sub topic. Although message ordering is not important, you need to be able to aggregate events across disjoint hourly intervals before loading the results to BigQuery for analysis. What technology should you use to process and load this data to BigQuery while ensuring that it will scale with large volumes of events?

A. Create a Cloud Function to perform the necessary data processing that executes using the Pub/Sub trigger every time a new message is published to the topic.

B. Schedule a Cloud Function to run hourly, pulling all available messages from the Pub/Sub topic and performing the necessary aggregations.

C. Schedule a batch Dataflow job to run hourly, pulling all available messages from the Pub/Sub topic and performing the necessary aggregations.

D. Create a streaming Dataflow job that reads continually from the Pub/Sub topic and performs the necessary aggregations using tumbling windows. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Atnafu **Highly Voted** 3 years, 3 months ago

D TUMBLE=> fixed windows. HOP=> sliding windows. SESSION=> session windows.

musumusu **Highly Voted** 3 years, 1 month ago

why not c ? as data is arriving hourly why we can use batch processing rather than streaming with 1 hour fixed window?

MrMone 2 years, 11 months ago

"you need to be able to aggregate events across disjoint hourly intervals" does not means data is arriving hourly. however, it's tricky! Answer D

ga8our 2 years, 5 months ago

I second your question. Noone who suggests Dataflow streaming (D) has given an explanation why an hourly batch job is insufficient.

ga8our 2 years, 5 months ago

I second your question. Noone who suggests C has given an explanation why an hourly batch job is insufficient.

22c1725 10 months ago

Selected Answer: D

I would go with "D" not (C) for this: while ensuring that it will scale with large volumes of events. Since the main reason people would with streaming job is 4v conisdertion (Volume, Vairity..etc)

baimus 1 year, 6 months ago

Selected Answer: D

Just to provide clarity to people asking "why not C" - the source is a pub/sub. Pub/Sub has a limit of 10 MB or 1000 messages for a single batch publish request, which means that batch dataflow will not necessarily be able to retrieve all messages. If the question had said "there will always be less than 1000 messages and less than 10mb", only then would batch be acceptable.

mayankazyour 1 year, 6 months ago

Selected Answer: D

The question asks for future scalability for large volumes of events, its better to go with streaming dataflow job.

emmylou 2 years, 4 months ago

I just do not understand why this needs to be streamed. I understand that there might be a slight delay using batch processing but there is no indication this is critical data. Can someone please provide your thinking?

vamgcp 2 years, 8 months ago

We can use TUMBLE(1 HOUR) to create hourly windows, where each window contains events from a specific hour.

vamgcp 2 years, 8 months ago

Selected Answer: D

Option D : A streaming Dataflow job is the best way to process and load data from Pub/Sub to BigQuery in real time. This is because streaming Dataflow jobs can scale to handle large volumes of data, and they can perform aggregations using tumbling windows.

devaid 3 years, 5 months ago

Selected Answer: D

Answer D Tumbling Windows = Fixed Windows

TNT87 3 years, 6 months ago

Selected Answer: D

Answer D

AWSandeep 3 years, 7 months ago

Selected Answer: D

D. Create a streaming Dataflow job that reads continually from the Pub/Sub topic and performs the necessary aggregations using tumbling windows. A tumbling window represents a consistent, disjoint time interval in the data stream. Reference: https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#tumbling-windows

https://zhangxuetu.online

[202610]

Question #174 Topic 1

You work for a large financial institution that is planning to use Dialogflow to create a chatbot for the company's mobile app. You have reviewed old chat logs and tagged each conversation for intent based on each customer's stated intention for contacting customer service. About 70% of customer requests are simple requests that are solved within 10 intents. The remaining 30% of inquiries require much longer, more complicated requests. Which intents should you automate first?

A. Automate the 10 intents that cover 70% of the requests so that live agents can handle more complicated requests. **Most Voted**

B. Automate the more complicated requests first because those require more of the agents' time.

C. Automate a blend of the shortest and longest intents to be representative of all intents.

D. Automate intents in places where common words such as 'payment' appear only once so the software isn't confused.

Correct Answer: A

Community vote distribution

A (100%)

Comments

MaxNRG 1 year, 9 months ago

Selected Answer: A

This is the best approach because it follows the Pareto principle (80/20 rule). By automating the most common 10 intents that address 70% of customer requests, you free up the live agents to focus their time and effort on the more complex 30% of requests that likely require human insight/judgement. Automating the simpler high-volume requests first allows the chatbot to handle those easily, efficiently routing only the trickier cases to agents. This makes the best use of automation for high-volume simple cases and human expertise for lower-volume complex issues.

vamgcp 2 years, 2 months ago

Selected Answer: A

Option A : : By automating the intents that cover a significant majority (70%) of customer requests, you target the areas with the highest volume of interactions. This helps reduce the load on live agents, enabling them to focus on more complicated and time-consuming inquiries that require their expertise.

Takshashila 2 years, 3 months ago

Selected Answer: A

A is the answer.

SMASL 3 years ago

Correct answer: A As it states in the documentation: "If your agent will be large or complex, start by building a dialog that only addresses the top level requests. Once the basic structure is established, iterate on the conversation paths to ensure you're covering all of the possible routes an end-user may take." (https://cloud.google.com/dialogflow/cx/docs/concept/agent-design#build-iteratively) Therefore, you should initally automate the 70 % of the requests that are simpler before automating the more complicated ones.

AWSandeep 3 years ago

Selected Answer: A

A. Automate the 10 intents that cover 70% of the requests so that live agents can handle more complicated requests.

https://zhangxuetu.online

[202610]

Question #175 Topic 1

Your company is implementing a data warehouse using BigQuery, and you have been tasked with designing the data model. You move your on-premises sales data warehouse with a star data schema to BigQuery but notice performance issues when querying the data of the past 30 days. Based on Google's recommended practices, what should you do to speed up the query without increasing storage costs?

A. Denormalize the data.

B. Shard the data by customer ID.

C. Materialize the dimensional data in views.

D. Partition the data by transaction date. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

waiebdi **Highly Voted** 2 years, 7 months ago

Selected Answer: D

D is the right answer because it does not increase storage costs. A is not correct because denormalization typically increases the amount of storage needed.

Kimich 1 year, 10 months ago

"Agree with you, denormalize usually increases storage, which may lead to an increase in cost. As for speeding up the query without increasing storage costs, another method is to partition the data by transaction date."

Aman47 1 year, 9 months ago

Bro, you are playing with words now. Gotta read the question fully.

philv 2 years ago

Some might say that Star schema is already denormalized, but it is considered relationnal (hence kind of normalized) from Google's perspective: "BigQuery performs best when your data is denormalized. Rather than preserving a relational schema such as a star or snowflake schema, denormalize your data and take advantage of nested and repeated columns. Nested and repeated columns can maintain relationships without the performance impact of preserving a relational (normalized) schema." I would go for A https://cloud.google.com/bigquery/docs/nested-repeated#when_to_use_nested_and_repeated_columns

philv 1 year, 11 months ago

Changed my mind to D because of the "without increasing storage costs" part.

vamgcp 2 years, 2 months ago

Selected Answer: D

Option D - BigQuery supports partitioned tables, where the data is divided into smaller, manageable portions based on a chosen column (e.g., transaction date). By partitioning the data based on the transaction date, BigQuery can efficiently query only the relevant partitions that contain data for the past 30 days, reducing the amount of data that needs to be scanned.Partitioning does not increase storage costs. It organizes existing data in a more structured manner, allowing for better query performance without any additional storage expenses.

WillemHendr 2 years, 3 months ago

A is not a bad idea, but this questions is written around "please partition first on date", which is common best practice. The "storage" remark is hinting on we are not going to 'explode' the data for the sake of performance.

pcadolini 2 years, 9 months ago

Selected Answer: A

I think better option is [A] considering GCP Documentation: https://cloud.google.com/bigquery/docs/migration/schema-data-overview#denormalization "BigQuery supports both star and snowflake schemas, but its native schema representation is neither of those two. It uses nested and repeated fields instead for a more natural representation of the data ..... Changing your schema to use nested and repeated fields is an excellent evolutionary choice. It reduces the number of joins required for your queries, and it aligns your schema with the BigQuery internal data representation. Internally, BigQuery organizes data using the Dremel model and stores it in a columnar storage format called Capacitor."

NicolasN 2 years, 11 months ago

Selected Answer: D

A sneaky question. [D] Yes - Since data is queried with date criteria, partitioning by transaction date will surely speed it up without further cost. [A] Yes? - Star schema is a denormalized model but as user Reall01 pointed out, the option to use nested and repeated fields can be considered a further denormalization. And if the model hasn't frequently changing dimensions, this kind of denormalization will result in increased performance, according to https://cloud.google.com/bigquery/docs/loading-data#loading_denormalized_nested_and_repeated_data : "In some circumstances, denormalizing your data and using nested and repeated fields doesn't result in increased performance. Avoid denormalization in these use cases: - You have a star schema with frequently changing dimensions" I guess that the person who added this question, had in mind [D] as a correct answer. If the questioner had all the aforementioned under consideration, would state clearly if there are frequently changing dimensions in the schema.

josrojgra 2 years, 11 months ago

Selected Answer: D

Star schema is supported by Big Query but is not the most efficient form, if you should design a schema from scratch google recommend to use nested and repeated fields. In this case, you already have done a migration of the schema and data, so it sounds good and with less effort to do partitioning by transaction date than to redesign the schema. And other aspect to consider is that this is a data warehouse, so is sure that there is an ETL process and if you change the schema you must adapt the ETL process. I vote for D.

devaid 2 years, 12 months ago

Selected Answer: D

Star schema is not denormalized itself, but this assumes you already have moved ur data to big query, because you are querying. So, as BQ is not relational, the data already have been denormalized. I go with D.

learner2610 3 years ago

I think Denormalizing here means ,using big queries native data representation and that is using nested and repeated columns .Thats is the best practice in GCP https://cloud.google.com/bigquery/docs/nested-repeated#example

[Removed] 3 years ago

Selected Answer: D

https://cloud.google.com/bigquery/docs/migration/schema-data-overview#migrating_data_and_schema_from_on-premises_to_bigquery Star schema. This is a denormalized model, where a fact table collects metrics such as order amount, discount, and quantity, along with a group of keys. These keys belong to dimension tables such as customer, supplier, region, and so on. Graphically, the model resembles a star, with the fact table in the center surrounded by dimension tables. Star schema is already denormalized so partition makes more sense going with D

Reall01 3 years ago

If you drill down within that link and land at: https://cloud.google.com/architecture/bigquery-data-warehouse it mentions “ In some cases, you might want to use nested and repeated fields to denormalize your data.” under schema design. Feels like a poorly written question since all depends on what context you take things in as “denormalization”

NicolasN 2 years, 11 months ago

You bring up a valid point. According to denormalization best practices, there is a critical info missing in order to decide whether further denormalization with nested and repeated fields could help, if there are frequently changing dimensions. Here's a quote from https://cloud.google.com/bigquery/docs/loading-data#loading_denormalized_nested_and_repeated_data : "In some circumstances, denormalizing your data and using nested and repeated fields doesn't result in increased performance. Avoid denormalization in these use cases: - You have a star schema with frequently changing dimensions."

GabyB 2 years, 3 months ago

In some circumstances, denormalizing your data and using nested and repeated fields doesn't result in increased performance. For example, star schemas are typically optimized schemas for analytics, and as a result, performance might not be significantly different if you attempt to denormalize further. https://cloud.google.com/bigquery/docs/best-practices-performance-nested

AWSandeep 3 years, 1 month ago

D. Partition the data by transaction date. Star schema is already denormalized.

PhuocT 3 years, 1 month ago

Selected Answer: D

should be D, not A

https://zhangxuetu.online

[202610]

Question #176 Topic 1

You have uploaded 5 years of log data to Cloud Storage. A user reported that some data points in the log data are outside of their expected ranges, which indicates errors. You need to address this issue and be able to run the process again in the future while keeping the original data for compliance reasons. What should you do?

A. Import the data from Cloud Storage into BigQuery. Create a new BigQuery table, and skip the rows with errors.

B. Create a Compute Engine instance and create a new copy of the data in Cloud Storage. Skip the rows with errors.

C. Create a Dataflow workflow that reads the data from Cloud Storage, checks for values outside the expected range, sets the value to an appropriate default, and writes the updated records to a new dataset in Cloud Storage. **Most Voted**

D. Create a Dataflow workflow that reads the data from Cloud Storage, checks for values outside the expected range, sets the value to an appropriate default, and writes the updated records to the same dataset in Cloud Storage.

Correct Answer: C

Community vote distribution

C (100%)

Comments

AWSandeep Highly Voted 3 years, 1 month ago

Selected Answer: C

C. Create a Dataflow workflow that reads the data from Cloud Storage, checks for values outside the expected range, sets the value to an appropriate default, and writes the updated records to a new dataset in Cloud Storage. You can't filter out data using BQ load commands. You must imbed the logic to filter out data (i.e. time ranges) in another decoupled way (i.e. Dataflow, Cloud Functions, etc.). Therefore, A and B add additional complexity and deviates from the Data Lake design paradigm. D is wrong as the question strictly implies that the existing data set needs to be retained for compliance.

FP77 Highly Voted 2 years, 1 month ago

Strange answers... Since when does cloud storage have datasets? Lol Keeping this in mind, the answer must be C, but none is really correcg

Ben_oso 9 months, 1 week ago

Selected Answer: C

with C the user dont see the data with errors, all clean

ea2023 1 year, 7 months ago

why not D if the versioning is activated while creating your bucket ?

MaxNRG 1 year, 9 months ago

Selected Answer: C

Option C is the best approach in this situation. Here is why: Option A would remove data which may be needed for compliance reasons. Keeping the original data is preferred. Option B makes a copy of the data but still removes potentially useful records. Additional storage costs would be incurred as well. Option C uses Dataflow to clean the data by setting out of range values while keeping the original data intact. The fixed records are written to a new location for further analysis. This meets the requirements. Option D writes the fixed data back to the original location, overwriting the original data. This would violate the compliance needs to keep the original data untouched. So option C leverages Dataflow to properly clean the data while preserving the original data for compliance, at reasonable operational costs. This best achieves the stated requirements.

AzureDP900 2 years, 9 months ago

C. Create a Dataflow workflow that reads the data from Cloud Storage, checks for values outside the expected range, sets the value to an appropriate default, and writes the updated records to a new dataset in Cloud Storage.

PhuocT 3 years, 1 month ago

Selected Answer: C

C is correct

https://zhangxuetu.online

[202610]

Question #177 Topic 1

You want to rebuild your batch pipeline for structured data on Google Cloud. You are using PySpark to conduct data transformations at scale, but your pipelines are taking over twelve hours to run. To expedite development and pipeline run time, you want to use a serverless tool and SOL syntax. You have already moved your raw data into Cloud Storage. How should you build the pipeline on Google Cloud while meeting speed and processing requirements?

A. Convert your PySpark commands into SparkSQL queries to transform the data, and then run your pipeline on Dataproc to write the data into BigQuery.

B. Ingest your data into Cloud SQL, convert your PySpark commands into SparkSQL queries to transform the data, and then use federated quenes from BigQuery for machine learning.

C. Ingest your data into BigQuery from Cloud Storage, convert your PySpark commands into BigQuery SQL queries to transform the data, and then write the transformations to a new table. **Most Voted**

D. Use Apache Beam Python SDK to build the transformation pipelines, and write the data into BigQuery.

Correct Answer: C

Community vote distribution



Comments

devaid **Highly Voted** 2 years, 12 months ago

Selected Answer: C

The question is C but not because the SQL Syntax, as you can perfectly use SparkSQL on Dataproc reading files from GCS. It's because the "serverless" requirement.

GCP001 1 year, 8 months ago

Selected Answer: A

A) Looks more suitable , serverless approach for handling and performance.

MaxNRG 1 year, 9 months ago

Selected Answer: C

Option C is the best approach to meet the stated requirements. Here's why: BigQuery SQL provides a fast, scalable, and serverless method for transforming structured data, easier to develop than PySpark. Directly ingesting the raw Cloud Storage data into BigQuery avoids needing an intermediate processing cluster like Dataproc. Transforming the data via BigQuery SQL queries will be faster than PySpark, especially since the data is already loaded into BigQuery. Writing the transformed results to a new BigQuery table keeps the original raw data intact and provides a clean output. So migrating to BigQuery SQL for transformations provides a fully managed serverless architecture that can

significantly expedite development and reduce pipeline runtime versus PySpark. The ability to avoid clusters and conduct transformations completely within BigQuery is the most efficient approach here.

MoeHaydar 2 years, 2 months ago

Selected Answer: C

Note: Dataproc by itself is not serverless https://cloud.google.com/dataproc-serverless/docs/overview

Prudvi3266 2 years, 5 months ago

Selected Answer: C

because of serverless nature

musumusu 2 years, 7 months ago

Answer C: need to setup SQL based job means transformation in not very complex. And Biqquery sql are faster than spark sql context. (google claims) However, i will make a test by myself to check it.

maci_f 2 years, 8 months ago

Selected Answer: A

In the GCP Machine Learning Engineer practice question (Q4) there's the same question with similar answers and the correct answer is A since B "is incorrect, here transformation is done on Cloud SQL, which wouldn't scale the process" and C "is incorrect as this process wouldn't scale the data transformation routine. And, it is always better to transform data during ingestion": https://medium.com/@gcpguru/google-google-cloud-professional-machine-learning-engineer-practice-questions-part-1-3ee4a2b3f0a4

evanfebrianto 2 years, 4 months ago

Dataproc is not a serverless tool unless it mentions "Dataproc Serverless" explicitly.

Atnafu 2 years, 10 months ago

C D-is incorrect because you are rebuild your batch pipeline for structured data on Google Cloud.

Atnafu 2 years, 10 months ago

A could be answer if it was Dataproc serverless and no conversion of code. Dp serverless support: scala,pyspark,sparksql and SparkR

TNT87 3 years ago

Selected Answer: C

This same question is there on Google's Professional Machine Learning Engineer, Question 4 Answer is C.

Wasss123 3 years ago

Selected Answer: C

I choose C BigQuery SQL is more performant but more expensive. Here, it's a performance issue (time reduction)
Source : https://medium.com/paypal-tech/comparing-bigquery-processing-and-spark-dataproc-4c90c10e31ac

John_Pongthorn 3 years ago

C is the most likely , bigquery is severless and sql D is dataflow severless but it is wrong at using python sdk but using sql beam rthen it will be correct

TNT87 3 years ago

Answer C

ducc 3 years ago

Selected Answer: A

A - You have to maintain PySpark Code -> Proc

ducc 3 years ago

After thinking a while, I think the question is not clear enough. To be honest

ducc 3 years ago

A or C. I go for C because they said they want to use SQL syntax...

AWSandeep 3 years, 1 month ago

Selected Answer: C

C. Ingest your data into BigQuery from Cloud Storage, convert your PySpark commands into BigQuery SQL queries to transform the data, and then write the transformations to a new table. Keys: "Serverless" and "SQL"

AWSandeep 3 years, 1 month ago

Changing answer to A as this is a new question referring to Dataproc Serverless. Dataproc Serverless for Spark batch workloads supports Spark SQL. Why modify ETL to ELT and convert PySpark to BigQuery SQL when it can be similar to a lift-and-shift?

Atnafu 2 years, 10 months ago

Dataproc is diffrent than Dataproc Serveless. This question is talking about dataproc. By the way dp serverless support both pyspark and sparkSql no need of conversion. C is best answer

ducc 3 years ago

The question said "use SQL syntax" C might still correct

https://zhangxuetu.online

[202610]

Question #178 Topic 1

You are testing a Dataflow pipeline to ingest and transform text files. The files are compressed gzip, errors are written to a dead-letter queue, and you are using
SideInputs to join data. You noticed that the pipeline is taking longer to complete than expected; what should you do to expedite the Dataflow job?

A. Switch to compressed Avro files.

B. Reduce the batch size.

C. Retry records that throw an error.

D. Use CoGroupByKey instead of the SideInput. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

John_Pongthorn **Highly Voted** 3 years ago

Selected Answer: D

D: it is most likely. There are a lot of reference doc to tell about comparison between them
https://cloud.google.com/architecture/building-production-ready-data-pipelines-using-dataflow-developing-and-testing#choose_correctly_between_side_inputs_or_cogroupbykey_for_joins
https://cloud.google.com/blog/products/data-analytics/guide-to-common-cloud-dataflow-use-case-patterns-part-2
https://stackoverflow.com/questions/58080383/sideinput-i-o-kills-performance

MaxNRG 1 year, 9 months ago

Selected Answer: D

To expedite the Dataflow job that involves ingesting and transforming text files, especially if the pipeline is taking longer than expected, the most effective strategy would be: D. Use CoGroupByKey instead of the SideInput.

MaxNRG 1 year, 9 months ago

Here's why this approach is beneficial: 1. Efficiency in Handling Large Datasets: SideInputs are not optimal for large datasets because they require that the entire dataset be available to each worker. This can lead to performance bottlenecks, especially if the dataset is large. CoGroupByKey, on the other hand, is more efficient for joining large datasets because it groups elements by key and allows the pipeline to process each key-group separately. 2. Scalability: CoGroupByKey is more scalable than SideInputs for large-scale data processing. It distributes the workload more evenly across the Dataflow workers, which can significantly improve the performance of your pipeline. 3. Better Resource Utilization: By using CoGroupByKey, the Dataflow job can make better use of its resources, as it

doesn't need to replicate the entire dataset to each worker. This results in faster processing times and better overall efficiency.

MaxNRG 1 year, 9 months ago

The other options may not be as effective: • A (Switch to compressed Avro files): While Avro is a good format for certain types of data processing, simply changing the file format from gzip to Avro may not address the underlying issue causing the delay, especially if the problem is related to the way data is being joined or processed. • B (Reduce the batch size): Reducing the batch size could potentially increase overhead and might not significantly improve the processing time, especially if the bottleneck is due to the method of data joining. • C (Retry records that throw an error): Retrying errors could be useful in certain contexts, but it's unlikely to speed up the pipeline if the delay is due to inefficiencies in data processing methods like the use of SideInputs.

musumusu 2 years, 7 months ago

Answer: B, reducing the batch size improve the speed performance also improve cpu utilisation. Dead letter queues are generated for messages that are errorrly acknowledged and its good to use sideinputs for that to check small amount of errors in memory. Cogroupbykey is not necessary for error messages. I see only batch size that can be customized to improve the performance. In practical use case: you check these tools Stackdriver Monitoring and Logging, Cloud Trace, and Cloud Profiler. and try to find the cause, if its file type issue in compression, or batch size.

Atnafu 2 years, 9 months ago

D Flatten will just merge all results into a single PCollection. To join them you can use CoGroupByKey

TNT87 2 years, 11 months ago

Selected Answer: A

When optimizing for load speed, Avro file format is preferred. Avro is a binary row-based format which can be split and read in parallel by multiple slots including compressed files.

devaid 2 years, 11 months ago

that is for Big Query isn't?

TNT87 2 years, 8 months ago

datflow can use avro format sir. streaming or batching to bigquery in avro format it can

TNT87 2 years, 11 months ago

https://cloud.google.com/blog/topics/developers-practitioners/bigquery-explained-data-ingestion

devaid 2 years, 12 months ago

Selected Answer: D

D probably, side inputs have to fit in memory. If the p-collection in the side input doesn't fit well in memory it's better to use CoGroupByKey.

TNT87 3 years ago

Selected Answer: A

Answer A the same question is in number 70 you transform the files to Avro using Dataflow

KC_go_reply 2 years, 3 months ago

Avro requires the data to be at least semi-structured, because it wants a fixed schema. Text files are unstructured data, therefore it doesn't make sense to use Avro files for them

csd1fggfhfgvh234 3 years ago

A switching to avro. No serialisation

TNT87 3 years ago

Switch to Avro format Answer A

TNT87 3 years ago

https://docs.confluent.io/platform/current/schema-registry/serdes-develop/serdes-avro.html

YorelNation 3 years ago

Selected Answer: D

D probably, side inputs have to fit in memory. If the p-collection in the side input doesn't fit well in memory it's better to use CoGroupByKey.

AWSandeep 3 years, 1 month ago

Selected Answer: B

B. Reduce the batch size.

https://zhangxuetu.online

[202610]

Question #179 Topic 1

You are building a real-time prediction engine that streams files, which may contain PII (personal identifiable information) data, into Cloud Storage and eventually into BigQuery. You want to ensure that the sensitive data is masked but still maintains referential integrity, because names and emails are often used as join keys.
How should you use the Cloud Data Loss Prevention API (DLP API) to ensure that the PII data is not accessible by unauthorized individuals?

A. Create a pseudonym by replacing the PII data with cryptogenic tokens, and store the non-tokenized data in a locked-down button.

B. Redact all PII data, and store a version of the unredacted data in a locked-down bucket.

C. Scan every table in BigQuery, and mask the data it finds that has PII.

D. Create a pseudonym by replacing PII data with a cryptographic format-preserving token. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

ToiToi 1 year, 5 months ago

Selected Answer: D

Why other options are not as suitable: A (Cryptogenic tokens and locked-down bucket): While this provides some protection, storing the non-tokenized data in a separate bucket adds complexity and risk. B (Redaction and locked-down bucket): Redaction removes sensitive data entirely, which might limit its usefulness for analysis and other purposes. C (Scanning and masking in BigQuery): This approach might be less efficient than masking the data during the streaming process before it reaches BigQuery.

SamuelTsch 1 year, 5 months ago

Selected Answer: D

I would like to go with D. If the data could be deindentified later by the token, why should we store the data in a locked-down bucket?

GCP001 2 years, 2 months ago

Selected Answer: D

D> Looks more suitable as it will handle Referential integrity. https://cloud.google.com/dlp/docs/pseudonymization

pss111423 2 years, 4 months ago

answer A https://cloud.google.com/dlp/docs/transformations-reference Replaces an input value with a token, or surrogate value, of the same length using AES in Synthetic Initialization Vector mode (AES-SIV). This transformation

method, unlike format-preserving tokenization, has no limitation on supported string character sets, generates identical tokens for each instance of an identical input value, and uses surrogates to enable re-identification given the original encryption key.

akg001 2 years, 7 months ago

Selected Answer: D

D is correct.

cetanx 2 years, 9 months ago

Selected Answer: B

I've also asked to GPT but I had to remind the hard condition "names and emails are often used as join keys". It changed the answer to "B" after 3rd iteration. masking all PII data may not satisfy the requirement of using names and emails as join keys, as the data is obfuscated and cannot be used for accurate join operations. In this approach, you would redact or remove the sensitive PII data, such as names and emails, from the dataset that will be used for real-time processing and analysis. The redacted data would be stored in the primary dataset to ensure that sensitive information is not accessible. Additionally, you would create a copy of the original dataset with the PII data still intact, but this copy would be stored in a locked-down bucket with restricted access. This ensures that authorized individuals who need access to the unredacted data for specific purposes, such as join operations, can retrieve it from the secured location.

cetanx 2 years, 9 months ago

made a typo up there, it has to be A

Oleksandr0501 2 years, 11 months ago

gpt: The recommended approach for using the Cloud Data Loss Prevention API (DLP API) to protect sensitive PII data while maintaining referential integrity is to create pseudonyms by replacing the PII data with cryptographic format-preserving tokens. This approach ensures that sensitive data is not accessible by unauthorized individuals, while still preserving the format and length of the original data, which is essential for maintaining referential integrity. Replacing PII data with cryptogenic tokens, as mentioned in option A, is not recommended because cryptogenic tokens are not necessarily format-preserving, and this could affect the accuracy of data joins. Therefore, option D is the best approach for using the DLP API to ensure that PII data is not accessible by unauthorized individuals while still maintaining referential integrity.

loicrichonnier 2 years, 10 months ago

You shouldn't use ChatGPT as a source, the data used are not up to date and for such complex question a predicting text chatbot can help but, it's better to refer to the google documentation.

Oleksandr0501 2 years, 10 months ago

that`s why i always mark "gpt", when copy from there... i know, thx also, it might be A. Or D... Confusing question.

Prudvi3266 2 years, 11 months ago

Selected Answer: D

here catch is "cryptographic" key

musumusu 3 years, 1 month ago

Answer D, key word - "referential integrity" use format preserve option, it keeps same length of the value and last four digits of your value in column

tunstila 3 years, 2 months ago

Selected Answer: D

The answer is D

nkit 3 years, 3 months ago

Selected Answer: D

I believe "Format preserving token" in option D makes it easier choice for me

PrashantGupta1616 3 years, 3 months ago

Selected Answer: D

D looks right

jkhong 3 years, 3 months ago

Selected Answer: A

Question is super tricky, B and C are not the answers since they do not maintain referential integrity. For D, it does preserve the length of input. But since we are only concerned with referencing during joins, there is no point of maintaining the length anyway. Also, characters must be encoded as ASCII, this means that the name and email must be within the 256 character set. which is further limited to the alphabet characters, i.e. 94 characters. (https://cloud.google.com/dlp/docs/transformations-reference#crypto) Names nowadays do not just have ASCII characters but unicode as well, so D will not necessarily work all the time.

Atnafu 3 years, 3 months ago

D is the answer Pseudonymization is a de-identification technique that replaces sensitive data values with cryptographically generated tokens. Keywords: You want to ensure that the sensitive data is masked but still maintains referential integrity Part1- data is masked-Create a pseudonym by replacing PII data with a cryptographic token Part 2 still maintains referential integrity- with a cryptographic format-preserving token A Not an answer because the locked-down button does not seem to google cloud word

juliobs 3 years ago

"button" is just a typo for "bucket"

dish11dish 3 years, 4 months ago

Selected Answer: D

Though both option A nad D maintains referential integrity,question is why you wnat to keep untokenize data in GCS,best way is option D which even support Reversible feature which is not supported by option A refer chart in reference document. reference:- https://cloud.google.com/dlp/docs/pseudonymization

cloudmon 3 years, 4 months ago

Selected Answer: D

It's D. "You want to ensure that the sensitive data is masked but still maintains referential integrity." They don't ask you to also keep the original data (which answer A relates to). Also, format-preservation is important in this case.

cloudmon 3 years, 4 months ago

And, answer A does not include format preservation, which would lose referential integrity.

NicolasN 3 years, 4 months ago

I think that this isn't true. Look at the table https://cloud.google.com/dlp/docs/transformations-reference#transformation_methods and notice the 6th line "Pseudonymization by replacing input value with cryptographic hash" (which refers to the case of answer [A]). Referential integrity is preserved.

NicolasN 3 years, 4 months ago

Selected Answer: A

[B] and [C] aren't correct since they don't preserve referential integrity. [A] describes, in other words, Cryptographic hashing, where the sensitive data is replaced with a hashed value. The hashed value can't be reversed (https://cloud.google.com/dlp/docs/transformations-reference#crypto-hashing) so the phrase "store the non-tokenized

data in a locked-down button (bucket)" ensures that data can be restored if needed. [D] seems to be a valid option too. However, in https://cloud.google.com/dlp/docs/pseudonymization#fpe-ffx, there is a warning: "FPE provides fewer security guarantees compared to other deterministic encryption methods such as AES- SIV ... ... For these reasons, Google strongly recommends using deterministic encryption with AES-SIV instead of FPE for all security sensitive use cases" Since there is no option to select Deterministic Encryption, and the question doesn't require to preserve the format of the data (keep the same length of data), I choose [A] as a more secure approach.

NicolasN 3 years, 4 months ago

The table in https://cloud.google.com/dlp/docs/transformations-reference#transformation_methods shows which transformation preserves referential integrity

wan2three 2 years, 7 months ago

From here I see if A really meant Cryptographic hashing then it also satisfy referential integrity https://cloud.google.com/dlp/docs/pseudonymization#:~:text=following%20the%20table.-,Deterministic%20encryption%20using%20AES%2DSIV,-Format%20preserving%20encryption However, I cant see why A means Crpyptographic Hashing, no defition I can find online at all.

https://zhangxuetu.online

[202610]

Question #180 Topic 1

You are migrating an application that tracks library books and information about each book, such as author or year published, from an on-premises data warehouse to BigQuery. In your current relational database, the author information is kept in a separate table and joined to the book information on a common key. Based on Google's recommended practice for schema design, how would you structure the data to ensure optimal speed of queries about the author of each book that has been borrowed?

A. Keep the schema the same, maintain the different tables for the book and each of the attributes, and query as you are doing today.

B. Create a table that is wide and includes a column for each attribute, including the author's first name, last name, date of birth, etc.

C. Create a table that includes information about the books and authors, but nest the author fields inside the author column. **Most Voted**

D. Keep the schema the same, create a view that joins all of the tables, and always query the view.

Correct Answer: C

Community vote distribution



Comments

musumusu **Highly Voted** 1 year, 7 months ago

C if data is time based or sequential, find partition and cluster option if data is not time based, always look for denomalize / nesting option.

AzureDP900 1 year, 9 months ago

C. Create a table that includes information about the books and authors, but nest the author fields inside the author column.

Atnafu 1 year, 10 months ago

C Best practice: Use nested and repeated fields to denormalize data storage and increase query performance.

dish11dish 1 year, 10 months ago

Selected Answer: C

Use nested and repeated fields to denormalize data storage which will increase query performance.BigQuery doesn't require a completely flat denormalization. You can use nested and repeated fields to maintain relationships

Thobm 2 years ago

Selected Answer: C

https://cloud.google.com/bigquery/docs/best-practices-performance-nested

ducc 2 years ago

Selected Answer: C

C is correct

AWSandeep 2 years, 1 month ago

Selected Answer: C

C. Create a table that includes information about the books and authors, but nest the author fields inside the author column.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #181 Topic 1

You need to give new website users a globally unique identifier (GUID) using a service that takes in data points and returns a GUID. This data is sourced from both internal and external systems via HTTP calls that you will make via microservices within your pipeline. There will be tens of thousands of messages per second and that can be multi-threaded. and you worry about the backpressure on the system. How should you design your pipeline to minimize that backpressure?

A. Call out to the service via HTTP.

B. Create the pipeline statically in the class definition.

C. Create a new object in the startBundle method of DoFn.

D. Batch the job into ten-second increments. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

John_Pongthorn **Highly Voted** 3 years ago

Selected Answer: D

D: I have insisted on this choice all aling. please read find the keyword massive backpressure https://cloud.google.com/blog/products/data-analytics/guide-to-common-cloud-dataflow-use-case-patterns-part-1 if the call takes on average 1 sec, that would cause massive backpressure on the pipeline. In these circumstances you should consider batching these requests, instead.

NicolasN 2 years, 10 months ago

Thanks for sharing, you found exactly the same problem! The document defitely proposes batching for this scenario. I'm quoting another part from the same example that would be useful for a similar question with different conditions: - If you're using a client in the DoFn that has heavy instantiation steps, rather than create that object in each DoFn call: * If the client is thread-safe and serializable, create it statically in the class definition of the DoFn. * If it's not thread-safe, create a new object in the startBundle method of DoFn. By doing so, the client will be reused across all elements of a bundle, saving initialization time.

Atnafu 2 years, 10 months ago

By the way if you see the shared Pseudocode, it's talking about start bundle and finish bundle of DoFn. The question is which one to choose to avoid back pressure? you can see why you need to choose bundle instead of batching in below link Batching introduces some processing overhead as well as the need for a magic number to determine the key space. Instead, use the StartBundle and FinishBundle lifecycle elements to batch your data. With these options, no shuffling is needed. https://cloud.google.com/dataflow/docs/tutorials/ecommerce-java#micro-batch-calls

NicolasN 2 years, 10 months ago

Valid points. but I don't change my mind, regarding the requirements of this particular question: - multi-threaded ability - no mention of heavy initialization steps or a lot of disk I/O (where shuffling might be a problem). And especially the excerpt: "if the call takes on average 1 sec, that would cause massive backpressure on the pipeline. In these circumstances you should consider batching these requests, instead" It's like the guys that authored the question had this sentence in front of their eyes.

John_Pongthorn Highly Voted 3 years ago

Selected Answer: D

D All guys ,pls read carefully on Pattern: Calling external services for data enrichment https://cloud.google.com/blog/products/data-analytics/guide-to-common-cloud-dataflow-use-case-patterns-part-1 A , B , C all of them are solution for norma case but if you need to stand for backpressure, in last sector in Note : Note: When using this pattern, be sure to plan for the load that's placed on the external service and any associated backpressure. For example, imagine a pipeline that's processing tens of thousands of messages per second in steady state. If you made a callout per element, you would need the system to deal with the same number of API calls per second. Also, if the call takes on average 1 sec, that would cause massive backpressure on the pipeline. In these circumstances, you should consider batching these requests, instead. Anyone can share ideas to debate with me.

MaxNRG 1 year, 9 months ago

Selected Answer: D

Option D is the best approach to minimize backpressure in this scenario. By batching the jobs into 10-second increments, you can throttle the rate at which requests are made to the external GUID service. This prevents too many simultaneous requests from overloading the service. Option A would not help with backpressure since it just makes synchronous HTTP requests as messages arrive. Similarly, options B and C don't provide any inherent batching or throttling mechanism. Batching into time windows is a common strategy in stream processing to deal with high velocity data. The 10-second windows allow some buffering to happen, rather than making a call immediately for each message. This provides a natural throttling that can be tuned based on the external service's capacity.

MaxNRG 1 year, 9 months ago

To design a pipeline that minimizes backpressure, especially when dealing with tens of thousands of messages per second in a multi-threaded environment, it's important to consider how each option affects system performance and scalability. Let's examine each of your options:

MaxNRG 1 year, 9 months ago

A. Call out to the service via HTTP: Making HTTP calls to an external service for each message can introduce significant latency and backpressure, especially at high throughput. This is due to the overhead of establishing a connection, waiting for the response, and handling potential network delays or failures.

MaxNRG 1 year, 9 months ago

B. Create the pipeline statically in the class definition: While this approach can improve initialization time and reduce overhead during execution, it doesn't directly address the issue of backpressure caused by high message throughput.

MaxNRG 1 year, 9 months ago

C. Create a new object in the startBundle method of DoFn: This approach is typically used in Apache Beam to initialize resources before processing a bundle of elements. While it can optimize resource usage and performance within each bundle, it doesn't inherently solve the backpressure issue caused by high message rates.

MaxNRG 1 year, 9 months ago

D. Batch the job into ten-second increments: Batching messages can be an effective way to reduce backpressure. By grouping multiple messages into larger batches, you can reduce the frequency of external calls and distribute the

processing load more evenly over time. This can lead to more efficient use of resources and potentially lower latency, as the system spends less time waiting on external services.

MaxNRG 1 year, 9 months ago

Given these considerations, option D (Batch the job into ten-second increments) seems to be the most effective strategy for minimizing backpressure in your scenario. By batching messages, you can reduce the strain on your pipeline and external services, making the system more resilient and scalable under high load. However, the exact batch size and interval should be fine-tuned based on the specific characteristics of your workload and the capabilities of the external systems you are interacting with. Additionally, it's important to consider other strategies in conjunction with batching, such as implementing efficient error handling, load balancing, and potentially using asynchronous I/O for external HTTP calls to further optimize performance and minimize backpressure.

izekc 2 years, 5 months ago

Selected Answer: D

Option C is not correct because it does not address the issue of backpressure. Creating a new object in the startBundle method of DoFn will not help to reduce the number of calls that are made to the service, which can lead to backpressure. Here are some reasons why C is not correct: Creating a new object in the startBundle method of DoFn is not a scalable solution. As the number of messages increases, the number of objects that need to be created will also increase. This can lead to performance problems and memory usage issues. Creating a new object in the startBundle method of DoFn does not address the issue of backpressure. The service may still experience backpressure if the number of messages exceeds the service's capacity. A better solution would be to use batching to reduce the number of calls that are made to the service. This can help to improve performance and reduce backpressure.

Oleksandr0501 2 years, 5 months ago

gpt: Option C is a better approach as it allows for object creation to occur in a more controlled manner within the DoFn, potentially reducing the pressure on the system. However, it could still create a large number of objects depending on the rate of incoming messages. Option D of batching the job into ten-second increments can also be a good solution to reduce backpressure on the system. This way, you can limit the number of messages being processed at any given time, which can help prevent bottlenecks and reduce the likelihood of backpressure. Therefore, the best approach would be to combine options C and D, creating a new object in the startBundle method of a DoFn, and batching the job into smaller time increments, such as 10 seconds. This way, you can control the rate of object creation and processing, which can help minimize backpressure on the system.

Oleksandr0501 2 years, 5 months ago

another vague question, as we see... so, i`ll choose D... if i get this test "However, depending on the specifics of your use case, one option may be better suited than the other. For example, if you have a high volume of incoming messages with occasional spikes, option D of batching the job into smaller time increments may be more effective in managing the load. On the other hand, if the incoming messages are more evenly distributed over time, option C of creating a new object in the startBundle method of DoFn may be a better option. Ultimately, it may be necessary to experiment with both approaches and determine which one works best for your specific use case."

juliobs 2 years, 6 months ago

Selected Answer: D

D works. Could be C, but who said that the pipeline is in Dataflow/Beam?

musumusu 2 years, 7 months ago

Answer C batch increment in 10 sec, can improve load balancing, but overall back pressure (messages are generating more than consuming or publishing) in this case startBundle in DoFn or find other options in future like caching, load shedding (prioritising message flow), messege queuing These options handle backpressure.. If your cpu is performing bad then go with change in batch increment timing

maci_f 2 years, 8 months ago

Selected Answer: D

I was hesitating between C and D, but then I realised this: https://cloud.google.com/blog/products/data-analytics/guide-to-common-cloud-dataflow-use-case-patterns-part-1 Here is says "If it's not thread-safe, create a new object in the startBundle method of DoFn." The task explicitly says "There will be tens of thousands of messages per second and that can be multi-threaded." Correct me if I'm wrong, but multi-threaded == thread-safe. Therefore, no need to go for the C approach.

Atnafu 2 years, 10 months ago

C C is an answer because First of all, no doubt that we should avoid single call of element that's why we use multi-threading else it overwhelm an external service endpoint.To avoid this issue, batch calls to external systems. Batch calls has also issue:GroupByKey transform or Apache Beam Timer API. these approaches both require shuffling, which introduces some processing overhead as well as the need for a magic number to determine the key space. Instead, use the StartBundle and FinishBundle lifecycle elements to batch your data. With these options, no shuffling is needed. Source: https://cloud.google.com/dataflow/docs/tutorials/ecommerce-java#micro-batch-calls https://cloud.google.com/blog/products/data-analytics/guide-to-common-cloud-dataflow-use-case-patterns-part-1 Summary: StartBundle and FinishBundle do batch with no shuffling

AHUI 3 years ago

Ans C: reference https://cloud.google.com/architecture/e-commerce/patterns/batching-external-calls

SMASL 3 years ago

Selected Answer: C

Based on the answers in this discussion thread, I would go for C. The most important link to support this choice is as following: https://cloud.google.com/architecture/e-commerce/patterns/batching-external-calls

Thobm 3 years ago

Selected Answer: D

Beam docs recommend batching https://beam.apache.org/documentation/patterns/grouping-elements-for-efficient-external-service-calls/

John_Pongthorn 3 years ago

C , It is straight foward , You can take a look at https://cloud.google.com/blog/products/data-analytics/guide-to-common-cloud-dataflow-use-case-patterns-part-1 Pattern : Calling external services for data enrichmen

TNT87 3 years ago

Answer C https://cloud.google.com/blog/products/data-analytics/guide-to-common-cloud-dataflow-use-case-patterns-part-1

TNT87 3 years ago

https://cloud.google.com/architecture/e-commerce/patterns/batching-external-calls To support choice C

YorelNation 3 years ago

Selected Answer: C

i think you are right gg

nwk 3 years ago

How about C? https://cloud.google.com/architecture/e-commerce/patterns/batching-external-calls

AWSandeep 3 years ago

Selected Answer: D

D. Batch the job into ten-second increments.

https://zhangxuetu.online

[202610]

Question #182 Topic 1

You are migrating your data warehouse to Google Cloud and decommissioning your on-premises data center. Because this is a priority for your company, you know that bandwidth will be made available for the initial data load to the cloud. The files being transferred are not large in number, but each file is 90 GB.
Additionally, you want your transactional systems to continually update the warehouse on Google Cloud in real time. What tools should you use to migrate the data and ensure that it continues to write to your warehouse?

A. Storage Transfer Service for the migration; Pub/Sub and Cloud Data Fusion for the real-time updates

B. BigQuery Data Transfer Service for the migration; Pub/Sub and Dataproc for the real-time updates

C. gsutil for the migration; Pub/Sub and Dataflow for the real-time updates **Most Voted**

D. gsutil for both the migration and the real-time updates

Correct Answer: C

Community vote distribution

C (82%) A (18%)

Comments

AWSandeep **Highly Voted** 3 years, 1 month ago

Selected Answer: C

C. gsutil for the migration; Pub/Sub and Dataflow for the real-time updates Use Gsutil when there is enough bandwidth to meet your project deadline for less than 1 TB of data. Storage Transfer Service is for much larger volumes for migration. Moreover, Cloud Data Fusion and Dataproc are not ideal for real-time updates. BigQuery Data Transfer Service does not support all on-prem sources.

shangning007 1 year, 3 months ago

Selected Answer: A

According to the latest documentation, "Generally, you should use gcloud storage commands instead of gsutil commands. The gsutil tool is a legacy Cloud Storage CLI and minimally maintained." We should remove the presence of gsutil in the questions.

TVH_Data_Engineer 1 year, 9 months ago

Selected Answer: C

Considering the requirement for handling large files and the need for real-time data integration, Option C (gsutil for the migration; Pub/Sub and Dataflow for the real-time updates) seems to be the most appropriate. gsutil will effectively handle the large file transfers, while Pub/Sub and Dataflow provide a robust solution for real-time data capture and processing, ensuring continuous updates to your warehouse on Google Cloud.

MaxNRG 1 year, 9 months ago

Selected Answer: C

Option C is the best choice given the large file sizes for the initial migration and the need for real-time updates after migration. Specifically: gsutil can transfer large files in parallel over multiple TCP connections to maximize bandwidth. This works well for the 90GB files during initial migration. Pub/Sub allows real-time messaging of updates that can then be streamed into Cloud Dataflow. Dataflow provides scalable stream processing to handle transforming and writing those updates into BigQuery or other sinks.

MaxNRG 1 year, 9 months ago

Option A is incorrect because Storage Transfer Service is better for scheduled batch transfers, not ad hoc large migrations. Option B is incorrect because BigQuery Data Transfer Service is more focused on scheduled replication jobs, not ad hoc migrations. Option D would not work well for real-time updates after migration is complete. So option C leverages the right Google cloud services for the one-time migration and ongoing real-time processing.

xiangbobopopo 1 year, 11 months ago

Selected Answer: C

agree with C

TNT87 3 years ago

https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#gsutil_for_smaller_transfers_of_on-premises_data Answer C

YorelNation 3 years ago

Selected Answer: C

C seems legit

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #183 Topic 1

You are using Bigtable to persist and serve stock market data for each of the major indices. To serve the trading application, you need to access only the most recent stock prices that are streaming in. How should you design your row key and tables to ensure that you can access the data with the simplest query?

A. Create one unique table for all of the indices, and then use the index and timestamp as the row key design. **Most Voted**

B. Create one unique table for all of the indices, and then use a reverse timestamp as the row key design.

C. For each index, have a separate table and use a timestamp as the row key design.

D. For each index, have a separate table and use a reverse timestamp as the row key design.

Correct Answer: A

Community vote distribution



Comments

John_Pongthorn **Highly Voted** 3 years, 6 months ago

This is special case , plese Take a look carefully the below link and read at last paragraph at the bottom of this comment, let everyone share idea, We will go with B, C https://cloud.google.com/bigtable/docs/schema-design#time-based Don't use a timestamp by itself or at the beginning of a row key, because this will cause sequential writes to be pushed onto a single node, creating a hotspot. If you usually retrieve the most recent records first, you can use a reversed timestamp in the row key by subtracting the timestamp from your programming language's maximum value for long integers (in Java, java.lang.Long.MAX_VALUE). With a reversed timestamp, the records will be ordered from most recent to least recent.

Mcloudgirl 3 years, 4 months ago

I agree, based on the docs, B. Leading with a non-reversed timestamp will lead to hotspotting, reversed is the way to go.

Ryannn23 1 year, 1 month ago

According to the link you provided: If you usually retrieve the most recent records first in your queries, a pattern to consider is using reversed timestamps in the row key. This pattern causes rows to be ordered from most recent to least recent, so more recent data is earlier in the table. ---------- READ CAREFULY ---------- As with any timestamp, avoid starting a row key with a reversed timestamp so that you don't cause hotspots. ---------- You can get a reversed timestamp by subtracting the timestamp from your programming language's maximum value for long integers (in Java, java.lang.Long.MAX_VALUE). Hence, starting row key with timestamp should be avoided (normal or reversed). That leads to answer A, which is a best practice.

arien_chen Highly Voted 2 years, 7 months ago

Selected Answer: D

Option B using reverse timestamp only, this is not the answer. the right answer should be using the index and revers timestamp as the row key. So, Option D is the only answer, because not A,B,C .

imrane1995 8 months, 2 weeks ago

Selected Answer: D

When designing for Bigtable, particularly for time-series data like stock prices, your row key design directly impacts read efficiency and data locality. You want to: Access latest data quickly Avoid hotspots Keep queries simple and fast

Ryannn23 1 year, 1 month ago

Selected Answer: A

Vote A, as explained by Augustax: Agree A is the best option because: 1. Multi-tenancy solution 2. As with any timestamp, avoid starting a row key with a reversed timestamp so that you don't cause hotspots.

Augustax 1 year, 2 months ago

Selected Answer: A

Agree A is the best option because: 1. Multi-tenancy solution 2. As with any timestamp, avoid starting a row key with a reversed timestamp so that you don't cause hotspots.

shangning007 1 year, 3 months ago

Selected Answer: D

I don't think any answer is correct. A lot people upvote for B, but based on https://cloud.google.com/bigtable/docs/schema-design#time-based, "As with any timestamp, avoid starting a row key with a reversed timestamp so that you don't cause hotspots."

ToiToi 1 year, 5 months ago

Selected Answer: D

Why other options are not as suitable: A and B (One table for all indices): Storing all indices in a single table can lead to performance issues as the table grows larger. It also makes it harder to scale individual indices independently. C (Timestamp as row key): Using a regular timestamp would place the most recent data at the end of the table, making it less efficient to retrieve the latest prices.

SamuelTsch 1 year, 5 months ago

Selected Answer: D

Option B and Option D are both from my point of view correct. It depens on the situation. If there is need to get the information from each stock index, then D is more suitable. Otherwise B.

mayankazyour 1 year, 6 months ago

Selected Answer: D

1. Reverse Timestamp for most recent stock prices 2. Having different table for each stock is more efficient, improves the query performance and option B doesn't specify stock in row key.

iooj 1 year, 8 months ago

Selected Answer: A

Row keys that start with a timestamp (irrespective reversed or not) causes sequential writes to be pushed onto a single node, creating a hotspot. If you put a timestamp in a row key, precede it with a high-cardinality value (index in our case) to avoid hotspots. The ideal option would be: "use the index and reversed timestamp as the row key design".

datapassionate 2 years, 2 months ago

Selected Answer: B

B is a correct answer because "you need to access only the most recent stock prices" "If you usually retrieve the most recent records first, you can use a reversed timestamp in the row key by subtracting the timestamp from your programming language's maximum value for long integers (in Java, java.lang.Long.MAX_VALUE). With a reversed timestamp, the records will be ordered from most recent to least recent."
https://cloud.google.com/bigtable/docs/schema-design#time-based

TVH_Data_Engineer 2 years, 3 months ago

Selected Answer: B

B. One unique table for all indices, reverse timestamp as row key: A single table for all indices keeps the structure simple. Using a reverse timestamp as part of the row key ensures that the most recent data comes first in the sorted order. This design is beneficial for quickly accessing the latest data. For example: you can convert the timestamp to a string and format it in reverse order, like "yyyyMMddHHmmss", ensuring newer dates and times are sorted lexicographically before older ones.

kshehadyx 2 years, 6 months ago

Correct Is B

Lanro 2 years, 8 months ago

Selected Answer: B

https://cloud.google.com/bigtable/docs/schema-design#row-keys - If you usually retrieve the most recent records first, you can use a reversed timestamp B it is.

Chom 2 years, 8 months ago

Selected Answer: A

A is the answer

vaga1 2 years, 9 months ago

Selected Answer: B

the answer relieves on whether the application need to access the whole indexes at the same time or not. If yes then is B, if no is A. in mind the answer is yes, so B makes more sense: I retrieve all the list at the same time.

ajdf 2 years, 9 months ago

Selected Answer: B

https://cloud.google.com/bigtable/docs/schema-design#time-based If you usually retrieve the most recent records first, you can use a reversed timestamp in the row key by subtracting the timestamp from your programming language's maximum value for long integers (in Java, java.lang.Long.MAX_VALUE). With a reversed timestamp, the records will be ordered from most recent to least recent.

https://zhangxuetu.online

[202610]

Question #184 Topic 1

You are building a report-only data warehouse where the data is streamed into BigQuery via the streaming API. Following Google's best practices, you have both a staging and a production table for the data. How should you design your data loading to ensure that there is only one master dataset without affecting performance on either the ingestion or reporting pieces?

A. Have a staging table that is an append-only model, and then update the production table every three hours with the changes written to staging.

B. Have a staging table that is an append-only model, and then update the production table every ninety minutes with the changes written to staging.

C. Have a staging table that moves the staged data over to the production table and deletes the contents of the staging table every three hours. **Most Voted**

D. Have a staging table that moves the staged data over to the production table and deletes the contents of the staging table every thirty minutes.

Correct Answer: C

Community vote distribution



Comments

NicolasN **Highly Voted** 3 years, 4 months ago

Selected Answer: C

[C] I found the correct answer based on a real case, where Google's Solutions Architect team decided to move an internal process to use BigQuery. The related doc is here: https://cloud.google.com/blog/products/data-analytics/moving-a-publishing-workflow-to-bigquery-for-new-data-insights

NicolasN 3 years, 4 months ago

The interesting excerpts: "Following common extract, transform, load (ETL) best practices, we used a staging table and a separate production table so that we could load data into the staging table without impacting users of the data. The design we created based on ETL best practices called for first deleting all the records from the staging table, loading the staging table, and then replacing the production table with the contents." "When using the streaming API, the BigQuery streaming buffer remains active for about 30 to 60 minutes or more after use, which means that you can’t delete or change data during that time. Since we used the streaming API, we scheduled the load every three hours to balance getting data into BigQuery quickly and being able to subsequently delete the data from the staging table during the load process."

squishy_fishy 2 years, 5 months ago

I second this. At my work, I run into this exact steaming buffer thing, it will not let me delete the data until after 60 minutes.

AzureDP900 3 years, 3 months ago

Agreed C is right

nwk Highly Voted 3 years, 6 months ago

Vote B - "Some recently streamed rows might not be available for table copy typically for a few minutes. In rare cases, this can take up to 90 minutes" https://cloud.google.com/bigquery/docs/streaming-data-into-bigquery#dataavailability

jkhong 3 years, 3 months ago

Aren't there other aspects of data pipelining that we should be aware of? other than merely referring to the number of 'recommended' minutes stated in docs. B doesn't address how the appended data is subsequently deleted, since the table is append-only, the size will constantly grow, and so the user may unnecessarily incur more storage costs.

YorelNation 3 years, 6 months ago

They don't seems concerned too much by data accuracy in the question

devaid 3 years, 5 months ago

A and B are discarded because the UPDATE statement, is not performance efficient. Neither appending more and more values to the stagging table. It's better cleaning the stagging table, and merging with the master dataset.

MaxNRG 2 years, 3 months ago

You can use BigQuery's features like MERGE to efficiently update the production table with only the new or changed data from the staging table, reducing processing time and costs.

Feisar 2 months, 3 weeks ago

Selected Answer: A

Option a. You shouldn´t really delete from staging and append only means BQ doesn´t have to worry about updates etc. 3 hours so the streaming buffer is definitely written to disk

AdriHubert 10 months ago

Selected Answer: A

Read the MaxNRG comment

SamuelTsch 1 year, 5 months ago

Selected Answer: A

deleting data from my point of view is not a good practice to build datawarehouse solutions. So, C and D are excluded. according to the official documentation, the updating/merging process could last till 90 minutes. 3 hours could be enough.

TVH_Data_Engineer 1 year, 10 months ago

Selected Answer: A

An append-only staging table ensures that all incoming data is captured without risk of data loss or overwrites, which is crucial for maintaining data integrity in a streaming ingestion scenario. Three-Hour Update Interval: Updating the production table every three hours strikes a good balance between minimizing the latency of data availability for reporting and reducing the frequency of potentially resource-intensive update operations. This interval is frequent enough to keep the production table relatively up-to-date for reporting purposes while ensuring that the performance of both ingestion and reporting processes is not significantly impacted. Frequent updates (like every ninety minutes or every thirty minutes) could introduce unnecessary overhead and contention, especially if the dataset is large or if there are complex transformations involved.

MaxNRG 2 years, 3 months ago

Selected Answer: A

Not C nor D. Moving and deleting: Deleting data from the staging table every 3 or 30 minutes could lead to data loss if the production table update fails, and it also requires more frequent and potentially resource-intensive operations. Options C and D cause rebuilding of the staging table, which slows down ingestion, and may lose data if errors occur during recreation. A or B

MaxNRG 2 years, 3 months ago

When designing a report-only data warehouse in BigQuery, where data is streamed in and you have both staging and production tables, the key is to balance the frequency of updates with the performance needs of both the ingestion and reporting processes. Let's evaluate each option:

MaxNRG 2 years, 3 months ago

A. Staging table as append-only, updating production every three hours: This approach allows for a consistent flow of data into the staging table without interruptions. Updating the production table every three hours strikes a balance between having reasonably fresh data and not overloading the system with too frequent updates. However, this may not be suitable if your reporting requirements demand more up-to-date data. B. Staging table as append-only, updating production every ninety minutes: This is similar to option A but with a more frequent update cycle. This could be more appropriate if your reporting needs require more current data. However, more frequent updates can impact performance, especially during the update windows.

MaxNRG 2 years, 3 months ago

C. Staging table moves data to production and clears staging every three hours: Moving data from staging to production and then clearing the staging table ensures that there is only one master dataset. However, this method might lead to more significant interruptions in data availability, both during the move and the clearing process. This might not be ideal if continuous access to the latest data is required. D. Staging table moves data to production and clears staging every thirty minutes: This option provides the most up-to-date data in the production table but could significantly impact performance. Such frequent data transfers and deletions might lead to more overhead and could interrupt both the ingestion and reporting processes.

MaxNRG 2 years, 3 months ago

Considering these options, A (Staging table as append-only, updating production every three hours) seems to be the most balanced approach. It provides a good compromise between having up-to-date data in the production environment and maintaining system performance. However, the exact frequency should be fine-tuned based on the specific performance characteristics of your system and the timeliness requirements of your reports. It's also important to implement efficient mechanisms for transferring data from staging to production to minimize the impact on system performance. Techniques like partitioning and clustering in BigQuery can be used to optimize query performance and manage large datasets more effectively.

Aman47 2 years, 3 months ago

Neither. In the current scenario, DataStream (a new google resource) captures the CDC data and uses Dataflow to Replicate the changes to big query.

hauhau 3 years, 3 months ago

Selected Answer: B

Vote B C : the doc says streaming data can be used up to 90 minutes not 3 hours B : correct , insert staging table first with append and use merge from staging into production table

hauhau 3 years, 3 months ago

B just say "update", not specificlly mention DML. update can be merge

MaxNRG 2 years, 3 months ago

You can use BigQuery's features like MERGE to efficiently update the production table with only the new or changed data from the staging table, reducing processing time and costs.

Atnafu 3 years, 4 months ago

C Following common extract, transform, load (ETL) best practices, we used a staging table and a separate production table so that we could load data into the staging table without impacting users of the data. The design we created based on ETL best practices called for first deleting all the records from the staging table, loading the staging table, and then replacing the production table with the contents. When using the streaming API, the BigQuery streaming buffer remains active for about 30 to 60 minutes or more after use, which means that you can't delete or change data during that time. Since we used the streaming API, we scheduled the load every three hours to balance getting data into BigQuery quickly and being able to subsequently delete the data from the staging table during the load process. Building a script with BigQuery on the back end https://cloud.google.com/blog/products/data-analytics/moving-a-publishing-workflow-to-bigquery-for-new-data-insights

John_Pongthorn 3 years, 5 months ago

Selected Answer: C

D : read more on Streaming inserts and timestamp-aware queries as the following link it is the same as this question exactly, but it is quite similar. https://cloud.google.com/blog/products/bigquery/performing-large-scale-mutations-in-bigquery read carefully in the content below. When using timestamps to keep track of updated and deleted records, it's a good idea to periodically delete stale entries. To illustrate, the following pair of DML statements can be used to remove older versions as well as deleted records. You'll notice that the above DELETE statements don't attempt to remove records that are newer than 3 hours. This is because data in BigQuery's streaming buffer is not immediately available for UPDATE, DELETE, or MERGE operations, as described in DML Limitations. These queries assume that the actual values for RecordTime roughly match the actual ingestion time.

John_Pongthorn 3 years, 5 months ago

https://cloud.google.com/architecture/database-replication-to-bigquery-using-change-data-capture#prune_merged_data https://cloud.google.com/bigquery/docs/reference/standard-sql/data-manipulation-language#limitations

John_Pongthorn 3 years, 6 months ago

Either C or D But When will we delete stale data on staging table ? Every xxx???? https://cloud.google.com/architecture/database-replication-to-bigquery-using-change-data-capture#prune_merged_data

Oleksandr0501 2 years, 10 months ago

gpt: "Overall, deleting the staging table every 30 minutes is a better choice than every 3 hours because it reduces the risk of data inconsistencies and performance issues."

TNT87 3 years, 6 months ago

Selected Answer: D

D. Have a staging table that moves the staged data over to the production table and deletes the contents of the staging table every thirty minutes.

TNT87 3 years, 6 months ago

Ans D D. Have a staging table that moves the staged data over to the production table and deletes the contents of the staging table every thirty minutes.

AWSandeep 3 years, 7 months ago

Selected Answer: D

D. Have a staging table that moves the staged data over to the production table and deletes the contents of the staging table every thirty minutes.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #185 Topic 1

You issue a new batch job to Dataflow. The job starts successfully, processes a few elements, and then suddenly fails and shuts down. You navigate to the
Dataflow monitoring interface where you find errors related to a particular DoFn in your pipeline. What is the most likely cause of the errors?

A. Job validation

B. Exceptions in worker code **Most Voted**

C. Graph or pipeline construction

D. Insufficient permissions

Correct Answer: B

Community vote distribution

B (100%)

Comments

AWSandeep **Highly Voted** 3 years, 1 month ago

Selected Answer: B

B. Exceptions in worker code While your job is running, you might encounter errors or exceptions in your worker code. These errors generally mean that the DoFns in your pipeline code have generated unhandled exceptions, which result in failed tasks in your Dataflow job. Exceptions in user code (for example, your DoFn instances) are reported in the Dataflow monitoring interface. Reference (Lists all answer choices and when to pick each one): https://cloud.google.com/dataflow/docs/guides/troubleshooting-your-pipeline#Causes

MaxNRG 1 year, 9 months ago

Selected Answer: B

The most likely cause of the errors you're experiencing in Dataflow, particularly if they are related to a particular DoFn (Dataflow's parallel processing operation), is B. Exceptions in worker code. When a Dataflow job processes a few elements successfully before failing, it suggests that the overall job setup, permissions, and pipeline graph are likely correct, as the job was able to start and initially process data. However, if it fails during execution and the errors are associated with a specific DoFn, this points towards issues in the code that executes within the workers. This could include: 1. Runtime exceptions in the code logic of the DoFn. 2. Issues handling specific data elements that might not be correctly managed by the DoFn code (e.g., unexpected data formats, null values, etc.). 3. Resource constraints or timeouts if the DoFn performs operations that are resource-intensive or long-running.

MaxNRG 1 year, 9 months ago

To resolve these issues, you should: 1. Inspect the stack traces and error messages in the Dataflow monitoring interface for details on the exception. 2. Test the DoFn with a variety of data inputs, especially edge cases, to ensure

robust error handling. 3. Review the resource usage and performance characteristics of the DoFn if the issue is related to resource constraints.

vaga1 2 years, 4 months ago

Selected Answer: B

A. Job validation - since it started successfully, it must have been validated. B. Exceptions in worker code - possible C. Graph or pipeline construction - same as A. D. Insufficient permissions - no elements to say that, and it should led to invalidation.

Atnafu 2 years, 10 months ago

C Code error

pluiedust 3 years ago

Selected Answer: B

B is correct

ducc 3 years ago

Selected Answer: B

B is correct

https://zhangxuetu.online

[202610]

Question #186 Topic 1

Your new customer has requested daily reports that show their net consumption of Google Cloud compute resources and who used the resources. You need to quickly and efficiently generate these daily reports. What should you do?

A. Do daily exports of Cloud Logging data to BigQuery. Create views filtering by project, log type, resource, and user. Most Voted

B. Filter data in Cloud Logging by project, resource, and user; then export the data in CSV format.

C. Filter data in Cloud Logging by project, log type, resource, and user, then import the data into BigQuery.

D. Export Cloud Logging data to Cloud Storage in CSV format. Cleanse the data using Dataprep, filtering by project, resource, and user.

Correct Answer: A

Community vote distribution



Comments

AWSandeep Highly Voted 3 years, 1 month ago

A. Do daily exports of Cloud Logging data to BigQuery. Create views filtering by project, log type, resource, and user. You cannot import custom or filtered billing criteria into BigQuery. There are three types of Cloud Billing data tables with a fixed schema that must further drilled-down via BigQuery views. Reference: https://cloud.google.com/billing/docs/how-to/export-data-bigquery#setup

22c1725 10 months ago

Selected Answer: A

Sadly, it's unclear what: "Do daily exports of Cloud Logging data to BigQuery" Does that means creating job? or you would be creating a sink into destination?

MaxNRG 1 year, 9 months ago

Selected Answer: A

For generating daily reports that show net consumption of Google Cloud compute resources and user details, the most efficient approach would be: A. Do daily exports of Cloud Logging data to BigQuery. Create views filtering by project, log type, resource, and user.

MaxNRG 1 year, 9 months ago

Here's why this option is the most effective: Integration with BigQuery: BigQuery is a powerful tool for analyzing large datasets. By exporting Cloud Logging data directly to BigQuery, you can leverage its fast querying capabilities and advanced analysis features. Automated Daily Exports: Setting up automated daily exports to BigQuery streamlines the

reporting process, ensuring that data is consistently and efficiently transferred. Creating Views for Specific Filters: By creating views in BigQuery that filter data by project, log type, resource, and user, you can tailor the reports to the specific needs of your customer. Views also simplify repeated analysis by encapsulating complex SQL queries. Efficiency and Scalability: This method is highly efficient and scalable, handling large volumes of data without the manual intervention required for CSV exports and data cleansing.

MaxNRG 1 year, 9 months ago

Option B (exporting data in CSV format) and Option D (using Cloud Storage and Dataprep) are less efficient due to the additional steps and manual handling involved. Option C is similar to A but lacks the specificity of creating views directly in BigQuery for filtering, which is a more streamlined approach.

Aman47 1 year, 9 months ago

You can choose a sink in which you want Cloud logging to continously send Logging data. You can choose which columns you want to see (filter).

Aman47 1 year, 9 months ago

Option C

vaga1 2 years, 4 months ago

Selected Answer: A

B, C, D do no generate a daily scalable solution.

Siant_137 2 years, 5 months ago

Selected Answer: C

I see A as quite inefficient as you are exporting ALL logs (hundreds of thousands) to bq and the filtering them with views. I would go for C, assuming that it does not involve doing it manually but rather creating a SINK with the correct filters and then using BQ Dataset as sink destination. But a lot of assumptions are taking place here as I believe the questions does not provide much context.

midgoo 2 years, 6 months ago

Selected Answer: A

I almost got it wrong by choosing C. By doing C, that means we will manually filter first one by one. We should just import them all and filter using BigQuery

maci_f 2 years, 8 months ago

Selected Answer: A

B and D do not consider the log type field. C looks good and I would go for it. However, A looks equally good and I've found a CloudSkillsBoost lab that is exactly describing what answer A does, i.e. exporting logs to BQ and then creating a VIEW. https://www.cloudskillsboost.google/focuses/6100?parent=catalog I think the advantage of exporting complete logs (i.e. filtering them after they reach BQ) is that in case we would want to adjust the reporting in the future, we would have the complete logs with all fields available, whereas with C we would need to take extra steps.

Atnafu 2 years, 10 months ago

A Bad exporting data in csv or json due lack of some data so export is best practice 1:10 https://www.youtube.com/watch?v=ZyMO9XabUUM

Atnafu 2 years, 9 months ago

You need to quickly and efficiently generate these daily reports by using Materialized view /View A materialized view is the best solution and having filtered value with a view is good solution so A is an answer

hauhau 2 years, 10 months ago

Selected Answer: A

A because you filter data daily by view not just once by cloud logging

devaid 2 years, 12 months ago

Selected Answer: A

A. The D isn't filtering by log type. B and C are discarded because you need to drill down the exported loggs in Big Query or other.

devaid 2 years, 11 months ago

2nd tought: Definitely A. If you go to google documentation for export billing, you see a message that "Exporting to JSON or CSV is obsolet. Use Big Query instead". Also why A? Look https://cloud.google.com/billing/docs/how-to/export-data-bigquery https://cloud.google.com/billing/docs/how-to/bq-examples#total-costs-on-invoice You can make a fast report template al Data Studio that read a Big Query view.

NicolasN 2 years, 10 months ago

A comment regarding the links you provided (and not the correctness of the selected answer). Using Cloud Billing is something different than detecting compute consumption data from Cloud Logging. In fact, manual exporting to CSV (and JSON) is possible through the Logs Explorer interface (I think without user data break-down):
https://cloud.google.com/logging/docs/view/logs-explorer-interface#download_logs

AHUI 3 years ago

Selected Answer: D

The Google Cloud Storage bucket where you would like your reports to be delivered. You can select any Cloud Storage bucket for which you are an owner, including buckets that are from different projects. This bucket must exist before you can start exporting reports and you must have owner access to the bucket. Google Cloud Storage charges for usage, so you should review the Cloud Storage pricesheet for information on how you might incur charges for the service. https://cloud.google.com/compute/docs/logging/usage-export

TNT87 3 years ago

Ans is C https://cloud.google.com/logging/docs/export/aggregated_sinks D isn't correct because Cloud storage is used as a sink when logs are in json format not csv.
https://cloud.google.com/logging/docs/export/aggregated_sinks#supported-destinations

TNT87 3 years ago

On the other hand Ans A makes sense https://cloud.google.com/logging/docs/export/bigquery#overview

jkhong 2 years, 10 months ago

The question explicitly mentions daily generation of data so this would highlight, B and C seems that it is only suggesting a one-off filtering

TNT87 2 years, 8 months ago

so wjhaytsa your argument about daily generartion of data??

changsu 3 years ago

Selected Answer: D

Quickly and efficiently! It's a flag to guide to DataPrep. And importing data to Bigquery does not mean a report.

pluiedust 3 years ago

Why not C?

Wasss123 3 years ago

why not D ?

Remi2021 3 years ago

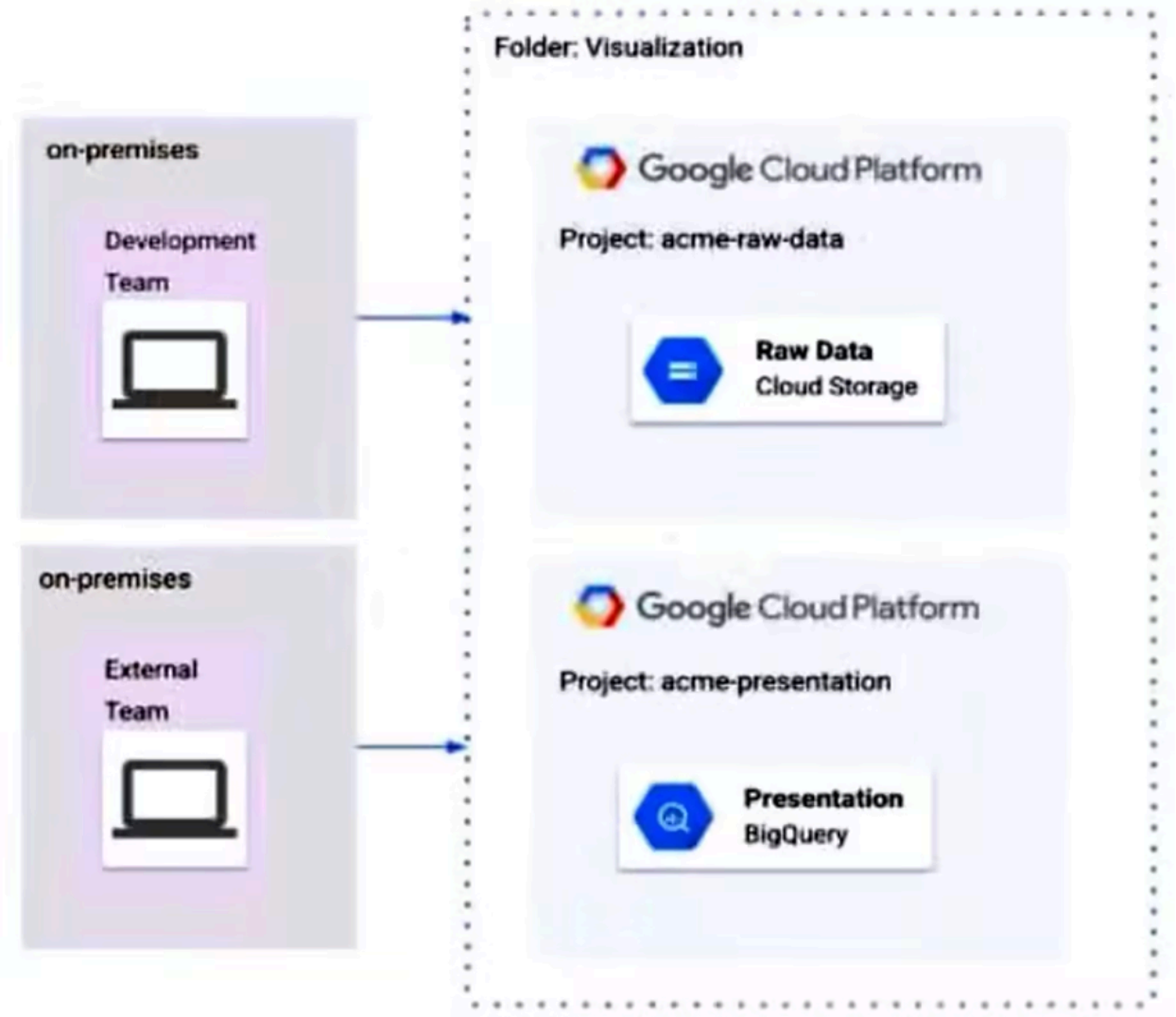
Challenging. B is right one but with B you do not automate wich makes it hard, with A you ensure automation but there is no SQL support being mentioned which also makes me think that A is not the best choice.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

Question #187 Topic 1

The Development and External teams have the project viewer Identity and Access Management (IAM) role in a folder named Visualization. You want the
Development Team to be able to read data from both Cloud Storage and BigQuery, but the External Team should only be able to read data from BigQuery. What should you do?



A. Remove Cloud Storage IAM permissions to the External Team on the acme-raw-data project.

B. Create Virtual Private Cloud (VPC) firewall rules on the acme-raw-data project that deny all ingress traffic from the External Team CIDR range.

C. Create a VPC Service Controls perimeter containing both projects and BigQuery as a restricted API. Add the External Team users to the perimeter's Access Level.

D. Create a VPC Service Controls perimeter containing both projects and Cloud Storage as a restricted API. Add the Development Team users to the perimeter's Access Level. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

AWSandeep **Highly Voted** 3 years, 1 month ago

Selected Answer: D

D. Create a VPC Service Controls perimeter containing both projects and Cloud Storage as a restricted API. Add the Development Team users to the perimeter's Access Level. Reveal Solution

TNT87 3 years ago

WHy do you have to put the development team at the access perimeter???

Oleksandr0501 2 years, 4 months ago

no, https://cloud.google.com/blog/products/serverless/cloud-run-gets-enterprise-grade-network-security-with-vpc-sc?utm_source=youtube&utm_medium=unpaidsoc&utm_campaign=CDR_pri_gcp_m0v4tedeiao_ThisWeekInCloud_082621&utm_content=description

Oleksandr0501 2 years, 4 months ago

damn, i am confused anyway. Can be D.

Oleksandr0501 2 years, 4 months ago

should be D, as i think now, because we create a "magic bulb" around around Cloud storage and Dev team, and it will be protected from external influence like a human cell. Meantime Dev team will still be able to acess Bigquery. But external team will not manage to access Cloud storage.

maci_f **Highly Voted** 2 years, 8 months ago

Selected Answer: D

"The grouping of GCP Project(s) and Service API(s) in the Service Perimeter result in restricting unauthorized access outside of the Service Perimeter to Service API endpoint(s) referencing resources inside of the Service Perimeter." https://scalesec.com/blog/vpc-service-controls-in-plain-english/ Development team: needs to access both Cloud Storage and BQ -> therefore we put the Development team inside a perimeter so it can access both the Cloud Storage and the BQ External team: allowed to access only BQ -> therefore we put Cloud Storage behind the restricted API and leave the external team outside of the perimeter, so it can access BQ, but is prohibited from accessing the Cloud Storage

josech 1 year, 4 months ago

Selected Answer: A

It is not a network issue but a IAM permissions issue. https://cloud.google.com/iam/docs/deny-overview#inheritance

Aman47 1 year, 9 months ago

Comments are saying it correct its C

Mamko 2 years ago

It's D for sure

techabhi2_0 2 years ago

A - Simple and straight forward

wan2three 2 years, 1 month ago

Why not B, I think CD will cause one of the team can not reach one or two of those DBs. A is not correct either

izekc 2 years, 5 months ago

Selected Answer: C

C is correct

midgoo 2 years, 6 months ago

Selected Answer: D

D sounds more correct, but if the project is already in the Service Control, would External people can access the BigQuery dataset in that project?

Atnafu 2 years, 10 months ago

c Extend perimeters to authorized VPN or Cloud Interconnect You can configure private communication to Google Cloud resources from VPC networks that span hybrid environments with Private Google Access on-premises extensions. A VPC network must be part of a service perimeter for VMs on that network to privately access managed Google Cloud resources within that service perimeter. https://cloud.google.com/vpc-service-controls/docs/overview#internet

Atnafu 2 years, 10 months ago

I meant D Not C

cloudmon 2 years, 11 months ago

Selected Answer: D

D makes the most sense to me

cloudmon 2 years, 11 months ago

Because "You want the Development Team to be able to read data from both Cloud Storage and BigQuery, but the External Team should only be able to read data from BigQuery."

cloudmon 2 years, 11 months ago

Therefore, Cloud Storage should be the restricted API, and you add the Development Team users to the perimeter's Access Level to allow them to access the restricted API.

yu_ 2 years, 11 months ago

why C? I thought the development team would not be able to access BigQuery as I would include BigQuery in the service perimeter and add External Team to the access level

jkhong 2 years, 10 months ago

Exactly, why would we need to consider BigQuery as a restricted service when it can already be accessed by both Dev and External team. The restricted service we are concerned with is Cloud Storage. If we go with C, we are only adding the external team into the access level... this means that the development team still wouldn't be able to access it

josrojgra 2 years, 11 months ago

Selected Answer: C

Answer C https://cloud.google.com/vpc-service-controls/docs/overview#isolate

TNT87 2 years, 11 months ago

Selected Answer: C

Answer C https://cloud.google.com/vpc-service-controls/docs/overview#isolate

Wasss123 3 years ago

Shoud be C https://cloud.google.com/vpc-service-controls/docs/vpc-accessible-services When configuring VPC accessible services for a perimeter, you can specify a list of individual services, as well as include the RESTRICTED-SERVICES value, which automatically includes all of the services protected by the perimeter. To ensure access to the expected services is fully limited, you must: Configure the perimeter to protect the same set of services that you want to make accessible. Configure VPCs in the perimeter to use the restricted VIP. Use layer 3 firewalls.

TNT87 3 years ago

Ans C https://cloud.google.com/vpc-service-controls/docs/overview#isolate

nwk 3 years ago

Vote C https://cloud.google.com/vpc-service-controls/docs/vpc-accessible-services

https://zhangxuetu.online

[202610]

Question #188 Topic 1

Your startup has a web application that currently serves customers out of a single region in Asia. You are targeting funding that will allow your startup to serve customers globally. Your current goal is to optimize for cost, and your post-funding goal is to optimize for global presence and performance. You must use a native JDBC driver. What should you do?

A. Use Cloud Spanner to configure a single region instance initially, and then configure multi-region Cloud Spanner instances after securing funding. **Most Voted**

B. Use a Cloud SQL for PostgreSQL highly available instance first, and Bigtable with US, Europe, and Asia replication after securing funding.

C. Use a Cloud SQL for PostgreSQL zonal instance first, and Bigtable with US, Europe, and Asia after securing funding.

D. Use a Cloud SQL for PostgreSQL zonal instance first, and Cloud SQL for PostgreSQL with highly available configuration after securing funding.

Correct Answer: A

Community vote distribution



Comments

AWSandeep Highly Voted 3 years, 7 months ago

Selected Answer: A

A. Use Cloud Spanner to configure a single region instance initially, and then configure multi-region Cloud Spanner instances after securing funding. When you create a Cloud Spanner instance, you must configure it as either regional (that is, all the resources are contained within a single Google Cloud region) or multi-region (that is, the resources span more than one region). You can change the instance configuration to multi-regional (or global) at anytime.

izekc Highly Voted 2 years, 11 months ago

Selected Answer: D

Although A is good, but concerning about the cost. Then D will be much more suitable

b2aaace 4 months, 2 weeks ago

Selected Answer: D

Native JDBC → eliminates Bigtable (B & C) Cost optimization initially → eliminates Spanner (A) Global expansion later → Cloud SQL can add read replicas

SamuelTsch 1 year, 5 months ago

Selected Answer: D

I go to D. Cloud SQL is usually used for web application (CRM) (https://cloud.google.com/blog/topics/developers-practitioners/your-google-cloud-database-options-explained?hl=en)

4a8ffd7 1 year, 6 months ago

Selected Answer: D

Although A is good, but concerning about the cost. Then D will be much more suitable

CGS22 1 year, 12 months ago

Selected Answer: D

Although A is good, but concerning about the cost. Then D will be much more suitable

Ronn27 1 year, 2 months ago

While a zonal instance is cost-effective, transitioning to a highly available Cloud SQL instance does not support global replication. Cloud SQL lacks the scalability and global presence needed for your post-funding goals. So I beleive Spanner is the right answer

tibuenoc 2 years, 2 months ago

Selected Answer: D

I think is D The best for Web app is Cloud SQL, and Spanner is the best for data more than 30GB

Ronn27 1 year, 2 months ago

While a zonal instance is cost-effective, transitioning to a highly available Cloud SQL instance does not support global replication. Cloud SQL lacks the scalability and global presence needed for your post-funding goals.

MaxNRG 2 years, 3 months ago

Selected Answer: A

A - This option allows for optimization for cost initially with a single region Cloud Spanner instance, and then optimization for global presence and performance after funding with multi-region instances. Cloud Spanner supports native JDBC drivers and is horizontally scalable, providing very high performance. A single region instance minimizes costs initially. After funding, multi-region instances can provide lower latency and high availability globally. Cloud SQL does not scale as well and has higher costs for multiple high availability regions. Bigtable does not support JDBC drivers natively. Therefore, Spanner is the best choice here for optimizing both for cost initially and then performance and availability globally post-funding.

lucaluca1982 3 years ago

Spanner has some limitations with JDBC. Maybe the quetion wants to help us tp choose Cloud SQL

musumusu 3 years, 1 month ago

Answer D: Cost effective transactional database Cloud SQL. Spanner is good case for data more than 30 GB

odacir 3 years, 3 months ago

Selected Answer: A

B and C has no sense because of the driver. D looks like a good option, but HA it's not to improve performance or global presence: The purpose of an HA configuration is to reduce downtime when a zone or instance becomes unavailable. This might happen during a zonal outage, or when an instance runs out of memory. With HA, your data continues to be available to client applications. So the best option is A.

TNT87 3 years, 6 months ago

Selected Answer: A

https://cloud.google.com/spanner/docs/jdbc-drivers Ans A https://cloud.google.com/spanner/docs/instance-configurations#tradeoffs_regional_versus_multi-region_configurations The last part of the question makes it easy

TNT87 3 years, 6 months ago

Yes Spanner is expensive , but the question expresslty states that "after securing funding you want to have a global presence" the word globally is repeatedly stated there. Answer is A.

TNT87 3 years, 6 months ago

https://cloud.google.com/spanner/docs/instance-configurations#tradeoffs_regional_versus_multi-region_configurations Ans A

badrisrinivas9 3 years, 6 months ago

Selected Answer: D

Spanner is expensive, they haven't mentioned the size of db... optimize for cost then option is Cloud SQL which cost effective and highly available in case of multi region.

Quevedo 3 years, 6 months ago

Selected Answer: A

A is the best option. It is globally scalable and it also meets the cost goal as it says that initially it will be configurated as single region wich is cheaper than multi region.

YorelNation 3 years, 6 months ago

Selected Answer: D

Spanner is expensive can't be A Would choose D

YorelNation 3 years, 6 months ago

Actually maybe C as you don't really need relational database for a webapp and BigTable is super performant and highly available

TNT87 3 years, 5 months ago

no its A

TNT87 3 years, 6 months ago

The fact that its global cloud spanner is the answer. Secondly Option D, the fact that it has to be highly avaible and multi regional its already more expensive than Cloud spanner Regional instance. https://cloud.google.com/spanner/docs/instance-configurations#tradeoffs_regional_versus_multi-region_configurations

ducc 3 years, 7 months ago

Selected Answer: A

Spanner still support JDBC https://cloud.google.com/spanner/docs/jdbc-drivers

https://zhangxuetu.online

[202610]

Question #189 Topic 1

You need to migrate 1 PB of data from an on-premises data center to Google Cloud. Data transfer time during the migration should take only a few hours. You want to follow Google-recommended practices to facilitate the large data transfer over a secure connection. What should you do?

A. Establish a Cloud Interconnect connection between the on-premises data center and Google Cloud, and then use the Storage Transfer Service. **Most Voted**

B. Use a Transfer Appliance and have engineers manually encrypt, decrypt, and verify the data.

C. Establish a Cloud VPN connection, start gcloud compute scp jobs in parallel, and run checksums to verify the data.

D. Reduce the data into 3 TB batches, transfer the data using gsutil, and run checksums to verify the data.

Correct Answer: A

Community vote distribution



Comments

devaid **Highly Voted** 3 years, 5 months ago

Selected Answer: A

Well it doesn't mentions anything about not enough bandwidth to meet your project deadline. I guess you can assume they have 200GBps+ of bandwith, otherwise it shouldn't take only a few hours.

iooj 1 year, 8 months ago

One who wanted to use Transfer Appliance to migrate data in a few hours, you should live near Google office and run really fast :D

MaxNRG 2 years, 3 months ago

Selected Answer: A

Cloud Interconnect provides a dedicated private connection between on-prem and Google Cloud for high bandwidth (up to 100 Gbps) and low latency. This facilitates large, fast data transfers. Storage Transfer Service supports parallel data transfers over Cloud Interconnect. It can transfer petabyte-scale datasets faster by transferring objects in parallel. Storage Transfer Service uses HTTPS encryption in transit and at rest by default for secure data transfers. It follows Google-recommended practices for large data migrations vs ad hoc methods like gsutil or scp. The other options would take too long for a 1 PB transfer (VPN capped at 3 Gbps, manual transfers) or introduce extra steps like batching and checksums. Cloud Interconnect + Storage Transfer is the recommended Google solution.

LanaOjisan 2 years, 5 months ago

It is believed that A. One reason is that for "secure" and "in a few hours," the communication can be done securely using a direct physical line without going through an ISP. Also, depending on the case, in the case of "Dedicated Interconnect," the maximum transfer can be as high as 200 Gbps, and the fastest data transfer of 1 PB can be completed in 11 hours. Therefore, A.

arien_chen 2 years, 7 months ago

Selected Answer: A

A https://cloud.google.com/storage-transfer/docs/transfer-options#:~:text=Transferring%20more%20than%201%20TB%20from%20on%2Dpremises

knith66 2 years, 8 months ago

Selected Answer: A

Dedicated Interconnect provides direct physical connections between your on-premises network and Google's network. Dedicated Interconnect enables you to transfer large amounts of data between networks, which can be more cost-effective than purchasing additional bandwidth over the public internet. https://cloud.google.com/network-connectivity/docs/interconnect/concepts/dedicated-overview

knith66 2 years, 8 months ago

This link has additional clarity https://cloud.google.com/network-connectivity/docs/interconnect/concepts/terminology

vaga1 2 years, 9 months ago

Selected Answer: B

1 PB and "few hours". It is clearly referring to Transfer Appliance https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#time

knith66 2 years, 8 months ago

Transfer Appliance is a slow process. wont be able to do in few hours

Oleksandr0501 2 years, 11 months ago

Selected Answer: A

gpt: Based on security and speed, if the data is highly sensitive and security is the top priority, then option B (using a Transfer Appliance) may be a better choice. Transfer Appliance uses hardware encryption to transfer data and is designed to securely transfer large amounts of data. However, if speed is the primary concern, then option A (using Cloud Interconnect and Storage Transfer Service) may be a better choice as it allows for faster transfer speeds over a dedicated and secure connection. It ultimately depends on the specific needs and priorities of the organization. A vague treaky question. Bad author of it... B is also good. As were said in discuss. by smb, a question asks "safe connection", so - a Cloud Interconnect (A)

midgoo 3 years ago

Selected Answer: B

Either this question is very tricky or very poor written. It says 'Data transfer time during the migration should take only a few hours'. We should not add the 20days for overhead time for Appliance into the total time of migration. If 'a few hours' = 30hours or more, A will be good enough. If 'a few hours' = 10 or less, B is the only way (with multiple devices to copy at the same time)

spicebits 2 years, 4 months ago

B can't be the answer - You have to wait 25 days to receive the appliance and another 25 days to get the appliance back and data loaded to cloud storage: https://cloud.google.com/transfer-appliance/docs/4.0/overview#transfer-speeds

Nandhu95 3 years ago

Selected Answer: A

Expected time via transfer appliance is around 20 days , and achieving the same using Storage transfer service with highest bandwidth of 100GPS is 30 hrs, so hence its been asked for hrs .. its A Acquiring a Transfer Appliance is straightforward. In the Google Cloud console, you request a Transfer Appliance, indicate how much data you have, and then Google ships one or more appliances to your requested location. You're given a number of days to transfer your data to the appliance ("data capture") and ship it back to Google. The expected turnaround time for a network appliance to be shipped, loaded with your data, shipped back, and rehydrated on Google Cloud is 20 days. If your online transfer timeframe is calculated to be substantially more than this timeframe, consider Transfer Appliance. The total cost for the 300 TB device process is less than $2,500.

vaga1 2 years, 9 months ago

it says data transfer during the migration. It mean from when the migration is "activated", which means from when the Transfer Appliace device is plugged and ready to be used

wjtb 3 years ago

Selected Answer: B

Even with 100gbps bandwith, you will not reach a data transfer time within the range of "hours" for 1PB. Transfer appliance is the way to go. https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#time

musumusu 3 years, 1 month ago

Answer A, One time transfer is cheaper and less secure always using Transfer Appliance. you need to do it in faster way, set up Interconnect speed limit is 50mbps - 10GBps and Transfer Appliance speed can goes up to 40GBps. I am choosing A for security concern only.

AzureDP900 3 years, 3 months ago

A is right

AzureDP900 3 years, 3 months ago

A. Establish a Cloud Interconnect connection between the on-premises data center and Google Cloud, and then use the Storage Transfer Service. Most Voted

Atnafu 3 years, 4 months ago

B https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#:~:text=Few%20things%20in,not%20be%20obtained.

Atnafu 3 years, 4 months ago

B It takes 30hrs with 100Gbps bandwidth- more than a day to transfer https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#:~:text=addresses%20or%20NATs.-,Online%20versus%20offline%20transfer,A%20certain%20amount%20of%20management%20overhead%20is%20built%20into%20these%20calculations.,-As%20noted%20earlier

pluiedust 3 years, 6 months ago

Selected Answer: A

A is correct.

bigquery1102 3 years, 6 months ago

Selected Answer: A

A is correct https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#transfer_appliance_for_larger_transfers

MounicaN 3 years, 6 months ago

Selected Answer: A

Huge data can be migrated over cloud interconnect

https://zhangxuetu.online

[202610]

Question #190 Topic 1

You are loading CSV files from Cloud Storage to BigQuery. The files have known data quality issues, including mismatched data types, such as STRINGs and
INT64s in the same column, and inconsistent formatting of values such as phone numbers or addresses. You need to create the data pipeline to maintain data quality and perform the required cleansing and transformation. What should you do?

A. Use Data Fusion to transform the data before loading it into BigQuery. **Most Voted**

B. Use Data Fusion to convert the CSV files to a self-describing data format, such as AVRO, before loading the data to BigQuery.

C. Load the CSV files into a staging table with the desired schema, perform the transformations with SQL, and then write the results to the final destination table.

D. Create a table with the desired schema, load the CSV files into the table, and perform the transformations in place using SQL.

Correct Answer: A

Community vote distribution



Comments

saurabhsingh4k **Highly Voted** 2 years, 9 months ago

Selected Answer: A

I'm kinda inclined towards C as SQL seems a powerful option to treat this kind of use case. Also, I didn't get how the transformations mentioned on this page will help to clean the data (https://cloud.google.com/data-fusion/docs/concepts/transformation-pushdown#supported_transformations) But I guess using Wrangler plugin, this kind of stuff can be done on DataFusion, also the question talks about an pipeline, so A is the final choice.

MaxNRG 1 year, 9 months ago

Selected Answer: A

Data Fusion's advantages: Visual interface: Offers a user-friendly interface for designing data pipelines without extensive coding, making it accessible to a wider range of users. Built-in transformations: Includes a wide range of pre-built transformations to handle common data quality issues, such as: Data type conversions Data cleansing (e.g., removing invalid characters, correcting formatting) Data validation (e.g., checking for missing values, enforcing constraints) Data enrichment (e.g., adding derived fields, joining with other datasets) Custom transformations: Allows for custom transformations using SQL or Java code for more complex cleaning tasks. Scalability: Can handle large datasets efficiently, making it suitable for processing CSV files with potential data quality issues. Integration with BigQuery: Integrates seamlessly with BigQuery, allowing for direct loading of transformed data.

MaxNRG 1 year, 9 months ago

Why other options are less suitable: B. Converting to AVRO: While AVRO is a self-describing format, it doesn't inherently address data quality issues. Transformations would still be needed, and Data Fusion provides a more comprehensive environment for this. C. Staging table: Requires manual SQL transformations, which can be time-consuming and error-prone for large datasets with complex data quality issues. D. Transformations in place: Modifying data directly in the destination table can lead to data loss or corruption if errors occur. It's generally safer to keep raw data intact and perform transformations separately. By using Data Fusion, you can create a robust and efficient data pipeline that addresses data quality issues upfront, ensuring that only clean and consistent data is loaded into BigQuery for accurate analysis and insights.

squishy_fishy 1 year, 11 months ago

The answer is C. That is what we do at work. We have landing/staging table, sort table and deliver table,

squishy_fishy 1 year, 11 months ago

Okay, second thought, it is asking for a pipeline, so the answer should be A. At work, we use dataflow inside the composer to build a pipeline injecting data into landing/staging table, then transform/clean data in the sort table, then send the cleaned data to deliver table.

phidelics 2 years, 3 months ago

Selected Answer: A

Keyword: Data Pipeline

mialll 2 years, 4 months ago

Selected Answer: A

same as @saurabhsingh4k

Adswerve 2 years, 5 months ago

Selected Answer: C

C is the right answer. Do ELT in BigQuery. Data Fusion is not the right too for this job.

musumusu 2 years, 7 months ago

Answer C, Datafusion is costly and current transformation is just a cast transformation in a column. I guess no one wanna pay for datafusion for this little transformation and Staging table processing handles such minor cleaning.

maci_f 2 years, 8 months ago

Selected Answer: A

Data Fusion enables changing the data type directly as shown in this lab: https://www.cloudskillsboost.google/focuses/25335?parent=catalog Wrangler is the feature to enable that, as already mentioned: https://stackoverflow.com/questions/58699872/google-cloud-data-fusion-how-to-change-datatype-from-string-to-date

Mike422 2 years, 9 months ago

Apparently chatGPT thinks C is the correct answer just sayin (for the same reason that @saurabhsingh4k wrote)

Atnafu 2 years, 9 months ago

A https://cloud.google.com/data-fusion/docs/concepts/overview#:~:text=The%20Cloud%20Data%20Fusion%20web%20UI%20lets%20you%20to%20build%20scalable%20data%20integration%20solutions%20to%20clean%2C%20prepare%2C%20blend%2C%20transfer%2C%20and%20transform%20data%2C%20without%20having%20to%20manage%20the%20infrastructure.

samirzubair 2 years, 10 months ago

The Correct Ans is C

jkhong 2 years, 9 months ago

although this is my preferred answer. this doesn't satisfy how this becomes a pipeline.

hiromi 2 years, 10 months ago

Selected Answer: A

Data Fusion

TNT87 3 years ago

Selected Answer: A

Ans A https://cloud.google.com/data-fusion/docs/concepts/transformation-pushdown#supported_transformations

ducc 3 years ago

Selected Answer: A

A is correct for me

AWSandeep 3 years, 1 month ago

Selected Answer: A

A. Use Data Fusion to transform the data before loading it into BigQuery.

https://zhangxuetu.online

[202610]

Question #191 Topic 1

You are developing a new deep learning model that predicts a customer's likelihood to buy on your ecommerce site. After running an evaluation of the model against both the original training data and new test data, you find that your model is overfitting the data. You want to improve the accuracy of the model when predicting new data. What should you do?

A. Increase the size of the training dataset, and increase the number of input features.

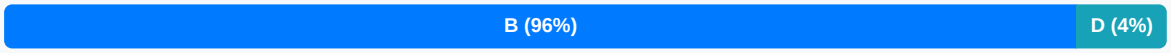
B. Increase the size of the training dataset, and decrease the number of input features. **Most Voted**

C. Reduce the size of the training dataset, and increase the number of input features.

D. Reduce the size of the training dataset, and decrease the number of input features.

Correct Answer: B

Community vote distribution



Comments

John_Pongthorn **Highly Voted** 3 years ago

Selected Answer: B

There 2 parts and they are relevant to each other 1. Overfit is fixed by decreasing the number of input features (select only essential features) 2. Accuracy is improved by increasing the amount of training data examples.

John_Pongthorn 3 years ago

https://docs.aws.amazon.com/machine-learning/latest/dg/model-fit-underfitting-vs-overfitting.html

Matt_108 1 year, 8 months ago

Selected Answer: B

Option B, the model learned to listen to too much stuff/noise. We need to reduce it, by decreasing the number of input feature, and we need to give the model more data, by increasing the amount of training data

NeoNitin 2 years, 2 months ago

Increase the size of the training dataset: By adding more diverse examples of customers and their buying behavior to the training data, the model will have a broader understanding of different scenarios and be better equipped to generalize to new customers. Increase the number of input features: Providing the model with more relevant information about customers can help it identify meaningful patterns and make better predictions. These input features could include things like the customer's age, past purchase history, browsing behavior, or any other relevant data that might impact their buying likelihood.

vaga1 2 years, 4 months ago

Selected Answer: B

A. can be a solution for a specific case, but it is not the academic answer as we do not know the quantity and proportion between them of n and k added. More records and more variables together can lead to even more overfitting due also to the curse of dimensionality. Adding a variable is much more impactful than records. B. just more records can lead to a more robust estimation and fewer variables certainly lead to at most the same estimation, but potentially reduce the fit on the training set. C. reduce n in favor of k is never a choice. it is against logic and it will lead to more overfitting. D. decrease both will reduce overfitting for sure but at the price of losing robustness on the model predictive power

AzureDP900 2 years, 9 months ago

B. Increase the size of the training dataset, and decrease the number of input features.

pluiedust 3 years ago

Selected Answer: B

B is correct

TNT87 3 years ago

Answer B https://machinelearningmastery.com/impact-of-dataset-size-on-deep-learning-model-skill-and-performance-estimates/

HarshKothari21 3 years ago

Selected Answer: B

Option B Feature selection is the one the ways to resolve overfitting. Which means reducing the features when the size of the training data is small, then the network tends to have greater control over the training data. so increasing the size of data would help.

YorelNation 3 years ago

Selected Answer: B

Best option is not mentioned: generalize you neural net by decreasing the complexity of it's structure. A part from that I guess you could remove some features and increase the size of the training dataset ==> B

AWSandeep 3 years ago

Selected Answer: B

B. Increase the size of the training dataset, and decrease the number of input features. Sorry, B is right. Read through extensive best-practices on ML.

ducc 3 years ago

Selected Answer: D

D is correct

AWSandeep 3 years ago

D. Reduce the size of the training dataset, and decrease the number of input features. Reveal Solution

ducc 3 years ago

Selected Answer: B

B. Increase the size of the training dataset, and decrease the number of input features.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #192 Topic 1

You are implementing a chatbot to help an online retailer streamline their customer service. The chatbot must be able to respond to both text and voice inquiries.
You are looking for a low-code or no-cade option, and you want to be able to easily train the chatbot to provide answers to keywords. What should you do?

A. Use the Cloud Speech-to-Text API to build a Python application in App Engine.

B. Use the Cloud Speech-to-Text API to build a Python application in a Compute Engine instance.

C. Use Dialogflow for simple queries and the Cloud Speech-to-Text API for complex queries.

D. Use Dialogflow to implement the chatbot, defining the intents based on the most common queries collected. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

PhuocT **Highly Voted** 3 years, 1 month ago

Selected Answer: D

D is correct: https://cloud.google.com/dialogflow/es/docs/how/detect-intent-tts#:~:text=Dialogflow%20can%20use%20Cloud%20Text,to%2Dspeech%2C%20or%20TTS.

MaxNRG 1 year, 9 months ago

Selected Answer: D

The best option would be to use Dialogflow to implement the chatbot, defining the intents based on the most common queries collected. Dialogflow is a conversational AI platform that allows for easy implementation of chatbots without needing to code. It has built-in integration for both text and voice input via APIs like Cloud Speech-to-Text. Defining intents and entity types allows you to map common queries and keywords to responses. This would provide a low/no-code way to quickly build and iteratively improve the chatbot capabilities. Option A and B would require more heavy coding to handle speech input/output. Option C still requires coding the complex query handling. Only option D leverages the full capabilities of Dialogflow to enable no-code chatbot development and ongoing improvements as more conversational data is collected. Hence, option D is the best approach given the requirements.

Lanro 2 years, 2 months ago

Selected Answer: D

Low-code or no-cade requirement makes it easy to decide.

Atnafu 2 years, 10 months ago

D https://cloud.google.com/dialogflow/es/docs/how/detect-intent-tts#:~:text=Dialogflow%20can%20use%20Cloud%20Text,to%2Dspeech%2C%20or%20TTS.

devaid 2 years, 11 months ago

Selected Answer: D

D definitely, as the documentation says (specially that you can call the detect Intect method for audio inputs): https://cloud.google.com/dialogflow/es/docs/how/detect-intent-tts Also Speech-To-Text API does nothing more than translate.

TNT87 3 years ago

Answer D https://cloud.google.com/dialogflow/es/docs/how/detect-intent-tts

nwk 3 years ago

https://cloud.google.com/dialogflow/es/docs/how/detect-intent-stream Vote D

ducc 3 years ago

Selected Answer: C

C. Use Dialogflow for simple queries and the Cloud Speech-to-Text API for complex queries. This seem the best answer here but not the best answer in real world. But with the Question, the answer must be the combination of both Diagflow and Speech API

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #193 Topic 1

An aerospace company uses a proprietary data format to store its flight data. You need to connect this new data source to BigQuery and stream the data into
BigQuery. You want to efficiently import the data into BigQuery while consuming as few resources as possible. What should you do?

A. Write a shell script that triggers a Cloud Function that performs periodic ETL batch jobs on the new data source.

B. Use a standard Dataflow pipeline to store the raw data in BigQuery, and then transform the format later when the data is used.

C. Use Apache Hive to write a Dataproc job that streams the data into BigQuery in CSV format.

D. Use an Apache Beam custom connector to write a Dataflow pipeline that streams the data into BigQuery in Avro format. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

beanz00 **Highly Voted** 2 years, 11 months ago

This has to be D. How could it even be B? The source is a proprietary format. Dataflow wouldn't have a built-in template to ead the file. You will have to create something custom.

devaid **Highly Voted** 2 years, 11 months ago

Selected Answer: D

For me it's clearly D It's between B and D, but read B, store raw data in Big Query? Use a Dataflow pipeline just to store raw data into Big Query, and transform later? You'd need to do another pipeline for that, and is not efficient.

MaxNRG 1 year, 9 months ago

Selected Answer: D

Option D is the best approach given the constraints - use an Apache Beam custom connector to write a Dataflow pipeline that streams the data into BigQuery in Avro format. The key reasons: • Dataflow provides managed resource scaling for efficient stream processing • Avro format has schema evolution capabilities and efficient serialization for flight telemetry data • Apache Beam connectors avoid having to write much code to integrate proprietary data sources • Streaming inserts data efficiently compared to periodic batch jobs In contrast, option A uses Cloud Functions which lack native streaming capabilities. Option B stores data in less efficient JSON format. Option C uses Dataproc which

requires manual cluster management. So leveraging Dataflow + Avro + Beam provides the most efficient way to stream proprietary flight data into BigQuery while using minimal resources.

Aman47 1 year, 9 months ago

Its talking about streaming? none of the options talk about triggering a load to begin. We need a trigger or schedule to run first.

AjoseO 1 year, 11 months ago

Selected Answer: D

Option D allows you to use a custom connector to read the proprietary data format and write the data to BigQuery in Avro format.

sergiomujica 2 years ago

Selected Answer: D

the keyword is streaming

knith66 2 years, 2 months ago

Between B and D. Firstly transformation is not mentioned in the question, So B is less probable. Then Efficient import is mentioned in the question, Converting to Avro will consume less space. I am going with D

musumusu 2 years, 7 months ago

Answer is D , Why not B, changing data format before uploading to bigquery is good approach.

cetanx 2 years, 8 months ago

Selected Answer: B

I believe keyword here is "An aerospace company uses a proprietary data format" So if we list the connectors available in Apache Beam, we are listed with these options; https://beam.apache.org/documentation/io/connectors/ So I believe, we have to create our own custom connector to read from the proprietary data format hence the answer should be B

cetanx 2 years, 8 months ago

sorry the answer should be D

AzureDP900 2 years, 9 months ago

D is right

hauhau 2 years, 10 months ago

Selected Answer: B

B is the most efficient

TNT87 2 years, 12 months ago

https://cloud.google.com/spanner/docs/change-streams/use-dataflow#core-concepts

TNT87 3 years ago

Ans B https://cloud.google.com/architecture/streaming-avro-records-into-bigquery-using-dataflow Is there a reason to use apache beam connector yet there is dataflow which is a standard solution for that scenario?

TNT87 3 years ago

https://cloud.google.com/blog/topics/developers-practitioners/bigquery-explained-data-ingestion

learner2610 3 years ago

Can standard dataflow be used to ingest any proprietary format of file ? shouldn't we use custom apache beam connector ? So I think it is D ,though it isn't simple ,But in this scenario they have asked to use less resources to import data

TNT87 3 years ago

Option D streams, thats not cost effective. We need something that is cost effectictive, hence B is the option

TNT87 3 years ago

I mean that consumes fewer resources

TNT87 3 years ago

Do you mind reading the links i provided and revisiting the question, then you will understand why D isnt the best. Why use Apache beam yet there is Dataflow

John_Pongthorn 2 years, 12 months ago

D: just have your team develop custom connector. https://cloud.google.com/architecture/bigquery-data-warehouse#storage_management Internally, BigQuery stores data in a proprietary columnar format called Capacitor, which has a number of benefits for data warehouse workloads. BigQuery uses a proprietary format I suppose this matter , it mean BQ use proprietary format by itself to work internally but the question means data as proprietary format as input for ingesting into BQ.

AWSandeep 3 years ago

Selected Answer: D

D. Use an Apache Beam custom connector to write a Dataflow pipeline that streams the data into BigQuery in Avro format. Reveal Solution

ducc 3 years ago

Selected Answer: B

B is the most efficient for me.

ducc 3 years ago

Sorry, D is correct

https://zhangxuetu.online

[202610]

Question #194 Topic 1

An online brokerage company requires a high volume trade processing architecture. You need to create a secure queuing system that triggers jobs. The jobs will run in Google Cloud and call the company's Python API to execute trades. You need to efficiently implement a solution. What should you do?

A. Use a Pub/Sub push subscription to trigger a Cloud Function to pass the data to the Python API. **Most Voted**

B. Write an application hosted on a Compute Engine instance that makes a push subscription to the Pub/Sub topic.

C. Write an application that makes a queue in a NoSQL database.

D. Use Cloud Composer to subscribe to a Pub/Sub topic and call the Python API.

Correct Answer: A

Community vote distribution



Comments

lucaluca1982 **Highly Voted** 3 years ago

A and D are both good. I go for A because we have high volume and easy to scale and optmize cost

kajitsu 1 year, 9 months ago

Selected Answer: D

D is the answer.

nadavw 1 year, 7 months ago

There is no need for a composer to call a Python API only - it's an overkill.

squishy_fishy 2 years, 5 months ago

Answer is D, at work we use solution A for low volume of Pub/Sub messages and Cloud function, and using D Composer for high volume Pub/Sub messages.

musumusu 3 years, 1 month ago

Answer A: assume, Company wants to buy immediately in same second if stock goes down or up. Somehow, it is connected to PubSub as SINK connector, then immediately there is PUSH to subcriber (cloud function) that is connected to their python API (internal application) that makes the purchase.

AzureDP900 3 years, 3 months ago

A. Use a Pub/Sub push subscription to trigger a Cloud Function to pass the data to the Python API.

GCPCloudArchitectUser 3 years, 4 months ago

Selected Answer: A

Because trading platform requires securely transmission to queuing If you use cloud compose then we need some other job to trigger composer … would that be cloud composer api or cloud function …

TNT87 3 years, 6 months ago

https://cloud.google.com/functions/docs/calling/pubsub

TNT87 3 years, 6 months ago

Selected Answer: A

Ans A https://cloud.google.com/functions/docs/calling/pubsub#deployment

YorelNation 3 years, 6 months ago

Selected Answer: A

A because D is stupidly high latency

nwk 3 years, 6 months ago

Vote A, can't see the need for composer

soichirokawa 3 years, 6 months ago

A might be enough. Cloud composer will be an overkill

AWSandeep 3 years, 7 months ago

A. Use a Pub/Sub push subscription to trigger a Cloud Function to pass the data to the Python API.

ducc 3 years, 7 months ago

Selected Answer: D

D is a more recommend way by Google, IMO.

squishy_fishy 2 years, 5 months ago

I agree, at work use solution A for low volume of Pub/Sub messages and function, and using Composer for high volume Pub/Sub messages.

PhuocT 3 years, 7 months ago

A. more sense to me.

ducc 3 years, 7 months ago

Composer support exception and retry for complex pipeline. D might be correct

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #195 Topic 1

Your company wants to be able to retrieve large result sets of medical information from your current system, which has over 10 TBs in the database, and store the data in new tables for further query. The database must have a low-maintenance architecture and be accessible via SQL. You need to implement a cost-effective solution that can support data analytics for large result sets. What should you do?

A. Use Cloud SQL, but first organize the data into tables. Use JOIN in queries to retrieve data.

B. Use BigQuery as a data warehouse. Set output destinations for caching large queries. **Most Voted**

C. Use a MySQL cluster installed on a Compute Engine managed instance group for scalability.

D. Use Cloud Spanner to replicate the data across regions. Normalize the data in a series of tables.

Correct Answer: B

Community vote distribution

B (100%)

Comments

AWSandeep **Highly Voted** 3 years ago

Selected Answer: B

B. Use BigQuery as a data warehouse. Set output destinations for caching large queries.

MaxNRG 1 year, 9 months ago

Selected Answer: B

Option B is the best approach - use BigQuery as a data warehouse, and set output destinations for caching large queries. The key reasons why BigQuery fits the requirements: It is a fully managed data warehouse built to scale to handle massive datasets and perform fast SQL analytics It has a low maintenance architecture with no infrastructure to manage SQL capabilities allow easy querying of the medical data Output destinations allow configurable caching for fast retrieval of large result sets It provides a very cost-effective solution for these large scale analytics use cases In contrast, Cloud Spanner and Cloud SQL would not scale as cost effectively for 10TB+ data volumes. Self-managed MySQL on Compute Engine also requires more maintenance. Hence, leveraging BigQuery as a fully managed data warehouse is the optimal solution here.

AzureDP900 2 years, 9 months ago

B. Use BigQuery as a data warehouse. Set output destinations for caching large queries. Most Voted

TNT87 3 years ago

Answer B. https://cloud.google.com/bigquery/docs/query-overview

ducc 3 years ago

Selected Answer: B

B is correct

https://zhangxuetu.online

[202610]

Question #196 Topic 1

You have 15 TB of data in your on-premises data center that you want to transfer to Google Cloud. Your data changes weekly and is stored in a POSIX-compliant source. The network operations team has granted you 500 Mbps bandwidth to the public internet. You want to follow Google-recommended practices to reliably transfer your data to Google Cloud on a weekly basis. What should you do?

A. Use Cloud Scheduler to trigger the gsutil command. Use the -m parameter for optimal parallelism.

B. Use Transfer Appliance to migrate your data into a Google Kubernetes Engine cluster, and then configure a weekly transfer job.

C. Install Storage Transfer Service for on-premises data in your data center, and then configure a weekly transfer job. **Most Voted**

D. Install Storage Transfer Service for on-premises data on a Google Cloud virtual machine, and then configure a weekly transfer job.

Correct Answer: C

Community vote distribution

C (100%)

Comments

musumusu **Highly Voted** 1 year, 7 months ago

answer C, To avoid confustion: Install Storage Transfer Service is always on EXTERNAL OR NON GOOGLE service or data centre to connect google service.

Prudvi3266 1 year, 5 months ago

Selected Answer: C

C is the Answer as we need weekly run Storage transfer service has the feature to schedule.

NicolasN 1 year, 10 months ago

Selected Answer: C

The fact that it's about a POSIX source makes necessary the set up of Storage Transfer Service agents. This detail limits [C] to be the correct answer, since it's the data center hosting the files where the agent must be installed. -- Some excerpts: (an older version of documentation was definite) "The following is a high-level overview of how Transfer service for on-premises data works: 1.Install Docker and run a small piece of software, called an agent, in your private data center. " Source: https://web.archive.org/web/20210529161414/https://cloud.google.com/storage-transfer/docs/on-prem-overview -- "Storage Transfer Service agents are applications running inside a Docker container, that coordinate with Storage Transfer Service to read data from POSIX file system sources, and/or write

data to POSIX file system sinks. If your transfer does not involve a POSIX file system, you do not need to set up agents." Source: https://cloud.google.com/storage-transfer/docs/managing-on-prem-agents

Atnafu 1 year, 10 months ago

C Storage Transfer Service agents are applications running inside a Docker container, that coordinate with Storage Transfer Service to read data from POSIX file system sources, and/or write data to POSIX file system sinks. https://cloud.google.com/storage-transfer/docs/managing-on-prem-agents#:~:text=Storage%20Transfer%20Service%20agents,agents%20on%20your%20servers.

namo621 2 years ago

why can't it be D?

Wasss123 2 years ago

Selected Answer: C

I vote for C

TNT87 2 years ago

Ans C https://cloud.google.com/storage-transfer/docs/overview

MounicaN 2 years ago

can you help with difference between c and d ?

gudiking 1 year, 10 months ago

If you install the software for on-premise data center on a Google Cloud VM, then it's not on-premise, it's on GCP, so it can't access your on-premise data.

AWSandeep 2 years ago

Selected Answer: C

C. Install Storage Transfer Service for on-premises data in your data center, and then configure a weekly transfer job.

ducc 2 years ago

Selected Answer: C

C is correct

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #197 Topic 1

You are designing a system that requires an ACID-compliant database. You must ensure that the system requires minimal human intervention in case of a failure.
What should you do?

A. Configure a Cloud SQL for MySQL instance with point-in-time recovery enabled.

B. Configure a Cloud SQL for PostgreSQL instance with high availability enabled. **Most Voted**

C. Configure a Bigtable instance with more than one cluster.

D. Configure a BigQuery table with a multi-region configuration.

Correct Answer: B

Community vote distribution

B (96%) D (4%)

Comments

NicolasN **Highly Voted** 3 years, 4 months ago

Selected Answer: B

We exclude [C[as non ACID and [D] for being invalid (location is configured on Dataset level, not Table). Then, let's focus on "minimal human intervention in case of a failure" requirement in order to eliminate one answer among [A] and [B]. Basically, we have to compare point-in-time recovery with high availability. It doesn't matter whether it's about MySQL or PostgreSQL since both databases support those features. - Point-in-time recovery logs are created automatically, but restoring an instance in case of failure requires manual steps (described here: https://cloud.google.com/sql/docs/mysql/backup-recovery/pitr#perform-pitr) - High availability, in case of failure requires no human intervention: "If an HA-configured instance becomes unresponsive, Cloud SQL automatically switches to serving data from the standby instance." (from https://cloud.google.com/sql/docs/postgres/high-availability#failover-overview) So answer [B] wins.

Mcloudgirl 3 years, 4 months ago

Your explanation is perfect, thanks

squishy_fishy 2 years, 5 months ago

Will you change your answer if the answer D says dataset instead of table?

edre 1 year, 8 months ago

Selected Answer: B

Its B because of HA cant be A because point in time recovery still requires human intervention

MaxNRG 2 years, 3 months ago

Selected Answer: B

The best option to meet the ACID compliance and minimal human intervention requirements is to configure a Cloud SQL for PostgreSQL instance with high availability enabled. Key reasons: Cloud SQL for PostgreSQL provides full ACID compliance, unlike Bigtable which provides only atomicity and consistency guarantees. Enabling high availability removes the need for manual failover as Cloud SQL will automatically failover to a standby replica if the leader instance goes down. Point-in-time recovery in MySQL requires manual intervention to restore data if needed. BigQuery does not provide transactional guarantees required for an ACID database. Therefore, a Cloud SQL for PostgreSQL instance with high availability meets the ACID and minimal intervention requirements best. The automatic failover will ensure availability and uptime without administrative effort.

[Removed] 2 years, 7 months ago

Selected Answer: D

I vote for D - BigQuery with multi region configuration. According to https://cloud.google.com/bigquery/docs/introduction , BigQuery support ACID and automatically replicated for high availability. """BigQuery stores data using a columnar storage format that is optimized for analytical queries. BigQuery presents data in tables, rows, and columns and provides full support for database transaction semantics (ACID). BigQuery storage is automatically replicated across multiple locations to provide high availability."""

vamgcp 2 years, 8 months ago

Selected Answer: B

Option B

musumusu 3 years, 1 month ago

Answer B, ACID -compliant database are Spanner and CloudSQL Option A could be the answer if they setup a secondary or failure replicas and auto maintenance window that could trigger in non business hours. Option B, does not explain about extra replica but in postgresql Highavailablity option means the same extra replicas instances are available for emergency.

AzureDP900 3 years, 3 months ago

B. Configure a Cloud SQL for PostgreSQL instance with high availability enabled.

samirzubair 3 years, 4 months ago

I voted for B

John_Pongthorn 3 years, 6 months ago

Selected Answer: B

B it is exact anwer.

TNT87 3 years, 6 months ago

Selected Answer: B

Ans B Postgres is highly ACID compliant as compared to Mysql

Remi2021 3 years, 6 months ago

Selected Answer: B

cloud sql with high availability enabled is enough

AWSandeep 3 years, 7 months ago

Selected Answer: B

B. Configure a Cloud SQL for PostgreSQL instance with high availability enabled.

ducc 3 years, 7 months ago

Selected Answer: B

I voted for B

https://zhangxuetu.online

[202610]

Question #198 Topic 1

You are implementing workflow pipeline scheduling using open source-based tools and Google Kubernetes Engine (GKE). You want to use a Google managed service to simplify and automate the task. You also want to accommodate Shared VPC networking considerations. What should you do?

A. Use Dataflow for your workflow pipelines. Use Cloud Run triggers for scheduling.

B. Use Dataflow for your workflow pipelines. Use shell scripts to schedule workflows.

C. Use Cloud Composer in a Shared VPC configuration. Place the Cloud Composer resources in the host project.

D. Use Cloud Composer in a Shared VPC configuration. Place the Cloud Composer resources in the service project. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

AWSandeep **Highly Voted** 3 years, 7 months ago

Selected Answer: D

D. Use Cloud Composer in a Shared VPC configuration. Place the Cloud Composer resources in the service project. Shared VPC requires that you designate a host project to which networks and subnetworks belong and a service project, which is attached to the host project. When Cloud Composer participates in a Shared VPC, the Cloud Composer environment is in the service project. Reference: https://cloud.google.com/composer/docs/how-to/managing/configuring-shared-vpc

loki82 1 year, 2 months ago

Selected Answer: D

Using the host project to deploy services is a bad practice and should only be used if the service doesn't support shared VPC. It may be easier networking wise, but that's why it's the wrong answer. Cloud Composer supports Shaved VPC, so it should go in it's own project.

8284a4c 1 year, 4 months ago

Selected Answer: C

Place in host project for network connectivity

ToiToi 1 year, 5 months ago

Selected Answer: C

The recommended approach is to place Cloud Composer resources in the host project of the Shared VPC. This centralizes network management, simplifies connectivity, and enhances security by adhering to the principle of least privilege.

TVH_Data_Engineer 1 year, 10 months ago

Selected Answer: C

Placing Cloud Composer resources in the service project can lead to more complex network configurations and management overhead compared to placing them in the host project, which is designed to manage Shared VPC resources.

vamgcp 2 years, 8 months ago

Please correct if I am wrong.. I think it is Option C coz I feel Option D is incorrect because placing the Cloud Composer resources in the service project would not allow you to access resources in the host project.

spicebits 2 years, 4 months ago

https://cloud.google.com/composer/docs/composer-2/configure-shared-vpc#shared-vpc-guidelines

Ender_H 2 years, 10 months ago

Selected Answer: A

 Use Cloud Composer in a Shared VPC configuration. Place the Cloud Composer resources in the service project. - Cloud Composer is a managed Apache Airflow service. It is an open-source tool that programmatically author, schedule, and monitor pipelines, which fits your needs perfectly. - In a Shared VPC configuration, Cloud Composer resources should be placed in the service project. This provides network isolation while still allowing the Cloud Composer environment to communicate with resources in the host project. - With Shared VPC, the host project's network (including its subnets and secondary IP ranges) is shared by other service projects, which promotes network peering, and it's compliant with the networking considerations of GKE.

ckanaar 2 years, 6 months ago

That's answer D though.

[Removed] 3 years, 4 months ago

Selected Answer: D

D is the answer. https://cloud.google.com/composer/docs/how-to/managing/configuring-shared-vpc Shared VPC enables organizations to establish budgeting and access control boundaries at the project level while allowing for secure and efficient communication using private IPs across those boundaries. In the Shared VPC configuration, Cloud Composer can invoke services hosted in other Google Cloud projects in the same organization without exposing services to the public internet.

AzureDP900 3 years, 3 months ago

Agreed

Oleksandr0501 2 years, 11 months ago

agreed to what..

Oleksandr0501 2 years, 11 months ago

D it is, as per doc link, provided by users. thx

Atnafu 3 years, 4 months ago

D Shared VPC requires that you designate a host project to which networks and subnetworks belong and a service project, which is attached to the host project. https://cloud.google.com/composer/docs/how-to/managing/configuring-shared-vpc#:~:text=This%20page%20describes,the%20service%20project.

ducc 3 years, 7 months ago

Selected Answer: D

D according to documentation Shared VPC requires that you designate a host project to which networks and subnetworks belong and a service project, which is attached to the host project. When Cloud Composer participates in a Shared VPC, the Cloud Composer environment is in the service project.
https://cloud.google.com/composer/docs/how-to/managing/configuring-shared-vpc#set_up_shared_vpc_and_attach_the_service_project

https://zhangxuetu.online

[202610]

Question #199 Topic 1

You are using BigQuery and Data Studio to design a customer-facing dashboard that displays large quantities of aggregated data. You expect a high volume of concurrent users. You need to optimize the dashboard to provide quick visualizations with minimal latency. What should you do?

A. Use BigQuery BI Engine with materialized views. **Most Voted**

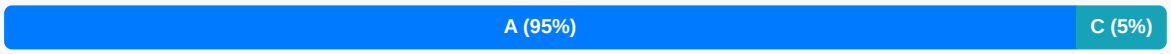
B. Use BigQuery BI Engine with logical views.

C. Use BigQuery BI Engine with streaming data.

D. Use BigQuery BI Engine with authorized views.

Correct Answer: A

Community vote distribution



Comments

AWSandeep **Highly Voted** 3 years ago

Selected Answer: A

A. Use BigQuery BI Engine with materialized views.

maic01234 1 year, 7 months ago

Option A is the better one. But keep in mind for real life: https://cloud.google.com/bigquery/docs/bi-engine-preferred-tables Limitations BI Engine preferred tables have the following limitations: You cannot add views into the preferred tables reservation list. BI Engine preferred tables only support tables. Queries to materialized views are only accelerated if both the materialized views and their base tables are in the preferred tables list.

vamgcp 2 years, 2 months ago

Selected Answer: A

Materialized views are precomputed query results that are stored in memory, allowing for faster retrieval of aggregated data. When you create a materialized view in BigQuery, it stores the results of a query as a table, and subsequent queries that can leverage this materialized view can be significantly faster compared to computing them on the fly.

sporch08 2 years, 1 month ago

If we take minimal latency into consideration, I am not sure a materialized view will be the right answer since the user gets data from the cache but is not up to date.

phidelics 2 years, 3 months ago

Selected Answer: A

periodically cache the results for perfomance

LPIT 2 years, 11 months ago

Selected Answer: A

A. https://cloud.google.com/bigquery/docs/materialized-views-intro In BigQuery, materialized views are precomputed views that periodically cache the results of a query for increased performance and efficiency

Julionga 3 years ago

Selected Answer: A

I vote A https://cloud.google.com/bigquery/docs/bi-engine-intro#:~:text=Materialized%20views%20%2D%20Materialized%20views%20in%20BigQuery%20perform%20precomputation%2C%20thereby%20reducing%20query%20time.%20You%20should%20create%20materialized%20views%20to%20improve%20performance%20and%20to%20reduce%20processed%20data%20by%20using%20aggregations%2C%20filters%2C%20inner%20joins%2C%20and%20unnests.

MounicaN 3 years ago

Selected Answer: A

use materialized views is better option here

ducc 3 years ago

Selected Answer: C

By integrating BI Engine with BigQuery streaming, you can perform real-time data analysis over streaming data without sacrificing write speeds or data freshness. https://cloud.google.com/bigquery/docs/bi-engine-intro

ducc 3 years ago

Sorry, A is correct As AWSandeep mention

https://zhangxuetu.online

[202610]

Question #200 Topic 1

Government regulations in the banking industry mandate the protection of clients' personally identifiable information (PII). Your company requires PII to be access controlled, encrypted, and compliant with major data protection standards. In addition to using Cloud Data Loss Prevention (Cloud DLP), you want to follow Google-recommended practices and use service accounts to control access to PII. What should you do?

A. Assign the required Identity and Access Management (IAM) roles to every employee, and create a single service account to access project resources.

B. Use one service account to access a Cloud SQL database, and use separate service accounts for each human user.

C. Use Cloud Storage to comply with major data protection standards. Use one service account shared by all users.

D. Use Cloud Storage to comply with major data protection standards. Use multiple service accounts attached to IAM groups to grant the appropriate access to each group. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

NicolasN **Highly Voted** 2 years, 10 months ago

Selected Answer: A

[A] is the only acceptable answer. [B] rejected (no need to elaborate) [C] and [D] rejected. Why should we be obliged to use Cloud Storage? Other storage options in Google Cloud aren't compliant with "major data protection standards"? ========== [D] has another rejection reason, the following quotes: From <https://cloud.google.com/iam/docs/service-accounts>: "You can add service accounts to a Google group, then grant roles to the group. However, adding service accounts to groups is not a best practice. Service accounts are used by applications, and each application is likely to have its own access requirements" From <https://cloud.google.com/iam/docs/best-practices-service-accounts#groups>: "Avoid using groups for granting service accounts access to resources"

KC_go_reply 2 years, 3 months ago

Rejecting C + D solely based on Cloud Storage, which CAN be used in this scenario, is not sound reasoning.

MaxNRG 1 year, 9 months ago

A single shared service account or granting every employee direct access violates security best practices, so not [A].

cetanx **Highly Voted** 2 years, 8 months ago

Selected Answer: D

for A: please refer to this link below which suggests "Sharing a single service account across multiple applications can complicate the management of the service account" - meaning it's not a best practice. https://cloud.google.com/iam/docs/best-practices-service-accounts#single-purpose Also, what if we have hundreds of users, does it really make sense to manage each user's IAM individually? for D: it's indeed not one of the best practices but I believe it's much more managable and better than A

skhaire 1 year ago

Selected Answer: D

Without Cloud storage, I believe just DLP does not provide encryption. DLP can redact or mask data, not encrypt it. Only on Cloud storage, encryption can be performed. So seems like option D is the closest choice, though service accounts should NOT be attached to IAM groups.

MaxNRG 1 year, 9 months ago

Selected Answer: D

To align with Google's recommended practices for managing access to personally identifiable information (PII) in compliance with banking industry regulations, let's analyze the options: A. Assign the required IAM roles to every employee, and create a single service account to access project resources: While assigning specific IAM roles to employees is a good practice for access control, using a single service account for all access to PII is not ideal. Service accounts should be used for applications and automated processes, not as a shared account for multiple users or employees.

MaxNRG 1 year, 9 months ago

B. Use one service account to access a Cloud SQL database, and use separate service accounts for each human user: Again, service accounts are intended for automated tasks or applications, not for individual human users. Assigning separate service accounts to each human user is not a recommended practice and does not align with the principle of least privilege.

MaxNRG 1 year, 9 months ago

C. Use Cloud Storage to comply with major data protection standards. Use one service account shared by all users: Using Cloud Storage can indeed help comply with data protection standards, especially when configured correctly with encryption and access controls. However, sharing a single service account among all users is not a best practice. It goes against the principle of least privilege and does not provide adequate granularity for access control.

MaxNRG 1 year, 9 months ago

D. Use Cloud Storage to comply with major data protection standards. Use multiple service accounts attached to IAM groups to grant the appropriate access to each group: This approach is more aligned with best practices. Using Cloud Storage can ensure compliance with data protection standards. Creating multiple service accounts, each with specific access controls attached to different IAM groups, allows for more granular and controlled access to PII. This setup adheres to the principle of least privilege, ensuring that each service (or group of services) only has access to the resources necessary for its function. Based on these considerations, option D is the most appropriate choice. It ensures compliance with data protection standards, uses Cloud Storage for secure data management, and employs multiple service accounts tied to IAM groups for granular access control, aligning well with Google-recommended practices and regulatory requirements in the banking industry.

vamgcp 2 years, 2 months ago

Selected Answer: D

Option D - Using multiple service accounts attached to IAM groups helps enforce the principle of least privilege. Each group can be assigned only the necessary permissions, reducing the risk of unauthorized access to sensitive data.

MoeHaydar 2 years, 2 months ago

Selected Answer: D

Google Cloud Storage is designed to comply with major data protection standards. Creating multiple service accounts and attaching them to IAM groups provides granular control over who has access to the data. This approach is aligned with the principle of least privilege, a security best practice where a user is given the minimum levels of access necessary to complete their tasks.

KC_go_reply 2 years, 3 months ago

Selected Answer: D

It´s not A because 1. assigning IAM roles to single users instead of groups is not Google best practice, and 2. the question explicitly states that we want to use multiple service accounts.

Ender_H 2 years, 3 months ago

Selected Answer: D

D. Use Cloud Storage to comply with major data protection standards. Use multiple service accounts attached to IAM groups to grant the appropriate access to each group. - Google Cloud Storage is built for secure and compliant data storage. It supports compliance with major data protection standards, which is essential in the banking industry where data protection regulations are stringent. - Service accounts in Google Cloud represent non-human users (applications or services) that need to authenticate and be authorized to access specific Google Cloud resources. - Creating multiple service accounts attached to IAM groups allows you to manage access control in a granular manner. This follows the principle of least privilege, providing each group with only the permissions they need to perform their tasks, which is a recommended practice for managing access to sensitive data like PII.

Ender_H 2 years, 3 months ago

 D. Use Cloud Storage to comply with major data protection standards. Use one service account shared by all users. - Sharing one service account among all users is not a secure practice. It goes against the principle of least privilege and does not allow for granular control over access permissions. If the shared service account were to be compromised, all resources accessible by the account would be at risk.

juliobs 2 years, 6 months ago

Why are so many questions like this? None of the answers is best practice.

alfguemat 1 year, 10 months ago

I would like to ask your question to those who decide the questions on the exams. I don't understand what they're trying to do, many of the questions cause divided responses, because they don't have a clear answer. The certification process is a waste of time.

SuperVee 2 years, 6 months ago

Selected Answer: D

I could be wrong but I think the wording in D caused this confusion, so it is an English problem. -- "Use multiple service accounts attached to IAM groups to grant the appropriate access to each group" I believe what D really means is that you can create a group for a bunch of people who only need access to resource A, so attach a Service account to the group and service account only have access to A. Then you create another group for another bunch of people who only need access to resource B, so attach a service account to this group. this service account can only access to B. So each group/service account has a very specific access target, and purpose of the group is very narrowly defined which is allowed by best practice. However, wording in option D merged all these into one sentence causing confusions. Option A is an administrative nightmare to manage IAM for a larger user population which is actually also against GCP best practices.

Aamir185 2 years, 7 months ago

Selected Answer: D

D it is

AzureDP900 2 years, 9 months ago

D is right

Amar2022 2 years, 9 months ago

Selected Answer: A

A is the correct one

jkhong 2 years, 9 months ago

Selected Answer: A

Agree with NicolasN, D is bad practice. For D this may result in permission creep, where a group is granted access to an increasing number of resources. Only grant service accounts specific access to resources.

odacir 2 years, 9 months ago

Selected Answer: A

A is the answer, as NicalsN says. https://cloud.google.com/iam/docs/service-accounts#groups

Andrix2405 2 years, 9 months ago

Selected Answer: A

Avoid using groups for granting service accounts access to resources -> D

Andrix2405 2 years, 9 months ago

Sorry A

hauhau 2 years, 10 months ago

Why GCS?

https://zhangxuetu.online

[202610]

Question #201 Topic 1

You need to migrate a Redis database from an on-premises data center to a Memorystore for Redis instance. You want to follow Google-recommended practices and perform the migration for minimal cost, time and effort. What should you do?

A. Make an RDB backup of the Redis database, use the gsutil utility to copy the RDB file into a Cloud Storage bucket, and then import the RDB file into the Memorystore for Redis instance. **Most Voted**

B. Make a secondary instance of the Redis database on a Compute Engine instance and then perform a live cutover.

C. Create a Dataflow job to read the Redis database from the on-premises data center and write the data to a Memorystore for Redis instance.

D. Write a shell script to migrate the Redis data and create a new Memorystore for Redis instance.

Correct Answer: A

Community vote distribution

A (100%)

Comments

AWSandeep **Highly Voted** 2 years, 7 months ago

Selected Answer: A

A. Make an RDB backup of the Redis database, use the gsutil utility to copy the RDB file into a Cloud Storage bucket, and then import the RDB file into the Memorystore for Redis instance. The import and export feature uses the native RDB snapshot feature of Redis to import data into or export data out of a Memorystore for Redis instance. The use of the native RDB format prevents lock-in and makes it very easy to move data within Google Cloud or outside of Google Cloud. Import and export uses Cloud Storage buckets to store RDB files. Reference: https://cloud.google.com/memorystore/docs/redis/import-export-overview

vamgcp 1 year, 8 months ago

Selected Answer: A

Option A

AzureDP900 2 years, 3 months ago

A. Make an RDB backup of the Redis database, use the gsutil utility to copy the RDB file into a Cloud Storage bucket, and then import the RDB file into the Memorystore for Redis instance.

gudiking 2 years, 4 months ago

Selected Answer: A

A https://cloud.google.com/memorystore/docs/redis/import-data

Atnafu 2 years, 4 months ago

A Import and export uses Cloud Storage buckets to store RDB files.
https://cloud.google.com/memorystore/docs/redis/about-importing-exporting#:~:text=Import%20and%20export%20uses%20Cloud%20Storage%20buckets%20to%20store%20RDB%20files.

ducc 2 years, 7 months ago

Selected Answer: A

A https://cloud.google.com/memorystore/docs/redis/general-best-practices

https://zhangxuetu.online

[202610]

Question #202 Topic 1

Your platform on your on-premises environment generates 100 GB of data daily, composed of millions of structured JSON text files. Your on-premises environment cannot be accessed from the public internet. You want to use Google Cloud products to query and explore the platform data. What should you do?

A. Use Cloud Scheduler to copy data daily from your on-premises environment to Cloud Storage. Use the BigQuery Data Transfer Service to import data into BigQuery.

B. Use a Transfer Appliance to copy data from your on-premises environment to Cloud Storage. Use the BigQuery Data Transfer Service to import data into BigQuery.

C. Use Transfer Service for on-premises data to copy data from your on-premises environment to Cloud Storage. Use the BigQuery Data Transfer Service to import data into BigQuery. Most Voted

D. Use the BigQuery Data Transfer Service dataset copy to transfer all data into BigQuery.

Correct Answer: C

Community vote distribution

C (89%) B (11%)

Comments

muhusman Highly Voted 2 years, 11 months ago

Therefore, the correct option is C. Use Transfer Service for on-premises data to copy data from your on-premises environment to Cloud Storage. Use the BigQuery Data Transfer Service to import data into BigQuery. Option A is incorrect because Cloud Scheduler is not designed for data transfer, but rather for scheduling the execution of Cloud Functions, Cloud Run, or App Engine applications. Option B is incorrect because Transfer Appliance is designed for large-scale data transfers from on-premises environments to Google Cloud and is not suitable for transferring data on a daily basis. Option D is also incorrect because the BigQuery Data Transfer Service dataset copy feature is designed for copying datasets between BigQuery projects and not suitable for copying data from on-premises environments to BigQuery.

datapassionate 2 years, 2 months ago

With BigQuery Data Transfer Service we can copy files not only from other BigQuery, but also a bunch of cloud services listed here: https://cloud.google.com/bigquery/docs/dts-introduction But you are right. It wont work with on-premises.

cetanx Highly Voted 2 years, 9 months ago

Selected Answer: C

"Your on-premises environment cannot be accessed from the public internet" statement suggests that inbound traffic from internet is NOT allowed however, it doesn't mean that outbound internet connectivity from on-prem resources is

not possible. Any on-prem system with outbound internet access can copy/transfer the CSV files. CSV files are located on a filesystem, therefore you cannot copy them with BQ Transfer Service. Leaving only possible option; first copy CSVs to cloud storage then run BQ Transfer Service pls refer to https://cloud.google.com/bigquery/docs/dts-introduction#supported_data_sources

desertlotus1211 1 year ago

Selected Answer: C

I'm torn on this question. Okay no access from public internet... does that mean they don't have private lines (e.g. Ded/Partner interconnects)? Poorly worded. IMO it can either be: Answer B or C based on interpretation of Public Internet.

marlon.andrei 1 year, 2 months ago

Selected Answer: B

I vote B, because in "Your on-premises environment cannot be accessed from the public internet.", it would only allow data to be extracted internally within the company. So Transfer Appliance is the most appropriate tool.

namesgeo 1 year, 3 months ago

Selected Answer: C

Transfer Service for on-premises data is designed specifically for this scenario. It uses a private, secure agent-based approach to move data from on-premises environments to Google Cloud Storage.

namesgeo 1 year, 3 months ago

https://cloud.google.com/blog/products/storage-data-transfer/introducing-storage-transfer-service-for-on-premises-data?hl=en

baimus 1 year, 5 months ago

They don't define "cannot be accessed from the public internet" - does this mean no incoming traffic, or no traffic or any kind regardless of the initiation point? We simply do not know, and so are left guessing. C? Probably, but could be B, just depending.

Takshashila 2 years, 9 months ago

Selected Answer: C

the correct option is C

wjtb 3 years ago

I would say B. It is the ONLY option that is possible without data being accessible over the public (unless we assume that a direct interconnect is already set up, which seems farfetched). Also, nowhere does it say how up-to-date the data needs to be that we are querying or how often we need to query, only that the data increases in size by 100gb per day (indicating that its going to be a lot of data)

musumusu 3 years, 1 month ago

Answer C, What is wrong with B ? Key words = Daily transfer .. so no to transfer appliance,

Atnafu 3 years, 4 months ago

C D-no answer because bq transfer service don't support from on-prem

Atnafu 3 years, 4 months ago

B-is not answer because you want transfer appliance for one time bulk transfer but the question is You want to use Google Cloud products to query and explore the platform data. query and explore is the key

John_Pongthorn 3 years, 5 months ago

Selected Answer: C

Transfer Service for on-premises is optimal for on-premises google (large files (< 1 TB) and bandwidth available and scheduling) https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#transfer-options https://cloud.google.com/blog/products/storage-data-transfer/introducing-storage-transfer-service-for-on-premises-data BigQuery Data Transfer Service is good for gcs to bigquery https://cloud.google.com/bigquery/docs/cloud-storage-transfer

John_Pongthorn 3 years, 5 months ago

Sorry I am wrong (large files > 1 TB + bandwidth available on internal IP address communication + daily scheduling)

John_Pongthorn 3 years, 5 months ago

Your on-premises environment cannot be accessed from the public internet. It signifies that we can apply private connection like Cloud Interconnect https://cloud.google.com/network-connectivity/docs/interconnect/concepts/overview

Wasss123 3 years, 6 months ago

Selected Answer: C

I will go with C

MounicaN 3 years, 6 months ago

I will g with C https://cloud.google.com/architecture/migration-to-google-cloud-transferring-your-large-datasets#transfer-options

John_Pongthorn 3 years, 6 months ago

C is correct, b is suitable for weekly . https://cloud.google.com/transfer-appliance/docs/4.0/overview

John_Pongthorn 3 years, 5 months ago

C Your on-premises environment cannot be accessed from the public internet. It signifies that we can apply private connection like Cloud Interconnect https://cloud.google.com/network-connectivity/docs/interconnect/concepts/overview

TNT87 3 years, 6 months ago

Selected Answer: C

Ans C https://cloud.google.com/storage-transfer/docs/on-prem-agent-best-practices

HarshKothari21 3 years, 6 months ago

I would go with option C. You need a service to transfer data from on-premises to cloud storage. so "Transfer service" is the best option & additionally you can easily configure the network so that data flows through private network. cloud scheduler on other hand is used mostly for automation. You can schedule a service but in my view cannot be used solo to transfer data.

nwk 3 years, 6 months ago

Data is generated daily. Unlikely to ship Transfer Appliance every day. Vote for C instead. "Transfer Service for on-premises data is a free Google Cloud service that's intended to streamline the process of uploading data into Google Cloud Storage buckets" https://cloud.google.com/blog/products/storage-data-transfer/introducing-storage-transfer-service-for-on-premises-data

https://zhangxuetu.online

[202610]

Question #203 Topic 1

A TensorFlow machine learning model on Compute Engine virtual machines (n2-standard-32) takes two days to complete training. The model has custom TensorFlow operations that must run partially on a CPU. You want to reduce the training time in a cost-effective manner. What should you do?

A. Change the VM type to n2-highmem-32.

B. Change the VM type to e2-standard-32.

C. Train the model using a VM with a GPU hardware accelerator. **Most Voted**

D. Train the model using a VM with a TPU hardware accelerator.

Correct Answer: C

Community vote distribution

C (100%)

Comments

jkhong **Highly Voted** 2 years, 9 months ago

Selected Answer: C

Cost effective - among the choices, it is cheaper to have a temporary accelerator instead of increasing our VM cost for an indefinite amount of time D -> TPU accelerator cannot support custom operations

MaxNRG **Highly Voted** 1 year, 9 months ago

Selected Answer: C

The best way to reduce the TensorFlow training time in a cost-effective manner is to use a VM with a GPU hardware accelerator. TensorFlow can take advantage of GPUs to significantly speed up training time for many models. Specifically, option C is the best choice. Changing the VM to another standard type like n2-highmem-32 or e2-standard-32 (options A and B) may provide some improvement, but likely not a significant speedup. Using a TPU (option D) could speed up training, but TPUs are more costly than GPUs. For a cost-effective solution, GPU acceleration provides the best performance per dollar. Since the model must run partially on CPUs, a VM instance with GPUs added will allow TensorFlow to offload appropriate operations to the GPUs while keeping CPU-specific operations on the CPU. This can provide a significant reduction in training time for many common TensorFlow models while keeping costs reasonable

wences 1 year, 4 months ago

Selected Answer: C

key pjrse is "run partially on a CPU" from https://cloud.google.com/tpu/docs/intro-to-tpu#when_to_use_tpus refers to GPU

spicebits 1 year, 11 months ago

Selected Answer: C

https://cloud.google.com/tpu/docs/intro-to-tpu#when_to_use_tpus

AzureDP900 2 years, 9 months ago

C. Train the model using a VM with a GPU hardware accelerator.

Atnafu 2 years, 10 months ago

C

https://cloud.google.com/tpu/docs/tpus#when_to_use_tpus:~:text=Models%20with%20a%20significant%20number%20of%20custom%20TensorFlow%20operations%20that%20must%20run%20at%20least%20partially%20on%20CPUs

Atnafu 2 years, 9 months ago

The model has custom TensorFlow operations that must run partially on a CPU. is the key for GPU

gudiking 2 years, 10 months ago

Selected Answer: C

I agree with C, for choosing a GPU one of the cases says: "Models with a significant number of custom TensorFlow operations that must run at least partially on CPUs" https://cloud.google.com/tpu/docs/tpus#when_to_use_tpus

gudiking 2 years, 10 months ago

C is not cost-effective, so I stand corrected. I do not know the answer.

https://zhangxuetu.online

[202610]

Question #204 Topic 1

You want to create a machine learning model using BigQuery ML and create an endpoint for hosting the model using Vertex AI. This will enable the processing of continuous streaming data in near-real time from multiple vendors. The data may contain invalid values. What should you do?

A. Create a new BigQuery dataset and use streaming inserts to land the data from multiple vendors. Configure your BigQuery ML model to use the "ingestion" dataset as the framing data.

B. Use BigQuery streaming inserts to land the data from multiple vendors where your BigQuery dataset ML model is deployed.

C. Create a Pub/Sub topic and send all vendor data to it. Connect a Cloud Function to the topic to process the data and store it in BigQuery.

D. Create a Pub/Sub topic and send all vendor data to it. Use Dataflow to process and sanitize the Pub/Sub data and stream it to BigQuery. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

anyone_99 1 year, 8 months ago

Why is the answer A? After paying $44 I am getting wrong answers.

987af6b 1 year, 8 months ago

The discussion is where the real answer is.

Matt_108 2 years, 2 months ago

Selected Answer: D

Option D

vamgcp 2 years, 8 months ago

Selected Answer: D

Option D -Dataflow provides a scalable and flexible way to process and clean the incoming data in real-time before loading it into BigQuery.

AzureDP900 3 years, 3 months ago

D. Create a Pub/Sub topic and send all vendor data to it. Use Dataflow to process and sanitize the Pub/Sub data and stream it to BigQuery.

odacir 3 years, 3 months ago

Selected Answer: D

D is the best option to sanitize the data to its D

jkhong 3 years, 3 months ago

Selected Answer: D

Better to use pubsub for streaming and reading message data Dataflow ParDo can perform filtering of data

vidts 3 years, 4 months ago

Selected Answer: D

It's D

Atnafu 3 years, 4 months ago

Answer is D

https://zhangxuetu.online

[202610]

Question #205 Topic 1

You have a data processing application that runs on Google Kubernetes Engine (GKE). Containers need to be launched with their latest available configurations from a container registry. Your GKE nodes need to have GPUs, local SSDs, and 8 Gbps bandwidth. You want to efficiently provision the data processing infrastructure and manage the deployment process. What should you do?

A. Use Compute Engine startup scripts to pull container images, and use gcloud commands to provision the infrastructure.

B. Use Cloud Build to schedule a job using Terraform build to provision the infrastructure and launch with the most current container images. **Most Voted**

C. Use GKE to autoscale containers, and use gcloud commands to provision the infrastructure.

D. Use Dataflow to provision the data pipeline, and use Cloud Scheduler to run the job.

Correct Answer: B

Community vote distribution

B (100%)

Comments

MaxNRG **Highly Voted** 2 years, 3 months ago

Selected Answer: B

B is the best option to efficiently provision and manage the deployment process for this data processing application on GKE:

MaxNRG 2 years, 3 months ago

• Cloud Build allows you to automate the building, testing, and deployment of your application using Docker containers. • Using Terraform with Cloud Build provides Infrastructure as Code capabilities to provision the GKE cluster with GPUs, SSDs, and network bandwidth. • Terraform can be configured to pull the latest container images from the registry when deploying. • Cloud Build triggers provide event-based automation to rebuild and redeploy when container images are updated. • This provides an automated CI/CD pipeline to launch the application on GKE using the desired infrastructure and latest images. • Dataflow and Cloud Scheduler don't directly provide infrastructure provisioning or deployment orchestration for GKE. • gcloud commands can be used but don't provide the same automation benefits as Cloud Build + Terraform.

MaxNRG 2 years, 3 months ago

So using Cloud Build with Terraform templates provides the most efficient way to provision and deploy this data processing application on GKE.

SamuelTsch 1 year, 5 months ago

Selected Answer: B

I would go to option B. That is from my point of view a CI/CD question. Only B covers the deployement and set up the latest container image.

anyone_99 1 year, 8 months ago

another wrong answer?

raaad 2 years, 3 months ago

Selected Answer: B

- Dataflow is a fully managed service for stream and batch data processing and is well-suited for real-time data processing tasks like identifying longtail and outlier data points. - Using BigQuery as a sink allows to efficiently store the cleansed and processed data for further analysis and serving it to AI models.

spicebits 2 years, 4 months ago

Selected Answer: B

I don't really like B or C... but given the choices I would go with B. B-Use Cloud Build to schedule a job using Terraform build to provision the infrastructure and launch with the most current container images. {The Terraform command is Terraform Apply and not Terraform build, but also why not use gcloud container command instead of introducing 3rd party builder image?)... I don't like this choice but it is the best one. C. Use GKE to autoscale containers, and use gcloud commands to provision the infrastructure. {This doesn't handle the building of the infra, or the deployment of the latest images, this one is clearly wrong, not sure why it is marked as the right choice}

vamgcp 2 years, 8 months ago

Selected Answer: B

B is correct

whorillo 2 years, 11 months ago

Selected Answer: B

B is correct

charline 3 years, 1 month ago

Selected Answer: B

b is ok

AzureDP900 3 years, 3 months ago

B. Use Cloud Build to schedule a job using Terraform build to provision the infrastructure and launch with the most current container images.

hauhau 3 years, 3 months ago

Selected Answer: B

Maybe B ref: https://cloud.google.com/architecture/managing-infrastructure-as-code

Atnafu 3 years, 4 months ago

C is correct answer

Atnafu 3 years, 4 months ago

Sorry I meant B

https://zhangxuetu.online

[202610]

Question #206 Topic 1

You need ads data to serve AI models and historical data for analytics. Longtail and outlier data points need to be identified. You want to cleanse the data in near-real time before running it through AI models. What should you do?

A. Use Cloud Storage as a data warehouse, shell scripts for processing, and BigQuery to create views for desired datasets.

B. Use Dataflow to identify longtail and outlier data points programmatically, with BigQuery as a sink. Most Voted

C. Use BigQuery to ingest, prepare, and then analyze the data, and then run queries to create views.

D. Use Cloud Composer to identify longtail and outlier data points, and then output a usable dataset to BigQuery.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Y___ash 1 year, 6 months ago

Selected Answer: B

Dataflow for Real-Time Processing: Dataflow allows you to process data in near-real time, making it well-suited for identifying longtail and outlier data points as they occur. You can use Dataflow to implement custom data cleansing and outlier detection algorithms that operate on streaming data. BigQuery as a Sink: Using BigQuery as a sink allows you to store the cleaned and processed data efficiently for further analysis or use in AI models. Dataflow can write the cleaned data to BigQuery tables, enabling seamless integration with downstream processes.

JyoGCP 1 year, 7 months ago

Selected Answer: B

B. Use Dataflow to identify longtail and outlier data points programmatically, with BigQuery as a sink.

datapassionate 1 year, 8 months ago

Selected Answer: B

B. Use Dataflow to identify longtail and outlier data points programmatically, with BigQuery as a sink.

Matt_108 1 year, 8 months ago

Selected Answer: B

B: Dataflow, solves exactly the use case described

MaxNRG 1 year, 8 months ago

Selected Answer: B

B is the best option for cleansing the ads data in near real-time before running it through AI models. The key reasons are: • Dataflow allows for stream processing of data in near real-time. This allows you to identify and cleanse longtail and outlier data points as the data is streamed in. • Dataflow has built-in capabilities for detecting and handling outliers and anomalies in streaming data. This makes it well-suited for programmatically identifying longtail and outlier data points. • Using BigQuery as the output sink allows the cleansed data to be immediatley available for analysis and serving to AI models. BigQuery can act as a serving layer for the models. • Options A, C, and D either don't provide real-time processing (A and C) or don't easily integrate with BigQuery for analysis and serving (D).

MaxNRG 1 year, 8 months ago

So B is the best architecture here to meet the needs of near real-time cleansing, identification of longtail/outlier data points, and integration with BigQuery for serving AI models.

raaad 1 year, 9 months ago

Selected Answer: B

- Dataflow is a fully managed service for stream and batch data processing and is well-suited for real-time data processing tasks like identifying longtail and outlier data points. - Using BigQuery as a sink allows to efficiently store the cleansed and processed data for further analysis and serving it to AI models.

e70ea9e 1 year, 9 months ago

Selected Answer: B

Real-time Data Processing: Dataflow excels at handling large-scale, streaming data with low latency, enabling near-real-time cleansing. Scalability: Easily scales to handle growing data volumes and processing needs. Programmatic Data Cleaning: Allows you to write custom logic in Apache Beam for identifying longtail and outlier data points accurately and efficiently. Integration with BigQuery: Seamless integration with BigQuery enables you to store cleansed data for AI model training and historical analytics. Cost-Effective: Dataflow's pay-as-you-go model optimizes costs for real-time data processing.

https://zhangxuetu.online

[202610]

Question #207 Topic 1

You are collecting IoT sensor data from millions of devices across the world and storing the data in BigQuery. Your access pattern is based on recent data, filtered by location_id and device_version with the following query:

```
SELECT
    MAX(temperature)
FROM
    acme_iot_data.sensors
WHERE
    create_date > DATE_SUB(CURRENT_DATE(), INTERVAL 7 day)
    AND location_id = "SW1W9TQ"
    AND device_version = "202007r3"
```

You want to optimize your queries for cost and performance. How should you structure your data?

A. Partition table data by create_date, location_id, and device_version.

B. Partition table data by create_date, cluster table data by location_id, and device_version. **Most Voted**

C. Cluster table data by create_date, location_id, and device_version.

D. Cluster table data by create_date, partition by location_id, and device_version.

Correct Answer: B

Community vote distribution

B (100%)

Comments

JyoGCP 1 year, 7 months ago

Selected Answer: B

B. Partition table data by create_date, cluster table data by location_id, and device_version.

datapassionate 1 year, 8 months ago

Selected Answer: B

B. Partition table data by create_date, cluster table data by location_id, and device_version.

Matt_108 1 year, 8 months ago

Selected Answer: B

B: Partitioning makes date-related querying efficient, clustering will keep relevant data close together and optimize the performance of filters for the cluster columns

MaxNRG 1 year, 8 months ago

Selected Answer: B

1. Partitioning the data by create_date will allow BigQuery to prune partitions that are not relevant to the query by date. 2. Clustering the data by location_id and device_version within each partition will keep related data close together and optimize the performance of filters on those columns. This provides both the pruning benefits of partitioning and locality benefits of clustering for filters on multiple columns. The query provided indicates that the access pattern is primarily based on the most recent data (within the last 7 days), filtered by location_id and device_version. Given this pattern, you would want to optimize your table structure in such a way that queries scanning through the data will process the least amount of data possible to reduce costs and improve performance.

Smakyel79 1 year, 8 months ago

Selected Answer: B

Only correct answer is B, you can only partition by one field, and you can only cluster on partitioned tables

raaad 1 year, 9 months ago

Selected Answer: B

Answer is B: - Partitioning the table by create_date allows us to efficiently query data based on time, which is common in access patterns that prioritize recent data. - Clustering the table by location_id and device_version further organizes the data within each partition, making queries filtered by these columns more efficient and cost-effective.

e70ea9e 1 year, 9 months ago

Selected Answer: B

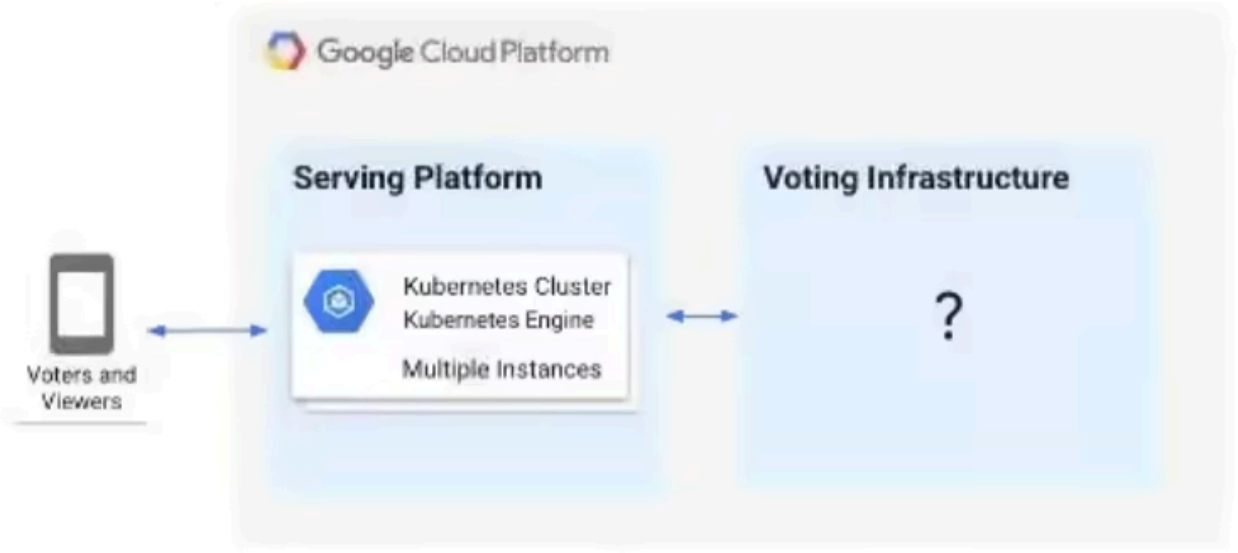
The best answer is B. Partition table data by create_date, cluster table data by location_id, and device_version. Here's a breakdown of why this structure is optimal: Partitioning by create_date: Aligns with query pattern: Filters for recent data based on create_date, so partitioning by this column allows BigQuery to quickly narrow down the data to scan, reducing query costs and improving performance. Manages data growth: Partitioning effectively segments data by date, making it easier to manage large datasets and optimize storage costs. Clustering by location_id and device_version: Enhances filtering: Frequently filtering by location_id and device_version, clustering physically co-locates related data within partitions, further reducing scan time and improving performance.

https://zhangxuetu.online

[202610]

Question #208 Topic 1

A live TV show asks viewers to cast votes using their mobile phones. The event generates a large volume of data during a 3-minute period. You are in charge of the "Voting infrastructure" and must ensure that the platform can handle the load and that all votes are processed. You must display partial results while voting is open. After voting closes, you need to count the votes exactly once while optimizing cost. What should you do?



A. Create a Memorystore instance with a high availability (HA) configuration.

B. Create a Cloud SQL for PostgreSQL database with high availability (HA) configuration and multiple read replicas.

C. Write votes to a Pub/Sub topic and have Cloud Functions subscribe to it and write votes to BigQuery.

D. Write votes to a Pub/Sub topic and load into both Bigtable and BigQuery via a Dataflow pipeline. Query Bigtable for real-time results and BigQuery for later analysis. Shut down the Bigtable instance when voting concludes. Most Voted

Correct Answer: D

Community vote distribution

D (82%) C (18%)

Comments

MaxNRG Highly Voted 1 year, 8 months ago

Selected Answer: D

Since cost optimization and minimal latency are key requirements, option D is likely the best choice to meet all the needs: The key reasons option D works well: Using Pub/Sub to ingest votes provides scalable, reliable transport. Loading into Bigtable and BigQuery provides both: Low latency reads from Bigtable for real-time results. Cost effective storage in BigQuery for longer term analysis. Shutting down Bigtable after voting concludes reduces costs. BigQuery remains available for cost-optimized storage and analysis. So you are correct that option D combines the best of real-time performance for queries using Bigtable, with cost-optimized storage in BigQuery. The only additional consideration may be if 3 minutes of Bigtable usage still incurs higher charges than ingesting directly into BigQuery. But for minimizing latency while optimizing cost, option D is likely the right architectural choice given the requirements.

f74ca0c 1 year, 3 months ago

Selected Answer: C

Modern Capabilities: BigQuery's advancements make it suitable for both real-time and historical querying. Cost Efficiency: No need to spin up and shut down a Bigtable instance. Simplified Workflow: Real-time and post-event data are stored in the same system, reducing the need to synchronize or transfer data between systems.

JyoGCP 1 year, 7 months ago

Selected Answer: D

D. Write votes to a Pub/Sub topic and load into both Bigtable and BigQuery via a Dataflow pipeline. Query Bigtable for real-time results and BigQuery for later analysis. Shut down the Bigtable instance when voting concludes.

Matt_108 1 year, 8 months ago

Selected Answer: D

D, i do agree with everything MaxNRG said.

Smakyel79 1 year, 8 months ago

Selected Answer: C

Pub/Sub for sure, and Cloud Functions + BigQuery Streaming seems a good solution. Won't use BigTable as need at least 100GB of data (don't thing a voting system could arrive to that amount of data) and needs to "heat" to work right for >10 minutes... and would be $$$ over C solution

raaad 1 year, 9 months ago

Selected Answer: D

Answer is D: - Google Cloud Pub/Sub can manage the high-volume data ingestion. - Google Cloud Dataflow can efficiently process and route data to both Bigtable and BigQuery. - Bigtable is excellent for handling high-throughput writes and reads, making it suitable for real-time vote tallying. - BigQuery is ideal for exact vote counting and deeper analysis once voting concludes.

e70ea9e 1 year, 9 months ago

Selected Answer: D

Handling High-Volume Data Ingestion: Pub/Sub: Decouples vote collection from processing, ensuring scalability and resilience under high load. Dataflow: Efficiently ingests and processes large data streams, scaling as needed. Real-Time Results with Exactly-Once Processing: Bigtable: Optimized for low-latency, high-throughput reads and writes, ideal for real-time partial results. Exactly-Once Semantics: Dataflow guarantees each vote is processed only once, ensuring accurate counts. Cost Optimization: Temporary Bigtable Instance: Running Bigtable only during voting minimizes costs. BigQuery Storage: Cost-effective for long-term storage and analysis.

https://zhangxuetu.online

[202610]

Question #209 Topic 1

A shipping company has live package-tracking data that is sent to an Apache Kafka stream in real time. This is then loaded into BigQuery. Analysts in your company want to query the tracking data in BigQuery to analyze geospatial trends in the lifecycle of a package. The table was originally created with ingest-date partitioning. Over time, the query processing time has increased. You need to copy all the data to a new clustered table. What should you do?

A. Re-create the table using data partitioning on the package delivery date.

B. Implement clustering in BigQuery on the package-tracking ID column. **Most Voted**

C. Implement clustering in BigQuery on the ingest date column.

D. Tier older data onto Cloud Storage files and create a BigQuery table using Cloud Storage as an external data source.

Correct Answer: B

Community vote distribution

B (100%)

Comments

desertlotus1211 11 months, 3 weeks ago

Selected Answer: B

Almost the same at #166:

apoio.certificacoes.closer 1 year, 3 months ago

Selected Answer: B

You don't need to recreate a table to cluster it, contrary to partitioning, where you have to create a new table with the old data (migration) > If you alter an existing non-clustered table to be clustered, the existing data is not automatically clustered. Only new data that's stored using the clustered columns is subject to automatic reclustering. https://cloud.google.com/bigquery/docs/clustered-tables#limitations

JyoGCP 1 year, 7 months ago

Selected Answer: B

B. Implement clustering in BigQuery on the package-tracking ID column.

datapassionate 1 year, 8 months ago

Selected Answer: B

B. Implement clustering in BigQuery on the package-tracking ID column.

Matt_108 1 year, 8 months ago

Selected Answer: B

Definitely B

MaxNRG 1 year, 8 months ago

Selected Answer: B

This looks like Question #166 Option B, implementing clustering in BigQuery on the package-tracking ID column, seems the most appropriate. It directly addresses the query slowdown issue by reorganizing the data in a way that aligns with the analysts' query patterns, leading to more efficient and faster query execution.

raaad 1 year, 9 months ago

Selected Answer: B

Answer is B

e70ea9e 1 year, 9 months ago

Selected Answer: B

Query Focus: Analysts are interested in geospatial trends within individual package lifecycles. Clustering by package-tracking ID physically co-locates related data, significantly improving query performance for these analyses. Addressing Slow Queries: Clustering addresses the query slowdown issue by optimizing data organization for the specific query patterns. Partitioning vs. Clustering: Partitioning: Divides data into segments based on a column's values, primarily for managing large datasets and optimizing query costs. Clustering: Organizes data within partitions for faster querying based on specific columns.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #210 Topic 1

You are designing a data mesh on Google Cloud with multiple distinct data engineering teams building data products. The typical data curation design pattern consists of landing files in Cloud Storage, transforming raw data in Cloud Storage and BigQuery datasets, and storing the final curated data product in BigQuery datasets. You need to configure Dataplex to ensure that each team can access only the assets needed to build their data products. You also need to ensure that teams can easily share the curated data product. What should you do?

A. 1. Create a single Dataplex virtual lake and create a single zone to contain landing, raw, and curated data.
2. Provide each data engineering team access to the virtual lake.

B. 1. Create a single Dataplex virtual lake and create a single zone to contain landing, raw, and curated data.
2. Build separate assets for each data product within the zone.
3. Assign permissions to the data engineering teams at the zone level.

C. 1. Create a Dataplex virtual lake for each data product, and create a single zone to contain landing, raw, and curated data.
2. Provide the data engineering teams with full access to the virtual lake assigned to their data product.

D. 1. Create a Dataplex virtual lake for each data product, and create multiple zones for landing, raw, and curated data.
2. Provide the data engineering teams with full access to the virtual lake assigned to their data product. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

f74ca0c 1 year, 3 months ago

Selected Answer: D

Create a Dataplex lake that acts as the domain for your data mesh. Add zones to your lake that represents individual teams within each domain and provide managed data contracts. Attach assets that map to data stored in Cloud Storage. https://cloud.google.com/transfer-appliance/docs/4.0/overview?_gl=1*8pbq1*_up*MQ..&gclid=CjwKCAiAg8S7BhATEiwAO2-R6gOYtlc2FJa7zE4lhz3-2f00x9F3hwgul9lYjfJs2cAprxOIeXq_NhoCw-8QAvD_BwE&gclsrc=aw.ds

SamuelTsch 1 year, 5 months ago

Selected Answer: D

just like MaxNRG said

JyoGCP 2 years, 1 month ago

Selected Answer: D

Answer D

datapassionate 2 years, 2 months ago

Selected Answer: D

D. 1. Create a Dataplex virtual lake for each data product, and create multiple zones for landing, raw, and curated data. 2. Provide the data engineering teams with full access to the virtual lake assigned to their data product. Lake: A logical construct representing a data domain or business unit. For example, to organize data based on group usage, you can set up a lake for each department (for example, Retail, Sales, Finance). Zone: A subdomain within a lake, which is useful to categorize data by the following: Stage: For example, landing, raw, curated data analytics, and curated data science.

datapassionate 2 years, 2 months ago

https://cloud.google.com/dataplex/docs/introduction

Matt_108 2 years, 2 months ago

Selected Answer: D

D: 1 virtual lake per Data Product (which stands for domain basically), zones to split data by "status". Each Data Eng team can access their own data exclusively and in a data mesh compliant way

MaxNRG 2 years, 2 months ago

Selected Answer: D

The best approach is to create a Dataplex virtual lake for each data product, with multiple zones for landing, raw, and curated data. Then provide the data engineering teams with access only to the zones they need within the virtual lake assigned to their product. To enable teams to easily share curated data products, you should use cross-lake sharing in Dataplex. This allows curated zones to be shared across virtual lakes while maintaining data isolation for other zones.

MaxNRG 2 years, 2 months ago

So the steps would be: 1. Create a Dataplex virtual lake for each data product. 2. Within each lake, create separate zones for landing, raw, and curated data. 3. Provide each data engineering team with access only to the zones they need within their assigned virtual lake. 4. Configure cross-lake sharing on the curated data zones to share curated data products between teams. This provides isolation and access control between teams for raw data while enabling easy sharing of curated data products. https://cloud.google.com/dataplex/docs/introduction#a_domain-centric_data_mesh

Smakyel79 2 years, 2 months ago

I believe the answer is B, but there is a misspelling in the answer, should say "create multiple zones"

Helinia 2 years, 2 months ago

Selected Answer: D

Each lake should be created per data product since data product sounds like a domain in this question. Since we have landing, raw, curated data, we should create different zones. "Zones are of two types: raw and curated. Raw zone: Contains data that is in its raw format and not subject to strict type-checking. Curated zone: Contains data that is cleaned, formatted, and ready for analytics. The data is columnar, Hive-partitioned, and stored in Parquet, Avro, Orc files, or BigQuery tables. Data undergoes type-checking- for example, to prohibit the use of CSV files because they don't perform as well for SQL access." Ref: https://cloud.google.com/dataplex/docs/introduction#terminology

Jordan18 2 years, 2 months ago

why not B?

Sofiia98 2 years, 2 months ago

Why not B?

tibuenoc 2 years, 2 months ago

Because it's the best practice is separated zones. One zone for landing, raw and curated. The answer B - has this part that excluded it "create a single zone to contain landing" The correct awser is D

Ed_Kim 2 years, 3 months ago

Selected Answer: D

The answer is D

e70ea9e 2 years, 3 months ago

Selected Answer: C

Virtual Lake per Data Product: Each virtual lake acts as a self-contained domain for a specific data product, aligning with the data mesh principle of decentralized ownership and responsibility. Team Autonomy: Teams have full control over their virtual lake, enabling independent development, management, and sharing of their data products.

https://zhangxuetu.online

[202610]

Question #211 Topic 1

You are using BigQuery with a multi-region dataset that includes a table with the daily sales volumes. This table is updated multiple times per day. You need to protect your sales table in case of regional failures with a recovery point objective (RPO) of less than 24 hours, while keeping costs to a minimum. What should you do?

A. Schedule a daily export of the table to a Cloud Storage dual or multi-region bucket. **Most Voted**

B. Schedule a daily copy of the dataset to a backup region.

C. Schedule a daily BigQuery snapshot of the table.

D. Modify ETL job to load the data into both the current and another backup region.

Correct Answer: A

Community vote distribution



Comments

MaxNRG **Highly Voted** 2 years, 2 months ago

Selected Answer: A

Why not C: A table snapshot must be in the same region, and under the same organization, as its base table. https://cloud.google.com/bigquery/docs/table-snapshots-intro#limitations

MaxNRG 2 years, 2 months ago

Based on the information provided and the need to avoid data loss in the case of a hard regional failure in BigQuery, which could result in the destruction of all data in that region, the focus should be on creating backups in a geographically distinct region. Considering this scenario, the most suitable option would be Option A Here's why this option is the most appropriate:

MaxNRG 2 years, 2 months ago

• Cross-Region Backup: Exporting the data to a Google Cloud Storage bucket that is either dual or multi-regional ensures that your backups are stored in a different geographic location. This is critical for protecting against hard regional failures. • Data Durability: Cloud Storage provides high durability for stored data, making it a reliable option for backups in the case of regional disasters. • Cost-Effectiveness: While there are costs associated with storage and data transfer, this method can be more cost-effective compared to maintaining active replicas of the data in multiple regions, especially if the data is large.

MaxNRG 2 years, 2 months ago

• Flexibility and Automation: The export process can be automated and scheduled to occur daily, aligning with your RPO of less than 24 hours. This ensures that the most recent data is always backed up. • Recovery Process: In the

event of a hard regional failure, the data can be restored from the Cloud Storage backup to another operational BigQuery region, ensuring continuity of operations.

MaxNRG 2 years, 2 months ago

The other options, while viable in certain scenarios, do not provide the same level of protection against a hard regional failure: • Option B (Copy to Backup Region) and Option D (Modify ETL to Load into Backup Region) do not address the possibility of a hard regional failure adequately, as they do not necessarily imply storing data in a geographically distinct region. • Option C (BigQuery Snapshot) is useful for point-in-time recovery but does not inherently protect against hard regional failures since the snapshots are within the same BigQuery service. Focusing on a robust disaster recovery strategy is crucial. Option A provides a balance between ensuring data availability in the event of a regional disaster and managing costs, aligning with best practices for data management in the cloud.

raaad Highly Voted 2 years, 3 months ago

Selected Answer: C

Option C provides cost-effective way. - BigQuery table snapshots are a feature that allows you to capture the state of a table at a particular point in time. - Snapshots are incremental, so they only store the data that has changed, making them more cost-effective than full table copies. - In the event of a regional failure, you can quickly restore the table from a snapshot.

ToiToi 1 year, 5 months ago

Selected Answer: A

Why other options are not as suitable: B (Copy of the dataset): Copying the entire dataset daily is more expensive and less efficient than exporting just the table data. C (BigQuery snapshot): snapshots are within the same region and won't protect against a regional outage. D (Modify ETL job): This adds complexity to your ETL process and might not be the most efficient or cost-effective way to achieve your RPO.

desertlotus1211 1 year ago

What about the cost inplications? BigQuery exports are full-table exports, so you pay for every row scanned and written. Storage in multi-region buckets is a bit more expensive than single-region. Answer C is better suited

Priyal19 1 year, 7 months ago

A : BQ snapshot should be in same region, and if so the region fails so does the snapshot.

viciousjpjp 1 year, 7 months ago

Selected Answer: C

Why Option A is not suitable: Restoring data from Option A would require reloading it back into BigQuery, which is time-consuming. This process cannot guarantee a recovery point objective (RPO) of less than 24 hours.

meh_33 1 year, 7 months ago

Selected Answer: C

C seems correct and Raaad also saying same.

ostora 1 year, 11 months ago

Selected Answer: C

it is c

pandeyspecial 2 years, 1 month ago

Selected Answer: C

C. Schedule a daily BigQuery snapshot of the table. Here's why: Cost-effective: BigQuery snapshots are significantly cheaper than daily exports to Cloud Storage or copying the entire dataset to a backup region. They offer point-in-time backups with minimal storage costs. Fast recovery: Snapshots can be restored quickly, meeting your RPO

requirement of less than 24 hours. Multi-regional: By default, BigQuery snapshots are automatically stored in a different region from the source data, ensuring redundancy and disaster recovery.

Sergei_B 2 years ago

At the beginning I also thought that "C" is a correct answer, but futher I found documentation https://cloud.google.com/bigquery/docs/locations. According this documentation "Selecting a multi-region location does not provide cross-region replication or regional redundancy, so there is no increase in dataset availability in the event of a regional outage. Data is stored in a single region within the geographic location. Data located in the EU multi-region is only stored in the europe-west1 (Belgium) or europe-west4 (Netherlands) data centers" So, multi-region dataset just means locating data inside one of US regions, hence snapshot also will be stored in the same region what means that answer C is not correct

JyoGCP 2 years, 1 month ago

Selected Answer: A

A. Schedule a daily export of the table to a Cloud Storage dual or multi-region bucket.

GCP001 2 years, 2 months ago

Selected Answer: A

Option A. Check the ref for regional loss - https://cloud.google.com/bigquery/docs/reliability-intro#scenario_loss_of_region

datapassionate 2 years, 2 months ago

Selected Answer: A

A. Schedule a daily export of the table to a Cloud Storage dual or multi-region bucket.

Matt_108 2 years, 2 months ago

Selected Answer: A

A: MaxNRG and Helinia cleared the reasons very well

Helinia 2 years, 2 months ago

Selected Answer: A

"BigQuery does not offer durability or availability in the extraordinarily unlikely and unprecedented event of physical region loss. This is true for both "regions and multi-region" configurations. Hence maintaining durability and availability under such a scenario requires customer planning." "To avoid data loss in the face of destructive regional loss, you need to back up data to another geographic location. For example, you could periodically export a snapshot of your data to Google Cloud Storage in another geographically distinct region." Ref: https://cloud.google.com/bigquery/docs/reliability-intro#scenario_loss_of_region

MaxNRG 2 years, 2 months ago

Option A (Export to Cloud Storage): While exporting to Cloud Storage is a viable backup strategy, it can be more expensive and less efficient than using snapshots, especially if the table is large and updated frequently.

MaxNRG 2 years, 2 months ago

I agree, It's A: A table snapshot must be in the same region, and under the same organization, as its base table. https://cloud.google.com/bigquery/docs/table-snapshots-intro#limitations

Helinia 2 years, 2 months ago

Why not C: "BigQuery also supports the ability to snapshot tables. With this feature you can explicitly backup data within the same region for longer than the 7 day time travel window. A snapshot is purely a metadata operation and results in no additional storage bytes. While this can add protection against accidental deletion, it does not increase

the durability of the data." https://cloud.google.com/bigquery/docs/reliability-intro#scenario_accidental_deletion_or_data_corruption

qq589539483084gfrgrgfr 2 years, 2 months ago

Option A

qq589539483084gfrgrgfr 2 years, 2 months ago

https://cloud.google.com/bigquery/docs/reliability-intro

e70ea9e 2 years, 3 months ago

Selected Answer: D

Automatically replicates data to a backup region upon each update, ensuring an RPO of less than 24 hours, even with multiple daily updates.

raaad 2 years, 3 months ago

Option D: Doubles the write load and storage costs since you are maintaining two live datasets.

https://zhangxuetu.online

[202610]

Question #212 Topic 1

You are troubleshooting your Dataflow pipeline that processes data from Cloud Storage to BigQuery. You have discovered that the Dataflow worker nodes cannot communicate with one another. Your networking team relies on Google Cloud network tags to define firewall rules. You need to identify the issue while following Google-recommended networking security practices. What should you do?

A. Determine whether your Dataflow pipeline has a custom network tag set.

B. Determine whether there is a firewall rule set to allow traffic on TCP ports 12345 and 12346 for the Dataflow network tag. **Most Voted**

C. Determine whether there is a firewall rule set to allow traffic on TCP ports 12345 and 12346 on the subnet used by Dataflow workers.

D. Determine whether your Dataflow pipeline is deployed with the external IP address option enabled.

Correct Answer: B

Community vote distribution

B (100%)

Comments

MaxNRG **Highly Voted** 1 year, 8 months ago

Selected Answer: B

The best approach would be to check if there is a firewall rule allowing traffic on TCP ports 12345 and 12346 for the Dataflow network tag. Dataflow uses TCP ports 12345 and 12346 for communication between worker nodes. Using network tags and associated firewall rules is a Google-recommended security practice for controlling access between Compute Engine instances like Dataflow workers. So the key things to check would be: 1. Ensure your Dataflow pipeline is using the Dataflow network tag on the worker nodes. This tag is applied by default unless overridden. 2. Check if there is a firewall rule allowing TCP 12345 and 12346 ingress and egress traffic for instances with the Dataflow network tag. If not, add the rule. Options A, C and D relate to other networking aspects but do not directly address the Google recommended practice of using network tags and firewall rules.

JyoGCP 1 year, 7 months ago

Selected Answer: B

B. Determine whether there is a firewall rule set to allow traffic on TCP ports 12345 and 12346 for the Dataflow network tag.

Matt_108 1 year, 8 months ago

Selected Answer: B

B, check if there is a firewall rule allowing traffic on TCP ports 12345 and 12346 for the Dataflow network tag.

Smakyel79 1 year, 8 months ago

Selected Answer: B

Because network tags are used and Dataflow uses TCP ports 12345 and 12346 as stated on https://cloud.google.com/dataflow/docs/guides/routes-firewall

raaad 1 year, 9 months ago

Selected Answer: B

This option focuses directly on ensuring that the firewall rules are set up correctly for the network tags used by Dataflow worker nodes. It specifically addresses the potential issue of worker nodes not being able to communicate due to restrictive firewall rules blocking the necessary ports.

e70ea9e 1 year, 9 months ago

Selected Answer: B

Focus on Network Tags: Adheres to the recommended practice of using network tags for firewall configuration, enhancing security and flexibility. Avoids targeting specific subnets, which can be less secure and harder to manage.

https://zhangxuetu.online

[202610]

Question #213 Topic 1

Your company's customer_order table in BigQuery stores the order history for 10 million customers, with a table size of 10 PB. You need to create a dashboard for the support team to view the order history. The dashboard has two filters, country_name and username. Both are string data types in the BigQuery table. When a filter is applied, the dashboard fetches the order history from the table and displays the query results. However, the dashboard is slow to show the results when applying the filters to the following query:

```
SELECT date, order, status FROM customer_order
WHERE country = '<country_name>' AND username = '<username>'
```

How should you redesign the BigQuery table to support faster access?

A. Cluster the table by country and username fields. **Most Voted**

B. Cluster the table by country field, and partition by username field.

C. Partition the table by country and username fields.

D. Partition the table by _PARTITIONTIME.

Correct Answer: A

Community vote distribution

A (95%) B (5%)

Comments

Ryannn23 1 year, 2 months ago

Selected Answer: A

Partition on String is not available in BQ - excludes B and C Partition by ingest time not useful as the query is filtering on 2 other columns - excludes D Correct answer A: cluster on both fields.

niujo 1 year, 6 months ago

Why not D? if u partition by date and int going to be the best option??

b3e59c2 1 year, 2 months ago

Because our query filtering is relating to country and user, and nothing to do with time. A partition by time will provide no performance increase in this case.

JyoGCP 1 year, 7 months ago

Selected Answer: A

If country is represented by an integer code, then partition by country and cluster by username would be a better solution. As country code is a string, available best solution is "A. Cluster the table by country and username fields."

datapassionate 1 year, 8 months ago

Selected Answer: A

Correct answer: A. Cluster the table by country and username fields. Why not B and C - > Intiger is required for partitioning https://cloud.google.com/bigquery/docs/partitioned-tables#integer_range

Matt_108 1 year, 8 months ago

Selected Answer: A

A: the fields are both strings, which are not supported for partitioning. Moreover, the fields are regularly used in filters, which is where clustering really improves performance

SanjeevRoy91 1 year, 6 months ago

Is not mandatory to have partitioning for clustering?

Takshashila 1 year, 8 months ago

Selected Answer: B

Clustering can also be done after partiton?

chambg 1 year, 6 months ago

Yes but the partition is done on username field which has 10 million values. Since a BQ table can only have 4000 it is not suitable

raaad 1 year, 9 months ago

Selected Answer: A

- Clustering organizes the data based on the specified columns (in this case, country_name and username). - When a query filters on these columns, BigQuery can efficiently scan only the relevant parts of the table

e70ea9e 1 year, 9 months ago

Selected Answer: A

country and username --> cluster

https://zhangxuetu.online

[202610]

Question #214 Topic 1

You have a Standard Tier Memorystore for Redis instance deployed in a production environment. You need to simulate a Redis instance failover in the most accurate disaster recovery situation, and ensure that the failover has no impact on production data. What should you do?

A. Create a Standard Tier Memorystore for Redis instance in the development environment. Initiate a manual failover by using the limited-data-loss data protection mode.

B. Create a Standard Tier Memorystore for Redis instance in a development environment. Initiate a manual failover by using the force-data-loss data protection mode. Most Voted

C. Increase one replica to Redis instance in production environment. Initiate a manual failover by using the force-data-loss data protection mode.

D. Initiate a manual failover by using the limited-data-loss data protection mode to the Memorystore for Redis instance in the production environment.

Correct Answer: B

Community vote distribution



Comments

MaxNRG Highly Voted 2 years, 2 months ago

Selected Answer: B

The best option is B - Create a Standard Tier Memorystore for Redis instance in a development environment. Initiate a manual failover by using the force-data-loss data protection mode. The key points are: • The failover should be tested in a separate development environment, not production, to avoid impacting real data. • The force-data-loss mode will simulate a full failover and restart, which is the most accurate test of disaster recovery. • Limited-data-loss mode only fails over reads which does not fully test write capabilities. • Increasing replicas in production and failing over (C) risks losing real production data. • Failing over production (D) also risks impacting real data and traffic. So option B isolates the test from production and uses the most rigorous failover mode to fully validate disaster recovery capabilities.

desertlotus1211 1 year ago

Answer A is best suited for an 'accurate disaster recovery' scenario... limited-data-loss mode is Google's recommended failover simulation mode: Promotes the replica. Attempts to minimize data loss (vs. force failover). Mimics a realistic Redis failover due to a zonal outage or instance crash.

Parandhaman_Margan 1 year ago

Selected Answer: D

Initiate a manual failover by using the limited-data-loss data protection mode to the Memorystore for Redis instance in the production environment.

desertlotus1211 1 year ago

did you read this part? ' ensure that the failover has no impact on production data'.... Answer D is wrong.

SamuelTsch 1 year, 5 months ago

Selected Answer: B

The question says "no impact on production data" Thus, the best practice is about simulating in a different environment. force-data-loss mode covers the most accurate disaster recovery situation. (https://cloud.google.com/memorystore/docs/redis/about-manual-failover)

mi_yulai 1 year, 5 months ago

D: "A standard tier Memorystore for Redis instance uses a replica node to back up the primary node. A normal failover occurs when the primary node becomes unhealthy, causing the replica to be designated as the new primary. A manual failover differs from a normal failover because you initiate it yourself." The limited-data-loss mode minimizes data loss by verifying that the difference in data between the primary and replica is below 30 MB before initiating the failover. The offset on the primary is incremented for each byte of data that must be synchronized to its replicas.

mayankazyour 1 year, 7 months ago

Selected Answer: A

We are trying to simulate the disaster recovery on a redis Instance and we want minimum data loss. Therefore, Option A - create a test Standard Tier Memorystore for Redis instance in Dev Environment and use the limited data loss data protection mode, seems to be the correct option here.

anyone_99 1 year, 8 months ago

D seems correct. We are required to simulate and not test in a different environment. "How data protection modes work The limited-data-loss mode minimizes data loss by verifying that the difference in data between the primary and replica is below 30 MB before initiating the failover. The offset on the primary is incremented for each byte of data that must be synchronized to its replicas. In the limited-data-loss mode, the failover will abort if the greatest offset delta between the primary and each replica is 30MB or greater. If you can tolerate more data loss and want to aggressively execute the failover, try setting the data protection mode to force-data-loss. The force-data-loss mode employs a chain of failover strategies to aggressively execute the failover. It does not check the offset delta between the primary and replicas before initiating the failover; you can potentially lose more than 30MB of data changes."

tibuenoc 2 years, 2 months ago

Selected Answer: B

https://cloud.google.com/memorystore/docs/redis/about-manual-failover

datapassionate 2 years, 2 months ago

Selected Answer: B

B. Create a Standard Tier Memorystore for Redis instance in a development environment. Initiate a manual failover by using the force-data-loss data protection mode

Matt_108 2 years, 2 months ago

Selected Answer: B

Best option is B - no impact on production env and forces a full failover

raaad 2 years, 3 months ago

Selected Answer: C

Increasing the number of replicas in a Redis instance in a production environment means that we will have additional copies of the same data and thats why failover will not impact the production data

MaxNRG 2 years, 2 months ago

"no impact on production data" - not C nor D

e70ea9e 2 years, 3 months ago

Selected Answer: C

Separate Development Environment: Isolates testing from production, preventing any impact on live data or services. Provides a safe and controlled environment for simulating failover scenarios.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #215 Topic 1

You are administering a BigQuery dataset that uses a customer-managed encryption key (CMEK). You need to share the dataset with a partner organization that does not have access to your CMEK. What should you do?

A. Provide the partner organization a copy of your CMEKs to decrypt the data.

B. Export the tables to parquet files to a Cloud Storage bucket and grant the storageinsights.viewer role on the bucket to the partner organization.

C. Copy the tables you need to share to a dataset without CMEKs. Create an Analytics Hub listing for this dataset. Most Voted

D. Create an authorized view that contains the CMEK to decrypt the data when accessed.

Correct Answer: C

Community vote distribution

C (100%)

Comments

JyoGCP 1 year, 7 months ago

Selected Answer: C

Analytics Hub

raaad 1 year, 9 months ago

Selected Answer: C

- Create a copy of the necessary tables into a new dataset that doesn't use CMEK, ensuring the data is accessible without requiring the partner to have access to the encryption key. - Analytics Hub can then be used to share this data securely and efficiently with the partner organization, maintaining control and governance over the shared data.

e70ea9e 1 year, 9 months ago

Selected Answer: C

Preserves Key Confidentiality: Avoids sharing your CMEK with the partner, upholding key security and control.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #216 Topic 1

You are developing an Apache Beam pipeline to extract data from a Cloud SQL instance by using JdbcIO. You have two projects running in Google Cloud. The pipeline will be deployed and executed on Dataflow in Project A. The Cloud SQL. instance is running in Project B and does not have a public IP address. After deploying the pipeline, you noticed that the pipeline failed to extract data from the Cloud SQL instance due to connection failure. You verified that VPC Service Controls and shared VPC are not in use in these projects. You want to resolve this error while ensuring that the data does not go through the public internet. What should you do?

A. Set up VPC Network Peering between Project A and Project B. Add a firewall rule to allow the peered subnet range to access all instances on the network. **Most Voted**

B. Turn off the external IP addresses on the Dataflow worker. Enable Cloud NAT in Project A.

C. Add the external IP addresses of the Dataflow worker as authorized networks in the Cloud SQL instance.

D. Set up VPC Network Peering between Project A and Project B. Create a Compute Engine instance without external IP address in Project B on the peered subnet to serve as a proxy server to the Cloud SQL database.

Correct Answer: A

Community vote distribution

A (50%) D (50%)

Comments

aoifneofi_ef **Highly Voted** 1 year, 7 months ago

Selected Answer: D

It is a tie between A and D. Option A will definitely provide necessary connectivity but is less secure as access is enabled to "all instances". Which i feel is unnecessary considering industry best practices. Option D provides the necessary connectivity but brings in the unnecessary overhead of managing an extra VM and introduces a bit of extra complexity. Since the question emphasises on data not going through public internet(which is satisfied in both options), i would prioritise security over simplicity and choose option D in this case.

clouditis **Highly Voted** 1 year, 3 months ago

Selected Answer: A

A looks to be the best out of the 4, D is complicated involving Compute Engine which is unnecessary making it cumbersome to address the problem

Ben_oso 9 months, 1 week ago

Selected Answer: A

VPC Peering allow to two private routes can communicate each other.

71083a7 1 year, 2 months ago

Selected Answer: D

"all instances" freaks me out

julydev82 1 year, 3 months ago

Selected Answer: A

vpc create a subnet beetwen sql Cloud and the dataflow. https://cloud.google.com/sql/docs/mysql/private-ip#multiple_vpc_connectivity

fadlkhafdofpew 1 year, 6 months ago

Selected Answer: A

The answer is A. While D might work, it adds unnecessary complexity. Setting up a proxy is an extra layer of infrastructure that isn't required

Saaaurabh 1 year, 7 months ago

Selected Answer: A

If properly implemented with the right routing and firewall rules, Option A can be the correct and most straightforward solution, as it leverages VPC Peering to maintain internal traffic without going through the public internet.

meh_33 1 year, 7 months ago

A is correct

kk1211 1 year, 8 months ago

still confused between A and D

Lenifia 1 year, 9 months ago

Selected Answer: A

A is correct

kajitsu 1 year, 9 months ago

Selected Answer: A

no proxy needed

Lestrang 1 year, 9 months ago

Selected Answer: A

People referencing "VPC Network Peering does not provide transitive routing. For example, if VPC networks net-a and net-b are connected using VPC Network Peering, and VPC networks net-a and net-c are also connected using VPC Network Peering, VPC Network Peering does not provide connectivity between net-b and net-c." the question states that cloud sql is running in project B. Which means the instance is already part of the VPC in project B, so with Network Peering workers from A can definitely access data in B. No proxy is needed.

fabiogoma 1 year, 10 months ago

Selected Answer: A

Why so many people are voting for D? There's no need for a proxy, the peering is enough to allow network traffic between subnets.

fabiogoma 1 year, 10 months ago

Now I see why, I put this on ChatGPT and it thinks the right answer is D. I'm pretty sure that's a hallucination.

Positron75 9 months, 3 weeks ago

A says to allow access to "all instances in the network", which is excessive and not good practice security-wise. This is yet another question where no answer is fully correct and it's up to you to choose which one is less wrong.

ccpmad 1 year, 10 months ago

Proxy? no, it is not necessary.. A

josech 1 year, 10 months ago

Selected Answer: D

https://cloud.google.com/sql/docs/mysql/connect-multiple-vpcs

chrissamharris 2 years, 1 month ago

Selected Answer: A

A - The requirement for a proxy is un-necessary: https://cloud.google.com/sql/docs/mysql/private-ip#multiple_vpc_connectivity

ML6 2 years, 1 month ago

Option D. Source: https://cloud.google.com/sql/docs/mysql/private-ip#multiple_vpc_connectivity

https://zhangxuetu.online

[202610]

Question #217 Topic 1

You have a BigQuery table that contains customer data, including sensitive information such as names and addresses. You need to share the customer data with your data analytics and consumer support teams securely. The data analytics team needs to access the data of all the customers, but must not be able to access the sensitive data. The consumer support team needs access to all data columns, but must not be able to access customers that no longer have active contracts. You enforced these requirements by using an authorized dataset and policy tags. After implementing these steps, the data analytics team reports that they still have access to the sensitive columns. You need to ensure that the data analytics team does not have access to restricted data. What should you do? (Choose two.)

A. Create two separate authorized datasets; one for the data analytics team and another for the consumer support team.

B. Ensure that the data analytics team members do not have the Data Catalog Fine-Grained Reader role for the policy tags. **Most Voted**

C. Replace the authorized dataset with an authorized view. Use row-level security and apply filter_expression to limit data access.

D. Remove the bigquery.dataViewer role from the data analytics team on the authorized datasets.

E. Enforce access control in the policy tag taxonomy.

Correct Answer: B

Community vote distribution



Comments

qq589539483084gfrgrgfr Highly Voted 2 years, 2 months ago

Option B & E

datapassionate Highly Voted 2 years, 2 months ago

Selected Answer: E

B& E https://cloud.google.com/bigquery/docs/column-level-security-intro

Kalai_1 2 months, 3 weeks ago

Selected Answer: B

B & E. As per Google recommended security best practices

Lenifia 1 year, 9 months ago

Selected Answer: A

the correct options are: A. Create two separate authorized datasets; one for the data analytics team and another for the consumer support team. C. Replace the authorized dataset with an authorized view. Use row-level security and apply filter_expression to limit data access.

Lenifia 1 year, 9 months ago

Explanation of why other options are incorrect: B. Ensure that the data analytics team members do not have the Data Catalog Fine-Grained Reader role for the policy tags: This role relates to viewing data in Data Catalog based on policy tags, not directly controlling access to BigQuery data. D. Remove the bigquery.dataViewer role from the data analytics team on the authorized datasets: Removing this role would block all access to the dataset, which is too restrictive if they still need access to non-sensitive columns. E. Enforce access control in the policy tag taxonomy: While policy tags are used to enforce access controls, simply enforcing controls in the taxonomy does not directly address the issue of sensitive data access in BigQuery.

GCP001 2 years, 2 months ago

Selected Answer: E

B & E B - It will ensure they don't have access to secure columns E- It will allow to enforce column level security Ref - https://cloud.google.com/bigquery/docs/column-level-security-intro

Matt_108 2 years, 2 months ago

Selected Answer: B

Option B& E to me

MaxNRG 2 years, 2 months ago

Selected Answer: A

A & B The current setup is not effective because the data analytics team still has access to the sensitive columns despite using an authorized dataset and policy tags. This indicates that the policy tags are not being enforced properly, and the data analytics team members are able to view the tags and gain access to the sensitive data. Separating the data into two distinct authorized datasets is a better approach because it isolates the sensitive data from the non-sensitive data. This prevents the data analytics team from accessing the sensitive columns directly, even if they have access to the authorized dataset for general customer data. Additionally, revoking the Data Catalog Fine-Grained Reader role from the data analytics team members ensures that they cannot view or modify the policy tags. This limits their ability to bypass the access control imposed by the authorized dataset and policy tags.

Matt_108 2 years, 2 months ago

Max I feel like it's more B&E. I do agree on the revoking Data Catalog Fine-grained reader role to avoid the data analytics team to read policy tags metadata, but if the tags are setup as stated, it's just missing the enforcement of the policy tags themselves. Creating 2 auth dataset is not efficient on big datasets and Data catalog+ policy tags are built to manage these situations. Don't you agree?

imiu 2 years, 2 months ago

And the second answer? One is option B and the other is option D maybe?

raaad 2 years, 2 months ago

Selected Answer: B

- The Data Catalog Fine-Grained Reader role allows users to read metadata that is restricted by policy tags. - If members of the data analytics team have this role, they might bypass the restrictions set by policy tags. - Ensuring they do not have this role will help enforce the restrictions intended by the policy tags.

e70ea9e 2 years, 3 months ago

Selected Answer: B

Prevents data analytics team members from viewing sensitive data, even if it's tagged. Restricts access to policy tags themselves, ensuring confidentiality of sensitive information.

https://zhangxuetu.online

[202610]

Question #218 Topic 1

You have a Cloud SQL for PostgreSQL instance in Region' with one read replica in Region2 and another read replica in Region3. An unexpected event in Region' requires that you perform disaster recovery by promoting a read replica in Region2. You need to ensure that your application has the same database capacity available before you switch over the connections. What should you do?

A. Enable zonal high availability on the primary instance. Create a new read replica in a new region.

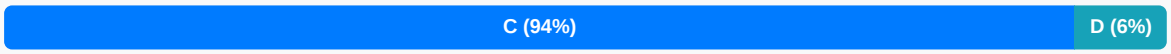
B. Create a cascading read replica from the existing read replica in Region3.

C. Create two new read replicas from the new primary instance, one in Region3 and one in a new region. **Most Voted**

D. Create a new read replica in Region1, promote the new read replica to be the primary instance, and enable zonal high availability.

Correct Answer: C

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

After promoting the read replica in Region2 to be the new primary instance, creating additional read replicas from it can help distribute the read load and maintain or increase the database's total capacity.

skhaire 1 year ago

Selected Answer: C

Corrected answer: C If the primary instance (db-a-0) becomes unavailable, you can promote the replica in region B to become the primary. To again have additional replicas in regions A and C, delete the old instances (the former primary instance in A, and the replica in C), and create new read replicas from the new primary instance in B.

skhaire 1 year ago

Selected Answer: D

Question is flawed but the closest answer would be D since C will result in 2 read replicas on Region 3 (original one and new now) Option C- Create two new read replicas from the new primary instance - contradicts with the requirements - You need to ensure that your application has the same database capacity available 'before you switch over the connections.' Option D- Create a new read replica in Region1, promote the new read replica to be the primary instance - contradicts with the requirement - 'requires that you perform disaster recovery by promoting a read replica in Region2.' How does this affect the answer choices?

josech 1 year, 10 months ago

Selected Answer: C

https://cloud.google.com/sql/docs/mysql/replication#cross-region-read-replicas

nadavw 1 year, 7 months ago

requires 2 new read replicas as the read replica that wasn't promoted isn't capable to be a replica any more as the primary isa gone

CGS22 1 year, 11 months ago

Selected Answer: C

The best option here is C. Create two new read replicas from the new primary instance, one in Region3 and one in a new region. Here's the breakdown: Capacity Restoration: Promoting the Region2 replica makes it the new primary. You need to replicate from this new primary to maintain redundancy and capacity. Creating two replicas (Region3, new region) accomplishes this. Geographic Distribution: Distributing replicas across regions ensures availability if another regional event occurs. Speed: Creating new replicas from the promoted primary is likely faster than promoting another existing replica.

BigDataBB 1 year, 11 months ago

Who said that i can use a 4° region? If have constraint that i can't go out from that 3 regions? By My opinion will be a right solution if the new replica will in another zona of the region 1 or 3. may be the best solution is the case D

BigDataBB 1 year, 11 months ago

https://cloud.google.com/sql/docs/postgres/replication/cross-region-replicas?hl=en

JyoGCP 2 years, 1 month ago

Selected Answer: C

Option C

e70ea9e 2 years, 3 months ago

Selected Answer: C

Immediate Failover: Promoting the read replica in Region2 quickly restores database operations in a different region, aligning with disaster recovery goals. Capacity Restoration: Creates two new read replicas from the promoted primary instance (formerly the read replica in Region2). This replaces the lost capacity in Region1 and adds a read replica in a new region for further redundancy.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #219 Topic 1

You orchestrate ETL pipelines by using Cloud Composer. One of the tasks in the Apache Airflow directed acyclic graph (DAG) relies on a third-party service. You want to be notified when the task does not succeed. What should you do?

A. Assign a function with notification logic to the on_retry_callback parameter for the operator responsible for the task at risk.

B. Configure a Cloud Monitoring alert on the sla_missed metric associated with the task at risk to trigger a notification.

C. Assign a function with notification logic to the on_failure_callback parameter tor the operator responsible for the task at risk. **Most Voted**

D. Assign a function with notification logic to the sla_miss_callback parameter for the operator responsible for the task at risk.

Correct Answer: C

Community vote distribution

C (100%)

Comments

jonty4gcp 1 year, 7 months ago

What is Task is long-running and in between stuck?

Anudeep58 1 year, 10 months ago

Selected Answer: C

https://airflow.apache.org/docs/apache-airflow/stable/administration-and-deployment/logging-monitoring/callbacks.html

JyoGCP 2 years, 1 month ago

Selected Answer: C

on_failure_callback

datapassionate 2 years, 2 months ago

Selected Answer: C

on_failure_callback is invoked when the task fails https://airflow.apache.org/docs/apache-airflow/stable/administration-and-deployment/logging-monitoring/callbacks.html

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C to me

raaad 2 years, 2 months ago

Selected Answer: C

- The on_failure_callback is a function that gets called when a task fails. - Assigning a function with notification logic to this parameter is a direct way to handle task failures. - When the task fails, this function can trigger a notification, making it an appropriate solution for the need to be alerted on task failures.

e70ea9e 2 years, 3 months ago

Selected Answer: C

Direct Trigger: The on_failure_callback parameter is specifically designed to invoke a function when a task fails, ensuring immediate notification. Customizable Logic: You can tailor the notification function to send emails, create alerts, or integrate with other notification systems, providing flexibility.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #220 Topic 1

You are migrating your on-premises data warehouse to BigQuery. One of the upstream data sources resides on a MySQL. database that runs in your on-premises data center with no public IP addresses. You want to ensure that the data ingestion into BigQuery is done securely and does not go through the public internet. What should you do?

A. Update your existing on-premises ETL tool to write to BigQuery by using the BigQuery Open Database Connectivity (ODBC) driver. Set up the proxy parameter in the simba.googlebigqueryodbc.ini file to point to your data center's NAT gateway.

B. Use Datastream to replicate data from your on-premises MySQL database to BigQuery. Set up Cloud Interconnect between your on-premises data center and Google Cloud. Use Private connectivity as the connectivity method and allocate an IP address range within your VPC network to the Datastream connectivity configuration. Use Server-only as the encryption type when setting up the connection profile in Datastream. **Most Voted**

C. Use Datastream to replicate data from your on-premises MySQL database to BigQuery. Use Forward-SSH tunnel as the connectivity method to establish a secure tunnel between Datastream and your on-premises MySQL database through a tunnel server in your on-premises data center. Use None as the encryption type when setting up the connection profile in Datastream.

D. Use Datastream to replicate data from your on-premises MySQL database to BigQuery. Gather Datastream public IP addresses of the Google Cloud region that will be used to set up the stream. Add those IP addresses to the firewall allowlist of your on-premises data center. Use IP Allowlisting as the connectivity method and Server-only as the encryption type when setting up the connection profile in Datastream.

Correct Answer: B

Community vote distribution

B (100%)

Comments

raaad Highly Voted 1 year, 9 months ago

Selected Answer: B

- Datastream is a serverless change data capture and replication service, which can be used to replicate data changes from MySQL to BigQuery. - Using Cloud Interconnect provides a private, secure connection between your on-premises environment and Google Cloud ==> This method ensures that data doesn't go through the public internet and is a recommended approach for secure, large-scale data migrations. - Setting up private connectivity with Datastream allows for secure and direct data transfer.

josech 1 year, 4 months ago

Selected Answer: B

https://cloud.google.com/datastream/docs/network-connectivity-options

datapassionate 1 year, 8 months ago

Selected Answer: B

Datastream is a seamless replication from relational databases directly to BigQuery. The source database can be hosted on-premises, on Google Cloud services such as Cloud SQL or Bare Metal Solution for Oracle, or anywhere else on any cloud. https://cloud.google.com/datastream-for-bigquery#benefits

datapassionate 1 year, 8 months ago

It is required that the data ingestion into BigQuery is done securely and does not go through the public internet. It can be done by Interconnect.

e70ea9e 1 year, 9 months ago

Selected Answer: B

Secure Private Connection: Cloud Interconnect establishes a direct, private connection between your on-premises network and Google Cloud, bypassing the public internet and ensuring data confidentiality. Datastream Integration: Datastream seamlessly replicates data from your MySQL database to BigQuery, handling the complexities of data transfer and synchronization.

https://zhangxuetu.online

[202610]

Question #221 Topic 1

You store and analyze your relational data in BigQuery on Google Cloud with all data that resides in US regions. You also have a variety of object stores across Microsoft Azure and Amazon Web Services (AWS), also in US regions. You want to query all your data in BigQuery daily with as little movement of data as possible. What should you do?

A. Use BigQuery Data Transfer Service to load files from Azure and AWS into BigQuery.

B. Create a Dataflow pipeline to ingest files from Azure and AWS to BigQuery.

C. Load files from AWS and Azure to Cloud Storage with Cloud Shell gsutil rsync arguments.

D. Use the BigQuery Omni functionality and BigLake tables to query files in Azure and AWS. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: D

- BigQuery Omni allows us to analyze data stored across Google Cloud, AWS, and Azure directly from BigQuery without having to move or copy the data. - It extends BigQuery's data analysis capabilities to other clouds, enabling cross-cloud analytics.

e70ea9e **Highly Voted** 1 year, 9 months ago

Selected Answer: D

Direct Querying: BigQuery Omni allows you to query data in Azure and AWS object stores directly without physically moving it to BigQuery, reducing data transfer costs and delays. BigLake Tables: Provide a unified view of both BigQuery tables and external object storage files, enabling seamless querying across multi-cloud data.

josech 1 year, 4 months ago

https://cloud.google.com/blog/products/data-analytics/introducing-bigquery-omni
https://cloud.google.com/bigquery/docs/omni-introduction

Ramon98 1 year, 7 months ago

Option A, B, and C all involve moving data, which is described as something that shouldn't happen.

JyoGCP 1 year, 7 months ago

Selected Answer: D

BigQuery Omni

https://zhangxuetu.online

[202610]

Question #222 Topic 1

You have a variety of files in Cloud Storage that your data science team wants to use in their models. Currently, users do not have a method to explore, cleanse, and validate the data in Cloud Storage. You are looking for a low code solution that can be used by your data science team to quickly cleanse and explore data within Cloud Storage. What should you do?

A. Provide the data science team access to Dataflow to create a pipeline to prepare and validate the raw data and load data into BigQuery for data exploration.

B. Create an external table in BigQuery and use SQL to transform the data as necessary. Provide the data science team access to the external tables to explore the raw data.

C. Load the data into BigQuery and use SQL to transform the data as necessary. Provide the data science team access to staging tables to explore the raw data.

D. Provide the data science team access to Dataprep to prepare, validate, and explore the data within Cloud Storage. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

raaad Highly Voted 1 year, 9 months ago

Selected Answer: D

- Dataprep is a serverless, no-code data preparation tool that allows users to visually explore, cleanse, and prepare data for analysis. - It's designed for business analysts, data scientists, and others who want to work with data without writing code. - Dataprep can directly access and transform data in Cloud Storage, making it a suitable choice for a team that prefers a low-code, user-friendly solution.

JyoGCP 1 year, 7 months ago

Selected Answer: D

Dataprep

JimmyBK 1 year, 8 months ago

Selected Answer: D

Goes without say

Matt_108 1 year, 8 months ago

Selected Answer: D

Option D - Low code and efficient way to explore and prep data

Alex3551 1 year, 8 months ago

why you message wrong answers correct is C

Matt_108 1 year, 8 months ago

The "Reveal Answer" button contains 90% of the time an incorrect answer. You should always check the community and the discussion during studying :)

e70ea9e 1 year, 9 months ago

Selected Answer: D

Low-Code Interface: Offers a visual, drag-and-drop interface that empowers users with varying technical skills to cleanse and explore data without extensive coding, aligning with the low-code requirement. Data Cleaning and Validation: Provides built-in tools for data profiling, cleaning, transformation, and validation, ensuring data quality and accuracy before model training. Direct Cloud Storage Access: Connects directly to Cloud Storage, allowing users to work with data in place without additional data movement or storage costs, optimizing efficiency.

https://zhangxuetu.online

[202610]

Question #223 Topic 1

You are building an ELT solution in BigQuery by using Dataform. You need to perform uniqueness and null value checks on your final tables. What should you do to efficiently integrate these checks into your pipeline?

A. Build BigQuery user-defined functions (UDFs).

B. Create Dataplex data quality tasks.

C. Build Dataform assertions into your code. **Most Voted**

D. Write a Spark-based stored procedure.

Correct Answer: C

Community vote distribution

C (100%)

Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: C

- Dataform provides a feature called "assertions," which are essentially SQL-based tests that you can define to verify the quality of your data. - Assertions in Dataform are a built-in way to perform data quality checks, including checking for uniqueness and null values in your tables.

JyoGCP 1 year, 7 months ago

Selected Answer: C

https://docs.dataform.co/guides/assertions

tibuenoc 1 year, 8 months ago

Selected Answer: C

https://cloud.google.com/dataform/docs/assertions

Alex3551 1 year, 8 months ago

Selected Answer: C

Agree with C

Alex3551 1 year, 8 months ago

agree with C

e70ea9e 1 year, 9 months ago

Selected Answer: C

Native Integration: Dataform assertions are designed specifically for data quality checks within Dataform pipelines, ensuring seamless integration and compatibility. They leverage Dataform's execution model and configuration, aligning with the existing workflow. Declarative Syntax: Assertions are defined using a simple, declarative syntax within Dataform code, making them easy to write and understand, even for users with less SQL expertise.

https://zhangxuetu.online

[202610]

Question #224 Topic 1

A web server sends click events to a Pub/Sub topic as messages. The web server includes an eventTimestamp attribute in the messages, which is the time when the click occurred. You have a Dataflow streaming job that reads from this Pub/Sub topic through a subscription, applies some transformations, and writes the result to another Pub/Sub topic for use by the advertising department. The advertising department needs to receive each message within 30 seconds of the corresponding click occurrence, but they report receiving the messages late. Your Dataflow job's system lag is about 5 seconds, and the data freshness is about 40 seconds. Inspecting a few messages show no more than 1 second lag between their eventTimestamp and publishTime. What is the problem and what should you do?

A. The advertising department is causing delays when consuming the messages. Work with the advertising department to fix this.

B. Messages in your Dataflow job are taking more than 30 seconds to process. Optimize your job or increase the number of workers to fix this.

G. Messages in your Dataflow job are processed in less than 30 seconds, but your job cannot keep up with the backlog in the Pub/Sub subscription. Optimize your job or increase the number of workers to fix this. Most Voted

D. The web server is not pushing messages fast enough to Pub/Sub. Work with the web server team to fix this.

Correct Answer: G

Community vote distribution

G (89%) B (11%)

Comments

e70ea9e Highly Voted 2 years, 3 months ago

Selected Answer: G

System Lag vs. Data Freshness: System lag is low (5 seconds), indicating that individual messages are processed quickly. However, data freshness is high (40 seconds), suggesting a backlog in the pipeline. Not Advertising's Fault: The issue is upstream of their consumption, as they're already receiving delayed messages. Not Web Server's Fault: The lag between eventTimestamp and publishTime is minimal (1 second), meaning the server is publishing messages promptly.

raaad Highly Voted 2 years, 2 months ago

Selected Answer: G

- It suggest a backlog problem. - It indicates that while individual messages might be processed quickly once they're handled, the job overall cannot keep up with the rate of incoming messages, causing a delay in processing the backlog.

datapassionate 2 years, 2 months ago

Why not B than?

RenePetersen 2 years, 1 month ago

I guess that's because it says in the text that "Your Dataflow job's system lag is about 5 seconds".

SajadAhm 1 month ago

Selected Answer: B

Low System Lag means no backlog problem. The bottleneck is in the actual processing logic.

desertlotus1211 1 year ago

Selected Answer: G

Answer is C lol

4a8ffd7 1 year, 6 months ago

Selected Answer: B

I don't know why you guys got the processing time is less than 30 sec. But I would consider the processing time with 40(freshness) - 5(system lag) = 35 sec. Even minus the publish time of Pub/sub which is less than 1 sec. The processing time still larger than 30 sec. I believe inspecting a few messages show no more than 1 sec lag is about pub/sub processing time. Not inspecting a few messages for dataflow. So I would choose B.

22c1725 10 months, 1 week ago

Your Dataflow job's system lag is about 5 seconds,

JyoGCP 2 years, 1 month ago

Selected Answer: G

Option C

Matt_108 2 years, 2 months ago

Selected Answer: G

Option C - low system lag (which identifies fast processing) but high data freshness (which identifies that the messages sit in the backlog a lot)

Alex3551 2 years, 2 months ago

Selected Answer: G

agree correct is C

https://zhangxuetu.online

[202610]

Question #225 Topic 1

Your organization stores customer data in an on-premises Apache Hadoop cluster in Apache Parquet format. Data is processed on a daily basis by Apache Spark jobs that run on the cluster. You are migrating the Spark jobs and Parquet data to Google Cloud. BigQuery will be used on future transformation pipelines so you need to ensure that your data is available in BigQuery. You want to use managed services, while minimizing ETL data processing changes and overhead costs. What should you do?

A. Migrate your data to Cloud Storage and migrate the metadata to Dataproc Metastore (DPMS). Refactor Spark pipelines to write and read data on Cloud Storage, and run them on Dataproc Serverless. **Most Voted**

B. Migrate your data to Cloud Storage and register the bucket as a Dataplex asset. Refactor Spark pipelines to write and read data on Cloud Storage, and run them on Dataproc Serverless.

C. Migrate your data to BigQuery. Refactor Spark pipelines to write and read data on BigQuery, and run them on Dataproc Serverless.

D. Migrate your data to BigLake. Refactor Spark pipelines to write and read data on Cloud Storage, and run them on Dataproc on Compute Engine.

Correct Answer: A

Community vote distribution



Comments

raaad Highly Voted 2 years, 2 months ago

Selected Answer: A

- This option involves moving Parquet files to Cloud Storage, which is a common and cost-effective storage solution for big data and is compatible with Spark jobs. - Using Dataproc Metastore to manage metadata allows us to keep Hadoop ecosystem's structural information. - Running Spark jobs on Dataproc Serverless takes advantage of managed Spark services without managing clusters. - Once the data is in Cloud Storage, you can also easily load it into BigQuery for further analysis.

skhaire Highly Voted 1 year, 1 month ago

Selected Answer: B

BigQuery Integration: The requirement is to make data available in BigQuery. Dataplex has built-in integration with BigQuery. It can automatically discover data in Cloud Storage and create external tables in BigQuery, making the data readily queryable. DPMS doesn't have this direct integration with BigQuery.

22c1725 10 months, 1 week ago

Selected Answer: A

Would Go with A

380e3c6 1 year, 1 month ago

Selected Answer: A

A is correctbecause it minimizes ETL changes, keeps Parquet data in Cloud Storage (cost-effective and Spark-compatible), and integrates with BigQuery via external tables. C is flawed** since moving directly to BigQuery requires refactoring Spark jobs, increasing complexity and costs. B adds unnecessary governance overhead, and D focuses on infrastructure instead of pipeline efficiency.

plum21 1 year, 1 month ago

Selected Answer: D

The requirement: "You want to use managed services" excludes Dataproc Serverless. Dataproc on Compute Engine remains. Next requirement: "BigQuery will be used on future transformation pipelines so you need to ensure that your data is available in BigQuery" -> BigLake Next requirement: "while minimizing ETL data processing changes and overhead costs" -> Refactor Spark pipelines to write and read data on Cloud Storage Notes 1. Dataproc Metastore (DPMS) could be used on Dataproc to read data from BQ but not the other way round.

Positron75 10 months, 2 weeks ago

How is Dataproc Serverless not a managed service, but running Dataproc on Compute Engine is? D is the first answer to rule out.

LP_PDE 1 year, 2 months ago

Selected Answer: A

Both Spark and BigQuery can directly access data in Cloud Storage.

hrishi19 1 year, 4 months ago

Selected Answer: C

The question states that the data should be available on BigQuery and only option C meets this requirement.

JamesKarianis 1 year, 7 months ago

Selected Answer: A

A is correct

Anudeep58 1 year, 9 months ago

Selected Answer: A

Option B: Registering the bucket as a Dataplex asset adds an additional layer of data governance and management. While useful, it may not be necessary for your immediate migration needs and can introduce additional complexity. Option C: Migrating data directly to BigQuery would require significant changes to your Spark pipelines since they would need to be refactored to read from and write to BigQuery instead of Parquet files. This approach could introduce higher costs due to BigQuery storage and querying. Option D: Using BigLake and Dataproc on Compute Engine is more complex and requires more management compared to Dataproc Serverless. Additionally, it might not be as cost-effective as leveraging Cloud Storage and Dataproc Serverless.

aoifneofi_ef 1 year, 7 months ago

Just adding further commentary on why A is correct while why other options are incorrect is explained above. Parquet files have schema engrained in them. Hence Spark pipelines on Hadoop Cluster may not have needed tables at all. Hence the simplest solution would be to move it to Cloud Storage instead of BigQuery and this way there would be minimal changes to the ETL pipelines - just change HDFS file system pointer to GCS file system for read writes and no need for any additional tables

josech 1 year, 10 months ago

Selected Answer: A

The question says "You want to use managed services, while minimizing ETL data processing changes and overhead costs". Dataproc is a managed service that doesn't need to refactor the data transformation Spark code you already have (you will have to refactor only the wrtie and read code), an it has a Big Query connector for future use. https://cloud.google.com/dataproc/docs/concepts/connectors/bigquery

52ed0e5 2 years ago

Selected Answer: C

Migrate your data directly to BigQuery. Refactor Spark pipelines to read from and write to BigQuery. Run the Spark jobs on Dataproc Serverless. The best choice for ensuring data availability in BigQuery. It allows seamless integration with BigQuery and minimizes ETL changes.

Ramon98 2 years, 1 month ago

Selected Answer: C

A tricky one, because of "you need to ensure that your data is available in BigQuery". The easiest and most straight forward migration seems answer A to me, and then you can use external tables to make the parquet data directly available in BigQuery. https://cloud.google.com/bigquery/docs/loading-data-cloud-storage-parquet However creating the external tables is an extra step? So therefore maybe C is the answer?

Moss2011 2 years, 1 month ago

Selected Answer: C

I think the key phrase here is "you need to ensure that your data is available in BigQuery" that's why I thing C it's the best option

JyoGCP 2 years, 1 month ago

Selected Answer: C

I think it's C. Dataproc can use BigQuery to read and write data. Dataproc's BigQuery connector is a library that allows Spark and Hadoop applications to process and write data from BigQuery. Here's how Dataproc can be used with BigQuery: Process large datasets: Use Spark to process data stored in BigQuery. Write results: Write the results back to BigQuery or other data storage for further analysis. Read data: The BigQuery connector can read data from BigQuery into a Spark DataFrame. Write data: The connector writes data to BigQuery by buffering all the data into a Cloud Storage temporary table.

JyoGCP 2 years, 1 month ago

As per question.. "BigQuery will be used on future transformation pipelines so you need to ensure that your data is available in BigQuery. You want to use managed services (DATAPROC), while minimizing ETL data processing changes and overhead costs."

matiijax 2 years, 1 month ago

Selected Answer: B

I think its B and the reason is that egistering the data as a Dataplex asset enables seamless integration with BigQuery later on. Dataplex simplifies data discovery and lineage tracking, making it easier to prepare your data for BigQuery transformations.

saschak94 2 years, 1 month ago

Why would I select A here? Why not moving the data to BigQuery and running Dataproc Serverless jobs accessing the data in BigQuery?

e70ea9e 2 years, 3 months ago

Selected Answer: A

Managed Services: Leverages Dataproc Serverless for a fully managed Spark environment, reducing overhead and administrative tasks. Minimal Data Processing Changes: Keeps Spark pipelines largely intact by working with Parquet files on Cloud Storage, minimizing refactoring efforts. BigQuery Integration: Dataproc Serverless can directly access BigQuery, enabling future transformation pipelines without additional data movement. Cost-Effective: Serverless model scales resources only when needed, optimizing costs for intermittent workloads.

https://zhangxuetu.online

[202610]

Question #226 Topic 1

Your organization has two Google Cloud projects, project A and project B. In project A, you have a Pub/Sub topic that receives data from confidential sources. Only the resources in project A should be able to access the data in that topic. You want to ensure that project B and any future project cannot access data in the project A topic. What should you do?

A. Add firewall rules in project A so only traffic from the VPC in project A is permitted.

B. Configure VPC Service Controls in the organization with a perimeter around project A. **Most Voted**

C. Use Identity and Access Management conditions to ensure that only users and service accounts in project A. can access resources in project A.

D. Configure VPC Service Controls in the organization with a perimeter around the VPC of project A.

Correct Answer: B

Community vote distribution



Comments

datapassionate **Highly Voted** 2 years, 2 months ago

Selected Answer: C

And I would agree with GPT. The question is about that who can do what within GCP environment. It's all about permissions and access management, not about networking.

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

Option B: -It allows us to create a secure boundary around all resources in Project A, including the Pub/Sub topic. - It prevents data exfiltration to other projects and ensures that only resources within the perimeter (Project A) can access the sensitive data. - VPC Service Controls are specifically designed for scenarios where you need to secure sensitive data within a specific context or boundary in Google Cloud.

22c1725 10 months, 1 week ago

Selected Answer: B

Since such case would not be stright forward as C

MithunDesai 1 year, 7 months ago

Selected Answer: B

The best solution to prevent project B and any future projects from accessing data in project A's Pub/Sub topic is B. Configure VPC Service Controls in the organization with a perimeter around project A.

meh_33 1 year, 7 months ago

Selected Answer: B

B is correct Raaad is always right

fabiogoma 1 year, 10 months ago

Selected Answer: B

Setting up a perimeter around project A is future proof, the question asks to "ensure that project B and any future project cannot access data in the project A topic", IAM is not future proof. Reference: https://cloud.google.com/vpc-service-controls/docs/overview#isolate p.s: VPC Service Controls is not the same thing as VPC, instead its a security layer on top of a VPC and it should be used together with IAM, not one or the other (https://cloud.google.com/vpc-service-controls/docs/overview#how-vpc-service-controls-works)

virat_kohli 1 year, 10 months ago

Selected Answer: C

C. Use Identity and Access Management conditions to ensure that only users and service accounts in project A. can access resources in project A. [SIMPLE!!!]

JyoGCP 2 years, 1 month ago

Selected Answer: B

I'll go with "B. Configure VPC Service Controls in the organization with a perimeter around project A."

datapassionate 2 years, 2 months ago

GPT: C. Use Identity and Access Management conditions to ensure that only users and service accounts in project A can access resources in project A. Analysis: This is the most appropriate option. IAM allows you to define who (which users or service accounts) has what access to your GCP resources. By setting IAM policies with conditions specific to Project A, you can ensure that only designated entities within Project A have access to its resources, including the Pub/Sub topic. D. Configure VPC Service Controls in the organization with a perimeter around the VPC of project A.

datapassionate 2 years, 2 months ago

A. Add firewall rules in project A so only traffic from the VPC in project A is permitted. Analysis: Firewall rules in GCP are used to control traffic to and from instances within Google Cloud Virtual Private Clouds (VPCs). However, they don't specifically control access to Pub/Sub resources. Pub/Sub access is managed through IAM, not VPC firewall rules.

datapassionate 2 years, 2 months ago

B. Configure VPC Service Controls in the organization with a perimeter around project A. Analysis: VPC Service Controls provide a security perimeter for your data, but they are more focused on preventing data exfiltration; this might be more complex and broader than necessary for the specific requirement of restricting access to a Pub/Sub topic.

datapassionate 2 years, 2 months ago

D. Configure VPC Service Controls in the organization with a perimeter around the VPC of project A. Analysis: Similar to option B, this is focused on securing network boundaries rather than specific resource access within GCP. While it could provide an additional layer of security, it's not the most direct way to control access to a specific Pub/Sub topic.

e70ea9e 2 years, 3 months ago

Selected Answer: B

VPC Service Controls enforce a security perimeter around entire projects, ensuring that resources within project A (including the Pub/Sub topic) are inaccessible from any other project, including project B and future projects. This aligns with the requirement to prevent cross-project access.

https://zhangxuetu.online

[202610]

Question #227 Topic 1

You stream order data by using a Dataflow pipeline, and write the aggregated result to Memorystore. You provisioned a Memorystore for Redis instance with Basic Tier, 4 GB capacity, which is used by 40 clients for read-only access. You are expecting the number of read-only clients to increase significantly to a few hundred and you need to be able to support the demand. You want to ensure that read and write access availability is not impacted, and any changes you make can be deployed quickly. What should you do?

A. Create a new Memorystore for Redis instance with Standard Tier. Set capacity to 4 GB and read replica to No read replicas (high availability only). Delete the old instance.

B. Create a new Memorystore for Redis instance with Standard Tier. Set capacity to 5 GB and create multiple read replicas. Delete the old instance. **Most Voted**

C. Create a new Memorystore for Memcached instance. Set a minimum of three nodes, and memory per node to 4 GB. Modify the Dataflow pipeline and all clients to use the Memcached instance. Delete the old instance.

D. Create multiple new Memorystore for Redis instances with Basic Tier (4 GB capacity). Modify the Dataflow pipeline and new clients to use all instances.

Correct Answer: B

Community vote distribution

B (100%)

Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- Upgrading to the Standard Tier and adding read replicas is an effective way to scale and manage increased read load. - The additional capacity (5 GB) provides more space for data, and read replicas help distribute the read load across multiple instances.

datapassionate 2 years, 2 months ago

Descrived here: https://cloud.google.com/memorystore/docs/redis/redis-tiers

SamuelTsch 1 year, 5 months ago

Selected Answer: B

I don't like any answer. It seems Option B makes more senses due to read replicas.

e70ea9e 2 years, 3 months ago

Selected Answer: B

Scalability for Read-Only Clients: Read replicas distribute read traffic across multiple instances, significantly enhancing read capacity to support a large number of clients without impacting write performance. High Availability: Standard Tier ensures high availability with automatic failover, minimizing downtime in case of instance failure. Minimal Code Changes: Redis clients can seamlessly connect to read replicas without requiring extensive code modifications, enabling a quick deployment.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #228 Topic 1

You have a streaming pipeline that ingests data from Pub/Sub in production. You need to update this streaming pipeline with improved business logic. You need to ensure that the updated pipeline reprocesses the previous two days of delivered Pub/Sub messages. What should you do? (Choose two.)

A. Use the Pub/Sub subscription clear-retry-policy flag

B. Use Pub/Sub Snapshot capture two days before the deployment.

C. Create a new Pub/Sub subscription two days before the deployment.

D. Use the Pub/Sub subscription retain-acked-messages flag. **Most Voted**

E. Use Pub/Sub Seek with a timestamp.

Correct Answer: D

Community vote distribution



Comments

tibuenoc **Highly Voted** 2 years, 2 months ago

Selected Answer: D

DE Another way to replay messages that have been acknowledged is to seek to a timestamp. To seek to a timestamp, you must first configure the subscription to retain acknowledged messages using retain-acked-messages. If retain-acked-messages is set, Pub/Sub retains acknowledged messages for 7 days. You only need to do this step if you intend to seek to a timestamp, not to a snapshot. https://cloud.google.com/pubsub/docs/replay-message

joao_01 1 year, 11 months ago

Its BE. By the way, you can seek to a snapshot yes: "Seeks an existing subscription to a point in time or to a given snapshot, whichever is provided in the request"
Link:https://cloud.google.com/pubsub/docs/reference/rest/v1/projects.subscriptions/seek

2ad2bc7 1 year, 3 months ago

Think about it - what you want to do - process last 2 days of messages. What does snapshot give you - it give you what were the un-acknowledged messages in pub/sub at that point in time 2 days ago. How will that help you process messages that were sent to pub/sub in the last 2 days (i.e. after the snapshot?)

nadavw 1 year, 7 months ago

this is correct as there are 2 options (timestamp and snapshot) and foreach there are 2 stages. Snapshot - create ('B') and seek Timestamp - configure 'retain' ('D') and seek ('E') as shown 'B' is missing the 'seek' operation

GCP001 **Highly Voted** 2 years, 2 months ago

Selected Answer: B

B and E, already tested at cloud console.

Kalai_1 2 months, 3 weeks ago

Selected Answer: E

D & E. Make use of Seek feature of Pub/Sub which is available for pre-defined configuration.

Ben_oso 9 months, 1 week ago

Selected Answer: D

DE. The questions says "Delivered Messages", so this are a acknowledged messages, this can only do the D. retain-acked-messages flag. Snapshot only save the unacknowledged messages, so, you miss the acknowledged to make a reprocess. And Letter E is mandatory to comeback in time.

desertlotus1211 1 year ago

Selected Answer: B

The only way to ensure that the updated pipeline reprocesses the previous two days of delivered Pub/Sub messages (current) is to use B&E.... for future D&E can work.

71083a7 1 year, 2 months ago

Selected Answer: D

Since E is seek to a timestamp only one to replay, I think D is correct answer and not B

AWSandeep 1 year, 3 months ago

D and E Please seem to not understand that the retain_acked_messages parameter needs to be enabled for any snapshot or seeking functionality to work.

SamuelTsch 1 year, 5 months ago

Selected Answer: B

After reading https://cloud.google.com/pubsub/docs/replay-overview#snapshot_overview carefully, B and E are correct.

SamuelTsch 1 year, 5 months ago

change my mind. D and E should be correct.

Preetmehta1234 1 year, 6 months ago

Selected Answer: D

First, read this document: https://cloud.google.com/pubsub/docs/replay-overview. Key Points: Seek to a Snapshot: Reprocesses only unacknowledged messages. Seek to a Timestamp: Reprocesses all messages (acknowledged and unacknowledged) after that time. Since the question asks for delivering all messages, option E is correct, as it includes both acknowledged and unacknowledged messages. Regarding Option D: Configuring a subscription with the retain_acked_messages property allows replaying previously acknowledged messages retained for up to 31 days. This satisfies the requirement to deliver all messages and retains them longer than the mentioned 2 days.

MithunDesai 1 year, 7 months ago

Selected Answer: B

B&D - based on Vertex AI feedback

Anudeep58 1 year, 9 months ago

Selected Answer: E

BE B. Use Pub/Sub Snapshot capture two days before the deployment. Pub/Sub Snapshot: Creating a snapshot captures the state of the subscription at a specific point in time. You can then seek to this snapshot to replay

messages from that point onwards. By capturing a snapshot two days before the deployment, you can ensure that your pipeline reprocesses messages from the past two days. E. Use Pub/Sub Seek with a timestamp. Pub/Sub Seek: This feature allows you to reset the subscription to a specific timestamp. Messages published to the topic after this timestamp are re-delivered. By seeking to the timestamp from two days ago, you can instruct Pub/Sub to start re-delivering messages from that point in time

virat_kohli 1 year, 10 months ago

Selected Answer: D

D. Use the Pub/Sub subscription retain-acked-messages flag. E. Use Pub/Sub Seek with a timestamp.

cuadradobertolinisebastiancami 2 years, 1 month ago

Selected Answer: D

E for sure, you need to seek from a timestamp. To accomplish to that you need to "Set the retain-acked-messages flag to true for the subscription." From google documentation: "Note: To seek to a previous time point, your subscription must be configured to retain acknowledged messages. You can change this setting by clicking Edit on the subscription details page, and checking the box for Retain acknowledged messages." https://cloud.google.com/pubsub/docs/replay-message

Tryolabs 2 years, 1 month ago

Selected Answer: E

D and E, https://cloud.google.com/pubsub/docs/replay-message

ML6 2 years, 1 month ago

Selected Answer: E

B and E: The seek feature extends subscriber capabilities by allowing you to alter the acknowledgement state of messages in bulk. For example, you can replay previously acknowledged messages or purge messages in bulk. In addition, you can copy the acknowledgement state of one subscription to another by using seek in combination with a snapshot. Source: https://cloud.google.com/pubsub/docs/replay-overview

Sofiia98 2 years, 2 months ago

Selected Answer: B

BE https://cloud.google.com/pubsub/docs/replay-overview

tibuenoc 2 years, 2 months ago

But There is a problem snapshot you shoudl seek by subscriptions not by timestamp

Matt_108 2 years, 2 months ago

Selected Answer: D

Option D and E

https://zhangxuetu.online

[202610]

Question #229 Topic 1

You currently use a SQL-based tool to visualize your data stored in BigQuery. The data visualizations require the use of outer joins and analytic functions. Visualizations must be based on data that is no less than 4 hours old. Business users are complaining that the visualizations are too slow to generate. You want to improve the performance of the visualization queries while minimizing the maintenance overhead of the data preparation pipeline. What should you do?

A. Create materialized views with the allow_non_incremental_definition option set to true for the visualization queries. Specify the max_staleness parameter to 4 hours and the enable_refresh parameter to true. Reference the materialized views in the data visualization tool. **Most Voted**

B. Create views for the visualization queries. Reference the views in the data visualization tool.

C. Create a Cloud Function instance to export the visualization query results as parquet files to a Cloud Storage bucket. Use Cloud Scheduler to trigger the Cloud Function every 4 hours. Reference the parquet files in the data visualization tool.

D. Create materialized views for the visualization queries. Use the incremental updates capability of BigQuery materialized views to handle changed data automatically. Reference the materialized views in the data visualization tool.

Correct Answer: A

Community vote distribution



Comments

baimus 1 year, 5 months ago

Selected Answer: A

Just a note, the question saying "data no less than 4 hours old" presumably means "no more than 4 hours old"

JamesKarianis 1 year, 7 months ago

Selected Answer: B

Unfortunately the correct answer is B due to the limitations of materialized views, doesn't support any other join than inner and no analytical function is supported

AWSandeep 1 year, 3 months ago

Yes, they do if they are non-incremental.

ricardovazz 2 years ago

Selected Answer: A

A https://cloud.google.com/bigquery/docs/materialized-views-create#non-incremental In scenarios where data staleness is acceptable, for example for batch data processing or reporting, non-incremental materialized views can improve query performance and reduce cost. allow_non_incremental_definition option. This option must be accompanied by the max_staleness option. To ensure a periodic refresh of the materialized view, you should also configure a refresh policy.

Matt_108 2 years, 2 months ago

Selected Answer: A

Option A is better than D, since it accounts for data staleness and is better suited for heavy querying, thanks to the allow_non_incremental_definition

Jordan18 2 years, 2 months ago

A seems right but whats wrong with option D, can anybody please explain?

datapassionate 2 years, 2 months ago

Seems like materialiazed views can use incremental updates only if data was not delated or updated in original table. Here the data changes so I think thats the reason why its not correct answer https://cloud.google.com/bigquery/docs/materialized-views-use#incremental_updates "BigQuery combines the cached view's data with new data to provide consistent query results while still using the materialized view. For single-table materialized views, this is possible if the base table is unchanged since the last refresh, or if only new data was added. For multi-table views, no more than one table can have appended data. If more than one of a multi-table view's base tables has changed, then the view cannot be incrementally updated."

raaad 2 years, 2 months ago

Selected Answer: A

- Materialized views in BigQuery precompute and store the result of a base query, which can speed up data retrieval for complex queries used in visualizations. - The max_staleness parameter allows us to specify how old the data can be, ensuring that the visualizations are based on data no less than 4 hours old. - The enable_refresh parameter ensures that the materialized view is periodically refreshed. - The allow_non_incremental_definition is used for enabling the creation of non-incrementally refreshable materialized views.

e70ea9e 2 years, 3 months ago

Selected Answer: A

Precomputed Results: Materialized views store precomputed results of complex queries, significantly accelerating subsequent query performance, addressing the slow visualization issue. Allow Non-Incremental Views: Using allow_non_incremental_definition circumvents the limitation of incremental updates for outer joins and analytic functions, ensuring views can be created for the specified queries. Near-Real-Time Data: Setting max_staleness to 4 hours guarantees data freshness within the acceptable latency for visualizations. Automatic Refresh: Enabling refresh with enable_refresh maintains view consistency with minimal maintenance overhead. Minimal Overhead: Materialized views automatically update as underlying data changes, reducing maintenance compared to manual exports or view definitions.

https://zhangxuetu.online

[202610]

Question #230 Topic 1

You need to modernize your existing on-premises data strategy. Your organization currently uses:
• Apache Hadoop clusters for processing multiple large data sets, including on-premises Hadoop Distributed File System (HDFS) for data replication.
• Apache Airflow to orchestrate hundreds of ETL pipelines with thousands of job steps.

You need to set up a new architecture in Google Cloud that can handle your Hadoop workloads and requires minimal changes to your existing orchestration processes. What should you do?

A. Use Bigtable for your large workloads, with connections to Cloud Storage to handle any HDFS use cases. Orchestrate your pipelines with Cloud Composer.

B. Use Dataproc to migrate Hadoop clusters to Google Cloud, and Cloud Storage to handle any HDFS use cases. Orchestrate your pipelines with Cloud Composer. **Most Voted**

C. Use Dataproc to migrate Hadoop clusters to Google Cloud, and Cloud Storage to handle any HDFS use cases. Convert your ETL pipelines to Dataflow.

D. Use Dataproc to migrate your Hadoop clusters to Google Cloud, and Cloud Storage to handle any HDFS use cases. Use Cloud Data Fusion to visually design and deploy your ETL pipelines.

Correct Answer: B

Community vote distribution



Comments

raaad Highly Voted 1 year, 9 months ago

Selected Answer: B

Straight forward

datasmg 1 year, 6 months ago

Selected Answer: B

You can use Dataproc for doing Apache Hadoop process, then Cloud Storage to replace the HDFS, and using Cloud Composer (built in Apache Airflow) for orchestrator.

cuadradobertolinisebastiancami 1 year, 7 months ago

Selected Answer: B

Airflow -> composer Minimum changes -> Dataproc

JyoGCP 1 year, 7 months ago

Selected Answer: B

Option B

Matt_108 1 year, 8 months ago

Selected Answer: B

definitely B

scaenruy 1 year, 9 months ago

Selected Answer: B

Cloud Composer -> Airflow

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #231 Topic 1

You recently deployed several data processing jobs into your Cloud Composer 2 environment. You notice that some tasks are failing in Apache Airflow. On the monitoring dashboard, you see an increase in the total workers memory usage, and there were worker pod evictions. You need to resolve these errors. What should you do? (Choose two.)

A. Increase the directed acyclic graph (DAG) file parsing interval.

B. Increase the Cloud Composer 2 environment size from medium to large.

C. Increase the maximum number of workers and reduce worker concurrency. **Most Voted**

D. Increase the memory available to the Airflow workers.

E. Increase the memory available to the Airflow triggerer.

Correct Answer: C

Community vote distribution



Comments

ML6 **Highly Voted** 1 year, 7 months ago

Selected Answer: D

If an Airflow worker pod is evicted, all task instances running on that pod are interrupted, and later marked as failed by Airflow. The majority of issues with worker pod evictions happen because of out-of-memory situations in workers. You might want to: - (D) Increase the memory available to workers. - (C) Reduce worker concurrency. In this way, a single worker handles fewer tasks at once. This provides more memory or storage to each individual task. If you change worker concurrency, you might also want to increase the maximum number of workers. In this way, the number of tasks that your environment can handle at once stays the same. For example, if you reduce worker Concurrency from 12 to 6, you might want to double the maximum number of workers. Source: https://cloud.google.com/composer/docs/composer-2/optimize-environments

af17139 4 months, 2 weeks ago

Selected Answer: D

D. Increase the memory available to the Airflow workers. Here's why: Directly Addresses the Root Cause: Worker pod evictions are happening because the pods are exceeding their memory limits. Increasing the memory allocation for each worker pod directly provides more resources to handle the demands of the tasks. Handles Memory-Intensive Tasks: If individual tasks themselves require a significant amount of memory, reducing concurrency (as in option C) might not prevent OOMs. A single task could still consume more memory than the worker has, even if it's the only task

running. Simpler Initial Step: Adjusting the memory per worker is often a more straightforward change to make and observe the impact.

imrane1995 8 months, 2 weeks ago

Selected Answer: C

BD B. Increase the Cloud Composer 2 environment size from medium to large. Why: Increasing the environment size automatically adjusts the GKE cluster's node resources (CPU and memory), which can help avoid pod evictions caused by memory pressure. Effect: This increases the capacity available to all components, including workers, helping to accommodate the memory needs of your jobs. D. Increase the memory available to the Airflow workers. Why: If workers are being evicted due to memory pressure, increasing their memory limit directly addresses the root cause. How: In Cloud Composer 2, this can be done via the resources settings under workloadsConfig (like worker.resources.limits.memory).

taka5094 1 year ago

Selected Answer: C

CD On the Monitoring dashboard, in the Workers section, observe the Worker Pods evictions graphs for your environment. The Total workers memory usage graph shows a total perspective of the environment. A single worker can still exceed the memory limit, even if the memory utilization is healthy at the environment level. According to your observations, you might want to: - Increase the memory available to workers. - Reduce worker concurrency. In this way, a single worker handles fewer tasks at once. This provides more memory or storage to each individual task. If you change worker concurrency, you might also want to increase the maximum number of workers. In this way, the number of tasks that your environment can handle at once stays the same. For example, if you reduce worker Concurrency from 12 to 6, you might want to double the maximum number of workers.

desertlotus1211 1 year ago

Answer is B,D

Anudeep58 1 year, 4 months ago

Selected Answer: D

Answer C,D According to your observations, you might want to: Increase the memory available to workers. Reduce worker concurrency. In this way, a single worker handles fewer tasks at once. This provides more memory or storage to each individual task. If you change worker concurrency, you might also want to increase the maximum number of workers. In this way, the number of tasks that your environment can handle at once stays the same. For example, if you reduce worker Concurrency from 12 to 6, you might want to double the maximum number of workers.

desertlotus1211 1 year ago

Reducing concurrency can reduce memory pressure per worker, but won't help if the memory limit per pod is too low

virat_kohli 1 year, 4 months ago

Selected Answer: D

C. Increase the maximum number of workers and reduce worker concurrency. Most Voted D. Increase the memory available to the Airflow workers.

ML6 1 year, 7 months ago

Selected Answer: D

If an Airflow worker pod is evicted, all task instances running on that pod are interrupted, and later marked as failed by Airflow. The majority of issues with worker pod evictions happen because of out-of-memory situations in workers. You might want to: - Increase the memory available to workers. - Reduce worker concurrency. In this way, a single worker handles fewer tasks at once. This provides more memory or storage to each individual task. If you change worker concurrency, you might also want to increase the maximum number of workers. In this way, the number of tasks that your environment can handle at once stays the same. For example, if you reduce worker Concurrency from 12 to 6,

you might want to double the maximum number of workers. Source: https://cloud.google.com/composer/docs/composer-2/optimize-environments

qq589539483084gfrgrgfr 1 year, 8 months ago

Selected Answer: C

CD It is clear

Matt_108 1 year, 8 months ago

Selected Answer: C

C & D to me

GCP001 1 year, 8 months ago

Selected Answer: C

C and D Check ref for memory optimization - https://cloud.google.com/composer/docs/composer-2/optimize-environments

AllenChen123 1 year, 8 months ago

Agree. Straightforward. https://cloud.google.com/composer/docs/composer-2/optimize-environments#monitor-scheduler -> Figure 3. Graph that displays worker pod evictions

qq589539483084gfrgrgfr 1 year, 8 months ago

Selected Answer: B

B&D See this- https://cloud.google.com/composer/docs/composer-2/troubleshooting-dags#task-fails-without-logs go through the suggested fixes for If there are airflow-worker pods that show Evicted

Jordan18 1 year, 8 months ago

Selected Answer: C

C and D

raaad 1 year, 9 months ago

Selected Answer: B

B&D: B : - Scaling up the environment size can provide more resources, including memory, to the Airflow workers. If worker pod evictions are occurring due to insufficient memory, increasing the environment size to allocate more resources could alleviate the problem and improve the stability of your data processing jobs. D: - Increase the memory available to the Airflow workers. - Directly increasing the memory allocation for Airflow workers can address the issue of high memory usage and worker pod evictions. More memory per worker means that each worker can handle more demanding tasks or a higher volume of tasks without running out of memory.

GCP001 1 year, 8 months ago

why not B) It s not decreasing concurrency which may cause issue again

https://zhangxuetu.online

[202610]

Question #232 Topic 1

You are on the data governance team and are implementing security requirements to deploy resources. You need to ensure that resources are limited to only the europe-west3 region. You want to follow Google-recommended practices.

What should you do?

A. Set the constraints/gcp.resourceLocations organization policy constraint to in:europe-west3-locations. **Most Voted**

B. Deploy resources with Terraform and implement a variable validation rule to ensure that the region is set to the europe-west3 region for all resources.

C. Set the constraints/gcp.resourceLocations organization policy constraint to in:eu-locations.

D. Create a Cloud Function to monitor all resources created and automatically destroy the ones created outside the europe-west3 region.

Correct Answer: A

Community vote distribution

A (100%)

Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: A

- The constraints/gcp.resourceLocations organization policy constraint is used to define where resources in the organization can be created. - Setting it to in:europe-west3-locations would specify that resources can only be created in the europe-west3 region.

SanjeevRoy91 2 years ago

I am new to this forum. In almost all the questions, the reveal solution is different than the once's discussed here??

b3e59c2 1 year, 2 months ago

Selected Answer: A

https://cloud.google.com/resource-manager/docs/organization-policy/defining-locations#location_types

chrissamharris 1 year, 6 months ago

B, D B - Increase the Cloud Composer 2 environment size from medium to large. Increasing the environment size will provide more resources (including memory) to the entire environment, which should help mitigate memory usage issues. This will also support scaling if the jobs demand more resources. D. Increase the memory available to the

Airflow workers. Increasing memory for Airflow workers will directly address the memory usage issue that's causing the pod evictions. By allocating more memory to the workers, they can handle larger tasks or more intensive workloads without failing due to memory constraints.

hrishi19 1 year, 4 months ago

this comment is for previous question, not this one.

JyoGCP 2 years, 1 month ago

Selected Answer: A

Option A

Matt_108 2 years, 2 months ago

Selected Answer: A

Option A

scaenruy 2 years, 2 months ago

Selected Answer: A

Set the constraints/gcp.resourceLocations organization policy constraint to in:europe-west3-locations.

https://zhangxuetu.online

[202610]

Question #233 Topic 1

You are a BigQuery admin supporting a team of data consumers who run ad hoc queries and downstream reporting in tools such as Looker. All data and users are combined under a single organizational project. You recently noticed some slowness in query results and want to troubleshoot where the slowdowns are occurring. You think that there might be some job queuing or slot contention occurring as users run jobs, which slows down access to results. You need to investigate the query job information and determine where performance is being affected. What should you do?

A. Use slot reservations for your project to ensure that you have enough query processing capacity and are able to allocate available slots to the slower queries.

B. Use Cloud Monitoring to view BigQuery metrics and set up alerts that let you know when a certain percentage of slots were used.

C. Use available administrative resource charts to determine how slots are being used and how jobs are performing over time. Run a query on the INFORMATION_SCHEMA to review query performance. **Most Voted**

D. Use Cloud Logging to determine if any users or downstream consumers are changing or deleting access grants on tagged resources.

Correct Answer: C

Community vote distribution

C (100%)

Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

- BigQuery provides administrative resource charts that show slot utilization and job performance, which can help identify patterns of heavy usage or contention. - Additionally, querying the INFORMATION_SCHEMA with the JOBS or JOBS_BY_PROJECT view can provide detailed information about specific queries, including execution time, slot usage, and whether they were queued.

datapassionate 2 years, 2 months ago

descrived here: https://cloud.google.com/blog/products/data-analytics/troubleshoot-bigquery-performance-with-these-dashboards

ToiToi 1 year, 5 months ago

Selected Answer: C

Without doubt, C!

JyoGCP 2 years, 1 month ago

Selected Answer: C

https://cloud.google.com/blog/topics/developers-practitioners/monitor-analyze-bigquery-performance-using-information-schema

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C

scaenruy 2 years, 2 months ago

Selected Answer: C

C. Use available administrative resource charts to determine how slots are being used and how jobs are performing over time. Run a query on the INFORMATION_SCHEMA to review query performance.

https://zhangxuetu.online

[202610]

Question #234 Topic 1

You migrated a data backend for an application that serves 10 PB of historical product data for analytics. Only the last known state for a product, which is about 10 GB of data, needs to be served through an API to the other applications. You need to choose a cost-effective persistent storage solution that can accommodate the analytics requirements and the API performance of up to 1000 queries per second (QPS) with less than 1 second latency. What should you do?

A. 1. Store the historical data in BigQuery for analytics.
2. Use a materialized view to precompute the last state of a product.
3. Serve the last state data directly from BigQuery to the API.

B. 1. Store the products as a collection in Firestore with each product having a set of historical changes.
2. Use simple and compound queries for analytics.
3. Serve the last state data directly from Firestore to the API.

C. 1. Store the historical data in Cloud SQL for analytics.
2. In a separate table, store the last state of the product after every product change.
3. Serve the last state data directly from Cloud SQL to the API.

D. 1. Store the historical data in BigQuery for analytics.
2. In a Cloud SQL table, store the last state of the product after every product change.
3. Serve the last state data directly from Cloud SQL to the API. Most Voted

Correct Answer: D

Community vote distribution



Comments

einchkrein Highly Voted 2 years, 2 months ago

Serve the last state data directly from Cloud SQL to the API. Here's why this option is most suitable: BigQuery for Analytics: BigQuery is an excellent choice for storing and analyzing large datasets like your 10 PB of historical product data. It is designed for handling big data analytics efficiently and cost-effectively. Cloud SQL for Last State Data: Cloud SQL is a fully managed relational database that can effectively handle the storage of the last known state of products. Storing this subset of data (about 10 GB) in Cloud SQL allows for optimized and faster query performance for your API needs. Cloud SQL can comfortably handle the requirement of up to 1000 QPS with sub-second latency. Separation of Concerns: This approach separates the analytics workload (BigQuery) from the operational query workload (Cloud SQL). This separation ensures that analytics queries do not interfere with the operational performance of the API and vice versa.

datapassionate Highly Voted 2 years, 2 months ago

Selected Answer: D

D. 1. Store the historical data in BigQuery for analytics. 2. In a Cloud SQL table, store the last state of the product after every product change. 3. Serve the last state data directly from Cloud SQL to the AP This approach leverages BigQuery's scalability and efficiency for handling large datasets for analytics. BigQuery is well-suited for managing the 10 PB of historical product data. Meanwhile, Cloud SQL provides the necessary performance to handle the API queries with the required low latency. By storing the latest state of each product in Cloud SQL, you can efficiently handle the high QPS with sub-second latency, which is crucial for the API's performance. This combination of BigQuery and Cloud SQL offers a balanced solution for both the large-scale analytics and the high-performance API needs.

zanhsieh 1 year, 1 month ago

Selected Answer: D

Why not A? Because BQ API 100 reqs per second for API method. Other possible limits do not meet the 1000 QPS requirement. Yes, max number of tabledata.list reqs per second is 1000 but we won't always call tabledata.list all time. https://cloud.google.com/bigquery/quotas#api_request_quotas

clouditis 1 year, 3 months ago

Selected Answer: A

A is the most plausible option - Cloud SQL can not retrieve results out with 1 second latency as the requirement here is, with BQ MV"s that could be a possibility as its pre-computed.

ToiToi 1 year, 5 months ago

Selected Answer: A

Why A? Because: Materialized View for API: A materialized view in BigQuery pre-computes the last known state of each product. This ensures that your API can quickly retrieve the latest product information without needing to query the entire historical dataset. BigQuery for API Serving: BigQuery can handle high query volumes with low latency, meeting your requirement of 1000 QPS with sub-second latency. Cost-Effectiveness: This solution avoids the need for a separate database like Cloud SQL, minimizing costs and management overhead. Why not D: While Cloud SQL is a good option for transactional workloads, it's not as cost-effective or scalable as BigQuery for analytical queries on 10 PB of data. It might also not be the ideal choice for serving high-volume API requests with low latency.

Rip696 6 months, 3 weeks ago

In option D, it says to store the historical data in BQ not in CSQL.

Anudeep58 1 year, 9 months ago

Selected Answer: D

Why not A: Serving data directly from BigQuery to the API may not meet the low latency requirements for high QPS operations, as BigQuery is optimized for analytical queries rather than transactional workloads.

Ryannn23 1 year, 2 months ago

what transactional workload? you just need to provide latest status for each product through an API. Select from a 10GB BQ MV will provide the result in 1 sec.

josech 1 year, 10 months ago

Selected Answer: A

Materialized views are precomputed views that periodically cache the results of a query for increased performance and efficiency. Materialized views can optimize queries with high computation cost and small dataset results. https://cloud.google.com/bigquery/docs/materialized-views-intro#use_cases https://cloud.google.com/bigquery/docs/materialized-views-intro

CGS22 1 year, 11 months ago

Selected Answer: D

Why D is the best choice: Cost-Effective Analytics: BigQuery excels at handling large datasets (10 PB) and complex analytical queries. Its columnar storage and massively parallel processing make it ideal for analyzing historical product data. High-Performance API: Cloud SQL provides a managed relational database service optimized for transactional workloads. It can easily handle the 1000 QPS requirement with low latency, ensuring fast API responses. Separation of Concerns: Storing historical data in BigQuery and the last known state in Cloud SQL separates analytical and transactional workloads, optimizing performance and cost for each use case.

JyoGCP 2 years, 1 month ago

Selected Answer: D

Option D

ML6 2 years, 1 month ago

Selected Answer: D

BigQuery = data warehouse that is optimized for querying and analyzing large datasets using SQL. Can easily process petabytes of data. Cloud SQL = designed for transactional workloads and traditional relational database use cases, such as web applications, e-commerce platforms, and content management systems.

Matt_108 2 years, 2 months ago

Selected Answer: D

Option D is the right one, compared to option A, Cloud SQL is more efficient and cost effective for the amount of time the data needs to be accessed by the api

scaenruy 2 years, 2 months ago

Selected Answer: A

A. 1. Store the historical data in BigQuery for analytics. 2. Use a materialized view to precompute the last state of a product. 3. Serve the last state data directly from BigQuery to the API.

RenePetersen 2 years, 1 month ago

I believe the latency of BigQuery is too high to accommodate the sub-second latency requirement.

https://zhangxuetu.online

[202610]

Question #235 Topic 1

You want to schedule a number of sequential load and transformation jobs. Data files will be added to a Cloud Storage bucket by an upstream process. There is no fixed schedule for when the new data arrives. Next, a Dataproc job is triggered to perform some transformations and write the data to BigQuery. You then need to run additional transformation jobs in BigQuery. The transformation jobs are different for every table. These jobs might take hours to complete. You need to determine the most efficient and maintainable workflow to process hundreds of tables and provide the freshest data to your end users. What should you do?

A. 1. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Cloud Storage, Dataproc, and BigQuery operators.
2. Use a single shared DAG for all tables that need to go through the pipeline.
3. Schedule the DAG to run hourly.

B. 1. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Cloud Storage, Dataproc, and BigQuery operators.
2. Create a separate DAG for each table that needs to go through the pipeline.
3. Schedule the DAGs to run hourly.

C. 1. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Dataproc and BigQuery operators.
2. Use a single shared DAG for all tables that need to go through the pipeline.
3. Use a Cloud Storage object trigger to launch a Cloud Function that triggers the DAG.

D. 1. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Dataproc and BigQuery operators.
2. Create a separate DAG for each table that needs to go through the pipeline.
3. Use a Cloud Storage object trigger to launch a Cloud Function that triggers the DAG. Most Voted

Correct Answer: D

Community vote distribution

D (61%) C (39%)

Comments

cuadradobertolinisebastiancami Highly Voted 2 years, 1 month ago

D * Transformations are in Dataproc and BigQuery. So you don't need operators for GCS (A and B can be discard) * "There is no fixed schedule for when the new data arrives." so you trigger the DAG when a file arrives * "The transformation jobs are different for every table. " so you need a DAG for each table. Then, D is the most suitable answer

af17139 4 months, 2 weeks ago

Selected Answer: D

A & B: Hourly schedules don't fit the event-driven nature of data arrival. C: A single shared DAG for hundreds of tables with different transformations is not maintainable or scalable from a logic perspective. It would require complex branching and conditional logic within the DAG, making it hard to manage.

imrane1995 8 months, 2 weeks ago

Selected Answer: C

 C. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Dataproc and BigQuery operators. Use a single shared DAG for all tables that need to go through the pipeline. Use a Cloud Storage object trigger to launch a Cloud Function that triggers the DAG.

22c1725 10 months ago

Selected Answer: D

programmatic DAG generation.

22c1725 10 months, 1 week ago

Selected Answer: D

"maintainable" this is an clear. Whould creating one dag for each transformtion is better or having large code with thousend of lines would be better? there is no clear right or wrong here. But I would go with "D" becuse there is "Jobs" not "Job"

choprat1 1 year, 1 month ago

Selected Answer: D

managing indidivuals DAGs is the best way when they're too different

f74ca0c 1 year, 3 months ago

Selected Answer: C

A single shared DAG is efficient to manage, and table-specific transformations can be handled using parameters (e.g., passing table names and configurations dynamically). Triggering the DAG using a Cloud Storage object notification and a Cloud Function ensures the workflow starts immediately upon data arrival. Event-driven architecture minimizes delays and provides the freshest data to users. Efficient, maintainable, and event-driven.

8ad5266 1 year, 9 months ago

Selected Answer: C

This explains why it's not D: maintainable workflow to process hundreds of tables and provide the freshest data to your end users How is creating a DAG for each of the hundreds of tables maintainable?

plum21 1 year, 1 month ago

It's possible to generate multiple DAGs programatically. That's the reason for C. https://cloud.google.com/blog/products/data-analytics/optimize-cloud-composer-via-better-airflow-dags -> look at #5

JyoGCP 2 years, 1 month ago

Selected Answer: D

Option D

Matt_108 2 years, 2 months ago

Selected Answer: D

Option D, which gets triggered when the data comes in and accounts for the fact that each table has its own set of transformations

Jordan18 2 years, 2 months ago

why not C?

cuadradobertolinisebastiancami 2 years, 1 month ago

It says that the transformations for each table are very different

AllenChen123 2 years, 2 months ago

Same question, why not use single DAG to manage as there are hundreds of tables.

raaad 2 years, 2 months ago

Selected Answer: D

- Option D: Tailored handling and scheduling for each table; triggered by data arrival for more timely and efficient processing.

scaenruy 2 years, 2 months ago

Selected Answer: D

D. 1. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Dataproc and BigQuery operators. 2. Create a separate DAG for each table that needs to go through the pipeline. 3. Use a Cloud Storage object trigger to launch a Cloud Function that triggers the DAG.

https://zhangxuetu.online

[202610]

Question #236 Topic 1

You are deploying a MySQL database workload onto Cloud SQL. The database must be able to scale up to support several readers from various geographic regions. The database must be highly available and meet low RTO and RPO requirements, even in the event of a regional outage. You need to ensure that interruptions to the readers are minimal during a database failover. What should you do?

A. Create a highly available Cloud SQL instance in region Create a highly available read replica in region B. Scale up read workloads by creating cascading read replicas in multiple regions. Backup the Cloud SQL instances to a multi-regional Cloud Storage bucket. Restore the Cloud SQL backup to a new instance in another region when Region A is down.

B. Create a highly available Cloud SQL instance in region A. Scale up read workloads by creating read replicas in multiple regions. Promote one of the read replicas when region A is down.

C. Create a highly available Cloud SQL instance in region A. Create a highly available read replica in region B. Scale up read workloads by creating cascading read replicas in multiple regions. Promote the read replica in region B when region A is down. **Most Voted**

D. Create a highly available Cloud SQL instance in region A. Scale up read workloads by creating read replicas in the same region. Failover to the standby Cloud SQL instance when the primary instance fails.

Correct Answer: C

Community vote distribution



Comments

rohan.sahi Highly Voted 2 years, 1 month ago

Selected Answer: C

Option C: Because HA read replica in multiple regions. NotA: Coz restore from back up is time taking NotB: No HA in Multiple regions read replica Not D: Only one region mentioned.

raaad Highly Voted 2 years, 2 months ago

Selected Answer: C

- Combines high availability with geographic distribution of read workloads. - Promoting a highly available read replica can provide a quick failover solution, potentially meeting low RTO and RPO requirements. ===== Why not A: Restoring from backup to a new instance in another region during a regional outage might not meet low RTO and RPO requirements due to the time it takes to perform a restore.

AllenChen123 2 years, 2 months ago

Why not B?

datapassionate 2 years, 2 months ago

Why not B: While B option scales up read workloads across multiple regions, it doesn't specify high availability for the read replica in another region. In the event of a regional outage, promoting a non-highly available read replica might not provide the desired uptime and reliability.

mi_yulai 1 year, 5 months ago

Why C? Is it possible to have HA enable in different regions? How the synchronization in disk will wokr for HA?

tibuenoc 2 years, 2 months ago

Selected Answer: B

https://cloud.google.com/sql/docs/mysql/replication This option involves having read replicas in multiple regions, allowing you to promote one of them in the event of a failure in region A. While there may still be a brief interruption during the failover, it is likely to be less than the time required for the synchronization of cascading read replicas.

Matt_108 2 years, 2 months ago

Selected Answer: B

To me, it's B. it provides: High availability: The highly available Cloud SQL instance in region A will ensure that the database remains accessible even if one of the zones in the region becomes unavailable. Scalability: The read replicas in multiple regions will enable you to scale up the read capacity of the database to support the demands of readers from various geographic regions. Minimal interruptions: When region A is down, one of the read replicas in another region will be promoted to become the new primary instance. This will ensure that there is no interruption to the readers.

Matt_108 2 years, 2 months ago

Why not others: Approach A: This approach requires you to restore a backup from a different region, which could take some time. This could result in a significant RPO (Recovery Point Objective) for the database. Additionally, the restored instance may not be physically located in the same region as the readers, which could impact performance. Approach C: This approach requires you to promote the read replica in region B, which could result in a temporary interruption to the readers while the promotion is taking place. Additionally, the read replica in region B may not be able to handle the same level of read traffic as the primary instance in region A. Approach D: This approach does not provide the same level of scalability as the other approaches, as you are limited to read replicas in the same region. Additionally, failover to the standby instance could result in a temporary interruption to the readers.

Matt_108 2 years, 2 months ago

Ignore my previous messages, it's C :D

scaenruy 2 years, 2 months ago

Selected Answer: A

A. Create a highly available Cloud SQL instance in region Create a highly available read replica in region B. Scale up read workloads by creating cascading read replicas in multiple regions. Backup the Cloud SQL instances to a multi-regional Cloud Storage bucket. Restore the Cloud SQL backup to a new instance in another region when Region A is down.

https://zhangxuetu.online

[202610]

Question #237 Topic 1

You are planning to load some of your existing on-premises data into BigQuery on Google Cloud. You want to either stream or batch-load data, depending on your use case. Additionally, you want to mask some sensitive data before loading into BigQuery. You need to do this in a programmatic way while keeping costs to a minimum. What should you do?

A. Use Cloud Data Fusion to design your pipeline, use the Cloud DLP plug-in to de-identify data within your pipeline, and then move the data into BigQuery.

B. Use the BigQuery Data Transfer Service to schedule your migration. After the data is populated in BigQuery, use the connection to the Cloud Data Loss Prevention (Cloud DLP) API to de-identify the necessary data.

C. Create your pipeline with Dataflow through the Apache Beam SDK for Python, customizing separate options within your code for streaming, batch processing, and Cloud DLP. Select BigQuery as your data sink. Most Voted

D. Set up Datastream to replicate your on-premise data on BigQuery.

Correct Answer: C

Community vote distribution



Comments

raaad Highly Voted 2 years, 2 months ago

Selected Answer: C

- Programmatic Flexibility: Apache Beam provides extensive control over pipeline design, allowing for customization of data transformations, including integration with Cloud DLP for sensitive data masking. - Streaming and Batch Support: Beam seamlessly supports both streaming and batch data processing modes, enabling flexibility in data loading patterns. - Cost-Effective Processing: Dataflow offers a serverless model, scaling resources as needed, and only charging for resources used, helping optimize costs. - Integration with Cloud DLP: Beam integrates well with Cloud DLP for sensitive data masking, ensuring data privacy before loading into BigQuery.

qq589539483084gfrgrgfr 2 years, 2 months ago

In correct Option is A because you want a programatic way whereas datafusion is codeless solution and also dataflow is cost effective

AllenChen123 2 years, 2 months ago

You are saying Option C

JyoGCP 2 years, 1 month ago

Selected Answer: C

Option C

tibuenoc 2 years, 2 months ago

Selected Answer: C

C is correct. Using Dataflow as Python as programming and BQ as sink. A is incorrect - DataFusion is Code-free as the main propose

scaenruy 2 years, 2 months ago

Selected Answer: A

A. Use Cloud Data Fusion to design your pipeline, use the Cloud DLP plug-in to de-identify data within your pipeline, and then move the data into BigQuery.

ggg24 1 year, 5 months ago

Data Fusion support only Batch and Streaming is required

chrissamharris 1 year, 10 months ago

Incorrect, that's a low-code solution. Doesnt meet this specific requirement: "You need to do this in a programmatic way"

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #238 Topic 1

You want to encrypt the customer data stored in BigQuery. You need to implement per-user crypto-deletion on data stored in your tables. You want to adopt native features in Google Cloud to avoid custom solutions. What should you do?

A. Implement Authenticated Encryption with Associated Data (AEAD) BigQuery functions while storing your data in BigQuery. **Most Voted**

B. Create a customer-managed encryption key (CMEK) in Cloud KMS. Associate the key to the table while creating the table.

C. Create a customer-managed encryption key (CMEK) in Cloud KMS. Use the key to encrypt data before storing in BigQuery.

D. Encrypt your data during ingestion by using a cryptographic library supported by your ETL pipeline.

Correct Answer: A

Community vote distribution

A (100%)

Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: A

- AEAD cryptographic functions in BigQuery allow for encryption and decryption of data at the column level. - You can encrypt specific data fields using a unique key per user and manage these keys outside of BigQuery (for example, in your application or using a key management system). - By "deleting" or revoking access to the key for a specific user, you effectively make their data unreadable, achieving crypto-deletion. - This method provides fine-grained encryption control but requires careful key management and integration with your applications.

JyoGCP 1 year, 7 months ago

Selected Answer: A

https://cloud.google.com/bigquery/docs/aead-encryption-concepts
https://cloud.google.com/bigquery/docs/reference/standard-sql/aead_encryption_functions

scaenruy 1 year, 9 months ago

Selected Answer: A

A. Implement Authenticated Encryption with Associated Data (AEAD) BigQuery functions while storing your data in BigQuery.

https://zhangxuetu.online

[202610]

Question #239 Topic 1

The data analyst team at your company uses BigQuery for ad-hoc queries and scheduled SQL pipelines in a Google Cloud project with a slot reservation of 2000 slots. However, with the recent introduction of hundreds of new non time-sensitive SQL pipelines, the team is encountering frequent quota errors. You examine the logs and notice that approximately 1500 queries are being triggered concurrently during peak time. You need to resolve the concurrency issue. What should you do?

A. Increase the slot capacity of the project with baseline as 0 and maximum reservation size as 3000.

B. Update SQL pipelines to run as a batch query, and run ad-hoc queries as interactive query jobs. **Most Voted**

C. Increase the slot capacity of the project with baseline as 2000 and maximum reservation size as 3000.

D. Update SQL pipelines and ad-hoc queries to run as interactive query jobs.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- BigQuery allows you to specify job priority as either BATCH or INTERACTIVE. - Batch queries are queued and then started when idle resources are available, making them suitable for non-time-sensitive workloads. - Running ad-hoc queries as interactive ensures they have prompt access to resources.

judy_data 6 months, 2 weeks ago

Selected Answer: B

You can query BigQuery data by using one of the following query job types: Interactive query jobs. By default, BigQuery runs queries as interactive query jobs, which are intended to start executing as quickly as possible. Batch query jobs. Batch queries have lower priority than interactive queries. When a project or reservation is using all of its available compute resources, batch queries are more likely to be queued and remain in the queue. After a batch query starts running, the batch query runs the same as an interactive query. For more information, see query queues.

LP_PDE 1 year, 2 months ago

Selected Answer: B

By updating your SQL pipelines to run as batch queries you can reduce concurrency, avoid quota errors, and ensure that your analysts have the resources they need for their interactive queries.

ToiToi 1 year, 5 months ago

Selected Answer: B

This question has nothing to do with increasing slots, it is just confusing and misleading, therefore A and C do not make sense. D (All interactive queries): Running all queries as interactive would prioritize speed over cost-efficiency and might not be necessary for your non-time-sensitive SQL pipelines.

josech 1 year, 10 months ago

Selected Answer: C

You already have a 2000 slots consumption and sudden peaks, so you should use a baseline of 2000 slots and a maximum of 3000 to tackle the peak concurrent activity. https://cloud.google.com/bigquery/docs/slots-autoscaling-intro

CGS22 1 year, 11 months ago

Selected Answer: A

Why A is the best choice: Addresses Concurrency: Increasing the maximum reservation size to 3000 slots directly addresses the concurrency issue by providing more capacity for simultaneous queries. Since the current peak usage is 1500 queries, this increase ensures sufficient headroom. Cost Optimization: Setting the baseline to 0 means you only pay for the slots actually used, avoiding unnecessary costs for idle capacity. This is ideal for non-time-sensitive workloads where flexibility is more important than guaranteed instant availability.

JyoGCP 2 years, 1 month ago

Selected Answer: B

Option B

CGS22 1 year, 11 months ago

B: While batch queries are generally more cost-effective for large, non-interactive workloads, they don't solve the concurrency problem. If multiple batch queries are triggered simultaneously, they would still compete for the same limited slot pool.

scaenruy 2 years, 2 months ago

Selected Answer: B

B. Update SQL pipelines to run as a batch query, and run ad-hoc queries as interactive query jobs.

https://zhangxuetu.online

[202610]

Question #240 Topic 1

You are designing a data mesh on Google Cloud by using Dataplex to manage data in BigQuery and Cloud Storage. You want to simplify data asset permissions. You are creating a customer virtual lake with two user groups:

• Data engineers, which require full data lake access
• Analytic users, which require access to curated data

You need to assign access rights to these two groups. What should you do?

A. 1. Grant the dataplex.dataOwner role to the data engineer group on the customer data lake.
2. Grant the dataplex.dataReader role to the analytic user group on the customer curated zone. **Most Voted**

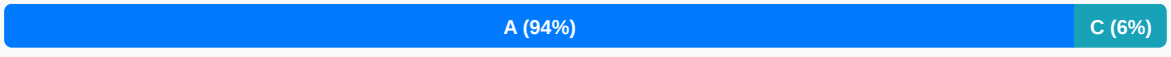
B. 1. Grant the dataplex.dataReader role to the data engineer group on the customer data lake.
2. Grant the dataplex.dataOwner to the analytic user group on the customer curated zone.

C. 1. Grant the bigquery.dataOwner role on BigQuery datasets and the storage.objectCreator role on Cloud Storage buckets to data engineers.
2. Grant the bigquery.dataViewer role on BigQuery datasets and the storage.objectViewer role on Cloud Storage buckets to analytic users.

D. 1. Grant the bigquery.dataViewer role on BigQuery datasets and the storage.objectViewer role on Cloud Storage buckets to data engineers.
2. Grant the bigquery.dataOwner role on BigQuery datasets and the storage.objectEditor role on Cloud Storage buckets to analytic users.

Correct Answer: A

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: A

- dataplex.dataOwner: Grants full control over data assets, including reading, writing, managing, and granting access to others. - dataplex.dataReader: Allows users to read data but not modify it.

AllenChen123 2 years, 2 months ago

Yes, https://cloud.google.com/dataplex/docs/lake-security#data-roles Dataplex maps its roles to the data roles for each underlying storage resource (Cloud Storage, BigQuery). ^ simplify the permissions.

josech 1 year, 10 months ago

Selected Answer: C

The quetion is for BigQuery AND Cloud Storage for a Data Lake, so you should assign IAM permissions for both of them. C is correct.

JyoGCP 2 years, 1 month ago

Selected Answer: A

Option A

qq589539483084gfrgrgfr 2 years, 2 months ago

Selected Answer: A

A correct answer

Matt_108 2 years, 2 months ago

Selected Answer: A

Option A clearly correct

scaenruy 2 years, 2 months ago

Selected Answer: A

A. 1. Grant the dataplex.dataOwner role to the data engineer group on the customer data lake. 2. Grant the dataplex.dataReader role to the analytic user group on the customer curated zone.

https://zhangxuetu.online

[202610]

Question #241 Topic 1

You are designing the architecture of your application to store data in Cloud Storage. Your application consists of pipelines that read data from a Cloud Storage bucket that contains raw data, and write the data to a second bucket after processing. You want to design an architecture with Cloud Storage resources that are capable of being resilient if a Google Cloud regional failure occurs. You want to minimize the recovery point objective (RPO) if a failure occurs, with no impact on applications that use the stored data. What should you do?

A. Adopt multi-regional Cloud Storage buckets in your architecture.

B. Adopt two regional Cloud Storage buckets, and update your application to write the output on both buckets.

C. Adopt a dual-region Cloud Storage bucket, and enable turbo replication in your architecture. **Most Voted**

D. Adopt two regional Cloud Storage buckets, and create a daily task to copy from one bucket to the other.

Correct Answer: C

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

- Dual-region buckets are a specific type of storage that automatically replicates data between two geographically distinct regions. - Turbo replication is an enhanced feature that provides faster replication between the two regions, thus minimizing RPO. - This option ensures that your data is resilient to regional failures and is replicated quickly, meeting the needs for low RPO and no impact on application performance.

therealsohail **Highly Voted** 2 years, 2 months ago

Selected Answer: C

Turbo replication provides faster redundancy across regions for data in your dual-region buckets, which reduces the risk of data loss exposure and helps support uninterrupted service following a regional outage.

petulda 1 year, 4 months ago

Why not A https://cloud.google.com/storage/docs/locations multi regional location has cross region redundancy

petulda 1 year, 4 months ago

Sorry, it is about minimizing RPO, where Turbo replication is a factor..

CGS22 1 year, 11 months ago

Selected Answer: A

A. Adopt multi-regional Cloud Storage buckets in your architecture. Why A is the best choice: Automatic Cross-Region Replication: Multi-regional buckets automatically replicate data across multiple geographically separated regions within a selected multi-region location (e.g., us). This ensures data redundancy and availability even if one region experiences an outage. Minimal RPO: Data written to a multi-regional bucket is synchronously replicated to at least two regions. This means that in the event of a regional failure, the RPO is essentially zero, as the data is already available in other regions. No Application Changes: Applications can continue reading and writing data to the multi-regional bucket without any modifications, as the cross-region replication is handled transparently by Cloud Storage

mdell 1 year, 3 months ago

Minimal yet, but not minimized as stated in the question. That's why C is correct. Turbo replication provides faster redundancy across regions for data in your dual-region buckets, which reduces the risk of data loss exposure and helps support uninterrupted service following a regional outage. When enabled, turbo replication is designed to replicate 100% of newly written objects to the two regions that constitute a dual-region within the recovery point objective of 15 minutes, regardless of object size. Note that even for default replication, most objects finish replication within minutes. https://cloud.google.com/storage/docs/availability-durability#turbo-replication

hanoverquay 2 years ago

Selected Answer: C

vote c

ricardovazz 2 years ago

Selected Answer: C

https://cloud.google.com/storage/docs/availability-durability#turbo-replication "Default replication in Cloud Storage is designed to provide redundancy across regions for 99.9% of newly written objects within a target of one hour and 100% of newly written objects within a target of 12 hours" "When enabled, turbo replication is designed to replicate 100% of newly written objects to both regions that constitute the dual-region within the recovery point objective of 15 minutes, regardless of object size." Thus, since they want to minimize RPO, should use turbo replication

JyoGCP 2 years, 1 month ago

Selected Answer: C

Option C

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C: https://cloud.google.com/storage/docs/dual-regions + https://cloud.google.com/storage/docs/managing-turbo-replication

scaenruy 2 years, 2 months ago

Selected Answer: A

A. Adopt multi-regional Cloud Storage buckets in your architecture.

datapassionate 2 years, 2 months ago

It wont be a correct answer. Correct is C. It is required "no impact on applications that use the stored data"

ashdam 2 years, 1 month ago

But multi-region is completely transparent for the application if one fails. it would need to fail all EU or US regions. I dont undertand why multi-region would have impact on that

datapassionate 2 years, 2 months ago

Whereas with multi-region " it can also introduce unpredictable latency into the response time and higher network egress charges for cloud workloads when multi-region data is read from remote regions"

https://cloud.google.com/blog/products/storage-data-transfer/choose-between-regional-dual-region-and-multi-region-cloud-storage

ashdam 2 years, 1 month ago

There is no requirment on latency, just RPO which it would be 0 since multi-region.

https://zhangxuetu.online

[202610]

Question #242 Topic 1

You have designed an Apache Beam processing pipeline that reads from a Pub/Sub topic. The topic has a message retention duration of one day, and writes to a Cloud Storage bucket. You need to select a bucket location and processing strategy to prevent data loss in case of a regional outage with an RPO of 15 minutes. What should you do?

A. 1. Use a dual-region Cloud Storage bucket.
2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs.
3. Seek the subscription back in time by 15 minutes to recover the acknowledged messages.
4. Start the Dataflow job in a secondary region.

B. 1. Use a multi-regional Cloud Storage bucket.
2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs.
3. Seek the subscription back in time by 60 minutes to recover the acknowledged messages.
4. Start the Dataflow job in a secondary region.

C. 1. Use a regional Cloud Storage bucket.
2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs.
3. Seek the subscription back in time by one day to recover the acknowledged messages.
4. Start the Dataflow job in a secondary region and write in a bucket in the same region.

D. 1. Use a dual-region Cloud Storage bucket with turbo replication enabled.
2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs.
3. Seek the subscription back in time by 60 minutes to recover the acknowledged messages.
4. Start the Dataflow job in a secondary region. Most Voted

Correct Answer: D

Community vote distribution



Comments

datapassionate Highly Voted 1 year, 8 months ago

Selected Answer: D

D. 1. Use a dual-region Cloud Storage bucket with turbo replication enabled. 2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs. 3. Seek the subscription back in time by 60 minutes to recover the acknowledged messages. 4. Start the Dataflow job in a secondary region. RPO of 15 minutes is guaranteed when turbo replication is used https://cloud.google.com/storage/docs/availability-durability

ashdam 1 year, 7 months ago

Why multi-region is not correct. There is no downtime in case a region goes down.

JyoGCP Highly Voted 1 year, 7 months ago

Selected Answer: D

Option D is correct. Not A, because dual-region bucket WITHOUT turbo replication takes atleast 1 hour to sync data between regions. SLA for 100% data sync is 12 hours as per google.

0dd4e0c 11 months ago

Selected Answer: D

it's D, keyword "Turbo replication" for RPO recovery

LP_PDE 1 year, 2 months ago

Selected Answer: D

Could be A or D. The choice between a 15-minute seek and a 60-minute seek depends on your specific requirements and priorities. If a very low RPO is critical, a 60-minute seek might be necessary to ensure data completeness.If minimizing cost and processing time is more important, a 15-minute seek might be sufficient, especially if you're confident in the reliability of Turbo Replication.

m_a_p_s 1 year, 3 months ago

Selected Answer: A

An RPO of 15 minutes seemingly suggests using Turbo Replication. But here's the thing - why would you want to seek the subscription back in time by 60 minutes and run the Dataflow job? Thus, if turbo replication is enabled, steps 3 & 4 are completely redundant and unnecessary. Which is why option A is correct. This was a tricky one!

SajadAhm 1 month, 2 weeks ago

It might take up to 1 day to replicate data across regions without turbo enabled. so without turbo, we should seek up to 1 day.

shangning007 1 year, 3 months ago

Selected Answer: A

I don't like answer D. If we have turbo replication can ensure that change within 15min can be replicated, why do we still need to seek the subscription back in time by 60min?

SVGoogle89 1 year, 4 months ago

D https://cloud.google.com/storage/docs/availability-durability#cross-region-redundancy

lipa31 1 year, 8 months ago

Selected Answer: D

https://cloud.google.com/storage/docs/availability-durability#turbo-replication says : "When enabled, turbo replication is designed to replicate 100% of newly written objects to both regions that constitute the dual-region within the recovery point objective of 15 minutes, regardless of object size." so seems D to me

raaad 1 year, 9 months ago

Selected Answer: A

- Low RPO: Dual-region buckets offer synchronous replication, ensuring data is immediately available in both regions, aligning with the 15-minute RPO. - Turbo Replication: enabling turbo replication can further reduce replication latency to near-real-time for even stricter RPO requirements. - Resilient Data Storage: Dual-region buckets ensure data availability even during regional outages, protecting processed data. - Fast Recovery: Reprocessing from the last 15 minutes of acknowledged messages minimizes data loss and downtime.

qq589539483084gfrgrgfr 1 year, 8 months ago

why not D then, if turbo replication improves RPO??

scaenruy 1 year, 9 months ago

Selected Answer: A

A. 1. Use a dual-region Cloud Storage bucket. 2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs. 3. Seek the subscription back in time by 15 minutes to recover the acknowledged messages. 4. Start the Dataflow job in a secondary region.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #243 **Topic 1**

You are preparing data that your machine learning team will use to train a model using BigQueryML. They want to predict the price per square foot of real estate. The training data has a column for the price and a column for the number of square feet. Another feature column called 'feature1' contains null values due to missing data. You want to replace the nulls with zeros to keep more data points. Which query should you use?

A.

```
SELECT * EXCEPT(feature1),
  IFNULL(feature1, 0) AS feature1_cleaned
FROM training_data;
```

B.

```
SELECT * EXCEPT(price, square_feet),
  price/square_feet AS price_per_sqft
FROM training_data
WHERE feature1 IS NOT NULL;
```

C.

```
SELECT * EXCEPT(price, square_feet, feature1),
  price/square_feet AS price_per_sqft,
  IFNULL(feature1, 0) AS feature1_cleaned
FROM training_data;
```

Most Voted

D.

```
SELECT *
FROM training_data
WHERE feature1 IS NOT NULL;
```

Correct Answer: C

Community vote distribution



Comments

52ed0e5 Highly Voted 2 years ago

Selected Answer: A

Option A is the correct choice because it retains all the original columns and specifically addresses the issue of null values in ‘feature1’ by replacing them with zeros, without altering any other columns or performing unnecessary calculations. This makes the data ready for use in BigQueryML without losing any important information. Option C is not the best choice because it includes the EXCEPT clause for the price and square_feet columns, which would exclude these columns from the results. This is not desirable since you need these columns for the machine learning model to predict the price per square foot

datapassionate Highly Voted 2 years, 2 months ago

Selected Answer: C

Correct answer is C. It both replace NULL with 0 and pass price per square foot of real estate.

George_Zhu 2 years, 1 month ago

Option C isn't a good practice. What if any 0 value is contained in the column of squre_feet, then price / 0 will throw an exception. IF(IFNULL(squre_feet, 0) = 0, 0, price/squre_feet).

baimus 1 year, 5 months ago

I think the assumption here is that no houses are zero feet in size. If they are, that should be caught in preprocessing, which is outside the short scope of this question. If the answer isn't C, then it's A, which would mean the question is suggesting you need an ML model to calculate price per square for data where you already have both price and square feet as features. In that instance you clearly need to only divide one by the other. Those columns must be intended to be the target, or the whole question is nonsense.

desertlotus1211 1 year ago

The question is asking about Null values for the feature1 column, no other column with Null values.

SahandJ 4 months, 3 weeks ago

Selected Answer: A

You shouldn't remove features from the dataset based on an assumption of what the ML Engineer wants. Clean the column that's needed and allow the ML Engineer to do any further processing if he so desires.

noki_nho 8 months, 3 weeks ago

Selected Answer: C

"You are preparing data that your machine learning team will use to train a model using BigQueryML" this detail makes answer C more eligible than answer A

22c1725 10 months, 1 week ago

Selected Answer: C

I would go with "C" becuse of this: "Another feature" so that implies there is a first feaure which would be doin calculation to prdict the price.

22c1725 10 months, 1 week ago

Selected Answer: A

 Final Answer: A is the most correct answer based on what is explicitly required. Option C would only be correct if the question said something like: “You need to prepare data that includes the price_per_sqft target column...” But it doesn’t — it only states what the model will predict, not what you are to calculate.

MarcoPellegrino 1 year, 1 month ago

Selected Answer: C

It's the only one that: - computes the price per square foot of real estate. Note that "the training data has a column for the price and a column for the number of square feet" only. - fills NAs with zeros

LP_PDE 1 year, 2 months ago

Selected Answer: C

Not worded well but the best answer I would think would be C since it has price per square foot but I understand the argument for A.

AWSandeep 1 year, 3 months ago

Selected Answer: C

Let's step away from GCP for a minute. If price & square feet are already features, then there is no need for BigQuery ML to do any prediction. You'd want to predict price per square foot based on other fields like location, weather, etc. The first sentence in the question indicates that a machine learning team is preparing the data for model training. Therefore, C's query a fantastic preparation step. If this query for any other use case, then A would've been the answer.

Robbing_the_hood 1 year, 3 months ago

Selected Answer: C

To people saying you need price and square footage to predict price/sq. feet: you do not need an ML model then, you need a calculator. C is the correct answer because you want to predict price/sq. feet from the features EXCLUDING price and sq. footage.

ToiToi 1 year, 5 months ago

Selected Answer: C

Gemini told me C Here's why it's the best of the limited choices: Calculates price_per_sqft: It includes the calculation for the target variable your model needs. Handles Nulls: It uses IFNULL(feature1, 0) to replace nulls in feature1 with 0, similar to COALESCE. Most Comprehensive: While it excludes the original price, square_feet, and feature1 columns, it still retains any other columns that might be present in the training_data table.

cloud_rider 1 year, 4 months ago

C is wrong as it excludes price and square feet values, what will model use to train model?

Robbing_the_hood 1 year, 3 months ago

The features? Why would you train an ML model if you have price and sq. feet available?

SamuelTsch 1 year, 5 months ago

Selected Answer: C

it should be C.

baimus 1 year, 5 months ago

Selected Answer: C

This must be C, though the wording isn't great. If price and square foot are included in the data, they are either intended to be the target, in which case you need to create that target as per C, or if they are genuinely features, you DO NOT need a machine learning model. If you already know price and square feet, price per square foot is just price/ft2. You don't need ML to predict that, it's just a division. The only context this makes sense in is if they mean

"price and square foot are the target, and feature1 is the predictive feature", which means C is correct. The removing nulls from feature1 and the creation of price per square foot is C.

47767f9 1 year, 9 months ago

Selected Answer: C

Font Cloude 3.5 and GPT 4o, in theoy is better to keep the less amount of features, then price_per_sqft and feature1 cleaned is the best option

srinidutt 1 year, 11 months ago

EXCEPT means it won't select that column.

demoro86 2 years, 1 month ago

Selected Answer: A

C is not a valid answer. You are introducing a redundant variable, that could be valid, but removing from the dataset 2 variables that exactly influence in the predictions you are trying to make.

demoro86 2 years, 1 month ago

C is not a valid answer. You are introducing a redundant variable, that could be valid, but removing from the dataset 2 variables that exactly influence in the predictions you are trying to make.

baimus 1 year, 5 months ago

Just to clarify, they don't "influence" the prediction, they are in fact the target. The model needs to predict price per square foot. If you have price, and square foot, they are either 1) the prediction target price/squarefoot, or if not then you absolutely do not need a machine learning model, you just device price by square foot.

https://zhangxuetu.online

[202610]

Question #244 Topic 1

Different teams in your organization store customer and performance data in BigQuery. Each team needs to keep full control of their collected data, be able to query data within their projects, and be able to exchange their data with other teams. You need to implement an organization-wide solution, while minimizing operational tasks and costs. What should you do?

A. Ask each team to create authorized views of their data. Grant the biquery.jobUser role to each team.

B. Create a BigQuery scheduled query to replicate all customer data into team projects.

C. Ask each team to publish their data in Analytics Hub. Direct the other teams to subscribe to them. **Most Voted**

D. Enable each team to create materialized views of the data they need to access in their projects.

Correct Answer: C

Community vote distribution

C (100%)

Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: C

- Analytics Hub allows organizations to create and manage exchanges where producers can publish their data and consumers can discover and subscribe to data products. - Asking each team to publish their data in Analytics Hub and having other teams subscribe to them is a scalable and controlled way of sharing data. - It minimizes operational tasks because data doesn't need to be duplicated or manually managed after setup, and teams can maintain full control over their datasets.

CGS22 1 year, 5 months ago

Selected Answer: C

Why C is the best choice: Centralized Data Exchange: Analytics Hub provides a unified platform for data sharing across teams and organizations. It simplifies the process of publishing, discovering, and subscribing to datasets, reducing operational overhead. Data Ownership and Control: Each team retains full control over their data, deciding which datasets to publish and who can access them. This ensures data governance and security. Cross-Project Querying: Once a team subscribes to a dataset in Analytics Hub, they can query it directly from their own BigQuery project, enabling seamless data access without data replication. Cost Efficiency: Analytics Hub eliminates the need for data duplication or complex ETL processes, reducing storage and processing costs.

hanoverquay 1 year, 6 months ago

Selected Answer: C

vote C

JyoGCP 1 year, 7 months ago

Selected Answer: C

C. Analytics Hub

Matt_108 1 year, 8 months ago

Selected Answer: C

that's what analytics hub is designed for

chickenwingz 1 year, 9 months ago

Selected Answer: C

Analytics hub to reduce operational overhead of creating/maintaining views permissions etc

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #245 Topic 1

You are developing a model to identify the factors that lead to sales conversions for your customers. You have completed processing your data. You want to continue through the model development lifecycle. What should you do next?

A. Use your model to run predictions on fresh customer input data.

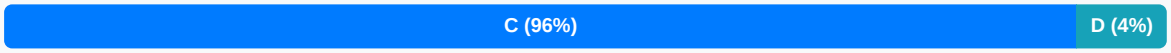
B. Monitor your model performance, and make any adjustments needed.

C. Delineate what data will be used for testing and what will be used for training the model. **Most Voted**

D. Test and evaluate your model on your curated data to determine how well the model performs.

Correct Answer: C

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

- Before you can train a model, you need to decide how to split your dataset.

Matt_108 **Highly Voted** 2 years, 2 months ago

Selected Answer: C

Option C - you've just concluded processing data, ending up with clean and prepared data for the model. Now you need to decide how to split the data for testing and for training. Only afterwards, you can train the model, evaluate it, fine tune it and, eventually, predict with it

desertlotus1211 1 year ago

Selected Answer: D

Anwser D is correct: the next step in the machine learning lifecycle is to evaluate the model Delineate test/train data should have been done BEFORE or during data processing

desertlotus1211 1 year ago

The machine learning life cycle typically involves planning, data preparation, model engineering, model evaluation, model deployment, and monitoring/maintenance.

meh_33 1 year, 7 months ago

Selected Answer: C

First ever time Exam Topic answers matching with users answer yoooo hoooooo. C

1919730 1 year, 6 months ago

Yoooo hoooooo.Yep !

JyoGCP 2 years, 1 month ago

Selected Answer: C

Option C

chickenwingz 2 years, 3 months ago

Selected Answer: C

Model doesn't seem to be trained yet

https://zhangxuetu.online

[202610]

Question #246 Topic 1

You have one BigQuery dataset which includes customers' street addresses. You want to retrieve all occurrences of street addresses from the dataset. What should you do?

A. Write a SQL query in BigQuery by using REGEXP_CONTAINS on all tables in your dataset to find rows where the word "street" appears.

B. Create a deep inspection job on each table in your dataset with Cloud Data Loss Prevention and create an inspection template that includes the STREET_ADDRESS infoType. **Most Voted**

C. Create a discovery scan configuration on your organization with Cloud Data Loss Prevention and create an inspection template that includes the STREET_ADDRESS infoType.

D. Create a de-identification job in Cloud Data Loss Prevention and use the masking transformation.

Correct Answer: B

Community vote distribution

B (100%)

Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: B

- Cloud Data Loss Prevention (Cloud DLP) provides powerful inspection capabilities for sensitive data, including predefined detectors for infoTypes such as STREET_ADDRESS. - By creating a deep inspection job for each table with the STREET_ADDRESS infoType, you can accurately identify and retrieve rows that contain street addresses.

josech 1 year, 4 months ago

Selected Answer: B

https://cloud.google.com/sensitive-data-protection/docs/learn-about-your-data#inspection

JyoGCP 1 year, 7 months ago

Selected Answer: B

Option B

AllenChen123 1 year, 8 months ago

Why not C? Discovery scan configuration can also help to identify risk/sensitivity fields.

datapassionate 1 year, 8 months ago

In the question we need to retrieve all occurances of street adresses from the dataset. In C you create discovery confiuration plan on whole organization. Its not needed.

mdell 1 year, 3 months ago

This. C scans EVERYTHING in the org when all we want is the dataset

Matt_108 1 year, 8 months ago

Selected Answer: B

Option B - you want to retrieve ALL occurrences within the dataset

scaenruy 1 year, 9 months ago

Selected Answer: B

B. Create a deep inspection job on each table in your dataset with Cloud Data Loss Prevention and create an inspection template that includes the STREET_ADDRESS infoType.

https://zhangxuetu.online

[202610]

Question #247 Topic 1

Your company operates in three domains: airlines, hotels, and ride-hailing services. Each domain has two teams: analytics and data science, which create data assets in BigQuery with the help of a central data platform team. However, as each domain is evolving rapidly, the central data platform team is becoming a bottleneck. This is causing delays in deriving insights from data, and resulting in stale data when pipelines are not kept up to date. You need to design a data mesh architecture by using Dataplex to eliminate the bottleneck. What should you do?

A. 1. Create one lake for each team. Inside each lake, create one zone for each domain.
2. Attach each of the BigQuery datasets created by the individual teams as assets to the respective zone.
3. Have the central data platform team manage all zones' data assets.

B. 1. Create one lake for each team. Inside each lake, create one zone for each domain.
2. Attach each of the BigQuery datasets created by the individual teams as assets to the respective zone.
3. Direct each domain to manage their own zone's data assets.

C. 1. Create one lake for each domain. Inside each lake, create one zone for each team.
2. Attach each of the BigQuery datasets created by the individual teams as assets to the respective zone.
3. Direct each domain to manage their own lake's data assets. Most Voted

D. 1. Create one lake for each domain. Inside each lake, create one zone for each team.
2. Attach each of the BigQuery datasets created by the individual teams as assets to the respective zone.
3. Have the central data platform team manage all lakes' data assets.

Correct Answer: C

Community vote distribution



Comments

raaad Highly Voted 1 year, 9 months ago

Selected Answer: C

- each domain should manage their own lake's data assets

AllenChen123 1 year, 8 months ago

Agree. https://cloud.google.com/dataplex/docs/introduction#a_domain-centric_data_mesh

hanoverquay 1 year, 6 months ago

Selected Answer: C

vote C

JyoGCP 1 year, 7 months ago

Selected Answer: C

Option C

Matt_108 1 year, 8 months ago

Selected Answer: C

Option C - create a lake for each domain, each team manages its own assets

task_7 1 year, 8 months ago

Selected Answer: B

Separate lakes for each team Zones within each lake dedicated to different domains

scaenruy 1 year, 9 months ago

Selected Answer: C

C. 1. Create one lake for each domain. Inside each lake, create one zone for each team. 2. Attach each of the BigQuery datasets created by the individual teams as assets to the respective zone. 3. Direct each domain to manage their own lake’s data assets.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #248 **Topic 1**

dataset.inventory_vm sample records:

Row	id	name	components.name	components.qty
1	vm02781	d-jp-kfk-02-02	vcpu	2
			memory	8
			boot_disk	10
			disk_1	50
2	vm11490	i-jp-kfk-02-07	vcpu	16
			memory	64
			boot_disk	10
			disk_1	200
3	vm18130	i-jp-kfk-02-08	vcpu	8
			memory	8
			boot_disk	10

You have an inventory of VM data stored in the BigQuery table. You want to prepare the data for regular reporting in the most cost-effective way. You need to exclude VM rows with fewer than 8 vCPU in your report. What should you do?

A. Create a view with a filter to drop rows with fewer than 8 vCPU, and use the UNNEST operator. **Most Voted**

B. Create a materialized view with a filter to drop rows with fewer than 8 vCPU, and use the WITH common table expression.

C. Create a view with a filter to drop rows with fewer than 8 vCPU, and use the WITH common table expression.

D. Use Dataflow to batch process and write the result to another BigQuery table.

Correct Answer: A

Community vote distribution

A (100%)

Comments

raaad Highly Voted 1 year, 9 months ago

Selected Answer: A

- The table structure shows that the vCPU data is stored in a nested field within the components column. - Using the UNNEST operator to flatten the nested field and apply the filter.

hanoverquay 1 year, 6 months ago

Selected Answer: A

option A

JyoGCP 1 year, 7 months ago

Selected Answer: A

Option A - UNNEST

Krauser59 1 year, 8 months ago

Selected Answer: A

A seems to be the correct answer because of the table structure and the UNNEST operator. However, i don’t understand why wouldn’t we chose a materialized view

Positron75 10 months, 2 weeks ago

Materialized views increase cost, which would go against the "most cost-effective way" part of the question.

Matt_108 1 year, 8 months ago

Selected Answer: A

Option A - The regular reporting doesn't justify a materialized view, since the frequency of access is not so high; a simple view would do the trick. Moreover, the vcpu data is in a nested field and requires Unnest.

scaenruy 1 year, 9 months ago

Selected Answer: A

A. Create a view with a filter to drop rows with fewer than 8 vCPU, and use the UNNEST operator.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #249 Topic 1

Your team is building a data lake platform on Google Cloud. As a part of the data foundation design, you are planning to store all the raw data in Cloud Storage. You are expecting to ingest approximately 25 GB of data a day and your billing department is worried about the increasing cost of storing old data. The current business requirements are:

• The old data can be deleted anytime.
• There is no predefined access pattern of the old data.
• The old data should be available instantly when accessed.
• There should not be any charges for data retrieval.

What should you do to optimize for cost?

A. Create the bucket with the Autoclass storage class feature. **Most Voted**

B. Create an Object Lifecycle Management policy to modify the storage class for data older than 30 days to nearline, 90 days to coldline, and 365 days to archive storage class. Delete old data as needed.

C. Create an Object Lifecycle Management policy to modify the storage class for data older than 30 days to coldline, 90 days to nearline, and 365 days to archive storage class. Delete old data as needed.

D. Create an Object Lifecycle Management policy to modify the storage class for data older than 30 days to nearline, 45 days to coldline, and 60 days to archive storage class. Delete old data as needed.

Correct Answer: A

Community vote distribution



Comments

Smakyel79 **Highly Voted** 2 years, 2 months ago

Selected Answer: A

https://cloud.google.com/storage/docs/autoclass

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: A

- Autoclass automatically moves objects between storage classes without impacting performance or availability, nor incurring retrieval costs. - It continuously optimizes storage costs based on access patterns without the need to set specific lifecycle management policies.

SamuelTsch 1 year, 5 months ago

Selected Answer: A

From the documentation https://cloud.google.com/storage/docs/autoclass

hussain.sain 1 year, 8 months ago

Selected Answer: B

the question clearly specifies there should not be any retrieval charges. so enabling autoclass is not recommended because we have to pay one time fees while retrieving the data. and usually soft delete is enable.

nadavw 1 year, 7 months ago

A one-time pay isn't considered a retrieval charge. A is correct

CGS22 1 year, 11 months ago

Selected Answer: B

Why B is the best choice: Cost Optimization: This option leverages Cloud Storage's different storage classes to significantly reduce costs for storing older data. Nearline, coldline, and archive storage classes are progressively cheaper than the standard storage class, with trade-offs in availability and retrieval times. Meets Requirements: Old data deletion: You can manually delete old data whenever needed, fulfilling the first requirement. No predefined access pattern: The policy automatically transitions data to cheaper storage classes based on age, regardless of access patterns. Instant availability: Nearline storage provides immediate access to data, meeting the third requirement. No retrieval charges: While there are retrieval charges for coldline and archive storage, nearline storage has no retrieval fees, satisfying the fourth requirement.

Sofiia98 2 years, 2 months ago

Selected Answer: A

For sure A, read the documentation

GCP001 2 years, 2 months ago

Selected Answer: A

autoclass is the correct way to handle all business cases

therealsohail 2 years, 2 months ago

Selected Answer: B

Create an Object Lifecycle Management policy to modify the storage class for data older than 30 days to nearline, 90 days to coldline, and 365 days to archive storage class. Delete old data as needed.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #250 Topic 1

Your company's data platform ingests CSV file dumps of booking and user profile data from upstream sources into Cloud Storage. The data analyst team wants to join these datasets on the email field available in both the datasets to perform analysis. However, personally identifiable information (PII) should not be accessible to the analysts. You need to de-identify the email field in both the datasets before loading them into BigQuery for analysts. What should you do?

A. 1. Create a pipeline to de-identify the email field by using recordTransformations in Cloud Data Loss Prevention (Cloud DLP) with masking as the de-identification transformations type.
2. Load the booking and user profile data into a BigQuery table.

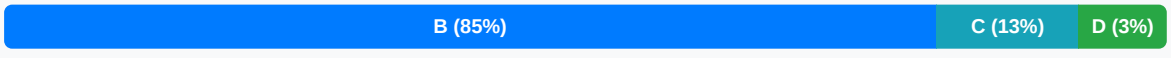
B. 1. Create a pipeline to de-identify the email field by using recordTransformations in Cloud DLP with format-preserving encryption with FFX as the de-identification transformation type.
2. Load the booking and user profile data into a BigQuery table. **Most Voted**

C. 1. Load the CSV files from Cloud Storage into a BigQuery table, and enable dynamic data masking.
2. Create a policy tag with the email mask as the data masking rule.
3. Assign the policy to the email field in both tables. A
4. Assign the Identity and Access Management bigquerydatapolicy.maskedReader role for the BigQuery tables to the analysts.

D. 1. Load the CSV files from Cloud Storage into a BigQuery table, and enable dynamic data masking.
2. Create a policy tag with the default masking value as the data masking rule.
3. Assign the policy to the email field in both tables.
4. Assign the Identity and Access Management bigquerydatapolicy.maskedReader role for the BigQuery tables to the analysts

Correct Answer: B

Community vote distribution



Comments

lipa31 **Highly Voted** 2 years, 2 months ago

Selected Answer: B

Format-preserving encryption (FPE) with FFX in Cloud DLP is a strong choice for de-identifying PII like email addresses. FPE maintains the format of the data and ensures that the same input results in the same encrypted output consistently. This means the email fields in both datasets can be encrypted to the same value, allowing for accurate joins in BigQuery while keeping the actual email addresses hidden.

Smakyel79 Highly Voted 2 years, 2 months ago

As it states "You need to de-identify the email field in both the datasets before loading them into BigQuery for analysts" data masking should not be an option as the data would stored unmasked in BigQuery?

Anudeep58 1 year, 9 months ago

Selected Answer: B

Option A: Masking: Simple masking might not preserve the uniqueness and joinability of the email field, making it difficult to perform accurate joins between datasets. Option C and D: Dynamic Data Masking: These options involve masking the email field dynamically within BigQuery, which does not address the requirement to de-identify data before loading into BigQuery. Additionally, dynamic masking does not prevent access to the actual email data before it is loaded into BigQuery, potentially exposing PII during the data ingestion process.

chrissamharris 1 year, 10 months ago

Selected Answer: B

format-preserving encryption with FFX is required as the analysts want to perform JOINs

JyoGCP 2 years, 1 month ago

Selected Answer: B

Option B https://cloud.google.com/sensitive-data-protection/docs/pseudonymization

ML6 2 years, 1 month ago

Selected Answer: B

A) masking = replace with a surrogate character like # or * = output not unique, so cannot apply joins C and D) question specifies to de-identify BEFORE loading into BQ, whereas these options perform dynamic masking IN BigQuery. Therefore, only valid option is B.

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C. The need is to just mask the data to Analyst, without modifying the underlying data. Moreover, it's stored on 2 separate tables and the analysts need to be able to perform joins based on the masked data. Dynamic masking is the right module and the right masking rule is email mask (https://cloud.google.com/bigquery/docs/column-data-masking-intro#masking_options) which guarantees the join capabilities join

task_7 2 years, 2 months ago

Selected Answer: B

A wouldn't preserve the email format C&D maskedReader roles still grant access to the underlying values. the only option is B

alfguemat 2 years, 2 months ago

I dont't know why preserve email format is necessary to perform the join. A could be valid.

dduenas 2 years, 1 month ago

masking only replace by specific characters, doing the field not unique and not ready for joins.

Sofiia98 2 years, 2 months ago

Selected Answer: C

I will go for C, because there is a separate type of masking for emails, so whe to use the dafault? https://cloud.google.com/bigquery/docs/column-data-masking-intro#masking_options

GCP001 2 years, 2 months ago

Selected Answer: C

data masking with BQ is correct with email masking rule. Ref - https://cloud.google.com/bigquery/docs/column-data-masking-intro

tibuenoc 2 years, 1 month ago

should be correct if they want to access tables and it's not valid for datasets

Jordan18 2 years, 2 months ago

why not B?

raaad 2 years, 2 months ago

Selected Answer: C

- The reason option C works well is that dynamic data masking in BigQuery allows the underlying data to remain unaltered (thus preserving the ability to join on this field), while also preventing analysts from viewing the actual PII. - The analysts can query and join the data as needed for their analysis, but when they access the data, the email field will be masked according to the policy tag, and they will only see the masked version.

scaenruy 2 years, 2 months ago

Selected Answer: D

D. 1. Load the CSV files from Cloud Storage into a BigQuery table, and enable dynamic data masking. 2. Create a policy tag with the default masking value as the data masking rule. 3. Assign the policy to the email field in both tables. 4. Assign the Identity and Access Management bigquerydatapolicy.maskedReader role for the BigQuery tables to the analysts

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #251 Topic 1

You have important legal hold documents in a Cloud Storage bucket. You need to ensure that these documents are not deleted or modified. What should you do?

A. Set a retention policy. Lock the retention policy. **Most Voted**

B. Set a retention policy. Set the default storage class to Archive for long-term digital preservation.

C. Enable the Object Versioning feature. Add a lifecycle rule.

D. Enable the Object Versioning feature. Create a copy in a bucket in a different region.

Correct Answer: A

Community vote distribution

A (100%)

Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: A

- Setting a retention policy on a Cloud Storage bucket prevents objects from being deleted for the duration of the retention period. - Locking the policy makes it immutable, meaning that the retention period cannot be reduced or removed, thus ensuring that the documents cannot be deleted or overwritten until the retention period expires.

AllenChen123 1 year, 8 months ago

Agree. https://cloud.google.com/storage/docs/bucket-lock#overview

JyoGCP 1 year, 7 months ago

Selected Answer: A

Option A

Matt_108 1 year, 8 months ago

Selected Answer: A

Option A - set retention policy to prevent deletion, lock it to make it immutable (not subject to edits)

scaenruy 1 year, 9 months ago

Selected Answer: A

A. Set a retention policy. Lock the retention policy.

https://zhangxuetu.online

[202610]

Question #252 Topic 1

You are designing a data warehouse in BigQuery to analyze sales data for a telecommunication service provider. You need to create a data model for customers, products, and subscriptions. All customers, products, and subscriptions can be updated monthly, but you must maintain a historical record of all data. You plan to use the visualization layer for current and historical reporting. You need to ensure that the data model is simple, easy-to-use, and cost-effective. What should you do?

A. Create a normalized model with tables for each entity. Use snapshots before updates to track historical data.

B. Create a normalized model with tables for each entity. Keep all input files in a Cloud Storage bucket to track historical data.

C. Create a denormalized model with nested and repeated fields. Update the table and use snapshots to track historical data.

D. Create a denormalized, append-only model with nested and repeated fields. Use the ingestion timestamp to track historical data. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

raaad **Highly Voted** 1 year, 9 months ago

- A denormalized, append-only model simplifies query complexity by eliminating the need for joins. - Adding data with an ingestion timestamp allows for easy retrieval of both current and historical states. - Instead of updating records, new records are appended, which maintains historical information without the need to create separate snapshots.

JyoGCP 1 year, 7 months ago

Selected Answer: D

Option D

JimmyBK 1 year, 8 months ago

Selected Answer: D

Straight forward, good for costs

Sofiia98 1 year, 8 months ago

Selected Answer: D

D looks logical

GCP001 1 year, 8 months ago

Selected Answer: D

Easy, cost effective and no cpmpexity

scaenruy 1 year, 9 months ago

Selected Answer: D

D. Create a denormalized, append-only model with nested and repeated fields. Use the ingestion timestamp to track historical data.

https://zhangxuetu.online

[202610]

Question #253 Topic 1

You are deploying a batch pipeline in Dataflow. This pipeline reads data from Cloud Storage, transforms the data, and then writes the data into BigQuery. The security team has enabled an organizational constraint in Google Cloud, requiring all Compute Engine instances to use only internal IP addresses and no external IP addresses. What should you do?

A. Ensure that your workers have network tags to access Cloud Storage and BigQuery. Use Dataflow with only internal IP addresses.

B. Ensure that the firewall rules allow access to Cloud Storage and BigQuery. Use Dataflow with only internal IPs.

C. Create a VPC Service Controls perimeter that contains the VPC network and add Dataflow, Cloud Storage, and BigQuery as allowed services in the perimeter. Use Dataflow with only internal IP addresses.

D. Ensure that Private Google Access is enabled in the subnetwork. Use Dataflow with only internal IP addresses. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: D

- Private Google Access for services allows VM instances with only internal IP addresses in a VPC network or on-premises networks (via Cloud VPN or Cloud Interconnect) to reach Google APIs and services. - When you launch a Dataflow job, you can specify that it should use worker instances without external IP addresses if Private Google Access is enabled on the subnetwork where these instances are launched. - This way, your Dataflow workers will be able to access Cloud Storage and BigQuery without violating the organizational constraint of no external IPs.

Jordan18 1 year, 8 months ago

why not C?

GCP001 1 year, 8 months ago

Even if you create VPC service control, your dataflow worker will run on google compute engine instances with private ips only after policy enforcement. Without external IP addresses, you can still perform administrative and monitoring tasks. You can access your workers by using SSH through the options listed in the preceding list. However, the pipeline cannot access the internet, and internet hosts cannot access your Dataflow workers.

GCP001 1 year, 8 months ago

ref - https://cloud.google.com/dataflow/docs/guides/routes-firewall

BIGQUERY_ALT_ALT 1 year, 8 months ago

VPC Service Controls are typically used to define and enforce security perimeters around APIs and services, restricting their access to a specified set of Google Cloud projects. In this scenario, the security constraint is focused on Compute Engine instances used by Dataflow, and VPC Service Controls might be considered a bit heavy-handed for just addressing the internal IP address requirement.

Lestrang 1 year, 3 months ago

Selected Answer: D

No way it is C. Like the use case for Google VPC Service Controls perimeter is not to establish secure connectivity on its own but rather to control connectivity, like allowing vms within x premise to access, and blocking vms outside premise even if in same VPC from access. D on the other hand is completely sensical.

Moss2011 1 year, 7 months ago

Selected Answer: C

According to this documentation: https://cloud.google.com/vpc-service-controls/docs/overview I think the correct answer is C. Take into account the phrase "organizational constraint" and the VPC Service Control allow you to do that.

Tryolabs 1 year, 7 months ago

Selected Answer: D

https://cloud.google.com/vpc/docs/private-google-access "VM instances that only have internal IP addresses (no external IP addresses) can use Private Google Access. They can reach the external IP addresses of Google APIs and services."

pandeyspecial 1 year, 8 months ago

Selected Answer: C

It should be C

Matt_108 1 year, 8 months ago

Selected Answer: C

Option D, as GCP001 said

Matt_108 1 year, 8 months ago

Missclicked the answer <.<

GCP001 1 year, 8 months ago

Selected Answer: D

https://cloud.google.com/dataflow/docs/guides/routes-firewall

scaenruy 1 year, 9 months ago

Selected Answer: C

C. Create a VPC Service Controls perimeter that contains the VPC network and add Dataflow, Cloud Storage, and BigQuery as allowed services in the perimeter. Use Dataflow with only internal IP addresses.

BIGQUERY_ALT_ALT 1 year, 8 months ago

C is wrong. Option D is simple and straight forward. VPC Service Controls are typically used to define and enforce security perimeters around APIs and services, restricting their access to a specified set of Google Cloud projects. In this scenario, the security constraint is focused on Compute Engine instances used by Dataflow, and VPC Service Controls might be considered a bit heavy-handed for just addressing the internal IP address requirement.

https://zhangxuetu.online

[202610]

Question #254 Topic 1

You are running a Dataflow streaming pipeline, with Streaming Engine and Horizontal Autoscaling enabled. You have set the maximum number of workers to 1000. The input of your pipeline is Pub/Sub messages with notifications from Cloud Storage. One of the pipeline transforms reads CSV files and emits an element for every CSV line. The job performance is low, the pipeline is using only 10 workers, and you notice that the autoscaler is not spinning up additional workers. What should you do to improve performance?

A. Enable Vertical Autoscaling to let the pipeline use larger workers.

B. Change the pipeline code, and introduce a Reshuffle step to prevent fusion. **Most Voted**

C. Update the job to increase the maximum number of workers.

D. Use Dataflow Prime, and enable Right Fitting to increase the worker resources.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- Fusion optimization in Dataflow can lead to steps being "fused" together, which can sometimes hinder parallelization. - Introducing a Reshuffle step can prevent fusion and force the distribution of work across more workers. - This can be an effective way to improve parallelism and potentially trigger the autoscaler to increase the number of workers.

meh_33 1 year, 7 months ago

Selected Answer: B

https://cloud.google.com/dataflow/docs/pipeline-lifecycle#prevent_fusion

Lestrang 1 year, 9 months ago

Selected Answer: C

Right fitting is for declaration, declaring the correct resources will not help. Reshuffling step is what can prevent fusion which can lead to unused workers.

ML6 2 years, 1 month ago

Selected Answer: B

Fusion occurs when multiple transformations are fused into a single stage, which can limit parallelism and hinder performance, especially in streaming pipelines. By introducing a Reshuffle step, you break fusion and allow for better parallelism.

srivastavas08 2 years, 1 month ago

https://cloud.google.com/dataflow/docs/guides/right-fitting

GCP001 2 years, 2 months ago

Selected Answer: B

Problem is performnace and not using all workers properly, https://cloud.google.com/dataflow/docs/pipeline-lifecycle#fusion_optimization

scaenruy 2 years, 2 months ago

Selected Answer: D

D. Use Dataflow Prime, and enable Right Fitting to increase the worker resources.

judy_data 6 months, 3 weeks ago

Right fitting is used in batch pipelines and not streaming https://cloud.google.com/dataflow/docs/guides/enable-dataflow-prime?hl=en#features

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #255 Topic 1

You have an Oracle database deployed in a VM as part of a Virtual Private Cloud (VPC) network. You want to replicate and continuously synchronize 50 tables to BigQuery. You want to minimize the need to manage infrastructure. What should you do?

A. Deploy Apache Kafka in the same VPC network, use Kafka Connect Oracle Change Data Capture (CDC), and Dataflow to stream the Kafka topic to BigQuery.

B. Create a Pub/Sub subscription to write to BigQuery directly. Deploy the Debezium Oracle connector to capture changes in the Oracle database, and sink to the Pub/Sub topic.

C. Deploy Apache Kafka in the same VPC network, use Kafka Connect Oracle change data capture (CDC), and the Kafka Connect Google BigQuery Sink Connector.

D. Create a Datastream service from Oracle to BigQuery, use a private connectivity configuration to the same VPC network, and a connection profile to BigQuery. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: D

- Datastream is a serverless and easy-to-use change data capture (CDC) and replication service. - You would create a Datastream service that sources from your Oracle database and targets BigQuery, with private connectivity configuration to the same VPC. - This option is designed to minimize the need to manage infrastructure and is a fully managed service.

Zek 7 months, 2 weeks ago

Selected Answer: D

https://cloud.google.com/data-fusion/docs/tutorials/replicating-data/oracle-to-bigquery This tutorial shows you how to deploy a job that continuously replicates changed data from an Oracle database to a BigQuery dataset, using Cloud Data Fusion Replication. This feature is powered by Datastream.

JyoGCP 1 year, 7 months ago

Selected Answer: D

D. Datastream

scaenruy 1 year, 9 months ago

Selected Answer: D

D. Create a Datastream service from Oracle to BigQuery, use a private connectivity configuration to the same VPC network, and a connection profile to BigQuery.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #256 Topic 1

You are deploying an Apache Airflow directed acyclic graph (DAG) in a Cloud Composer 2 instance. You have incoming files in a Cloud Storage bucket that the DAG processes, one file at a time. The Cloud Composer instance is deployed in a subnetwork with no Internet access. Instead of running the DAG based on a schedule, you want to run the DAG in a reactive way every time a new file is received. What should you do?

A. 1. Enable Private Google Access in the subnetwork, and set up Cloud Storage notifications to a Pub/Sub topic.
2. Create a push subscription that points to the web server URL.

B. 1. Enable the Cloud Composer API, and set up Cloud Storage notifications to trigger a Cloud Function.
2. Write a Cloud Function instance to call the DAG by using the Cloud Composer API and the web server URL.
3. Use VPC Serverless Access to reach the web server URL.

C. 1. Enable the Airflow REST API, and set up Cloud Storage notifications to trigger a Cloud Function instance.
2. Create a Private Service Connect (PSC) endpoint.
3. Write a Cloud Function that connects to the Cloud Composer cluster through the PSC endpoint. **Most Voted**

D. 1. Enable the Airflow REST API, and set up Cloud Storage notifications to trigger a Cloud Function instance.
2. Write a Cloud Function instance to call the DAG by using the Airflow REST API and the web server URL.
3. Use VPC Serverless Access to reach the web server URL.

Correct Answer: C

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

- Enable Airflow REST API: In Cloud Composer, enable the "Airflow web server" option. - Set Up Cloud Storage Notifications: Create a notification for new files, routing to a Cloud Function. - Create PSC Endpoint: Establish a PSC endpoint for Cloud Composer. - Write Cloud Function: Code the function to use the Airflow REST API (via PSC endpoint) to trigger the DAG. ======== Why not Option D - Using the web server URL directly wouldn't work without internet access or a direct path to the web server.

AllenChen123 2 years, 2 months ago

Why not B, use Cloud Composer API

STEVE_PEGLEG Highly Voted 1 year, 7 months ago

Selected Answer: A

This is the guidance how to use method in A: https://cloud.google.com/composer/docs/composer-2/triggering-gcf-pubsub "In this specific example, you create a Cloud Function and deploy two DAGs. The first DAG pulls Pub/Sub messages and triggers the second DAG according to the Pub/Sub message content." For C & D, this guidance says it can't be done when you have Private or VPS Service Controls set up: https://cloud.google.com/composer/docs/composer-2/triggering-with-gcf#check_your_environments_networking_configuration "This solution does not work in Private IP and VPC Service Controls configurations because it is not possible to configure connectivity from Cloud Functions to the Airflow web server in these configurations."

hola_prep 1 month, 1 week ago

Selected Answer: D

D. 1. Enable the Airflow REST API, and set up Cloud Storage notifications to trigger a Cloud Function instance.Write a Cloud Function instance to call the DAG by using the Airflow REST API and the web server URL.Use VPC Serverless Access to reach the web server URL

Zek 7 months, 1 week ago

Selected Answer: C

Accessing the Airflow REST API on Cloud Composer without internet access typically involves configuring a Private IP environment and utilizing Private Service Connect. https://cloud.google.com/composer/docs/composer-2/private-ip-environments#private_ip_environment

56d02cd 9 months ago

Selected Answer: B

I don't think the PSC endpoint is really needed. The cloud function can connect to the Airflow webserver through VPC serverless access, which is easier to configure than a PSC endpoint.

aditya_ali 11 months ago

Selected Answer: C

PSC is the only secure way to reach the Airflow REST API privately from a serverless service in a VPC-restricted Cloud Composer environment. Therefore, Option C provides the most secure and functional architecture.

Augustax 1 year, 1 month ago

Selected Answer: B

Option B is the only viable solution because: It uses the Cloud Composer API, which is compatible with Private IP configurations. It leverages VPC Serverless Access to allow Cloud Functions to securely access the Airflow web server within the subnetwork. It avoids the limitations of the Airflow REST API in Private IP environments.

baimus 1 year, 5 months ago

Selected Answer: A

This is A, as steve_pegleg says, there is no way to connect the cloud function to the Airflow instance, without first enabling private access. The pubsub pattern makes sense in this context.

josech 1 year, 10 months ago

Selected Answer: A

C is not correct because "this solution does not work in Private IP and VPC Service Controls configurations because it is not possible to configure connectivity from Cloud Functions to the Airflow web server in these configurations". https://cloud.google.com/composer/docs/how-to/using/triggering-with-gcf The correct answer is A using Pub/Sub https://cloud.google.com/composer/docs/composer-2/triggering-gcf-pubsub

chrissamharris 2 years ago

Selected Answer: D

Why not Option C? C involves creating a Private Service Connect (PSC) endpoint, which, while viable for creating private connections to Google services, adds complexity and might not be required when simpler solutions like VPC Serverless Access (as in Option D) can suffice.

chrissamharris 2 years ago

https://cloud.google.com/vpc/docs/serverless-vpc-access: Serverless VPC Access makes it possible for you to connect directly to your Virtual Private Cloud (VPC) network from serverless environments such as Cloud Run, App Engine, or Cloud Functions

d11379b 2 years ago

Selected Answer: D

The answer should be D Serverless VPC Access makes it possible for you to connect directly to your Virtual Private Cloud (VPC) network from serverless environments such as Cloud Run, App Engine, or Cloud Functions. Configuring Serverless VPC Access allows your serverless environment to send requests to your VPC network by using internal DNS and internal IP addresses (as defined by RFC 1918 and RFC 6598). The responses to these requests also use your internal network. You can use Serverless VPC Access to access Compute Engine VM instances, Memorystore instances, and any other resources with internal DNS or internal IP address. (Reference: https://cloud.google.com/vpc/docs/serverless-vpc-access) When you use Airflow Rest API to tigger the job, the url is based on the private IP address of Cloud Composer Instance, so you need to use Serverless VPC Access for it.

d11379b 2 years ago

Why not C: The reference here (https://cloud.google.com/vpc/docs/private-service-connect#published-services) limits the available use cases: Private Service Connect supports access to the following types of managed services: Published VPC-hosted services, which include the following: Google published services, such as Apigee or the GKE control plane Third-party published services provided by Private Service Connect partners Intra-organization published services, where the consumer and producer might be two different VPC networks within the same company Google APIs, such as Cloud Storage or BigQuery Unfortunately your airflow Rest API is not published as a service in the list, so you can not use it This is also one of the reasons why you should reject A

d11379b 2 years ago

B is not appropriate while Cloud Composer API can really execute Airflow command，but It’s not via web server Url to run a DAG in this case, and I doubt if it is really possible

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C, raaad explained well why

scaenruy 2 years, 2 months ago

Selected Answer: C

C. 1. Enable the Airflow REST API, and set up Cloud Storage notifications to trigger a Cloud Function instance. 2. Create a Private Service Connect (PSC) endpoint. 3. Write a Cloud Function that connects to the Cloud Composer cluster through the PSC endpoint.

https://zhangxuetu.online

[202610]

Question #257 Topic 1

You are planning to use Cloud Storage as part of your data lake solution. The Cloud Storage bucket will contain objects ingested from external systems. Each object will be ingested once, and the access patterns of individual objects will be random. You want to minimize the cost of storing and retrieving these objects. You want to ensure that any cost optimization efforts are transparent to the users and applications. What should you do?

A. Create a Cloud Storage bucket with Autoclass enabled. **Most Voted**

B. Create a Cloud Storage bucket with an Object Lifecycle Management policy to transition objects from Standard to Coldline storage class if an object age reaches 30 days.

C. Create a Cloud Storage bucket with an Object Lifecycle Management policy to transition objects from Standard to Coldline storage class if an object is not live.

D. Create two Cloud Storage buckets. Use the Standard storage class for the first bucket, and use the Coldline storage class for the second bucket. Migrate objects from the first bucket to the second bucket after 30 days.

Correct Answer: A

Community vote distribution

A (100%)

Comments

desertlotus1211 1 year ago

Selected Answer: A

By chance is this a repeat question?

iooj 1 year, 8 months ago

Selected Answer: A

Thanks to you guys, I found out about this feature :D The feature was released on November 3, 2023. Note that enabling Autoclass on an existing bucket incurs additional charges.

JyoGCP 2 years, 1 month ago

Selected Answer: A

A. Autoclass

Matt_108 2 years, 2 months ago

Selected Answer: A

Option A

raaad 2 years, 2 months ago

Selected Answer: A

- Autoclass automatically analyzes access patterns of objects and automatically transitions them to the most cost-effective storage class within Standard, Nearline, Coldline, or Archive. - This eliminates the need for manual intervention or setting specific age thresholds. - No user or application interaction is required, ensuring transparency.

scaenruy 2 years, 2 months ago

Selected Answer: A

A. Create a Cloud Storage bucket with Autoclass enabled.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #258 Topic 1

You have several different file type data sources, such as Apache Parquet and CSV. You want to store the data in Cloud Storage. You need to set up an object sink for your data that allows you to use your own encryption keys. You want to use a GUI-based solution. What should you do?

A. Use Storage Transfer Service to move files into Cloud Storage.

B. Use Cloud Data Fusion to move files into Cloud Storage. **Most Voted**

C. Use Dataflow to move files into Cloud Storage.

D. Use BigQuery Data Transfer Service to move files into BigQuery.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- Cloud Data Fusion is a fully managed, code-free, GUI-based data integration service that allows you to visually connect, transform, and move data between various sources and sinks. - It supports various file formats and can write to Cloud Storage. - You can configure it to use Customer-Managed Encryption Keys (CMEK) for the buckets where it writes data.

AllenChen123 2 years, 2 months ago

Agree. https://cloud.google.com/data-fusion/docs/how-to/customer-managed-encryption-keys#create-instance

Helinia **Highly Voted** 2 years, 2 months ago

Selected Answer: B

Even though storage transfer service can be used in GUI, it does not support CMEK which is required in this question. "Storage Transfer Service does not encrypt data on your behalf, such as in customer-managed encryption keys (CMEK). We only encrypt data in transit." Ref: https://cloud.google.com/storage-transfer/docs/on-prem-security

aaaaaaaasdasdasfs 11 months, 1 week ago

Selected Answer: A

A. Use Storage Transfer Service • □ GUI-based: Yes, can be set up in the Google Cloud Console. • □ Can move data from on-premises, AWS, or even between GCS buckets. • □ Works with different file types, including CSV, Parquet. • □ Supports writing into buckets protected with CMEK. • □ Best suited when you’re moving or syncing raw files into Cloud Storage

vishavpreet 1 year, 1 month ago

Selected Answer: B

Cloud Data Fusion is a fully-managed, cloud native, enterprise data integration service for quickly building and managing data pipelines. Graphically, no coding solution.

hussain.sain 1 year, 3 months ago

Selected Answer: B

B is answer as requirement is for GUI and sink

baimus 1 year, 5 months ago

I acknolwedge that the question wants the answer to be B, so I'm calling it as B. I don't like this though, as can't we just create a bucket with a CMEK upfront and then use the storage transfer service? It would be easier and cheaper, and achieve the same thing. The language of "sink" strongly suggests to me they intend this to be B though, as datafusion uses that terminology, and the CMEK thing is probably indicating that datafusion can encrypt for you with CMEK.

hanoverquay 2 years ago

Selected Answer: B

option 8

ashdam 2 years, 1 month ago

Selected Answer: B

B just because tranfer service does not support CMEK A: * GUI + encryption but no CMEK. B: * Its GUI ETL + support CMEK but not sure why you need an ETL tool for transfering something once? (No scheduling or event-driven trigger is mentioned)

casadocc 2 years, 1 month ago

B if data fusion creates the bucket. We could create the bucket and associate, in this case A is better.

task_7 2 years, 2 months ago

Selected Answer: A

A. Use Storage Transfer Service to move files into Cloud Storage. move files into Cloud Storage should be Storage Transfer Service Cloud Data Fusion is like using a tank to kill an ant

scaenruy 2 years, 2 months ago

Selected Answer: B

B. Use Cloud Data Fusion to move files into Cloud Storage.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #259 Topic 1

Your business users need a way to clean and prepare data before using the data for analysis. Your business users are less technically savvy and prefer to work with graphical user interfaces to define their transformations. After the data has been transformed, the business users want to perform their analysis directly in a spreadsheet. You need to recommend a solution that they can use. What should you do?

A. Use Dataprep to clean the data, and write the results to BigQuery. Analyze the data by using Connected Sheets. **Most Voted**

B. Use Dataprep to clean the data, and write the results to BigQuery. Analyze the data by using Looker Studio.

C. Use Dataflow to clean the data, and write the results to BigQuery. Analyze the data by using Connected Sheets.

D. Use Dataflow to clean the data, and write the results to BigQuery. Analyze the data by using Looker Studio.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Sofiia98 Highly Voted 1 year, 8 months ago

Selected Answer: A

If only all the questions were like this...

raaad Highly Voted 1 year, 9 months ago

Selected Answer: A

- Allow business users to perform their analysis in a familiar spreadsheet interface via Connected Sheets.

josech 1 year, 4 months ago

Selected Answer: A

https://cloud.google.com/bigquery/docs/connected-sheets https://cloud.google.com/dataprep

hanoverquay 1 year, 6 months ago

Selected Answer: A

vote A

Matt_108 1 year, 8 months ago

Selected Answer: A

Clearly option A

scaenruy 1 year, 9 months ago

Selected Answer: A

A. Use Dataprep to clean the data, and write the results to BigQuery. Analyze the data by using Connected Sheets.

https://zhangxuetu.online

[202610]

Question #260 Topic 1

You have two projects where you run BigQuery jobs:
• One project runs production jobs that have strict completion time SLAs. These are high priority jobs that must have the required compute resources available when needed. These jobs generally never go below a 300 slot utilization, but occasionally spike up an additional 500 slots.
• The other project is for users to run ad-hoc analytical queries. This project generally never uses more than 200 slots at a time. You want these ad-hoc queries to be billed based on how much data users scan rather than by slot capacity.

You need to ensure that both projects have the appropriate compute resources available. What should you do?

A. Create a single Enterprise Edition reservation for both projects. Set a baseline of 300 slots. Enable autoscaling up to 700 slots.

B. Create two reservations, one for each of the projects. For the SLA project, use an Enterprise Edition with a baseline of 300 slots and enable autoscaling up to 500 slots. For the ad-hoc project, configure on-demand billing. Most Voted

C. Create two Enterprise Edition reservations, one for each of the projects. For the SLA project, set a baseline of 300 slots and enable autoscaling up to 500 slots. For the ad-hoc project, set a reservation baseline of 0 slots and set the ignore idle slots flag to False.

D. Create two Enterprise Edition reservations, one for each of the projects. For the SLA project, set a baseline of 800 slots. For the ad-hoc project, enable autoscaling up to 200 slots.

Correct Answer: B

Community vote distribution



Comments

raaad Highly Voted 2 years, 2 months ago

Selected Answer: B

- The SLA project gets a dedicated reservation with autoscaling to handle spikes, ensuring it meets its strict completion time SLAs. - The ad-hoc project uses on-demand billing, which means it will be billed based on the amount of data scanned rather than slot capacity, fitting the billing preference for ad-hoc queries.

ce9e395 2 years ago

Critical jobs can spike up to 800 slots, making option B wrong

barrru 2 years ago

in this context, "enable autoscaling up to 500 slots" means that the system can add up to 500 slots beyond the baseline

chrissamharris 2 years ago

I dont think thats correct. "up to 500 slots" means the maximum limit is 500 slots - it doesnt specify autoscaling an additional 500 slots.

JyoGCP Highly Voted 2 years, 1 month ago

Selected Answer: B

Option B. Not D because "In Project-2, ad-hoc queries need to be billed based on how much data users scan rather than by slot capacity."

22c1725 10 months ago

Selected Answer: B

Go With (B)

Abizi 1 year ago

Selected Answer: B

answer B, because of the pay as you go for the Adhoc projet

ToiToi 1 year, 5 months ago

Selected Answer: B

100% B I work with BQ on a daily basis, did the transition from falt rate to editions last year, have been configuring this for so many customers. Billing for data analysed rather than slots is on demand - so no other option other than B makes sense.

SamuelTsch 1 year, 5 months ago

Selected Answer: B

https://cloud.google.com/bigquery/docs/slots-autoscaling-intro#using_reservations_with_baseline_and_autoscaling_slots says clearly, "Autoscaling slots are only added after all of the baseline slots (and idle slots if applicable) are consumed."

CGS22 1 year, 11 months ago

Selected Answer: B

Separate Reservations: This approach provides tailored resource allocation and billing models to match the distinct needs of each project. SLA Project Reservation: Enterprise Edition: Guarantees consistent slot availability for your production jobs. Baseline of 300 slots: Ensures resources are always available to meet your core usage at a predictable cost. Autoscaling up to 500 slots: Accommodates bursts in workload while controlling costs. Ad-hoc Project On-demand: On-demand billing: Charges based on data scanned, ideal for unpredictable and variable query patterns by your ad-hoc users.

chrissamharris 2 years ago

Selected Answer: D

Note, Option A states autoscale "up to" (not an additional) 500 slots, whereas the requirement is 800 slots. Making option D the only viable option.

chrissamharris 2 years ago

Scratch this - Option B: https://cloud.google.com/bigquery/docs/slots-autoscaling-intro#using_reservations_with_baseline_and_autoscaling_slots Baseline Slots and AutoScaling Slots are treated as two different entities in the documentation. Therefore B is right despite the horrific wording of the answers.

ToiToi 1 year, 5 months ago

"You want these ad-hoc queries to be billed based on how much data users scan rather than by slot capacity." this is the only thing you need to read out of the question. Having spikes of 500 slots does not mean you should set a baseline of 800, it is WAY too expensive to do that for spikes.

potatoKiller 2 years ago

"You want these ad-hoc queries to be billed based on how much data users scan rather than by slot capacity." So D is out. Choose B

barrru 2 years ago

B "enable autoscaling up to 500 slots" means that the system can add up to 500 slots beyond the baseline as needed, effectively allowing for a total of 800 slots (300 baseline + 500 autoscaled) during peak usage.

hanoverquay 2 years ago

Selected Answer: D

500 (additional) +300 = 800, so answer is D

danisxp 2 years, 2 months ago

Selected Answer: D

Considering the emphasis on strict completion time SLA's.I go with option D. However I think both B and D are not the best solution here.

Matt_108 2 years, 2 months ago

Selected Answer: B

Option B - first project works well with dedicated reservation and autoscaling. The second one requires on demand billing, as per question requires.

ElenaL 2 years, 2 months ago

Selected Answer: D

"These jobs generally never go below a 300 slot utilization, but occasionally spike up an additional 500 slots." -> if it spikes up an ADITIONAL 500 slots, on top of the regular 300, shouldn't we reserve at a minimum 800? open to explanations as to why this is not the case.

scaenruy 2 years, 2 months ago

Selected Answer: B

B. Create two reservations, one for each of the projects. For the SLA project, use an Enterprise Edition with a baseline of 300 slots and enable autoscaling up to 500 slots. For the ad-hoc project, configure on-demand billing.

https://zhangxuetu.online

[202610]

Question #261 Topic 1

You want to migrate your existing Teradata data warehouse to BigQuery. You want to move the historical data to BigQuery by using the most efficient method that requires the least amount of programming, but local storage space on your existing data warehouse is limited. What should you do?

A. Use BigQuery Data Transfer Service by using the Java Database Connectivity (JDBC) driver with FastExport connection. **Most Voted**

B. Create a Teradata Parallel Transporter (TPT) export script to export the historical data, and import to BigQuery by using the bq command-line tool.

C. Use BigQuery Data Transfer Service with the Teradata Parallel Transporter (TPT) tbuild utility.

D. Create a script to export the historical data, and upload in batches to Cloud Storage. Set up a BigQuery Data Transfer Service instance from Cloud Storage to BigQuery.

Correct Answer: A

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: A

- Reduced Local Storage: By using FastExport, data is directly streamed from Teradata to BigQuery without the need for local storage, addressing your storage limitations. - Minimal Programming: BigQuery Data Transfer Service offers a user-friendly interface, eliminating the need for extensive scripting or coding.

AllenChen123 2 years, 2 months ago

Agree. https://cloud.google.com/bigquery/docs/migration/teradata-overview#extraction_method Extraction using a JDBC driver with FastExport connection. If there are constraints on the local storage space available for extracted files, or if there is some reason you can't use TPT, then use this extraction method.

chickenwingz **Highly Voted** 2 years, 3 months ago

Selected Answer: A

https://cloud.google.com/bigquery/docs/migration/teradata-overview#extraction_method Lack of local storage pushes this to JDBC driver

desertlotus1211 1 year ago

Selected Answer: C

BigQuery Data Transfer Service (DTS) supports Teradata via Teradata Parallel Transporter (TPT) in combination with the tbuild utility, which is designed for high-performance parallel data exports. This is Google's recommended approach for Teradata migrations when local disk space is constrained and high throughput is desired.

Parandhaman_Margan 1 year ago

Selected Answer: C

Use BigQuery Data Transfer Service with the Teradata Parallel Transporter (TPT) tbuild utility...minimal coding

ToiToi 1 year, 5 months ago

Selected Answer: C

BigQuery Data Transfer Service (DTS): DTS automates data movement from various sources (including Teradata) to BigQuery. It handles schema conversion, data transfer, and scheduling, minimizing manual effort and programming. Teradata Parallel Transporter (TPT) tbuild: TPT is a powerful utility for high-performance data extraction from Teradata. The tbuild operator specifically creates optimized external data files. Efficiency: Combining DTS with TPT tbuild allows you to efficiently extract large volumes of data from Teradata and load it into BigQuery with minimal coding. Limited Local Storage: This approach streams data directly from Teradata to Cloud Storage, minimizing the need for temporary storage on your Teradata system.

kurayish 1 year, 5 months ago

Selected Answer: C

Using TPT with the tbuild utility ensures that you can efficiently move large volumes of data directly from Teradata to BigQuery without requiring significant local storage space or extensive custom programming. This method leverages Teradata's optimized export capabilities and integrates with Google Cloud's tools for seamless data transfer. JDBC driver with FastExport can be used, it typically requires more programming and manual setup compared to the TPT solution, and may not be as optimized for large-scale data transfers

CGS22 1 year, 11 months ago

Selected Answer: A

Extraction using a JDBC driver with FastExport connection. If there are constraints on the local storage space available for extracted files, or if there is some reason you can't use TPT, then use this extraction method. https://cloud.google.com/bigquery/docs/migration/teradata-overview#extraction_method

Matt_108 2 years, 2 months ago

Selected Answer: A

Option A, the JDBC driver is the key to solve the limited local storage

https://zhangxuetu.online

[202610]

Question #262 Topic 1

You are on the data governance team and are implementing security requirements. You need to encrypt all your data in BigQuery by using an encryption key managed by your team. You must implement a mechanism to generate and store encryption material only on your on-premises hardware security module (HSM). You want to rely on Google managed solutions. What should you do?

A. Create the encryption key in the on-premises HSM, and import it into a Cloud Key Management Service (Cloud KMS) key. Associate the created Cloud KMS key while creating the BigQuery resources.

B. Create the encryption key in the on-premises HSM and link it to a Cloud External Key Manager (Cloud EKM) key. Associate the created Cloud KMS key while creating the BigQuery resources. **Most Voted**

C. Create the encryption key in the on-premises HSM, and import it into Cloud Key Management Service (Cloud HSM) key. Associate the created Cloud HSM key while creating the BigQuery resources.

D. Create the encryption key in the on-premises HSM. Create BigQuery resources and encrypt data while ingesting them into BigQuery.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- Cloud EKM allows you to use encryption keys managed in external key management systems, including on-premises HSMs, while using Google Cloud services. - This means that the key material remains in your control and environment, and Google Cloud services use it via the Cloud EKM integration. - This approach aligns with the need to generate and store encryption material only on your on-premises HSM and is the correct way to integrate such keys with BigQuery. ====== Why not Option C - Cloud HSM is a fully managed service by Google Cloud that provides HSMs for your cryptographic needs. However, it's a cloud-based solution, and the keys generated or managed in Cloud HSM are not stored on-premises. This option doesn't align with the requirement to use only on-premises HSM for key storage.

meh_33 1 year, 7 months ago

Selected Answer: B

Option B, I agree with Raaad on the approach

f74ca0c 1 year, 10 months ago

Selected Answer: B

https://cloud.google.com/kms/docs/ekm#ekm-management-mode

f74ca0c 1 year, 10 months ago

B- https://cloud.google.com/kms/docs/ekm#ekm-management-mode Coordinated external keys are made possible by EKM via VPC connections that use EKM key management from Cloud KMS. If your EKM supports the Cloud EKM control plane, then you can enable EKM key management from Cloud KMS for your EKM via VPC connections to create coordinated external keys. With EKM key management from Cloud KMS enabled, Cloud EKM can request the following changes in your EKM:

Matt_108 2 years, 2 months ago

Selected Answer: B

Option B, I agree with Raaad on the approach

scaenruy 2 years, 2 months ago

Selected Answer: C

C. Create the encryption key in the on-premises HSM, and import it into Cloud Key Management Service (Cloud HSM) key. Associate the created Cloud HSM key while creating the BigQuery resources.

https://zhangxuetu.online

[202610]

Question #263 Topic 1

You maintain ETL pipelines. You notice that a streaming pipeline running on Dataflow is taking a long time to process incoming data, which causes output delays. You also noticed that the pipeline graph was automatically optimized by Dataflow and merged into one step. You want to identify where the potential bottleneck is occurring. What should you do?

A. Insert a Reshuffle operation after each processing step, and monitor the execution details in the Dataflow console. **Most Voted**

B. Insert output sinks after each key processing step, and observe the writing throughput of each block.

C. Log debug information in each ParDo function, and analyze the logs at execution time.

D. Verify that the Dataflow service accounts have appropriate permissions to write the processed data to the output sinks.

Correct Answer: A

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: A

- The Reshuffle operation is used in Dataflow pipelines to break fusion and redistribute elements, which can sometimes help improve parallelization and identify bottlenecks. - By inserting Reshuffle after each processing step and observing the pipeline's performance in the Dataflow console, you can potentially identify stages that are disproportionately slow or stalled. - This can help in pinpointing the step where the bottleneck might be occurring.

srivastavas08 2 years, 1 month ago

ince we don't know for sure if fusion is the culprit, detailed debug logging is still the top choice to find the precise slow operation(s).

Blackstile 1 year ago

Selected Answer: A

Reshuffle is the key.

m_a_p_s 1 year, 3 months ago

Selected Answer: A

Looks like A. However, this option does not provide any option of identifying the underlying cause.
https://cloud.google.com/dataflow/docs/pipeline-lifecycle#prevent_fusion

f6bc4a0 1 year, 5 months ago

Selected Answer: B

B identifies where the problem lies.

JyoGCP 2 years, 1 month ago

Selected Answer: A

Option A

srivastavas08 2 years, 1 month ago

It should be C

tibuenoc 2 years, 2 months ago

Selected Answer: B

The best option is B Because create additional output to capturing and processing error data, will get error each step that allows you to observe the writing throughput of each block, which can help identify specific processing steps causing bottlenecks. Option A also is valid but can not directly address all bottlenecks, especially if the graph was merged.

Sofiia98 2 years, 2 months ago

Selected Answer: A

From the Dataflow documentation: "There are a few cases in your pipeline where you may want to prevent the Dataflow service from performing fusion optimizations. These are cases in which the Dataflow service might incorrectly guess the optimal way to fuse operations in the pipeline, which could limit the Dataflow service's ability to make use of all available workers. You can insert a Reshuffle step. Reshuffle prevents fusion, checkpoints the data, and performs deduplication of records. Reshuffle is supported by Dataflow even though it is marked deprecated in the Apache Beam documentation."

scaenruy 2 years, 2 months ago

Selected Answer: A

A. Insert a Reshuffle operation after each processing step, and monitor the execution details in the Dataflow console.

https://zhangxuetu.online

[202610]

Question #264 Topic 1

You are running your BigQuery project in the on-demand billing model and are executing a change data capture (CDC) process that ingests data. The CDC process loads 1 GB of data every 10 minutes into a temporary table, and then performs a merge into a 10 TB target table. This process is very scan intensive and you want to explore options to enable a predictable cost model. You need to create a BigQuery reservation based on utilization information gathered from BigQuery Monitoring and apply the reservation to the CDC process. What should you do?

A. Create a BigQuery reservation for the dataset.

B. Create a BigQuery reservation for the job.

C. Create a BigQuery reservation for the service account running the job.

D. Create a BigQuery reservation for the project. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

Matt_108 **Highly Voted** 2 years, 2 months ago

Selected Answer: D

Option D, reservation can't be applied to resources lower than projects (only to Org, folders or projects)

AllenChen123 2 years, 2 months ago

Seems correct. https://cloud.google.com/bigquery/docs/reservations-intro#understand_workload_management

task_7 **Highly Voted** 2 years, 2 months ago

Selected Answer: B

Reserve assignments To use the slot capacity you purchased, assign projects, folders, or organizations to a reservation. When a job in a project runs, it uses slots from the assigned reservation. Resources can inherit roles from their parents in the resource hierarchy. Even if a project is not assigned to a reservation, it inherits the assignment from the parent folder or organization, if any. If a project does not have an assigned or inherited reservation, the job uses on-demand pricing. For more information about the resource hierarchy, see Organizing BigQuery Resources .

Positron75 1 year, 4 months ago

The first sentence of that text already points towards D, not B. You can't assign jobs to a reservation, only projects, folders or organizations.

apoio.certificacoes.closer 1 year, 3 months ago

Best that could be done is a specify this a batch job, but the slot capacity is still at the project level.

judy_data 6 months, 3 weeks ago

Selected Answer: D

Reservations are assigned at project level https://cloud.google.com/bigquery/docs/reservations-tasks

desertlotus1211 1 year ago

Selected Answer: C

Answer C provides the granularity you need. Answer D is for the entire project... however the question ONLY wants it for the CDC process - so service account is better. In BigQuery, reservations are assigned to "assignments", which can be: Project Folder Organization Service account

m_a_p_s 1 year, 3 months ago

Selected Answer: D

"When you create commitments and reservations, they are associated with a Google Cloud project." - https://cloud.google.com/bigquery/docs/reservations-intro#admin-project

Positron75 1 year, 4 months ago

Selected Answer: D

https://cloud.google.com/bigquery/docs/reservations-intro#assignments

SamuelTsch 1 year, 5 months ago

Selected Answer: B

Option B provides a more granularity solution

viciousjpjp 1 year, 7 months ago

Selected Answer: B

The correct answer is B. Create a BigQuery reservation at the job level. Create a BigQuery reservation at the job level: This is the most suitable option. By creating a job-level reservation, you can allocate resources specifically to the CDC process and improve the accuracy of cost forecasting. Steps to create and apply a BigQuery reservation: Identify the job: Clearly identify the job that executes the CDC process. Create a reservation: Use the BigQuery console or API to create a reservation, specifying the job's label, query text, and other details. Apply the reservation: Assign the created reservation to the job that executes the CDC process.

Positron75 1 year, 4 months ago

Jobs cannot be assigned to a reservation. You should read the documentation instead of relying on AI answers which are wrong half the time. https://cloud.google.com/bigquery/docs/reservations-intro#understand_workload_management

Positron75 1 year, 4 months ago

Sorry, this is the actual link: https://cloud.google.com/bigquery/docs/reservations-intro#assignments

meh_33 1 year, 7 months ago

Are these questions really useful andd coming in GDE Exam? Anyone appeared recently and passed ?

Anudeep58 1 year, 10 months ago

Selected Answer: D

https://cloud.google.com/blog/products/data-analytics/manage-bigquery-costs-with-custom-quotas. Quotas can be applied on Project or User Level

f74ca0c 1 year, 10 months ago

Selected Answer: D

D- choose the correct project and apply the task type
background:https://cloud.google.com/bigquery/docs/reservations-intro?hl=fr#assignments

JyoGCP 2 years, 1 month ago

Selected Answer: D

Option D

GCP001 2 years, 2 months ago

D. Reservation is on project, folder or organisation level.

raaad 2 years, 2 months ago

Selected Answer: D

C or D ?? Option C (service account) allows you to target the reservation specifically to the CDC process or any other jobs run by that service account. This is particularly useful if you have multiple processes running in the project with different performance or cost requirements. Option D (project) applies the reservation across all jobs in the project, which is a broader approach. If the CDC process is the primary or sole job running in the project and you want all jobs to share the same reservation, then this option might be more straightforward.

scaenruy 2 years, 2 months ago

Selected Answer: D

D. Create a BigQuery reservation for the project.

https://zhangxuetu.online

[202610]

Question #265 Topic 1

You are designing a fault-tolerant architecture to store data in a regional BigQuery dataset. You need to ensure that your application is able to recover from a corruption event in your tables that occurred within the past seven days. You want to adopt managed services with the lowest RPO and most cost-effective solution. What should you do?

A. Access historical data by using time travel in BigQuery. **Most Voted**

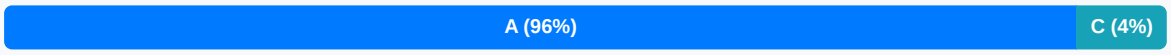
B. Export the data from BigQuery into a new table that excludes the corrupted data

C. Create a BigQuery table snapshot on a daily basis.

D. Migrate your data to multi-region BigQuery buckets.

Correct Answer: A

Community vote distribution



Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: A

- Lowest RPO: Time travel offers point-in-time recovery for the past seven days by default, providing the shortest possible recovery point objective (RPO) among the given options. You can recover data to any state within that window. - No Additional Costs: Time travel is a built-in feature of BigQuery, incurring no extra storage or operational costs. - Managed Service: BigQuery handles time travel automatically, eliminating manual backup and restore processes.

srivastavas08 1 year, 7 months ago

BigQuery's time travel feature typically retains history up to 7 days. However, if the corruption affects the underlying data for an extended period, the 7-day window might not be long enough.

CGS22 1 year, 5 months ago

Selected Answer: C

Meets Recovery Needs: Table snapshots provide point-in-time copies of your data, allowing you to restore data from any point within the last seven days, effectively addressing the corruption event recovery requirement. Low RPO: With daily snapshots, your Recovery Point Objective (RPO) is at most 24 hours, satisfying the need for a low RPO. Managed Service: Table snapshots are a fully managed service within BigQuery, aligning with your preference. Cost-Effective: Snapshots only store the changes from the base table, minimizing storage costs compared to full table copies.

hanoverquay 1 year, 6 months ago

Selected Answer: A

vote for A

JyoGCP 1 year, 7 months ago

Selected Answer: A

Option A

Matt_108 1 year, 8 months ago

Selected Answer: A

Option A, raaad explanation is perfect

scaenruy 1 year, 9 months ago

Selected Answer: A

A. Access historical data by using time travel in BigQuery.

https://zhangxuetu.online

[202610]

Question #266 Topic 1

You are building a streaming Dataflow pipeline that ingests noise level data from hundreds of sensors placed near construction sites across a city. The sensors measure noise level every ten seconds, and send that data to the pipeline when levels reach above 70 dBA. You need to detect the average noise level from a sensor when data is received for a duration of more than 30 minutes, but the window ends when no data has been received for 15 minutes. What should you do?

A. Use session windows with a 15-minute gap duration. **Most Voted**

B. Use session windows with a 30-minute gap duration.

C. Use hopping windows with a 15-minute window, and a thirty-minute period.

D. Use tumbling windows with a 15-minute window and a fifteen-minute .withAllowedLateness operator.

Correct Answer: A

Community vote distribution



Comments

datapassionate **Highly Voted** 2 years, 2 months ago

Selected Answer: A

to detect average noise levels from sensors, the best approach is to use session windows with a 15-minute gap duration (Option A). Session windows are ideal for cases like this where the events (sensor data) are sporadic. They group events that occur within a certain time interval (15 minutes in your case) and a new window is started if no data is received for the duration of the gap. This matches your requirement to end the window when no data is received for 15 minutes, ensuring that the average noise level is calculated over periods of continuous data

ashdam 2 years, 1 month ago

But you are not fulfilling this requirement "You need to detect the average noise level from a sensor when data is received for a duration of more than 30 minutes". I would say C

saschak94 **Highly Voted** 2 years, 1 month ago

Selected Answer: A

You need a window that start when data for a sensor arrives and end when there's a gap in data. That would rule out hopping and tumbling windows. - > Windows need to stay open as long as there's data arriving - 30+ mins -> Window Should close when no data has been received for 15 mins -> Gap 15 mins

shanks_t 1 year, 7 months ago

Selected Answer: A

The problem requires detecting average noise levels when data is received for more than 30 minutes, but the window should end when no data has been received for 15 minutes. Session windows are ideal for this scenario because: They are designed to capture bursts of activity followed by periods of inactivity. They dynamically size based on the data received, which fits well with the variable duration of noise events. The gap duration can be set to define when a session ends. The 15-minute gap duration aligns perfectly with the requirement to end the window when no data has been received for 15 minutes. Session windows will naturally extend beyond 30 minutes if data keeps coming in, satisfying the requirement to detect levels for durations of more than 30 minutes.

viciousjpjp 1 year, 7 months ago

Selected Answer: D

D. Using a 15-minute window with a 15-minute tumbling window withAllowedLateness is the most suitable option for the following reasons: Flexibility: By allowing a 15-minute delay, it can accommodate various situations such as network latency or sensor failures. Processing efficiency: Using a fixed window improves processing efficiency. Compliance with conditions: The window ends if no data is received for 15 minutes, meeting the specified condition. Implementation points: .withAllowedLateness: This operator allows delayed events to be included in the current window. Trigger: When 30 minutes of data is collected, a trigger event is generated, and the average value is calculated based on this event. Watermark: By setting a watermark, processing of old data can be terminated.

JamesKarianis 1 year, 7 months ago

Selected Answer: A

Without a doubt A: https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#session-windows

iooj 1 year, 8 months ago

Selected Answer: A

The requirements are - recieve data for a duration of MORE than 30 minutes - end the window based on inactivity

josech 1 year, 10 months ago

Correct answer: A. Use a session Window to cature data and create an aggregation when the Session is larger than 30 minutes. https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#session-windows https://beam.apache.org/releases/javadoc/2.6.0/org/apache/beam/sdk/transforms/windowing/Sessions.html

f74ca0c 1 year, 10 months ago

Selected Answer: C

C- Running average: https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#hopping-windows

joao_01 1 year, 11 months ago

Guys, I think its B. I was considering C, but C is calculating every 30 min, the 15min window gap data.Thats not what the questions wants. The questions wants a solution to get the average data of a 30 min window. So Its B. Look at this relating to C: "Use hopping windows with a 15-minute window, and a thirty-minute period" --> Wrong (IS DIFFERENT THEN) "Use hopping windows with a 30-minute window, and a 15-minute period" --> Right. Thats why I think the B is the right answer.

joao_01 1 year, 11 months ago

Actually I think with B, the window will never be closed because the probability of having a period of inactivity of 30 minutes is very low. In that case I think the option A is the more correct one.

342f1c6 2 years ago

Selected Answer: C

To take running averages of data, use hopping windows. You can use one-minute hopping windows with a thirty-second period to compute a one-minute running average every thirty seconds.

kck6ra4214wm 2 years, 1 month ago

Selected Answer: A

https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#session-windows A session window contains elements within a gap duration of another element. The gap duration is an interval between new data in a data stream. If data arrives after the gap duration, the data is assigned to a new window. So, we need

imiu 2 years, 2 months ago

Selected Answer: C

https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#hopping-windows To take running averages of data, use hopping windows. You can use one-minute hopping windows with a thirty-second period to compute a one-minute running average every thirty seconds.

Matt_108 2 years, 2 months ago

Selected Answer: D

Option D to me, It aligns with the specified criteria for detecting the average noise level within a 30-minute duration and handling the end of the window when no data is received for 15 minutes.

AllenChen123 2 years, 2 months ago

Agree D. Data comes -> 30 mts duration. Data didn't come in 15 mts -> 15 mts duration

BIGQUERY_ALT_ALT 2 years, 2 months ago

Selected Answer: D

OPTION D is correct for the specific scenario where we want to detect the average noise level for a duration of more than 30 minutes but end the window when no data has been received for 15 minutes. Explanation: - Tumbling windows are non-overlapping windows, and in this case, you want to capture data continuously for 30-minute intervals. - Using a tumbling window with a 15-minute window size aligns with your requirement to detect the average noise level for a duration of more than 30 minutes. - Adding a .withAllowedLateness operator with a duration of fifteen minutes ensures that the window will still consider late-arriving data within that time frame. After fifteen minutes of no data, the window will be closed, and any late-arriving data will not be considered. Option A and B invalid as they capture fixed logic with 15 or 30 mins. Option C captures only 15 min average with 30 min trigger hence not suitable.

Sofiia98 2 years, 2 months ago

Selected Answer: C

Hopping windows (called sliding windows in Apache Beam). To take running averages of data, use hopping windows.

scaenruy 2 years, 2 months ago

Selected Answer: C

C. Use hopping windows with a 15-minute window, and a thirty-minute period.

https://zhangxuetu.online

[202610]

Question #267 Topic 1

You are creating a data model in BigQuery that will hold retail transaction data. Your two largest tables, sales_transaction_header and sales_transaction_line, have a tightly coupled immutable relationship. These tables are rarely modified after load and are frequently joined when queried. You need to model the sales_transaction_header and sales_transaction_line tables to improve the performance of data analytics queries. What should you do?

A. Create a sales_transaction table that holds the sales_transaction_header information as rows and the sales_transaction_line rows as nested and repeated fields. **Most Voted**

B. Create a sales_transaction table that holds the sales_transaction_header and sales_transaction_line information as rows, duplicating the sales_transaction_header data for each line.

C. Create a sales_transaction table that stores the sales_transaction_header and sales_transaction_line data as a JSON data type.

D. Create separate sales_transaction_header and sales_transaction_line tables and, when querying, specify the sales_transaction_line first in the WHERE clause.

Correct Answer: A

Community vote distribution

A (100%)

Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: A

- In BigQuery, nested and repeated fields can significantly improve performance for certain types of queries, especially joins, because the data is co-located and can be read efficiently. - - This approach is often used in data warehousing scenarios where query performance is a priority, and the data relationships are immutable and rarely modified.

josech 1 year, 4 months ago

Selected Answer: A

Option A https://cloud.google.com/bigquery/docs/best-practices-performance-nested

hanoverquay 1 year, 6 months ago

Selected Answer: A

optional A

JyoGCP 1 year, 7 months ago

Selected Answer: A

Option A

Matt_108 1 year, 8 months ago

Selected Answer: A

Option A

scaenruy 1 year, 9 months ago

Selected Answer: A

A. Create a sales_transaction table that holds the sales_transaction_header information as rows and the sales_transaction_line rows as nested and repeated fields.

https://zhangxuetu.online

[202610]

Question #268 Topic 1

You created a new version of a Dataflow streaming data ingestion pipeline that reads from Pub/Sub and writes to BigQuery. The previous version of the pipeline that runs in production uses a 5-minute window for processing. You need to deploy the new version of the pipeline without losing any data, creating inconsistencies, or increasing the processing latency by more than 10 minutes. What should you do?

A. Update the old pipeline with the new pipeline code.

B. Snapshot the old pipeline, stop the old pipeline, and then start the new pipeline from the snapshot.

C. Drain the old pipeline, then start the new pipeline. **Most Voted**

D. Cancel the old pipeline, then start the new pipeline.

Correct Answer: C

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

- Graceful Data Transition: Draining the old pipeline ensures it processes all existing data in its buffers and watermarks before shutting down, preventing data loss or inconsistencies. - Minimal Latency Increase: The latency increase will be limited to the amount of time it takes to drain the old pipeline, typically within the acceptable 10-minute threshold.

AlizCert **Highly Voted** 2 years, 1 month ago

I don't think C is correct, as it will immediately fire the window: "Draining can result in partially filled windows. In that case, if you restart the drained pipeline, the same window might fire a second time, which can cause issues with your data. " https://cloud.google.com/dataflow/docs/guides/stopping-a-pipeline#effects Maybe "A" means launching a replacement job? https://cloud.google.com/dataflow/docs/guides/updating-a-pipeline#Launching

SamuelTsch 1 year, 5 months ago

we don't restart the drained pipeline.

d11379b 2 years ago

So why not B it is the better choice to save intermediate state and easy to use

Kadhem 6 months ago

Selected Answer: B

Draining can result in partially filled windows. In that case, if you restart the drained pipeline, the same window might fire a second time, which can cause issues with your data. For example, in the following scenario, files might have conflicting names, and data might be overwritten. https://cloud.google.com/dataflow/docs/guides/stopping-a-pipeline

nimo9977 6 months ago

Selected Answer: B

Drain (C) Avoids in-flight loss, but can create inconsistent results due to early window closure. This violates “no inconsistencies.” Snapshot (B) Captures state + offsets, lets new job continue without gaps or duplicates. Meets “no data loss / no inconsistencies.” Latency bump stays well under 10 minutes in practice.

judy_data 6 months, 4 weeks ago

Selected Answer: B

A snapshot captures the exact ack state of the subscription. Starting the new pipeline from that snapshot guarantees no data loss and no duplicates, preserving window consistency even if your windowing/logic changed. (C) Draining waits for in-flight windows/watermarks to complete; with a 5-minute window this can easily exceed the allowed +10 minutes latency and still risks a gap while you switch pipelines.

petulda 1 year, 4 months ago

Why not B? https://cloud.google.com/dataflow/docs/guides/upgrade-guide#stop-and-replace

STEVE_PEGLEG 1 year, 7 months ago

Selected Answer: C

There is requirement to avoid data loss. https://cloud.google.com/dataflow/docs/guides/upgrade-guide#stop-and-replace "To avoid data loss, in most cases, draining is the preferred action."

Ouss_123 1 year, 9 months ago

Selected Answer: C

- Draining the old pipeline ensures that it finishes processing all in-flight data before stopping, which prevents data loss and inconsistencies. - After draining, you can start the new pipeline, which will begin processing new data from where the old pipeline left off. - This approach maintains a smooth transition between the old and new versions, minimizing latency increases and avoiding data gaps or overlaps. ==> Other options, such as updating, snapshotting, or canceling, might not provide the same level of consistency and could lead to data loss or increased latency beyond the acceptable 10-minute window. Draining is the safest method to ensure a seamless transition.

d11379b 2 years ago

Selected Answer: B

I would choose B as mentioned by Alizcert, a simple drain may cause problem Dataflow snapshots save the state of a streaming pipeline, which lets you start a new version of your Dataflow job without losing state. Snapshots are useful for backup and recovery, testing and rolling back updates to streaming pipelines, and other similar scenarios.

hanoverquay 2 years ago

Selected Answer: C

C option

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C, draining the old pipeline solves all requests

scaenruy 2 years, 2 months ago

Selected Answer: C

C. Drain the old pipeline, then start the new pipeline.

https://zhangxuetu.online

[202610]

Question #269 Topic 1

Your organization's data assets are stored in BigQuery, Pub/Sub, and a PostgreSQL instance running on Compute Engine. Because there are multiple domains and diverse teams using the data, teams in your organization are unable to discover existing data assets. You need to design a solution to improve data discoverability while keeping development and configuration efforts to a minimum. What should you do?

A. Use Data Catalog to automatically catalog BigQuery datasets. Use Data Catalog APIs to manually catalog Pub/Sub topics and PostgreSQL tables.

B. Use Data Catalog to automatically catalog BigQuery datasets and Pub/Sub topics. Use Data Catalog APIs to manually catalog PostgreSQL tables. **Most Voted**

C. Use Data Catalog to automatically catalog BigQuery datasets and Pub/Sub topics. Use custom connectors to manually catalog PostgreSQL tables.

D. Use customer connectors to manually catalog BigQuery datasets, Pub/Sub topics, and PostgreSQL tables.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- It utilizes Data Catalog's native support for both BigQuery datasets and Pub/Sub topics. - For PostgreSQL tables running on a Compute Engine instance, you'd use Data Catalog APIs to create custom entries, as Data Catalog does not automatically discover external databases like PostgreSQL.

AllenChen123 2 years, 2 months ago

Agree. https://cloud.google.com/data-catalog/docs/concepts/overview#catalog-non-google-cloud-assets

datapassionate **Highly Voted** 2 years, 2 months ago

Selected Answer: C

Data Catalog is the best choice. But for catalogging PostgreSQL it is better to use a connector when available, instead of using API. https://cloud.google.com/data-catalog/docs/integrate-data-sources#integrate_unsupported_data_sources

7787de3 1 year, 6 months ago

I agree. On the linked page: If you can't find a connector for your data source, you can still manually integrate it by creating entry groups and custom entries. As we can find a connector there, it should be used.

tibuenoc 2 years, 2 months ago

Agree. If it doesn't have a connector, it must be manually built on the Data Catalog API. As PostgreSQL already has a connector it's the best option is C

af17139 5 months, 1 week ago

Selected Answer: C

Automatic Cataloging: Data Catalog (now part of Dataplex) has built-in integrations to automatically discover and register metadata from Google Cloud managed services like BigQuery and Pub/Sub. This requires little to no configuration. Custom Connector for PostgreSQL: For the PostgreSQL instance running on Compute Engine, you'll need to develop a "custom connector." This typically involves writing a script or application that: Connects to the PostgreSQL database. Extracts schema information (tables, columns, types, etc.), potentially from information_schema. Transforms this information into the format required by the Data Catalog API. Uses the Data Catalog APIs to create or update entries for the PostgreSQL tables. This connector can be run on a schedule (e.g., using Cloud Scheduler and Cloud Functions) to keep the metadata in Data Catalog synchronized with the PostgreSQL instance.

aaaaaaaasdasdasfs 11 months, 3 weeks ago

Selected Answer: B

This is the correct approach. Data Catalog provides automatic discovery and metadata extraction for both BigQuery datasets and Pub/Sub topics as native Google Cloud services. You'll only need to use the Data Catalog APIs to manually catalog the PostgreSQL tables running on Compute Engine, as this is a non-native data source. This maximizes automation while minimizing development effort.

Abizi 1 year ago

Selected Answer: C

Why C is Correct? BigQuery datasets → ✅ Automatically cataloged in Data Catalog Pub/Sub topics → ✅ Automatically cataloged in Data Catalog PostgreSQL on Compute Engine → ❌ Not automatically cataloged Requires a custom connector to extract metadata and push it to Data Catalog. Option B (using Data Catalog APIs manually) is not enough because PostgreSQL metadata isn't natively supported. Why Not B? Option B suggests using Data Catalog APIs manually for PostgreSQL. However, Data Catalog does not natively support PostgreSQL metadata extraction. You need a custom connector to first extract PostgreSQL schema information, then push it to Data Catalog.

AWSandeep 1 year, 3 months ago

Selected Answer: B

This section explains it clearly: https://cloud.google.com/data-catalog/docs/integrate-data-sources#integrate_unsupported_data_sources.

baimus 1 year, 5 months ago

Selected Answer: C

This is C. To clarify some issues below with B, the links provided by supporters of B actually do say that it's preferable to use a community connector where available, and to only use the API when the case is genuinely not supported by community connectors. In this case it's Postgresql, so it's supported, see here for full list: https://cloud.google.com/data-catalog/docs/integrate-data-sources#integrate_on-premises_data_sources So this would be B if it was something like Q+ or some genuinely unsupported database, but postgres is supported for community connector.

shanks_t 1 year, 7 months ago

Selected Answer: B

Data Catalog automatically catalogs metadata from Google Cloud sources such as BigQuery, Vertex AI, Pub/Sub, Spanner, Bigtable, and more. To catalog metadata from non-Google Cloud systems in your organization, you can use the following: Community-contributed connectors to multiple popular on-premises data sources Manually build on the Data Catalog APIs for custom entries

shanks_t 1 year, 7 months ago

C. While similar to B, using custom connectors for PostgreSQL might involve more development effort than using the Data Catalog APIs directly.

meh_33 1 year, 7 months ago

raaad mostly correct and we can check his description supporting his answer so we can go with it .Cheers mate

987af6b 1 year, 8 months ago

Selected Answer: C

I'm voting for C because the documentation states that Postgres is a custom connector developed by the community.

987af6b 1 year, 8 months ago

Changed my mind. B. -This is not on premise, so the custom connector should not be applicable -Question says keep manual dev and config to a minimum

fitri001 1 year, 9 months ago

Selected Answer: B

BigQuery Datasets and Pub/Sub Topics: Google Data Catalog can automatically catalog metadata from BigQuery and Pub/Sub, making it easy to discover and manage these data assets without additional development effort. PostgreSQL Tables: While Data Catalog does not have built-in connectors for PostgreSQL, you can use the Data Catalog APIs to manually catalog the PostgreSQL tables. This requires some custom development but is manageable compared to creating custom connectors for everything.

virat_kohli 1 year, 10 months ago

Selected Answer: B

B. Use Data Catalog to automatically catalog BigQuery datasets and Pub/Sub topics. Use Data Catalog APIs to manually catalog PostgreSQL tables.

Cassim 1 year, 10 months ago

Selected Answer: B

Option B leverages Data Catalog to automatically catalog BigQuery datasets and Pub/Sub topics, which streamlines the process and reduces manual effort. Using Data Catalog APIs to manually catalog PostgreSQL tables ensures consistency across all data assets while minimizing development and configuration efforts.

LaxmanTiwari 1 year, 11 months ago

Selected Answer: C

I vote for c as per Integrate on-premises data sources To integrate on-premises data sources, you can use the corresponding Python connectors contributed by the community: under the link https://cloud.google.com/data-catalog/docs/integrate-data-sources

LaxmanTiwari 1 year, 11 months ago

data catalog api will come into effect if custom connectors are not available via community repos.

joao_01 1 year, 11 months ago

In the opction C, the expression "Use custom connectors to manually catalog PostgreSQL tables." is refering to the use case of Google when you want to use "Community-contributed connectors to multiple popular on-premises data

sources". As you can see, this connectors are for ON-PREMISSES data sources ONLY. In this case the Postgres is in a VM in the cloud. Thus, the option correct is B.

joao_01 1 year, 11 months ago

Link: https://cloud.google.com/data-catalog/docs/concepts/overview#catalog-non-google-cloud-assets

hanoverquay 2 years ago

Selected Answer: B

option B, there's no need to build a custom connector now, postgreSQL is now supported https://github.com/GoogleCloudPlatform/datacatalog-connectors-rdbms/tree/master/google-datacatalog-postgresql-connector

d11379b 2 years ago

I think “custom connector” here may just infer that this is not official tools? as the doc mentioned “ connectors contributed by the community” And should not be B as “manually catalog by API “ this is a way even more basic than using connector

Y___ash 2 years ago

Selected Answer: B

Use Data Catalog to automatically catalog BigQuery datasets and Pub/Sub topics. Use Data Catalog APIs to manually catalog PostgreSQL tables.

https://zhangxuetu.online

[202610]

Question #270 Topic 1

You need to create a SQL pipeline. The pipeline runs an aggregate SQL transformation on a BigQuery table every two hours and appends the result to another existing BigQuery table. You need to configure the pipeline to retry if errors occur. You want the pipeline to send an email notification after three consecutive failures. What should you do?

A. Use the BigQueryUpsertTableOperator in Cloud Composer, set the retry parameter to three, and set the email_on_failure parameter to true.

B. Use the BigQueryInsertJobOperator in Cloud Composer, set the retry parameter to three, and set the email_on_failure parameter to true. **Most Voted**

C. Create a BigQuery scheduled query to run the SQL transformation with schedule options that repeats every two hours, and enable email notifications.

D. Create a BigQuery scheduled query to run the SQL transformation with schedule options that repeats every two hours, and enable notification to Pub/Sub topic. Use Pub/Sub and Cloud Functions to send an email after three failed executions.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 1 year, 9 months ago

Selected Answer: B

- It provides a direct and controlled way to manage the SQL pipeline using Cloud Composer (Apache Airflow). - The BigQueryInsertJobOperator is well-suited for running SQL jobs in BigQuery, including aggregate transformations and handling of results. - The retry and email_on_failure parameters align with the requirements for error handling and notifications. - Cloud Composer requires more setup than using BigQuery's scheduled queries directly, but it offers robust workflow management, retry logic, and notification capabilities, making it suitable for more complex and controlled data pipeline requirements.

SuperVan 1 year, 7 months ago

The prompt wants an email notification sent after three failed attempts. Is there any concern that the retry parameter is set to 3, wouldn't this mean that the email is sent after 4 failed attempts (1 original + 3 retries)?

lmch 3 months, 2 weeks ago

Selected Answer: B

This requirement describes a standard orchestration workflow involving a schedule (every two hours), a transformation (SQL aggregation), an append operation, and custom error handling (retries and conditional notifications). Cloud Composer (managed Apache Airflow) is the standard tool for this level of logic.

56d02cd 9 months ago

Selected Answer: D

The retries in A and B are only for task failures/retries within the DAG. The question asks for a notification after three consecutive failures of the DAG. IOW ... the retries parameter is correct for handling failures within a single task instance. email_on_failure would typically fire after the final retry of that single task instance fails, not after three consecutive DAG runs fail.

22c1725 10 months ago

Selected Answer: B

There is no retry in (D) https://cloud.google.com/bigquery/docs/scheduling-queries

gabbferreira 11 months, 2 weeks ago

Selected Answer: D

It´s D "Notifications on operator failure Set email_on_failure to True to send an email notification when an operator in the DAG fails. To send email notifications from a Cloud Composer environment, you must configure your environment to use SendGrid." Setting email_on_failure = True will send the email after the FIRST failure https://cloud.google.com/composer/docs/composer-2/write-dags#notifications_on_operator_failure

gabbferreira 11 months, 2 weeks ago

Selected Answer: D

chatpgt and gemini said it is D

MarcoPellegrino 1 year, 1 month ago

Selected Answer: D

A) Wrong, Upsert is not for appending B) Wrong, doesn't mention the 2 hours scheduling C) Wrong, doesn't mention the emailing D) Correct

Augustax 1 year, 2 months ago

Selected Answer: D

The retry times in B and clearly mentioned 2 hours in D make me think D is the better option..

plum21 1 year, 2 months ago

Selected Answer: D

"You want the pipeline to send an email notification after three consecutive failures" - it is not about retries which are configurable via Composer operator - it is about 3 consecutive executions which could be for different hours.

b3e59c2 1 year, 2 months ago

Selected Answer: D

Terrible options as usual. Whilst B is the most elegant, it doesn't explicitly address the 2 hour scheduling (you can schedule within Composer, but the answer doesn't mention it). If we take these answers on the surface level, D is the only option that actually achieves our goal.

e593506 1 year, 3 months ago

Selected Answer: D

The prompt wants an email notification sent after three failed attempts Option B does not meet that condition

josech 1 year, 4 months ago

Selected Answer: B

https://airflow.apache.org/docs/apache-airflow-providers-google/stable/_api/airflow/providers/google/cloud/operators/bigquery/index.html#airflow.providers.google.cloud.operators.bigquery.BigQueryInsertJobOperator https://cloud.google.com/composer/docs/composer-2/write-dags#notifications_on_operator_failure

joao_01 1 year, 5 months ago

It s B (however for me its a incomplete answers cause it does not address the schedule of every 2 hours). Its not C or D because BigQuery scheduled queries by default does not retries the queries when error occurs. Link: https://cloud.google.com/bigquery/docs/scheduling-queries

JyoGCP 1 year, 7 months ago

Selected Answer: B

Option B

datapassionate 1 year, 8 months ago

Selected Answer: D

D. Create a BigQuery scheduled query to run the SQL transformation with schedule options that repeats every two hours, and enable notification to Pub/Sub topic. Use Pub/Sub and Cloud Functions to send an email after three failed executions This method utilizes BigQuery's native scheduling capabilities for running the SQL job and leverages Pub/Sub and Cloud Functions for customized notification handling, including the specific requirement of sending an email after three consecutive failures.

RenePetersen 1 year, 7 months ago

Option D mentions nothing about how the job retrying is put in place, so for that reason I don't think this is the correct option.

scaenruy 1 year, 9 months ago

Selected Answer: B

B. Use the BigQueryInsertJobOperator in Cloud Composer, set the retry parameter to three, and set the email_on_failure parameter to true.

https://zhangxuetu.online

[202610]

Question #271 Topic 1

You are monitoring your organization's data lake hosted on BigQuery. The ingestion pipelines read data from Pub/Sub and write the data into tables on BigQuery. After a new version of the ingestion pipelines is deployed, the daily stored data increased by 50%. The volumes of data in Pub/Sub remained the same and only some tables had their daily partition data size doubled. You need to investigate and fix the cause of the data increase. What should you do?

A. 1. Check for duplicate rows in the BigQuery tables that have the daily partition data size doubled.
2. Schedule daily SQL jobs to deduplicate the affected tables.
3. Share the deduplication script with the other operational teams to reuse if this occurs to other tables.

B. 1. Check for code errors in the deployed pipelines.
2. Check for multiple writing to pipeline BigQuery sink.
3. Check for errors in Cloud Logging during the day of the release of the new pipelines.
4. If no errors, restore the BigQuery tables to their content before the last release by using time travel.

C. 1. Check for duplicate rows in the BigQuery tables that have the daily partition data size doubled.
2. Check the BigQuery Audit logs to find job IDs.
3. Use Cloud Monitoring to determine when the identified Dataflow jobs started and the pipeline code version.
4. When more than one pipeline ingests data into a table, stop all versions except the latest one. Most Voted

D. 1. Roll back the last deployment.
2. Restore the BigQuery tables to their content before the last release by using time travel.
3. Restart the Dataflow jobs and replay the messages by seeking the subscription to the timestamp of the release.

Correct Answer: C

Community vote distribution



Comments

raaad Highly Voted 2 years, 2 months ago

Selected Answer: C

- Detailed Investigation of Logs and Jobs Checking for duplicate rows targets the potential immediate cause of the issue. - Checking the BigQuery Audit logs helps identify which jobs might be contributing to the increased data volume. - Using Cloud Monitoring to correlate job starts with pipeline versions helps identify if a specific version of the

pipeline is responsible. - Managing multiple versions of pipelines ensures that only the intended version is active, addressing any versioning errors that might have occurred during deployment. ======= Why not B While it addresses the symptom (excess data), it doesn't necessarily stop the problem from recurring. (The questions asked to investigate and fix)

mi_yulai 1 year, 5 months ago

Why not D?

judy_data 6 months, 4 weeks ago

Because option D doesn't investigate the root cause of the issue and doesn't even check the code.

SamuelTsch 1 year, 5 months ago

Selected Answer: B

No idea which one to choose. Option C miss a step - to restore the tables.

Ryannn23 1 year, 2 months ago

" You need to investigate and fix the cause of the data increase. " - fixing the target tables was not required.

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C - agree with Raaad on the reasons

task_7 2 years, 2 months ago

Selected Answer: B

B. Check for code errors in the deployed pipelines, multiple writing to pipeline BigQuery sink, errors in Cloud Logging, and if necessary, restore tables using time travel. Check for code errors Check for multiple writes Check Cloud Logging Restore tables if necessary:

RenePetersen 2 years, 1 month ago

This does not fix the error, it basically assumes that the error is not really there.

https://zhangxuetu.online

[202610]

Question #272 Topic 1

You have a BigQuery dataset named “customers”. All tables will be tagged by using a Data Catalog tag template named “gdpr”. The template contains one mandatory field, “has_sensitive_data”, with a boolean value. All employees must be able to do a simple search and find tables in the dataset that have either true or false in the “has_sensitive_data’ field. However, only the Human Resources (HR) group should be able to see the data inside the tables for which “has_sensitive data” is true. You give the all employees group the bigquery.metadataViewer and bigquery.connectionUser roles on the dataset. You want to minimize configuration overhead. What should you do next?

A. Create the “gdpr” tag template with private visibility. Assign the bigquery.dataViewer role to the HR group on the tables that contain sensitive data.

B. Create the “gdpr” tag template with private visibility. Assign the datacatalog.tagTemplateViewer role on this tag to the all employees group, and assign the bigquery.dataViewer role to the HR group on the tables that contain sensitive data.

C. Create the “gdpr” tag template with public visibility. Assign the bigquery.dataViewer role to the HR group on the tables that contain sensitive data. **Most Voted**

D. Create the “gdpr” tag template with public visibility. Assign the datacatalog.tagTemplateViewer role on this tag to the all employees group, and assign the bigquery.dataViewer role to the HR group on the tables that contain sensitive data.

Correct Answer: C

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

- The most straightforward solution with minimal configuration overhead. - By creating the "gdpr" tag template with public visibility, you ensure that all employees can search and find tables based on the "has_sensitive_data" field. - Assigning the bigquery.dataViewer role to the HR group on tables with sensitive data ensures that only they can view the actual data in these tables.

ML6 2 years, 1 month ago

Wouldn't employees still need the roles/datacatalog.tagTemplateViewer role to view private AND public tags? To get the permissions that you need to view public and private tags on Bigtable resources, ask your administrator to grant

you the following IAM roles: - roles/datacatalog.tagTemplateViewer - roles/bigtable.viewer Source: https://cloud.google.com/bigtable/docs/manage-data-assets-using-data-catalog#permissions-view-tags

ML6 2 years, 1 month ago

Ignore the last reply. The correct answer would be C. Tags = Custom metadata fields that you can attach to a data entry to provide context. Tag templates = Reusable structures that you can use to rapidly create new tags. In short, the employees do not need a tagTemplateViewer role because it pertains to the tag templates, not the tags themselves.

NickForDiscussions 2 months ago

Selected Answer: D

"All employees must be able to do a simple search and find tables in the dataset that have either true or false in the “has_sensitive_data’ field." To be able to search for values in the tags you need the role roles/datacatalog.tagTemplateViewer. Meaning option D is correct.

meh_33 1 year, 7 months ago

Selected Answer: C

This Guy Raasd is mostly correct with explanation thanks mate.

iooj 1 year, 8 months ago

A - employees cannot use the tag B - increases the configuration overhead C - exactly what we need D - unnecessary role assignment, the tag template is already visibile

d11379b 2 years ago

Selected Answer: C

While D works well, it is not obligated to give all employees the role of tagTemplateViewer, as it will give them the view permission for tag templates as well as the tags created by the template. However, Tags are a type of business metadata. Adding tags to a data entry helps provide meaningful context to anyone who needs to use the asset.And public tags provide less strict access control for searching and viewing the tag as compared to private tags. Any user who has the required view permissions for a data entry can view all the public tags associated with it. View permissions for public tags are only required when you perform a search in Data Catalog using the tag: syntax or when you view an unattached tag template.

d11379b 2 years ago

As all employees have the role “ bigquery.metadataViewer” they are already capable to see tags on BigQuery then

JyoGCP 2 years, 1 month ago

Selected Answer: C

I'll go with raaad's answer

tibuenoc 2 years, 2 months ago

Selected Answer: B

If you working with PII, We can't granted public access. So Private Visibility for the Tag Template its the best option. Check it https://cloud.google.com/data-catalog/docs/tags-and-tag-templates

scaenruy 2 years, 2 months ago

Selected Answer: D

D. Create the “gdpr” tag template with public visibility. Assign the datacatalog.tagTemplateViewer role on this tag to the all employees group, and assign the bigquery.dataViewer role to the HR group on the tables that contain sensitive data.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #273 Topic 1

You are creating the CI/CD cycle for the code of the directed acyclic graphs (DAGs) running in Cloud Composer. Your team has two Cloud Composer instances: one instance for development and another instance for production. Your team is using a Git repository to maintain and develop the code of the DAGs. You want to deploy the DAGs automatically to Cloud Composer when a certain tag is pushed to the Git repository. What should you do?

A. 1. Use Cloud Build to copy the code of the DAG to the Cloud Storage bucket of the development instance for DAG testing.
2. If the tests pass, use Cloud Build to copy the code to the bucket of the production instance. **Most Voted**

B. 1. Use Cloud Build to build a container with the code of the DAG and the KubernetesPodOperator to deploy the code to the Google Kubernetes Engine (GKE) cluster of the development instance for testing.
2. If the tests pass, use the KubernetesPodOperator to deploy the container to the GKE cluster of the production instance.

C. 1. Use Cloud Build to build a container and the KubernetesPodOperator to deploy the code of the DAG to the Google Kubernetes Engine (GKE) cluster of the development instance for testing.
2. If the tests pass, copy the code to the Cloud Storage bucket of the production instance.

D. 1. Use Cloud Build to copy the code of the DAG to the Cloud Storage bucket of the development instance for DAG testing.
2. If the tests pass, use Cloud Build to build a container with the code of the DAG and the KubernetesPodOperator to deploy the container to the Google Kubernetes Engine (GKE) cluster of the production instance.

Correct Answer: A

Community vote distribution

A (100%)

Comments

BIGQUERY_ALT_ALT **Highly Voted** 2 years, 2 months ago

Selected Answer: A

The Answer is A. Given that there are two instances (development and production) already available, and the goal is to deploy DAGs to Cloud Composer not entire composer infra build. Explanation: - This approach leverages Cloud Build to manage the deployment process. - It first deploys the code to the Cloud Storage bucket of the development instance for testing purposes. - If the tests are successful in the development environment, the same Cloud Build process is used to copy the code to the Cloud Storage bucket of the production instance. B. GKE-based approach is

not standard for Cloud Composer. C. GKE used for testing is unconventional for DAG deployments. D. Involves unnecessary GKE deployment for production. Testing DAGs should use Composer instances directly, not Kubernetes containers in GKE.

meh_33 1 year, 7 months ago

Selected Answer: A

Most confusing question to confuse us why GKE needed its already mentioned they have 2 composer environment

JyoGCP 2 years, 1 month ago

Selected Answer: A

Option A

Matt_108 2 years, 2 months ago

Selected Answer: A

Option A, DAGs are routinely stored in cloud storage buckets, Cloud Build act as a trigger for both the deployment process to test env and the test itslef https://cloud.google.com/composer/docs/dag-cicd-integration-guide

Sofiia98 2 years, 2 months ago

Selected Answer: A

I vote fore A

GCP001 2 years, 2 months ago

C. It looks the correct choice, first build, test and verify everything on dev enviornment and then just copy the files on prod bucket. https://cloud.google.com/composer/docs/dag-cicd-integration-guide

Sofiia98 2 years, 2 months ago

But why do we need the Google Kubernetes Engine (GKE) cluster for this?

GCP001 2 years, 2 months ago

Yea, it should be A

Smakyel79 2 years, 2 months ago

Selected Answer: A

This approach is straightforward and leverages Cloud Build to automate the deployment process. It doesn't require containerization, making it simpler and possibly quicker.

https://zhangxuetu.online

[202610]

Question #274 Topic 1

You have a BigQuery table that ingests data directly from a Pub/Sub subscription. The ingested data is encrypted with a Google-managed encryption key. You need to meet a new organization policy that requires you to use keys from a centralized Cloud Key Management Service (Cloud KMS) project to encrypt data at rest. What should you do?

A. Use Cloud KMS encryption key with Dataflow to ingest the existing Pub/Sub subscription to the existing BigQuery table.

B. Create a new BigQuery table by using customer-managed encryption keys (CMEK), and migrate the data from the old BigQuery table. **Most Voted**

C. Create a new Pub/Sub topic with CMEK and use the existing BigQuery table by using Google-managed encryption key.

D. Create a new BigQuery table and Pub/Sub topic by using customer-managed encryption keys (CMEK), and migrate the data from the old BigQuery table.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- New BigQuery Table with CMEK: This option involves creating a new BigQuery table configured to use a CMEK from Cloud KMS. It directly addresses the need to use a CMEK for data at rest in BigQuery. - Migrate Data: Migrating data from the old table (encrypted with a Google-managed key) to the new table (encrypted with CMEK) ensures that all existing data complies with the new policy.

Matt_108 2 years, 2 months ago

But also pub/sub has some data at rest, e.g. messages with retention period. To comply with the organisation policy, we need to adapt also pub/sub

AllenChen123 2 years, 2 months ago

No, "The ingested data is encrypted with a Google-managed encryption key", target is ingested data in BigQuery.

ML6 2 years, 1 month ago

Correct, but the question states 'use keys from a centralized Cloud KMS project', so only D is correct.

cloud_rider 1 year, 4 months ago

PubSub is an application and holds data on the fly, this data does not mean data at rest. The data that is ingested in GSC only means data at rest so B is the right answer.

Blackstile 1 year, 1 month ago

The question did not say anything about the retention policy. Therefore, the correct answer is B. A tip for the exam: never answer what was not asked.

Matt_108 Highly Voted 2 years, 2 months ago

Selected Answer: D

Option D - I get the discussion about B and D, but also pub/sub has some data at rest, e.g. messages with retention period. To comply with the organisation policy, we need to adapt also pub/sub

Blackstile 1 year, 1 month ago

The question did not say anything about the retention policy. Therefore, the correct answer is B. A tip for the exam: never answer what was not asked.

plum21 1 year, 2 months ago

Selected Answer: D

There is data at rest in Pub/Sub, which is stated here in the docs: https://cloud.google.com/pubsub/docs/encryption At rest data -> Application layer -> CMEK encryption

m_a_p_s 1 year, 3 months ago

Selected Answer: B

B. You don't need to create a new topic in order to use the new CMEK. Existing topic can be updated to use the new key: https://cloud.google.com/pubsub/docs/encryption#update_cmek_for_a_topic

SamuelTsch 1 year, 5 months ago

Selected Answer: B

should be B. Pub/Sub is not designed for storing data at rest.

gr3yWind 1 year, 5 months ago

Selected Answer: B

Agree with raaad

shanks_t 1 year, 7 months ago

Selected Answer: D

Requirement for Cloud KMS keys: The new organization policy requires using keys from a centralized Cloud KMS project for encrypting data at rest. This necessitates the use of customer-managed encryption keys (CMEK). BigQuery table encryption: The existing BigQuery table is encrypted with a Google-managed key. To meet the new policy, a new table needs to be created with CMEK. Pub/Sub topic encryption: Since the data is ingested directly from a Pub/Sub subscription, the Pub/Sub topic also needs to use CMEK to ensure end-to-end encryption with customer-managed keys. Data migration: The existing data in the old BigQuery table needs to be migrated to the new CMEK-encrypted table to ensure all data complies with the new policy

carmltekai 1 year, 8 months ago

Selected Answer: B

"The best solution here is B. Create a new BigQuery table by using customer-managed encryption keys (CMEK), and migrate the data from the old BigQuery table. Here's why: Customer-Managed Encryption Keys (CMEK): CMEKs allow you to have granular control over your encryption keys, complying with the organization's policy to use keys from a centralized Cloud KMS project. Data Migration: Since the data in the existing table is already encrypted with a

Google-managed key, you cannot retroactively change the encryption key for that table. Migrating the data to a new table with the correct encryption is the most efficient way to meet compliance.

carmltekai 1 year, 8 months ago

Why other options aren't suitable: A: Dataflow can't retroactively change the encryption of data that's already in BigQuery. C: Creating a new Pub/Sub topic with CMEK wouldn't address the data that's already in BigQuery. D: While creating a new Pub/Sub topic might be useful in the long run, it's not necessary for solving the immediate compliance issue with the existing data."

iooj 1 year, 8 months ago

You have some data in Pub/Sub at rest as well which is immediate compliance issue.

Anudeep58 1 year, 9 months ago

Selected Answer: D

D. Create a new BigQuery table and Pub/Sub topic by using customer-managed encryption keys (CMEK), and migrate the data from the old BigQuery table. This approach comprehensively addresses the requirement to use CMEK from a centralized Cloud KMS project for encrypting data at rest: Create a new Pub/Sub topic configured to use CMEK from the centralized Cloud KMS project. Create a new BigQuery table with CMEK enabled, using the same centralized Cloud KMS project. Update the ingestion process to use the new Pub/Sub topic to feed data into the new BigQuery table. Migrate existing data from the old BigQuery table to the new BigQuery table to ensure all data complies with the new encryption policy.

AlizCert 1 year, 10 months ago

Selected Answer: B

B, been there, done that...

AlizCert 1 year, 10 months ago

sry, I mean D

josech 1 year, 10 months ago

Selected Answer: D

BigQuery and Pub/Sub shall be encrypted using CMEK using new versions of each one.
https://cloud.google.com/pubsub/docs/encryption#using-cmek

chrissamharris 1 year, 10 months ago

Selected Answer: B

Data at rest in requirement = Big Query ONLY. Pub/Sub is data in movement - overkill for the solution

f74ca0c 1 year, 10 months ago

Selected Answer: D

D- BigQuery and Pub/sub are automatically encrypted but here we need to apply a more secured policy by using CMEK so we need to use it for bigquery and pub/sub to meet this policy

LaxmanTiwari 1 year, 11 months ago

Selected Answer: B

B. Create a new BigQuery table by using customer-managed encryption keys (CMEK), and migrate the data from the old BigQuery table. Most Voted

LaxmanTiwari 1 year, 11 months ago

it should be B as the data in pub sub is already encrypted , please read it carefully and use Copilot or chat gpt to have confirmation.

amanbawa96 1 year, 12 months ago

Selected Answer: B

BigQuery allows you to encrypt data at rest using either Google-managed encryption keys or customer-managed encryption keys (CMEK) from Cloud KMS. Since the new policy requires using keys from a centralized Cloud KMS project, you need to create a new BigQuery table that is configured to use CMEK for encryption. After creating the new table with CMEK, you can migrate the data from the old table (encrypted with Google-managed keys) to the new table (encrypted with CMEK). This approach ensures that the data in the BigQuery table is encrypted using the required CMEK while preserving the existing data. Creating a new BigQuery table and Pub/Sub topic with CMEK is not necessary because the focus is on encrypting the data at rest in BigQuery. The existing Pub/Sub subscription can still be used to ingest data into the new BigQuery table.

Izzyt99 2 years ago

D - 'as new organization policy that requires you to use keys from a centralized Cloud Key Management Service (Cloud KMS) project to encrypt data at rest.' Therefore, the Pub/Sub default Google-managed encryption key is not sufficient as the organization requires it's own CMEK that is to be generated from a centralized Cloud KMS project.

https://zhangxuetu.online

[202610]

Question #275 Topic 1

You created an analytics environment on Google Cloud so that your data scientist team can explore data without impacting the on-premises Apache Hadoop solution. The data in the on-premises Hadoop Distributed File System (HDFS) cluster is in Optimized Row Columnar (ORC) formatted files with multiple columns of Hive partitioning. The data scientist team needs to be able to explore the data in a similar way as they used the on-premises HDFS cluster with SQL on the Hive query engine. You need to choose the most cost-effective storage and processing solution. What should you do?

A. Import the ORC files to Bigtable tables for the data scientist team.

B. Import the ORC files to BigQuery tables for the data scientist team.

C. Copy the ORC files on Cloud Storage, then deploy a Dataproc cluster for the data scientist team.

D. Copy the ORC files on Cloud Storage, then create external BigQuery tables for the data scientist team. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: D

- It leverages the strengths of BigQuery for SQL-based exploration while avoiding additional costs and complexity associated with data transformation or migration. - The data remains in ORC format in Cloud Storage, and BigQuery's external tables feature allows direct querying of this data.

nadavw 1 year, 7 months ago

There is a requirement to use a 'hive query engine'', and BQ is using only the hive metastore and his own engine, so 'D' seems a better fit here.

kaisarfarel **Highly Voted** 2 years ago

I think C is the correct answer, DS want to explore the data in a "similar way as they used the on-premises HDFS cluster with SQL on the Hive query engine". Dataproc can help to create clusters quickly with the Hadoop cluster. CMIIW

apoio.certificacoes.closer 1 year, 3 months ago

I think "Similar" is doing a lot of heavy lift on the confusion. If it was equal, I'd say C. Since it similar, it can be GoogleSQL (Bigquery).

56d02cd 9 months ago

Selected Answer: C

It says that scientists need to "explore the data with SQL on the Hive query engine". That excludes BigQuery.

SamuelTsch 1 year, 5 months ago

Selected Answer: B

using external tables have always limitations - affecting performance, no preview of the data and no cost estimation. So, why option D is correct?

hanoverquay 2 years ago

Selected Answer: D

option d

0725f1f 2 years ago

Selected Answer: C

it is talking about partition as well

JyoGCP 2 years, 1 month ago

Selected Answer: D

Option D

Matt_108 2 years, 2 months ago

Selected Answer: D

Option D - leverages BigQuery for SQL-based exploration on direct querying to cloud storage

Smakyel79 2 years, 2 months ago

Selected Answer: D

This approach leverages BigQuery's powerful analytics capabilities without the overhead of data transformation or maintaining a separate cluster, while also allowing your team to use SQL for data exploration, similar to their experience with the on-premises Hadoop/Hive environment.

https://zhangxuetu.online

掌学兔 ZHANGXUETU
[202610]

Question #276 Topic 1

You are designing a Dataflow pipeline for a batch processing job. You want to mitigate multiple zonal failures at job submission time. What should you do?

A. Submit duplicate pipelines in two different zones by using the --zone flag.

B. Set the pipeline staging location as a regional Cloud Storage bucket.

C. Specify a worker region by using the --region flag. **Most Voted**

D. Create an Eventarc trigger to resubmit the job in case of zonal failure when submitting the job.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Matt_108 **Highly Voted** 1 year, 8 months ago

Selected Answer: C

Option C: https://cloud.google.com/dataflow/docs/guides/pipeline-workflows#zonal-failures

raaad **Highly Voted** 1 year, 8 months ago

Selected Answer: C

- Specifying a worker region (instead of a specific zone) allows Google Cloud's Dataflow service to manage the distribution of resources across multiple zones within that region

JyoGCP 1 year, 7 months ago

Selected Answer: C

Option C

Sofiia98 1 year, 8 months ago

Selected Answer: C

https://cloud.google.com/dataflow/docs/guides/pipeline-workflows#zonal-failures

scaenruy 1 year, 9 months ago

Selected Answer: C

C. Specify a worker region by using the --region flag.

https://zhangxuetu.online

[202610]

Question #277 Topic 1

You are designing a real-time system for a ride hailing app that identifies areas with high demand for rides to effectively reroute available drivers to meet the demand. The system ingests data from multiple sources to Pub/Sub, processes the data, and stores the results for visualization and analysis in real-time dashboards. The data sources include driver location updates every 5 seconds and app-based booking events from riders. The data processing involves real-time aggregation of supply and demand data for the last 30 seconds, every 2 seconds, and storing the results in a low-latency system for visualization. What should you do?

A. Group the data by using a tumbling window in a Dataflow pipeline, and write the aggregated data to Memorystore.

B. Group the data by using a hopping window in a Dataflow pipeline, and write the aggregated data to Memorystore. **Most Voted**

C. Group the data by using a session window in a Dataflow pipeline, and write the aggregated data to BigQuery.

D. Group the data by using a hopping window in a Dataflow pipeline, and write the aggregated data to BigQuery.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- Hopping Window: Hopping windows are fixed-sized, overlapping intervals. - Aggregate data over the last 30 seconds, every 2 seconds, as hopping windows allow for overlapping data analysis. - Memorystore: Ideal for low-latency access required for real-time visualization and analysis.

anushree09 1 year, 11 months ago

Hopping windows are sliding windows. It makes sense to use that over tumbling (fixed) window because the ask is to collect last 30 seconds of data every 5 second

Jeyaraj 1 year, 8 months ago

OPTION A. (IGNORE MY Previous Comment) Tumbling windows are the best choice for this ride-hailing app because they provide accurate 2-second aggregations without the complexities of overlapping data. This is crucial for real-time decision-making and ensuring accurate visualization of supply and demand. Hopping windows introduce potential

inaccuracies and complexity, making them less suitable for this scenario. While they can be useful in other situations, they are not the optimal choice for real-time aggregation with strict accuracy requirements.

Jeyaraj 1 year, 8 months ago

Option B. Tumbling windows are the best choice for this ride-hailing app because they provide accurate 2-second aggregations without the complexities of overlapping data. This is crucial for real-time decision-making and ensuring accurate visualization of supply and demand. Hopping windows introduce potential inaccuracies and complexity, making them less suitable for this scenario. While they can be useful in other situations, they are not the optimal choice for real-time aggregation with strict accuracy requirements.

JyoGCP 2 years, 1 month ago

Selected Answer: B

Option B

ashdam 2 years, 1 month ago

hopping window is clear but memorystore vs bigquery?? Why memorystore and not bigquery?

ML6 2 years, 1 month ago

Memory store is an in-memory key-value database for use cases such as real-time application.

ea2023 1 year, 11 months ago

Let me complete your answer MS vs BQ in this case is a matter of low latence where MS is the winner but if precision were stated about a large amount of data BQ then would"ve been the best choice.

Jordan18 2 years, 2 months ago

why not D?

RenePetersen 2 years, 1 month ago

Because BigQuery is not a low latency system...

scaenruy 2 years, 2 months ago

Selected Answer: B

B. Group the data by using a hopping window in a Dataflow pipeline, and write the aggregated data to Memorystore.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #278 Topic 1

Your car factory is pushing machine measurements as messages into a Pub/Sub topic in your Google Cloud project. A Dataflow streaming job, that you wrote with the Apache Beam SDK, reads these messages, sends acknowledgment to Pub/Sub, applies some custom business logic in a DoFn instance, and writes the result to BigQuery. You want to ensure that if your business logic fails on a message, the message will be sent to a Pub/Sub topic that you want to monitor for alerting purposes. What should you do?

A. Enable retaining of acknowledged messages in your Pub/Sub pull subscription. Use Cloud Monitoring to monitor the subscription/num_retained_acked_messages metric on this subscription.

B. Use an exception handling block in your Dataflow's DoFn code to push the messages that failed to be transformed through a side output and to a new Pub/Sub topic. Use Cloud Monitoring to monitor the topic/num_unacked_messages_by_region metric on this new topic. Most Voted

C. Enable dead lettering in your Pub/Sub pull subscription, and specify a new Pub/Sub topic as the dead letter topic. Use Cloud Monitoring to monitor the subscription/dead_letter_message_count metric on your pull subscription.

D. Create a snapshot of your Pub/Sub pull subscription. Use Cloud Monitoring to monitor the snapshot/num_messages metric on this snapshot.

Correct Answer: B

Community vote distribution

B (100%)

Comments

raaad Highly Voted 2 years, 2 months ago

Selected Answer: B

- Exception Handling in DoFn: Implementing an exception handling block within DoFn in Dataflow to catch failures during processing is a direct way to manage errors. - Side Output to New Topic: Using a side output to redirect failed messages to a new Pub/Sub topic is an effective way to isolate and manage these messages. - Monitoring: Monitoring the num_unacked_messages_by_region on the new topic can alert you to the presence of failed messages.

chrissamharris 1 year, 6 months ago

Option C - dead letter topic is built in and requires no changes https://cloud.google.com/pubsub/docs/handling-failures Enable dead lettering in your Pub/Sub pull subscription, and specify a new Pub/Sub topic as the dead letter topic. Use Cloud Monitoring to monitor the subscription/dead_letter_message_count metric on your pull subscription.

Positron75 1 year, 4 months ago

Dead lettering is used to handle messages that have not been acknowledged, but that's unrelated to the processing that Dataflow does, which takes place later in the chain. A message could still be acknowledged and fail processing for whatever reason, so it would not be sent to the dead letter topic. Also, Google advises against using dead lettering with Dataflow anyway: https://cloud.google.com/dataflow/docs/concepts/streaming-with-cloud-pubsub#dead-letter-topics Correct answer is B. The error handling has to be written into the Dataflow pipeline itself.

7787de3 1 year, 6 months ago

Selected Answer: B

See here: https://cloud.google.com/dataflow/docs/concepts/streaming-with-cloud-pubsub#unsupported-features It's not recommended to use Pub/Sub dead-letter topics with Dataflow (...) Instead, implement the dead-letter pattern explicitly in the pipeline

Jeyaraj 1 year, 8 months ago

Option B. Here's why: Side Output for Failed Messages: Dataflow allows you to use side outputs to handle messages that fail processing. In your DoFn , you can catch exceptions and write the failed messages to a separate PCollection . This PCollection can then be written to a new Pub/Sub topic. New Pub/Sub Topic for Monitoring: Creating a dedicated Pub/Sub topic for failed messages allows you to monitor it specifically for alerting purposes. This provides a clear view of any issues with your business logic. topic/num_unacked_messages_by_region Metric: This Cloud Monitoring metric tracks the number of unacknowledged messages in a Pub/Sub topic. By monitoring this metric on your new topic, you can identify when messages are failing to be processed correctly.

joao_01 1 year, 11 months ago

I would like to know why isn't anyone considering the option C.

joao_01 1 year, 11 months ago

I think that C is not right anyways: In order to use dead_letter feature, the message CANNOT be acknowledge (somehow) by the subscriber. In this question it says that the messages are first acknowledge and then applied the business logic. So, if there are error in the business logic we cannot use the feature dead_letter, beacuse the message was already acknowledge. Thus, option B is the right one.

hanoverquay 2 years ago

Selected Answer: B

option B

JyoGCP 2 years, 1 month ago

Selected Answer: B

Option B

Matt_108 2 years, 2 months ago

Selected Answer: B

Option B - Raaad explanation is complete

scaenruy 2 years, 2 months ago

Selected Answer: B

B. Use an exception handling block in your Dataflow's DoFn code to push the messages that failed to be transformed through a side output and to a new Pub/Sub topic. Use Cloud Monitoring to monitor the topic/num_unacked_messages_by_region metric on this new topic.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #279 Topic 1

You want to store your team's shared tables in a single dataset to make data easily accessible to various analysts. You want to make this data readable but unmodifiable by analysts. At the same time, you want to provide the analysts with individual workspaces in the same project, where they can create and store tables for their own use, without the tables being accessible by other analysts. What should you do?

A. Give analysts the BigQuery Data Viewer role at the project level. Create one other dataset, and give the analysts the BigQuery Data Editor role on that dataset.

B. Give analysts the BigQuery Data Viewer role at the project level. Create a dataset for each analyst, and give each analyst the BigQuery Data Editor role at the project level.

C. Give analysts the BigQuery Data Viewer role on the shared dataset. Create a dataset for each analyst, and give each analyst the BigQuery Data Editor role at the dataset level for their assigned dataset. **Most Voted**

D. Give analysts the BigQuery Data Viewer role on the shared dataset. Create one other dataset and give the analysts the BigQuery Data Editor role on that dataset.

Correct Answer: C

Community vote distribution

C (100%)

Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

- Data Viewer on Shared Dataset: Grants read-only access to the shared dataset. - Data Editor on Individual Datasets: Giving each analyst Data Editor role on their respective dataset creates private workspaces where they can create and store personal tables without exposing them to other analysts.

meh_33 1 year, 7 months ago

Selected Answer: C

Will GO with C

hanoverquay 2 years ago

Selected Answer: C

voted C

JyoGCP 2 years, 1 month ago

Selected Answer: C

Option C

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C

Sofiia98 2 years, 2 months ago

Selected Answer: C

option C, because analysts can not see the individual datasets of other analysts

scaenruy 2 years, 2 months ago

Selected Answer: C

C. Give analysts the BigQuery Data Viewer role on the shared dataset. Create a dataset for each analyst, and give each analyst the BigQuery Data Editor role at the dataset level for their assigned dataset.

https://zhangxuetu.online

[202610]

Question #280 Topic 1

You are running a streaming pipeline with Dataflow and are using hopping windows to group the data as the data arrives. You noticed that some data is arriving late but is not being marked as late data, which is resulting in inaccurate aggregations downstream. You need to find a solution that allows you to capture the late data in the appropriate window. What should you do?

A. Use watermarks to define the expected data arrival window. Allow late data as it arrives. **Most Voted**

B. Change your windowing function to tumbling windows to avoid overlapping window periods.

C. Change your windowing function to session windows to define your windows based on certain activity.

D. Expand your hopping window so that the late data has more time to arrive within the grouping.

Correct Answer: A

Community vote distribution

A (100%)

Comments

raaad **Highly Voted** 1 year, 8 months ago

Selected Answer: A

- Watermarks: Watermarks in a streaming pipeline are used to specify the point in time when Dataflow expects all data up to that point to have arrived. - Allow Late Data: configure the pipeline to accept and correctly process data that arrives after the watermark, ensuring it's captured in the appropriate window.

m_a_p_s 1 year, 3 months ago

Selected Answer: A

A - https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#watermarks

JyoGCP 1 year, 7 months ago

Selected Answer: A

Option A

Matt_108 1 year, 8 months ago

Selected Answer: A

Option A - https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#watermarks

Sofiia98 1 year, 8 months ago

Selected Answer: A

https://cloud.google.com/dataflow/docs/concepts/streaming-pipelines#watermarks

scaenruy 1 year, 9 months ago

Selected Answer: A

A. Use watermarks to define the expected data arrival window. Allow late data as it arrives.

https://zhangxuetu.online

[202610]

Question #281 Topic 1

You work for a large ecommerce company. You store your customer's order data in Bigtable. You have a garbage collection policy set to delete the data after 30 days and the number of versions is set to 1. When the data analysts run a query to report total customer spending, the analysts sometimes see customer data that is older than 30 days. You need to ensure that the analysts do not see customer data older than 30 days while minimizing cost and overhead. What should you do?

A. Set the expiring values of the column families to 29 days and keep the number of versions to 1.

B. Use a timestamp range filter in the query to fetch the customer's data for a specific range. **Most Voted**

C. Schedule a job daily to scan the data in the table and delete data older than 30 days.

D. Set the expiring values of the column families to 30 days and set the number of versions to 2.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Matt_108 Highly Voted 1 year, 8 months ago

Selected Answer: B

Agree with others https://cloud.google.com/bigtable/docs/garbage-collection

AllenChen123 1 year, 8 months ago

Agree. https://cloud.google.com/bigtable/docs/garbage-collection#data-removed "Because it can take up to a week for expired data to be deleted, you should never rely solely on garbage collection policies to ensure that read requests return the desired data. Always apply a filter to your read requests that excludes the same values as your garbage collection rules. You can filter by limiting the number of cells per column or by specifying a timestamp range."

cuadradobertolinisebastiancami Highly Voted 1 year, 7 months ago

Selected Answer: B

Agree with MAtt_108 and AllenChen 123. "Garbage collection is a continuous process in which Bigtable checks the rules for each column family and deletes expired and obsolete data accordingly. In general, it can take up to a week from the time that data matches the criteria in the rules for the data to actually be deleted. You are not able to change the timing of garbage collection." "Always apply a filter to your read requests that exclude the same values as your garbage collection rules. " Ref: https://cloud.google.com/bigtable/docs/garbage-collection#data-removed

Ben_oso 9 months, 1 week ago

Selected Answer: B

Its "B", but the filter responsability transfer to user, so, this dont ensure that him filter the data.

m_a_p_s 1 year, 3 months ago

Selected Answer: B

"Because it can take up to a week for expired data to be deleted, you should never rely solely on garbage collection policies to ensure that read requests return the desired data. Always apply a filter to your read requests that excludes the same values as your garbage collection rules. You can filter by limiting the number of cells per column or by specifying a timestamp range." https://cloud.google.com/bigtable/docs/garbage-collection#data-removed

Sofiia98 1 year, 8 months ago

Selected Answer: B

I will go for B too

GCP001 1 year, 8 months ago

B. Use a timestamp range filter in the query to fetch the customer's data for a specific range. Always use query filter as garbage collectore runs on it's way - https://cloud.google.com/bigtable/docs/garbage-collection

scaenruy 1 year, 9 months ago

Selected Answer: B

B. Use a timestamp range filter in the query to fetch the customer's data for a specific range.

https://zhangxuetu.online

[202610]

Question #282 Topic 1

You are using a Dataflow streaming job to read messages from a message bus that does not support exactly-once delivery. Your job then applies some transformations, and loads the result into BigQuery. You want to ensure that your data is being streamed into BigQuery with exactly-once delivery semantics. You expect your ingestion throughput into BigQuery to be about 1.5 GB per second. What should you do?

A. Use the BigQuery Storage Write API and ensure that your target BigQuery table is regional.

B. Use the BigQuery Storage Write API and ensure that your target BigQuery table is multiregional. **Most Voted**

C. Use the BigQuery Streaming API and ensure that your target BigQuery table is regional.

D. Use the BigQuery Streaming API and ensure that your target BigQuery table is multiregional.

Correct Answer: B

Community vote distribution



Comments

AlizCert **Highly Voted** 1 year, 10 months ago

Selected Answer: B

It should B, Storage Write API has "3 GB per second throughput in multi-regions; 300 MB per second in regions"

rajshiv 11 months, 3 weeks ago

B is incorrect. Multiregional tables are not supported by the Storage Write API for exactly-once delivery. This option is invalid.

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: A

- BigQuery Storage Write API: This API is designed for high-throughput, low-latency writing of data into BigQuery. It also provides tools to prevent data duplication, which is essential for exactly-once delivery semantics. - Regional Table: Choosing a regional location for the BigQuery table could potentially provide better performance and lower latency, as it would be closer to the Dataflow job if they are in the same region.

AllenChen123 2 years, 2 months ago

Agree. https://cloud.google.com/bigquery/docs/write-api#advantages

22c1725 10 months, 1 week ago

Max throughput for regional currently only 300 MB/s as for the dcos

Kalai_1 3 months, 1 week ago

Selected Answer: A

BigQuery is Regional component and not Multiregional. Hence Option:A

50336e5 3 months, 1 week ago

Selected Answer: A

API BigQuery Storage Write with region because lower latency.

lmch 3 months, 2 weeks ago

Selected Answer: B

To achieve high-scale, exactly-once ingestion in BigQuery: API: Always use the Storage Write API. Mode: Use Committed Streams (for exactly-once) or the Default Stream (for at-least-once). Scalability: Monitor the "Bytes per second" quota. If you are operating at "Gbps" levels, multiregional locations are the architectural standard due to higher throughput limits.

imrane1995 8 months, 2 weeks ago

Selected Answer: A

Why A is correct: BigQuery Storage Write API: Supports exactly-once semantics (deduplication via stream offsets). Designed for high-throughput streaming ingestion (scales much better than the legacy Streaming Insert API). Can handle your throughput requirement of 1.5 GB/s. Regional table: Required for exactly-once delivery guarantees with the Storage Write API.

Ben_oso 9 months, 1 week ago

Selected Answer: B

3 GB per second throughput in multi-regions; 300 MB per second in regions

22c1725 10 months ago

Selected Answer: B

Go With (B) Not (A), Max throughput for regional currently only 300 MB/s as for the dcos

22c1725 10 months, 1 week ago

Selected Answer: A

Honestly, I think exam topic should do better job. the answers only make more mislead.

aditya_ali 11 months ago

Selected Answer: A

You need a write latency of 1.5 GBs per second. Given the high throughput requirement, a regional BigQuery table (Option A) is generally preferred over a multi-regional table due to potentially lower write latency in multi-region. Simple.

Aungshuman 11 months ago

Selected Answer: B

As per GCP document multi-region meets the troughput requirement.

gabbferreira 11 months, 1 week ago

Selected Answer: A

It’s A

Siahara 1 year, 1 month ago

Selected Answer: A

A. Implement the BigQuery Storage Write API and guarantee that the target BigQuery table is regional. Here's the breakdown: Why Option A is Superior Exactly-Once Delivery: The BigQuery Storage Write API intrinsically supports exactly-once delivery using stream offsets. This guarantees that each message is written to BigQuery exactly one time, even in the case of retries due to the lack of native exactly-once support in your message bus. High Throughput: The Storage Write API is optimized for high-throughput scenarios. It can handle the expected ingestion throughput of 1.5 GB per second. Regional Tables: Using a regional BigQuery table aligns with best practices when utilizing the Storage Write API, as it helps to minimize latency and reduce potential cross-region communication costs.

gord_nat 1 year ago

Has to be multi-regional (B) Max throughput for regional currently only 300 MB/s
https://cloud.google.com/bigquery/quotas

juliorevk 1 year, 2 months ago

Selected Answer: B

- BigQuery Storage Write API: This API is designed for high-throughput, low-latency writing of data into BigQuery. It also provides tools to prevent data duplication, which is essential for exactly-once delivery semantics. - The multiregional table ensures that your data is highly available and can be streamed into BigQuery across multiple regions. It is better suited for high-throughput and low-latency workloads, as it provides distributed write capabilities that can handle large data volumes, such as the 1.5 GB per second you expect to stream.

hussain.sain 1 year, 3 months ago

Selected Answer: B

B is correct. When aiming for exactly-once delivery in a Dataflow streaming job, the key is to use the BigQuery Storage Write API, as it provides the capability to handle large-scale data ingestion with the correct semantics, including exactly-once delivery.

himadri1983 1 year, 3 months ago

Selected Answer: B

3 GB per second throughput in multi-regions; 300 MB per second in regions
https://cloud.google.com/bigquery/quotas#write-api-limits

m_a_p_s 1 year, 3 months ago

Selected Answer: B

streamed into BigQuery with exactly-once delivery semantics >>> Storage Write API ingestion throughput into BigQuery to be about 1.5 GB per second >>> multiregional (check throughput rate here >>> https://cloud.google.com/bigquery/quotas#write-api-limits)

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #283 Topic 1

You have created an external table for Apache Hive partitioned data that resides in a Cloud Storage bucket, which contains a large number of files. You notice that queries against this table are slow. You want to improve the performance of these queries. What should you do?

A. Change the storage class of the Hive partitioned data objects from Coldline to Standard.

B. Create an individual external table for each Hive partition by using a common table name prefix. Use wildcard table queries to reference the partitioned data.

C. Upgrade the external table to a BigLake table. Enable metadata caching for the table. **Most Voted**

D. Migrate the Hive partitioned data objects to a multi-region Cloud Storage bucket.

Correct Answer: C

Community vote distribution



Comments

raaad **Highly Voted** 1 year, 8 months ago

Selected Answer: C

- BigLake Table: BigLake allows for more efficient querying of data lakes stored in Cloud Storage. It can handle large datasets more effectively than standard external tables. - Metadata Caching: Enabling metadata caching can significantly improve query performance by reducing the time taken to read and process metadata from a large number of files.

AllenChen123 1 year, 8 months ago

Agree. https://cloud.google.com/bigquery/docs/biglake-intro#metadata_caching_for_performance

AllenChen123 1 year, 8 months ago

And https://cloud.google.com/bigquery/docs/external-data-cloud-storage#upgrade-external-tables-to-biglake-tables

hanoverquay 1 year, 6 months ago

Selected Answer: C

vote C

JyoGCP 1 year, 7 months ago

Selected Answer: C

Option C

Matt_108 1 year, 8 months ago

Selected Answer: C

Option C

Sofiia98 1 year, 8 months ago

Selected Answer: C

agree with C

GCP001 1 year, 8 months ago

C. Upgrade the external table to a BigLake table. Enable metadata caching for the table. Check ref - https://cloud.google.com/bigquery/docs/biglake-intro

https://zhangxuetu.online

[202610]

Question #284 Topic 1

You have a network of 1000 sensors. The sensors generate time series data: one metric per sensor per second, along with a timestamp. You already have 1 TB of data, and expect the data to grow by 1 GB every day. You need to access this data in two ways. The first access pattern requires retrieving the metric from one specific sensor stored at a specific timestamp, with a median single-digit millisecond latency. The second access pattern requires running complex analytic queries on the data, including joins, once a day. How should you store this data?

A. Store your data in BigQuery. Concatenate the sensor ID and timestamp, and use it as the primary key.

B. Store your data in Bigtable. Concatenate the sensor ID and timestamp and use it as the row key. Perform an export to BigQuery every day. **Most Voted**

C. Store your data in Bigtable. Concatenate the sensor ID and metric, and use it as the row key. Perform an export to BigQuery every day.

D. Store your data in BigQuery. Use the metric as a primary key.

Correct Answer: B

Community vote distribution

B (100%)

Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- Bigtable excels at incredibly fast lookups by row key, often reaching single-digit millisecond latencies. - Constructing the row key with sensor ID and timestamp enables efficient retrieval of specific sensor readings at exact timestamps. - Bigtable's wide-column design effectively stores time series data, allowing for flexible addition of new metrics without schema changes. - Bigtable scales horizontally to accommodate massive datasets (petabytes or more), easily handling the expected data growth.

22c1725 10 months, 1 week ago

Selected Answer: B

If anyone in the future answers other then B I wouldn't be suprise.

fitri001 1 year, 9 months ago

Selected Answer: B

agree with raaad

hanoverquay 2 years ago

Selected Answer: B

voted b

JyoGCP 2 years, 1 month ago

Selected Answer: B

Option B

Matt_108 2 years, 2 months ago

Selected Answer: B

Option B - agree with raaad

scaenruy 2 years, 2 months ago

Selected Answer: B

B. Store your data in Bigtable. Concatenate the sensor ID and timestamp and use it as the row key. Perform an export to BigQuery every day.

Smakyel79 2 years, 2 months ago

Based on your requirements, Option B seems most suitable. Bigtable's design caters to the low-latency access of time-series data (your first requirement), and the daily export to BigQuery enables complex analytics (your second requirement). The use of sensor ID and timestamp as the row key in Bigtable would facilitate efficient access to specific sensor data at specific times.

https://zhangxuetu.online

[202610]

Question #285 Topic 1

You have 100 GB of data stored in a BigQuery table. This data is outdated and will only be accessed one or two times a year for analytics with SQL. For backup purposes, you want to store this data to be immutable for 3 years. You want to minimize storage costs. What should you do?

A. 1. Create a BigQuery table clone.
2. Query the clone when you need to perform analytics.

B. 1. Create a BigQuery table snapshot.
2. Restore the snapshot when you need to perform analytics.

C. 1. Perform a BigQuery export to a Cloud Storage bucket with archive storage class.
2. Enable versioning on the bucket.
3. Create a BigQuery external table on the exported files.

D. 1. Perform a BigQuery export to a Cloud Storage bucket with archive storage class.
2. Set a locked retention policy on the bucket.
3. Create a BigQuery external table on the exported files. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: D

Straight Forward

judy_data 7 months ago

Selected Answer: D

exporting the table to a bucket is the most straight forward and cost effective solution. C is not suited because it mentions enable versioning for a table that wont change and is immutable for 3 years. and it doesn't mention retention

fitri001 1 year, 9 months ago

Selected Answer: D

For data keeping till last 3 years, use bucket lock with rentention policy

hanoverquay 2 years ago

Selected Answer: D

voted D

JyoGCP 2 years, 1 month ago

Selected Answer: D

Option D

Matt_108 2 years, 2 months ago

Selected Answer: D

Option D, clearly

GCP001 2 years, 2 months ago

D. For data keeping till last 3 years, use bucket lock with rentention policy

https://zhangxuetu.online

[202610]

Question #286 Topic 1

You have thousands of Apache Spark jobs running in your on-premises Apache Hadoop cluster. You want to migrate the jobs to Google Cloud. You want to use managed services to run your jobs instead of maintaining a long-lived Hadoop cluster yourself. You have a tight timeline and want to keep code changes to a minimum. What should you do?

A. Move your data to BigQuery. Convert your Spark scripts to a SQL-based processing approach.

B. Rewrite your jobs in Apache Beam. Run your jobs in Dataflow.

C. Copy your data to Compute Engine disks. Manage and run your jobs directly on those instances.

D. Move your data to Cloud Storage. Run your jobs on Dataproc. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

hussain.sain 1 year, 3 months ago

Selected Answer: D

D is correct. Dataproc is the most suitable choice for migrating your existing Apache Spark jobs to Google Cloud because it is a fully managed service that supports Apache Spark and Hadoop workloads with minimal changes to your existing code. Moving your data to Cloud Storage and running jobs on Dataproc offers a fast, efficient, and scalable solution for your needs.

meh_33 1 year, 7 months ago

Selected Answer: D

option D, minimum code changes

hanoverquay 2 years ago

Selected Answer: D

option D, minimum code changes

JyoGCP 2 years, 1 month ago

Selected Answer: D

Option D

ML6 2 years, 1 month ago

Selected Answer: D

D) That is what Dataproc is made for. It is a fully managed and highly scalable service for running Apache Hadoop, Apache Spark, etc.

Matt_108 2 years, 2 months ago

Selected Answer: D

Clearly D

Sofiia98 2 years, 2 months ago

Selected Answer: D

of course D

GCP001 2 years, 2 months ago

D. Move your data to Cloud Storage. Run your jobs on Dataproc. Dataproc is managed service and not needed much code changes.

scaenruy 2 years, 2 months ago

Selected Answer: D

D. Move your data to Cloud Storage. Run your jobs on Dataproc.

https://zhangxuetu.online

[202610]

Question #287 Topic 1

You are administering shared BigQuery datasets that contain views used by multiple teams in your organization. The marketing team is concerned about the variability of their monthly BigQuery analytics spend using the on-demand billing model. You need to help the marketing team establish a consistent BigQuery analytics spend each month. What should you do?

A. Create a BigQuery Enterprise reservation with a baseline of 250 slots and autoscaling set to 500 for the marketing team, and bill them back accordingly.

B. Establish a BigQuery quota for the marketing team, and limit the maximum number of bytes scanned each day.

C. Create a BigQuery reservation with a baseline of 500 slots with no autoscaling for the marketing team, and bill them back accordingly. **Most Voted**

D. Create a BigQuery Standard pay-as-you go reservation with a baseline of 0 slots and autoscaling set to 500 for the marketing team, and bill them back accordingly.

Correct Answer: C

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

Reservations guarantee a fixed number of slots (computational resources) for BigQuery queries, ensuring a predictable monthly cost, addressing the marketing team's concern about variability.

AllenChen123 2 years, 2 months ago

Why 500 slots?

AllenChen123 2 years, 2 months ago

But seems only C makes sense. https://cloud.google.com/bigquery/quotas#query_jobs "There is no limit to the number of bytes that can be processed by queries in a project."

datapassionate 2 years, 2 months ago

"However, you can set limits on the amount of data users can query by creating custom quotas to control query usage per day or query usage per day per user." https://cloud.google.com/blog/products/data-analytics/manage-bigquery-costs-with-custom-quotas B would be correct

saschak94 2 years, 1 month ago

If you use B - the marketing team wouldn't be able to run their queries when the quota is reached, which could harm the business. Having a reservation for 500 slots and no autoscaling gives you exact predictable cost for each month without harming the business or have variable cost with autoscaling So C should be the right answer

22c1725 10 months ago

Selected Answer: C

I would go with (C) consistent is required. I have read some comment any arugment aginst (C) could be said about (A) while losing the consistent part.

22c1725 10 months, 1 week ago

Selected Answer: C

I would go with raaad

22c1725 10 months, 1 week ago

Selected Answer: C

C: "You need to help the marketing team establish a consistent BigQuery analytics spend each month" He wants it just give him what he want. Don't over think it.

gabbferreira 11 months, 1 week ago

Selected Answer: C

It’s C

desertlotus1211 1 year ago

Selected Answer: C

Answer A - Autoscaling introduces variable costs — which defeats the goal of cost consistency Answer B: it doesn’t convert to predictable costs — on-demand billing still applies per scan. Answer C is best.

MarcoPellegrino 1 year, 1 month ago

Selected Answer: B

The input doesn't specify the consistent monthly spent. hence, A, C, and D can't be used

desertlotus1211 1 year ago

Answer A - Autoscaling introduces variable costs — which defeats the goal of cost consistency Answer B: it doesn’t convert to predictable costs — on-demand billing still applies per scan. Answer C is best.

desertlotus1211 1 year ago

'The marketing team is concerned about the variability of their monthly BigQuery analytics spend using the on-demand billing model' so yes - this implies wanting consistent spend.

Augustax 1 year, 1 month ago

Selected Answer: A

Estimate a consistent spending doesn't mean overpay...

Maxd 1 year, 1 month ago

A because allow flexibility and scaling, so setting a baseline with autoscaling ensures that the marketing team can handle their queries without large fluctuations in cost.

b3e59c2 1 year, 2 months ago

Selected Answer: C

C seems much more robust and reliable than B. We can keep spend consistent whilst not sacrificing on performance (if we do B, once the byte scan limit has been reached, users will not be able to perform any analysis which could be

detrimental to business)

himadri1983 1 year, 3 months ago

Selected Answer: C

This is trick question. The answer B is setting quota on bytes but it does not address the cost variability. The C will give the predictable monthly cost.

m_a_p_s 1 year, 3 months ago

Selected Answer: C

Answer appears to be C. Check the example from docs: https://cloud.google.com/bigquery/docs/reservations-workload-management#managing_your_workloads_and_departments_using_reservations

CloudAdrMX 1 year, 3 months ago

It's a treaking question but it's C, they are asking for establish a consistent Bigquery analytics spend each month, if you put 500 slots as baseline and with no autoscaling, each month they'll get the a consistent Bigquery analytics spend.

cloud_rider 1 year, 4 months ago

Selected Answer: B

A, C and D talks about slot counts, whereas question does not talk about any such requirement, we should not make assumption on slots required or not required. Option B provides the visibility of cost to the team and can be revised as needed. So B is the right option.

8284a4c 1 year, 5 months ago

Selected Answer: A

The correct answer is: A. Create a BigQuery Enterprise reservation with a baseline of 250 slots and autoscaling set to 500 for the marketing team, and bill them back accordingly. Here's the rationale: Consistent Spend with Reservation: Creating a BigQuery Enterprise reservation provides the marketing team with dedicated slots, which can help stabilize and predict their monthly costs. By having a reservation baseline of 250 slots, they are guaranteed a certain level of performance and cost each month. Autoscaling for Flexibility: The autoscaling up to 500 slots allows the team to handle spikes in demand without being constrained by the fixed slot count. Autoscaling in this scenario enables some flexibility while still providing predictable spending due to the baseline. Billing Back: The reservation model allows for internal chargeback by department based on slot usage, helping the marketing team plan a predictable budget.

mi_yulai 1 year, 5 months ago

Answer is B. Custom quotas are a powerful feature that allow you to set hard limits on specific resource usage. In the case of BigQuery, quotas allow you to control query usage (number of bytes processed) at a project- or user-level. Project-level custom quotas limit the aggregate usage of all users in that project, while user-level custom quotas are separately applied to each user or service account within a project. Custom quotas are relevant when you are using BigQuery's on-demand pricing model, which charges for the number of bytes processed by each query. When you are using the capacity pricing model, you are charged for compute capacity (measured in slots) used to run queries, so limiting the number of bytes processed is less useful. By setting custom quotas, you can control the amount of query usage by different teams, applications, or users within your organization, preventing unexpected spikes in usage and costs.

baimus 1 year, 5 months ago

Selected Answer: C

Just to clarify a point of confusion: setting a quota does not affect variability (as specified in the question). It means there is a limit to the maximum but it can still vary anywhere between zero and that maximum each month. It would

also prevent the marking team actually performing the queries if set too low. C is the only one that makes sense, though the question "why 500" is a valid one, all the other answers simply do not deliver the requirements.

https://zhangxuetu.online

[202610]

Question #288 Topic 1

You are part of a healthcare organization where data is organized and managed by respective data owners in various storage services. As a result of this decentralized ecosystem, discovering and managing data has become difficult. You need to quickly identify and implement a cost-optimized solution to assist your organization with the following:

• Data management and discovery
• Data lineage tracking
• Data quality validation

How should you build the solution?

A. Use BigLake to convert the current solution into a data lake architecture.

B. Build a new data discovery tool on Google Kubernetes Engine that helps with new source onboarding and data lineage tracking.

C. Use BigQuery to track data lineage, and use Dataprep to manage data and perform data quality validation.

D. Use Dataplex to manage data, track data lineage, and perform data quality validation. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

987af6b 1 year, 8 months ago

Selected Answer: D

D. Dataplex

fitri001 1 year, 9 months ago

Selected Answer: D

Option D, no doubt

hanoverquay 2 years ago

Selected Answer: D

Option D, no doubt

JyoGCP 2 years, 1 month ago

Selected Answer: D

Option D

Matt_108 2 years, 2 months ago

Selected Answer: D

Clearly D

Sofiia98 2 years, 2 months ago

Selected Answer: D

Agree with Dataplex option

raaad 2 years, 2 months ago

Selected Answer: D

Straight forward

scaenruy 2 years, 2 months ago

Selected Answer: D

D. Use Dataplex to manage data, track data lineage, and perform data quality validation.

https://zhangxuetu.online

[202610]

Question #289 Topic 1

You have data located in BigQuery that is used to generate reports for your company. You have noticed some weekly executive report fields do not correspond to format according to company standards. For example, report errors include different telephone formats and different country code identifiers. This is a frequent issue, so you need to create a recurring job to normalize the data. You want a quick solution that requires no coding. What should you do?

A. Use Cloud Data Fusion and Wrangler to normalize the data, and set up a recurring job. **Most Voted**

B. Use Dataflow SQL to create a job that normalizes the data, and that after the first run of the job, schedule the pipeline to execute recurrently.

C. Create a Spark job and submit it to Dataproc Serverless.

D. Use BigQuery and GoogleSQL to normalize the data, and schedule recurring queries in BigQuery.

Correct Answer: A

Community vote distribution

A (83%) D (17%)

Comments

Matt_108 **Highly Voted** 2 years, 2 months ago

Selected Answer: A

Definitely A, cloud data fusion and wrangler to setup the clean up pipeline with no coding required

22c1725 10 months, 1 week ago

Selected Answer: A

A. Use Cloud Data Fusion and Wrangler to normalize the data, and set up a recurring job.

marlon.andrei 1 year, 2 months ago

Selected Answer: D

The question say "You want a quick solution that requires no coding.". The data is in BQ, then is most easy normalize the data, and schedule recurring queries in BigQuery.

22c1725 10 months ago

No code, SQL sure is coding.

987af6b 1 year, 8 months ago

Selected Answer: A

A. Use Cloud Data Fusion and Wrangler to normalize the data, and set up a recurring job. Explanation No Coding Required: Cloud Data Fusion's Wrangler offers a no-code interface for data transformation tasks. You can visually

design data normalization workflows without writing any code. Recurring Jobs: Cloud Data Fusion allows you to schedule these data normalization tasks to run on a recurring basis, meeting your need for automation.

carmltekai 1 year, 8 months ago

Selected Answer: D

The best solution here is D. Use BigQuery and GoogleSQL to normalize the data, and schedule recurring queries in BigQuery. Here's why: * No-code solution: BigQuery's built-in capabilities and GoogleSQL offer a no-code way to transform and standardize data. You can leverage functions like REGEXP_REPLACE to normalize phone numbers and FORMAT to ensure consistent formatting across fields. * Recurring jobs: BigQuery allows you to schedule queries to run regularly, which is perfect for maintaining data consistency over time. * Quick and efficient: BigQuery is designed for large-scale data processing, making it fast and efficient for normalization tasks.

carmltekai 1 year, 8 months ago

Why other options aren't as suitable: A. Cloud Data Fusion and Wrangler: While powerful, these tools might be overkill for a simple normalization task and could involve a steeper learning curve. B. Dataflow SQL: Dataflow is primarily for stream processing and might not be the most efficient for batch transformations on data already in BigQuery. C. Dataproc Serverless: This involves using a Spark job, which requires coding and might be more complex than necessary for this task.

fitri001 1 year, 9 months ago

Selected Answer: A

https://cloud.google.com/data-fusion/docs

SohiniV 2 years, 1 month ago

As per chatGPT, Option D allows you to utilize BigQuery's SQL capabilities to write queries that normalize the data according to company standards. You can then schedule these queries to run on a recurring basis using BigQuery's scheduled queries feature. This feature allows you to specify a schedule (e.g., weekly) for executing SQL queries automatically. This approach requires no additional setup or coding outside of BigQuery, making it a quick and straightforward solution to address the issue of data normalization.

SohiniV 2 years, 1 month ago

Any views on this ?

RenePetersen 2 years, 1 month ago

Wouldn't writing the SQL transformation be considered coding? The question specifically states that a solution requiring no coding is needed.

jreale64 2 years ago

While Cloud Data Fusion with Wrangler offers a visual interface for data wrangling, it requires setting up the environment and potentially writing code for ransformations. So it its not appropriate. I think D

JyoGCP 2 years, 1 month ago

Selected Answer: A

Option A

Sofiia98 2 years, 2 months ago

Selected Answer: A

Cloud Data Fusion and Wrangler

scaenruy 2 years, 2 months ago

Selected Answer: A

A. Use Cloud Data Fusion and Wrangler to normalize the data, and set up a recurring job.

https://zhangxuetu.online

[202610]

Question #290 Topic 1

You are designing a messaging system by using Pub/Sub to process clickstream data with an event-driven consumer app that relies on a push subscription. You need to configure the messaging system that is reliable enough to handle temporary downtime of the consumer app. You also need the messaging system to store the input messages that cannot be consumed by the subscriber. The system needs to retry failed messages gradually, avoiding overloading the consumer app, and store the failed messages after a maximum of 10 retries in a topic. How should you configure the Pub/Sub subscription?

A. Increase the acknowledgement deadline to 10 minutes.

B. Use immediate redelivery as the subscription retry policy, and configure dead lettering to a different topic with maximum delivery attempts set to 10.

C. Use exponential backoff as the subscription retry policy, and configure dead lettering to the same source topic with maximum delivery attempts set to 10.

D. Use exponential backoff as the subscription retry policy, and configure dead lettering to a different topic with maximum delivery attempts set to 10. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

raaad **Highly Voted** 1 year, 8 months ago

Selected Answer: D

- Exponential Backoff: This retry policy gradually increases the delay between retries, which helps to avoid overloading the consumer app. - Dead Lettering to a Different Topic: Configuring dead lettering sends messages that couldn't be processed after the specified number of delivery attempts (10 in this case) to a separate topic. This allows for handling of failed messages without interrupting the regular flow of new messages. - Maximum Delivery Attempts Set to 10: This setting ensures that the system retries each message up to 10 times before considering it a failure and moving it to the dead letter topic.

JyoGCP 1 year, 7 months ago

Selected Answer: D

Option D

Matt_108 1 year, 8 months ago

Selected Answer: D

Option D - agree with other comments explanation

GCP001 1 year, 8 months ago

D. Use exponential backoff as the subscription retry policy, and configure dead lettering to a different topic with maximum delivery attempts set to 10 Best suitable options for graceful retry and storing failed messages

scaenruy 1 year, 9 months ago

Selected Answer: D

D. Use exponential backoff as the subscription retry policy, and configure dead lettering to a different topic with maximum delivery attempts set to 10.

Smakyel79 1 year, 8 months ago

Exponential backoff will help in managing the load on the consumer app by gradually increasing the delay between retries. Configuring dead lettering to a different topic after a maximum of 10 delivery attempts ensures that undeliverable messages are stored separately, preventing them from being retried endlessly and cluttering the main message flow.

https://zhangxuetu.online

[202610]

Question #291 Topic 1

You designed a data warehouse in BigQuery to analyze sales data. You want a self-serving, low-maintenance, and cost- effective solution to share the sales dataset to other business units in your organization. What should you do?

A. Create an Analytics Hub private exchange, and publish the sales dataset. **Most Voted**

B. Enable the other business units' projects to access the authorized views of the sales dataset.

C. Create and share views with the users in the other business units.

D. Use the BigQuery Data Transfer Service to create a schedule that copies the sales dataset to the other business units' projects.

Correct Answer: A

Community vote distribution

A (100%)

Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: A

Analytics Hub offers a centralized platform for managing data sharing and access within the organization. This simplifies access control management.

cloud_rider 1 year, 4 months ago

What is wrong with the option B?

Ryannn23 1 year, 2 months ago

I assume "low-maintenance" is the main problem on B

987af6b 1 year, 8 months ago

Selected Answer: A

A. is the answer I select

JyoGCP 2 years, 1 month ago

Selected Answer: A

Analytics Hub

Matt_108 2 years, 2 months ago

Selected Answer: A

Definitely A

scaenruy 2 years, 2 months ago

Selected Answer: A

A. Create an Analytics Hub private exchange, and publish the sales dataset.

https://zhangxuetu.online

[202610]

Question #292 Topic 1

You have terabytes of customer behavioral data streaming from Google Analytics into BigQuery daily. Your customers' information, such as their preferences, is hosted on a Cloud SQL for MySQL database. Your CRM database is hosted on a Cloud SQL for PostgreSQL instance. The marketing team wants to use your customers' information from the two databases and the customer behavioral data to create marketing campaigns for yearly active customers. You need to ensure that the marketing team can run the campaigns over 100 times a day on typical days and up to 300 during sales. At the same time, you want to keep the load on the Cloud SQL databases to a minimum. What should you do?

A. Create BigQuery connections to both Cloud SQL databases. Use BigQuery federated queries on the two databases and the Google Analytics data on BigQuery to run these queries.

B. Create a job on Apache Spark with Dataproc Serverless to query both Cloud SQL databases and the Google Analytics data on BigQuery for these queries.

C. Create streams in Datastream to replicate the required tables from both Cloud SQL databases to BigQuery for these queries. **Most Voted**

D. Create a Dataproc cluster with Trino to establish connections to both Cloud SQL databases and BigQuery, to execute the queries.

Correct Answer: C

Community vote distribution

C (100%)

Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: C

- Datastream: It's a fully managed, serverless service for real-time data replication. It allows to stream data from various sources, including Cloud SQL, into BigQuery. - Reduced Load on Cloud SQL: By replicating the required tables from both Cloud SQL databases into BigQuery, you minimize the load on the Cloud SQL instances. The marketing team's queries will be run against BigQuery, which is designed to handle high-volume analytics workloads. - Frequency of Queries: BigQuery can easily handle the high frequency of queries (100 times daily, up to 300 during sales events) due to its powerful data processing capabilities. - Combining Data Sources: Once the data is in BigQuery, you can efficiently combine it with the Google Analytics data for comprehensive analysis and campaign planning.

SanjeevRoy91 2 years ago

Why not A ? Federrated queries will downgrade Cloud SQL perf?

Blackstile 1 year ago

Selected Answer: C

To Replication data, use datastream

987af6b 1 year, 8 months ago

Selected Answer: C

Initially I said A, but this question was how I learned about Datastream, which I think would be the better solution in this scenario. So my answer is C

AlizCert 1 year, 10 months ago

Selected Answer: C

C, noting that federated queries on read replicas would be the ideal solution

joao_01 1 year, 11 months ago

Its option C. "Performance. A federated query is likely to not be as fast as querying only BigQuery storage. BigQuery needs to wait for the source database to execute the external query and temporarily move data from the external data source to BigQuery. Also, the source database might not be optimized for complex analytical queries." So, it will load the Cloud SQL external sources with the queries, impacting performance on those. Link: https://cloud.google.com/bigquery/docs/federated-queries-intro

datasmg 2 years ago

Selected Answer: C

C is make sense

JyoGCP 2 years, 1 month ago

Selected Answer: C

Option C

scaenruy 2 years, 2 months ago

Selected Answer: C

C. Create streams in Datastream to replicate the required tables from both Cloud SQL databases to BigQuery for these queries.

Smakyel79 2 years, 2 months ago

Datastream is a serverless, easy-to-use change data capture (CDC) and replication service. By replicating the necessary tables from the Cloud SQL databases to BigQuery, you can offload the query load from the Cloud SQL databases. The marketing team can then run their queries directly on BigQuery, which is designed for large-scale data analytics. This approach seems to balance both efficiency and performance, minimizing load on the Cloud SQL instances.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #293 Topic 1

Your organization is modernizing their IT services and migrating to Google Cloud. You need to organize the data that will be stored in Cloud Storage and BigQuery. You need to enable a data mesh approach to share the data between sales, product design, and marketing departments. What should you do?

A. 1. Create a project for storage of the data for each of your departments.
2. Enable each department to create Cloud Storage buckets and BigQuery datasets.
3. Create user groups for authorized readers for each bucket and dataset.
4. Enable the IT team to administer the user groups to add or remove users as the departments' request.

B. 1. Create multiple projects for storage of the data for each of your departments' applications.
2. Enable each department to create Cloud Storage buckets and BigQuery datasets.
3. Publish the data that each department shared in Analytics Hub.
4. Enable all departments to discover and subscribe to the data they need in Analytics Hub.

C. 1. Create a project for storage of the data for your organization.
2. Create a central Cloud Storage bucket with three folders to store the files for each department.
3. Create a central BigQuery dataset with tables prefixed with the department name.
4. Give viewer rights for the storage project for the users of your departments.

D. 1. Create multiple projects for storage of the data for each of your departments' applications.
2. Enable each department to create Cloud Storage buckets and BigQuery datasets.
3. In Dataplex, map each department to a data lake and the Cloud Storage buckets, and map the BigQuery datasets to zones.
4. Enable each department to own and share the data of their data lakes. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: D

- Decentralized ownership: Each department controls its data lake, aligning with the core principle of data ownership in a data mesh. - Self-service data access: Departments can create and manage their own Cloud Storage buckets and BigQuery datasets within their data lakes, enabling self-service data access. - Interdepartmental sharing: Dataplex facilitates data sharing by enabling departments to publish their data products from their data lakes, making it easily discoverable and usable by other departments.

daed09 1 year ago

It says we want to work with GCS data assets, thus Dataplex is a better option, so it's option D! For those who says B as correct answer, I'd say that question states data assets in both GCS and BQ, however Analytics Hub is focused primarily for BQ assets.

plum21 1 year, 1 month ago

Selected Answer: D

D because it looks like B is impossible due to the lack of GCS support in Analytics Hub

marlon.andrei 1 year, 2 months ago

Selected Answer: B

In "a data mesh approach to share the data between sales, product design, and marketing departments", Analytics Hub is the solution.

Nandababy 1 year, 8 months ago

B is better option as organization is migrating to google cloud, that means teams doesnt have much hands on, analytical hub is more ease to use and solved the purpose as compared to dataplex were setup itself if very complex.

987af6b 1 year, 8 months ago

Selected Answer: B

For a straightforward data mesh approach where the focus is on decentralizing data management while enabling easy data sharing and discovery, Analytics Hub is often the more appropriate choice due to its simplicity and directness. It facilitates the core objectives of a data mesh—decentralized data ownership and accessible data sharing—without the added complexity of managing data lakes and advanced governance features.

joao_01 1 year, 11 months ago

I think its B. I know since we are talking about Datamesh we want to go to the Dataplex service suddenly. However, in Dataplex a Lake can only have assets (bq tables etc) that are in the same project as the Dataplex service. Example: There is bq table in project A and B. I want to to create a Lake in Dataplex in Project A that contains tables of project B. I can´t do that, i can only host tables of the Project A, since the Lake is in project A. With this said, I think the best option is B, because the datamesh approach is related to "to share the data between sales, product design, and marketing departments". So the question is focusing only in the sharing part of the datamesh. Option B fits just fine.

joao_01 1 year, 11 months ago

I was wrong in my explanation guys. Look at this link: https://cloud.google.com/dataplex/docs/add-zone "A lake can include one or more zones. While a zone can only be part of one lake, it may contain assets that point to resources that are part of projects outside of its parent project." So, option D seems good.

JyoGCP 2 years, 1 month ago

Selected Answer: D

Option D

Matt_108 2 years, 2 months ago

Selected Answer: D

that's pure data mesh, which is what dataplex has been built for

Sofiia98 2 years, 2 months ago

Selected Answer: D

For me, Dataplex looks more logical

GCP001 2 years, 2 months ago

D. Dataplex looks more suitable for data mesh approach, Check the ref - https://cloud.google.com/dataplex/docs/introduction

https://zhangxuetu.online

[202610]

Question #294 Topic 1

You work for a large ecommerce company. You are using Pub/Sub to ingest the clickstream data to Google Cloud for analytics. You observe that when a new subscriber connects to an existing topic to analyze data, they are unable to subscribe to older data. For an upcoming yearly sale event in two months, you need a solution that, once implemented, will enable any new subscriber to read the last 30 days of data. What should you do?

A. Create a new topic, and publish the last 30 days of data each time a new subscriber connects to an existing topic.

B. Set the topic retention policy to 30 days. **Most Voted**

C. Set the subscriber retention policy to 30 days.

D. Ask the source system to re-push the data to Pub/Sub, and subscribe to it.

Correct Answer: B

Community vote distribution

B (100%)

Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- Topic Retention Policy: This policy determines how long messages are retained by Pub/Sub after they are published, even if they have not been acknowledged (consumed) by any subscriber. - 30 Days Retention: By setting the retention policy of the topic to 30 days, all messages published to this topic will be available for consumption for 30 days. This means any new subscriber connecting to the topic can access and analyze data from the past 30 days.

hussain.sain 1 year, 3 months ago

Selected Answer: B

B is correct. By setting the topic retention policy to 30 days, any new subscriber will be able to access the data for the past 30 days, regardless of when they connect. This solution is both cost-effective and efficient for your use case.

romain773 1 year, 5 months ago

Option B is wrong i think (topic retention) because it only makes unconsumed messages available for 30 days. I propose option A Option A (creating a new topic and republishing the last 30 days of data for each new subscriber) is actually a better solution to ensure that new subscribers have access to the full 30-day history.

romain773 1 year, 5 months ago

Option B is wrong (topic retention) because it only makes unconsumed messages available for 30 days. Option A (creating a new topic and republishing the last 30 days of data for each new subscriber) is actually a better solution to ensure that new subscribers have access to the full 30-day history.

joao_01 1 year, 11 months ago

Its B. It could be C as well because subscription has message retention. However, in the subscription there is a maximum value for it: 7 days. Link:https://cloud.google.com/pubsub/docs/subscription-properties

joao_01 1 year, 11 months ago

In a topic the maximum value is 31 days. Link: https://cloud.google.com/pubsub/docs/topic-properties

Matt_108 2 years, 2 months ago

Selected Answer: B

Definitely B

Sofiia98 2 years, 2 months ago

Selected Answer: B

https://cloud.google.com/blog/products/data-analytics/pubsub-gains-topic-retention-feature

scaenruy 2 years, 2 months ago

Selected Answer: B

B. Set the topic retention policy to 30 days.

https://zhangxuetu.online

[202610]

Question #295 Topic 1

You are designing the architecture to process your data from Cloud Storage to BigQuery by using Dataflow. The network team provided you with the Shared VPC network and subnetwork to be used by your pipelines. You need to enable the deployment of the pipeline on the Shared VPC network. What should you do?

A. Assign the compute.networkUser role to the Dataflow service agent.

B. Assign the compute.networkUser role to the service account that executes the Dataflow pipeline. **Most Voted**

C. Assign the dataflow.admin role to the Dataflow service agent.

D. Assign the dataflow.admin role to the service account that executes the Dataflow pipeline.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: A

- Dataflow service agent is the one responsible for setting up and managing the network resources that Dataflow requires. - By granting the compute.networkUser role to this service agent, we are enabling it to provision the necessary network resources within the Shared VPC for your Dataflow job.

saschak94 **Highly Voted** 2 years, 1 month ago

Selected Answer: A

All projects that have used the resource Dataflow Job have a Dataflow Service Account, also known as the Dataflow service agent. Make sure the Shared VPC subnetwork is shared with the Dataflow service account and has the Compute Network User role assigned on the specified subnet.

pamrona 1 month, 1 week ago

Selected Answer: B

If VMs need to attach to a subnet : Grant compute.networkUser to the runtime service account, not the service agent

50336e5 3 months, 1 week ago

Selected Answer: A

A Dataflow service agent is the one responsible for setting up and managing the network resources that Dataflow requires.

lmch 3 months, 2 weeks ago

Selected Answer: A

Key Technical Takeaway When working with Dataflow and Shared VPC, always remember: Service Agent = Network Permissions (Host Project) and Worker Service Account = Data Permissions (Service Project).

judy_data 6 months, 4 weeks ago

Selected Answer: B

Make sure the Shared VPC subnetwork is shared with the Dataflow service account and has the Compute Network User role assigned on the specified subnet. The Compute Network User role must be assigned to the Dataflow service account in the host project. https://cloud.google.com/dataflow/docs/guides/specifying-networks?utm_source=chatgpt.com#shared

gabbferreira 11 months, 1 week ago

Selected Answer: A

It's a

desertlotus1211 1 year ago

Answer is B: The Dataflow service agent manages Dataflow internals, but does not launch pipeline worker VMs. So Answer A is incorrect

loki82 1 year, 2 months ago

Selected Answer: A

https://cloud.google.com/dataflow/docs/concepts/security-and-permissions#df-service-account

gord_nat 1 year ago

There is no compute.networkUser role in df agent, only in IAM. Answer is B

desertlotus1211 1 year ago

The Dataflow service agent manages Dataflow internals, but does not launch pipeline worker VMs.

SamuelTsch 1 year, 5 months ago

Selected Answer: A

From https://cloud.google.com/dataflow/docs/guides/specifying-networks, it says "Make sure the Shared VPC subnetwork is shared with the Dataflow service account and has the Compute Network User role assigned on the specified subnet. The Compute Network User role must be assigned to the Dataflow service account in the host project."

ach5 1 year, 4 months ago

service account - it's B

Preetmehta1234 1 year, 6 months ago

Selected Answer: B

If you see in the comments, A was answer by people around 8 months ago but recent ones have answered B with the documentation. The GCP documentation evolves with time

Preetmehta1234 1 year, 6 months ago

Selected Answer: B

service account that executes the Dataflow pipeline It's straight forward

Preetmehta1234 1 year, 6 months ago

Selected Answer: B

Assign the compute.networkUser role to the service account that executes the Dataflow pipeline

Jeyaraj 1 year, 8 months ago

The correct answer is B. Assign the compute.networkUser role to the service account that executes the Dataflow pipeline. Here's why: Shared VPC and Network Access: When using a Shared VPC, you need to grant specific permissions to service accounts in the service project (where your Dataflow pipeline runs) to access resources in the host project's network. compute.networkUser Role: This role grants the necessary permissions for a service account to use the network resources in the Shared VPC. This includes accessing subnets, creating instances, and communicating with other services within the network. Service Account for Pipeline Execution: The service account that executes your Dataflow pipeline is the one that needs these network permissions. This is because the Dataflow service uses this account to create and manage worker instances within the Shared VPC network.

extraego 1 year, 9 months ago

Selected Answer: B

Dataflow service agent is a role that is assigned to a service account. So is compute.networkUser. https://cloud.google.com/dataflow/docs/concepts/access-control#example

josech 1 year, 10 months ago

Selected Answer: B

Option B https://cloud.google.com/knowledge/kb/dataflow-job-in-shared-vpc-xpn-permissions-000004261

chrissamharris 1 year, 11 months ago

Selected Answer: B

I believe the answer is B. All authentication documentation points to Service Accounts. https://cloud.google.com/dataflow/docs/concepts/authentication#on-gcp Dataflow service agent typically manages general interactions with the Dataflow service but does not execute the actual jobs.

https://zhangxuetu.online

[202610]

Question #296 Topic 1

Your infrastructure team has set up an interconnect link between Google Cloud and the on-premises network. You are designing a high-throughput streaming pipeline to ingest data in streaming from an Apache Kafka cluster hosted on- premises. You want to store the data in BigQuery, with as minimal latency as possible. What should you do?

A. Setup a Kafka Connect bridge between Kafka and Pub/Sub. Use a Google-provided Dataflow template to read the data from Pub/Sub, and write the data to BigQuery.

B. Use a proxy host in the VPC in Google Cloud connecting to Kafka. Write a Dataflow pipeline, read data from the proxy host, and write the data to BigQuery.

C. Use Dataflow, write a pipeline that reads the data from Kafka, and writes the data to BigQuery. **Most Voted**

D. Setup a Kafka Connect bridge between Kafka and Pub/Sub. Write a Dataflow pipeline, read the data from Pub/Sub, and write the data to BigQuery.

Correct Answer: C

Community vote distribution



Comments

rajshiv 11 months, 3 weeks ago

Selected Answer: A

I think it is A and not C. While Dataflow can read from Kafka directly, it works best for Kafka clusters hosted in Google Cloud. Reading from an on-prem Kafka over Interconnect directly from Dataflow is not recommended due to latency, firewall/NAT issues, and network complexity. Most important this option is Not optimal for performance and reliability across hybrid environments.

meh_33 1 year, 7 months ago

Going with C

Anudeep58 1 year, 8 months ago

Selected Answer: C

Latency: Option C, with direct integration between Kafka and Dataflow, offers lower latency by eliminating intermediate steps. Flexibility: Custom Dataflow pipelines (Option C) provide more control over data processing and optimization compared to using a pre-built template.

anushree09 1 year, 11 months ago

per the text below at https://cloud.google.com/dataflow/docs/kafka-dataflow - "Alternatively, you might have an existing Kafka cluster that resides outside of Google Cloud. For example, you might have an existing workload that is deployed on-premises or in another public cloud."

Moss2011 2 years, 1 month ago

Selected Answer: C

From my point of view, the best option is C taking into account this doc: https://cloud.google.com/dataflow/docs/kafka-dataflow

MaxNRG 2 years, 1 month ago

Selected Answer: D

Based on the key requirements highlighted: • Interconnect link between GCP and on-prem Kafka • High throughput streaming pipeline • Minimal latency • Data to be stored in BigQuery D - The key reasons this meets the requirements: • Kafka connect provides a reliable bridge to Pub/Sub over the interconnect • Reading from Pub/Sub minimizes latency vs reading directly from Kafka • Dataflow provides a high throughput streaming engine • Writing to BigQuery gives scalable data storage By leveraging these fully managed GCP services over the dedicated interconnect, a low latency streaming pipeline from on-prem Kafka into BigQuery can be implemented rapidly. Options A/B/C have higher latencies or custom code requirements, so do not meet the minimal latency criteria as well as option D.

MaxNRG 2 years, 1 month ago

Why choose option D over A? The key advantage with option D is that by writing a custom Dataflow pipeline rather than using a Google provided template, there is more flexibility to customize performance tuning and optimization for lowest latency. • Some potential optimizations: • Fine tuning number of workers, machine types to meet specific throughput targets • Custom data parsing/processing logic if applicable • Experimenting with autoscaling parameters or triggers

MaxNRG 2 years, 1 month ago

The Google template may be easier to set up initially, but a custom pipeline provides more control over optimizations specifically for low latency requirements stated in the question. That being said, option A would still work reasonably well - but option D allows squeezing out that extra bit of performance if low millisecond latency is absolutely critical in the pipeline through precise tuning. So in summary, option A is easier to implement but option D provides more optimization flexibility for ultra low latency streaming requirements.

MaxNRG 2 years, 1 month ago

Why not C: At first option C (using a Dataflow pipeline to directly read from Kafka and write to BigQuery) seems reasonable. However, the key requirement stated in the question is to have minimal latency for the streaming pipeline. By reading directly from Kafka within Dataflow, there can be additional latency and processing overhead compared to reading from Pub/Sub, for a few reasons: 1. Pub/Sub acts as a buffer and handles scaling/reliability of streaming data automatically. This reduces processing burden on the pipeline. 2. Network latency can be lower by leveraging Pub/Sub instead of making constant pull requests for data from Kafka within the streaming pipeline. 3. Any failures have to be handled within the pipeline code itself when reading directly from Kafka. With Pub/Sub, reliability is built-in.

MaxNRG 2 years, 1 month ago

So in summary, while option C is technically possible, option D introduces Pub/Sub as a streaming buffer which reduces overall latency for the pipeline, allowing the key requirement of minimal latency to be better satisfied.

SanjeevRoy91 2 years ago

You are adding an intermediate hop in between on prem kafka and Dataflow (pubsub). Why won't this add additional latency.

JyoGCP 2 years, 1 month ago

A Vs C -- Not sure which one would have low latency. Points related to option C: "Yes, Dataflow can read events from Kafka. Dataflow is a fully-managed, serverless streaming analytics service that supports both batch and stream processing. It can read events from Kafka, process them, and write the results to a BigQuery table for further analysis. " "Dataflow supports Kafka support, which was added to Apache Beam in 2016. Google provides a Dataflow template that configures a Kafka-to-BigQuery pipeline. The template uses the BigQueryIO connector provided in the Apache Beam SDK. "

JyoGCP 2 years, 1 month ago

Going with C

DarkLord2104 2 years, 1 month ago

Final???

T2Clubber 2 years, 1 month ago

Selected Answer: C

Option C makes more sense to me because of the "minimal latency as possible". I would have chosen option A if it were "less CODING as possible".

Matt_108 2 years, 2 months ago

Selected Answer: A

Option A, leverage dataflow template for Kafka https://cloud.google.com/dataflow/docs/kafka-dataflow

AllenChen123 2 years, 2 months ago

Agree. "Google provides a Dataflow template that configures a Kafka-to-BigQuery pipeline. The template uses the BigQueryIO connector provided in the Apache Beam SDK."

ML6 2 years, 1 month ago

But it includes setting up a Kafka Connect bridge while an interconnect link has already been set up. https://cloud.google.com/dataflow/docs/kafka-dataflow#connect_to_an_external_cluster

scaenruy 2 years, 2 months ago

Selected Answer: C

C. Use Dataflow, write a pipeline that reads the data from Kafka, and writes the data to BigQuery.

chickenwingz 2 years, 3 months ago

Selected Answer: C

Dataflow has templates to read from Kafka. Other options are too complicated
https://cloud.google.com/dataflow/docs/kafka-dataflow

Sofiia98 2 years, 2 months ago

so, this is the answer A, whe C?

Matt_108 2 years, 2 months ago

Yeah, the answer is A. C requires you to develop the pipeline yourself and ensure minimal latency, which means that you perform better than a pre-built template from Google...not really the case most of the times :)

saschak94 2 years, 2 months ago

but Option A introduces additional replication into Pub/Sub and the question states with minimal latency. In my opinion subscribing to Kafka via Dataflow has a lower latency than replicating the messages first to Pub/Sub and subscribing with Dataflow to it.

https://zhangxuetu.online

[202610]

Question #297 Topic 1

You migrated your on-premises Apache Hadoop Distributed File System (HDFS) data lake to Cloud Storage. The data scientist team needs to process the data by using Apache Spark and SQL. Security policies need to be enforced at the column level. You need a cost-effective solution that can scale into a data mesh. What should you do?

A. 1. Deploy a long-living Dataproc cluster with Apache Hive and Ranger enabled.
2. Configure Ranger for column level security.
3. Process with Dataproc Spark or Hive SQL.

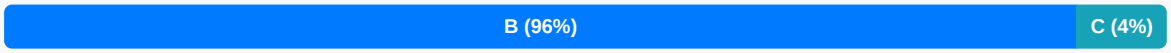
B. 1. Define a BigLake table.
2. Create a taxonomy of policy tags in Data Catalog.
3. Add policy tags to columns.
4. Process with the Spark-BigQuery connector or BigQuery SQL. **Most Voted**

C. 1. Load the data to BigQuery tables.
2. Create a taxonomy of policy tags in Data Catalog.
3. Add policy tags to columns.
4. Process with the Spark-BigQuery connector or BigQuery SQL.

D. 1. Apply an Identity and Access Management (IAM) policy at the file level in Cloud Storage.
2. Define a BigQuery external table for SQL processing.
3. Use Dataproc Spark to process the Cloud Storage files.

Correct Answer: B

Community vote distribution



Comments

raaad **Highly Voted** 1 year, 8 months ago

Selected Answer: B

- BigLake Integration: BigLake allows you to define tables on top of data in Cloud Storage, providing a bridge between data lake storage and BigQuery's powerful analytics capabilities. This approach is cost-effective and scalable. - Data Catalog for Governance: Creating a taxonomy of policy tags in Google Cloud's Data Catalog and applying these tags to specific columns in your BigLake tables enables fine-grained, column-level access control. - Processing with Spark and SQL: The Spark-BigQuery connector allows data scientists to process data using Apache Spark directly against BigQuery (and BigLake tables). This supports both Spark and SQL processing needs. - Scalability into a Data Mesh:

BigLake and Data Catalog are designed to scale and support the data mesh architecture, which involves decentralized data ownership and governance.

JyoGCP 1 year, 7 months ago

Selected Answer: B

Going with 'B' based on the comments

Matt_108 1 year, 8 months ago

Selected Answer: B

Option B, agree with comments explanation

Jordan18 1 year, 8 months ago

Selected Answer: B

BigLake leverages existing Cloud Storage infrastructure, eliminating the need for a dedicated Dataproc cluster, reducing costs significantly.

scaenruy 1 year, 9 months ago

Selected Answer: C

C. 1. Load the data to BigQuery tables. 2. Create a taxonomy of policy tags in Data Catalog. 3. Add policy tags to columns. 4. Process with the Spark-BigQuery connector or BigQuery SQL.

raaad 1 year, 8 months ago

- Option B offers a serverless approach that integrates Cloud Storage (as a data lake), BigLake (for table definition), Data Catalog (for data mesh), and BigQuery (for analytics), all of which are essential components of a flexible, scalable, and secure data platform.

https://zhangxuetu.online

[202610]

Question #298 Topic 1

One of your encryption keys stored in Cloud Key Management Service (Cloud KMS) was exposed. You need to re- encrypt all of your CMEK-protected Cloud Storage data that used that key, and then delete the compromised key. You also want to reduce the risk of objects getting written without customer-managed encryption key (CMEK) protection in the future. What should you do?

A. Rotate the Cloud KMS key version. Continue to use the same Cloud Storage bucket.

B. Create a new Cloud KMS key. Set the default CMEK key on the existing Cloud Storage bucket to the new one.

C. Create a new Cloud KMS key. Create a new Cloud Storage bucket. Copy all objects from the old bucket to the new one bucket while specifying the new Cloud KMS key in the copy command.

D. Create a new Cloud KMS key. Create a new Cloud Storage bucket configured to use the new key as the default CMEK key. Copy all objects from the old bucket to the new bucket without specifying a key. Most Voted

Correct Answer: D

Community vote distribution



Comments

raaad Highly Voted 1 year, 8 months ago

Selected Answer: D

- New Key Creation: A new Cloud KMS key ensures a secure replacement for the compromised one. - New Bucket: A separate bucket prevents potential conflicts with existing objects and configurations. - Default CMEK: Setting the new key as default enforces encryption for all objects in the bucket, reducing the risk of unencrypted data. - Copy Without Key Specification: Copying objects without specifying a key leverages the default key, simplifying the process and ensuring consistent encryption. - Old Key Deletion: After copying, the compromised key can be safely deleted.

chickenwingz Highly Voted 1 year, 9 months ago

Selected Answer: D

Wrong: A - rotating external key doesn't trigger re-encryption of data already in GCS: https://cloud.google.com/kms/docs/rotate-key#rotate-external-coordinated C - Setting key during copy doesn't take care of objects that are later uploaded to the bucket, that will still use the default key

judy_data 6 months, 4 weeks ago

Selected Answer: C

passing the cloud KMS key in the command is more secure and explicit https://cloud.google.com/storage/docs/encryption/customer-managed-keys?utm_source=chatgpt.com#key-replacement

desertlotus1211 1 year ago

Selected Answer: C

If no key is specified, and the bucket's default CMEK key is used, there's a risk that some objects might fall back to Google-managed encryption, especially if misconfigured

JyoGCP 1 year, 7 months ago

Selected Answer: D

Option D

ML6 1 year, 7 months ago

Selected Answer: D

The correct answer is D. Rotating the key does not seem to re-encrypt: In the event that a key is compromised, regular rotation (!!) limits the number of actual messages vulnerable to compromise (!!). If you suspect that a key version is compromised, disable it and revoke access to it as soon as possible. Source: https://cloud.google.com/kms/docs/key-rotation#why_rotate_keys

ML6 1 year, 7 months ago

Note: When you rotate a key, data encrypted with previous key versions is not automatically re-encrypted with the new key version. You can learn more about re-encrypting data. Source: https://cloud.google.com/kms/docs/key-rotation#how_often_to_rotate_keys

Medmah 1 year, 8 months ago

I don't understand why only Matt select A https://cloud.google.com/sdk/gcloud/reference/kms/keys/update This seems to do the job, am I wrong ?

Matt_108 1 year, 8 months ago

Selected Answer: A

Definitely A

ML6 1 year, 7 months ago

Rotating does not mean you re-encrypt data.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #299 Topic 1

You have an upstream process that writes data to Cloud Storage. This data is then read by an Apache Spark job that runs on Dataproc. These jobs are run in the us-central1 region, but the data could be stored anywhere in the United States. You need to have a recovery process in place in case of a catastrophic single region failure. You need an approach with a maximum of 15 minutes of data loss (RPO=15 mins). You want to ensure that there is minimal latency when reading the data. What should you do?

A. 1. Create two regional Cloud Storage buckets, one in the us-central1 region and one in the us-south1 region.
2. Have the upstream process write data to the us-central1 bucket. Use the Storage Transfer Service to copy data hourly from the us-central1 bucket to the us-south1 bucket.
3. Run the Dataproc cluster in a zone in the us-central1 region, reading from the bucket in that region.
4. In case of regional failure, redeploy your Dataproc clusters to the us-south1 region and read from the bucket in that region instead.

B. 1. Create a Cloud Storage bucket in the US multi-region.
2. Run the Dataproc cluster in a zone in the us-central1 region, reading data from the US multi-region bucket.
3. In case of a regional failure, redeploy the Dataproc cluster to the us-central2 region and continue reading from the same bucket.

C. 1. Create a dual-region Cloud Storage bucket in the us-central1 and us-south1 regions.
2. Enable turbo replication.
3. Run the Dataproc cluster in a zone in the us-central1 region, reading from the bucket in the us-south1 region.
4. In case of a regional failure, redeploy your Dataproc cluster to the us-south1 region and continue reading from the same bucket.

D. 1. Create a dual-region Cloud Storage bucket in the us-central1 and us-south1 regions.
2. Enable turbo replication.
3. Run the Dataproc cluster in a zone in the us-central1 region, reading from the bucket in the same region.
4. In case of a regional failure, redeploy the Dataproc clusters to the us-south1 region and read from the same bucket. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

raaad Highly Voted 1 year, 8 months ago

Selected Answer: D

- Rapid Replication: Turbo replication ensures near-real-time data synchronization between regions, achieving an RPO of 15 minutes or less. - Minimal Latency: Dataproc clusters can read from the bucket in the same region, minimizing data transfer latency and optimizing performance. - Disaster Recovery: In case of regional failure, Dataproc clusters can seamlessly redeploy to the other region and continue reading from the same bucket, ensuring business continuity.

JyoGCP 1 year, 7 months ago

Selected Answer: D

Option D

Matt_108 1 year, 8 months ago

Selected Answer: D

Option D, answers all needs from the request

scaenruy 1 year, 9 months ago

Selected Answer: D

D. 1. Create a dual-region Cloud Storage bucket in the us-central1 and us-south1 regions. 2. Enable turbo replication. 3. Run the Dataproc cluster in a zone in the us-central1 region, reading from the bucket in the same region. 4. In case of a regional failure, redeploy the Dataproc clusters to the us-south1 region and read from the same bucket.

https://zhangxuetu.online

[202610]

Question #300 Topic 1

You currently have transactional data stored on-premises in a PostgreSQL database. To modernize your data environment, you want to run transactional workloads and support analytics needs with a single database. You need to move to Google Cloud without changing database management systems, and minimize cost and complexity. What should you do?

A. Migrate and modernize your database with Cloud Spanner.

B. Migrate your workloads to AlloyDB for PostgreSQL. **Most Voted**

C. Migrate to BigQuery to optimize analytics.

D. Migrate your PostgreSQL database to Cloud SQL for PostgreSQL.

Correct Answer: B

Community vote distribution



Comments

8ad5266 **Highly Voted** 1 year, 9 months ago

Selected Answer: D

Minimize cost. https://cloud.google.com/alloydb?hl=en AlloyDB offers superior performance, 4x faster than standard PostgreSQL for transactional workloads. That does not come without cost.

MvazM 7 months, 1 week ago

Selected Answer: D

D is clean

Ben_oso 9 months, 1 week ago

Selected Answer: D

D is clean

duers 11 months, 3 weeks ago

Selected Answer: D

AlloyDB for PostgreSQL is a fully managed, PostgreSQL-compatible database service offered by Google Cloud. It's designed for high-performance transactional and analytical workloads and offers performance and scalability benefits over standard PostgreSQL. While it meets the requirement of not changing the database system in a broad sense (as it's PostgreSQL-compatible), it's a different service than standard PostgreSQL and might introduce a level of complexity and cost beyond simply migrating to Cloud SQL for PostgreSQL.

aaaaaaaasdasdasfs 11 months, 3 weeks ago

Selected Answer: B

The correct answer is B. Migrate your workloads to AlloyDB for PostgreSQL. Here's why: Your requirements are: Run both transactional and analytics workloads in a single database Stay with PostgreSQL (don't change database systems) Minimize cost and complexity AlloyDB for PostgreSQL is specifically designed for this scenario - it's fully PostgreSQL-compatible but optimized for both transactional and analytical workloads. It offers: PostgreSQL compatibility (minimizing migration complexity) Enhanced analytics capabilities with column store indexes Better performance for mixed workloads

rajshiv 11 months, 3 weeks ago

Selected Answer: D

While B looks good too but It's more expensive than Cloud SQL and better suited when you need advanced analytics and heavy transactional performance. I think it is an Overkill if you're looking to minimize cost/complexity which the question states.

mednoun 1 year, 1 month ago

Selected Answer: B

The question specifies that the analytical needs need to reside in a single database. This can't be done using Cloud SQL. The database that supports all of that is AlloyDB that's why I will go with the B answer.

plum21 1 year, 1 month ago

Selected Answer: B

"support analytics needs" -> columnar storage -> AlloyDB

juliorevk 1 year, 2 months ago

Selected Answer: D

Cloud SQL natively supports PostgreSQL AlloyDB for PostgreSQL is a great option if you're specifically looking for high performance in both transactional and analytical workloads. However, it might be more complex and costly than Cloud SQL

joelcaro 1 year, 3 months ago

Selected Answer: B

B AlloyDB es la mejor opción para modernizar el entorno, mantener compatibilidad con PostgreSQL y manejar tanto cargas transaccionales como analíticas en un único sistema, minimizando costos y complejidad.

baimus 1 year, 5 months ago

Selected Answer: B

In real life clearly how performant it needed to be would be a massive factor. AlloyDB is more expensive (see https://cloud.google.com/alloydb/pricing, vs https://cloud.google.com/sql/pricing), but when they say "minimise cost" is that per query, or is it per year assuming similar instance size. There's no way for us to know, we have to guess. I'm guessing AlloyDB, as the question seem to be telegraphing that, but it could just as easily be CloudSQL postgres based on the cheaper costs. We simply cannot know.

Antmal 1 year, 8 months ago

Selected Answer: B

Because AlloyDB is optimised for hybrid transactional and analytical processing (HTAP), meaning you can run both transactional workloads and analytics on the same database with excellent performance.

Anudeep58 1 year, 8 months ago

Selected Answer: B

AlloyDB

finixd 1 year, 9 months ago

Selected Answer: B

It's a little complicated, considering it says minimize costs (Cloud SQL) and run transactional workloads and support analytics needs (AlloyDB). I consider B. because you can minimize costs in the long-term instead of doing it immediately with possible extra costs in the long-term. Think about it

extraego 1 year, 9 months ago

Selected Answer: D

AlloyDB is for large scale and more expensive. We want to minimize cost and complexity, so the answer is D.

virat_kohli 1 year, 10 months ago

Selected Answer: B

B. Migrate your workloads to AlloyDB for PostgreSQL.

virat_kohli 1 year, 10 months ago

Sorry its D. Migrate your PostgreSQL database to Cloud SQL for PostgreSQL.

omkarr24 2 years ago

Selected Answer: D

They currently have transactional data stored on-premises in a PostgreSQL database and they want to modernize their database that supports transactional workloads and analytics .If they select cloud Sql (postgreSQL) it will minimize the cost and complexity. and for analytics purpose they can create federated queries over cloudSql(postgreSql) https://cloud.google.com/bigquery/docs/federated-queries-intro This approach will minimze the cost

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #301 Topic 1

You are architecting a data transformation solution for BigQuery. Your developers are proficient with SQL and want to use the ELT development technique. In addition, your developers need an intuitive coding environment and the ability to manage SQL as code. You need to identify a solution for your developers to build these pipelines. What should you do?

A. Use Dataform to build, manage, and schedule SQL pipelines. **Most Voted**

B. Use Dataflow jobs to read data from Pub/Sub, transform the data, and load the data to BigQuery.

C. Use Data Fusion to build and execute ETL pipelines.

D. Use Cloud Composer to load data and run SQL pipelines by using the BigQuery job operators.

Correct Answer: A

Community vote distribution

A (100%)

Comments

raaad **Highly Voted** 1 year, 8 months ago

Selected Answer: A

- Aligns with ELT Approach: Dataform is designed for ELT (Extract, Load, Transform) pipelines, directly executing SQL transformations within BigQuery, matching the developers' preference. -SQL as Code: It enables developers to write and manage SQL transformations as code, promoting version control, collaboration, and testing. - Intuitive Coding Environment: Dataform provides a user-friendly interface and familiar SQL syntax, making it easy for SQL-proficient developers to adopt. - Scheduling and Orchestration: It includes built-in scheduling capabilities to automate pipeline execution, simplifying pipeline management.

JyoGCP 1 year, 7 months ago

Selected Answer: A

Option A

Matt_108 1 year, 8 months ago

Selected Answer: A

Definitely A

scaenruy 1 year, 9 months ago

Selected Answer: A

A. Use Dataform to build, manage, and schedule SQL pipelines.

chickenwingz 1 year, 9 months ago

Selected Answer: A

Dataform = transformations in SQL

https://zhangxuetu.online

[202610]

Question #302 Topic 1

You work for a farming company. You have one BigQuery table named sensors, which is about 500 MB and contains the list of your 5000 sensors, with columns for id, name, and location. This table is updated every hour. Each sensor generates one metric every 30 seconds along with a timestamp, which you want to store in BigQuery. You want to run an analytical query on the data once a week for monitoring purposes. You also want to minimize costs. What data model should you use?

A. 1. Create a metrics column in the sensors table.
2. Set RECORD type and REPEATED mode for the metrics column.
3. Use an UPDATE statement every 30 seconds to add new metrics.

B. 1. Create a metrics column in the sensors table.
2. Set RECORD type and REPEATED mode for the metrics column.
3. Use an INSERT statement every 30 seconds to add new metrics.

C. 1. Create a metrics table partitioned by timestamp.
2. Create a sensorId column in the metrics table, that points to the id column in the sensors table.
3. Use an INSERT statement every 30 seconds to append new metrics to the metrics table.
4. Join the two tables, if needed, when running the analytical query. Most Voted

D. 1. Create a metrics table partitioned by timestamp.
2. Create a sensorId column in the metrics table, which points to the id column in the sensors table.
3. Use an UPDATE statement every 30 seconds to append new metrics to the metrics table.
4. Join the two tables, if needed, when running the analytical query.

Correct Answer: C

Community vote distribution



Comments

raaad Highly Voted 2 years, 2 months ago

Partitioned Metrics Table: Creating a separate metrics table partitioned by timestamp is a standard practice for time-series data like sensor readings. Partitioning by timestamp allows for more efficient querying, especially when you're only interested in a specific time range (like weekly monitoring). Reference to Sensors Table: Including a sensorId column that references the id column in the sensors table allows you to maintain a relationship between the metrics and the sensors without duplicating sensor information. INSERT Every 30 Seconds: Using an INSERT statement every 30 seconds to the partitioned metrics table is a standard approach for time-series data ingestion in BigQuery. It

allows for efficient data storage and querying. Join for Analysis: When you need to analyze the data, you can join the metrics table with the sensors table based on the sensorId, allowing for comprehensive analysis with sensor details.

rajshiv 11 months, 3 weeks ago

Selected Answer: C

C is the best answer. It cannot be A and B - as Embedding a RECORD type (nested structure) in the sensors table and updating it every 30 seconds is inefficient and expensive. Moreover, BigQuery is not designed for frequent updates or modifying nested fields repeatedly. Plus It increases storage and write costs significantly. It cannot be D - Even though it uses a good table design (separate metrics table with timestamp partition), using UPDATE every 30 seconds to append data is inefficient as BigQuery is not optimized for UPDATE-heavy workloads.

plum21 1 year, 1 month ago

Selected Answer: C

C. B is not feasible – update on the metrics column will be required in such a case or an insert with all sensor data with one-element array of metrics which does not make any sense.

7787de3 1 year, 6 months ago

Selected Answer: C

Because "Minimize costs" was requested, i would go for C. Storage cost will be lower for partitions where no writes took place for a certain amount of time, see https://cloud.google.com/bigquery/pricing#storage Partitioning by timestamp can be configured to use daily, hourly, monthly, or yearly partitioning - so if you choose daily partitioning, the number of partitions should not be an issue. Working with RECORDS (A,B) would be an option if performance was in focus.

dac9215 1 year, 8 months ago

Option C will not violate partitioning limit of 4000 as the lowest grain of partitioning is hourly

vbrege 1 year, 9 months ago

Selected Answer: B

Here's my logic (some people have already said same thing) Cannot be C and D - Total 5000 sensors are sending new timestamp every 30 seconds. If you partition this table with timestamp, you are getting partitions above 4000 (single job) or 10000 (partition limit) so option C and D don't look correct - For C and D, also need to consider that BigQuery best practices advise to avoid JOINs and use STRUCT and RECORD types to solve the parent-child join issue. Now coming back to A and B, we will be adding sensor readings for every sensor. I don't think this is a transactional type database where you need to update data. You will add new data for more accurate analysis later so A is discarded. BigQuery best practices also advise to avoid UPDATE statements since its an Analytical columnar database B is the correct option.

apoio.certificacoes.closer 1 year, 3 months ago

Avoid Joins when tables are large. The sensors table is 500mb, hardly anything. The only watchout is for multiplication of columns when joining.

Gloups 1 year, 10 months ago

Selected Answer: A

Since BigQuery tables are limited to 4000 partitions, options C & D are discarded. Option B is wrong as insertion is invalid too. So option A.

gabrielosluz 1 year, 2 months ago

I also thought about this limitation. But researching about partitioning with timestamp, I found this in the documentation: "For TIMESTAMP and DATETIME columns, the partitions can have either hourly, daily, monthly, or

yearly granularity. For DATE columns, the partitions can have daily, monthly, or yearly granularity." In other words, I believe that even with timestamp partitioning, it would not reach this limit. What do you think?

anushree09 1 year, 11 months ago

I'm in favor of Option B Reason: BQ has nested columns feature specifically to address these scenarios where a join would be needed in a traditional/ relational data model. Nesting field will reduce the need to join tables, performance will be high and design will be simple

96f3bfa 2 years, 1 month ago

Selected Answer: C

Option C

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C

SanjeevRoy91 2 years ago

Why C. Partitioning by timestamp could breach the 4000 cap of number of partitions easily. And with soo much less data, why partitioning is required in the first place. Ans should be B

raaad 2 years, 2 months ago

Selected Answer: C

Option C

scaenruy 2 years, 2 months ago

Selected Answer: C

C. 1. Create a metrics table partitioned by timestamp. 2. Create a sensorId column in the metrics table, that points to the id column in the sensors table. 3. Use an INSERT statement every 30 seconds to append new metrics to the metrics table. 4. Join the two tables, if needed, when running the analytical query.

https://zhangxuetu.online

[202610]

Question #303 Topic 1

You are managing a Dataplex environment with raw and curated zones. A data engineering team is uploading JSON and CSV files to a bucket asset in the curated zone but the files are not being automatically discovered by Dataplex. What should you do to ensure that the files are discovered by Dataplex?

A. Move the JSON and CSV files to the raw zone. **Most Voted**

B. Enable auto-discovery of files for the curated zone.

C. Use the bg command-line tool to load the JSON and CSV files into BigQuery tables.

D. Grant object level access to the CSV and JSON files in Cloud Storage.

Correct Answer: A

Community vote distribution



Comments

GCP001 **Highly Voted** 2 years, 2 months ago

Selected Answer: A

Should be A. Curated zone need Parquet, Avro, ORC format not CSV or JSON. Check the ref - https://cloud.google.com/dataplex/docs/add-zone#curated-zones

Positron75 10 months ago

Agreed. This link makes it more explicit: https://cloud.google.com/dataplex/docs/discover-data?hl=en#invalid_data_format "Invalid data format in curated zones (data not in Avro, Parquet, or ORC formats)."

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- Auto-Discovery Feature: Dataplex has an auto-discovery feature that, when enabled, automatically discovers and catalogs data assets within a zone. - Appropriate for Both Raw and Curated Zones: This feature is applicable to both raw and curated zones, and it should be tailored to the specific data governance and cataloging needs of the organization.

cloud_rider 1 year, 4 months ago

A is correct, Auto-Discovery features works on both curated and raw zones, but to keep JSON and CSV in curated zone, they must be kept along with the specification. Whereas in RAW zone, the discovery of these files happens even without specification file. refer to this link->> https://cloud.google.com/dataplex/docs/discover-data#discovery-configuration

22c1725 10 months, 2 weeks ago

Selected Answer: B

Still if you go with "A" you need to do "B". the question is not about best practice.

Positron75 10 months ago

This isn't about best practice. The documentation outright states that data *not* in Avro, Parquet, or ORC formats within curated zones is considered invalid for the purposes of discovery: https://cloud.google.com/dataplex/docs/discover-data?hl=en#invalid_data_format

rajshiv 11 months, 3 weeks ago

Selected Answer: B

We can store JSON and CSV in the curated zone if those files represent curated data. Usually we store JSON/CSV files in the raw zone if they are straight from source. But nowhere in the question is any of that detail mentioned. So I think the correct answer is : B - Dataplex automatically discovers and catalogs data in the zones only if auto-discovery is enabled for the zone or asset. In this scenario - JSON and CSV files are being uploaded to a curated zone, which is fine. But if files are not being discovered, it's likely because auto-discovery is not enabled for that zone.

MBNR 1 year ago

Selected Answer: A

Answer is A Data Format supported: Data in curated zones is typically columnar, Hive-partitioned, and stored in formats like Parquet, Avro, or ORC Restrictions: Dataplex does NOT allow users to create CSV files within a "curated zone

desertlotus1211 1 year ago

Auto-Discovery is the better option

juliorevk 1 year, 2 months ago

Selected Answer: A

- Raw zones store structured data, semi-structured data such as CSV files and JSON files, and unstructured data in any format from external sources. Raw zones are useful for staging raw data before performing any transformations. Data can be stored in Cloud Storage buckets or BigQuery datasets. - Curated Zones do not support JSON / CSV

SamuelTsch 1 year, 5 months ago

Selected Answer: A

Raw zones store structured data, semi-structured data such as CSV files and JSON files, and unstructured data in any format from external sources. Curated zones store structured data. Data can be stored in Cloud Storage buckets or BigQuery datasets. Supported formats for Cloud Storage buckets include Parquet, Avro, and ORC.

rajnairds 1 year, 7 months ago

Selected Answer: B

Discovery configuration Discovery is enabled by default when you create a new zone or asset. You can disable Discovery at the zone or asset level. For each Dataplex asset with Discovery enabled, Dataplex does the following: Scans the data associated with the asset. Groups structured and semi-structured files into tables. Collects technical metadata, such as table name, schema, and partition definition. For unstructured data, such as images and videos, Dataplex Discovery automatically detects and registers groups of files sharing media type as filesets. For example, if gs://images/group1 contains GIF images, and gs://images/group2 contains JPEG images, Dataplex Discovery detects and registers two filesets. For structured data, such as Avro, Discovery detects files only if they are located in folders that contain the same data format and schema. Reference : https://cloud.google.com/dataplex/docs/discover-data#exclude-files-from-Discovery

hussain.sain 1 year, 9 months ago

Selected Answer: B

While JSON and CSV can technically be stored in curated zones, it is not a common practice due to the reasons mentioned above. no where in the mention link its mention that there is a restriction.

Anudeep58 1 year, 9 months ago

Selected Answer: A

While none of the original options (A, B, C, or D) directly address the issue, the closest solution is: Move the JSON and CSV files to a raw zone. (This was previously marked as the most voted option, but it's not ideal due to data organization disruption) Here's why this approach might be necessary (but not ideal): Dataplex curated zones currently don't support native processing of JSON and CSV formats. They are designed for structured data formats like Parquet, Avro, or ORC.

chrissamharris 1 year, 11 months ago

Selected Answer: A

Option A https://cloud.google.com/dataplex/docs/add-zone#raw-zones Raw zones are the only zones that support CSV & JSON

joao_01 1 year, 11 months ago

Its B guys, i encounter this in my job, and I had to do B to make it work

joao_01 1 year, 11 months ago

Actually I did this in a Raw zone, not Curated.

joao_01 1 year, 11 months ago

Its A :)

demoro86 2 years ago

Selected Answer: A

GCP001 agree with him

Moss2011 2 years, 1 month ago

Selected Answer: A

The answer can be found reading a common config of Dataplex in this URL: https://medium.com/google-cloud/google-cloud-dataplex-part-1-lakes-zones-assets-and-discovery-5f288486cb2f

kck6ra4214wm 2 years, 1 month ago

Selected Answer: A

Dataplex does not allow users to create CSV files within a “curated zone”

daidai75 2 years, 1 month ago

Selected Answer: B

According to this URL: https://cloud.google.com/dataplex/docs/discover-data, the auto-discovery can support CSV and Json in both Raw-Zone and Curated-Zone. I also open a console the verify it, both Raw and Curated zone can set up csv&json auto-discovery.

dungct 2 years, 1 month ago

Selected Answer: B

Discovery raises the following administrator actions whenever data-related issues are detected during scans : Inconsistent data format in a table. For example, files of different formats exist with the same table prefix. Inconsistent data format in a table. For example, files of different formats exist with the same table prefix.

dungct 2 years, 1 month ago

https://cloud.google.com/dataplex/docs/discover-data#invalid_data_format

https://zhangxuetu.online

[202610]

Question #304 Topic 1

You have a table that contains millions of rows of sales data, partitioned by date. Various applications and users query this data many times a minute. The query requires aggregating values by using AVG, MAX, and SUM, and does not require joining to other tables. The required aggregations are only computed over the past year of data, though you need to retain full historical data in the base tables. You want to ensure that the query results always include the latest data from the tables, while also reducing computation cost, maintenance overhead, and duration. What should you do?

A. Create a materialized view to aggregate the base table data. Include a filter clause to specify the last one year of partitions. **Most Voted**

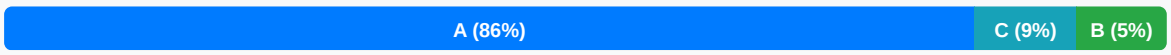
B. Create a materialized view to aggregate the base table data. Configure a partition expiration on the base table to retain only the last one year of partitions.

C. Create a view to aggregate the base table data. Include a filter clause to specify the last year of partitions.

D. Create a new table that aggregates the base table data. Include a filter clause to specify the last year of partitions. Set up a scheduled query to recreate the new table every hour.

Correct Answer: A

Community vote distribution



Comments

raaad **Highly Voted** 1 year, 8 months ago

Selected Answer: A

- Materialized View: Materialized views in BigQuery are precomputed views that periodically cache the result of a query for increased performance and efficiency. They are especially beneficial for heavy and repetitive aggregation queries. - Filter for Recent Data: Including a clause to focus on the last year of partitions ensures that the materialized view is only storing and updating the relevant data, optimizing storage and refresh time. - Always Up-to-date: Materialized views are maintained by BigQuery and automatically updated at regular intervals, ensuring they include the latest data up to a certain freshness point.

22c1725 10 months, 2 weeks ago

Selected Answer: A

Would go with "A".

MBNR 1 year ago

Selected Answer: A

Answer : A Question has below three requirements , it did NOT talk about STORAGE cost Reducing computation cost: Using Materialized views in BigQuery, query costs can be lower due to faster performance maintenance overhead : Bigquery takes care of data updates and duration: Since the results are precomputed and stored , it takes very less time for the query output

JyoGCP 1 year, 7 months ago

Selected Answer: A

Option A

et2137 1 year, 7 months ago

Selected Answer: C

materialized view requires refreshing so it might not fulfill the requirement: "results always include the latest data from the tables". Opt. C will give you the newest data every time you execute the query but it will have to be computed every time

d11379b 1 year, 6 months ago

Agree, these questions always play with words, making many of options seem plausible

d11379b 1 year, 6 months ago

But materialized view always returns fresh data Fresh data. Materialized views return fresh data. If changes to base tables might invalidate the materialized view, then data is read directly from the base tables. If the changes to the base tables don't invalidate the materialized view, then rest of the data is read from the materialized view and only the changes are read from the base tables. https://cloud.google.com/bigquery/docs/materialized-views-intro

casadocc 1 year, 8 months ago

A We can do aggregations, bit if not specified table will not be partitioned on the view. B partition expiration is not possible, as expiration is the same as base table C It might be the right one, although not specific savings vs the original query, but here we would guarantee accessing only last year data. D not a good one in any sense A and C might be equally good solutions depending on some understandings. Would probably opt for A

Matt_108 1 year, 8 months ago

Selected Answer: A

. Create a materialized view to aggregate the base table data. Include a filter clause to specify the last one year of partitions.

Sofiia98 1 year, 8 months ago

Selected Answer: A

To preserve the historical data

scaenruy 1 year, 9 months ago

Selected Answer: B

B. Create a materialized view to aggregate the base table data. Configure a partition expiration on the base table to retain only the last one year of partitions.

raaad 1 year, 8 months ago

Why not B - Configuring partition expiration on the BASE TABLE is a way to manage storage and costs by automatically dropping old data. However, the question specifies the need to retain full historical data, making this approach not suitable since it doesnt keep all historical records.

Sofiia98 1 year, 8 months ago

Don't agree, it is said thad that we need to store the historical data, so answer A is correct

https://zhangxuetu.online

[202610]

Question #305 Topic 1

Your organization uses a multi-cloud data storage strategy, storing data in Cloud Storage, and data in Amazon Web Services' (AWS) S3 storage buckets. All data resides in US regions. You want to query up-to-date data by using BigQuery, regardless of which cloud the data is stored in. You need to allow users to query the tables from BigQuery without giving direct access to the data in the storage buckets. What should you do?

A. Setup a BigQuery Omni connection to the AWS S3 bucket data. Create BigLake tables over the Cloud Storage and S3 data and query the data using BigQuery directly. **Most Voted**

B. Set up a BigQuery Omni connection to the AWS S3 bucket data. Create external tables over the Cloud Storage and S3 data and query the data using BigQuery directly.

C. Use the Storage Transfer Service to copy data from the AWS S3 buckets to Cloud Storage buckets. Create BigLake tables over the Cloud Storage data and query the data using BigQuery directly.

D. Use the Storage Transfer Service to copy data from the AWS S3 buckets to Cloud Storage buckets. Create external tables over the Cloud Storage data and query the data using BigQuery directly.

Correct Answer: A

Community vote distribution



Comments

raaad **Highly Voted** 2 years, 2 months ago

Selected Answer: A

- BigQuery Omni: This is an extension of BigQuery that allows you to analyze data across Google Cloud, AWS, and Azure without having to manage the infrastructure or move data across clouds. It's suitable for querying data stored in AWS S3 buckets directly. - BigLake: Allows you to create a logical abstraction (table) over data stored in Cloud Storage and S3, so you can query data using BigQuery without moving it. - Unified Querying: By setting up BigQuery Omni to connect to AWS S3 and creating BigLake tables over both Cloud Storage and S3 data, you can query all data using BigQuery directly.

AllenChen123 2 years, 2 months ago

Agree. https://cloud.google.com/bigquery/docs/omni-introduction "To run BigQuery analytics on your external data, you first need to connect to Amazon S3 or Blob Storage. If you want to query external data, you would need to create a BigLake table that references Amazon S3 or Blob Storage data."

ML6 2 years, 1 month ago

I wonder, why BigLake tables (A) over external tables (B)?

aoifneofi_ef 1 year, 7 months ago

external tables can be created only on data residing in Cloud Storage, BigTable or Google Drive: https://cloud.google.com/bigquery/docs/external-tables. Hence creating external tables WITHOUT BQ Omni is not an option

Ben_oso 9 months, 1 week ago

Selected Answer: B

B, Biglake dont support to create table from S3

Ben_oso 9 months, 1 week ago

Selected Answer: A

Biglake dont support create tables from aws S3

JyoGCP 2 years, 1 month ago

Selected Answer: A

Option A

Matt_108 2 years, 2 months ago

Selected Answer: A

Option A - clearly explained in comments

chickenwingz 2 years, 3 months ago

Selected Answer: A

A - BigLake tables work for S3 and GCS

chickenwingz 2 years, 3 months ago

https://cloud.google.com/bigquery/docs/external-data-sources#external_data_source_feature_comparison

https://zhangxuetu.online

[202610]

Question #306 Topic 1

You are preparing an organization-wide dataset. You need to preprocess customer data stored in a restricted bucket in Cloud Storage. The data will be used to create consumer analyses. You need to comply with data privacy requirements.

What should you do?

A. Use Dataflow and the Cloud Data Loss Prevention API to mask sensitive data. Write the processed data in BigQuery. **Most Voted**

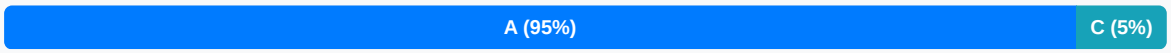
B. Use customer-managed encryption keys (CMEK) to directly encrypt the data in Cloud Storage. Use federated queries from BigQuery. Share the encryption key by following the principle of least privilege.

C. Use the Cloud Data Loss Prevention API and Dataflow to detect and remove sensitive fields from the data in Cloud Storage. Write the filtered data in BigQuery.

D. Use Dataflow and Cloud KMS to encrypt sensitive fields and write the encrypted data in BigQuery. Share the encryption key by following the principle of least privilege.

Correct Answer: A

Community vote distribution



Comments

raaad **Highly Voted** 1 year, 8 months ago

Selected Answer: A

- Prioritizes Data Privacy: It protects sensitive information by masking it, reducing the risk of exposure in case of unauthorized access or accidental leaks. - Reduces Data Sensitivity: Masking renders sensitive data unusable for attackers, even if they gain access to it. - Preserves Data Utility: Masked data can still be used for consumer analyses, as patterns and relationships are often preserved, allowing meaningful insights to be derived.

dungct 1 year, 7 months ago

why not d ?

ML6 1 year, 7 months ago

Data in Cloud Storage is encrypted by default.

22c1725 10 months, 2 weeks ago

Selected Answer: A

this is repeated question.

desertlotus1211 1 year ago

Selected Answer: C

If I had to choose... I choose C or A... A can still leave partial sensitive data available.

desertlotus1211 1 year ago

For data privacy, removing data through a DLP (Data Loss Prevention) system is generally considered better than masking, as it permanently eliminates sensitive information, whereas masking only conceals it, potentially leaving traces or vulnerabilities

AlizCert 1 year, 3 months ago

Selected Answer: A

What made me decide on A instead of C was the "The data will be used to create consumer analyses" sentence. Having all the PIIs completely redacted from the records, we were unable to distinguish between the individual customers.

Matt_108 1 year, 8 months ago

Selected Answer: A

Option A, agree with raaad explanation

scaenruy 1 year, 9 months ago

Selected Answer: A

A. Use Dataflow and the Cloud Data Loss Prevention API to mask sensitive data. Write the processed data in BigQuery.

https://zhangxuetu.online

[202610]

Question #307 Topic 1

You need to connect multiple applications with dynamic public IP addresses to a Cloud SQL instance. You configured users with strong passwords and enforced the SSL connection to your Cloud SQL instance. You want to use Cloud SQL public IP and ensure that you have secured connections. What should you do?

A. Add CIDR 0.0.0.0/0 network to Authorized Network. Use Identity and Access Management (IAM) to add users.

B. Add all application networks to Authorized Network and regularly update them.

C. Leave the Authorized Network empty. Use Cloud SQL Auth proxy on all applications. **Most Voted**

D. Add CIDR 0.0.0.0/0 network to Authorized Network. Use Cloud SQL Auth proxy on all applications.

Correct Answer: C

Community vote distribution



Comments

raaad **Highly Voted** 1 year, 8 months ago

Selected Answer: C

- Using the Cloud SQL Auth proxy is a recommended method for secure connections, especially when dealing with dynamic IP addresses. - The Auth proxy provides secure access to your Cloud SQL instance without the need for Authorized Networks or managing IP addresses. - It works by encapsulating database traffic and forwarding it through a secure tunnel, using Google's IAM for authentication. - Leaving the Authorized Networks empty means you're not allowing any direct connections based on IP addresses, relying entirely on the Auth proxy for secure connectivity. This is a secure and flexible solution, especially for applications with dynamic IPs.

FreshMind 1 year ago

In question "You want to use Cloud SQL public IP", how this could be if "Leaving the Authorized Networks empty means you're not allowing any direct connections based on IP addresses" ?

JyoGCP 1 year, 7 months ago

Selected Answer: C

Option C

Pukapuiz 1 year, 8 months ago

Selected Answer: C

The Cloud SQL Auth Proxy is a Cloud SQL connector that provides secure access to your instances without a need for Authorized networks or for configuring SSL. https://cloud.google.com/sql/docs/mysql/sql-proxy

Matt_108 1 year, 8 months ago

Selected Answer: C

always use Cloud SQL Auth proxy if possible

Sofiia98 1 year, 8 months ago

https://stackoverflow.com/questions/27759356/how-to-authorize-my-dynamic-ip-network-address-in-google-cloud-sql
https://stackoverflow.com/questions/24749810/how-to-make-a-google-cloud-sql-instance-accessible-for-any-ip-address

JyoGCP 1 year, 7 months ago

Links also say not to go with option D. 0.0.0.0/0 which includes all possible IP Addresses is not recommended for security reasons. You have to keep access as restricted as possible.

Sofiia98 1 year, 8 months ago

Selected Answer: D

As for me, after reading documentation, option D looks appropriate

BennyXu 1 year, 5 months ago

Save your shxx answer in your dxxb head.

scaenruy 1 year, 9 months ago

Selected Answer: C

C. Leave the Authorized Network empty. Use Cloud SQL Auth proxy on all applications.

https://zhangxuetu.online

[202610]

Question #308 Topic 1

You are migrating a large number of files from a public HTTPS endpoint to Cloud Storage. The files are protected from unauthorized access using signed URLs. You created a TSV file that contains the list of object URLs and started a transfer job by using Storage Transfer Service. You notice that the job has run for a long time and eventually failed. Checking the logs of the transfer job reveals that the job was running fine until one point, and then it failed due to HTTP 403 errors on the remaining files. You verified that there were no changes to the source system. You need to fix the problem to resume the migration process. What should you do?

A. Set up Cloud Storage FUSE, and mount the Cloud Storage bucket on a Compute Engine instance. Remove the completed files from the TSV file. Use a shell script to iterate through the TSV file and download the remaining URLs to the FUSE mount point.

B. Renew the TLS certificate of the HTTPS endpoint. Remove the completed files from the TSV file and rerun the Storage Transfer Service job.

C. Create a new TSV file for the remaining files by generating signed URLs with a longer validity period. Split the TSV file into multiple smaller files and submit them as separate Storage Transfer Service jobs in parallel. Most Voted

D. Update the file checksums in the TSV file from using MD5 to SHA256. Remove the completed files from the TSV file and rerun the Storage Transfer Service job.

Correct Answer: C

Community vote distribution

C (100%)

Comments

raaad Highly Voted 2 years, 2 months ago

Selected Answer: C

- It addresses the likely issue: that the signed URLs have expired or are otherwise invalid. By creating a new TSV file with freshly generated signed URLs (with a longer validity period), you're ensuring that the Storage Transfer Service has valid authorization to access the files. - Splitting the TSV file and running parallel jobs might help in managing the workload more efficiently and overcoming any limitations related to the number of files or transfer speed.

srivastavas08 Highly Voted 2 years, 1 month ago

C. Create a new TSV file for the remaining files by generating signed URLs with a longer validity period. Split the TSV file into multiple smaller files and submit them as separate Storage Transfer Service jobs in parallel. Here's why: HTTP 403 errors: These errors indicate unauthorized access, but since you verified the source system and signed URLs, the issue likely lies with expired signed URLs. Renewing the URLs with a longer validity period prevents this issue for the

remaining files. Separate jobs: Splitting the file into smaller chunks and submitting them as separate jobs improves parallelism and potentially speeds up the transfer process. Avoid manual intervention: Options A and D require manual intervention and complex setups, which are less efficient and might introduce risks. Longer validity: While option B addresses expired URLs, splitting the file offers additional benefits for faster migration.

iooj 1 year, 8 months ago

Selected Answer: C

got this one on the exam, aug 2024, passed

Matt_108 2 years, 2 months ago

Selected Answer: C

Option C - agree with Raaad

scaenruy 2 years, 2 months ago

Selected Answer: C

C. Create a new TSV file for the remaining files by generating signed URLs with a longer validity period. Split the TSV file into multiple smaller files and submit them as separate Storage Transfer Service jobs in parallel.

https://zhangxuetu.online

[202610]

Question #309 Topic 1

You work for an airline and you need to store weather data in a BigQuery table. Weather data will be used as input to a machine learning model. The model only uses the last 30 days of weather data. You want to avoid storing unnecessary data and minimize costs. What should you do?

A. Create a BigQuery table where each record has an ingestion timestamp. Run a scheduled query to delete all the rows with an ingestion timestamp older than 30 days.

B. Create a BigQuery table partitioned by datetime value of the weather date. Set up partition expiration to 30 days. Most Voted

C. Create a BigQuery table partitioned by ingestion time. Set up partition expiration to 30 days.

D. Create a BigQuery table with a datetime column for the day the weather data refers to. Run a scheduled query to delete rows with a datetime value older than 30 days.

Correct Answer: B

Community vote distribution



Comments

iooj Highly Voted 1 year, 8 months ago

Selected Answer: B

got this one on the exam, aug 2024, passed

AllenChen123 Highly Voted 2 years, 2 months ago

Selected Answer: B

Partitioned based on weather date, with partition expiration set

desertlotus1211 1 year ago

Selected Answer: C

Partitioning by ingestion time is simpler and sufficient if data retention is based on load time, not the data's internal timestamp

juliorevk 1 year, 2 months ago

Selected Answer: B

B BQ partitioning with partition expiration of 30 days allows you to only filter for the last 30 days and delete days that are beyond 30 days.

d11379b 2 years ago

https://cloud.google.com/bigquery/docs/partitioned-tables Here it mentions “ For TIMESTAMP and DATETIME columns, the partitions can have either hourly, daily, monthly, or yearly granularity.l So you should not calculate the amount of partitions on second granularity

chambg 2 years ago

Selected Answer: D

Skeptical about Option B as maximum partitions in a BQ table is 4000.Since Datetime value is a timestamp it will have more than 4000 values in a duration for 30 days (30*24*60*60 = 259,200). So Option D is right imo

ce9e395 2 years ago

This is a good point

joao_01 1 year, 11 months ago

It's not a good point. The granularity goes to DAYs, not SECONDs. So, the right answer is B.

JyoGCP 2 years, 1 month ago

Selected Answer: B

Option B

Sofiia98 2 years, 1 month ago

Selected Answer: B

We need the last 30 days, we don't care about ingestion time

https://zhangxuetu.online

[202610]

Question #310 Topic 1

You need to look at BigQuery data from a specific table multiple times a day. The underlying table you are querying is several petabytes in size, but you want to filter your data and provide simple aggregations to downstream users. You want to run queries faster and get up-to-date insights quicker. What should you do?

A. Run a scheduled query to pull the necessary data at specific intervals dally.

B. Use a cached query to accelerate time to results.

C. Limit the query columns being pulled in the final result.

D. Create a materialized view based off of the query being run. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

AllenChen123 Highly Voted 1 year, 8 months ago

Selected Answer: D

Create a materialized view as query source. Materialized views are precomputed views that periodically cache the results of a query for increased performance and efficiency.

Shenbasekhar 1 year, 7 months ago

Selected Answer: D

Option D. Materialized view

Sofiia98 1 year, 8 months ago

Selected Answer: D

materialized view

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #311 Topic 1

Your chemical company needs to manually check documentation for customer order. You use a pull subscription in Pub/Sub so that sales agents get details from the order. You must ensure that you do not process orders twice with different sales agents and that you do not add more complexity to this workflow. What should you do?

A. Use a Deduplicate PTransform in Dataflow before sending the messages to the sales agents.

B. Create a transactional database that monitors the pending messages.

C. Use Pub/Sub exactly-once delivery in your pull subscription. **Most Voted**

D. Create a new Pub/Sub push subscription to monitor the orders processed in the agent's system.

Correct Answer: C

Community vote distribution



Comments

JimmyBK **Highly Voted** 2 years, 2 months ago

Selected Answer: C

I remember seeing this in the exam.

Jordan18 2 years, 2 months ago

how many questions were from here?

iooj 1 year, 8 months ago

also got this one. about 70%

22c1725 10 months, 2 weeks ago

Selected Answer: C

"that you do not add more complexity to this workflow". I would go with C

22c1725 10 months, 2 weeks ago

Selected Answer: C

I would go with pup/sup even thou ack might be cuse message to be send twice.

cien91 1 year, 8 months ago

Selected Answer: A

Why not C - Exactly-once delivery in Pub/Sub guarantees that a message is delivered to a subscriber exactly once. However, it doesn't prevent multiple subscribers from processing the same message.

JyoGCP 2 years, 1 month ago

Selected Answer: C

Option C

Sofiia98 2 years, 1 month ago

Selected Answer: C

C, of course

AllenChen123 2 years, 2 months ago

Selected Answer: C

Straightforward. https://cloud.google.com/pubsub/docs/exactly-once-delivery

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #312 Topic 1

You are migrating your on-premises data warehouse to BigQuery. As part of the migration, you want to facilitate cross-team collaboration to get the most value out of the organization's data. You need to design an architecture that would allow teams within the organization to securely publish, discover, and subscribe to read-only data in a self-service manner. You need to minimize costs while also maximizing data freshness. What should you do?

A. Use Analytics Hub to facilitate data sharing. **Most Voted**

B. Create authorized datasets to publish shared data in the subscribing team's project.

C. Create a new dataset for sharing in each individual team's project. Grant the subscribing team the bigquery.dataViewer role on the dataset.

D. Use BigQuery Data Transfer Service to copy datasets to a centralized BigQuery project for sharing.

Correct Answer: A

Community vote distribution

A (100%)

Comments

22c1725 10 months, 2 weeks ago

Selected Answer: A

Analytics Hub is designed to enable secure and scalable data sharing across organizational boundaries

FireAtMe 1 year, 3 months ago

Selected Answer: A

Analytics Hub is a fully managed data sharing platform provided by Google Cloud. It allows organizations to publish, discover, and subscribe to datasets securely and efficiently. It facilitates collaboration across teams or even across organizations by enabling self-service access to shared data without duplicating or moving it.

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #313 Topic 1

You want to migrate an Apache Spark 3 batch job from on-premises to Google Cloud. You need to minimally change the job so that the job reads from Cloud Storage and writes the result to BigQuery. Your job is optimized for Spark, where each executor has 8 vCPU and 16 GB memory, and you want to be able to choose similar settings. You want to minimize installation and management effort to run your job. What should you do?

A. Execute the job as part of a deployment in a new Google Kubernetes Engine cluster.

B. Execute the job from a new Compute Engine VM.

C. Execute the job in a new Dataproc cluster.

D. Execute as a Dataproc Serverless job. **Most Voted**

Correct Answer: D

Community vote distribution

D (52%) C (48%)

Comments

chicity_de **Highly Voted** 1 year, 3 months ago

Selected Answer: D

Priority is "minimize installation and management effort" which is done via Dataproc Serverless. Furthermore, with Dataproc serverless you can still specify resource settings for your job, such as the number of vCPUs and memory per executor (https://cloud.google.com/dataproc-serverless/docs/concepts/properties)

22c1725 10 months, 2 weeks ago

Selected Answer: C

I would go with (C) : "and you want to be able to choose similar settings" not applicable for (D)

Positron75 10 months ago

Dataproc Serverless allows configuring those parameters: https://cloud.google.com/dataproc-serverless/docs/concepts/properties

gabbferreira 11 months, 1 week ago

Selected Answer: C

" where each executor has 8 vCPU and 16 GB memory, and you want to be able to choose similar settings." minimize effort by using dataproc not GKE or vms..

rajshiv 11 months, 3 weeks ago

Selected Answer: D

D is the best answer - It allows Dataproc serverless allows to specify executor configurations like vCPU and memory settings (e.g., executor cores and memory) to match the current setup as is specified. C is a valid but sub-optimal choice — while we can specify vCPUs and memory similar to our on-prem setup but it requires provisioning and managing clusters, which we want to avoid. it requires slightly more effort compared to Dataproc Serverless.

gabazzzo 1 year, 1 month ago

I agree that minimizing installation and management means using Dataproc Serverles. Also, Serverles can be configured with up to 16 VPU and up to 29696m of memory in for the premium tier. https://cloud.google.com/dataproc-serverless/docs/concepts/properties#:~:text=Total%20driver%20memory%20per%20driver%20core%2C%20including%20driver%20memory%20overhead%2C%20which%20must%20be%20between%201024m%20and%207424m%20for%20the%20Standard%20compute%20tier%20(24576m%20for%20the%20Premium%20compute%20tier).%20For%20example%2C%20if%20spark.driver.cores%20%3D%204%2C%20then%204096m%20%3C%3D%20spark.driver.memory%20%2B%20spark.driver.memoryOverhead%20%3C%3D%2029696m.

a494e30 1 year, 1 month ago

Selected Answer: C

Needs to be able to configure "similar settings"

plum21 1 year, 1 month ago

Selected Answer: C

It's not possible to specify a machine type using Dataproc Serverless

marlon.andrei 1 year, 2 months ago

Selected Answer: C

I choice "C", just: "where each executor has 8 vCPU and 16 GB memory, and you want to be able to choose similar settings"

mcdaley 1 year, 3 months ago

Selected Answer: C

Dataproc supports Spark 3, ensuring compatibility with your existing job. It also allows you to customize the cluster configuration, including the number of executors, vCPUs, and memory per executor, to match your on-premises setup (8 vCPU and 16 GB memory)

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #314 Topic 1

You are configuring networking for a Dataflow job. The data pipeline uses custom container images with the libraries that are required for the transformation logic preinstalled. The data pipeline reads the data from Cloud Storage and writes the data to BigQuery. You need to ensure cost-effective and secure communication between the pipeline and Google APIs and services. What should you do?

A. Disable external IP addresses from worker VMs and enable Private Google Access. **Most Voted**

B. Leave external IP addresses assigned to worker VMs while enforcing firewall rules.

C. Disable external IP addresses and establish a Private Service Connect endpoint IP address.

D. Enable Cloud NAT to provide outbound internet connectivity while enforcing firewall rules.

Correct Answer: A

Community vote distribution

A (100%)

Comments

m_a_p_s 1 year, 3 months ago

Selected Answer: A

While option C is technically implementable, option A is a straightforward and a simpler solution.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #315 Topic 1

You are using Workflows to call an API that returns a 1KB JSON response, apply some complex business logic on this response, wait for the logic to complete, and then perform a load from a Cloud Storage file to BigQuery. The Workflows standard library does not have sufficient capabilities to perform your complex logic, and you want to use Python's standard library instead. You want to optimize your workflow for simplicity and speed of execution. What should you do?

A. Create a Cloud Composer environment and run the logic in Cloud Composer.

B. Create a Dataproc cluster, and use PySpark to apply the logic on your JSON file.

C. Invoke a Cloud Function instance that uses Python to apply the logic on your JSON file. **Most Voted**

D. Invoke a subworkflow in Workflows to apply the logic on your JSON file.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Arvi1984 7 months, 1 week ago

Selected Answer: C

find alternative of "does not have sufficient capabilities to perform your complex logic" in GCP which is cloud function

22c1725 10 months, 2 weeks ago

Selected Answer: C

"A" not possible since you will be runing the same logic inside of airflow nothing else. dataproc is unneeded.

FireAtMe 1 year, 3 months ago

Selected Answer: C

Cloud Functions is a serverless compute service ideal for executing lightweight, event-driven tasks with low latency.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #316 Topic 1

You are administering a BigQuery on-demand environment. Your business intelligence tool is submitting hundreds of queries each day that aggregate a large (50 TB) sales history fact table at the day and month levels. These queries have a slow response time and are exceeding cost expectations. You need to decrease response time, lower query costs, and minimize maintenance. What should you do?

A. Build authorized views on top of the sales table to aggregate data at the day and month level.

B. Enable BI Engine and add your sales table as a preferred table.

C. Build materialized views on top of the sales table to aggregate data at the day and month level. **Most Voted**

D. Create a scheduled query to build sales day and sales month aggregate tables on an hourly basis.

Correct Answer: C

Community vote distribution

C (100%)

Comments

hussain.sain 1 year, 3 months ago

Selected Answer: C

C is the answer. Materialized Views: Materialized views in BigQuery are precomputed views that store the results of a query, allowing for much faster query execution because BigQuery doesn't need to recompute the results each time the query is run. The results are stored in a persistent table, which significantly improves performance for repeated queries that aggregate the same data. In this case, you can create materialized views that aggregate the sales data at the day and month levels. This will reduce the amount of data that needs to be processed for each query and speed up the response time. Materialized views also lower costs because BigQuery only scans the precomputed data in the materialized view, rather than the full 50 TB sales history table.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #317 Topic 1

You have several different unstructured data sources, within your on-premises data center as well as in the cloud. The data is in various formats, such as Apache Parquet and CSV. You want to centralize this data in Cloud Storage. You need to set up an object sink for your data that allows you to use your own encryption keys. You want to use a GUI-based solution. What should you do?

A. Use BigQuery Data Transfer Service to move files into BigQuery.

B. Use Storage Transfer Service to move files into Cloud Storage

C. Use Dataflow to move files into Cloud Storage

D. Use Cloud Data Fusion to move files into Cloud Storage. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

22c1725 10 months, 2 weeks ago

Selected Answer: D

I would go with "D" since GUI is required.

apoio.certificacoes.closer 1 year, 3 months ago

Selected Answer: D

I have read in previous questions that Transfer Services only uses CMEK in-transit. https://cloud.google.com/storage-transfer/docs/on-prem-security#in-flight

m_a_p_s 1 year, 3 months ago

Selected Answer: D

D - only Cloud Data Fusion is a GUI-based solution.

skycracker 1 year, 3 months ago

Selected Answer: D

data fusion allows encryption

noiz 1 year, 3 months ago

Selected Answer: B

Is B incorrect? Transfer service + CloudKMS

22c1725 10 months, 2 weeks ago

There is no GUI.

https://zhangxuetu.online

[202610]

Question #318 Topic 1

You are using BigQuery with a regional dataset that includes a table with the daily sales volumes. This table is updated multiple times per day. You need to protect your sales table in case of regional failures with a recovery point objective (RPO) of less than 24 hours, while keeping costs to a minimum. What should you do?

A. Schedule a daily export of the table to a Cloud Storage dual or multi-region bucket. **Most Voted**

B. Schedule a daily copy of the dataset to a backup region.

C. Schedule a daily BigQuery snapshot of the table.

D. Modify ETL job to load the data into both the current and another backup region.

Correct Answer: A

Community vote distribution



Comments

22c1725 10 months, 2 weeks ago

Selected Answer: A

Not 'C' snapchot are stored in the same region.

desertlotus1211 11 months, 3 weeks ago

Selected Answer: A

almost the same as 211, 211 says multi-region vs regional...

Parandhaman_Margan 1 year ago

Selected Answer: C

Meets the RPO requirement (< 24 hours) Cost-effective solution Quick recovery from regional failures

gabbferreira 11 months, 1 week ago

snapshots are stored in the same region, so it dont protect from regional failure

MarcoPellegrino 1 year, 1 month ago

Selected Answer: D

https://cloud.google.com/blog/topics/developers-practitioners/backup-disaster-recovery-strategies-bigquery Google presents both A and D Why A: - Cost: Lower. GCS storage is significantly cheaper than BigQuery storage. You pay for storage in GCS and minimal egress charges when exporting. - Complexity: Simpler. You schedule a daily export job. Restoring involves importing from GCS to BigQuery in another region. - Consistency: Easier to manage. The export process creates a consistent snapshot of the data at the time of export. You might have some latency (up to 24 hours in this scenario), but the data within the export is consistent. - RPO: Meets the requirement. A daily export ensures an

RPO of less than 24 hours. - RTO: Depends on the restore process from GCS to BigQuery. You can pre-provision slots in the backup region to minimize restore time.

FireAtMe 1 year, 3 months ago

Selected Answer: A

Both A and B works. But it is cheaper to save data in GCS.

joelcaro 1 year, 3 months ago

Selected Answer: D

Opción D: Modify ETL job to load the data into both the current and another backup region Evaluación: Ajustar el ETL para escribir en dos tablas (una en la región principal y otra en una región de respaldo) asegura que los datos estén disponibles en ambas ubicaciones casi en tiempo real. Esto garantiza un RPO de menos de 24 horas, ya que las actualizaciones intradía se reflejan en ambas regiones. Aunque podría aumentar los costos de almacenamiento por duplicar los datos, es la solución más efectiva y directa para proteger contra fallos regionales.

mdell 1 year, 3 months ago

Selected Answer: B

In most cases, it is cheaper to copy a BigQuery dataset to a new region directly rather than exporting it to a Cloud Storage bucket and then loading it into a new BigQuery dataset in the desired region, as you only pay for data transfer costs when copying within BigQuery, while exporting to a bucket incurs additional storage charges for the exported data in Cloud Storage, even if it's only temporary. Key points to consider: No extra storage cost for copying: When copying a BigQuery dataset to a new region, you only pay for the data transfer cost, not the storage of the data in a separate location. Storage cost for exporting: Exporting data to a Cloud Storage bucket means you are charged for the storage of that data in the bucket until you delete it, even if you are just temporarily storing it for transfer.

HectorLeon2099 1 year, 3 months ago

Selected Answer: A

Option A is the most cost efficient: https://cloud.google.com/blog/topics/developers-practitioners/backup-disaster-recovery-strategies-bigquery

mdell 1 year, 3 months ago

Additionally it only mentions backing up the sales table and not the entire dataset

https://zhangxuetu.online

[202610]

Question #319 Topic 1

You are preparing an organization-wide dataset. You need to preprocess customer data stored in a restricted bucket in Cloud Storage. The data will be used to create consumer analyses. You need to follow data privacy requirements, including protecting certain sensitive data elements, while also retaining all of the data for potential future use cases. What should you do?

A. Use the Cloud Data Loss Prevention API and Dataflow to detect and remove sensitive fields from the data in Cloud Storage. Write the filtered data in BigQuery.

B. Use customer-managed encryption keys (CMEK) to directly encrypt the data in Cloud Storage. Use federated queries from BigQuery. Share the encryption key by following the principle of least privilege.

C. Use Dataflow and the Cloud Data Loss Prevention API to mask sensitive data. Write the processed data in BigQuery. **Most Voted**

D. Use Dataflow and Cloud KMS to encrypt sensitive fields and write the encrypted data in BigQuery. Share the encryption key by following the principle of least privilege.

Correct Answer: C

Community vote distribution

C (100%)

Comments

HectorLeon2099 Highly Voted 1 year, 3 months ago

Selected Answer: C

It's C. "A" removes data and retaining all is a requirement.

22c1725 10 months, 2 weeks ago

Selected Answer: C

Not A since "while also retaining all of the data" is required.

22c1725 10 months, 2 weeks ago

Selected Answer: C

Removing data will lead to unability to do study & consumer analyses. since it's likely all of consumer data is under PII.

EIM_Exam_Topics 1 week, 2 days ago

While I agree with C, removing data does not lead to the inability to do consumer analyses. Analyses such as segmentation and clustering don't require PII data.

Nagamanikanta 1 year ago

Selected Answer: C

option C we can simply mask the data and process in biguery

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #320 Topic 1

Your company uses Looker Studio connected to BigQuery for reporting. Users are experiencing slow dashboard load times due to complex queries on a large table. The queries involve aggregations and filtering on several columns. You need to optimize query performance to decrease the dashboard load times. What should you do?

A. Configure Looker Studio to use a shorter data refresh interval to ensure fresh data is always displayed.

B. Create a materialized view in BigQuery that pre-calculates the aggregations and filters used in the Looker Studio dashboards. **Most Voted**

C. Implement row-level security in BigQuery to restrict data access and reduce the amount of data processed by the queries.

D. Use BigQuery BI Engine to accelerate query performance by caching frequently accessed data. **Most Voted**

Correct Answer: BD

Community vote distribution

D (100%)

Comments

5523042 1 week, 6 days ago

Selected Answer: D

It is D

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #321 Topic 1

Your company has data assets across multiple Cloud Storage buckets and BigQuery datasets containing raw and processed data. The requirement is to establish a unified data governance framework that allows for centralized metadata discovery, data quality monitoring, and consistent security policy application across these various data stores without physically moving or duplicating the data. You need to implement a solution to achieve this federated governance. What should you do?

A. 1. Export metadata out of Dataplex Universal Catalog by running a metadata export job.
2. Implement Dataproc Metastore to manage table schemas and Apache Hive metastore for metadata discovery.
3. Manage security using a combination of BigQuery row-level security and Cloud Storage policies.

B. 1. Use Dataplex to organize the BigQuery datasets and Cloud Storage buckets into lakes and zones.
2. Use Dataplex for automated metadata discovery, centralized security policy management, data profiling, and data quality tasks. Most Voted

C. 1. Deploy a centralized Cloud SQL database to store metadata extracted from BigQuery and Cloud Storage using custom scripts.
2. Integrate the database with Looker Studio for data discovery and visualization.
3. Implement a custom policy engine using Cloud Run functions triggered by changes in IAM policies to enforce consistent security across projects.

D. 1. Create a Looker Studio dashboard on BigQuery INFORMATION_SCHEMA views to visualize and monitor data quality.
2. Manage security using IAM policies at the project level, supplemented by BigQuery authorized views for granular access control.

Correct Answer: B

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #322 Topic 1

Your company is planning to migrate a large on-premises data warehouse to BigQuery. The data is currently stored in a proprietary, vendor-specific format. You need to perform a batch migration of this data to BigQuery. What should you do?

A. Use the bq command-line tool to load the data directly from the on-premises data warehouse.

B. Export the data to CSV files, upload the files to Cloud Storage, then load the files into BigQuery. Most Voted

C. Use the BigQuery Data Transfer Service.

D. Use Datastream to replicate the data in real time.

Correct Answer: B

https://zhangxuetu.online

Question #323 Topic 1

Your company uses separate Google Cloud projects for development, staging, and production. Developers need edit access in the development project, read-only access in the staging project, and no access in the production project. You need an effective and manageable solution to assign and enforce these permissions according to Google-recommended practices. What should you do?

A. Create separate VPC Service Controls perimeters for each project and use access levels.

B. Grant individual developers specific IAM roles directly on each project.

C. Configure firewall rules within each project's VPC to block traffic from unauthorized developer machines.

D. Create Google Groups for access levels, assign developers to groups, and grant the groups the appropriate IAM roles on each project. Most Voted

Correct Answer: D

Question #324 Topic 1

Your financial services company has a critical daily reconciliation process that involves several distinct steps: fetching data from an external SFTP server, decrypting the files, loading them into Cloud Storage, and finally running a series of BigQuery SQL transformations. Each step has strict dependencies, and the entire process should notify you if not completed by 7:00 AM. Manual intervention for failures is costly and delays compliance reporting. You need a highly observable and robust solution that supports easy re-runs of individual steps if errors occur. What should you do?

A. Develop a Cloud Composer DAG that includes a single PythonOperator to execute a Python script that runs each step sequentially, incorporating error handling and retries. Upload the scripts to a Cloud Composer environments DAGs folder, and configure it to run daily.

B. Implement a Cloud Composer DAG, with each step defined as a separate task using appropriate Airflow operators, and schedule the DAG for daily execution. Most Voted

C. Define a Cloud Composer DAG to orchestrate the SFTP fetch and decryption steps, and then use Cloud Scheduler to trigger a separate Dataflow job that handles the Cloud Storage load and BigQuery transformations and schedule to run daily.

D. Create a Cloud Composer DAG that includes a single BashOperator to execute a top-level shell script, which in turn calls individual scripts for each pipeline step. Upload the scripts to a Cloud Composer environment's DAGs folder, and configure it to run daily.

Correct Answer: B

掌学兔
ZHANGXUETU
[202610]

Question #325 Topic 1

Your large e-commerce company uses BigQuery for its data warehousing needs. They have two primary types of workloads:
• Analyst dashboards: Interactive queries run by business analysts throughout the day, requiring low latency for real-time insights.
• ETL jobs: Large, complex batch queries that load and transform data overnight. These jobs are cost-sensitive and can tolerate longer execution times, and they do not have the same performance requirements as the dashboards.

Analysts are reporting inconsistent performance of their dashboards, particularly during peak business hours. You need to ensure reliable performance for both workload types while optimizing costs. What should you do?

A. Use BigQuery editions to create an autoscaling reservation with dedicated slots for analyst dashboards, while running ETL jobs with a separate reservation or using the on-demand billing model. **Most Voted**

B. Implement a custom autoscaling solution using Cloud Run functions to dynamically adjust available BigQuery slots based on time of day.

C. Migrate all ETL jobs to Cloud SQL to reduce BigQuery's workload.

D. Run all analyst dashboard queries as interactive queries and configure ETL jobs to run as batch queries.

Correct Answer: A

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #326 Topic 1

Your team runs a complex analytical query daily that processes terabytes of data. Recently, after running for 20 minutes, the query fails with a “Resources exceeded” error. You need to resolve this issue. What should you do?

A. Move from BigQuery on-demand to slot reservations. **Most Voted**

B. Analyze the SQL syntax for errors.

C. Increase the maximum table size limit.

D. Increase your project’s BigQuery API request quota.

Correct Answer: A

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #327 **Topic 1**

Your company wants to implement a Retrieval-Augmented Generation (RAG) system to allow employees to query an extensive knowledge base of internal documents, such as policy manuals and project reports. You need to prepare this unstructured text for embedding to be used in the RAG system. What should you do to ensure the system can retrieve the most relevant information?

A. Use Cloud Data Loss Prevention (Cloud DLP) to scan and redact sensitive information within the documents before processing.

B. Store the documents as compressed files in a traditional relational database to enable more efficient storage and retrieval.

C. Convert the unstructured documents into high-dimensional numerical vectors that capture the semantic meaning and relationships of the text. Most Voted

D. Index each word from the documents into a search engine to enable keyword-based search.

Correct Answer: C

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #328 Topic 1

You need to load a dataset with multiple terabytes of clickstream data into BigQuery. The data arrives each day as compressed JSON files in a Cloud Storage bucket. You need a low-cost, programmatic, and scalable solution to load the data into BigQuery. What should you do?

A. Create an external table in BigQuery pointing to the Cloud Storage bucket and run the INSERT INTO ... SELECT * FROM external_table command. **Most Voted**

B. Use the BigQuery Data Transfer Service from Cloud Storage. **Most Voted**

C. Create a Cloud Run function to run a Python script to read and parse each JSON file, and use the BigQuery streaming insert API.

D. Use Cloud Data Fusion to create a pipeline to load the JSON files into BigQuery.

Correct Answer: A

Community vote distribution



Comments

hola_prep 1 month, 1 week ago

Selected Answer: A

why option A is correct Low-cost Scalable Programmatic

NickForDiscussions 2 months, 1 week ago

Selected Answer: B

Should be B. When you use BigQuery Data Transfer Service you don't pay for the load job, you only pay for the storage in BigQuery. If you go with Option A you are paying for the query plus the storage.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #329 Topic 1

Your Cloud Storage data lake has raw, processed, and historical data in different buckets. Data older than two years is rarely accessed, and all data must be retained for no longer than seven years. You are concerned about rising storage costs. How should you control costs for the historical data bucket?

A. Write a script on a Compute Engine instance, triggered daily by Cloud Scheduler, to scan all objects and delete any older than seven years.

B. Configure an Object Lifecycle Management rule to transition objects older than two years to the Archive storage class and eventually delete them after seven years. Most Voted

C. Enable the Autoclass feature on your Cloud Storage buckets and select Opt-in to object transitions to Coldline and Archive storage classes.

D. Replicate the buckets to a different region with lower storage costs and configure an Object Lifecycle Management rule to delete objects after seven years.

Correct Answer: B

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #330 Topic 1

Your data science team needs to perform interactive SQL queries on large datasets stored in Apache Parquet format within a Cloud Storage bucket. The team is familiar with Apache Hive and wants to leverage existing HiveQL queries. You need to provide an environment for the team to run their interactive HiveQL queries directly against the data in Cloud Storage. You want to keep operational overhead to a minimum. What should you do?

A. Install and configure an Apache Hadoop and Hive cluster manually on a group of Compute Engine instances.

B. Load the Parquet data into a BigQuery native table and use the BigQuery Connector for Hive to run the queries.

C. Configure BigQuery with an external table definition pointing to the Parquet files.

D. Deploy a Dataproc cluster with Hive services enabled. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

hola_prep 1 month, 1 week ago

Selected Answer: D

HiveQL compatibility Direct access to Parquet in Cloud Storage: Interactive queries: Low operational overhead: Dataproc is a managed service. You don’t need to manually install, configure, or maintain Hadoop/Hive clusters Scalability: Dataproc clusters can scale up or down as needed

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #331 Topic 1

You analyze user clickstream data to personalize content recommendations. The data arrives continuously and needs to be processed with low latency, including transformations such as sessionization (grouping clicks by user within a time window) and aggregation of user activity. You need to identify a scalable solution to handle millions of events each second and be resilient to late-arriving data. What should you do?

A. Use Firebase Realtime Database for ingestion and storage, and Cloud Run functions for processing and analytics.

B. Use Cloud Data Fusion for ingestion and transformation, and Cloud SQL for storage and analytics.

C. Use Pub/Sub for ingestion, Dataflow with Apache Beam for processing, and BigQuery for storage and analytics. Most Voted

D. Use Cloud Storage for ingestion, Dataproc with Apache Spark for batch processing, and BigQuery for storage and analytics.

Correct Answer: C

Community vote distribution

C (100%)

Comments

hola_prep 1 month, 1 week ago

Selected Answer: C

C. Use Pub/Sub for ingestion, Dataflow with Apache Beam for processing, and BigQuery for storage and analytics.

https://zhangxuetu.online

Question #332 Topic 1

You are designing BigQuery tables for large volumes of clickstream event data. Your data analyst team will most frequently query by specific event date ranges and filter by the user ID UUID. You want to optimize table structure for query cost and performance. What should you do?

A. Partition the table by the user ID column and cluster the table by the event date column.

B. Create an ingestion-time partitioned table and cluster it by the user ID column.

C. Cluster the table by both the event date and the user ID columns.

D. Partition the table by the event date column and cluster the table by user ID column. Most Voted

Correct Answer: D

掌学兔
ZHANGXUETU
[202610]

Question #333 Topic 1

You are building a Dataflow pipeline to ingest customer feedback. Before loading to your data warehouse, you must validate email addresses and enrich unstructured comment strings with a generative AI sentiment classification. Invalid records need to be routed for manual review. How should you implement this pipeline?

A. Configure the data source system to pre-validate data before sending it to Dataflow and use the BigQuery ML.GENERATE_TEXT command to assign a sentiment score.

B. Use Dataflow to load all data into BigQuery and execute a SQL MERGE statement to flag invalid records and BigQuery ML to assign sentiment scores.

C. After the Dataflow load completes, execute a Cloud Run function to scan for invalid entries and call Vertex AI to assign sentiment scores.

D. Apply a ParDo transform in Dataflow to validate each element, use a RunInference transform to assign sentiment scores, and use side outputs to route valid/invalid records. Most Voted

Correct Answer: D

https://zhangxuetu.online

[202610]

Question #334 Topic 1

You need to orchestrate a pipeline with several Google Cloud services: a batch Dataflow job, then a BigQuery query job followed by a Vertex AI batch prediction. The logic is sequential. You want a lightweight, serverless orchestration solution with minimal operational overhead. What service should you use?

A. Select Cloud Composer.

B. Select Compute Engine with cron.

C. Select Dataproc with Apache Oozie.

D. Select Cloud Workflows. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

EIM_Exam_Topics 1 week ago

Selected Answer: A

Cloud Composer is better suited to chain GCP services that handle data processing activities. Cloud Workflows is designed for HTTPS based services, and chaining together microservices. Reference: https://docs.cloud.google.com/workflows/docs/choose-orchestration

hola_prep 1 month, 1 week ago

Selected Answer: D

For a lightweight, serverless, low-overhead orchestration solution across Dataflow, BigQuery, and Vertex AI, the best service is Cloud Workflows (Option D).

https://zhangxuetu.online

[202610]

Question #335 Topic 1

You have a data analyst team member who needs to analyze data by using BigQuery. The data analyst wants to create a data pipeline that would load 200 CSV files with an average size of 15MB from a Cloud Storage bucket into BigQuery daily. The data needs to be ingested and transformed before being accessed in BigQuery for analysis. You need to recommend a fully managed, no-code solution for the data analyst. What should you do?

A. Create a Cloud Run function and schedule it to run daily using Cloud Scheduler to load the data into BigQuery.

B. Use the BigQuery Data Transfer Service to load files from Cloud Storage to BigQuery, create a BigQuery job which transforms the data using BigQuery SQL and schedule it to run daily.

C. Build a custom Apache Beam pipeline and run it on Dataflow to load the file from Cloud Storage to BigQuery and schedule it to run daily using Cloud Composer.

D. Create a pipeline by using BigQuery pipelines and schedule it to load the data into BigQuery daily. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

jreale64 1 week, 6 days ago

Selected Answer: D

BigQuery Pipelines is fully managed, serverless, and no-code/low-code, designed explicitly for analysts. It allows your analyst to: Build ingestion pipelines from Cloud Storage into BigQuery Perform transformations inside BigQuery using SQL or point-and-click steps Schedule daily runs Avoid writing code, managing servers, or handling orchestration tools Keep everything within the BigQuery interface analysts already use https://docs.cloud.google.com/bigquery/docs/pipelines-introduction?hl=es-419

mpuche3 1 month, 1 week ago

Selected Answer: D

BigQuery Pipelines is a real, generally available product. It was first released in September 2024 under the name "BigQuery Workflows," then renamed to "BigQuery Pipelines" when it reached General Availability in March 2025.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #336 Topic 1

You are designing a stateful data processing pipeline that reads data from a Cloud Storage bucket and writes transformed data to a BigQuery table. The pipeline must be highly available and resilient to zonal failures within the us-central1 region. You need to configure a Dataflow pipeline ensuring minimal disruption during a zonal outage. What should you do?

A. Launch the Dataflow job with the --region=us-central1 parameter. **Most Voted**

B. Deploy the Dataflow job to a single zone within us-central1 and configure it to use a regional persistent disk to store its state.

C. Deploy the Dataflow job to a single zone within us-central1 and use a multi-regional Cloud Storage bucket to store its state.

D. Launch the Dataflow job with the --zone=us-central1a parameter.

Correct Answer: A

Community vote distribution

A (100%)

Comments

hola_prep 1 month, 1 week ago

Selected Answer: A

To ensure high availability and resilience against zonal failures with minimal disruption, you should deploy the Dataflow job regionally using --region=us-central1 (Option A).

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #337 Topic 1

Your team has created several BigQuery curated datasets containing anonymized industry benchmark data. You want to make these datasets easily discoverable and accessible for querying by external partner companies within their own Google Cloud projects. You need a secure and scalable solution. What should you do?

A. Publish the datasets as listings within BigQuery sharing (Analytics Hub). **Most Voted**

B. Export the datasets to partner-specific Cloud Storage buckets.

C. Create authorized views for each dataset and grant access to each partner.

D. Grant the roles/bigquery.dataViewer IAM role to the partner group email addresses on the datasets.

Correct Answer: A

Community vote distribution

A (100%)

Comments

hola_prep 1 month, 1 week ago

Selected Answer: A

A. Publish the datasets as listings within BigQuery sharing (Analytics Hub).

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #338 Topic 1

Your organization stores highly personal data in BigQuery and needs to comply with strict data privacy regulations. You need to ensure that sensitive data values are rendered unreadable whenever an employee leaves the organization. What should you do?

A. Use column-level access controls with policy tags and revoke viewer permissions when employees leave the organization.

B. Use dynamic data masking and revoke viewer permissions when employees leave the organization.

C. Use customer-managed encryption keys (CMEK) and delete keys when employees leave the organization.

D. Use AEAD functions and delete keys when employees leave the organization. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

DerfelCardarn 2 months, 1 week ago

Selected Answer: D

C is too complex

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #339 Topic 1

You monitor and optimize the BigQuery instance for your team. You notice that a particular daily report that uses a large JOIN operation is consistently slow. You want to examine the query's execution plan to identify potential performance bottlenecks within the JOIN as quickly as possible. What should you do?

A. Use the bg query --dry_run command to review the estimated number of bytes read and review query syntax.

B. Run a query on the INFORMATION_SCHEMA.JOBS_BY_PROJECT view filtering by the job_ID and analyze total_bytes_processed.

C. Review the BigQuery audit logs in Cloud Logging.

D. Leverage BigQuery's Query History view and analyze the execution graph. Most Voted

Correct Answer: D

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #340 Topic 1

Your company stores vital operational sales data in a BigQuery dataset in us-central1. Your company requires a disaster recovery plan to restore this data to us-east1 with a recovery point objective (RPO) of 24 hours and a recovery time objective (RTO) of 4 hours if us-central1 experiences an outage. You need to implement the disaster recovery plan while keeping costs and complexity to a minimum. What should you do?

A. Set up continuous queries and Pub/Sub to stream data changes from BigQuery tables in us-central1 to us-east1.

B. Manually export data to a CSV file in a multi-regional Cloud Storage bucket daily and use bq load to restore to us-east1. Most Voted

C. Configure BigQuery cross-region dataset replication from ns-central1 to us-east1. Most Voted

D. Take daily BigQuery table snapshots in us-central1.

Correct Answer: B

Community vote distribution

C (100%)

Comments

5523042 5 days, 9 hours ago

Selected Answer: C

Es la correcta

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #341 Topic 1

You are integrating your legacy on-premises MySQL database into a new Google Cloud data warehouse. The database contains historical customer loyalty data which needs to be extracted daily and loaded into BigQuery. You want a managed integration solution that is secure and supports incremental loads. What should you do?

A. Implement a data pipeline using Pub/Sub for streaming the data changes from MySQL.

B. Configure a scheduled transfer in BigQuery Data Transfer Service to ingest data from the MySQL database with a scheduled transfer configuration. Most Voted

C. Leverage Database Migration Service for continuous replication of the entire database to Cloud SQL.

D. Use Cloud SQL federated queries to directly query the on-premises MySQL database from BigQuery.

Correct Answer: B

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #342 Topic 1

You are developing a fraud detection model using BigQuery ML. You have a raw transaction dataset and need to create new features such as the average_transaction_amount_last_24_hours and time_since_last_transaction. These features require aggregation and time-window calculations on the existing data. The goal is to ensure that these features are consistently applied during both model training and prediction without manual intervention. You need to prepare these features efficiently for your model. What should you do?

A. Implement a Cloud Run function that triggers on new transactions, calculates the features, and inserts them into a feature store before model serving.

B. Export the BigQuery data to Cloud Storage, perform feature engineering using a custom Python script in a Dataflow job, and then re-import the engineered features into BigQuery.

C. Use the TRANSFORM clause within the CREATE MODEL statement, leveraging SQL functions for aggregations and time-based calculations. Most Voted

D. Create a separate BigQuery table containing pre-computed features using complex SQL queries and join this table with the raw data during model training and serving. Most Voted

Correct Answer: D

Community vote distribution

C (100%)

Comments

NS1425 1 week, 3 days ago

Selected Answer: C

Agree on C, as it ensures same transformation at the training/inference time.

maggiemhanna 3 weeks ago

Selected Answer: C

D is too complex

https://zhangxuetu.online

掌学兔
ZHANGXUETU

[202610]

Question #343 Topic 1

Your organization has thousands of images stored in Cloud Storage. The organization's data analysts need to categorize these images upon analyzing the image content and store the results in a structured format for integration with other BigQuery datasets. You need a solution that allows the analysts to perform this task with the least amount of coding. What should you do?

A. Use the Gemini 2.5 Pro model in Vertex AI to analyze the images and store the results in BigQuery.

B. Use the ML.UNDERSTAND_TEXT function along with object tables to analyze the images, and store the JSON output in a Cloud Storage bucket.

C. Use Vertex AI Vision API to analyze the images and store the results in BigQuery. Most Voted

D. Use the ML.GENERATE_TEXT function along with object tables to analyze the images and store the JSON output in a BigQuery table.

Correct Answer: C

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #344 Topic 1

You are building a system to process financial transactions. The system must handle a high throughput of concurrent user operations and each operation requires low-latency reads and writes to individual records. You need to identify a storage solution that guarantees ACID compliance for the processed transactions. You want a Google Cloud managed service. What should you do?

A. Select Bigtable.

B. Select BigQuery.

C. Select Cloud Storage.

D. Select Spanner. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

jreale64 2 weeks, 4 days ago

Selected Answer: D

D: High performance for concurrent user operations: Capable of handling many operations simultaneously. Low-latency reads and writes to individual records: Optimized for fast transactional operations on specific records. Ensures ACID compliance: This is crucial for financial transactions (Atomicity, Consistency, Isolation, Durability). Google Cloud managed service: A serverless solution with minimal user management.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #345 Topic 1

You are preparing data to serve a sales demand prediction model. The training data undergoes several pre-processing steps, including scaling numerical features and one-hot encoding categorical features. The model is deployed on Vertex AI Endpoints. You need to prevent training-serving skew and ensure accurate predictions in production. You want a solution that is easy to implement.

What should you do?

A. Implement a custom handler within the Vertex AI Endpoint to automatically perform data transformations before the model makes a prediction.

B. Replicate the exact same pre-processing logic in the inference pipeline that was used during model training. **Most Voted**

C. Store the raw, unprocessed data in a separate Cloud Storage bucket exclusively for serving.

D. Ensure the serving data is a smaller, random sample of the training data.

Correct Answer: B

Community vote distribution

B (100%)

Comments

jreale64 2 weeks, 4 days ago

Selected Answer: B

B: Training-serving skew occurs when the data used to train your model differs in its characteristics or processing from the data used for making predictions in production. The problem explicitly mentions preprocessing steps like scaling numerical features and one-hot encoding categorical features. If these transformations are not applied identically during inference as they were during training, the model will receive inconsistent input, leading to inaccurate predictions.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #346 Topic 1

Your retail company is concerned about their BigQuery analytics spend. The company runs several queries that require the use of the same aggregation for the store ID and real-time sales volume. You need to implement the optimal solution that minimizes analytics spend and returns faster results. What should you do?

A. Create a new table from a CSV file with the repeated aggregation for the other queries to reference for faster processing.

B. Create a materialized view to minimize repetitive computations. **Most Voted**

C. Use join acceleration with primary and foreign keys to increase query joining to live data.

D. Leverage partitioning to minimize the number of bytes read.

Correct Answer: B

Community vote distribution

B (100%)

Comments

jreale64 2 weeks, 4 days ago

Selected Answer: B

B: Materialized views in BigQuery store the precomputed results of a query. Unlike logical views, which execute the query every time they are referenced, materialized views store the result set and automatically update as data in the base tables changes.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #347 Topic 1

You are defining the data governance strategy for a new BigQuery table with medical and financial data. You want a scalable solution that ensures the clinical researchers can access patient medical data without financial information, while allowing the accounting team to access only financial data with minimal patient identifiers. What should you do?

A. Implement column-level security policies in BigQuery tables with IAM permissions.

B. Create separate tables for personally identifiable information (PII), financial data, and anonymized medical data. Use IAM permissions to control access to each table.

C. Implement row-level security policies in BigQuery tables with IAM permissions.

D. Create separate datasets with authorized views exposing only approved data. Most Voted

Correct Answer: D

Community vote distribution

A (100%)

Comments

jreale64 2 weeks, 4 days ago

Selected Answer: A

A: It allows you to restrict access to specific columns within a table, even if the user has access to the entire table. You can tag columns containing medical information, those with financial information, and those with patient identifiers. Clinical researchers would be granted access to the policy tags for medical data and minimal necessary identifiers but denied access to financial data. The accounting team would be granted access to the policy tags for financial data and minimal identifiers but denied access to sensitive medical data.

https://zhangxuetu.online

掌学兔
ZHANGXUETU
[202610]

Question #348 Topic 1

You are designing a data lake on Google Cloud to store vast amounts of customer interaction data from various sources, such as websites, mobile apps, and social media. You need to ensure that this data, which arrives in different formats, is consistently cataloged and easy for data analysts to discover and use. You also want to perform basic data quality checks and transformations before the data is consumed by downstream applications. You need an automated and managed data governance solution. What should you do?

A. Use Cloud Storage as the central repository. Use Vertex AI to classify and process the data and perform data quality checks.

B. Stream all the data directly into BigQuery, where it is automatically cataloged and governed.

C. Use Cloud Storage and BigQuery as repositories. Use Dataplex Universal Catalog for metadata discovery, data quality checks, and transformations. **Most Voted**

D. Use Cloud Storage as the central repository. Use a Cloud Run function to catalog, transform the data, and perform data quality checks.

Correct Answer: C

Community vote distribution

C (100%)

Comments

jreale64 2 weeks, 4 days ago

Selected Answer: C

C: Dataplex: It is a fully managed Google Cloud data fabric platform, specifically designed to unify data management, governance, and analytics across a data lake environment. It automates data classification, metadata discovery, data quality, and provides a consistent governance layer.

https://zhangxuetu.online

掌学兔 ZHANGXUETU
[202610]

Question #349 Topic 1

You are using Dataflow to build a streaming data pipeline to analyze user website click activity from Pub/Sub. You need to calculate the number of clicks for each user site visit. A site visit is defined as a period of activity followed by 30 minutes of inactivity for a specific user. What should you do?

A. Use tumbling windows with a 30-minute window.

B. Use hopping windows with a 30-minute window, and a 1-minute period.

C. Use hopping windows with a 30-minute window, and a 30-minute period.

D. Use session windows with a 30-minute gap duration. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

jreale64 2 weeks, 4 days ago

Selected Answer: D

D: Session Windows: These are designed to group data elements that occur together over a period of time, separating those groups by a period of inactivity. The "gap duration" defines how much inactivity is required before the current session window is considered complete.

https://zhangxuetu.online